

A SELF-STUDY BOOK FOR HIRING TESTS

Quant, Aptitude & Reasoning for Placements

Every chapter climbs three tiers of difficulty

- TIER 1** Indian service companies — TCS NQT, Accenture, Infosys, Wipro, Capgemini, Cognizant
- TIER 2** Singapore & Thailand — Grab, Sea/Shopee, GIC, DBS, Agoda, SCB, LINE MAN, Razer
- TIER 3** Global product companies — Google, Amazon, Microsoft, Goldman Sachs, D. E. Shaw, Adobe, Uber

24 chapters · quantitative aptitude · logical reasoning · data interpretation · 5 full mock papers

A NOTE ON WHERE THESE PROBLEMS COME FROM

Every problem in this book is **original**. Each one was written for this book.

Nothing here is copied from any published book, question bank, or exam paper.

No company's actual test paper is reproduced, quoted, or paraphrased. Real hiring tests are private property and are not public documents.

When a problem is labelled, for example, "TCS NQT pattern" or "Grab · pattern", that label means one thing only: the problem was written to match the **publicly documented shape** of that company's test — its topic mix, its usual difficulty, and its time pressure. The label is never a claim that this question, or anything like it, appeared on a real test.

Company names are used only to describe that pattern. This book is not affiliated with, endorsed by, or connected to any company named in it. All trademarks belong to their owners.

Every answer has been worked twice. If you still find a mistake, trust your own working and re-check it — that habit is worth more than any answer key.

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How to use this book

Read this page once. Then start Chapter 1.

1 • What the three tiers mean

Every chapter has the same four blocks. You climb them in order, inside the chapter.

Tier	Companies	What the questions feel like	Time / Q
0 Warm-up	nobody — it is you	Bare mechanics. One idea per question. You are building the reflex, not being tested.	30 s
1 Service	TCS NQT, Accenture, Infosys, Wipro, Capgemini, Cognizant	Direct formula use. Light two-step word problems. Clean numbers. Speed beats depth.	45–75 s
2 SE Asia	Grab, Sea/Shopee, GIC, DBS, Agoda, SCB, LINE MAN, Razer	Two or three steps, mixed topics in one question. Fares, orders, FX, logistics. Often no options given.	90–120 s
3 Product	Google, Amazon, Microsoft, Goldman Sachs, D. E. Shaw, Adobe, Uber	Needs an insight, not recall. Counting arguments, invariants, expectation, bounds, estimation. Often asked aloud in an interview.	3–6 min

Target the tier that matches the companies you are applying to — but always work Tier 1 first, even for a product-company interview. The Tier 3 questions are built on the Tier 1 and Tier 2 ones that come before them in the same chapter.

2 • The five rules

1	Never read a solution before you have attempted the problem. Give it a full, honest try with a pen. A solution you read before trying teaches you nothing — it only feels like learning.
2	If you are stuck for more than the tier's time limit, stop. Write down where you got stuck. Then read the solution and find that exact line.
3	Work every practice set on paper, timed, with the answer key covered. Mark it only after the timer ends.
4	Redo every question you got wrong — three days later, from a blank page. Not the same day. Three days later.
5	Do the arithmetic in your head wherever the appendix tables let you. The calculator is not there on test day, and neither is long division time.

3 • The revision cards

Each chapter ends with a one-page **revision card**: every formula, every shortcut, and the top five traps.

While studying Ignore it. Learn from the worked examples instead.

After finishing a chapter Cover the card. Write it out from memory. Compare. The gaps are your revision list.

Last week before a test Read only the 24 cards — about 40 minutes for the whole book. Then attempt one mock paper per day from Chapter 24.

Last hour Cards of your three weakest chapters, plus the speed-arithmetic tables at the back. Nothing else.

4 • The 12-week plan

Two chapters a week. About 90 minutes a day, six days a week. Day 7 is the checkpoint.

Week	Chapters	Daily target	Checkpoint (day 7)
1	1, 2	Warm-up + Tier 1 of one chapter	Redo every Tier 1 question you got wrong
2	3, 4	Warm-up + Tier 1, then Tier 2	Mixed set of Ch 1–4, timed, 60 s/Q
3	5, 6	Tier 1 and Tier 2 fully	Write out the Ch 1–6 revision cards from memory
4	7, 8	Tier 1, Tier 2, then Tier 3	Mixed sets of Ch 5–8, timed
5	9, 10	All tiers, one chapter per 3 days	Tier 3 questions of Ch 7–10, no time limit
6	11, 12	All tiers	Half-book check: 30 mixed questions from Ch 1–12
7	13, 14	Warm-up + Tier 1 daily, Tier 2 on alternate days	Mixed set of Ch 13–14
8	15, 16	One full puzzle set per day	Re-solve two seating puzzles from a blank grid
9	17, 18	Tier 1 and Tier 2 daily	Mixed sets of Ch 15–18, timed
10	19, 20	All tiers	Revision cards of Ch 13–20, written from memory
11	21, 22	One full DI set per day, timed	Mixed sets of Ch 21–22, strict timing
12	23, 24	One mock paper per day from Ch 24	Score every paper. List your three weakest chapters and redo their cards.

5 • The 4-week crash plan

Use this only if your test is inside a month. It skips nothing — it just cuts the practice volume. About 3 hours a day.

Week	Chapters	Daily target	Checkpoint (day 7)
1	1–8	Worked examples of Tier 1 and Tier 2 only. Skip Tier 3 for now. Two chapters a day, days 1–4; practice sets on days 5–6.	Mixed sets of Ch 1–8, timed at 60 s/Q
2	9–14	Same method. Add the Tier 3 examples of Ch 9, 10 and 11 if you are targeting a product company.	Revision cards of Ch 1–14, written from memory
3	15–22	Reasoning and DI. One full puzzle set and one full DI set every day, always timed.	Mixed sets of Ch 15–22
4	23, 24	Data sufficiency on days 1–2. Then one mock paper per day, days 3–6, scored properly.	Redo the two lowest-scoring papers. Read all 24 revision cards.

TRAP

The crash plan works only if you **time everything from day one**. Untimed practice will make you feel ready and then fail you on the clock.

Numbers & Number System

Divisibility, HCF–LCM, remainders, digits, factorials

What you need to know

WHAT YOU NEED TO KNOW

Divisibility tests.

By	Test	By	Test
2	last digit even	9	digit sum divisible by 9
3	digit sum divisible by 3	10	last digit 0
4	last 2 digits divisible by 4	11	(sum of odd-place digits) – (sum of even-place digits) divisible by 11
5	last digit 0 or 5	12	divisible by 3 and by 4
6	divisible by 2 and by 3	25	last 2 digits divisible by 25
7	drop last digit, subtract twice it, repeat	8	last 3 digits divisible by 8

Factors. If $N = p^a \cdot q^b \cdot r^c$ (distinct primes p, q, r):

- Number of factors = $(a + 1)(b + 1)(c + 1)$
- Sum of factors = $\left(\frac{p^{a+1}-1}{p-1}\right) \cdot \left(\frac{q^{b+1}-1}{q-1}\right) \cdot \left(\frac{r^{c+1}-1}{r-1}\right)$
- Product of all factors = $N^{d/2}$, where d = number of factors
- Odd factors: count with the power of 2 removed. Even factors = total – odd.
- Factors that are perfect squares: count only even exponents 0, 2, 4, ...

HCF and LCM.

- For two numbers: $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$
- HCF of fractions = $\frac{\text{HCF of numerators}}{\text{LCM of denominators}}$
- LCM of fractions = $\frac{\text{LCM of numerators}}{\text{HCF of denominators}}$
- Largest number dividing a, b, c leaving the **same** remainder = $\text{HCF}(b - a, c - b, c - a)$
- Largest number dividing a, b leaving remainders r_1, r_2 = $\text{HCF}(a - r_1, b - r_2)$
- Smallest number leaving remainder r with each of a, b, c = $\text{LCM}(a, b, c) + r$

Remainders.

- $(a \times b) \bmod n = ((a \bmod n) \times (b \bmod n)) \bmod n$
- Use negatives: if $a \equiv -1 \pmod{n}$ then $a^{\text{even}} \equiv 1$ and $a^{\text{odd}} \equiv -1 \equiv n - 1$
- Fermat: if p is prime and p does not divide a , then $a^{p-1} \equiv 1 \pmod{p}$
- Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ when $\text{HCF}(a, n) = 1$; $\varphi(n) = n \prod \left(1 - \frac{1}{p}\right)$

Unit digit (cyclicity).

Last digit	0, 1, 5, 6	4, 9	2	3, 7	8
Cycle length	1	2	4	4	4

Rule: divide the power by 4. If remainder is 1, 2, 3 take that term of the cycle. If remainder is **0**, take the **4th** term.

Last two digits.

- Ends in 1: tens digit of answer = (tens digit of base $\times$ unit digit of power) $\bmod 10$; unit digit is 1.

- $2^{20} \equiv 76, 76^k \equiv 76 \pmod{100}$ for every $k \geq 1$.
- $7^4 = 2401 \rightarrow$ last two digits of 7^n repeat with period 4.

Factorials.

- Highest power of a prime p in $n! = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \dots$
- Trailing zeros of $n! = \left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{25} \right\rfloor + \left\lfloor \frac{n}{125} \right\rfloor + \dots$
- For a composite $m = p^\alpha q^\beta$, find each prime power, divide by its exponent, take the minimum.

Base systems. $(d_k \dots d_1 d_0)_b = d_k b^k + \dots + d_1 b + d_0$. To convert a decimal number to base b , divide repeatedly by b and read remainders bottom-up.

Useful counts. Numbers from 1 to N divisible by d is $\left\lfloor \frac{N}{d} \right\rfloor$. Divisible by a or b is $\left\lfloor \frac{N}{a} \right\rfloor + \left\lfloor \frac{N}{b} \right\rfloor - \left\lfloor \frac{N}{\text{LCM}(a,b)} \right\rfloor$.

Warm-up

WARM-UP

build the reflex

Example 1

Is 453,672 divisible by 8?

SOLUTION

Test for 8: look at the last three digits only. Last three digits = 672. $672 \div 8 = 84$, exact. So yes.

Answer: Yes

Example 2

Find the unit digit of 7^{35} .

SOLUTION

Cycle of 7: 7, 9, 3, 1 then repeat. Cycle length 4. $35 \div 4 = 8$ remainder 3. Remainder 3 $\rightarrow$ take the 3rd term of the cycle = 3.

Answer: 3

Example 3

Find the HCF and the LCM of 24 and 36.

SOLUTION

$24 = 2^3 \times 3$ $36 = 2^2 \times 3^2$ HCF = lowest power of each common prime = $2^2 \times 3 = 12$. LCM = highest power of each prime = $2^3 \times 3^2 = 8 \times 9 = 72$. Check: $12 \times 72 = 864$ and $24 \times 36 = 864$. Matches.

Answer: HCF = 12, LCM = 72

Example 4

How many factors does 360 have?

SOLUTION

$360 = 36 \times 10 = (2^2 \times 3^2) \times (2 \times 5) = 2^3 \times 3^2 \times 5^1$. Number of factors = $(3+1)(2+1)(1+1) = 4 \times 3 \times 2 = 24$.

Answer: 24

Example 5

How many zeros are at the end of $30!$?

SOLUTION

Count the 5s. $\lfloor \frac{30}{5} \rfloor = 6$ $\lfloor \frac{30}{25} \rfloor = 1$ $\lfloor \frac{30}{125} \rfloor = 0$, stop. Total = $6 + 1 = 7$.

Answer: 7

Example 6

Find the remainder when 2^{10} is divided by 7.

SOLUTION

$2^3 = 8 = 7 + 1$, so $2^3 \equiv 1 \pmod{7}$. $2^{10} = 2^9 \times 2 = (2^3)^3 \times 2$. $(2^3)^3 \equiv 1^3 = 1$. So $2^{10} \equiv 1 \times 2 = 2 \pmod{7}$.

Answer: 2

Example 7

Write the decimal number 45 in base 2.

SOLUTION

$45 \div 2 = 22$ remainder 1 $22 \div 2 = 11$ remainder 0 $11 \div 2 = 5$ remainder 1 $5 \div 2 = 2$ remainder 1 $2 \div 2 = 1$ remainder 0 $1 \div 2 = 0$ remainder 1 Read remainders bottom-up: 101101. Check: $32 + 0 + 8 + 4 + 0 + 1 = 45$. Correct.

Answer: $(101101)_2$

Example 8

Is 100,001 divisible by 11?

SOLUTION

Digits, from the right: 1, 0, 0, 0, 0, 1. Odd places (1st, 3rd, 5th from right): $1 + 0 + 0 = 1$. Even places (2nd, 4th, 6th): $0 + 0 + 1 = 1$. Difference = $1 - 1 = 0$, and 0 is divisible by 11. So yes. Check: $11 \times 9091 = 100,001$.

Answer: Yes

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

What is the smallest number that must be added to 5,432 to make it divisible by 9?

(a) 2

(b) 4

(c) 5

(d) 7

SOLUTION

Digit sum of 5432 = $5 + 4 + 3 + 2 = 14$. The next multiple of 9 after 14 is 18. $18 - 14 = 4$. So add 4. New number = 5436, digit sum = $5 + 4 + 3 + 6 = 18$, divisible by 9.

Answer: (b) 4

Example 10 [Infosys pattern]

Three bells ring at intervals of 12, 18 and 30 minutes. They ring together at 9:00 am. At what time do they next ring together?

SOLUTION

They meet again after $\text{LCM}(12, 18, 30)$ minutes. $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $30 = 2 \times 3 \times 5$. $\text{LCM} = 2^2 \times 3^2 \times 5 = 4 \times 9 \times 5 = 180$ minutes. 180 minutes = 3 hours. 9 : 00 am + 3 h = 12 : 00 noon.

Answer: 12:00 noon

Example 11 [Accenture pattern]

Find the largest number that divides 400 and 556 leaving remainders 4 and 6 respectively.

SOLUTION

If the number divides 400 leaving remainder 4, it divides $400 - 4 = 396$ exactly. If it divides 556 leaving remainder 6, it divides $556 - 6 = 550$ exactly. So we want $\text{HCF}(396, 550)$. $396 = 2^2 \times 3^2 \times 11$, $550 = 2 \times 5^2 \times 11$. Common primes: 2^1 and 11^1 . $\text{HCF} = 2 \times 11 = 22$. Check: $400 = 22 \times 18 + 4$. And $556 = 22 \times 25 + 6$. Both correct.

Answer: 22

SHORTCUT

“Leaves remainder r ” $\rightarrow$ subtract r first, then take the HCF. “Leaves the **same** unknown remainder” $\rightarrow$ take the HCF of the **differences** of the numbers. You never need to know the remainder.

Example 12 [Wipro pattern]

How many factors of 1080 are odd? How many are even?

SOLUTION

$1080 = 8 \times 135 = 2^3 \times (27 \times 5) = 2^3 \times 3^3 \times 5^1$. Total factors = $(3+1)(3+1)(1+1) = 4 \times 4 \times 2 = 32$. Odd factors have no 2 in them. Remove the 2s: odd part is $3^3 \times 5$. Odd factors = $(3+1)(1+1) = 4 \times 2 = 8$. Even factors = $32 - 8 = 24$.

Answer: 8 odd, 24 even

Example 13 [TCS NQT pattern]

Find the unit digit of $137^{59} \times 242^{43}$.

SOLUTION

Only the last digits matter: $7^{59} \times 2^{43}$. Step 1: 7^{59} . Cycle of 7 is 7, 9, 3, 1 (length 4). $59 \div 4 = 14$ remainder 3 $\rightarrow$ 3rd term = 3. Step 2: 2^{43} . Cycle of 2 is 2, 4, 8, 6 (length 4). $43 \div 4 = 10$ remainder 3 $\rightarrow$ 3rd term = 8. Step 3: multiply the unit digits: $3 \times 8 = 24$, unit digit 4.

Answer: 4

TRAP

Cyclicity remainder 0 does **not** mean “take the 0th term”. It means the power lands exactly at the **end** of a cycle, so take the **4th** term. For 2^{40} : $40 \div 4$ gives remainder 0, so the unit digit is 6, not 2.

Example 14 [Capgemini pattern]

Find the remainder when 2^{89} is divided by 7.

SOLUTION

$2^3 = 8$, and $8 = 7 \times 1 + 1$, so $2^3 \equiv 1 \pmod{7}$. Write 89 in terms of 3: $89 = 3 \times 29 + 2$. $2^{89} = (2^3)^{29} \times 2^2$. $(2^3)^{29} \equiv 1^{29} = 1$. So $2^{89} \equiv 1 \times 4 = 4 \pmod{7}$.

Answer: 4

Example 15 [Cognizant pattern]

How many zeros does 125! end with?

SOLUTION

Count the number of 5s in 125!. $\left\lfloor \frac{125}{5} \right\rfloor = 25$, $\left\lfloor \frac{25}{5} \right\rfloor = 5$, $\left\lfloor \frac{5}{5} \right\rfloor = 1$, $\left\lfloor \frac{1}{5} \right\rfloor = 0$, stop. Total = $25 + 5 + 1 = 31$. There are far more 2s than 5s, so 5s decide the answer.

Answer: 31

TRAP

A common slip is to answer $\left\lfloor \frac{125}{5} \right\rfloor = 25$ and stop. Every multiple of 25 gives an **extra** 5, and 125 gives one more. Keep dividing until the quotient is 0.

Example 16 [Infosys pattern]

Find the HCF and the LCM of $\frac{3}{4}$, $\frac{9}{10}$ and $\frac{6}{7}$.

SOLUTION

HCF of fractions = HCF of numerators $\div$ LCM of denominators. $\text{HCF}(3, 9, 6) = 3$. $\text{LCM}(4, 10, 7)$: $4 = 2^2$, $10 = 2 \times 5$, $7 = 7$. $\text{LCM} = 2^2 \times 5 \times 7 = 140$. So $\text{HCF} = \frac{3}{140}$. LCM of fractions = LCM of numerators $\div$ HCF of denominators. $\text{LCM}(3, 9, 6) = 18$. $\text{HCF}(4, 10, 7) = 1$. So $\text{LCM} = \frac{18}{1} = 18$.

Answer: HCF = $\frac{3}{140}$, LCM = 18

TRAP

Do not swap the two rules. HCF of fractions has HCF **on top**. LCM of fractions has LCM **on top**. The answer must satisfy: HCF is the smallest of the three or smaller, LCM is the largest or larger. Use that as your check.

Example 17 [TCS NQT pattern]

Find the smallest number which, when divided by 12, 15 and 18, leaves remainder 7 in each case.

SOLUTION

First find the smallest number divisible by all three. $12 = 2^2 \times 3$, $15 = 3 \times 5$, $18 = 2 \times 3^2$. $\text{LCM} = 2^2 \times 3^2 \times 5 = 180$. Now add the common remainder: $180 + 7 = 187$. Check: $187 = 12 \times 15 + 7$. $187 = 15 \times 12 + 7$. $187 = 18 \times 10 + 7$. All correct.

Answer: 187

Example 18 [Accenture pattern]

Convert $(1101101)_2$ to decimal, then write that decimal number in base 8.

SOLUTION

Step 1 — binary to decimal. Place values from the right: 1, 2, 4, 8, 16, 32, 64. Digits from the right: 1, 0, 1, 1, 0, 1, 1. $= 1 \times 1 + 0 \times 2 + 1 \times 4 + 1 \times 8 + 0 \times 16 + 1 \times 32 + 1 \times 64 = 1 + 0 + 4 + 8 + 0 + 32 + 64 = 109$. Step 2 — decimal to base 8. $109 \div 8 = 13$ remainder 5, $13 \div 8 = 1$ remainder 5, $1 \div 8 = 0$ remainder 1. Read bottom-up: 155. Check: $1 \times 64 + 5 \times 8 + 5 = 64 + 40 + 5 = 109$. Correct.

Answer: 109 in decimal; $(155)_8$

SHORTCUT

Binary to octal in one move: group the binary digits in 3s from the right. $1101101 \rightarrow 1 \mid 101 \mid 101 \rightarrow 1, 5, 5 \rightarrow (155)_8$. No decimal step needed.

Example 19 [Wipro pattern]

Find the unit digit of $1! + 2! + 3! + \dots + 100!$.

SOLUTION

$5! = 120$ ends in 0. Every factorial after that contains both a 2 and a 5, so all of $5!, 6!, \dots, 100!$ end in 0. They add nothing to the unit digit. So only the first four matter: $1! = 1$ $2! = 2$ $3! = 6$ $4! = 24$ $1 + 2 + 6 + 24 = 33$. Unit digit = 3.

Answer: 3**Example 20** [Capgemini pattern]

The four-digit number $523x$ is divisible by 6. Find all possible digits x .

SOLUTION

Divisible by 6 means divisible by 2 **and** by 3. By 2: x must be even, so $x \in \{0, 2, 4, 6, 8\}$. By 3: digit sum = $5 + 2 + 3 + x = 10 + x$ must be a multiple of 3. $10 + x \in \{12, 15, 18\} \rightarrow x \in \{2, 5, 8\}$. Both conditions: $x \in \{2, 8\}$. Check: $5232 \div 6 = 872$. $5238 \div 6 = 873$. Both exact.

Answer: $x = 2$ or $x = 8$ **PRACTICE · TIER 1** target: 60 s/Q

- Find the unit digit of 43^{17} .
- Find the HCF of 96 and 144.
- Find the LCM of 16, 24 and 40.
- How many factors does 540 have?
- How many zeros does $60!$ end with?
- What is the smallest number that must be subtracted from 3, 105 to make it divisible by 11?
- Find the remainder when 5^{40} is divided by 6.
- Two numbers are in the ratio 3 : 4 and their LCM is 84. Find the two numbers and their HCF.
- Find the sum of all factors of 200.
- Convert $(237)_8$ to decimal.
- Find the least number which when divided by 8, 12 and 16 leaves remainder 3 in each case.
- For which digits x is the four-digit number $234x$ divisible by 6?
- Find the unit digit of $24^{15} \times 33^{12}$.
(a) 2 (b) 4 (c) 6 (d) 8
- Find the largest four-digit number divisible by 36.
- Find the HCF of $\frac{2}{3}$, $\frac{8}{9}$ and $\frac{16}{81}$.

ANSWER KEY

3. Cycle of 3 is 3, 9, 7, 1; $17 \div 4$ leaves remainder 1, so take the 1st term.
48. $96 = 2^5 \times 3$, $144 = 2^4 \times 3^2$; HCF = $2^4 \times 3 = 48$.
240. $16 = 2^4$, $24 = 2^3 \times 3$, $40 = 2^3 \times 5$; LCM = $2^4 \times 3 \times 5 = 240$.
24. $540 = 2^2 \times 3^3 \times 5$, so $(2+1)(3+1)(1+1) = 3 \times 4 \times 2 = 24$.
14. $\left\lfloor \frac{60}{5} \right\rfloor = 12$, $\left\lfloor \frac{60}{25} \right\rfloor = 2$; $12 + 2 = 14$.
3. Alternating sum from the right = $5 - 0 + 1 - 3 = 3$, so the remainder is 3. $3105 - 3 = 3102 = 11 \times 282$.
1. $5 \equiv -1 \pmod{6}$, and the power 40 is even, so $(-1)^{40} = 1$.
- 21 and 28; HCF = 7. Let the numbers be $3k$ and $4k$; then HCF = k and LCM = $12k = 84$, so $k = 7$.
465. $200 = 2^3 \times 5^2$; sum = $(1 + 2 + 4 + 8)(1 + 5 + 25) = 15 \times 31 = 465$.
159. $2 \times 64 + 3 \times 8 + 7 = 128 + 24 + 7 = 159$.
51. LCM(8, 12, 16) = 48; $48 + 3 = 51$.
- $x = 0$ or $x = 6$. Digit sum $9 + x$ must be a multiple of 3, giving $x \in \{0, 3, 6, 9\}$; x must also be even.
- (b) 4. 4^{15} : cycle 4, 6, odd power gives 4. 3^{12} : $12 \div 4$ leaves remainder 0, so take the 4th term, 1. Then $4 \times 1 = 4$. Choosing (c) 6 means you used the even-power value of 4; choosing (a) 2 means you multiplied the bases' unit digits $4 \times 3 = 12$ and stopped; choosing (d) 8 means you used the cycle of 2 (2, 4, 8, 6) instead of the cycle of 4.

14. $9972.9999 \div 36 = 277$ remainder 27, so $277 \times 36 = 9972$.
 15. $\frac{2}{81}$. HCF of numerators = 2; LCM of denominators = $\text{LCM}(3, 9, 81) = 81$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A fleet hub services three vehicle groups on fixed cycles: every 36 days, every 48 days and every 60 days. All three were serviced together on day 0. In the next 730 days, how many more times are all three serviced on the same day?

SOLUTION

All three coincide after $\text{LCM}(36, 48, 60)$ days. $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$, $60 = 2^2 \times 3 \times 5$. Take the highest power of each prime: $2^4 \times 3^2 \times 5 = 16 \times 9 \times 5 = 16 \times 45 = 720$. So they coincide on day 720, day 1440, day 2160, and so on. Inside 730 days only day 720 qualifies.

Answer: Once, on day 720

Example 22 [Shopee · pattern]

Order IDs run from 1 to 2000. How many IDs are divisible by 3 or by 5, but **not** by 15?

SOLUTION

Step 1 — divisible by 3: $\lfloor \frac{2000}{3} \rfloor = 666$. Step 2 — divisible by 5: $\lfloor \frac{2000}{5} \rfloor = 400$. Step 3 — divisible by both, i.e. by 15: $\lfloor \frac{2000}{15} \rfloor = 133$. Step 4 — divisible by 3 or 5 = $666 + 400 - 133 = 933$. Step 5 — remove the ones divisible by 15: $933 - 133 = 800$.

Answer: 800

SHORTCUT

“By 3 or 5 but not by 15” = (count by 3) + (count by 5) $- 2 \times$ (count by 15). Here $666 + 400 - 266 = 800$. The multiples of 15 are subtracted twice: once to fix the double count, once to throw them out.

Example 23 [DBS · pattern]

Find the remainder when $1234 \times 5678 \times 9101$ is divided by 7.

SOLUTION

Reduce each factor first. That keeps the numbers small. $1234 \div 7: 7 \times 176 = 1232$, remainder 2. $5678 \div 7: 7 \times 811 = 5677$, remainder 1. $9101 \div 7: 7 \times 1300 = 9100$, remainder 1. So the product $\equiv 2 \times 1 \times 1 = 2 \pmod{7}$.

Answer: 2

Example 24 [GIC · pattern]

Find the highest power of 12 that divides $100!$.

SOLUTION

$12 = 2^2 \times 3$. We need both pieces. Power of 2 in $100!$: $\lfloor \frac{100}{2} \rfloor = 50$, $\lfloor \frac{100}{4} \rfloor = 25$, $\lfloor \frac{100}{8} \rfloor = 12$, $\lfloor \frac{100}{16} \rfloor = 6$, $\lfloor \frac{100}{32} \rfloor = 3$, $\lfloor \frac{100}{64} \rfloor = 1$. Total = $50 + 25 + 12 + 6 + 3 + 1 = 97$. Power of 3 in $100!$: $\lfloor \frac{100}{3} \rfloor = 33$, $\lfloor \frac{100}{9} \rfloor = 11$, $\lfloor \frac{100}{27} \rfloor = 3$, $\lfloor \frac{100}{81} \rfloor = 1$. Total = $33 + 11 + 3 + 1 = 48$. Each 12 needs **two** 2s: from 97 twos we get $\lfloor \frac{97}{2} \rfloor = 48$ copies. Each 12 needs **one** 3: from 48 threes we get 48 copies. Answer is the smaller: $\min(48, 48) = 48$.

Answer: 12^{48}

TRAP

Do not stop at “power of 2 is 97, power of 3 is 48, so answer 48” by luck. Divide each prime count by its exponent in the composite **first**. For the highest power of 8 in $100!$ the answer is $\lfloor \frac{97}{3} \rfloor = 32$, not 97.

Example 25 [Agoda · pattern]

Find the last two digits of 7^{2024} .

SOLUTION

Work modulo 100. $7^1 = 7$ $7^2 = 49$ $7^3 = 343 \rightarrow 43$ $7^4 = 43 \times 7 = 301 \rightarrow 01$ So the pattern 07, 49, 43, 01 repeats with period 4. $2024 \div 4 = 506$ remainder 0. Remainder 0 means we land on the 4th term of the cycle, which is 01.

Answer: 01

Example 26 [SCB · pattern]

How many factors of 21,600 are perfect squares?

SOLUTION

Factorise. $21600 = 216 \times 100$. $216 = 6^3 = 2^3 \times 3^3$. $100 = 2^2 \times 5^2$. So $21600 = 2^{3+2} \times 3^3 \times 5^2 = 2^5 \times 3^3 \times 5^2$. A factor is a perfect square only if **every** exponent in it is even. For 2: choose from 0, 2, 4 $\rightarrow$ 3 ways. For 3: choose from 0, 2 $\rightarrow$ 2 ways. For 5: choose from 0, 2 $\rightarrow$ 2 ways. Total = $3 \times 2 \times 2 = 12$.

Answer: 12

Example 27 [LINE MAN · pattern]

A Bangkok warehouse holds 1,344 packets of rice, 2,016 packets of oil and 1,680 packets of sauce. They must go into identical crates, each crate holding one product only, with no packet left over and no crate part-filled. What is the largest crate size, and how many crates are needed?

SOLUTION

Crate size must divide all three counts, so take the HCF. $1344 = 2^6 \times 3 \times 7$ (since $1344 = 64 \times 21$) $2016 = 2^5 \times 3^2 \times 7$ (since $2016 = 32 \times 63$) $1680 = 2^4 \times 3 \times 5 \times 7$ (since $1680 = 16 \times 105$) Lowest power of each common prime: $2^4, 3^1, 7^1$. (5 is missing from two of them.) HCF = $16 \times 3 \times 7 = 336$. Crates needed: rice = $1344 \div 336 = 4$ oil = $2016 \div 336 = 6$ sauce = $1680 \div 336 = 5$ Total = $4 + 6 + 5 = 15$.

Answer: 336 packets per crate; 15 crates

Example 28 [Sea · pattern]

A legacy pricing system stores numbers in an unknown base b . In that system a price is written 144, and its decimal value is 49. Find b , then write 49 in base 6.

SOLUTION

Step 1 — set up the place values. $(144)_b = 1 \times b^2 + 4 \times b + 4$. So $b^2 + 4b + 4 = 49$. The left side is a perfect square: $b^2 + 4b + 4 = (b + 2)^2$. $(b + 2)^2 = 49 \rightarrow b + 2 = 7 \rightarrow b = 5$. Check the digits are legal: the digits used are 1 and 4, both less than 5. Good. Check the value: $(144)_5 = 25 + 20 + 4 = 49$. Correct. Step 2 — write 49 in base 6. $49 \div 6 = 8$ remainder 1 $8 \div 6 = 1$ remainder 2 $1 \div 6 = 0$ remainder 1 Read bottom-up: 121. Check: $1 \times 36 + 2 \times 6 + 1 = 36 + 12 + 1 = 49$. Correct.

Answer: $b = 5$; $49 = (121)_6$

TRAP

In base b , every digit must be **less than** b . If an answer gives $b = 3$ but the number contains the digit 4, the answer is wrong. Always run this check.

Example 29 [Razer · pattern]

Find the remainder when 3^{100} is divided by 13.

SOLUTION

13 is prime and does not divide 3, so Fermat applies: $3^{12} \equiv 1 \pmod{13}$. Write $100 = 12 \times 8 + 4$. $3^{100} = (3^{12})^8 \times 3^4 \equiv 1^8 \times 3^4 = 3^4$. $3^4 = 81$. $81 \div 13$: $13 \times 6 = 78$, remainder 3.

Answer: 3

PRACTICE · TIER 2 target: 100 s/Q

- How many integers from 1 to 500 are divisible by 4 or by 6?
- Find the highest power of 7 that divides $200!$.
- Find the last two digits of 3^{2026} .
- How many factors of 14,400 are perfect squares?
- Find the remainder when 2^{31} is divided by 5.
- In base b , the number 121 equals 144 in decimal. Find b .
- Find the largest number that divides 1,305, 4,665 and 6,905 leaving the same remainder in each case. State that remainder.
- Find the remainder when $1! + 2! + 3! + \dots + 50!$ is divided by 15.
- A shop must pack 1,020 chocolates and 1,632 candies into identical boxes, one product per box, nothing left over. Find the largest box size and the total number of boxes.
- How many zeros are at the end of $1^1 \times 2^2 \times 3^3 \times \dots \times 10^{10}$?
- Find the remainder when 7^{77} is divided by 100.

ANSWER KEY

- 167.** $\left\lfloor \frac{500}{4} \right\rfloor = 125$, $\left\lfloor \frac{500}{6} \right\rfloor = 83$, $\left\lfloor \frac{500}{12} \right\rfloor = 41$; $125 + 83 - 41 = 167$.
- 7³².** $\left\lfloor \frac{200}{7} \right\rfloor = 28$, $\left\lfloor \frac{200}{49} \right\rfloor = 4$, $\left\lfloor \frac{200}{343} \right\rfloor = 0$; $28 + 4 = 32$.
- 29.** Powers of 3 mod 100 repeat with period 20 (since $3^{20} \equiv 01$). $2026 \div 20$ leaves remainder 6, and $3^6 = 729 \rightarrow 29$.
- 16.** $14400 = 2^6 \times 3^2 \times 5^2$. Even exponents: 2 gives 0, 2, 4, 6 (4 ways), 3 gives 0, 2 (2 ways), 5 gives 0, 2 (2 ways); $4 \times 2 \times 2 = 16$.
- 3.** Cycle of 2 mod 5 has length 4; $31 \div 4$ leaves remainder 3, and $2^3 = 8 \equiv 3$.
- 11.** $b^2 + 2b + 1 = 144 \rightarrow (b+1)^2 = 144 \rightarrow b = 11$. Digits 1 and 2 are both below 11, so it is valid.
- 1,120; remainder 185.** Differences: $4665 - 1305 = 3360$, $6905 - 4665 = 2240$, $6905 - 1305 = 5600$. HCF (3360, 2240, 5600) = 1120. Then $1305 = 1120 \times 1 + 185$.
- 3.** From $5!$ onward every term is a multiple of 15, so only $1 + 2 + 6 + 24 = 33$ matters, and $33 \div 15$ leaves remainder 3.
- 204 per box; 13 boxes.** $1020 = 2^2 \times 3 \times 5 \times 17$, $1632 = 2^5 \times 3 \times 17$, HCF = $2^2 \times 3 \times 17 = 204$. Boxes = $5 + 8 = 13$.
- 15.** Power of 5 comes from 5^5 and 10^{10} : $5 + 10 = 15$. Power of 2 comes from $2^2, 4^4, 6^6, 8^8, 10^{10}$: $2 + 8 + 6 + 24 + 10 = 50$. The smaller count wins.
- 7.** The last two digits of 7^n cycle as 07, 49, 43, 01; $77 \div 4$ leaves remainder 1, so the last two digits are 07, i.e. remainder 7.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 30 [Google · pattern]

Show that no factorial ends in exactly 5 zeros. Then find the next number of zeros that is also impossible.

SOLUTION

Insight: the zero-count function $Z(n)$ climbs in steps of 1 most of the time, but it jumps by 2 at every multiple of 25, skipping a value.

$$\text{Let } Z(n) = \left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{25} \right\rfloor + \left\lfloor \frac{n}{125} \right\rfloor + \dots$$

Compute around 25: $Z(24) = \left\lfloor \frac{24}{5} \right\rfloor + \left\lfloor \frac{24}{25} \right\rfloor = 4 + 0 = 4$. $Z(25) = \left\lfloor \frac{25}{5} \right\rfloor + \left\lfloor \frac{25}{25} \right\rfloor = 5 + 1 = 6$. $Z(n)$ never decreases, and it changes only when n crosses a multiple of 5. Between $n = 24$ and $n = 25$ it went straight from 4 to 6. So no n gives $Z(n) = 5$. A factorial ending in exactly 5 zeros does not exist.

Now the next skip. The next multiple of 25 is 50. $Z(49) = \left\lfloor \frac{49}{5} \right\rfloor + \left\lfloor \frac{49}{25} \right\rfloor = 9 + 1 = 10$. $Z(50) = \left\lfloor \frac{50}{5} \right\rfloor + \left\lfloor \frac{50}{25} \right\rfloor = 10 + 2 = 12$. So 11 is skipped too.

The pattern: a value is skipped at every multiple of 25. The skipped values are 5, 11, 17, 23, 29, ..., rising by 6 each time, until 125 is reached (where the jump is 3 and two values are skipped at once).

Answer: 5 is impossible; the next impossible value is 11

Example 31 [Amazon · pattern]

How many integers from 1 to 1000 share no common factor with 1000 other than 1? What is their sum?

SOLUTION

Insight: the count comes from Euler's φ ; the sum comes from pairing k with $1000 - k$, because one is coprime to 1000 exactly when the other is.

Step 1 — the count. $1000 = 2^3 \times 5^3$, so the only primes to avoid are 2 and 5. $\varphi(1000) = 1000 \times (1 - \frac{1}{2}) \times (1 - \frac{1}{5}) = 1000 \times \frac{1}{2} \times \frac{4}{5} = 400$. So 400 integers qualify. (1000 itself is not one of them, since $\text{HCF}(1000, 1000) = 1000$.)

Step 2 — the sum. If $\text{HCF}(k, 1000) = 1$, then any common factor of $1000 - k$ and 1000 would also divide k . So $\text{HCF}(1000 - k, 1000) = 1$ as well. This pairs the 400 numbers into $400 \div 2 = 200$ pairs, each pair adding to 1000. No number is paired with itself, because $k = 500$ is not coprime to 1000. Sum = $200 \times 1000 = 200,000$.

Answer: 400 numbers; sum = 200,000

Example 32 [Goldman Sachs · pattern]

Find the last two digits of 2^{2024} .

SOLUTION

Insight: 100 splits as 4×25 . Solve mod 4 and mod 25 separately, then glue the two answers together.

Mod 4: 2^{2024} has far more than two factors of 2, so $2^{2024} \equiv 0 \pmod{4}$.

Mod 25: $\text{HCF}(2, 25) = 1$, and $\varphi(25) = 25 \times (1 - \frac{1}{5}) = 20$. So $2^{20} \equiv 1 \pmod{25}$. $2024 \div 20 = 101$ remainder 4. $2^{2024} = (2^{20})^{101} \times 2^4 \equiv 1 \times 16 = 16 \pmod{25}$.

Glue: we need a number below 100 that is $\equiv 16 \pmod{25}$ and $\equiv 0 \pmod{4}$. Numbers $\equiv 16 \pmod{25}$: 16, 41, 66, 91. Which are multiples of 4? $16 \div 4 = 4$ exactly. 41, 66, 91 are not. So the last two digits are 16.

Cross-check with the 76-trick: $2^{20} \equiv 76 \pmod{100}$ and $76 \times 76 \equiv 76$. $2^{2024} = (2^{20})^{101} \times 2^4 \equiv 76 \times 16 = 1216 \rightarrow 16$. Same answer.

Answer: 16

Example 33 [D. E. Shaw · pattern]

How many positive integers less than 1000 have exactly 3 factors?

SOLUTION

Insight: the factor count $(a + 1)(b + 1) \dots$ equals 3 only if it is the single term 3, so the number must be p^2 for a prime p .

3 is prime, so the product $(a + 1)(b + 1) \dots = 3$ forces exactly one prime in the factorisation with $a + 1 = 3$, i.e. $a = 2$. So the number is p^2 . Condition: $p^2 < 1000$. $31^2 = 961 < 1000$. $32^2 = 1024 > 1000$. So $p \leq 31$. Primes up to 31: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31. Count them: 2, 3, 5, 7 (4), 11, 13, 17, 19 (4 more, total 8), 23, 29, 31 (3 more, total 11).

Answer: 11

Example 34 [Microsoft · pattern]

Find the smallest positive integer n such that $n!$ is divisible by 2^{30} .

SOLUTION

Insight: the power of 2 in $n!$ is n minus the number of 1s in the binary form of n . So the count grows almost like n , and the answer sits just above 30.

Power of 2 in $n! = \lfloor \frac{n}{2} \rfloor + \lfloor \frac{n}{4} \rfloor + \lfloor \frac{n}{8} \rfloor + \dots$

Try $n = 31$: $\lfloor \frac{31}{2} \rfloor = 15$ $\lfloor \frac{31}{4} \rfloor = 7$ $\lfloor \frac{31}{8} \rfloor = 3$ $\lfloor \frac{31}{16} \rfloor = 1$ $\lfloor \frac{31}{32} \rfloor = 0$ Total = $15 + 7 + 3 + 1 = 26$. That is less than 30, so $31!$ fails.

Try $n = 32$: $\lfloor \frac{32}{2} \rfloor = 16$ $\lfloor \frac{32}{4} \rfloor = 8$ $\lfloor \frac{32}{8} \rfloor = 4$ $\lfloor \frac{32}{16} \rfloor = 2$ $\lfloor \frac{32}{32} \rfloor = 1$ Total = $16 + 8 + 4 + 2 + 1 = 31$. That is at least 30, so $32!$ works.

Since the power of 2 never decreases as n grows, 32 is the smallest. Sanity check with the binary shortcut: $31 = (11111)_2$ has five 1s, giving $31 - 5 = 26$. And $32 = (100000)_2$ has one 1, giving $32 - 1 = 31$. Both match.

Answer: $n = 32$

Example 35 [Uber · pattern]

Find the smallest positive integer that leaves remainder 2 when divided by 3, remainder 3 when divided by 5, and remainder 4 when divided by 7. Then describe all such integers.

SOLUTION

Insight: build the answer one modulus at a time. Satisfy the largest modulus condition last, so you test the fewest candidates.

Step 1 — satisfy “remainder 3 when divided by 5”. Candidates: 3, 8, 13, 18, 23, 28, ... (step 5).

Step 2 — among those, keep the ones with remainder 2 when divided by 3. $3 \div 3$ leaves 0. No. $8 \div 3$ leaves 2. Yes. So the first is 8, and the family now steps by $\text{LCM}(3, 5) = 15$: 8, 23, 38, 53, 68, ...

Step 3 — among those, keep the ones with remainder 4 when divided by 7. $8 \div 7$ leaves 1. No. $23 \div 7$: $7 \times 3 = 21$, leaves 2. No. $38 \div 7$: $7 \times 5 = 35$, leaves 3. No. $53 \div 7$: $7 \times 7 = 49$, leaves 4. Yes.

So the smallest is 53. Full check: $53 = 3 \times 17 + 2$. $53 = 5 \times 10 + 3$. $53 = 7 \times 7 + 4$. All three hold.

All solutions: once one works, adding $\text{LCM}(3, 5, 7) = 105$ keeps all three remainders. So the numbers are 53, 158, 263, 368, ..., that is $53 + 105k$ for $k = 0, 1, 2, \dots$

Answer: 53; in general $53 + 105k$

Example 36 [Adobe · pattern]

Find the smallest positive integer with exactly 15 factors. Then find the smallest **odd** integer with exactly 15 factors.

SOLUTION

Insight: 15 factors means the exponents plus one must multiply to 15. List every way to write 15 as a product, then attach the biggest exponent to the smallest prime.

Ways to write 15 as a product of integers each at least 2: $15 = 15 \rightarrow$ one prime, exponent $15 - 1 = 14$, shape p^{14} $15 = 5 \times 3 \rightarrow$ two primes, exponents 4 and 2, shape $p^4 q^2$ $15 = 3 \times 5$ is the same shape with the exponents swapped.

Now minimise using the smallest primes. Shape p^{14} : smallest is $2^{14} = 16384$. Shape p^4q^2 with the bigger exponent on the smaller prime: $2^4 \times 3^2 = 16 \times 9 = 144$. Shape p^2q^4 (bigger exponent on the bigger prime): $2^2 \times 3^4 = 4 \times 81 = 324$. Larger. Smallest overall = 144. Check: $144 = 2^4 \times 3^2$, so it has $(4+1)(2+1) = 5 \times 3 = 15$ factors. Correct.

Odd case — drop the prime 2, so the smallest primes available are 3 and 5. Shape p^{14} : 3^{14} , enormous. Shape p^4q^2 : $3^4 \times 5^2 = 81 \times 25 = 2025$. Shape p^2q^4 : $3^2 \times 5^4 = 9 \times 625 = 5625$. Larger. Smallest odd = 2025. Check: $2025 = 3^4 \times 5^2$, factor count = $5 \times 3 = 15$. Correct.

Answer: 144; smallest odd is 2025

PRACTICE · TIER 3 target: 4 min/Q

- Find the smallest n for which $n!$ ends in exactly 20 zeros.
- Find the last two digits of 13^{2025} .
- Find the smallest positive integer with exactly 18 factors.
- Find the remainder when $2^{2024} + 3^{2024}$ is divided by 5.
- How many zeros are at the end of the product $100 \times 200 \times 300 \times \dots \times 1000$?
- Find the sum of all positive integers below 60 that share no factor with 60 except 1.
- Find the smallest positive integer that leaves remainder 1 when divided by each of 2, 3, 4, 5 and 6, and is exactly divisible by 7.

ANSWER KEY

- 85.** $Z(84) = 16 + 3 = 19$ and $Z(85) = 17 + 3 = 20$, so 85 is the first. Note $Z(86) = Z(87) = Z(88) = Z(89) = 20$ as well, but 85 is the smallest. The value 20 is reachable, unlike 5 or 11.
- 93.** Work mod 100: $13^2 \equiv 69$, $13^4 \equiv 69^2 = 4761 \equiv 61$, $13^5 \equiv 61 \times 13 = 793 \equiv 93$, $13^{10} \equiv 93^2 = 8649 \equiv 49$, $13^{20} \equiv 49^2 = 2401 \equiv 01$. Period 20, and $2025 \div 20$ leaves remainder 5, so the answer equals $13^5 \equiv 93$.
- 180.** $18 = 2 \times 3 \times 3$ gives exponents $(2, 2, 1)$, so $2^2 \times 3^2 \times 5 = 180$. Other shapes: $18 = 3 \times 6$ gives $2^5 \times 3^2 = 288$; $18 = 2 \times 9$ gives $2^8 \times 3 = 768$; $18 = 18$ gives 2^{17} . 180 is smallest, and it does have $3 \times 3 \times 2 = 18$ factors.
- 2.** Unit-digit cycles have length 4 for both. $2024 \div 4$ leaves remainder 0, so take the 4th terms: $2^4 = 16 \equiv 1 \pmod{5}$ and $3^4 = 81 \equiv 1 \pmod{5}$. Sum $\equiv 1 + 1 = 2$.
- 22.** The product is $(100 \times 1)(100 \times 2)\dots(100 \times 10) = 100^{10} \times 10!$. Now $100^{10} = 10^{20}$ gives 20 zeros, and $10! = 3,628,800$ ends in 2 zeros. Total = $20 + 2 = 22$.
- 480.** $60 = 2^2 \times 3 \times 5$, so $\varphi(60) = 60 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} = 16$. Those 16 numbers pair as $(k, 60 - k)$, giving 8 pairs each summing to 60, so the sum is $8 \times 60 = 480$.
- 301.** Remainder 1 with all of 2, 3, 4, 5, 6 means the number is $\text{LCM}(2, 3, 4, 5, 6) + 1 = 60k + 1$. Now need 7 to divide $60k + 1$. Since $60 \equiv 4 \pmod{7}$, we need $4k + 1 \equiv 0 \pmod{7}$, i.e. $4k \equiv 6 \pmod{7}$. Multiply both sides by 2 (because $4 \times 2 = 8 \equiv 1$): $k \equiv 12 \equiv 5 \pmod{7}$. Smallest $k = 5$ gives $n = 301 = 7 \times 43$.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 90 s/Q · 18 questions in 27 minutes

- Find the unit digit of 1234^{567} .
- Find the HCF of 72, 120 and 192.
- How many zeros does $200!$ end with?
- Find the remainder when 3^{1000} is divided by 7.
- How many factors of 720 are even?
- Write the decimal number 300 in base 7.
- Find the smallest four-digit number divisible by 15, 20 and 24.
- Find the last two digits of 2^{100} .
- How many integers from 1 to 300 are divisible by neither 2 nor 3?

10. What is the smallest positive integer n for which 7^n ends in the digit 3?
11. Find the product of all the factors of 36.
12. Find the remainder when $96^{96} + 1$ is divided by 97.
13. Find the smallest positive integer that leaves remainder 1 when divided by 3 and remainder 2 when divided by 5.
14. For which digits x is the four-digit number $51x3$ divisible by 9?
15. Find the highest power of 5 that divides $1000!$.
16. How many factors of 1,800 are multiples of 10?
17. Find the remainder when 2^{2024} is divided by 9.
18. A number N leaves remainder 63 when divided by 899. What remainder does N leave when divided by 29?

ANSWER KEY

1. **4.** Only the last digit 4 matters. Cycle of 4 is 4, 6; the power 567 is odd, so the answer is 4.
2. **24.** $72 = 2^3 \times 3^2$, $120 = 2^3 \times 3 \times 5$, $192 = 2^6 \times 3$; HCF = $2^3 \times 3 = 24$.
3. **49.** $\left\lfloor \frac{200}{5} \right\rfloor = 40$, $\left\lfloor \frac{200}{25} \right\rfloor = 8$, $\left\lfloor \frac{200}{125} \right\rfloor = 1$; total $40 + 8 + 1 = 49$.
4. **4.** 7 is prime, so $3^6 \equiv 1 \pmod{7}$. $1000 \div 6$ leaves remainder 4, and $3^4 = 81 = 7 \times 11 + 4$.
5. **24.** $720 = 2^4 \times 3^2 \times 5$; total factors = $5 \times 3 \times 2 = 30$; odd factors = $3 \times 2 = 6$; even = $30 - 6 = 24$.
6. **(606)₇.** $300 \div 7 = 42 \text{ r } 6$; $42 \div 7 = 6 \text{ r } 0$; $6 \div 7 = 0 \text{ r } 6$. Check: $6 \times 49 + 0 + 6 = 300$.
7. **1080.** LCM(15, 20, 24) = 120. $1000 \div 120 = 8 \text{ r } 40$, so the next multiple is $9 \times 120 = 1080$.
8. **76.** Mod 25: $2^{20} \equiv 1$ and $100 \div 20$ leaves remainder 0, so $2^{100} \equiv 1 \pmod{25}$. Candidates below 100: 1, 26, 51, 76. The answer must also be a multiple of 4, and only 76 is.
9. **100.** Divisible by 2: 150. By 3: 100. By 6: 50. By 2 or 3: $150 + 100 - 50 = 200$. Neither: $300 - 200 = 100$.
10. **3.** Cycle of 7 is 7, 9, 3, 1, and 3 is the 3rd term.
11. **10, 077, 696.** $36 = 2^2 \times 3^2$ has $3 \times 3 = 9$ factors, so the product is $36^{9/2} = 6^9 = 10,077,696$.
12. **2.** $96 \equiv -1 \pmod{97}$, and 96 is an even power, so $96^{96} \equiv 1$. Then $1 + 1 = 2$.
13. **7.** Numbers with remainder 2 mod 5: 2, 7, 12, ... The first with remainder 1 mod 3 is 7, since $7 = 3 \times 2 + 1$.
14. **$x = 0$ or $x = 9$.** Digit sum = $5 + 1 + x + 3 = 9 + x$, which must be a multiple of 9.
15. **5²⁴⁹.** $\left\lfloor \frac{1000}{5} \right\rfloor = 200$, $\left\lfloor \frac{1000}{25} \right\rfloor = 40$, $\left\lfloor \frac{1000}{125} \right\rfloor = 8$, $\left\lfloor \frac{1000}{625} \right\rfloor = 1$; total = 249.
16. **18.** A factor that is a multiple of 10 has the form $10 \times m$ where m divides $1800 \div 10 = 180 = 2^2 \times 3^2 \times 5$. Count of such m is $3 \times 3 \times 2 = 18$.
17. **4.** $\varphi(9) = 6$, so $2^6 \equiv 1 \pmod{9}$. $2024 \div 6 = 337 \text{ r } 2$, so $2^{2024} \equiv 2^2 = 4$.
18. **5.** $899 = 29 \times 31$, so $N = 899k + 63$ gives $N \equiv 63 \pmod{29}$ because $899k$ is a multiple of 29. Then $63 = 29 \times 2 + 5$.

ONE-PAGE REVISION CARD

Divisibility, fast. 2: last digit even · 3: digit sum · 4: last 2 digits · 5: ends 0 or 5 · 6: 2 and 3 · 8: last 3 digits · 9: digit sum · 11: alternating digit sum · 12: 3 and 4 · 25: last 2 digits.

Factors. For $N = p^a q^b r^c$: count $= (a+1)(b+1)(c+1)$; sum $= \prod \left(\frac{p^{a+1}-1}{p-1} \right)$; product $= N^{d/2}$. Odd factors: ignore the 2s. Even factors: total – odd. Square factors: even exponents only. Exactly 3 factors $\rightarrow$ the number is p^2 . Exactly 4 factors $\rightarrow$ it is p^3 or pq .

HCF / LCM. $\text{HCF} \times \text{LCM} = \text{product}$ (two numbers only). Fractions: $\text{HCF} = \text{HCF}(\text{num})/\text{LCM}(\text{den})$; $\text{LCM} = \text{LCM}(\text{num})/\text{HCF}(\text{den})$. Same unknown remainder $\rightarrow$ HCF of differences. Given remainders $\rightarrow$ subtract them first. Smallest number with common remainder $r \rightarrow \text{LCM} + r$.

Remainders. Reduce every factor before multiplying. Use -1 whenever possible. Fermat: $a^{p-1} \equiv 1 \pmod{p}$. Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$. Split a composite modulus, e.g. $100 = 4 \times 25$, solve both, then glue.

Unit digits. Cycle length 4 for 2, 3, 7, 8; length 2 for 4, 9; length 1 for 0, 1, 5, 6. Divide the power by 4; remainder 0 means take the 4th term.

Last two digits. $7^4 \equiv 01 \cdot 2^{20} \equiv 76$ and $76 \times 76 \equiv 76 \cdot 3^{20} \equiv 01 \cdot 13^{20} \equiv 01$.

Factorials. Power of prime p in $n! = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \dots$. Trailing zeros = the 5-count. For p^k divide the prime count by k . For a composite, take the minimum across its primes.

Bases. $(d_k \dots d_0)_b = \sum d_i b^i$. Decimal to base b : divide repeatedly, read remainders bottom-up. Binary to octal: group in 3s; binary to hex: group in 4s. Every digit must be less than the base.

Top 5 traps.

1. Cyclicity remainder 0 means the **4th** term, not the 1st and not zero.
2. Trailing zeros: keep dividing past $\frac{n}{5}$ — 25, 125, 625 each add more.
3. HCF/LCM of fractions: do not swap the two formulas. Check that $\text{HCF} \leq \text{LCM}$.
4. Highest power of a **composite**: divide each prime's count by its exponent **before** taking the minimum.
5. A remainder is never negative. If your work gives $-1 \pmod{7}$, the remainder is 6.

Percentages

Multipliers, successive change, growth, error, savings

What you need to know

WHAT YOU NEED TO KNOW

The basics.

- $x\% = \frac{x}{100}$. “Percent” means “out of 100”.
- $x\%$ of $N = \left(\frac{x}{100}\right) \times N$
- A as a percent of $B = \left(\frac{A}{B}\right) \times 100\%$
- Percentage change $= \frac{\text{new} - \text{old}}{\text{old}} \times 100\%$. The **old** value is always the base.

The multiplier method — use this for everything.

- Increase by $x\% \rightarrow$ multiply by $\left(1 + \frac{x}{100}\right)$
- Decrease by $x\% \rightarrow$ multiply by $\left(1 - \frac{x}{100}\right)$
- Two changes in a row $\rightarrow$ multiply the two multipliers. Order does not matter.
- Net percent of $+a\%$ then $+b\% = a + b + \frac{ab}{100}$ (use a negative sign for a fall)

Must-know conversions.

$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$	$\frac{1}{8}$	$\frac{1}{9}$
50%	$33\frac{1}{3}\%$	25%	20%	$16\frac{2}{3}\%$	$14\frac{2}{7}\%$	12.5%	$11\frac{1}{9}\%$

$\frac{1}{10}$	$\frac{1}{11}$	$\frac{1}{12}$	$\frac{1}{15}$	$\frac{1}{16}$	$\frac{1}{20}$	$\frac{1}{25}$	$\frac{1}{40}$
10%	$9\frac{1}{11}\%$	$8\frac{1}{3}\%$	$6\frac{2}{3}\%$	6.25%	5%	4%	2.5%

Also: $\frac{2}{3} = 66\frac{2}{3}\%$, $\frac{3}{4} = 75\%$, $\frac{3}{8} = 37.5\%$, $\frac{5}{8} = 62.5\%$, $\frac{7}{8} = 87.5\%$, $\frac{3}{5} = 60\%$.

The two “reverse” rules. (Both come from a change of base.)

- If A is $x\%$ **more** than B , then B is $\frac{100x}{100+x}\%$ **less** than A .
- If A is $x\%$ **less** than B , then B is $\frac{100x}{100-x}\%$ **more** than A .

Product constancy. If price $\times$ quantity must stay fixed and price rises $x\%$, quantity must fall by $\frac{100x}{100+x}\%$. If price falls $x\%$, quantity may rise by $\frac{100x}{100-x}\%$.

Growth and depreciation.

- After n years at $r\%$ growth per year: $P\left(1 + \frac{r}{100}\right)^n$
- After n years at $r\%$ depreciation per year: $P\left(1 - \frac{r}{100}\right)^n$
- Different rates each year: multiply the separate multipliers.

Percentage error. $\% \text{ error} = \frac{|\text{wrong value} - \text{correct value}|}{\text{correct value}} \times 100\%$

- If a measured length is $x\%$ too large, the area is $\left(\left(1 + \frac{x}{100}\right)^2 - 1\right) \times 100\%$ too large, and the volume is $\left(\left(1 + \frac{x}{100}\right)^3 - 1\right) \times 100\%$ too large.

Income, expenditure, savings.

- Savings = Income – Expenditure
- If expenditure is $e\%$ of income, savings are $(100 - e)\%$ of income.
- Work with income = 100. It removes all the fractions.

Mixing two groups (weighted percent). If a fraction w of a total changes by $a\%$ and the rest $(1 - w)$ changes by $b\%$, the whole changes by $wa + (1 - w)b$ percent.

Two-set percentages. $n(A \text{ or } B) = n(A) + n(B) - n(A \text{ and } B)$, and $(\text{neither}) = 100\% - n(A \text{ or } B)$.

Warm-up

WARM-UP

build the reflex

Example 1

Find 35% of 640.

SOLUTION

$$35\% = \frac{35}{100}. 640 \times \frac{35}{100} = \frac{640 \times 35}{100}. 640 \times 35 = 640 \times 30 + 640 \times 5 = 19200 + 3200 = 22400. 22400 \div 100 = 224.$$

Answer: 224

Example 2

45 is what percent of 180?

SOLUTION

$$\left(\frac{45}{180}\right) \times 100. \frac{45}{180} = \frac{1}{4}. \frac{1}{4} \times 100 = 25.$$

Answer: 25%

Example 3

Write $\frac{5}{8}$ as a percent.

SOLUTION

$$\frac{5}{8} \times 100 = \frac{500}{8} = 62.5.$$

Answer: 62.5%

Example 4

Increase 250 by 12%.

SOLUTION

$$\text{Multiplier} = 1 + \frac{12}{100} = 1.12. 250 \times 1.12 = 250 + 250 \times 0.12 = 250 + 30 = 280.$$

Answer: 280

Example 5

Decrease 480 by 15%.

SOLUTION

$$15\% \text{ of } 480 = 480 \times \frac{15}{100} = \frac{480 \times 15}{100} = \frac{7200}{100} = 72. 480 - 72 = 408.$$

Answer: 408

Example 6

60 is what percent more than 48?

SOLUTION

$$\text{Rise} = 60 - 48 = 12. \text{ Base is the } \mathbf{old} \text{ number, } 48. \left(\frac{12}{48}\right) \times 100 = \left(\frac{1}{4}\right) \times 100 = 25.$$

Answer: 25%

Example 7

If 20% of a number is 70, find the number.

SOLUTION

Let the number be N . $0.20N = 70$ $N = \frac{70}{0.20} = \frac{700}{2} = 350$. Check: 20% of 350 = 70. Correct.

Answer: 350

Example 8

A quantity of 100 is increased by 10%, then the result is increased by 20%. Find the final value.

SOLUTION

After the first rise: $100 \times 1.10 = 110$. After the second rise: $110 \times 1.20 = 132$.

Answer: 132

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

In an exam a student scores 36% of the total marks and fails by 24 marks. The pass mark is 40% of the total. Find the total marks.

SOLUTION

The gap between the pass mark and the score is 24 marks. In percent, that gap is $40\% - 36\% = 4\%$ of the total. So 4% of total = 24. 1% of total = $\frac{24}{4} = 6$. 100% of total = $6 \times 100 = 600$. Check: 36% of 600 = 216; 40% of 600 = 240; $240 - 216 = 24$. Correct.

Answer: 600 marks

TRAP

“Fails by 24 marks” means score +24 = pass mark. It does **not** mean the student scored 24 marks, and it does not mean 24% of the paper. Always turn the sentence into an equation before touching numbers.

Example 10 [Infosys pattern]

A salary is increased by 20% and then decreased by 20%. What is the net percent change?

SOLUTION

Use multipliers. Start with 100. Rise: $\times 1.20 \rightarrow 100 \times 1.20 = 120$. Fall: $\times 0.80 \rightarrow 120 \times 0.80 = 96$. Change = $96 - 100 = -4$. Percent change = $\frac{-4}{100} \times 100 = -4\%$.

Answer: 4% decrease

SHORTCUT

Up $x\%$ then down $x\%$ is **always** a fall of $\frac{x^2}{100}$ percent. Here $\frac{20^2}{100} = 4\%$. For 10% it is 1%; for 30% it is 9%; for 50% it is 25%. It never returns to the start.

Example 11 [Accenture pattern]

The price of rice rises by 25%. By what percent must a family cut its consumption so that the money spent on rice stays the same?

SOLUTION

Spending = price $\times$ consumption, and spending must not change. Take price = 100 and consumption = 100, so spending = 10000. New price = 125. New consumption = $10000 \div 125 = 80$. Drop = $100 - 80 = 20$. Percent drop = $\left(\frac{20}{100}\right) \times 100 = 20\%$. Formula check: $\frac{100 \times 25}{100 + 25} = \frac{2500}{125} = 20$. Same.

Answer: 20%

Example 12 [Wipro pattern]

A's income is 25% more than B's income. B's income is what percent less than A's?

SOLUTION

Let B = 100. A = $100 + 25 = 125$. B is less than A by $125 - 100 = 25$. Now the base is **A**, not B. $\left(\frac{25}{125}\right) \times 100 = \frac{2500}{125} = 20$.

Answer: 20%

TRAP

The base changes when you flip the comparison. "A is 25% more than B" and "B is 25% less than A" are **not** the same statement. The first uses B as base, the second uses A.

Example 13 [Capgemini pattern]

The population of a town is 50,000 and grows by 10% every year. Find the population after 3 years.

SOLUTION

Multiplier per year = 1.10. Year 1: $50000 \times 1.10 = 55000$. Year 2: $55000 \times 1.10 = 60500$. Year 3: $60500 \times 1.10 = 66550$. One-shot check: $50000 \times (1.1)^3 = 50000 \times 1.331 = 66550$.

Answer: 66,550

Example 14 [Cognizant pattern]

A machine costs Rs. 80,000 and loses 15% of its value each year. Find its value after 2 years.

SOLUTION

Multiplier per year = $1 - 0.15 = 0.85$. Year 1: $80000 \times 0.85 = 80000 - 80000 \times 0.15 = 80000 - 12000 = 68000$. Year 2: $68000 \times 0.85 = 68000 - 68000 \times 0.15 = 68000 - 10200 = 57800$.

Answer: Rs. 57,800

Example 15 [TCS NQT pattern]

In a town, 60% of the people are men. 40% of the men and 25% of the women can read. What percent of the whole town can read?

SOLUTION

Take the town as 100 people. Men = 60. Women = $100 - 60 = 40$. Men who can read = 40% of 60 = $0.40 \times 60 = 24$. Women who can read = 25% of 40 = $0.25 \times 40 = 10$. Total who can read = $24 + 10 = 34$. Out of 100 people, so 34%.

Answer: 34%

SHORTCUT

For “percent of a percent” questions, set the total to 100. Every answer then falls out as a plain count, and the final count is the percent. No fractions, no decimals.

Example 16 [Infosys pattern]

A man spends 75% of his income. Next year his income rises by 20% and his spending rises by 10%. By what percent do his savings rise?

SOLUTION

Take income = 100. Expenditure = 75. Savings = $100 - 75 = 25$. New income = $100 \times 1.20 = 120$. New expenditure = $75 \times 1.10 = 82.5$. New savings = $120 - 82.5 = 37.5$. Rise in savings = $37.5 - 25 = 12.5$. Percent rise = $\left(\frac{12.5}{25}\right) \times 100 = 50$.

Answer: 50%

Example 17 [Wipro pattern]

In an election between two candidates, the winner got 55% of the valid votes and won by 1,200 votes. Find the number of valid votes.

SOLUTION

Winner = 55%. Only two candidates, so loser = $100\% - 55\% = 45\%$. Margin in percent = $55\% - 45\% = 10\%$. So 10% of valid votes = 1200. 1% of valid votes = $\frac{1200}{10} = 120$. 100% of valid votes = $120 \times 100 = 12000$. Check: winner = 6600, loser = 5400, margin = 1200. Correct.

Answer: 12,000 valid votes

Example 18 [Accenture pattern]

A student had to multiply a number by $\frac{5}{3}$, but multiplied it by $\frac{3}{5}$ instead. Find the percentage error.

SOLUTION

Pick a convenient number. Take 15 (it divides by both 3 and 5). Correct answer = $15 \times \frac{5}{3} = 25$. Wrong answer = $15 \times \frac{3}{5} = 9$. Error = $25 - 9 = 16$. Percentage error = $\left(\frac{16}{25}\right) \times 100 = \frac{1600}{25} = 64$.

Answer: 64%

TRAP

Percentage error is always divided by the **correct** value, not the wrong one. Dividing by 9 here gives about 178%, which is a different (and wrong) quantity.

Example 19 [TCS NQT pattern]

The price of a product is cut by 20%. By what percent must the new price be raised to bring it back to the original?

SOLUTION

Take the original price = 100. New price = $100 - 20 = 80$. To get back to 100 we must add $100 - 80 = 20$. Base is now the **new** price, 80. $\left(\frac{20}{80}\right) \times 100 = \frac{2000}{80} = 25$.

Answer: 25%

Example 20 [Capgemini pattern]

In a class of 400 students, 45% are girls. 20% of the girls and 10% of the boys wear glasses. How many students wear glasses?

SOLUTION

Girls = 45% of 400 = $0.45 \times 400 = 180$. Boys = $400 - 180 = 220$. Girls with glasses = 20% of 180 = $0.20 \times 180 = 36$. Boys with glasses = 10% of 220 = $0.10 \times 220 = 22$. Total = $36 + 22 = 58$.

Answer: 58 students

PRACTICE · TIER 1 target: 60 s/Q

- Find 18% of 2,500.
- If 15% of a number is 54, find the number.
- 84 is what percent of 350?
- A number increased by 25% becomes 375. Find the original number.
- A quantity rises by 20% and then by 25%. Find the net percent rise.
- A salary is cut by 10% and then raised by 10%. Find the net percent change.
- If A is 40% less than B, then B is what percent more than A?
- A town's population is 25,000 and falls by 4% each year. Find the population after 2 years.
- A student scores 30% and fails by 45 marks. The pass mark is 45%. Find the total marks.
- A man spends 80% of his income. His income rises 25% and his spending rises 20%. Find the percent rise in savings.
- Write $\frac{7}{16}$ as a percent.
- In an election with two candidates, the loser got 42% of the votes and lost by 4,800 votes. Find the total number of votes.
- The price of sugar rises by 20%. By what percent must consumption fall so that spending is unchanged?
(a) $16\frac{2}{3}\%$ (b) 20% (c) 25% (d) $83\frac{1}{3}\%$
- If 30% of A equals 45% of B, find $A : B$.
- The price of a phone falls from Rs. 24,000 to Rs. 19,200. Find the percent fall.

ANSWER KEY

- 450.** $2500 \times \frac{18}{100} = 25 \times 18 = 450$.
- 360.** $0.15N = 54 \rightarrow N = \frac{54}{0.15} = 360$.
- 24%.** $\left(\frac{84}{350}\right) \times 100 = 0.24 \times 100 = 24$.
- 300.** $1.25N = 375 \rightarrow N = \frac{375}{1.25} = 300$.
- 50%.** $1.20 \times 1.25 = 1.50$. Or $20 + 25 + \frac{20 \times 25}{100} = 45 + 5 = 50$.
- 1% decrease.** $0.90 \times 1.10 = 0.99$. Same as the $\frac{x^2}{100}$ rule with $x = 10$.
- $66\frac{2}{3}\%$.** A = 60 when B = 100; B exceeds A by 40, and $\left(\frac{40}{60}\right) \times 100 = 66\frac{2}{3}$.
- 23,040.** $25000 \times 0.96 = 24000$; $24000 \times 0.96 = 23040$.
- 300.** Gap = $45\% - 30\% = 15\%$, so 15% of total = 45 and total = $45 \times \frac{100}{15} = 300$.
- 45%.** Income 100, spend 80, save 20. New: income 125, spend 96, save 29. Rise = $\frac{9}{20} = 45\%$.
- 43.75%.** $\frac{7}{16} \times 100 = \frac{700}{16} = 43.75$.
- 30,000.** Winner = 58%, margin = $58 - 42 = 16\%$, so 16% of total = 4800 and total = 30000.
- (a) $16\frac{2}{3}\%$.** $\frac{100 \times 20}{100 + 20} = \frac{2000}{120} = 16\frac{2}{3}$. Choosing (b) 20% means you used the same 20% for the cut, forgetting the base changed. Choosing (c) 25% means you divided by $100 - 20$ instead of $100 + 20$. Choosing (d) $83\frac{1}{3}\%$ gives the consumption that is **left**, not the size of the cut.
- 3 : 2.** $0.30A = 0.45B \rightarrow \frac{A}{B} = \frac{0.45}{0.30} = \frac{3}{2}$.
- 20%.** Fall = $24000 - 19200 = 4800$, and $\left(\frac{4800}{24000}\right) \times 100 = 20$. The base is the **old** price.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A ride has a base fare of S\$12.00. A surge raises the fare by 35%. A promo code then takes 20% off the surged fare. Find the amount paid and the net percent change from the base fare.

SOLUTION

Surged fare = 12×1.35 . $12 \times 1.35 = 12 + 12 \times 0.35 = 12 + 4.20 = 16.20$. After the promo = 16.20×0.80 . $16.20 \times 0.80 = 16.20 - 16.20 \times 0.20 = 16.20 - 3.24 = 12.96$. Net multiplier = $1.35 \times 0.80 = 1.08$. Net change = $1.08 - 1 = 0.08 \rightarrow +8\%$. Check: $12 \times 1.08 = 12.96$. Matches.

Answer: S\$12.96, a rise of 8%

SHORTCUT

Never carry the middle number. Multiply the multipliers first: $1.35 \times 0.80 = 1.08$. One multiplication instead of two, and no rounding builds up.

Example 22 [Shopee · pattern]

A seller lists an item at THB 2,500. Before a festival she raises the price by 12%, then during the festival she cuts the new price by 12%. Find the final price and the net percent change.

SOLUTION

Net multiplier = 1.12×0.88 . $1.12 \times 0.88 = 1.12 - 1.12 \times 0.12 = 1.12 - 0.1344 = 0.9856$. Final price = 2500×0.9856 . $2500 \times 0.9856 = 2500 - 2500 \times 0.0144 = 2500 - 36 = 2464$. Net change = $0.9856 - 1 = -0.0144 \rightarrow -1.44\%$. Shortcut check: $\frac{x^2}{100} = \frac{12^2}{100} = \frac{144}{100} = 1.44\%$ fall. Matches.

Answer: THB 2,464; a fall of 1.44%

Example 23 [DBS · pattern]

A bank's loan book grew 8% in year 1, shrank 5% in year 2, and grew 12% in year 3. Find the net percent change over the three years.

SOLUTION

Multiply the three multipliers. 1.08×0.95 : $1.08 \times 0.95 = 1.08 - 1.08 \times 0.05 = 1.08 - 0.054 = 1.026$. Now 1.026×1.12 : $1.026 \times 1.12 = 1.026 + 1.026 \times 0.12 = 1.026 + 0.12312 = 1.14912$. Net change = $1.14912 - 1 = 0.14912$. As a percent: 14.912%, about 14.91% growth.

Answer: About 14.91% growth

TRAP

Do not add $8 - 5 + 12 = 15\%$. Adding percent changes only works when they all sit on the same base, and successive changes never do. Here the true answer is 14.912%, not 15%.

Example 24 [GIC · pattern]

A fund holds 60% of its money in equity and 40% in bonds. Over a year equity returns +15% and bonds return -5%. Find the fund's overall percent return.

SOLUTION

Take the fund as 100 units. Equity = 60, bonds = 40. Equity becomes $60 \times 1.15 = 60 + 9 = 69$. Bonds become $40 \times 0.95 = 40 - 2 = 38$. Total = $69 + 38 = 107$. Change = $107 - 100 = 7$ out of 100, so +7%. Weighted-average check: $0.6 \times 15 + 0.4 \times (-5) = 9 - 2 = 7$. Matches.

Answer: +7%

Example 25 [Agoda · pattern]

A hotel ran at 80% occupancy with an average rate of S\$180 per night. The next year occupancy fell to 72% but the average rate rose to S\$207. Find the percent change in revenue per available room.

SOLUTION

Revenue per available room = occupancy $\times$ rate. Old = $0.80 \times 180 = 144$. New = 0.72×207 . $0.72 \times 207 = 0.72 \times 200 + 0.72 \times 7 = 144 + 5.04 = 149.04$. Change = $149.04 - 144 = 5.04$. Percent change = $\left(\frac{5.04}{144}\right) \times 100$. $\frac{5.04}{144} = 0.035$. $0.035 \times 100 = 3.5$.

Answer: +3.5%

Example 26 [SCB · pattern]

A Thai exporter's revenue in baht rose by 20% this year. Over the same period the value of one baht in US dollars fell by 10%. Find the percent change in the exporter's revenue measured in US dollars.

SOLUTION

Revenue in USD = (revenue in THB) $\times$ (USD per THB). THB multiplier = 1.20. Rate multiplier = 0.90. Combined = $1.20 \times 0.90 = 1.08$. Change = $1.08 - 1 = 0.08 \rightarrow +8\%$.

Answer: +8%

Example 27 [LINE MAN · pattern]

In March, 25% of a platform's orders came from Zone A and the rest from Zone B. In April, Zone A orders rose 40% and Zone B orders fell 8%. Find the percent change in total orders.

SOLUTION

Take March total = 100 orders. Zone A = 25. Zone B = $100 - 25 = 75$. April Zone A = $25 \times 1.40 = 35$. April Zone B = 75×0.92 . $75 \times 0.92 = 75 - 75 \times 0.08 = 75 - 6 = 69$. April total = $35 + 69 = 104$. Change = $104 - 100 = 4$ out of 100, so +4%.

Answer: +4%

Example 28 [Sea · pattern]

A marketplace deducts a 15% commission from the list price, then a 5% payment fee on whatever is left. A seller received THB 8,075. Find the list price.

SOLUTION

Let the list price be L . After commission: $L \times 0.85$. After the payment fee on that amount: $L \times 0.85 \times 0.95$. $0.85 \times 0.95 = 0.85 - 0.85 \times 0.05 = 0.85 - 0.0425 = 0.8075$. So $0.8075L = 8075$. $L = \frac{8075}{0.8075} = \frac{8075000}{8075} = 10000$. Check: $10000 \times 0.85 = 8500$; $8500 \times 0.95 = 8075$. Correct.

Answer: THB 10,000

TRAP

A 15% cut then a 5% cut is not a 20% cut. The 5% is charged on the reduced amount, so the true total cut is $100 - 80.75 = 19.25\%$. Reading "20%" here gives a list price of THB 10,093.75, which is wrong.

Example 29 [Razer · pattern]

A checkout system should apply a single 15% discount on an order of S\$240. By mistake it applies the 15% discount twice, one after the other. By how many dollars, and by what percent, is the charged amount below the correct amount?

SOLUTION

Correct amount = 240×0.85 . $240 \times 0.85 = 240 - 240 \times 0.15 = 240 - 36 = 204$. Charged amount = $204 \times 0.85 = 204 - 204 \times 0.15 = 204 - 30.60 = 173.40$. Shortfall = $204 - 173.40 = 30.60$. Percent below the correct amount, base = 204: $\left(\frac{30.60}{204}\right) \times 100 = 15$. That is expected: the extra step is one more 15% cut.

Answer: S\$30.60 short, which is 15% below the correct amount

PRACTICE · TIER 2 target: 100 s/Q

1. A price rises 8% and then rises 8% again. Find the net percent rise.
2. S\$500 falls by 12% and then rises by 12%. Find the final amount and the net percent change.
3. A portfolio is 70% bonds returning +3% and 30% equity returning −10%. Find the overall percent return.
4. An online seller receives THB 6,460 after a 15% commission is taken off the list price. Find the list price.
5. A shop's revenue rose 25% while the number of units sold fell 10%. Find the percent change in the price per unit.
6. A hotel moved from 85% occupancy at S\$120 per night to 78% occupancy at S\$135 per night. Find the percent change in revenue per available room.
7. A value in baht rose 15%, and the US dollar value of one baht fell 5%. Find the percent change in the US dollar value.
8. A company has 45% of its staff in Singapore and the rest in Bangkok. Singapore headcount grows 20% and Bangkok headcount falls 10%. Find the percent change in total headcount.
9. After three successive cuts of 10% each, what percent of the original price remains?
10. A restaurant bill of S\$250 has a 9% service charge added, and then 8% tax is charged on the new total. Find the final amount.
11. Ads make up 40% of a firm's revenue. Ad revenue grows 30% while the rest stays flat. Find the percent change in total revenue.

ANSWER KEY

1. **16.64%**. $1.08 \times 1.08 = 1.1664$. Or $8 + 8 + \frac{8 \times 8}{100} = 16 + 0.64$.
2. **S\$492.80; a 1.44% fall**. $0.88 \times 1.12 = 0.9856$, and $500 \times 0.9856 = 492.80$. The $\frac{x^2}{100}$ rule gives $\frac{144}{100} = 1.44\%$.
3. **−0.9%**. $0.7 \times 3 + 0.3 \times (-10) = 2.1 - 3 = -0.9$.
4. **THB 7,600**. $0.85L = 6460 \rightarrow L = \frac{6460}{0.85} = 7600$.
5. **About +38.89%**. Price = revenue $\div$ units, so the multiplier is $\frac{1.25}{0.90} = 1.3889$, a rise of 38.89%.
6. **About +3.24%**. Old = $0.85 \times 120 = 102$; new = $0.78 \times 135 = 105.30$; rise = 3.30, and $\frac{3.30}{102} = 0.03235$.
7. **+9.25%**. $1.15 \times 0.95 = 1.0925$.
8. **+3.5%**. $0.45 \times 1.20 + 0.55 \times 0.90 = 0.54 + 0.495 = 1.035$.
9. **72.9% remains, a fall of 27.1%**. $0.9^3 = 0.729$. Adding $10 + 10 + 10 = 30\%$ is wrong.
10. **S\$294.30**. $250 \times 1.09 = 272.50$; $272.50 \times 1.08 = 294.30$.
11. **+12%**. $0.40 \times 1.30 + 0.60 \times 1 = 0.52 + 0.60 = 1.12$.

Tier 3 — Product companies**TIER 3**

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 30 [Google · pattern]

A shop raises a price by $x\%$ and later lowers the new price by the same $x\%$. The final price is 9% below the original, and the drop in rupees is Rs. 270. Find x and the original price.

SOLUTION

Insight: an up-then-down move by the same percent is $(1 + x/100)(1 - x/100)$, which is the difference of two squares. The net loss is always $x^2/100$ percent, and it never depends on the price.

Net multiplier = $(1 + \frac{x}{100})(1 - \frac{x}{100}) = 1 - \frac{x^2}{10000}$.

So the loss as a fraction is $\frac{x^2}{10000}$, that is $\frac{x^2}{100}$ percent.

Set that equal to 9: $\frac{x^2}{100} = 9$ $x^2 = 900$ $x = 30$ (a percent change is taken positive here).

Now the original price P . The fall is 9% of P : $0.09P = 270$ $P = \frac{270}{0.09} = \frac{27000}{9} = 3000$.

| Check: $3000 \times 1.30 = 3900$; $3900 \times 0.70 = 2730$; drop $= 3000 - 2730 = 270$. Correct.

Answer: $x = 30$; original price Rs. 3,000

Example 31 [Amazon · pattern]

Revenue grew 10% in year 1 and fell $x\%$ in year 2, finishing 1% below where it started. Find x . Then find the single constant yearly rate that would produce the same two-year result.

SOLUTION

Insight: a chain of percent changes is a product of multipliers. Any question about a chain becomes one equation in multipliers.

Part 1. $1.10 \times \left(1 - \frac{x}{100}\right) = 0.99$ $1 - \frac{x}{100} = \frac{0.99}{1.10} = 0.90$ $\frac{x}{100} = 1 - 0.90 = 0.10$ $x = 10$. Check: $100 \times 1.10 = 110$; $110 \times 0.90 = 99$. That is 1% below 100. Correct.

Part 2. Let the constant yearly multiplier be $1 + g$. $(1 + g)^2 = 0.99$ $1 + g = \sqrt{0.99}$

First a quick estimate. For a small g , $(1 + g)^2 \approx 1 + 2g$, so $2g \approx -0.01$ and $g \approx -0.5\%$.

Now refine. Try $1 + g = 0.995$: $0.995^2 = 0.990025$. Slightly too big. Try $1 + g = 0.9949$: $0.9949^2 = 0.98982601$. Slightly too small. The true value sits between, at $1 + g \approx 0.994987$. So $g \approx -0.5013\%$, about a 0.5% fall each year.

Answer: $x = 10$; a constant rate of about -0.50% per year

Example 32 [Goldman Sachs · pattern]

A fund has exactly two sleeves. One returned $+15\%$, the other returned -2% , and the fund overall returned $+6\%$. What fraction of the money was in the first sleeve?

SOLUTION

Insight: the overall return is a weighted average of the two returns, so the weights split the gap between them in inverse ratio — this is the alligation rule.

Let the fraction in the first sleeve be w . Then $1 - w$ sits in the second. $15w + (-2)(1 - w) = 6$ $15w - 2 + 2w = 6$ $17w = 8$ $w = \frac{8}{17}$.

As a decimal: $\frac{8}{17} = 0.4706$, so about 47.06%.

Alligation check. Distances from the overall 6: first sleeve: $15 - 6 = 9$ second sleeve: $6 - (-2) = 8$ The weights are in the **inverse** ratio 8 : 9, so the first sleeve holds $\frac{8}{8+9} = \frac{8}{17}$. Matches.

Verify: $\frac{8}{17} \times 15 = \frac{120}{17}$ and $\frac{9}{17} \times (-2) = -\frac{18}{17}$. Sum $= \frac{102}{17} = 6$. Correct.

Answer: $\frac{8}{17}$, about 47.06%

Example 33 [D. E. Shaw · pattern]

A stock moves only in two ways: it rises 10% on an up-day and falls 10% on a down-day. Over 10 trading days there are exactly 5 up-days and 5 down-days, in some unknown order. Find the net percent change.

SOLUTION

Insight: multiplication does not care about order. So the unknown order is a red herring — the answer is fixed.

Net multiplier $= 1.1^5 \times 0.9^5 = (1.1 \times 0.9)^5 = 0.99^5$.

Compute step by step. $0.99^2 = 0.9801$ $0.99^4 = 0.9801^2$ $0.9801^2 = 0.9801 \times 0.9801 = 0.9801 - 0.9801 \times 0.0199 = 0.9801 - 0.01950399 = 0.96059601$ $0.99^5 = 0.96059601 \times 0.99 = 0.96059601 - 0.0096059601 = 0.9509900499$

Net change $= 0.9509900499 - 1 = -0.04900995$. As a percent: about -4.90% .

Sanity check with the pair rule: each up-down pair is a loss of $\frac{10^2}{100} = 1\%$. Five pairs give 0.99^5 , which is a little less than a 5% loss because the losses compound on a shrinking base. $4.90\% < 5\%$, as expected.

Answer: About a 4.90% fall

Example 34 [Microsoft · pattern]

In a survey of 1,000 people, 62% like tea, 47% like coffee and 15% like neither. What percent like both? If the “neither” figure were unknown, what is the largest and smallest the “both” figure could be?

SOLUTION

Insight: percentages of the same total behave exactly like counts, so the two-set formula applies directly.

Part 1. Like at least one = $100\% - 15\% = 85\%$. Two-set rule: $n(\text{tea or coffee}) = n(\text{tea}) + n(\text{coffee}) - n(\text{both})$ $85 = 62 + 47 - n(\text{both})$ $85 = 109 - n(\text{both})$ $n(\text{both}) = 109 - 85 = 24$. So 24%, that is $0.24 \times 1000 = 240$ people.

Part 2 — the range, with “neither” unknown. **Largest.** “Both” can never exceed the smaller group, so $n(\text{both}) \leq 47\%$. That happens when every coffee drinker also drinks tea. Then “at least one” = 62% and “neither” = 38%. Legal. **Smallest.** “At least one” can never exceed 100%, so $62 + 47 - n(\text{both}) \leq 100$ $109 - n(\text{both}) \leq 100$ $n(\text{both}) \geq 9$. At $n(\text{both}) = 9\%$, “at least one” = 100% and “neither” = 0%. Legal.

Answer: 24% like both; in general “both” lies between 9% and 47%

Example 35 [Uber · pattern]

A driver’s take-home pay equals (trips $\times$ fare per trip) minus the platform commission. Over one quarter, trips fell 12%, the fare per trip rose 20%, and the commission rose from 20% of gross to 25% of gross. Find the percent change in take-home pay.

SOLUTION

Insight: split the pay into two independent multipliers — one for gross earnings, one for the share the driver keeps. Then multiply.

Step 1 — gross earnings = trips $\times$ fare. Trips multiplier = 0.88. Fare multiplier = 1.20. Gross multiplier = 0.88×1.20 . $0.88 \times 1.20 = 0.88 + 0.88 \times 0.20 = 0.88 + 0.176 = 1.056$.

Step 2 — the share the driver keeps. Before: commission 20%, so the driver keeps $100\% - 20\% = 80\%$. After: commission 25%, so the driver keeps $100\% - 25\% = 75\%$. Share multiplier = $\frac{0.75}{0.80} = 0.9375$.

Step 3 — combine. Take-home multiplier = 1.056×0.9375 . $1.056 \times 0.9375 = 1.056 - 1.056 \times 0.0625 = 1.056 - 0.066 = 0.99$.

Change = $0.99 - 1 = -0.01 \rightarrow -1\%$.

Number check. Start with 100 trips at S\$10, gross = 1000, driver keeps 80% = 800. Now 88 trips at S\$12, gross = 1056, driver keeps 75% = 792. 792 against 800 is a fall of $\frac{8}{800} = 1\%$. Correct.

Answer: Take-home pay falls by 1%

Example 36 [Adobe · pattern]

The side of a square is measured 3% too long. Find the percentage error in the computed area. Then find the percentage error in the computed volume if the same 3% error is made on the edge of a cube. Why is the answer not exactly 6% and 9%?

SOLUTION

Insight: errors multiply, they do not add. Doubling or tripling the error is only the first term of the expansion; the leftover terms are what make the answer slightly larger.

Area. Side multiplier = 1.03, and area = side². Area multiplier = 1.03². $1.03^2 = 1.03 \times 1.03 = 1.03 + 1.03 \times 0.03 = 1.03 + 0.0309 = 1.0609$. Error = $1.0609 - 1 = 0.0609 \rightarrow 6.09\%$ too large.

Volume. Edge multiplier = 1.03, and volume = edge³. Volume multiplier = 1.03³. $1.03^3 = 1.0609 \times 1.03$. $1.0609 \times 1.03 = 1.0609 + 1.0609 \times 0.03 = 1.0609 + 0.031827 = 1.092727$. Error = $1.092727 - 1 = 0.092727 \rightarrow 9.2727\%$ too large.

Why not exactly 6% and 9%? Expand: $(1+x)^2 = 1 + 2x + x^2$. The $2x$ part gives the 6%. The extra $x^2 = 0.0009$ gives the extra 0.09%. Expand: $(1+x)^3 = 1 + 3x + 3x^2 + x^3$. The $3x$ gives 9%; the $3x^2 = 0.0027$ and $x^3 = 0.000027$ give the extra 0.2727%. For small errors the rough rule “ n times the error” is close, but it always understates a positive error.

Answer: Area 6.09% too large; volume 9.2727% too large

PRACTICE · TIER 3 target: 4 min/Q

1. A price rises $x\%$ and then falls $x\%$, finishing 6.25% below the original. Find x .
2. A stock rises 20% on day 1. By what percent must it fall on day 2 to finish 4% below where it started?
3. In a class, 70% passed Maths and 60% passed Physics, and 45% passed both. What percent failed both?
4. Two sleeves of a fund returned +18% and −6%, and the fund overall returned +4%. Find the fraction held in the first sleeve.
5. The radius of a circle is measured 4% too small. Find the percentage error in the computed area.
6. A driver's trips rise 15%, the fare per trip falls 10%, and the platform commission falls from 25% of gross to 20% of gross. Find the percent change in take-home pay.
7. A shopkeeper raises the price by 30% and then gives two successive discounts of 10% each. Find the net percent change in price.

ANSWER KEY

1. $x = 25$. Up then down by the same $x\%$ always loses $x^2/100$ percent, so $x^2/100 = 6.25$, giving $x^2 = 625$ and $x = 25$. Check: $1.25 \times 0.75 = 0.9375$, a 6.25% fall.
2. **20%**. Let the day-2 fall be f . Then $1.20(1-f) = 0.96$, so $1-f = \frac{0.96}{1.20} = 0.80$ and $f = 0.20$. Note the answer is **not** 24%, even though the stock must give back more than it gained.
3. **15%**. Passed at least one = $70 + 60 - 45 = 85\%$, so failed both = $100 - 85 = 15\%$.
4. $\frac{5}{12}$. With weight w in the first sleeve, $18w - 6(1-w) = 4$, so $24w = 10$ and $w = \frac{10}{24} = \frac{5}{12}$. Alligation check: the distances are $18 - 4 = 14$ and $4 - (-6) = 10$, and the inverse ratio $10 : 14 = 5 : 7$ gives $\frac{5}{12}$.
5. **7.84% too small**. Area multiplier = $0.96^2 = 0.9216$, so the error is $1 - 0.9216 = 0.0784$. The rough rule “twice the error” would say 8%, which overstates a negative error.
6. **+10.4%**. Gross multiplier = $1.15 \times 0.90 = 1.035$. The driver's share goes from 75% to 80%, a multiplier of $\frac{0.80}{0.75} = \frac{16}{15}$. Combined: $1.035 \times \frac{16}{15} = \frac{16.56}{15} = 1.104$.
7. **+5.3%**. $1.30 \times 0.90 \times 0.90 = 1.30 \times 0.81 = 1.053$. Adding $30 - 10 - 10 = 10\%$ is the common error.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 90 s/Q · 18 questions in 27 minutes

1. Find 12.5% of 4,800.
2. If 25% of x equals 40% of 150, find x .
3. A number decreased by 30% becomes 560. Find the original number.
4. A quantity rises 50% and then rises 20%. Find the net percent rise.
5. A salary of Rs. 45,000 is raised by 8%. Find the new salary.
6. The price of an item rises 60%. By what percent must consumption be cut so that spending is unchanged?
7. A town's population grew from 8,000 to 9,680 in two years at a constant yearly rate. Find the rate.
8. Write $\frac{9}{16}$ as a percent.
9. A man saves 15% of his income of Rs. 64,000. Find his savings and his expenditure.
10. In an election between two candidates, the winner won by a margin equal to 15% of the total votes. The margin was 4,500 votes. Find the total number of votes.
11. A machine costing Rs. 1,25,000 depreciates 20% each year. Find its value after 3 years.

12. In a college, 60% of the students are boys. 30% of the boys and 50% of the girls take the bus. What percent of all students take the bus?
13. A shop marks up its cost by 40% and then gives a 25% discount. Find the net percent change from cost.
14. A student multiplied a number by $\frac{4}{5}$ instead of $\frac{5}{4}$. Find the percentage error.
15. If A is 20% of B, and B is 30% of C, then A is what percent of C?
16. A town's population rose 10% in year 1 and 5% in year 2. Find the net percent rise over the two years.
17. A man's expenditure is 75% of his income. His income rises 20% and his expenditure rises 12%. Find the percent rise in savings.
18. Find 0.5% of 8,000.

ANSWER KEY

1. **600.** $12.5\% = \frac{1}{8}$, and $\frac{4800}{8} = 600$.
2. **240.** $0.25x = 0.40 \times 150 = 60$, so $x = \frac{60}{0.25} = 240$.
3. **800.** $0.70N = 560 \rightarrow N = \frac{560}{0.70} = 800$.
4. **80%.** $1.50 \times 1.20 = 1.80$.
5. **Rs. 48,600.** $45000 \times 1.08 = 45000 + 3600 = 48600$.
6. **37.5%.** $\frac{100 \times 60}{100 + 60} = \frac{6000}{160} = 37.5$.
7. **10%.** $\frac{9680}{8000} = 1.21 = (1 + r)^2$, so $1 + r = 1.1$ and $r = 10\%$.
8. **56.25%.** $\frac{9}{16} \times 100 = \frac{900}{16} = 56.25$.
9. **Savings Rs. 9,600; expenditure Rs. 54,400.** $0.15 \times 64000 = 9600$, and $64000 - 9600 = 54400$.
10. **30,000.** $0.15 \times \text{total} = 4500 \rightarrow \text{total} = \frac{4500}{0.15} = 30000$.
11. **Rs. 64,000.** $125000 \times 0.8^3 = 125000 \times 0.512 = 64000$.
12. **38%.** $0.60 \times 0.30 + 0.40 \times 0.50 = 0.18 + 0.20 = 0.38$.
13. **+5%.** $1.40 \times 0.75 = 1.05$. Adding $40 - 25 = 15\%$ is the trap.
14. **36%.** Take the number as 20: correct = 25, wrong = 16, error = 9, and $\frac{9}{25} = 0.36$.
15. **6%.** $A = 0.20B$ and $B = 0.30C$, so $A = 0.20 \times 0.30C = 0.06C$.
16. **15.5%.** $1.10 \times 1.05 = 1.155$. Adding $10 + 5 = 15\%$ misses the 0.5% cross term.
17. **44%.** Income 100, expenditure 75, savings 25. New income 120, new expenditure $75 \times 1.12 = 84$, new savings 36. Rise $= \frac{11}{25} = 0.44$.
18. **40.** $0.5\% = 0.005$, and $8000 \times 0.005 = 40$.

ONE-PAGE REVISION CARD

Core. $x\%$ of $N = \frac{xN}{100}$. A as a percent of $B = \left(\frac{A}{B}\right) \times 100$. change = (new – old) $\div$ old $\times 100$.

Multipliers — the whole chapter in one line. Up $x\% \rightarrow \times \left(1 + \frac{x}{100}\right)$. Down $x\% \rightarrow \times \left(1 - \frac{x}{100}\right)$. Chain them by multiplying. Order never matters. Two changes: net = $a + b + \frac{ab}{100}$.

Fraction table. $\frac{1}{2} = 50 \cdot \frac{1}{3} = 33\frac{1}{3} \cdot \frac{1}{4} = 25 \cdot \frac{1}{5} = 20 \cdot \frac{1}{6} = 16\frac{2}{3} \cdot \frac{1}{7} = 14\frac{2}{7} \cdot \frac{1}{8} = 12.5 \cdot \frac{1}{9} = 11\frac{1}{9} \cdot \frac{1}{10} = 10 \cdot \frac{1}{11} = 9\frac{1}{11} \cdot \frac{1}{12} = 8\frac{1}{3} \cdot \frac{1}{15} = 6\frac{2}{3} \cdot \frac{1}{16} = 6.25 \cdot \frac{1}{20} = 5 \cdot \frac{1}{25} = 4 \cdot \frac{1}{40} = 2.5$ (all in percent).

Reverse rules. A is $x\%$ more than $B \rightarrow B$ is $\frac{100x}{100+x}\%$ less than A . A is $x\%$ less than $B \rightarrow B$ is $\frac{100x}{100-x}\%$ more than A .

Product constancy. Price up $x\% \rightarrow$ consumption down $\frac{100x}{100+x}\%$ to hold spending fixed. Quick fraction form: price up $\frac{1}{n} \rightarrow$ consumption down $\frac{1}{n+1}$. (Up 25% = $\frac{1}{4} \rightarrow$ down $\frac{1}{5} = 20\%$.)

Up $x\%$ then down $x\%$. Always a fall of $\frac{x^2}{100}$ percent. $10 \rightarrow 1\% \cdot 20 \rightarrow 4\% \cdot 25 \rightarrow 6.25\% \cdot 30 \rightarrow 9\% \cdot 50 \rightarrow 25\%$.

Growth / depreciation. $P\left(1 \pm \frac{r}{100}\right)^n$. Mixed rates: multiply the separate multipliers.

Error. % error = $\frac{\text{wrong} - \text{correct}}{\text{correct}} \times 100$. Side off by $x\% \rightarrow$ area off by $\left(1 + \frac{x}{100}\right)^2 - 1$; volume off by $\left(1 + \frac{x}{100}\right)^3 - 1$.

Income. Always set income = 100. Savings = income – expenditure. Percent rise in savings uses the **old savings** as base.

Weighted percent. Whole change = $wa + (1 - w)b$. To find the weight from the result, use alligation: weights are in the inverse ratio of the distances from the overall figure.

Top 5 traps.

1. Never add successive percent changes. +8% then –5% is +2.6% net ($1.08 \times 0.95 = 1.026$), not +3%.
2. The base flips when the comparison flips. $x\%$ more one way is **not** $x\%$ less the other way.
3. Up $x\%$ then down $x\%$ never returns to the start. It always loses $\frac{x^2}{100}$ percent.
4. Percentage error divides by the **correct** value, never by the wrong one.
5. “Fails by m marks” means score $+m$ = pass mark. Read the sentence, then write the equation.

Ratio, Proportion & Partnership

Parts, not sizes. Then money split by parts.

What you need to know

FORMULA SHEET

Ratio. $a : b$ means $\frac{a}{b}$. Multiply or divide both terms by the same non-zero number and the ratio does not change. a is the **antecedent**, b the **consequent**.

Parts method. If two things are in ratio $a : b$, write them as ak and bk . One unknown, not two.

Splitting a total. Divide N in the ratio $a : b : c$:

$$\text{share of first} = N \times \frac{a}{a+b+c}, \quad \text{one part} = \frac{N}{a+b+c}$$

Chaining ratios. If $a : b = p : q$ and $b : c = r : s$, then

$$a : b : c = pr : qr : qs$$

Compounded ratio. $(a : b)$ compounded with $(c : d) = ac : bd$.

Powers of a ratio.

Name	of $a : b$	Name	of $a : b$
duplicate	$a^2 : b^2$	sub-duplicate	$\sqrt{a} : \sqrt{b}$
triplicate	$a^3 : b^3$	sub-triplicate	$\sqrt[3]{a} : \sqrt[3]{b}$

Proportion. $a : b = c : d$ means $ad = bc$. Here a, d are **extremes**, b, c are **means**.

- Fourth proportional to a, b, c : $x = \frac{bc}{a}$.
- Third proportional to a, b : $x = \frac{b^2}{a}$ (from $a : b = b : x$).
- Mean proportional between a and b : $x = \sqrt{ab}$ (from $a : x = x : b$).
- Continued proportion $a : b = b : c$ gives $b^2 = ac$.
- If $a : b = b : c = c : d = k$, then $\frac{a}{d} = k^3$.

Componendo–dividendo. If $\frac{a}{b} = \frac{c}{d}$ then

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Equal-ratio addition. If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = k$, then $\frac{a+c+e}{b+d+f} = k$.

Variation.

- Direct: $x \propto y \rightarrow x = ky \rightarrow \frac{x_1}{y_1} = \frac{x_2}{y_2}$.
- Inverse: $x \propto \frac{1}{y} \rightarrow xy = k \rightarrow x_1y_1 = x_2y_2$.
- Joint: $x \propto \frac{yz}{w} \rightarrow x = \frac{k y z}{w}$.

Partnership.

- Same time for everyone: profit ratio = capital ratio.
- Different times: profit ratio = capital $\times$ time (call it **capital-months**).
- Capital changed mid-year: add the pieces. $C_1t_1 + C_2t_2$.
- Working partner: take out his salary or commission **first**, then split what is left by the capital ratio.
- Sleeping partner: capital only, no salary.

Warm-up

WARM-UP

build the reflex

Example 1

Simplify the ratio $84 : 126$.

SOLUTION

HCF of 84 and 126 is 42. $84 \div 42 = 2$. $126 \div 42 = 3$.

Answer: $2 : 3$

Example 2

If $a : b = 3 : 4$ and $b : c = 6 : 7$, find $a : b : c$.

SOLUTION

Make b the same. LCM of 4 and 6 is 12.

$a : b = 3 : 4 = 9 : 12$ (multiply by 3). $b : c = 6 : 7 = 12 : 14$ (multiply by 2).

Answer: $9 : 12 : 14$

Example 3

Divide Rs. 960 in the ratio $5 : 7$.

SOLUTION

Total parts $= 5 + 7 = 12$. One part $= 960 \div 12 = 80$.

First $= 5 \times 80 = 400$. Second $= 7 \times 80 = 560$. Check: $400 + 560 = 960$.

Answer: Rs. 400 and Rs. 560

Example 4

Find the ratio compounded of $3 : 4$ and $8 : 9$.

SOLUTION

Multiply across: $(3 \times 8) : (4 \times 9) = 24 : 36$. Divide both by 12: $2 : 3$.

Answer: $2 : 3$

Example 5

Write (i) the duplicate ratio of $5 : 6$, (ii) the sub-duplicate ratio of $49 : 81$.

SOLUTION

(i) Square both: $5^2 : 6^2 = 25 : 36$.

(ii) Square-root both: $\sqrt{49} : \sqrt{81} = 7 : 9$.

Answer: (i) $25 : 36$ (ii) $7 : 9$

Example 6

Find the fourth proportional to 4, 6 and 14.

SOLUTION

$4 : 6 = 14 : x \rightarrow 4x = 6 \times 14 = 84 \rightarrow x = \frac{84}{4} = 21$.

Answer: 21

Example 7

Find (i) the mean proportional between 8 and 32, (ii) the third proportional to 9 and 12.

SOLUTION

(i) $x = \sqrt{8 \times 32} = \sqrt{256} = 16.$

(ii) $9 : 12 = 12 : x \rightarrow 9x = 144 \rightarrow x = 16.$

Answer: (i) 16 (ii) 16

Example 8

x varies inversely as y . $x = 12$ when $y = 5$. Find x when $y = 15$.

SOLUTION

Inverse means $xy = k$. So $k = 12 \times 5 = 60$.

When $y = 15$: $x = \frac{60}{15} = 4$.

Answer: $x = 4$

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Two numbers are in the ratio 5 : 8. If 9 is added to each, the ratio becomes 8 : 11. Find the numbers.

SOLUTION

Let the numbers be $5x$ and $8x$.

$$\frac{5x + 9}{8x + 9} = \frac{8}{11}$$

Cross-multiply: $11(5x + 9) = 8(8x + 9)$.

Left: $55x + 99$. Right: $64x + 72$.

$$55x + 99 = 64x + 72$$

$$99 - 72 = 64x - 55x$$

$$27 = 9x \rightarrow x = 3.$$

Numbers: $5 \times 3 = 15$ and $8 \times 3 = 24$.

Check: $(15 + 9) : (24 + 9) = 24 : 33 = 8 : 11$. Correct.

Answer: 15 and 24

SHORTCUT

Always use one letter, never two. A ratio 5 : 8 is $5x$ and $8x$ — that is one unknown. Students who write a and b and then $\frac{a}{b} = \frac{5}{8}$ need two equations and lose 30 seconds.

Example 10 [Infosys pattern]

Rs. 3,150 is divided among A, B and C so that $A : B = 2 : 3$ and $B : C = 4 : 5$. Find C's share.

SOLUTION

Make B the same in both. LCM of 3 and 4 is 12.

$$A : B = 2 : 3 = 8 : 12 \text{ (multiply by 4).}$$

$$B : C = 4 : 5 = 12 : 15 \text{ (multiply by 3).}$$

$$\text{So } A : B : C = 8 : 12 : 15.$$

Total parts = $8 + 12 + 15 = 35$.
 One part = $3150 \div 35 = 90$.
 $C = 15 \times 90 = 1350$.
 Check: $A = 720$, $B = 1080$; $720 + 1080 + 1350 = 3150$. Correct.

Answer: Rs. 1,350

Example 11 [Capgemini pattern]

The salaries of Arun and Bala are in the ratio 4 : 5. Each gets an increment of Rs. 3,000 and the new ratio is 5 : 6. Find Bala's new salary.

SOLUTION

Salaries: $4x$ and $5x$.

$$\frac{4x + 3000}{5x + 3000} = \frac{5}{6}$$

$$6(4x + 3000) = 5(5x + 3000)$$

$$24x + 18000 = 25x + 15000$$

$$18000 - 15000 = 25x - 24x$$

$$3000 = x.$$

$$\text{Bala's old salary} = 5 \times 3000 = 15000.$$

$$\text{Bala's new salary} = 15000 + 3000 = 18000.$$

$$\text{Check: Arun new} = 12000 + 3000 = 15000; 15000 : 18000 = 5 : 6. \text{ Correct.}$$

Answer: Rs. 18,000

Example 12 [TCS NQT pattern]

Rs. 8,400 is divided among P, Q and R so that P gets twice what Q gets, and Q gets three times what R gets. Find Q's share.

SOLUTION

Start from the smallest. Let $R = x$.

Q gets three times R $\rightarrow Q = 3x$.

P gets twice Q $\rightarrow P = 2 \times 3x = 6x$.

$$\text{Total: } 6x + 3x + x = 10x = 8400 \rightarrow x = 840.$$

$$Q = 3 \times 840 = 2520.$$

$$\text{Check: } P = 5040, R = 840; 5040 + 2520 + 840 = 8400. \text{ Correct.}$$

Answer: Rs. 2,520

TRAP

“P gets twice what Q gets” means $P = 2Q$, so $P : Q = 2 : 1$. Many students write 1 : 2. Read it as an equation first, then turn it into a ratio.

Example 13 [Cognizant pattern]

A 21-litre mixture has milk and water in the ratio 5 : 2. How much water must be added to make the ratio 5 : 3?

SOLUTION

$$\text{Parts} = 5 + 2 = 7. \text{ One part} = 21 \div 7 = 3 \text{ L.}$$

$$\text{Milk} = 5 \times 3 = 15 \text{ L. Water} = 2 \times 3 = 6 \text{ L.}$$

Adding water does not change the milk. Milk stays 15 L.

In the new ratio 5 : 3, milk is 5 parts = 15 L, so one part = 3 L.

$$\text{New water} = 3 \times 3 = 9 \text{ L.}$$

$$\text{Water to add} = 9 - 6 = 3 \text{ L.}$$

Answer: 3 litres

Example 14 [Infosys pattern]

A box has one-rupee, 50-paise and 25-paise coins. The **numbers** of the coins are in the ratio 5 : 6 : 8. The total value is Rs. 30. How many 50-paise coins are there?

SOLUTION

Let the counts be $5x$, $6x$, $8x$.

Value of one-rupee coins = $5x \times 1 = 5x$ rupees.

Value of 50-paise coins = $6x \times 0.50 = 3x$ rupees.

Value of 25-paise coins = $8x \times 0.25 = 2x$ rupees.

Total value = $5x + 3x + 2x = 10x = 30 \rightarrow x = 3$.

Number of 50-paise coins = $6 \times 3 = 18$.

Check: 15 coins of Re 1 = Rs. 15; 18 coins of 50p = Rs. 9; 24 coins of 25p = Rs. 6. Total Rs. 30. Correct.

Answer: 18 coins

TRAP

Ratio of coin numbers is not the ratio of coin values. Convert every count to money before you add. If the question had said “values are in the ratio 5 : 6 : 8”, the working would be completely different.

Example 15 [TCS NQT pattern]

Find the ratio compounded of 2 : 3, the duplicate ratio of 4 : 5, the sub-duplicate ratio of 36 : 49 and the triplicate ratio of 1 : 2.

SOLUTION

Write each ratio first.

Duplicate of 4 : 5 = 16 : 25.

Sub-duplicate of 36 : 49 = 6 : 7.

Triplicate of 1 : 2 = 1 : 8.

Now compound (multiply all the first terms, multiply all the second terms).

Numerator: $2 \times 16 \times 6 \times 1 = 192$.

Denominator: $3 \times 25 \times 7 \times 8 = 4200$.

192 : 4200. Divide both by 8: 24 : 525. Divide both by 3: 8 : 175.

Answer: 8 : 175

Example 16 [Wipro pattern]

x varies directly as y^2 and inversely as z . When $y = 4$ and $z = 6$, $x = 12$. Find x when $y = 6$ and $z = 9$.

SOLUTION

$$x = \frac{ky^2}{z}$$

Put in the known values: $12 = \frac{k \times 16}{6}$.

$12 \times 6 = 16k \rightarrow 72 = 16k \rightarrow k = \frac{72}{16} = 4.5$.

Now $y = 6$, $z = 9$:

$x = \frac{4.5 \times 36}{9} = 4.5 \times 4 = 18$.

Answer: $x = 18$

Example 17 [Accenture pattern]

A, B and C invest Rs. 45,000, Rs. 60,000 and Rs. 75,000 in a business for one full year. The year's profit is Rs. 36,000. Find B's share.

SOLUTION

Everyone invested for the same time, so profit splits in the capital ratio.

45000 : 60000 : 75000. Divide all by 15,000: 3 : 4 : 5.

Total parts = 12. One part = $36000 \div 12 = 3000$.

$B = 4 \times 3000 = 12000$.

Check: $A = 9000$, $C = 15000$; $9000 + 12000 + 15000 = 36000$. Correct.

Answer: Rs. 12,000

Example 18 [TCS NQT pattern]

A starts a business with Rs. 30,000. After 4 months B joins with Rs. 45,000. After 6 months from the start, C joins with Rs. 60,000. At the end of one year the profit is Rs. 49,500. Find each partner's share.

SOLUTION

Times invested, counting to the 12-month mark:

A: 12 months. B: $12 - 4 = 8$ months. C: $12 - 6 = 6$ months.

Capital-months (work in thousands to keep numbers small):

$A = 30 \times 12 = 360$.

$B = 45 \times 8 = 360$.

$C = 60 \times 6 = 360$.

Ratio = $360 : 360 : 360 = 1 : 1 : 1$.

So the profit splits equally: $49500 \div 3 = 16500$ each.

Answer: Rs. 16,500 each

SHORTCUT

Divide all capitals by a common factor before multiplying by time. Rs. 30,000 becomes 30. You are only after a ratio, so any common scaling is free. This kills most of the arithmetic.

Example 19 [Capgemini pattern]

A and B start a business with Rs. 24,000 and Rs. 36,000 for one year. A is the working partner and takes 15% of the profit for managing the business; the rest is divided in the capital ratio. If the profit is Rs. 24,000, what does A receive in total?

SOLUTION

Step 1 — A's management share.

15% of 24000 = $0.15 \times 24000 = 3600$.

Step 2 — what is left.

$24000 - 3600 = 20400$.

Step 3 — capital ratio.

$24000 : 36000 = 2 : 3$. Total parts = 5. One part = $20400 \div 5 = 4080$.

A's capital share = $2 \times 4080 = 8160$.

Step 4 — A's total.

$3600 + 8160 = 11760$.

Check: B gets $3 \times 4080 = 12240$; $11760 + 12240 = 24000$. Correct.

Answer: Rs. 11,760

Example 20 [Cognizant pattern]

If $\frac{5x+3y}{5x-3y} = \frac{7}{3}$, find $x : y$.

SOLUTION

Use componendo–dividendo. Add the two sides' parts and subtract them.

$$\frac{(5x + 3y) + (5x - 3y)}{(5x + 3y) - (5x - 3y)} = \frac{7 + 3}{7 - 3}$$

Numerator on the left: $5x + 3y + 5x - 3y = 10x$.

Denominator on the left: $5x + 3y - 5x + 3y = 6y$.

So $\frac{10x}{6y} = \frac{10}{4}$.

$$\frac{10x}{6y} = \frac{10}{4} \rightarrow \frac{x}{y} = \frac{10}{4} \times \frac{6}{10} = \frac{6}{4} = \frac{3}{2}.$$

Check with $x = 3, y = 2$: $\frac{15+6}{15-6} = \frac{21}{9} = \frac{7}{3}$. Correct.

Answer: $x : y = 3 : 2$

PRACTICE · TIER 1 target: 60 s/Q

- Simplify $1.2 : 0.8 : 2.0$.
- Divide Rs. 2,280 among A, B, C in the ratio $3 : 5 : 4$. Find B's share.
- If $a : b = 2 : 5$ and $b : c = 3 : 4$, find $a : c$.
- Two numbers are in the ratio $7 : 9$. If 12 is subtracted from each, the ratio becomes $5 : 7$. Find the larger number.
(a) 54 (b) 42 (c) 63 (d) 96
- Find the third proportional to 12 and 18.
- Find the mean proportional between 12.5 and 2.
- Find the ratio compounded of $5 : 6$ and the sub-duplicate ratio of $16 : 25$.
- A sum is divided in the ratio $4 : 7$ and the two shares differ by Rs. 2,100. Find the total sum.
- x varies inversely as $\sqrt{y}$. $x = 6$ when $y = 16$. Find y when $x = 4$.
- A and B invest Rs. 16,000 and Rs. 20,000. After 8 months A adds Rs. 8,000 more. The year's profit is Rs. 29,000. Find A's share.
- Three partners invest capitals in the ratio $3 : 4 : 5$ for times in the ratio $4 : 3 : 2$. The profit is Rs. 34,000. Find the third partner's share.
- A bag has Rs. 5, Rs. 2 and Re 1 coins whose numbers are in the ratio $3 : 4 : 5$. The total value is Rs. 336. How many Rs. 2 coins are there?
(a) 48 (b) 36 (c) 60 (d) 42
- In a college the ratio of students to teachers is $24 : 1$. If 40 more students join and no teacher is added, the ratio becomes $28 : 1$. How many teachers are there?
- If $a : b = 3 : 4$ and $b : c = 8 : 9$, find $(a + b + c) : c$.
- Rs. 2,700 is shared by A, B and C. A gets half of what B and C together get. B gets one-third of what A and C together get. Find C's share.
(a) Rs. 1,125 (b) Rs. 900 (c) Rs. 675 (d) Rs. 1,350

ANSWER KEY

- $3 : 2 : 5$ — multiply every term by 5 to clear decimals, then divide by 2.
- Rs. 950 — 12 parts, one part = 190, B = 5×190 .
- $3 : 10$ — make $b = 15$: $a : b = 6 : 15$, $b : c = 15 : 20$, so $a : c = 6 : 20$.
- (a) 54. $7(7x - 12) = 5(9x - 12) \rightarrow 49x - 84 = 45x - 60 \rightarrow 4x = 24 \rightarrow x = 6$; numbers 42 and 54. (b) 42 is the **smaller** number. (c) 63 comes from the slip $84 - 60 = 28$, giving $x = 7$ and $9x = 63$. (d) 96 is the **sum** $42 + 54$, not the larger number.
27. $x = \frac{18^2}{12} = \frac{324}{12}$.
5. $\sqrt{12.5 \times 2} = \sqrt{25}$.
- $2 : 3$ — sub-duplicate of $16 : 25$ is $4 : 5$; $(5 \times 4) : (6 \times 5) = 20 : 30$.
- Rs. 7,700 — difference is $7 - 4 = 3$ parts = 2100, so one part = 700 and total = 11×700 .

9. $y = 36$. $x\sqrt{y} = k = 6 \times 4 = 24$; $\sqrt{y} = \frac{24}{4} = 6$.

10. Rs. 14,000 — $A = 16 \times 8 + 24 \times 4 = 128 + 96 = 224$; $B = 20 \times 12 = 240$; ratio $224 : 240 = 14 : 15$, one part $= \frac{29000}{29} = 1000$.

11. Rs. 10,000 — profit ratio $= 3 \times 4 : 4 \times 3 : 5 \times 2 = 12 : 12 : 10 = 6 : 6 : 5$; one part $= \frac{34000}{17} = 2000$.

12. (a) 48 — value $= 15x + 8x + 5x = 28x = 336 \rightarrow x = 12$, so Rs. 2 coins $= 4x$. (b) 36 comes from using $x = 12$ with the Rs. 5 count. (c) 60 is the Re 1 count. (d) 42 comes from dividing 336 by 8 instead of 28.

13. 10 — let teachers $= t$, students $= 24t$; $24t + 40 = 28t \rightarrow 4t = 40$.

14. $23 : 9$. $a : b : c = 6 : 8 : 9$, sum $= 23$.

15. (a) Rs. 1,125 — $A = \frac{1}{3}$ of total $= 900$, $B = \frac{1}{4}$ of total $= 675$, $C = 2700 - 1575$. (b) 900 is A's share. (c) 675 is B's share. (d) 1,350 comes from treating "half of what B and C get" as "half the total".

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

On a ride-hailing app the fare of every trip is split between driver and platform in the ratio 4 : 1. The driver also pays a booking levy of S\$0.40 per trip out of his own share. On Monday a driver completed 25 trips with total fare S\$625. Find his net earning.

SOLUTION

Step 1 — driver's gross share.

$$\text{Total parts} = 4 + 1 = 5. \text{ One part} = 625 \div 5 = 125.$$

$$\text{Driver} = 4 \times 125 = 500.$$

Step 2 — levy.

$$25 \times 0.40 = 10.$$

Step 3 — net.

$$500 - 10 = 490.$$

Answer: S\$490

Example 22 [Shopee · pattern]

Three sellers share a warehouse bill of S\$15,000 in proportion to (space used $\times$ months stored). Seller A used 40 m³ for 6 months, B used 30 m³ for 7 months, C used 50 m³ for 3 months. Find each seller's bill.

SOLUTION

This is the partnership rule with space instead of money.

$$A = 40 \times 6 = 240.$$

$$B = 30 \times 7 = 210.$$

$$C = 50 \times 3 = 150.$$

$$\text{Total} = 240 + 210 + 150 = 600.$$

$$\text{One unit} = 15000 \div 600 = 25.$$

$$A = 240 \times 25 = 6000.$$

$$B = 210 \times 25 = 5250.$$

$$C = 150 \times 25 = 3750.$$

$$\text{Check: } 6000 + 5250 + 3750 = 15000. \text{ Correct.}$$

Answer: A S\$6,000, B S\$5,250, C S\$3,750

Example 23 [Agoda · pattern]

A Bangkok hotel has deluxe and standard rooms in the ratio 3 : 7. On one night 80% of the deluxe rooms and 60% of the standard rooms were occupied, giving 330 occupied rooms in all. How many rooms does the hotel have?

SOLUTION

Let deluxe = $3k$ and standard = $7k$.

Occupied deluxe = $0.80 \times 3k = 2.4k$.

Occupied standard = $0.60 \times 7k = 4.2k$.

Total occupied = $2.4k + 4.2k = 6.6k$.

$6.6k = 330 \rightarrow k = \frac{330}{6.6} = 50$.

Total rooms = $3k + 7k = 10k = 500$.

Check: deluxe 150, occupied 120; standard 350, occupied 210; $120 + 210 = 330$. Correct.

Answer: 500 rooms

Example 24 [DBS · pattern]

A fund holds Singapore dollars and Thai baht. The **values** of the two holdings are in the ratio 5 : 3. The exchange rate is 1 SGD = 25 THB. The baht holding is THB 1,875,000. Find the SGD holding.

SOLUTION

Step 1 — put both sides in the same currency. Convert the baht to SGD.

$1875000 \div 25 = 75000$ SGD.

Step 2 — use the ratio.

SGD holding : baht holding (in SGD) = 5 : 3.

3 parts = 75000 $\rightarrow$ one part = 25000.

SGD holding = $5 \times 25000 = 125000$.

Answer: S\$125,000

TRAP

Never form a ratio across two currencies without converting first. 5 : 3 compared “SGD value” with “SGD value”. Writing $x : 1875000 = 5 : 3$ gives a nonsense answer of 3.1 million.

Example 25 [Sea / Shopee · pattern]

Priya starts a shop with THB 240,000. After 5 months she withdraws THB 40,000. Wei joins 3 months after the start with THB 300,000 and 4 months later adds THB 60,000. At the end of 12 months the profit is THB 280,000. Split it.

SOLUTION

Work in thousands of baht.

Priya. 240 for the first 5 months, then $240 - 40 = 200$ for the remaining $12 - 5 = 7$ months.

$240 \times 5 = 1200$

$200 \times 7 = 1400$

Total = $1200 + 1400 = 2600$.

Wei. He joins at the end of month 3, so he is in for $12 - 3 = 9$ months. For the first 4 of those he has 300; for the remaining $9 - 4 = 5$ he has $300 + 60 = 360$.

$300 \times 4 = 1200$

$360 \times 5 = 1800$

Total = $1200 + 1800 = 3000$.

Ratio. $2600 : 3000 = 26 : 30 = 13 : 15$.

Total parts = 28. One part = $280000 \div 28 = 10000$.

Priya = $13 \times 10000 = 130000$. Wei = $15 \times 10000 = 150000$.

Check: $130000 + 150000 = 280000$. Correct.

Answer: Priya THB 130,000, Wei THB 150,000

SHORTCUT

Capital-months as a bar chart. Draw a 12-month line for each partner and mark where the capital changes. Each flat piece is one rectangle: height $\times$ width. Add the rectangles. You will never lose a segment again.

Example 26 [GIC · pattern]

The time to process a data batch varies directly as the number of records and inversely as the number of servers. 12 servers process 4.8 million records in 5 hours. How long will 18 servers take for 10.8 million records?

SOLUTION

$$t = \frac{kR}{S}$$

where R is records in millions and S is servers.

Put in the known values: $5 = \frac{k \times 4.8}{12}$.

$$5 \times 12 = 4.8k \rightarrow 60 = 4.8k \rightarrow k = \frac{60}{4.8} = 12.5.$$

Now $R = 10.8$, $S = 18$:

$$t = \frac{12.5 \times 10.8}{18}.$$

$$12.5 \times 10.8 = 135.$$

$$t = \frac{135}{18} = 7.5 \text{ hours.}$$

Answer: 7.5 hours

Example 27 [SCB · pattern]

Branch A has a book of THB 96 million with loans : deposits = 3 : 5. Branch B has a book of THB 144 million with loans : deposits = 5 : 7. Find the combined loans : deposits ratio for the two branches together.

SOLUTION

Branch A. Parts = $3 + 5 = 8$. One part = $96 \div 8 = 12$.

Loans = $3 \times 12 = 36$. Deposits = $5 \times 12 = 60$.

Branch B. Parts = $5 + 7 = 12$. One part = $144 \div 12 = 12$.

Loans = $5 \times 12 = 60$. Deposits = $7 \times 12 = 84$.

Combined. Loans = $36 + 60 = 96$. Deposits = $60 + 84 = 144$.

96 : 144. Divide both by 48: 2 : 3.

Answer: 2 : 3

TRAP

You cannot add ratios term by term. 3 : 5 and 5 : 7 do not combine into 8 : 12. Convert each ratio to real amounts, add the amounts, then re-form the ratio.

Example 28 [LINE MAN · pattern]

A delivery fleet has bikes and cars in the ratio 9 : 4. After 15 bikes are retired and 10 cars are added, the ratio becomes 3 : 2. How many bikes were there at the start?

SOLUTION

Let bikes = $9k$ and cars = $4k$.

$$\frac{9k - 15}{4k + 10} = \frac{3}{2}$$

$$2(9k - 15) = 3(4k + 10)$$

$$18k - 30 = 12k + 30$$

$$18k - 12k = 30 + 30$$

$$6k = 60 \rightarrow k = 10.$$

Bikes at the start = $9 \times 10 = 90$.

Check: bikes $90 - 15 = 75$; cars $40 + 10 = 50$; $75 : 50 = 3 : 2$. Correct.

Answer: 90 bikes

Example 29 [Razer · pattern]

The cost of a gaming headset splits into materials, labour and marketing in the ratio 7 : 4 : 9. Next year materials cost rises 20%, labour rises 25% and marketing falls 10%. Find the new cost ratio and the percent change in total cost.

SOLUTION

Take the three costs as 7, 4 and 9 units. Total = 20 units.

Materials: $7 \times 1.20 = 8.4$.

Labour: $4 \times 1.25 = 5.0$.

Marketing: $9 \times 0.90 = 8.1$.

New ratio = $8.4 : 5.0 : 8.1$. Multiply all by 10: $84 : 50 : 81$.

New total = $8.4 + 5.0 + 8.1 = 21.5$.

Change = $21.5 - 20 = 1.5$ units.

Percent change = $\frac{1.5}{20} \times 100 = 7.5\%$ increase.

Answer: 84 : 50 : 81; total cost up 7.5%

Example 30 [Grab · pattern]

In zone X the ratio of orders to riders is 12 : 1. In zone Y it is 9 : 1. The two zones are merged and the combined ratio of orders to riders is 10 : 1. Find the ratio of the number of riders in X to the number in Y.

SOLUTION

Let X have x riders and Y have y riders.

Orders in X = $12x$. Orders in Y = $9y$.

Combined riders = $x + y$. Combined orders = $12x + 9y$.

$$\frac{12x + 9y}{x + y} = \frac{10}{1}$$

$$12x + 9y = 10x + 10y$$

$$12x - 10x = 10y - 9y$$

$$2x = y.$$

$$\text{So } x : y = 1 : 2.$$

Check: take $x = 1, y = 2$. Orders = $12 + 18 = 30$; riders = 3; $30 : 3 = 10 : 1$. Correct.

Answer: 1 : 2

Example 30 is the **alligation** idea in disguise: the combined ratio 10 sits between 9 and 12, and it lands closer to 9, so zone Y must carry more weight. Chapter 4 makes this a one-line rule.

PRACTICE · TIER 2 target: 100 s/Q

1. A courier fare is split driver : platform in the ratio 7 : 3. A driver earns S\$420 in a day. Find the platform's revenue and the total fare collected.
2. A Bangkok café sells tea and coffee in the ratio 5 : 4. Next month tea sales rise 20% and coffee sales fall 25%. Find the new ratio.
3. A fund splits assets bonds : equity : cash in the ratio 8 : 11 : 1. Cash is S\$2.4 million. Find the total assets.
4. Meera invests Rs. 84,000 for 8 months and Raj invests Rs. 63,000 for 12 months. The profit is Rs. 1,02,000. Find each share.

- x varies inversely as the cube of y . $x = 5$ when $y = 2$. Find x when $y = 4$.
- A Phuket hotel has deluxe and suite rooms in the ratio 8 : 3, charged at THB 3,500 and THB 9,000 per night. If every room is sold, find the ratio of deluxe revenue to suite revenue.
- In a class the ratio of boys to girls is 5 : 3. After 10 boys leave and 6 girls join, the ratio is 4 : 3. Find the original class size.
- Three partners invest capitals in the ratio 5 : 7 : 8 for times in the ratio 6 : 4 : 3. The profit is S\$82,000. Split it.
- A logistics cost varies directly as distance and inversely as the load factor. The cost is THB 9,000 for 300 km at load factor 0.8. Find the cost for 450 km at load factor 0.9.
- Branch A has staff : customers = 1 : 40; branch B has 1 : 30. Together the ratio is 1 : 36. Find the ratio of staff in A to staff in B.
- If $\frac{2x+3y}{2x-3y} = \frac{11}{5}$, find $x : y$.
- A bonus pool of S\$2,40,000 is split in proportion to (headcount $\times$ months active). Team A: 10 people for 9 months. Team B: 12 people for 5 months. Team C: 15 people for 10 months. Find each team's bonus.

ANSWER KEY

- Platform S\$180, total fare S\$600 — 7 parts = 420, one part = 60.
- 2 : 1 — new tea = $5 \times 1.2 = 6$, new coffee = $4 \times 0.75 = 3$, so 6 : 3.
- S\$48 million — cash is 1 part = 2.4, total = 20 parts.
- Meera Rs. 48,000, Raj Rs. 54,000 — capital-months $84 \times 8 = 672$ and $63 \times 12 = 756$; $672 : 756 = 8 : 9$; one part = $\frac{102000}{17} = 6000$.
- $x = \frac{5}{8} = 0.625$. $k = xy^3 = 5 \times 8 = 40$; $x = \frac{40}{64}$.
- 28 : 27 — revenue = $8 \times 3500 = 28000$ and $3 \times 9000 = 27000$ per 11 rooms.
- 144 students. $3(5k - 10) = 4(3k + 6) \rightarrow 15k - 30 = 12k + 24 \rightarrow k = 18$; boys 90, girls 54.
- S\$30,000 / S\$28,000 / S\$24,000 — profit ratio = 30 : 28 : 24 = 15 : 14 : 12; one part = $\frac{82000}{41} = 2000$.
- THB 12,000. $C = k\frac{d}{L} \rightarrow 9000 = k \times \frac{300}{0.8} = 375k \rightarrow k = 24$; $C = 24 \times \frac{450}{0.9} = 24 \times 500$.
- 3 : 2. $40a + 30b = 36(a + b) \rightarrow 4a = 6b$.
- $x : y = 4 : 1$ — componendo-dividendo gives $\frac{4x}{6y} = \frac{16}{6}$, so $\frac{x}{y} = \frac{16}{6} \times \frac{6}{4} = 4$.
- A S\$72,000, B S\$48,000, C S\$1,20,000 — weights 90 : 60 : 150, total 300, one unit = 800.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Goldman Sachs · pattern]

Four positive numbers satisfy $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$. If $a = 27$ and $d = 8$, find $b + c$.

SOLUTION

Insight: in a continued proportion the ratio of the two ends is the common ratio cubed.

Let the common ratio be k , so $a = kb$, $b = kc$, $c = kd$.

Then

$$a = kb = k(kc) = k^2c = k^2(kd) = k^3d$$

So $\frac{a}{d} = k^3$.

$$k^3 = \frac{27}{8} \rightarrow k = \sqrt[3]{\frac{27}{8}} = \frac{3}{2}$$

Now walk back from d :

$$c = kd = \frac{3}{2} \times 8 = 12.$$

$$b = kc = \frac{3}{2} \times 12 = 18.$$

Check $a = kb = \frac{3}{2} \times 18 = 27$. Correct.

$$b + c = 18 + 12 = 30.$$

Answer: $b + c = 30$

Example 32 [Amazon · pattern]

A rectangle has length : breadth = 5 : 3. The same positive number is added to both the length and the breadth, and the new ratio is 7 : 5. By what percent does the area increase?

SOLUTION

Insight: adding the same amount to both terms of a ratio always pushes the ratio toward 1 : 1. So 7 : 5 is possible ($\frac{7}{5} < \frac{5}{3}$), and the amount added is forced.

Let length = $5k$, breadth = $3k$, and let x be added to each.

$$\frac{5k + x}{3k + x} = \frac{7}{5}$$

$$5(5k + x) = 7(3k + x)$$

$$25k + 5x = 21k + 7x$$

$$25k - 21k = 7x - 5x$$

$$4k = 2x \rightarrow x = 2k.$$

New length = $5k + 2k = 7k$. New breadth = $3k + 2k = 5k$. (Ratio 7 : 5. Correct.)

$$\text{Old area} = 5k \times 3k = 15k^2.$$

$$\text{New area} = 7k \times 5k = 35k^2.$$

$$\text{Increase} = 35k^2 - 15k^2 = 20k^2.$$

$$\text{Percent increase} = \frac{20k^2}{15k^2} \times 100 = \frac{4}{3} \times 100 = 133\frac{1}{3}\%.$$

Answer: $133\frac{1}{3}\%$ (that is, $\approx 133.33\%$)

Example 33 [D. E. Shaw · pattern]

A team's wins to losses stand at 5 : 8. The team then wins k more games in a row and loses none, after which the ratio is 5 : 6. Find the smallest possible number of games the team had played before this winning run.

SOLUTION

Insight: the ratio fixes k only up to a fraction. The real constraint is that wins, losses and k must all be whole numbers.

Let wins = $5m$ and losses = $8m$, with m a positive integer.

Losses do not change. After the run:

$$\frac{5m + k}{8m} = \frac{5}{6}$$

$$6(5m + k) = 5 \times 8m$$

$$30m + 6k = 40m$$

$$6k = 10m \rightarrow k = \frac{5m}{3}.$$

For k to be a whole number, m must be a multiple of 3. The smallest is $m = 3$.

Then wins = 15, losses = 24, and $k = 5$.

Games played before the run = $15 + 24 = 39$.

Check: after 5 more wins, $20 : 24 = 5 : 6$. Correct.

Answer: 39 games (with $k = 5$)

Example 34 [Google · pattern]

Two ad channels are compared on click-through rate.

	Mobile	CTR	Desktop	CTR
Channel P	63 of 90	70%	2 of 10	20%
Channel Q	8 of 10	80%	27 of 90	30%

Channel Q beats channel P on mobile and on desktop. Which channel has the better overall CTR?

SOLUTION

Insight: an overall rate is a weighted average of the sub-rates. The weights are the traffic sizes, and they can flip the comparison.

Channel P overall: clicks = $63 + 2 = 65$; impressions = $90 + 10 = 100$.

$$\text{CTR} = \frac{65}{100} = 65\%.$$

Channel Q overall: clicks = $8 + 27 = 35$; impressions = $10 + 90 = 100$.

$$\text{CTR} = \frac{35}{100} = 35\%.$$

So P wins overall, 65% to 35%, even though it loses both sub-groups.

Why: P put 90 of its 100 impressions on mobile, where rates are high. Q put 90 of its 100 impressions on desktop, where rates are low. The overall number measures the traffic mix as much as the channel quality.

Answer: Channel P, 65% against 35% — the sub-group winner is not the overall winner

TRAP

Never average two ratios by averaging their values. $\frac{70+20}{2} = 45\%$ for P is wrong; the true answer is 65%. Always go back to totals: add the numerators, add the denominators, then divide.

Example 35 [Adobe · pattern]

Three numbers are in the ratio 3 : 4 : 5 and their LCM is 2400. Find the numbers and their HCF.

SOLUTION

Insight: if the ratio terms are in lowest form, the numbers are $3x, 4x, 5x$ with $\text{HCF} = x$, and the LCM is x times the LCM of the ratio terms.

Let the numbers be $3x, 4x, 5x$.

Since 3, 4, 5 have HCF 1, the HCF of the numbers is x .

LCM of 3, 4, 5:

$$3 = 3, 4 = 2^2, 5 = 5, \text{ so } \text{LCM} = 2^2 \times 3 \times 5 = 60.$$

Therefore LCM of the numbers = $60x$.

$$60x = 2400 \rightarrow x = 40$$

Numbers: $3 \times 40 = 120, 4 \times 40 = 160, 5 \times 40 = 200$. HCF = 40.

Check: $120 = 2^3 \times 3 \times 5, 160 = 2^5 \times 5, 200 = 2^3 \times 5^2$.

$\text{LCM} = 2^5 \times 3 \times 5^2 = 32 \times 3 \times 25 = 2400$. Correct.

$\text{HCF} = 2^3 \times 5 = 40$. Correct.

Answer: 120, 160, 200; HCF = 40

Example 36 [Uber · pattern]

A city platform keeps a service rule: riders : drivers must stay at or below 40 : 1. Right now there are 42,000 riders and 1,200 drivers. Next month riders will grow 5% and 8% of the current drivers will quit. How many new drivers must be hired to keep the rule?

SOLUTION

Insight: do not compare the growth rates. Compute both sides next month and compare the needed count with the surviving count.

Step 1 — riders next month.

$$42000 \times 1.05 = 44100.$$

Step 2 — drivers still on the platform next month.

8% quit, so 92% remain: $1200 \times 0.92 = 1104$.

Step 3 — drivers **needed** next month.

The rule is $\frac{\text{riders}}{\text{drivers}} \leq 40$, so drivers $\geq \frac{44100}{40} = 1102.5$.

Drivers are whole people, so at least 1,103 are needed.

Step 4 — compare.

Surviving drivers = $1104 \geq 1103$. The rule already holds.

Check: $44100 \div 1104 = 39.95$, which is below 40. Correct.

Answer: No hiring needed — 0 new drivers

Riders grew 5% and drivers fell 8%, so the ratio did get worse. It just did not get worse **enough** to break the rule. Tier-3 questions reward finishing the arithmetic instead of guessing from the direction of change.

Example 37 [Microsoft · pattern]

Prove that if $\frac{a}{b} = \frac{c}{d}$ then

$$\frac{a^2 + c^2}{b^2 + d^2} = \frac{ac}{bd}$$

Then use it: if $\frac{a^2 + c^2}{b^2 + d^2} = \frac{16}{49}$, $\frac{a}{b} = \frac{c}{d}$ and $ac = 48$, find bd .

SOLUTION

Insight: when two fractions are equal, name the common value r and substitute. Every expression collapses to a power of r .

Let $\frac{a}{b} = \frac{c}{d} = r$. Then $a = rb$ and $c = rd$.

Left side.

$$a^2 + c^2 = r^2b^2 + r^2d^2 = r^2(b^2 + d^2)$$

$$\frac{a^2 + c^2}{b^2 + d^2} = \frac{r^2(b^2 + d^2)}{b^2 + d^2} = r^2$$

Right side.

$$ac = (rb)(rd) = r^2bd$$

$$\frac{ac}{bd} = \frac{r^2bd}{bd} = r^2$$

Both sides equal r^2 , so they are equal. Proved.

Now the numbers.

$$r^2 = \frac{16}{49}.$$

$$\frac{ac}{bd} = r^2 = \frac{16}{49}$$

$$\frac{48}{bd} = \frac{16}{49} \rightarrow 16 \times bd = 48 \times 49$$

$$bd = \frac{48 \times 49}{16} = 3 \times 49 = 147.$$

Check with $r = \frac{4}{7}$: take $b = 7, d = 21$, so $bd = 147$; then $a = 4, c = 12$ and $ac = 48$. Correct.

Answer: $bd = 147$

Example 38 [Goldman Sachs · pattern]

A invests Rs. 50,000 for 12 months, B invests Rs. 75,000 for 8 months and C invests Rs. 1,00,000 for 12 months. They agree that C is **guaranteed** at least Rs. 46,000 of the profit, and any shortfall is paid by A and B in their own profit ratio. The year's profit is Rs. 80,000. Find each partner's final share.

SOLUTION

Insight: a guarantee is a second step, not a new ratio. Split normally first, then transfer.

Step 1 — capital-months (in thousands).

$$A = 50 \times 12 = 600.$$

$$B = 75 \times 8 = 600.$$

$$C = 100 \times 12 = 1200.$$

$$\text{Ratio} = 600 : 600 : 1200 = 1 : 1 : 2.$$

Step 2 — normal split.

$$\text{Total parts} = 4. \text{ One part} = 80000 \div 4 = 20000.$$

$$A = 20000, B = 20000, C = 40000.$$

Step 3 — the guarantee.

C is promised 46,000 but the normal split gives 40,000.

$$\text{Shortfall} = 46000 - 40000 = 6000.$$

Step 4 — who pays.

A and B pay in their own ratio, which is 1 : 1. So each pays $6000 \div 2 = 3000$.

$$A = 20000 - 3000 = 17000.$$

$$B = 20000 - 3000 = 17000.$$

$$C = 40000 + 6000 = 46000.$$

$$\text{Check: } 17000 + 17000 + 46000 = 80000. \text{ Correct.}$$

Answer: A Rs. 17,000, B Rs. 17,000, C Rs. 46,000

PRACTICE · TIER 3 target: 4 min/Q

- Four positive numbers satisfy $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$. If $a = 81$ and $d = 24$, find $b + c$.
- Two positive integers are in the ratio 4 : 9. When 5 is added to both, the ratio becomes 9 : 14. Find the two integers, and explain why the answer is forced.
- Two teaching methods were tried on 100 students each. Method A: 45 of 50 passed in the strong section and 12 of 50 in the weak section. Method B: 19 of 20 passed in the strong section and 26 of 80 in the weak section. Method B has the higher pass rate in **both** sections. Which method has the higher overall pass rate, and why?
- Three numbers are in the ratio 2 : 3 : 5 and their HCF is 14. Find their LCM.
- X and Y invest Rs. 1,20,000 and Rs. 80,000 for 12 months. Z joins as a working partner with no capital and is guaranteed one-fifth of the profit; the rest is split in the capital ratio. The profit is Rs. 1,50,000. Find Y's share.
- A delivery platform has 900 couriers and 27,000 weekly orders. Its rule is orders : couriers $\leq 32 : 1$. Next month orders grow 12% and 5% of couriers quit. How many couriers must be hired?
- If $\frac{a}{b} = \frac{c}{d}$, $\frac{a^2+c^2}{b^2+d^2} = \frac{9}{25}$ and $bd = 200$, find ac .
- A batsman's runs to balls faced stand at 3 : 5, and he has scored 60 runs. He then scores n more runs off n more balls, and the ratio becomes 5 : 7. Find n , and say in one line why the ratio had to rise.

ANSWER KEY

- $b + c = 90$. The ratio of the ends is the common ratio cubed: $k^3 = \frac{a}{d} = \frac{81}{24} = \frac{27}{8}$, so $k = \frac{3}{2}$. Then $c = kd = \frac{3}{2} \times 24 = 36$ and $b = kc = \frac{3}{2} \times 36 = 54$. Check $a = \frac{3}{2} \times 54 = 81$. Sum $= 54 + 36 = 90$.
- The integers are 4 and 9. Write them as $4k$ and $9k$: $14(4k + 5) = 9(9k + 5) \rightarrow 56k + 70 = 81k + 45 \rightarrow 25 = 25k \rightarrow k = 1$. The answer is forced because adding a **fixed** number, not a fixed proportion, ties the ratio change to the actual size of the numbers — so only one size works.
- Method A, 57% against 45%. A passed $45 + 12 = 57$ of 100; B passed $19 + 26 = 45$ of 100. B won both sections but sent 80 of its 100 students into the weak section, where everyone does badly. The overall figure is a weighted average and the weights decided it.

4. $\text{LCM} = 420$. The numbers are $2 \times 14 = 28$, $3 \times 14 = 42$, $5 \times 14 = 70$. $\text{LCM} = \text{HCF} \times \text{LCM of } 2, 3, 5 = 14 \times 30 = 420$. Check: $28 = 2^2 \times 7$, $42 = 2 \times 3 \times 7$, $70 = 2 \times 5 \times 7$, so $\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$.
5. Y gets Rs. 48,000. Z first takes $\frac{1}{5} \times 150000 = 30000$. Remaining = 120000, split in $120000 : 80000 = 3 : 2$, so one part = 24000 and $Y = 2 \times 24000 = 48000$ (X gets 72,000). Check: $72000 + 48000 + 30000 = 150000$.
6. Hire 90 couriers. Orders next month = $27000 \times 1.12 = 30240$. Couriers left = $900 \times 0.95 = 855$. Couriers needed = $30240 \div 32 = 945$. Hire $945 - 855 = 90$. Note the rule is on a **future** pair of numbers, so both sides must be projected before comparing.
7. $ac = 72$. With $\frac{a}{b} = \frac{c}{d} = r$, both $\frac{a^2+c^2}{b^2+d^2}$ and $\frac{ac}{bd}$ equal r^2 . So $\frac{ac}{200} = \frac{9}{25} \rightarrow ac = 200 \times \frac{9}{25} = 72$.
8. $n = 40$. Runs = $3k = 60 \rightarrow k = 20$, so balls = $5k = 100$. Then $\frac{60+n}{100+n} = \frac{5}{7} \rightarrow 420 + 7n = 500 + 5n \rightarrow 2n = 80$. Check: $\frac{100}{140} = \frac{5}{7}$. The ratio had to rise because adding the **same** amount to a smaller top and a larger bottom always moves the fraction toward 1.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 25 minutes for all 18

- Simplify $0.75 : 1.25 : 2$.
- Rs. 5,400 is divided in the ratio $2 : 3 : 4$. Find the middle share.
- If $A : B = 5 : 6$ and $B : C = 8 : 9$, find $A : B : C$.
- Find the mean proportional between 4.5 and 8.
- If $x : y = 4 : 7$, find $(5x + 3y) : (2x - y)$.
- Two partners invest Rs. 36,000 and Rs. 54,000 for a full year. The profit is Rs. 45,000. Find the second partner's share.
- A varies inversely as B^2 . $A = 27$ when $B = 2$. Find A when $B = 6$.
- Two numbers are in the ratio $9 : 13$. Adding 6 to each makes the ratio $3 : 4$. Find the numbers.
- A bag has Rs. 10, Rs. 5 and Rs. 2 coins whose numbers are in the ratio $2 : 3 : 5$. The total value is Rs. 900. How many Rs. 5 coins are there?
- The incomes of Anil and Sunil are in the ratio $5 : 4$ and their expenditures are in the ratio $3 : 2$. Each saves Rs. 8,000. Find Anil's income.
- Find the ratio compounded of $3 : 5$, the duplicate ratio of $2 : 3$ and the sub-triplicate ratio of $27 : 64$.
- P invests Rs. 20,000 for 6 months, Q invests Rs. 15,000 for 8 months and R invests Rs. 12,000 for 5 months. The profit is Rs. 30,000. Find R's share.
- The ages of a father and son are in the ratio $7 : 2$. After 10 years the ratio will be $9 : 4$. Find the father's present age.
- Three numbers are in the ratio $4 : 5 : 6$ and the sum of their squares is 1,232. Find the largest number.
- If $\frac{7a-3b}{7a+3b} = \frac{1}{5}$, find $a : b$.
- A pool of S\$90,000 is split in proportion to (headcount $\times$ months): team 1 is 10 people for 6 months, team 2 is 8 people for 9 months, team 3 is 12 people for 4 months. Find the largest share.
- In a school boys : girls = $8 : 5$. After 60 more girls join, the ratio is $4 : 3$. How many boys are there?
- A and B start with Rs. 90,000 and Rs. 60,000. After 4 months A withdraws Rs. 30,000 and B adds Rs. 30,000. The year's profit is Rs. 1,26,000. Find A's share.

ANSWER KEY

- $3 : 5 : 8$ — multiply all by 4.
- Rs. 1,800 — 9 parts, one part = 600, middle = 3×600 .
- $20 : 24 : 27$ — make $B = 24$ (LCM of 6 and 8).
- $6. \sqrt{4.5 \times 8} = \sqrt{36}$.
- $41 : 1$ — put $x = 4$, $y = 7$: $(20 + 21) : (8 - 7)$.
- Rs. 27,000 — ratio $36 : 54 = 2 : 3$, one part = $\frac{45000}{5} = 9000$.

7. $A = 3$. $k = AB^2 = 27 \times 4 = 108$; $A = \frac{108}{36}$.
8. 18 and 26. $4(9k + 6) = 3(13k + 6) \rightarrow 36k + 24 = 39k + 18 \rightarrow k = 2$.
9. 60 — value $= 20x + 15x + 10x = 45x = 900 \rightarrow x = 20$; Rs. 5 coins $= 3x$.
10. Rs. 20,000. $5x - 3y = 8000$ and $4x - 2y = 8000$; the second gives $y = 2x - 4000$; substituting, $5x - 6x + 12000 = 8000 \rightarrow x = 4000$; income $= 5 \times 4000$.
11. $1 : 5$. $(3 : 5) \times (4 : 9) \times (3 : 4) = 36 : 180$.
12. Rs. 6,000 — capital-months $120 : 120 : 60 = 2 : 2 : 1$; one part $= \frac{30000}{5} = 6000$.
13. 35 years. $4(7k + 10) = 9(2k + 10) \rightarrow 28k + 40 = 18k + 90 \rightarrow k = 5$.
14. 24. $(16 + 25 + 36)x^2 = 77x^2 = 1232 \rightarrow x^2 = 16 \rightarrow x = 4$; largest $= 6 \times 4$.
15. $9 : 14$ — componendo-dividendo: $\frac{14a}{-6b} = \frac{6}{-4} \rightarrow \frac{a}{b} = \frac{6}{4} \times \frac{6}{14} = \frac{9}{14}$.
16. S\$36,000 — weights $60 : 72 : 48$, total 180, one unit $= 500$; largest $= 72 \times 500$.
17. 480 boys. $3 \times 8k = 4(5k + 60) \rightarrow 24k = 20k + 240 \rightarrow k = 60$.
18. Rs. 58,800 — $A = 90 \times 4 + 60 \times 8 = 360 + 480 = 840$; $B = 60 \times 4 + 90 \times 8 = 240 + 720 = 960$; ratio $7 : 8$; one part $= \frac{126000}{15} = 8400$; $A = 7 \times 8400$.

ONE-PAGE REVISION CARD

THE PARTS METHOD

$a : b \rightarrow$ write ak and bk . One unknown. Always.

Splitting N in $a : b : c$: one part = $\frac{N}{a+b+c}$.

Difference of two shares = $(a - b)$ parts. Sum = $(a + b)$ parts.

CHAINING AND COMPOUNDING

$a : b = p : q, b : c = r : s \rightarrow a : b : c = pr : qr : qs$ (make b match by LCM).

Compounded: $(a : b)$ with $(c : d) = ac : bd$.

Duplicate $a^2 : b^2$ · triplicate $a^3 : b^3$ · sub-duplicate $\sqrt{a} : \sqrt{b}$ · sub-triplicate $\sqrt[3]{a} : \sqrt[3]{b}$.

PROPORTION

$a : b = c : d \leftrightarrow ad = bc$.

Fourth proportional to a, b, c : $\frac{bc}{a}$. Third proportional to a, b : $\frac{b^2}{a}$. Mean proportional: $\sqrt{ab}$.

Continued proportion $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = k \rightarrow \frac{a}{d} = k^3$.

Componendo–dividendo: $\frac{a}{b} = \frac{c}{d} \rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$.

If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = k$ then $\frac{a+c+e}{b+d+f} = k$.

VARIATION

Direct $x = ky$ · Inverse $xy = k$ · Joint $x = kyz/w$.

Method: write the equation with k , plug in the first data set to get k , then plug in the second.

PARTNERSHIP

Same time $\rightarrow$ profit ratio = capital ratio.

Different times $\rightarrow$ profit ratio = capital $\times$ time (capital-months).

Capital changed mid-year $\rightarrow$ add the rectangles: $C_1t_1 + C_2t_2$.

Working partner $\rightarrow$ salary or commission comes out **first**; split the remainder.

Guarantee $\rightarrow$ split normally, then transfer the shortfall from the others in **their** ratio.

SHORTCUTS

1. Scale capitals down (30,000 $\rightarrow$ 30) before multiplying by time. Ratios do not care.
2. Adding the same number to both terms always moves a ratio toward 1 : 1. Use it to sanity-check any answer.
3. “Difference of shares is D ” $\rightarrow$ one part = $D/(\text{bigger term} - \text{smaller term})$.
4. Componendo–dividendo turns $\frac{mx+ny}{mx-ny} = \frac{p}{q}$ into $x : y$ in one line.
5. For chained ratios, the LCM of the two middle terms is the fastest bridge.

TOP 5 TRAPS

1. A ratio has no size. 2 : 3 does not mean the numbers are 2 and 3.
2. Ratios cannot be added term by term. Convert to real amounts, add, then re-form the ratio.
3. “A gets twice what B gets” is $A = 2B$, so $A : B = 2 : 1$ — not 1 : 2.
4. Ratio of coin **counts** is not the ratio of coin **values**. Convert counts to money first.
5. In partnership, if the times differ you must use capital $\times$ time. Capital alone is wrong.
6. Bonus trap: an overall rate is a weighted average of sub-rates. Never average two percentages unless the two group sizes are equal.

Averages, Mixtures & Alligation

One number for a group. Then blend two groups.

What you need to know

FORMULA SHEET

Average.

$$\text{Average} = \frac{\text{Sum of all values}}{\text{Number of values}}, \quad \text{Sum} = \text{Average} \times \text{Number}$$

Turning an average back into a sum is the single most useful move in this chapter.

Standard averages.

- First n natural numbers: $\frac{n+1}{2}$.
- First n odd numbers: n . First n even numbers: $n + 1$.
- Any set of numbers in arithmetic progression (evenly spaced): average = $\frac{\text{first} + \text{last}}{2}$. When the count is **odd** this is also the middle term.

Changing one member. If one value a is replaced by b in a set of n values:

$$\text{change in average} = \frac{b - a}{n}$$

Adding a member. n values average A . A new value x joins:

$$\text{new average} = \frac{nA + x}{n + 1}$$

If the average goes **up** by d when the $(n + 1)$ -th member joins, then $x = A + (n + 1)d$.

Weighted average. Groups of size n_1, n_2 with averages A_1, A_2 :

$$A = \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2}$$

Alligation. The same formula, read backwards, to get the **ratio of quantities**. Let C be the cheaper value, D the dearer value, M the mean value ($C < M < D$):

$$\frac{\text{quantity of cheaper}}{\text{quantity of dearer}} = \frac{D - M}{M - C}$$

Cross form:

C		D
	M	
$D - M$		$M - C$

It works for any quantity that averages: price, percent, speed, age, interest rate, score.

Repeated replacement. A vessel holds x units of pure liquid. Remove y units and top up with water. Repeat n times:

$$\text{pure liquid left} = x \left(1 - \frac{y}{x}\right)^n$$

The **fraction** of pure liquid after n steps is $\left(1 - \frac{y}{x}\right)^n$. Water left = x - pure left.

Average speed. Never average the speeds.

$$\text{Average speed} = \frac{\text{total distance}}{\text{total time}}$$

Equal distances at u and v :

$$\frac{2uv}{u+v}$$

Equal distances at u, v, w :

$$\frac{3}{\frac{1}{u} + \frac{1}{v} + \frac{1}{w}}$$

Mean, median, mode.

- Mean = sum $\div$ count. Pulled hard by one extreme value.
- Median = middle value after sorting. For an even count, the average of the two middle values. Not pulled by extremes.
- Mode = the most frequent value.

Warm-up

WARM-UP

build the reflex

Example 1

Find the average of 14, 22, 31, 45 and 63.

SOLUTION

Sum = $14 + 22 = 36$; $36 + 31 = 67$; $67 + 45 = 112$; $112 + 63 = 175$.

Average = $175 \div 5 = 35$.

Answer: 35

Example 2

Find the average of the first 20 natural numbers.

SOLUTION

Average = $\frac{n+1}{2} = \frac{21}{2} = 10.5$.

Answer: 10.5

Example 3

The average of 6 numbers is 42. Find their sum.

SOLUTION

Sum = average $\times$ count = $42 \times 6 = 252$.

Answer: 252

Example 4

The average of 8 numbers is 25. The number 18 is replaced by 42. Find the new average.

SOLUTION

Change in sum = $42 - 18 = 24$.
 Change in average = $24 \div 8 = 3$.
 New average = $25 + 3 = 28$.

Answer: 28

Example 5

The average of 5 numbers is 30. A sixth number, 48, is added. Find the new average.

SOLUTION

Old sum = $30 \times 5 = 150$.
 New sum = $150 + 48 = 198$.
 New average = $198 \div 6 = 33$.

Answer: 33

Example 6

4 kg of rice at Rs. 60/kg is mixed with 6 kg at Rs. 85/kg. Find the price per kg of the mixture.

SOLUTION

Cost = $4 \times 60 = 240$ and $6 \times 85 = 510$.
 Total cost = $240 + 510 = 750$ for $4 + 6 = 10$ kg.
 Price = $750 \div 10 = 75$.

Answer: Rs. 75 per kg

Example 7

In what ratio must goods costing Rs. 40 and Rs. 70 be mixed to cost Rs. 50 on average?

SOLUTION

$C = 40, D = 70, M = 50$.
 $\frac{\text{cheaper}}{\text{dearer}} = \frac{D-M}{M-C} = \frac{70-50}{50-40} = \frac{20}{10} = \frac{2}{1}$.

Answer: 2 : 1

Example 8

Find the mean and the median of 7, 3, 11, 9, 5, 15.

SOLUTION

Mean: sum = $7 + 3 + 11 + 9 + 5 + 15 = 50$; mean = $50 \div 6 = 8.33$ (2 d.p.).
 Median: sort them $\rightarrow 3, 5, 7, 9, 11, 15$. Count is even, so take the two middle values 7 and 9.
 Median = $\frac{7+9}{2} = 8$.

Answer: Mean ≈ 8.33 , median = 8

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

The average of 11 numbers is 36. The average of the first six is 34 and the average of the last six is 39. Find the sixth number.

SOLUTION

Total sum of all $11 = 36 \times 11 = 396$.

Sum of first six $= 34 \times 6 = 204$.

Sum of last six $= 39 \times 6 = 234$.

Add those two: $204 + 234 = 438$.

The first six and the last six together cover 12 slots, but there are only 11 numbers. The sixth number sits in both groups, so it has been counted twice.

Sixth number $= 438 - 396 = 42$.

Answer: 42

SHORTCUT

Overlap rule. First $k +$ last $k -$ total = the number counted twice, whenever the two groups together cover one more slot than the list has.

Example 10 [Infosys pattern]

The average age of 30 students is 14 years. When the teacher's age is included, the average rises by 1 year. Find the teacher's age.

SOLUTION

Old sum $= 14 \times 30 = 420$.

New count $= 31$. New average $= 14 + 1 = 15$.

New sum $= 15 \times 31 = 465$.

Teacher's age $= 465 - 420 = 45$ years.

Faster: the newcomer must carry the old average **plus** enough to lift all 31 heads by 1, so age $= 14 + 31 \times 1 = 45$.

Answer: 45 years

Example 11 [Wipro pattern]

The average weight of 4 boxes is 62 kg. A fifth box is added and the average falls to 59 kg. Find the weight of the fifth box.

SOLUTION

Sum of 4 boxes $= 62 \times 4 = 248$.

Sum of 5 boxes $= 59 \times 5 = 295$.

Fifth box $= 295 - 248 = 47$ kg.

Answer: 47 kg

Example 12 [Accenture pattern]

The average marks of a class of 40 students is 62. Later it is found that one student's mark of 78 was entered as 38. Find the correct average.

SOLUTION

The sum was too small by $78 - 38 = 40$.

Change in average $= 40 \div 40 = 1$.

Correct average $= 62 + 1 = 63$.

Answer: 63

TRAP

Decide the direction before you add. The entered mark was **lower** than the true mark, so the recorded average is **too low** and must go up. Students who subtract get 61 and it looks just as reasonable.

Example 13 [TCS NQT pattern]

A man drives 120 km at 40 km/h and returns over the same road at 60 km/h. Find his average speed for the whole journey.

SOLUTION

$$\text{Time out} = \frac{120}{40} = 3 \text{ hours.}$$

$$\text{Time back} = \frac{120}{60} = 2 \text{ hours.}$$

$$\text{Total distance} = 120 + 120 = 240 \text{ km. Total time} = 3 + 2 = 5 \text{ hours.}$$

$$\text{Average speed} = \frac{240}{5} = 48 \text{ km/h.}$$

$$\text{Formula check: } \frac{2uv}{u+v} = \frac{2 \times 40 \times 60}{40+60} = \frac{4800}{100} = 48.$$

Answer: 48 km/h

TRAP

Average speed is not $\frac{40+60}{2} = 50$. The slower leg eats more time, so it gets more weight. The true answer is always below the plain average of the two speeds.

Example 14 [Capgemini pattern]

The average of five consecutive even numbers is 36. Find the largest.

SOLUTION

Consecutive even numbers are evenly spaced, so the average is the middle one.

$$\text{Middle number} = 36.$$

The five numbers are 32, 34, 36, 38, 40.

$$\text{Check: } 32 + 34 + 36 + 38 + 40 = 180, \text{ and } 180 \div 5 = 36. \text{ Correct.}$$

$$\text{Largest} = 40.$$

Answer: 40

Example 15 [Cognizant pattern]

In what ratio must rice at Rs. 48/kg be mixed with rice at Rs. 72/kg so that the mixture is worth Rs. 56/kg?

SOLUTION

$$C = 48, D = 72, M = 56.$$

$$\frac{\text{cheap}}{\text{dear}} = \frac{D-M}{M-C} = \frac{72-56}{56-48} = \frac{16}{8} = \frac{2}{1}.$$

$$\text{Check with 2 kg and 1 kg: cost} = 2 \times 48 + 1 \times 72 = 96 + 72 = 168 \text{ for 3 kg, so } 168 \div 3 = 56. \text{ Correct.}$$

Answer: 2 : 1

Example 16 [TCS NQT pattern]

A shopkeeper mixes 30 kg of tea costing Rs. 260/kg with 20 kg costing Rs. 360/kg and sells the mixture at Rs. 330/kg. Find his profit percent.

SOLUTION

$$\text{Cost of first lot} = 30 \times 260 = 7800.$$

$$\text{Cost of second lot} = 20 \times 360 = 7200.$$

$$\text{Total cost} = 7800 + 7200 = 15000 \text{ for } 30 + 20 = 50 \text{ kg.}$$

$$\text{Cost price per kg} = 15000 \div 50 = 300.$$

$$\text{Profit per kg} = 330 - 300 = 30.$$

$$\text{Profit percent} = \frac{30}{300} \times 100 = 10\%.$$

Answer: 10%

Example 17 [Infosys pattern]

A vessel holds 60 litres of pure milk. 12 litres is drawn out and replaced with water. This is done once more. How much milk is left?

SOLUTION

Fraction removed each time = $\frac{12}{60} = \frac{1}{5}$.

Fraction of milk remaining each time = $1 - \frac{1}{5} = \frac{4}{5}$.

After two rounds, milk = $60 \times \left(\frac{4}{5}\right)^2$.

$\left(\frac{4}{5}\right)^2 = \frac{16}{25}$.

Milk = $60 \times \frac{16}{25} = \frac{960}{25} = 38.4$ litres.

Water = $60 - 38.4 = 21.6$ litres.

Slow check: after round 1, milk = $60 - 12 = 48$. In round 2 the 12 litres drawn is $\frac{12}{60} = \frac{1}{5}$ of the vessel, so it carries $48 \div 5 = 9.6$ litres of milk. Milk left = $48 - 9.6 = 38.4$. Correct.

Answer: 38.4 litres of milk

SHORTCUT

Use the formula $x\left(1 - \frac{y}{x}\right)^n$ only when the amount drawn each time is the same and the vessel is topped back to full. If the vessel is not refilled, the concentration does not change at all — only the volume does.

Example 18 [Accenture pattern]

A 40-litre mixture has milk and water in the ratio 3 : 1. How much water must be added to make the ratio 3 : 2?

SOLUTION

Parts = $3 + 1 = 4$. One part = $40 \div 4 = 10$ L.

Milk = $3 \times 10 = 30$ L. Water = $1 \times 10 = 10$ L.

Adding water leaves the milk unchanged at 30 L.

In the new ratio 3 : 2, milk is 3 parts = 30 L, so one part = 10 L.

New water = $2 \times 10 = 20$ L.

Water to add = $20 - 10 = 10$ L.

Answer: 10 litres

Example 19 [Wipro pattern]

In a test, 20 boys average 68 marks and 30 girls average 78 marks. Find the class average.

SOLUTION

Total marks of boys = $20 \times 68 = 1360$.

Total marks of girls = $30 \times 78 = 2340$.

Total marks = $1360 + 2340 = 3700$.

Total students = $20 + 30 = 50$.

Class average = $3700 \div 50 = 74$.

Alligation check: the mean 74 is 6 above 68 and 4 below 78, so boys : girls = $4 : 6 = 2 : 3$, which matches 20 : 30. Correct.

Answer: 74 marks

Example 20 [TCS NQT pattern]

Six support agents closed these numbers of tickets in a day: 7, 9, 11, 11, 13, 69. Find the mean, the median and the mode. Which one describes a typical agent best?

SOLUTION

Mean: sum = $7 + 9 = 16$; $16 + 11 = 27$; $27 + 11 = 38$; $38 + 13 = 51$; $51 + 69 = 120$.

Mean = $120 \div 6 = 20$.

Median: the list is already sorted. Count is even, so take the middle two, 11 and 11.

$$\text{Median} = \frac{11+11}{2} = 11.$$

Mode: 11 appears twice, every other value once, so mode = 11.

Five of the six agents closed 13 tickets or fewer, so the mean of 20 describes nobody. The single value 69 drags it up. The median of 11 is the honest summary.

Answer: Mean 20, median 11, mode 11 — the median describes a typical agent

TRAP

A mean can sit outside the bulk of the data. Whenever a question mentions one huge or tiny value, expect the mean and the median to disagree, and read carefully which one is asked for.

PRACTICE · TIER 1 target: 60 s/Q

- Find the average of 8, 12, 17, 23 and 30.
- The average of 9 numbers is 52. The average of the first five is 48 and of the last five is 56. Find the fifth number.
(a) 52 (b) 48 (c) 56 (d) 57.8
- The average age of 12 players is 25 years. The coach joins and the average becomes 26. Find the coach's age.
- The average of 40 numbers is 31. One number, 49, was read as 19. Find the correct average.
- A car covers 180 km at 45 km/h and another 180 km at 90 km/h. Find the average speed.
(a) 60 km/h (b) 67.5 km/h (c) 75 km/h (d) 30 km/h
- Find the average of the first 15 odd numbers.
- In what ratio must milk costing Rs. 54 per litre be mixed with water (free) to make the mixture cost Rs. 45 per litre?
- From 80 litres of pure juice, 16 litres is removed and replaced with water. This is done twice in all. How much juice is left?
- A 45-litre mixture has milk and water in the ratio 7 : 2. How much water must be added to make it 7 : 3?
- The average salary of 15 staff is Rs. 28,000. When the manager's salary is added the average becomes Rs. 29,500. Find the manager's salary.
- The average of 5 numbers is 27. Each number is multiplied by 3 and then 4 is subtracted from each. Find the new average.
- The mean of 6 observations is 15. Two more observations, 21 and 27, are added. Find the new mean.
- 12 kg of sugar at Rs. 42/kg is mixed with 8 kg at Rs. 52/kg. Find the price per kg of the mixture.
(a) Rs. 46 (b) Rs. 47 (c) Rs. 48 (d) Rs. 42
- Find the mean and the median of 6, 9, 12, 15, 18, 21, 87.
- Three sections have 30, 35 and 35 students with average marks 64, 70 and 76. Find the overall average.

ANSWER KEY

18. Sum = 90, and $90 \div 5 = 18$.
- (a) 52. Sum of all nine = 468; first five = 240, last five = 280, total 520; overlap = $520 - 468 = 52$. (b) 48 is the first-five average. (c) 56 is the last-five average. (d) 57.8 comes from dividing the combined 520 by 9 instead of subtracting the true total: $520 \div 9 = 57.8$.
- 38 years. Coach = $25 + 13 \times 1 = 38$; or new sum = $26 \times 13 = 338$ minus old sum 300.
- 31.75. The sum was short by $49 - 19 = 30$, so the average rises by $30 \div 40 = 0.75$.
- (a) 60 km/h. $\frac{2 \times 45 \times 90}{45+90} = \frac{8100}{135} = 60$. (b) 67.5 is the plain average of the speeds — wrong, because the slow leg takes longer. (c) 75 comes from weighting each speed by itself instead of by time: $\frac{45 \times 45 + 90 \times 90}{45+90} = \frac{10125}{135} = 75$. (d) 30 comes from using only one 180 km leg as the total distance: $180 \div 6 = 30$.
15. The average of the first n odd numbers is n itself.
- 5 : 1 (milk : water). Water costs 0, so $\frac{\text{water}}{\text{milk}} = \frac{54-45}{45-0} = \frac{9}{45} = \frac{1}{5}$.

8. 51.2 litres. $80 \times \left(1 - \frac{16}{80}\right)^2 = 80 \times (0.8)^2 = 80 \times 0.64$.
9. 5 litres. Milk = 35 L and water = 10 L; for 7 : 3 the water must be 15 L.
10. Rs. 52,000. Manager = $28000 + 16 \times 1500 = 28000 + 24000$.
11. 77. Multiplying every value by 3 multiplies the average by 3; subtracting 4 from every value subtracts 4 from the average. $27 \times 3 - 4 = 77$.
12. 17.25. New sum = $90 + 21 + 27 = 138$, new count = 8, and $138 \div 8 = 17.25$.
13. (a) Rs. 46. Total cost = $504 + 416 = 920$ for 20 kg. (b) 47 is the plain average of 42 and 52 — wrong, the weights differ. (c) 48 comes from swapping the weights: $\frac{8 \times 42 + 12 \times 52}{20} = \frac{960}{20} = 48$. (d) 42 is the price of the **larger** lot; the blend price must sit strictly between 42 and 52.
14. Mean 24, median 15. Sum = 168 and $168 \div 7 = 24$; the middle of seven sorted values is the 4th, which is 15.
15. 70.3. Total = $30 \times 64 + 35 \times 70 + 35 \times 76 = 1920 + 2450 + 2660 = 7030$ over 100 students.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A driver earned an average of S\$148 per day on the first 5 days of the week and S\$225 per day on the last 2 days. Find his average for the whole 7-day week.

SOLUTION

Earnings for the first 5 days = $5 \times 148 = 740$.

Earnings for the last 2 days = $2 \times 225 = 450$.

Total = $740 + 450 = 1190$.

Average = $1190 \div 7 = 170$.

Answer: S\$170 per day

Example 22 [Shopee · pattern]

In March a seller had 8,000 orders at an average order value of S\$36. In April the order count rose 25% and the average order value fell 8%. Find the average order value across the two months together.

SOLUTION

March.

Orders = 8000. AOV = 36.

Revenue = $8000 \times 36 = 288000$.

April.

Orders = $8000 \times 1.25 = 10000$.

AOV = $36 \times 0.92 = 33.12$.

Revenue = $10000 \times 33.12 = 331200$.

Combined.

Total revenue = $288000 + 331200 = 619200$.

Total orders = $8000 + 10000 = 18000$.

Combined AOV = $619200 \div 18000 = 34.40$.

Answer: S\$34.40

TRAP

Do not average 36 and 33.12 to get 34.56. April carried more orders, so it pulls the combined figure down past the midpoint. Always rebuild the two totals first.

Example 23 [Agoda · pattern]

Over 30 nights a Bangkok hotel's average room rate was THB 2,800. Over the first 18 of those nights it was THB 2,500. Find the average over the remaining 12 nights.

SOLUTION

$$\text{Total over 30 nights} = 2800 \times 30 = 84000.$$

$$\text{Total over the first 18 nights} = 2500 \times 18 = 45000.$$

$$\text{Total over the last 12 nights} = 84000 - 45000 = 39000.$$

$$\text{Average} = 39000 \div 12 = 3250.$$

Answer: THB 3,250 per night

Example 24 [DBS · pattern]

A bank blends two bond funds, one yielding 3.2% and one yielding 5.6%, into a S\$4.8 million portfolio with an overall yield of 4.1%. How much sits in each fund?

SOLUTION

Alligation on the yields. $C = 3.2$, $D = 5.6$, $M = 4.1$.

$$D - M = 5.6 - 4.1 = 1.5.$$

$$M - C = 4.1 - 3.2 = 0.9.$$

$$\frac{\text{amount at 3.2\%}}{\text{amount at 5.6\%}} = \frac{1.5}{0.9} = \frac{15}{9} = \frac{5}{3}.$$

Total parts = $5 + 3 = 8$. One part = $4.8 \div 8 = 0.6$ million.

At 3.2%: $5 \times 0.6 = 3.0$ million.

At 5.6%: $3 \times 0.6 = 1.8$ million.

Check: interest = $3.0 \times 0.032 + 1.8 \times 0.056 = 0.096 + 0.1008 = 0.1968$ million.

$0.1968 \div 4.8 = 0.041 = 4.1\%$. Correct.

Answer: S\$3.0 million at 3.2% and S\$1.8 million at 5.6%

SHORTCUT

Alligation with decimals: clear them first. Turn $1.5 : 0.9$ into $15 : 9$ and then $5 : 3$. You only ever need the ratio, so any common multiplier is free.

Example 25 [Sea / Shopee · pattern]

A 90-litre tank is full of pure syrup. 18 litres is drawn off and replaced with water. This is repeated twice more, three rounds in all. Find the final ratio of syrup to water.

SOLUTION

$$\text{Fraction drawn each round} = \frac{18}{90} = \frac{1}{5} = 0.2.$$

$$\text{Fraction of syrup kept each round} = 1 - 0.2 = 0.8.$$

$$\text{After 3 rounds the syrup fraction} = (0.8)^3.$$

$$(0.8)^2 = 0.64.$$

$$(0.8)^3 = 0.64 \times 0.8 = 0.512.$$

$$\text{Syrup} = 90 \times 0.512 = 46.08 \text{ litres.}$$

$$\text{Water} = 90 - 46.08 = 43.92 \text{ litres.}$$

Ratio = $46.08 : 43.92$. Multiply both by 1000 to clear decimals: $46080 : 43920$.

Both are $0.512 : 0.488$, that is $512 : 488$. Divide both by 8: $64 : 61$.

Answer: Syrup : water = 64 : 61 (46.08 L and 43.92 L)

TRAP

$(0.8)^3$ is the syrup fraction of the whole tank, not the syrup-to-water ratio. Convert to actual litres, or use $0.512 : (1 - 0.512)$. Writing the answer as $512 : 1000$ is a very common slip.

Example 26 [SCB · pattern]

The average age of 8 analysts in a team is 31 years. Two analysts aged 26 and 34 leave, and two new analysts join whose average age is 24. Find the new team average.

SOLUTION

Sum of ages now = $31 \times 8 = 248$.

The two leavers total $26 + 34 = 60$.

Sum after they leave = $248 - 60 = 188$ for 6 people.

The two joiners total $24 \times 2 = 48$.

New sum = $188 + 48 = 236$ for $6 + 2 = 8$ people.

New average = $236 \div 8 = 29.5$ years.

Answer: 29.5 years

Example 27 [LINE MAN · pattern]

A rider covers 12 km at 24 km/h, then 9 km at 18 km/h, then waits 10 minutes at a restaurant, then covers 15 km at 30 km/h. Find his average speed for the whole trip, counting the wait.

SOLUTION

Leg 1 time = $\frac{12}{24} = 0.5$ h.

Leg 2 time = $\frac{9}{18} = 0.5$ h.

Wait = 10 minutes = $\frac{10}{60} = \frac{1}{6}$ h.

Leg 3 time = $\frac{15}{30} = 0.5$ h.

Total time = $0.5 + 0.5 + 0.5 + \frac{1}{6} = 1.5 + 0.1667 = 1.6667$ h, which is exactly $\frac{5}{3}$ h.

Total distance = $12 + 9 + 15 = 36$ km.

Average speed = $36 \div \frac{5}{3} = 36 \times \frac{3}{5} = \frac{108}{5} = 21.6$ km/h.

Answer: 21.6 km/h

TRAP

Waiting time counts. It adds to the denominator and adds nothing to the numerator. Drop the 10-minute wait and you get 24 km/h, which is wrong by more than 10%.

Example 28 [GIC · pattern]

A S\$8 million portfolio returned 7% for the year. It holds only equities returning 12% and bonds returning 4%. How much is in equities?

SOLUTION

Alligation on returns. $C = 4$ (bonds), $D = 12$ (equities), $M = 7$.

$D - M = 12 - 7 = 5$.

$M - C = 7 - 4 = 3$.

$\frac{\text{bonds}}{\text{equities}} = \frac{5}{3}$.

Total parts = $5 + 3 = 8$. One part = $8 \div 8 = 1$ million.

Bonds = 5 million. Equities = 3 million.

Check: $3 \times 0.12 + 5 \times 0.04 = 0.36 + 0.20 = 0.56$ million on 8 million, and $0.56 \div 8 = 0.07 = 7\%$. Correct.

Answer: S\$3 million in equities

Example 29 [Razer · pattern]

A shop buys 25 kg of premium coffee at THB 900/kg and 35 kg of regular coffee at THB 540/kg, blends them, and wants a profit of 25% on the blend. Find the selling price per kg.

SOLUTION

Cost of premium = $25 \times 900 = 22500$.

Cost of regular = $35 \times 540 = 18900$.

Total cost = $22500 + 18900 = 41400$.

Total weight = $25 + 35 = 60$ kg.

Cost price per kg = $41400 \div 60 = 690$.

Selling price = 690×1.25 .

$690 \times 1.25 = 690 + 690 \times 0.25 = 690 + 172.50 = 862.50$.

Answer: THB 862.50 per kg

Example 30 [Grab · pattern]

Nine food deliveries took these times, in minutes: 18, 21, 22, 24, 25, 27, 29, 33, 125. The operations team must publish one number as the “typical delivery time”. Find the mean and the median and say which should be published.

SOLUTION

Mean: add in order.

$18 + 21 = 39$; $39 + 22 = 61$; $61 + 24 = 85$; $85 + 25 = 110$; $110 + 27 = 137$; $137 + 29 = 166$; $166 + 33 = 199$; $199 + 125 = 324$.

Mean = $324 \div 9 = 36$ minutes.

Median: the list is sorted and there are 9 values, so the median is the 5th one.

Median = 25 minutes.

Eight of the nine deliveries finished in 33 minutes or less. The 125-minute outlier alone lifts the mean by about 11 minutes.

Publish the median.

Answer: Mean 36 min, median 25 min — publish the median

PRACTICE · TIER 2 target: 100 s/Q

1. A driver earned S\$132, S\$145, S\$120, S\$168, S\$155 and S\$150 on six days. Find his average daily earning.
2. A seller's average monthly sales for January to April is S\$24,500, and for February to May it is S\$26,000. May's sales were S\$30,200. Find January's sales.
3. In what ratio must a 4.5% bond be mixed with a 7.5% bond to give a portfolio yield of 5.5%? If the portfolio is S\$9 million, how much is in the 7.5% bond?
4. From 120 litres of pure oil, 24 litres is drawn off and replaced with a cheaper oil. This is done three times in all. How much pure oil is left?
5. The average weight of 25 parcels is 8.4 kg. Two parcels weighing 12.5 kg and 13.5 kg are removed. Find the average weight of the remaining parcels.
6. A van covers 60 km at 30 km/h and the next 60 km at 60 km/h. Find the average speed.
7. A training intake has three groups: 24 trainees averaging 72, 36 averaging 78 and 40 averaging 65. Find the overall average.
8. Tea at THB 2,400/kg is mixed with tea at THB 1,600/kg. The blend sells at THB 2,310/kg at a profit of 10%. Find the mixing ratio.
9. A vessel has 54 litres of a milk-water mixture in the ratio 5 : 4. How much of the mixture must be drawn off and replaced with pure milk to make the ratio 2 : 1?
10. The average age of a 40-member team is 32 years. Eight members whose average age is 45 leave. Find the new average age.

11. A courier averaged 46 parcels a day over 20 working days, and 58 a day over the last 5 of those days. Find his average over the first 15 days.
12. Alloy A weighs 40 kg with copper : zinc = 5 : 3. Alloy B weighs 24 kg with copper : zinc = 3 : 5. They are melted together. Find the copper : zinc ratio of the new alloy.

ANSWER KEY

1. S\$145. Sum = 870 over 6 days.
2. S\$24,200. Feb+Mar+Apr = $4 \times 26000 - 30200 = 73800$; Jan = $4 \times 24500 - 73800$. Shortcut: Jan = May $-4 \times$ (difference of averages) = $30200 - 6000$.
3. 2 : 1, and S\$3 million in the 7.5% bond. $(7.5 - 5.5) : (5.5 - 4.5) = 2 : 1$, so 6 million and 3 million.
4. 61.44 litres. $120 \times \left(1 - \frac{24}{120}\right)^3 = 120 \times 0.8^3 = 120 \times 0.512$.
5. 8 kg. Sum = $8.4 \times 25 = 210$; remove $12.5 + 13.5 = 26$; $184 \div 23 = 8$.
6. 40 km/h. Equal distances, so $\frac{2 \times 30 \times 60}{30 + 60} = \frac{3600}{90}$.
7. 71.36. Total = $1728 + 2808 + 2600 = 7136$ over 100 trainees.
8. 3 : 5 (cheaper : dearer). Cost price of the blend = $2310 \div 1.10 = 2100$; then $(2400 - 2100) : (2100 - 1600) = 300 : 500$.
9. 13.5 litres. Milk = 30, water = 24. Drawing off a fraction f leaves water = $24(1 - f)$ and milk = $30(1 - f) + 54f$. Setting milk = $2 \times$ water gives $30 + 24f = 48 - 48f$, so $72f = 18$ and $f = 0.25$; $0.25 \times 54 = 13.5$.
10. 28.75 years. Sum = 1280; the leavers take $8 \times 45 = 360$; $920 \div 32 = 28.75$.
11. 42 parcels a day. Total = 920; last five = 290; $630 \div 15 = 42$.
12. 17 : 15. Alloy A: copper 25, zinc 15. Alloy B: copper 9, zinc 15. Totals 34 and 30.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Google · pattern]

Twenty **distinct** positive integers have an average of 30. What is the largest value the biggest of them can take?

SOLUTION

Insight: the sum is locked. To push one value as high as possible, squeeze every other value as low as it is allowed to go.

Sum of all twenty = $30 \times 20 = 600$.

The other nineteen must be distinct positive integers, all different from each other. The smallest such set is 1, 2, 3, ..., 19.

Their sum = $\frac{19 \times 20}{2} = 190$.

Largest possible value = $600 - 190 = 410$.

Check the rules: 410 is a positive integer and is different from every number in 1...19, and $190 + 410 = 600$, so the average is $600 \div 20 = 30$. Correct.

Answer: 410

Example 32 [Amazon · pattern]

In a class of 50 the average score is 64. Every score is a whole number from 0 to 100. At most how many students could have scored above 90?

SOLUTION

Insight: turn “at most how many” into a budget question. The total score is fixed, and every high scorer spends a large slice of it.

Total score = $64 \times 50 = 3200$.

“Above 90” with whole-number scores means at least 91.

Suppose k students scored 91 or more. They alone use at least $91k$ marks.

The other $50 - k$ students use at least 0 marks.

So $91k \leq 3200$.

$3200 \div 91 = 35.16\dots$

Therefore $k \leq 35$.

Now show 35 is actually reachable. Give 35 students exactly 91 marks: $35 \times 91 = 3185$.

Marks left = $3200 - 3185 = 15$ for the remaining $50 - 35 = 15$ students.

Give each of them 1 mark: $15 \times 1 = 15$.

Every score is a whole number between 0 and 100 and the total is $3185 + 15 = 3200$. Correct.

Answer: 35 students

Example 33 [Microsoft · pattern]

A tank holds 100 litres of pure acid. Every day 20 litres is drawn off and the tank is topped back to 100 litres with water. After how many days does the acid first fall below 50 litres?

SOLUTION

Insight: each day multiplies the acid by the same factor. So this is repeated multiplication, not repeated subtraction.

Fraction drawn each day = $\frac{20}{100} = 0.2$.

Fraction of acid kept each day = $1 - 0.2 = 0.8$.

After n days, acid = $100 \times (0.8)^n$.

We need $100 \times (0.8)^n < 50$, that is $(0.8)^n < 0.5$.

Day 1: $(0.8)^1 = 0.8$, acid = 80 L. Not below 50.

Day 2: $(0.8)^2 = 0.64$, acid = 64 L. Not below 50.

Day 3: $(0.8)^3 = 0.512$, acid = 51.2 L. Still not below 50.

Day 4: $(0.8)^4 = 0.4096$, acid = 40.96 L. Below 50.

Note how close day 3 is: 51.2 L. Guessing from the “20% a day” wording would give the wrong day.

Answer: On day 4, with 40.96 litres left

Example 34 [Goldman Sachs · pattern]

A fund gains 50% in year 1 and loses 40% in year 2. A report states that “the average annual return was +5%, so the fund is up”. Find the true two-year change, and explain the error in one line.

SOLUTION

Insight: percentages apply to different bases, so they multiply, they do not average. Turn each percent into a growth factor and multiply.

Start with 100 units.

Year 1 factor = $1 + 0.50 = 1.5$.

Value after year 1 = $100 \times 1.5 = 150$.

Year 2 factor = $1 - 0.40 = 0.6$.

Value after year 2 = $150 \times 0.6 = 90$.

True change = $90 - 100 = -10$ on a base of 100, that is -10% over two years.

The error: the +50% was earned on 100 but the -40% was charged on 150, so the loss is bigger in money terms even though it is smaller in percent terms.

$$1.5 \times 0.6 = 0.9, \quad \text{not} \quad \frac{1.5 + 0.6}{2} = 1.05$$

Answer: The fund is down 10% over the two years

Example 35 [D. E. Shaw · pattern]

Tokens numbered 1 to 50 sit in a bag. Every token whose number is a multiple of 5 is removed. Find the average of the tokens that remain.

SOLUTION

Insight: do not list 40 numbers. Work with sums — the sum of what remains is the sum of everything minus the sum of what left.

Sum of 1 to 50 = $\frac{50 \times 51}{2} = 1275$.

Multiples of 5 up to 50: 5, 10, 15, ..., 50. There are $50 \div 5 = 10$ of them.

Their sum = $5(1 + 2 + \dots + 10) = 5 \times \frac{10 \times 11}{2} = 5 \times 55 = 275$.

Sum of the tokens left = $1275 - 275 = 1000$.

Count of tokens left = $50 - 10 = 40$.

Average = $1000 \div 40 = 25$.

Sense check: the whole set averages $1275 \div 50 = 25.5$, and the group removed averaged $275 \div 10 = 27.5$, which is above 25.5. Removing an above-average group must pull the average down, and it did, to 25. Correct.

Answer: 25

Example 36 [Adobe · pattern]

Rice at Rs. 40, Rs. 50 and Rs. 70 per kg is mixed to sell at Rs. 55 per kg. The Rs. 40 and Rs. 50 varieties are used in the ratio 2 : 3. Find the ratio of all three.

SOLUTION

Insight: alligation handles only two things at a time. So first fuse the two varieties whose ratio is already fixed into one imaginary variety, then alligate that against the third.

Step 1 — price of the fused variety.

Take 2 kg at 40 and 3 kg at 50.

Cost = $2 \times 40 + 3 \times 50 = 80 + 150 = 230$ for 5 kg.

Price = $230 \div 5 = 46$ per kg.

Step 2 — alligate 46 against 70 at a mean of 55.

$D - M = 70 - 55 = 15$.

$M - C = 55 - 46 = 9$.

$\frac{\text{fused}}{\text{70-rice}} = \frac{15}{9} = \frac{5}{3}$.

Step 3 — unfold the fused part.

The fused variety is 5 parts, and inside it the split is 2 : 3, which also sums to 5. So the 5 parts are exactly 2 parts of Rs. 40 rice and 3 parts of Rs. 50 rice.

Ratio = 2 : 3 : 3.

Check: $\frac{2 \times 40 + 3 \times 50 + 3 \times 70}{2 + 3 + 3} = \frac{80 + 150 + 210}{8} = \frac{440}{8} = 55$. Correct.

Answer: 40 : 50 : 70 rice = 2 : 3 : 3

Example 37 [Uber · pattern]

A driver wants to average 60 km/h over a 60 km round trip, 30 km each way. He drives the first 30 km at 30 km/h. What speed must he use on the return leg?

SOLUTION

Insight: an average speed target is really a time budget. Check the budget before doing any algebra.

Time budget for the whole trip = $\frac{\text{total distance}}{\text{target speed}} = \frac{60}{60} = 1$ hour.

Time already used on the first leg = $\frac{30}{30} = 1$ hour.

Time left for the return leg = $1 - 1 = 0$ hours.

He must cover 30 km in zero time. No finite speed does that.

Algebra says the same thing. Let the return speed be v :

$$\frac{60}{1 + \frac{30}{v}} = 60 \rightarrow 1 + \frac{30}{v} = 1 \rightarrow \frac{30}{v} = 0$$

which has no solution for finite v .

The general rule: to double the average speed of the first half, the second half needs infinite speed, because the first half has already spent the entire time budget.

Answer: Impossible — no finite speed works

Example 38 [Google · pattern]

Seven **distinct** positive integers have a median of 12 and a mean of 15. What is the largest the biggest value can be?

SOLUTION

Insight: with seven sorted values the median pins the 4th value. Everything below it and the two values just above it must be pushed as low as the rules allow, and whatever is left goes to the 7th.

Write the sorted values as $a_1 < a_2 < a_3 < a_4 < a_5 < a_6 < a_7$.

Median is the 4th value, so $a_4 = 12$.

Sum = $15 \times 7 = 105$.

Now minimise everything except a_7 .

a_1, a_2, a_3 are distinct positive integers below 12. The smallest choice is 1, 2, 3, summing to 6.

a_5, a_6 are distinct integers above 12. The smallest choice is 13, 14, summing to 27.

Used so far = $6 + 12 + 27 = 45$.

$a_7 = 105 - 45 = 60$.

Check the list 1, 2, 3, 12, 13, 14, 60:

all distinct and increasing, so $a_7 = 60 > 14$. Correct.

Sum = $1 + 2 + 3 + 12 + 13 + 14 + 60 = 105$, so the mean is $105 \div 7 = 15$. Correct.

The 4th value is 12, so the median is 12. Correct.

Answer: 60

PRACTICE · TIER 3 target: 4 min/Q

1. Fifteen distinct positive integers have an average of 20. What is the largest the biggest of them can be?
2. In a class of 40 the average score is 70, and every score is a whole number from 0 to 100. At most how many students could have scored below 40?
3. A 200-litre tank is full of pure glycerin. Each cycle, 25 litres is drawn off and the tank is topped back up with water. After how many cycles does the glycerin first drop below 120 litres?
4. An investment rises 25% in year 1, falls 20% in year 2 and rises 25% in year 3. Find the net percent change over the three years.
5. Tokens numbered 1 to 60 are in a bag and every multiple of 4 is removed. Find the average of the tokens left.
6. Oil grades costing Rs. 80, Rs. 95 and Rs. 120 per litre are blended to sell at Rs. 100 per litre. The Rs. 80 and Rs. 95 grades are used in the ratio 1 : 2. Find the ratio of all three.
7. A cyclist must average 24 km/h over 48 km. He covers the first 24 km at 16 km/h. What speed does he need over the rest?
8. Six distinct positive integers have a median of 20 and a mean of 25. What is the largest the biggest value can be?

ANSWER KEY

1. 195. The total is $20 \times 15 = 300$. The other fourteen must be distinct positive integers, so their smallest possible sum is $1 + 2 + \dots + 14 = \frac{14 \times 15}{2} = 105$. The biggest value is then $300 - 105 = 195$, which is larger than 14, so the set is valid.
2. 19 students. Total = $70 \times 40 = 2800$. If k students score at most 39, the largest total the class can reach is $39k + 100(40 - k) = 4000 - 61k$. This must be at least 2800, so $61k \leq 1200$ and $k \leq 19.67$, giving $k \leq 19$. Now build a class that reaches 19. Give 19 students 39 marks each: $19 \times 39 = 741$. The other $40 - 19 = 21$ students must supply $2800 - 741 = 2059$ marks. Give 20 of them 100 marks each = 2000, and the last one $2059 - 2000 = 59$ marks. Every score is a

whole number from 0 to 100, the total is $741 + 2000 + 59 = 2800$, and exactly 19 students are below 40 (59 is not). So 19 is reached.

3. 4 cycles. Fraction kept each cycle $= 1 - \frac{25}{200} = 0.875$. Glycerin after n cycles $= 200 \times (0.875)^n$: 175.0, then 153.1, then 134.0, then 117.2. The first value below 120 comes on cycle 4. Note that cycle 3 leaves 134 L, so stopping at 3 is the trap.

4. Up 25%. Multiply the growth factors: $1.25 \times 0.80 \times 1.25$. First $1.25 \times 0.80 = 1.00$, then $1.00 \times 1.25 = 1.25$. So the value is 1.25 times the start, a 25% rise. Adding the percents would wrongly give +30%.

5. 30. Sum of 1 to 60 $= \frac{60 \times 61}{2} = 1830$. Multiples of 4 up to 60 are 4, 8, ..., 60, that is 15 numbers summing to $4 \times \frac{15 \times 16}{2} = 4 \times 120 = 480$. What is left: sum $= 1830 - 480 = 1350$ over $60 - 15 = 45$ tokens, so $1350 \div 45 = 30$.

6. 2 : 4 : 3. Fuse the first two in 1 : 2: cost $= \frac{1 \times 80 + 2 \times 95}{3} = \frac{270}{3} = 90$. Alligate 90 against 120 at mean 100: $(120 - 100) : (100 - 90) = 20 : 10 = 2 : 1$. The fused part is 2 units but splits in 1 : 2, which sums to 3, so multiply everything by 3: fused = 6 units splitting as 2 and 4, and the third grade = 3 units. Check: $\frac{2 \times 80 + 4 \times 95 + 3 \times 120}{9} = \frac{160 + 380 + 360}{9} = \frac{900}{9} = 100$.

7. 48 km/h. Time budget $= \frac{48}{24} = 2$ hours. Time already used $= \frac{24}{16} = 1.5$ hours. Time left = 0.5 hours for the remaining 24 km, so speed $= \frac{24}{0.5} = 48$ km/h. Unlike Example 37 the budget is not exhausted, so a finite answer exists.

8. 85. Sum $= 25 \times 6 = 150$. With six values the median is the average of the 3rd and 4th, so $a_3 + a_4 = 40$ with $a_3 < a_4$. To leave the most for a_6 , make a_4 as small as possible: $a_3 = 19$ and $a_4 = 21$. Then $a_1 = 1$, $a_2 = 2$ and $a_5 = 22$. Used $= 1 + 2 + 19 + 21 + 22 = 65$, so $a_6 = 150 - 65 = 85$. Check 1, 2, 19, 21, 22, 85: distinct, median $= \frac{19+21}{2} = 20$, sum = 150.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 25 minutes for all 18

- Find the average of 12, 18, 24, 30 and 36.
- The average of 20 numbers is 45. If 5 is added to every number, find the new average.
- The average of 6 numbers is 38. One number is removed and the average of the rest becomes 40. Find the number removed.
- 15 litres of a 20% acid solution is mixed with 25 litres of a 44% acid solution. Find the concentration of the mixture.
- A car covers three equal stretches at 30 km/h, 40 km/h and 60 km/h. Find the average speed for the whole run.
- From 72 litres of pure milk, 18 litres is drawn off and replaced with water, twice in all. How much milk is left?
- The average age of a family of 5 is 28 years. The youngest is 6 years old. Find the average age of the other four.
- In what ratio must goods at Rs. 36/kg and Rs. 60/kg be mixed to cost Rs. 45/kg?
- Find the average of the first 12 even numbers.
- A batsman's average after 16 innings is 36. He scores 87 in the 17th innings. Find his new average.
- A person's average monthly income for January to June is Rs. 42,000 and for July to December is Rs. 48,000. Find the average for the year.
- Alloy A weighs 60 kg with tin : lead = 3 : 2. Alloy B weighs 40 kg with tin : lead = 1 : 3. They are melted together. Find the tin : lead ratio.
- Daily sales at nine shops, in thousands of rupees, are 12, 14, 15, 15, 16, 18, 19, 21 and 95. Find the mean and the median.
- The average of 25 numbers is 54. Two numbers, 38 and 46, were wrongly entered as 83 and 64. Find the correct average.
- Forty distinct positive integers have an average of 50. What is the largest the biggest of them can be?
- A 60-litre mixture has milk and water in the ratio 7 : 3. How much water must be added to make it 3 : 2?
- The average age of 30 boys is 15 years. Ten new boys join and the average becomes 16 years. Find the average age of the new boys.
- A car travels from P to Q at 50 km/h and returns at 75 km/h. Find the average speed for the round trip.

ANSWER KEY

1. 24. Sum = 120 over 5 values; the numbers are evenly spaced so the middle one is the average.
2. 50. Adding the same amount to every value adds it to the average.
3. 28. Old sum = 228; the remaining five sum to $40 \times 5 = 200$; removed = $228 - 200$.
4. 35%. Acid = $0.20 \times 15 + 0.44 \times 25 = 3 + 11 = 14$ litres in 40 litres, and $\frac{14}{40} = 0.35$.
5. 40 km/h. Equal distances, so $3 \div \left(\frac{1}{30} + \frac{1}{40} + \frac{1}{60}\right) = 3 \div \frac{4+3+2}{120} = 3 \times \frac{120}{9}$.
6. 40.5 litres. $72 \times \left(1 - \frac{18}{72}\right)^2 = 72 \times (0.75)^2 = 72 \times 0.5625$.
7. 33.5 years. Total = 140; remove 6; $134 \div 4 = 33.5$.
8. 5 : 3. $(60 - 45) : (45 - 36) = 15 : 9$.
9. 13. The average of the first n even numbers is $n + 1$.
10. 39. Old sum = 576; new sum = 663; $663 \div 17 = 39$. Shortcut: the average rises by $\frac{87-36}{17} = 3$.
11. Rs. 45,000. Both halves have six months, so the plain average of 42,000 and 48,000 is correct here.
12. 23 : 27. Alloy A: tin 36, lead 24. Alloy B: tin 10, lead 30. Totals 46 and 54.
13. Mean 25, median 16. Sum = 225 over 9 shops; the median is the 5th of the nine sorted values.
14. 51.48. The entered sum was too big by $(83 - 38) + (64 - 46) = 45 + 18 = 63$; correct sum = $1350 - 63 = 1287$, and $1287 \div 25 = 51.48$.
15. 1,220. Total = 2000; the other 39 are at least $1 + 2 + \dots + 39 = \frac{39 \times 40}{2} = 780$; $2000 - 780 = 1220$.
16. 10 litres. Milk = 42 L, water = 18 L; for 3 : 2 the water must be $42 \times \frac{2}{3} = 28$ L.
17. 19 years. Old sum = 450; new sum = $40 \times 16 = 640$; the ten newcomers total 190, so their average is 19.
18. 60 km/h. $\frac{2 \times 50 \times 75}{50 + 75} = \frac{7500}{125}$.

ONE-PAGE REVISION CARD

THE ONE MOVE THAT SOLVES MOST QUESTIONS

Sum = average $\times$ count. Convert every average in the question into a sum, do the adding and subtracting on sums, then divide once at the end.

AVERAGE FACTS

First n naturals: $\frac{n+1}{2} \cdot$ first n odd: $n \cdot$ first n even: $n + 1$.

Evenly spaced list: average = (first + last) $\div$ 2; for an odd count this is the middle term.

Replace a by b in n values: average moves by $(b - a)/n$.

New member x joins n values averaging A : new average = $(nA + x)/(n + 1)$.

If the average rises by d when the $(n + 1)$ -th member joins: $x = A + (n + 1)d$.

Add c to every value $\rightarrow$ average $+c$. Multiply every value by $c \rightarrow$ average $\times c$.

WEIGHTED AVERAGE AND ALLIGATION

$$A = \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2}, \quad \frac{\text{cheaper}}{\text{dearer}} = \frac{D - M}{M - C}$$

Alligation works on any averaging quantity: price, percent, yield, speed, age, marks.

Three things? Fuse the two whose ratio is already known into one price, then alligate.

MIXTURES

Add water only $\rightarrow$ the milk stays fixed. Rebuild the new total from the fixed part.

Repeated replacement (draw y from x , top up, n times): pure left = $x(1 - y/x)^n$.

Draw off and replace with the **pure** liquid instead of water: milk = $M(1 - f) + xf$, water = $W(1 - f)$, where f is the fraction drawn.

SPEED

Average speed = total distance $\div$ total time. Count every stop.

Equal distances at u and v : $(2uv)/(u + v)$. Three legs: $3/(1/u + 1/v + 1/w)$.

Percent changes across years multiply as factors; they never average.

MEAN, MEDIAN, MODE

Mean follows the outliers. Median ignores them. Mode is the most frequent value.

Even count $\rightarrow$ median is the average of the two middle values.

TOP 5 TRAPS

1. Average speed is not the average of the speeds. Equal distances, not equal times, is the usual case.
2. Alligation returns a **ratio of quantities**, never the quantities. Multiply by the total afterwards.
3. $(1 - y/x)^n$ is the fraction of **pure liquid in the whole vessel**, not the pure-to-water ratio. Subtract to get the water.
4. When a wrong entry is corrected, decide the direction first. Entered too low $\rightarrow$ the average goes up.
5. Percent returns over several years multiply. +50% then -40% is -10%, not +5%.
6. Bonus trap: waiting time, rest stops and idle hours all count in the total time for average speed.

Profit, Loss & Discount

Cost price is the base. Marked price is the story.

What you need to know

WHAT YOU NEED TO KNOW

The three prices. **CP** = cost price (what the seller paid, all costs in). **SP** = selling price (what the buyer paid). **MP** = marked price / list price (the sticker, before any discount).

Profit and loss — always measured on CP

$$\text{Profit} = \text{SP} - \text{CP} \quad \text{Profit}\% = \frac{\text{Profit}}{\text{CP}} \times 100 \quad \text{Loss}\% = \frac{\text{Loss}}{\text{CP}} \times 100$$

$$\text{SP} = \text{CP} \times \frac{100 + p}{100} \quad \text{CP} = \text{SP} \times \frac{100}{100 + p}$$

For a loss of $l\%$, put $-l$ in place of p .

Discount — always measured on MP

$$\text{Discount}\% = \frac{\text{MP} - \text{SP}}{\text{MP}} \times 100 \quad \text{SP} = \text{MP} \times \frac{100 - d}{100}$$

Successive discounts. Multiply the remaining factors: $\text{SP} = \text{MP} (1 - d_1/100)(1 - d_2/100)(1 - d_3/100)$. For two discounts the single equivalent is $d = d_1 + d_2 - d_1 d_2/100$.

Markup and discount together. Mark up $m\%$ over CP, then give $d\%$ discount:

$$\text{Profit}\% = m - d - \frac{md}{100} \quad \frac{\text{SP}}{\text{CP}} = \frac{100 + m}{100} \times \frac{100 - d}{100} \quad \text{MP}(100 - d) = \text{CP}(100 + p)$$

Same SP, one at $+x\%$ and one at $-x\%$: always a **loss** of $x^2/100$ per cent.

Markup $m\%$ then discount $m\%$: always a **loss** of $m^2/100$ per cent.

Cost of a articles = SP of b articles ($a > b$): $\text{Profit}\% = (a - b)/b \times 100$.

False weight (dishonest dealer) selling at cost price, and the same dealer who also marks up m per cent:

$$\text{Gain}\% = \frac{\text{true weight} - \text{weight given}}{\text{weight given}} \times 100 \quad \frac{\text{SP}}{\text{CP}} = \frac{100 + m}{100} \times \frac{\text{true weight}}{\text{weight given}}$$

Markup needed to keep a profit of $p\%$ after a discount of $d\%$: $\text{MP}/\text{CP} = (100 + p)/(100 - d)$.

Profit as a share of SP. If profit is $p\%$ of CP, it is $p/(100 + p) \times 100$ per cent of SP.

Break-even. With contribution $c = \text{SP per unit} - \text{variable cost per unit}$: break-even units = fixed cost / c , and profit = units $\times c$ - fixed cost.

Warm-up

WARM-UP

build the reflex

Example 1

A shirt costs Rs. 480 and sells for Rs. 552. Find the profit and the profit%.

SOLUTION

$$\text{Profit} = 552 - 480 = 72, \text{ that is Rs. 72.}$$

$$\text{Profit\%} = \frac{72}{480} \times 100.$$

$$\frac{72}{480} = 0.15, \text{ and } 0.15 \times 100 = 15.$$

Answer: Profit Rs. 72, profit 15%

Example 2

A bag costs Rs. 1,250 and is sold at a 12% loss. Find the SP.

SOLUTION

$$\text{SP} = 1250 \times \frac{100-12}{100} = 1250 \times \frac{88}{100}.$$

$$1250 \times 88 = 110000. \text{ Then } \frac{110000}{100} = 1100.$$

Answer: Rs. 1,100

Example 3

An article is sold for Rs. 621 at a profit of 15%. Find the CP.

SOLUTION

$$\text{CP} = 621 \times \frac{100}{115}.$$

$$621 \times 100 = 62100. \text{ Then } \frac{62100}{115} = 540.$$

$$\text{Check: } 540 \times 1.15 = 621. \text{ Correct.}$$

Answer: Rs. 540

Example 4

A lamp is marked Rs. 900. A 20% discount is given. Find the SP.

SOLUTION

$$\text{SP} = 900 \times \frac{80}{100} = 900 \times 0.8 = 720.$$

Answer: Rs. 720

Example 5

MP is Rs. 1,600 and SP is Rs. 1,360. Find the discount%.

SOLUTION

$$\text{Discount} = 1600 - 1360 = 240.$$

$$\text{Discount\%} = \frac{240}{1600} \times 100 = 0.15 \times 100 = 15.$$

Answer: 15%

Example 6

MP is Rs. 2,000. Discounts of 20% and then 10% are given. Find the SP and the single equivalent discount.

SOLUTION

$$\text{After 20\%: } 2000 \times 0.8 = 1600.$$

$$\text{After 10\% on 1600: } 1600 \times 0.9 = 1440.$$

$$\text{Total discount} = 2000 - 1440 = 560.$$

$$\text{Single equivalent} = \frac{560}{2000} \times 100 = 28\%.$$

Answer: SP Rs. 1,440; single discount 28%

Example 7

CP is Rs. 250. The seller wants a 24% profit. Find the SP.

SOLUTION

$$SP = 250 \times \frac{124}{100}.$$

$$250 \times 124 = 31000. \text{ Then } \frac{31000}{100} = 310.$$

Answer: Rs. 310

Example 8

An item sold for Rs. 460 gives a loss of 8%. Find the CP.

SOLUTION

$$CP = 460 \times \frac{100}{92}.$$

$$460 \times 100 = 46000. \text{ Then } \frac{46000}{92} = 500.$$

$$\text{Check: } 500 \times 0.92 = 460. \text{ Correct.}$$

Answer: Rs. 500

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

A trader buys 24 notebooks for Rs. 1,080 and sells each at Rs. 54. Find the profit%.

SOLUTION

$$\text{CP of one notebook} = \frac{1080}{24} = 45.$$

$$\text{Profit on one} = 54 - 45 = 9.$$

$$\text{Profit\%} = \frac{9}{45} \times 100.$$

$$\frac{9}{45} = 0.2, \text{ so profit\%} = 20.$$

Answer: 20%

Example 10 [Accenture pattern]

A shopkeeper marks his goods 40% above cost and then allows a 25% discount. Find his profit%.

SOLUTION

Take CP = 100 (any number works; 100 is easiest).

$$MP = 100 + 40 = 140.$$

$$SP = 140 \times \frac{100-25}{100} = 140 \times 0.75 = 105.$$

$$\text{Profit} = 105 - 100 = 5 \text{ on a CP of } 100.$$

$$\text{Profit\%} = 5\%.$$

Answer: 5% profit

SHORTCUT

Markup $m\%$, discount $d\%$, in one line:

$$\text{Profit}\% = m - d - \frac{md}{100}$$

Here: $40 - 25 - \frac{40 \times 25}{100} = 40 - 25 - 10 = 5$. Same answer, no CP needed.

If the answer comes out negative, it is a loss of that much %.

Example 11 [Infosys pattern]

By selling an article for Rs. 810 a man loses 10%. At what price must he sell it to gain 20%?

SOLUTION

Step 1 — find CP.

$$\text{CP} = 810 \times \frac{100}{90}.$$

$$810 \times 100 = 81000. \text{ Then } \frac{81000}{90} = 900.$$

Step 2 — new SP for 20% gain.

$$\text{SP} = 900 \times \frac{120}{100} = 900 \times 1.2 = 1080.$$

Answer: Rs. 1,080

TRAP

Do **not** say “he lost 10%, so now add 30% to Rs. 810”. The 10% and the 20% are both on the **CP**, not on the old SP. You must go back to CP first, every time.

Example 12 [Wipro pattern]

Find the single discount equivalent to three successive discounts of 25%, 20% and 10%.

SOLUTION

Work with the price that remains, not the price removed.

$$\text{Remaining factor} = 0.75 \times 0.80 \times 0.90.$$

$$0.75 \times 0.80 = 0.60.$$

$$0.60 \times 0.90 = 0.54.$$

So 54% of MP is paid, which means 46% is taken off.

Answer: 46%

TRAP

Successive discounts are **never** added. $25 + 20 + 10 = 55$ is wrong; the true figure is 46%. Each later discount is charged on a smaller number.

Example 13 [Capgemini pattern]

A man sells two watches for Rs. 2,400 each. On one he gains 20%, on the other he loses 20%. Find his overall gain or loss%.

SOLUTION

$$\text{Watch 1 (20\% gain): } \text{CP}_1 = 2400 \times \frac{100}{120} = \frac{240000}{120} = 2000.$$

$$\text{Watch 2 (20\% loss): } \text{CP}_2 = 2400 \times \frac{100}{80} = \frac{240000}{80} = 3000.$$

$$\text{Total CP} = 2000 + 3000 = 5000.$$

$$\text{Total SP} = 2400 + 2400 = 4800.$$

$$\text{Loss} = 5000 - 4800 = 200.$$

$$\text{Loss\%} = \frac{200}{5000} \times 100 = 0.04 \times 100 = 4.$$

Answer: 4% loss

SHORTCUT

Same SP, one at $+x\%$ and one at $-x\%$: always a loss of $\frac{x^2}{100}\%$.

Here $x = 20$, so loss $= \frac{400}{100} = 4\%$. It is never a “no profit no loss” situation, because the losing item always had the bigger CP.

Example 14 [Cognizant pattern]

The cost price of 12 articles equals the selling price of 9 articles. Find the profit%.

SOLUTION

Let the CP of one article be Rs. 1.

CP of 12 articles = 12.

So SP of 9 articles = 12, giving SP of one article $= \frac{12}{9} = \frac{4}{3}$.

Profit on one $= \frac{4}{3} - 1 = \frac{1}{3}$.

Profit% $= \frac{1/3}{1} \times 100 = \frac{100}{3} = 33\frac{1}{3}\%$.

Answer: $33\frac{1}{3}\%$ profit

SHORTCUT

CP of a articles = SP of b articles $\rightarrow$ Profit% $= \frac{a-b}{b} \times 100$.

Here $\frac{12-9}{9} \times 100 = \frac{3}{9} \times 100 = 33\frac{1}{3}\%$. If $a < b$ it is a loss.

Example 15 [TCS NQT pattern]

A dealer allows a 10% discount on the marked price and still gains 17%. If the cost price is Rs. 900, find the marked price.

SOLUTION

Step 1 — SP from CP.

SP $= 900 \times \frac{117}{100} = 900 \times 1.17 = 1053$.

Step 2 — MP from SP.

SP is 90% of MP, so MP $= \frac{1053}{0.9}$.

$\frac{1053}{0.9} = \frac{10530}{9} = 1170$.

Check: $1170 \times 0.9 = 1053$, and $1053 - 900 = 153$; $\frac{153}{900} = 0.17$. Correct.

Answer: Rs. 1,170

Example 16 [Accenture pattern]

A shopkeeper sells goods at cost price but uses a weight of 900 g in place of 1 kg. Find his gain%.

SOLUTION

He charges the price of 1000 g but hands over 900 g.

His real cost is the cost of 900 g. His revenue is the price of 1000 g.

Gain (in grams of value) $= 1000 - 900 = 100$.

Gain% $= \frac{100}{900} \times 100 = \frac{10000}{900} = \frac{100}{9} = 11\frac{1}{9}\%$.

Answer: $11\frac{1}{9}\%$ (about 11.11%)

TRAP

The base is the **weight he actually gave**, not 1000 g. Writing $\frac{100}{1000} \times 100 = 10\%$ is the most common wrong answer here. He gained on what he parted with, and he parted with 900 g.

Example 17 [Infosys pattern]

An item is marked at Rs. 4,000. Two successive discounts of 20% and 5% are given. Find the SP and the total discount%.

SOLUTION

After 20%: $4000 \times 0.80 = 3200$.

After 5% on 3200: $3200 \times 0.95 = 3040$.

Total discount = $4000 - 3040 = 960$.

Discount% = $\frac{960}{4000} \times 100 = 0.24 \times 100 = 24$.

Check with the two-discount formula: $20 + 5 - \frac{20 \times 5}{100} = 25 - 1 = 24$. Correct.

Answer: SP Rs. 3,040; total discount 24%

Example 18 [Wipro pattern]

A fruit seller buys 100 kg of apples at Rs. 45 per kg. 10% of them rot and are thrown away. At what price per kg must he sell the rest to make a 20% profit on his total cost?

SOLUTION

Total CP = $100 \times 45 = 4500$.

Target SP = $4500 \times 1.20 = 5400$.

Apples left = $100 - 10 = 90$ kg.

Price per kg = $\frac{5400}{90} = 60$.

Answer: Rs. 60 per kg

Example 19 [Capgemini pattern]

A food stall has a fixed monthly cost of Rs. 36,000. Each meal costs Rs. 70 to make and sells for Rs. 130. (a) How many meals must it sell in a month to break even? (b) What is the profit if it sells 900 meals?

SOLUTION

Contribution per meal = $130 - 70 = 60$.

(a) Break-even meals = $\frac{36000}{60} = 600$.

(b) At 900 meals: total contribution = $900 \times 60 = 54000$.

Profit = $54000 - 36000 = 18000$.

Answer: (a) 600 meals (b) Rs. 18,000

Example 20 [Cognizant pattern]

A retailer buys an item listed at Rs. 2,500 at a 20% trade discount. He marks it at the list price and sells it to a customer at a 10% discount on the list price. Find his profit%.

SOLUTION

His CP = $2500 \times 0.80 = 2000$.

His SP = $2500 \times 0.90 = 2250$.

Profit = $2250 - 2000 = 250$.

Profit% = $\frac{250}{2000} \times 100 = 0.125 \times 100 = 12.5$.

Answer: 12.5%

PRACTICE · TIER 1 target: 60 s/Q

1. CP is Rs. 640 and SP is Rs. 736. Find the profit%.
2. An article sold for Rs. 1,485 gives a 10% loss. Find the CP.
3. An item is marked Rs. 3,200 and given discounts of 15% and then 20%. Find the SP.
4. A trader marks goods 60% above cost and gives a 25% discount. Find the profit%.
5. CP of 18 articles equals SP of 15 articles. Find the profit%.
6. Two shirts are sold at Rs. 1,800 each, one at 25% profit and one at 25% loss. Find the overall loss%.
7. A shopkeeper sells at cost price but uses an 800 g weight for 1 kg. Find his gain%.
8. An article costs Rs. 1,200. At what price must it be marked so that after a 20% discount a 20% profit still remains?
9. Find the single discount equivalent to 15%, 20% and 25% applied one after another.
10. A man buys 200 kg of rice at Rs. 42 per kg. He sells 120 kg at 20% profit and the rest at 5% loss. Find his overall profit%.
11. A workshop has a fixed cost of Rs. 48,000 per month, a variable cost of Rs. 55 per unit and a selling price of Rs. 95 per unit. Find the break-even number of units.
12. By selling at Rs. 900 a shopkeeper loses 25%. What SP would give him a 25% gain?
13. If a profit is 30% of the CP, what percentage of the SP is it?
14. A dealer marks 50% above CP and gives two successive discounts of 10% each. Find the profit%.
15. A laptop bought for Rs. 7,500 is sold at an 8% loss. Find the loss in rupees and the SP.

ANSWER KEY

1. 15%. $736 - 640 = 96$; $\frac{96}{640} = 0.15$.
2. Rs. 1,650. $\frac{1485}{0.9} = 1650$.
3. Rs. 2,176. $3200 \times 0.85 = 2720$; $2720 \times 0.8 = 2176$.
4. 20%. $60 - 25 - \frac{60 \times 25}{100} = 60 - 25 - 15 = 20$.
5. 20%. $\frac{18-15}{15} \times 100 = \frac{3}{15} \times 100 = 20$.
6. 6.25% loss. $\frac{x^2}{100} = \frac{625}{100} = 6.25$. (Check: CPs are 1440 and 2400; total CP 3840, total SP 3600, loss 240.)
7. 25%. $\frac{1000-800}{800} \times 100 = \frac{200}{800} \times 100 = 25$.
8. Rs. 1,800. Needed SP = $1200 \times 1.2 = 1440$; MP = $\frac{1440}{0.8} = 1800$.
9. 49%. $0.85 \times 0.80 \times 0.75 = 0.51$, so 49% off. (Adding to 60% is the trap.)
10. 10%. CP = 8400. First part: $120 \times 42 = 5040 \rightarrow$ SP 6048. Second: $80 \times 42 = 3360 \rightarrow$ SP 3192. Total SP = 9240; profit 840; $\frac{840}{8400} = 0.10$.
11. 1,200 units. Contribution = $95 - 55 = 40$; $\frac{48000}{40} = 1200$.
12. Rs. 1,500. CP = $\frac{900}{0.75} = 1200$; $1200 \times 1.25 = 1500$.
13. $23\frac{1}{13}\%$ (about 23.08%). Take CP 100, SP 130, profit 30; $\frac{30}{130} = \frac{3}{13}$.
14. 21.5%. $1.5 \times 0.9 \times 0.9 = 1.215$.
15. Loss Rs. 600, SP Rs. 6,900. $7500 \times 0.08 = 600$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A restaurant lists a meal set at THB 250 on a delivery app. The app keeps an 18% commission on the listed price. The restaurant's own cost per set is THB 160. Find the profit per set and the profit% on cost.

SOLUTION

$$\text{Money the restaurant keeps} = 250 \times \frac{100-18}{100} = 250 \times 0.82.$$

$$250 \times 0.82 = 205.$$

$$\text{Profit} = 205 - 160 = 45.$$

$$\text{Profit\%} = \frac{45}{160} \times 100.$$

$$\frac{45}{160} = 0.28125, \text{ so profit\%} = 28.125.$$

Answer: THB 45 per set; 28.125% profit

TRAP

A platform commission is a cut of the **listed price**, so it behaves exactly like a discount — it lowers SP. It is not a cost added to CP. If you had written $\text{CP} = 160 + 45 = 205$ you would get the wrong base.

Example 22 [Shopee · pattern]

A seller buys 300 phone cases at S\$3.20 each and lists them at S\$7.50. He sells 180 at the list price. To clear stock he then cuts the price by 40% and sells 100 more. The last 20 are written off as unsold. Find his overall profit%.

SOLUTION

$$\text{Total CP} = 300 \times 3.20 = 960.$$

$$\text{Full-price sales: } 180 \times 7.50 = 1350.$$

$$\text{Discounted price} = 7.50 \times 0.60 = 4.50.$$

$$\text{Clearance sales: } 100 \times 4.50 = 450.$$

$$\text{Unsold 20: revenue 0.}$$

$$\text{Total SP} = 1350 + 450 = 1800.$$

$$\text{Profit} = 1800 - 960 = 840.$$

$$\text{Profit\%} = \frac{840}{960} \times 100.$$

$$\frac{840}{960} = 0.875, \text{ so profit\%} = 87.5.$$

Answer: 87.5%

■ Unsold stock is **not** subtracted from CP. He paid for all 300; he only earned on 280. That is why write-offs hurt twice.

Example 23 [Sea · pattern]

A store marks its cost up by 80%. In a sale it gives 25% off, and members get a further 10% off the already reduced price. For a member sale, find (a) the profit%, and (b) the total discount% off the marked price.

SOLUTION

$$\text{Take CP} = 100.$$

$$\text{MP} = 180.$$

$$\text{After 25\%: } 180 \times 0.75 = 135.$$

$$\text{After a further 10\%: } 135 \times 0.90 = 121.5.$$

$$\text{(a) Profit} = 121.5 - 100 = 21.5 \text{ on a CP of 100, so profit\%} = 21.5.$$

$$\text{(b) Discount off MP} = 180 - 121.5 = 58.5.$$

$$\text{Discount\%} = \frac{58.5}{180} \times 100.$$

$$\frac{58.5}{180} = 0.325, \text{ so } 32.5\%.$$

$$\text{Check with the two-discount rule: } 25 + 10 - \frac{25 \times 10}{100} = 35 - 2.5 = 32.5. \text{ Correct.}$$

Answer: (a) 21.5% profit (b) 32.5% total discount

Example 24 [LINE MAN · pattern]

A grocer in Bangkok buys rice at THB 40 per kg and marks it up 25%. He also uses a 900 g weight whenever he sells “1 kg”. Find his actual profit%.

SOLUTION

For each “1 kg” sale:

He charges $40 \times 1.25 = 50$, that is THB 50.

He actually hands over 900 g, which cost him $0.9 \times 40 = 36$, that is THB 36.

Profit = $50 - 36 = 14$.

Profit% = $\frac{14}{36} \times 100 = \frac{1400}{36} = \frac{350}{9} = 38\frac{8}{9}\%$.

Answer: $38\frac{8}{9}\%$ (about 38.89%)

SHORTCUT

Two cheats multiply, they do not add.

$$\frac{SP}{CP} = \frac{100 + m}{100} \times \frac{\text{true weight}}{\text{weight given}}$$

Here $1.25 \times \frac{1000}{900} = 1.25 \times 1.1111 = 1.3889$, so about 38.89% profit. Adding $25 + 11.11 = 36.11$ is wrong.

Example 25 [GIC · pattern]

A Singapore trader imports 500 gadgets from India at Rs. 1,950 each when S\$1 = Rs. 65. Shipping and duty add S\$5 per unit. He sells each at S\$45.50. Find the profit% and the total profit.

SOLUTION

Cost of one gadget in SGD = $\frac{1950}{65} = 30$.

Add shipping and duty: $30 + 5 = 35$. This is the CP per unit.

SP per unit = 45.50.

Profit per unit = $45.50 - 35 = 10.50$.

Profit% = $\frac{10.50}{35} \times 100$.

$\frac{10.50}{35} = 0.30$, so profit% = 30.

Total profit = $500 \times 10.50 = 5250$.

Answer: 30% profit; total S\$5,250

Example 26 [Agoda · pattern]

A 40-room hotel has a fixed cost of THB 24,000 per night. Each occupied room costs THB 400 in cleaning and utilities, and the room rate is THB 1,200. Find (a) the break-even number of rooms, (b) the result on a night when 28 rooms are sold, (c) the occupancy% needed to break even.

SOLUTION

Contribution per room = $1200 - 400 = 800$.

(a) Break-even rooms = $\frac{24000}{800} = 30$.

(b) At 28 rooms: contribution = $28 \times 800 = 22400$.

$22400 - 24000 = -1600$. That is a loss of THB 1,600.

(c) Occupancy needed = $\frac{30}{40} \times 100 = 0.75 \times 100 = 75\%$.

Answer: (a) 30 rooms (b) loss of THB 1,600 (c) 75% occupancy

Below break-even every empty room still carries its share of the fixed cost. That is why hotels discount hard once the night is nearly over — any price above THB 400 helps.

Example 27 [DBS · pattern]

A wholesaler sells to a retailer at a 25% profit, and the retailer sells to a customer at a 20% profit. The customer pays Rs. 4,500. Find the wholesaler's cost price.

SOLUTION

Work backwards, one layer at a time.

$$\text{Retailer's CP} = \frac{4500}{1.20}.$$

$$\frac{4500}{1.2} = \frac{45000}{12} = 3750. \text{ This is also the wholesaler's SP.}$$

$$\text{Wholesaler's CP} = \frac{3750}{1.25}.$$

$$\frac{3750}{1.25} = \frac{375000}{125} = 3000.$$

Check forward: $3000 \times 1.25 = 3750$; $3750 \times 1.2 = 4500$. Correct.

Answer: Rs. 3,000

Example 28 [SCB · pattern]

By what percentage above cost must a seller mark his goods so that after allowing a 20% discount he still earns a 30% profit?

SOLUTION

Let CP = 100 and let the markup be $m\%$.

$$\text{MP} = 100 + m.$$

SP = (MP) $\times$ 0.80, and this must equal 130.

$$\text{So } (\text{MP}) \times 0.80 = 130.$$

$$\text{MP} = \frac{130}{0.80} = \frac{13000}{80} = 162.5.$$

$$\text{Markup} = 162.5 - 100 = 62.5.$$

Answer: 62.5% above cost

SHORTCUT

$$\text{Markup needed} = \left(\frac{100+p}{100-d} - 1 \right) \times 100.$$

Here $\frac{130}{80} = 1.625$, so 62.5%. Memorise the shape: **wanted profit factor divided by remaining-after-discount factor**.

Example 29 [LINE MAN · pattern]

A cloud kitchen buys 500 kg of chicken at THB 85 per kg. A freezer failure spoils 8% of it. The usable chicken is turned into meals — 1 kg makes 4 meals. All other costs (rent, staff, other ingredients) come to THB 77,500. Each meal sells for THB 75 and all meals are sold. Find the profit%.

SOLUTION

$$\text{Chicken cost} = 500 \times 85 = 42500.$$

$$\text{Total cost} = 42500 + 77500 = 120000.$$

$$\text{Usable chicken} = 500 \times \frac{100-8}{100} = 500 \times 0.92 = 460 \text{ kg.}$$

$$\text{Meals made} = 460 \times 4 = 1840.$$

$$\text{Revenue} = 1840 \times 75.$$

$$1840 \times 75 = 1840 \times 70 + 1840 \times 5 = 128800 + 9200 = 138000.$$

$$\text{Profit} = 138000 - 120000 = 18000.$$

$$\text{Profit\%} = \frac{18000}{120000} \times 100 = 0.15 \times 100 = 15.$$

Answer: 15%

Example 30 [Grab · pattern]

Three stores sell the same jacket, all marked S\$480. Store A gives 30% off then a further 10% off. Store B gives 20% off then a further 20% off. Store C gives a flat 37% off. Rank the three final prices.

SOLUTION

Store A: factor = $0.70 \times 0.90 = 0.63$.

$$480 \times 0.63 = 302.40.$$

Store B: factor = $0.80 \times 0.80 = 0.64$.

$$480 \times 0.64 = 307.20.$$

Store C: factor = $1 - 0.37 = 0.63$.

$$480 \times 0.63 = 302.40.$$

A and C are exactly equal at S\$302.40. B is dearer by $307.20 - 302.40 = \text{S\$}4.80$.

Answer: A = C = S\$302.40 (cheapest); B = S\$307.20

TRAP

“30% then 10%” is exactly 37%, not 40%. And “20% then 20%” is 36%, not 40%. Equal-looking discount pairs are not equal: for a fixed total $d_1 + d_2$, the pair that is **more lopsided** takes off more. Check: 30 + 10 gives 37% off, 20 + 20 gives 36% off.

PRACTICE · TIER 2 target: 100 s/Q

1. A merchant lists an item at S\$120 on a platform that charges a 15% commission. His cost is S\$80. Find his profit%.
2. A store marks up 75%, gives a 20% sale discount, then a further 5% member discount. Find the profit%.
3. A dealer uses a 960 g weight for 1 kg and also marks up 20% over cost. Find his actual profit%.
4. Goods pass from wholesaler to distributor at a 20% profit, distributor to shop at a 25% profit, and shop to customer at a 10% profit. The customer pays Rs. 3,300. Find the wholesaler's cost.
5. A subscription box has a fixed cost of S\$12,600 per month and a variable cost of S\$8 per box. Each box sells for S\$26. Find the break-even number of boxes.
6. By what percentage above cost must goods be marked so that a 25% discount still leaves a 20% gain?
7. A trader buys 400 kg of onions at THB 25 per kg. 20% spoil. He sells the rest at THB 40 per kg. Find his profit%.
8. Two laptops are sold at Rs. 3,600 each, one at a 20% gain and one at a 20% loss. Find the net loss in rupees.
9. A shop's selling price gives it a 25% profit. Its cost then rises by 20% but it keeps the same selling price. Find the new profit%.
10. A phone marked S\$800 is sold for S\$612 after two successive discounts. The first discount is 10%. Find the second.
11. A seller earns a 20% profit on cost. What percentage of his selling price is the profit?
12. A cafe sells a coffee at THB 90 and makes a 50% profit on cost. Bean prices rise and the cost goes up by THB 12. What new price keeps the profit at 50%?

ANSWER KEY

1. 27.5%. $\text{Kept} = 120 \times 0.85 = 102$; $\text{profit } 22; \frac{22}{80} = 0.275$.
2. 33%. $1.75 \times 0.80 \times 0.95 = 1.40 \times 0.95 = 1.33$.

3. 25%. $1.20 \times \frac{1000}{960} = 1.20 \times 1.041666... = 1.25$.
4. Rs. 2,000. $\frac{3300}{1.1} = 3000$; $\frac{3000}{1.25} = 2400$; $\frac{2400}{1.2} = 2000$.
5. 700 boxes. Contribution = $26 - 8 = 18$; $\frac{12600}{18} = 700$.
6. 60%. $\frac{1.20}{0.75} = 1.60$.
7. 28%. CP = $400 \times 25 = 10000$; usable 320 kg; SP = $320 \times 40 = 12800$; profit 2800; $\frac{2800}{10000} = 0.28$.
8. Rs. 300 loss (4%). $CP_1 = \frac{3600}{1.2} = 3000$, $CP_2 = \frac{3600}{0.8} = 4500$; total CP 7500, total SP 7200.
9. $4\frac{1}{6}\%$ (about 4.17%). CP 100 $\rightarrow$ SP 125. New CP 120, profit 5, $\frac{5}{120} = \frac{1}{24}$.
10. 15%. $800 \times 0.9 = 720$; $\frac{612}{720} = 0.85$.
11. $16\frac{2}{3}\%$. Take CP 100, SP 120, profit 20; $\frac{20}{120} = \frac{1}{6}$.
12. THB 108. Old cost = $\frac{90}{1.5} = 60$; new cost = 72; $72 \times 1.5 = 108$.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Goldman Sachs · pattern]

Two articles are sold at the same price, one at a profit of $x\%$ and one at a loss of $x\%$. Prove the net result is always a loss of $\frac{x^2}{100}\%$. Then find x if the net loss is 9%.

SOLUTION

Insight: the two cost prices are not equal. The loss-making article had the larger CP, so the loss is computed on a bigger base than the profit.

Let each selling price be S .

$$CP_1 = \frac{100S}{100+x} \quad CP_2 = \frac{100S}{100-x}$$

$$\text{Total CP} = 100S \left[\frac{1}{100+x} + \frac{1}{100-x} \right].$$

Add the fractions:

$$\frac{1}{100+x} + \frac{1}{100-x} = \frac{(100-x) + (100+x)}{(100+x)(100-x)} = \frac{200}{10000-x^2}$$

$$\text{So total CP} = \frac{20000S}{10000-x^2}.$$

$$\text{Total SP} = 2S.$$

$$\text{Loss} = \frac{20000S}{10000-x^2} - 2S = \frac{20000S - 2S(10000-x^2)}{10000-x^2} = \frac{2Sx^2}{10000-x^2}.$$

The numerator has $x^2 > 0$, so the loss is positive whatever x is. It is always a loss.

$$\text{Loss}\% = \frac{\text{Loss}}{\text{Total CP}} \times 100 = \frac{2Sx^2}{10000-x^2} \times \frac{10000-x^2}{20000S} \times 100.$$

The bracket $(10000 - x^2)$ and S both cancel:

$$\text{Loss}\% = \frac{2x^2}{20000} \times 100 = \frac{x^2}{100}$$

Now set $\frac{x^2}{100} = 9$, so $x^2 = 900$ and $x = 30$.

Answer: Net loss is always $\frac{x^2}{100}\%$; a 9% net loss means $x = 30$

Example 32 [Amazon · pattern]

A dealer cheats twice. When buying, his weight is set so that he receives 1,100 g for every 1,000 g he pays for. When selling, he gives only 900 g for every 1,000 g he charges for. He quotes exactly his cost price. Find his overall gain%.

SOLUTION

Insight: stop counting rupees and count grams. Each cheat is a clean multiplying factor on the grams that move.

Let the honest market rate be c per gram.

Buying. He pays for 1000 g, i.e. he pays $1000c$, and gets 1100 g.

His true cost per gram = $\frac{1000c}{1100} = \frac{10c}{11}$.

Selling. He charges the price of 1000 g, i.e. $1000c$, and gives 900 g.

His revenue per gram delivered = $\frac{1000c}{900} = \frac{10c}{9}$.

Profit per gram delivered = $\frac{10c}{9} - \frac{10c}{11} = 10c\left(\frac{1}{9} - \frac{1}{11}\right) = 10c \times \frac{11-9}{99} = \frac{20c}{99}$.

Gain% = $\frac{\frac{20c}{99}}{\frac{10c}{11}} \times 100 = \frac{20c}{99} \times \frac{11}{10c} \times 100$.

= $\frac{220}{990} \times 100 = \frac{2}{9} \times 100 = 22\frac{2}{9}\%$.

One-line check with factors:

$$\frac{SP}{CP} = \frac{1100}{1000} \times \frac{1000}{900} = 1.1 \times 1.111... = 1.2222...$$

which is a gain of $22\frac{2}{9}\%$.

Answer: $22\frac{2}{9}\%$ (about 22.22%)

Example 33 [Google · pattern]

A store sells 200 units a week at S\$50 each. Each unit costs S\$30. Market tests show that for every S\$1 cut in price, weekly sales rise by 20 units. What price maximises weekly profit, and what is that profit?

SOLUTION

Insight: profit is a quadratic in the size of the price cut. Write it out and take the vertex — do not guess prices one by one.

Let the price be cut by n dollars.

Price = $50 - n$.

Quantity = $200 + 20n$.

Margin per unit = $(50 - n) - 30 = 20 - n$.

Weekly profit

$$P(n) = (20 - n)(200 + 20n)$$

Expand:

$$P(n) = 20 \times 200 + 20 \times 20n - 200n - 20n^2 = 4000 + 400n - 200n - 20n^2$$

$$P(n) = 4000 + 200n - 20n^2$$

This is a downward parabola. Its peak is at $n = -\frac{b}{2a}$ with $a = -20$ and $b = 200$:

$$n = -\frac{200}{2 \times (-20)} = \frac{200}{40} = 5$$

So cut S\$5.

Price = $50 - 5 = 45$. Quantity = $200 + 100 = 300$. Margin = 15.

Profit = $300 \times 15 = 4500$.

Sanity check on either side:

$n = 4$: margin 16, quantity 280, profit = 4480.

$n = 6$: margin 14, quantity 320, profit = 4480.

Both are below 4500, so $n = 5$ is the peak. (At $n = 0$ the profit was only $200 \times 20 = 4000$.)

Answer: Sell at S\$45; maximum weekly profit S\$4,500

SHORTCUT

The midpoint rule for linear demand. Let k be the extra units sold per 1 unit of price cut. Define the **choke price** — the price at which sales would fall to zero:

$$\text{choke price} = \text{current price} + \frac{\text{current quantity}}{k}$$

Then the profit-maximising price is the **midpoint of the cost price and the choke price**:

$$\text{best price} = \frac{\text{cost} + \text{choke price}}{2}$$

Here $k = 20$, so choke = $50 + 200/20 = 60$, and best price = $(30 + 60)/2 = 45$. Same answer in one line.

Example 34 [D. E. Shaw · pattern]

A shopkeeper has 100 identical items. Each costs him Rs. 90, and he also has a fixed cost of Rs. 1,500 for the batch. He will sell every item, each at either Rs. 150 (full price) or Rs. 120 (clearance). He wants a profit of at least 25% on his total cost. What is the smallest number he must sell at full price?

SOLUTION

Insight: start from the worst case — everything at Rs. 120 — and note that each upgrade to full price adds exactly Rs. 30. The problem becomes a single division.

Total cost = $100 \times 90 + 1500 = 9000 + 1500 = 10500$.

Required SP = 10500×1.25 .

$10500 \times 1.25 = 10500 + 2625 = 13125$.

Let k items sell at Rs. 150 and $(100 - k)$ at Rs. 120.

$$\text{SP} = 150k + 120(100 - k) = 150k + 12000 - 120k = 12000 + 30k$$

Need $12000 + 30k \geq 13125$.

$30k \geq 1125$.

$k \geq \frac{1125}{30} = 37.5$.

k must be a whole number, so $k = 38$.

Verify both sides.

$k = 38$: SP = $12000 + 1140 = 13140$. Profit = $13140 - 10500 = 2640$. $\frac{2640}{10500} = 0.2514\dots$, i.e. 25.14%, which clears 25%.

$k = 37$: SP = $12000 + 1110 = 13110$. Profit = 2610. $\frac{2610}{10500} = 0.2486\dots$, i.e. 24.86%, which fails.

Answer: 38 items at full price

Example 35 [Microsoft · pattern]

A seller buys one unit for Rs. 500. With probability 0.6 it sells in week 1 at Rs. 800. If it does not sell, he must cut the price by 25% in week 2, and at that price it sells with probability 0.75. If it still does not sell, it is scrapped for Rs. 100. Find his expected profit and expected profit%.

SOLUTION

Insight: a discount decision is judged on expected revenue, not on the headline price. Build the tree and weigh every leaf.

Week-2 price = $800 \times 0.75 = 600$.

Leaf 1: sells in week 1. Probability 0.6, revenue 800.

Leaf 2: fails week 1, sells week 2. Probability $0.4 \times 0.75 = 0.30$, revenue 600.

Leaf 3: fails both. Probability $0.4 \times 0.25 = 0.10$, revenue 100.

Probabilities add to $0.6 + 0.3 + 0.1 = 1$. Good.

Expected revenue:

$$E = 0.6 \times 800 + 0.30 \times 600 + 0.10 \times 100$$

$$E = 480 + 180 + 10 = 670$$

Expected profit = $670 - 500 = 170$.

Expected profit% = $\frac{170}{500} \times 100 = 0.34 \times 100 = 34$.

Answer: Expected profit Rs. 170, i.e. 34% on cost

Compare: if he refused ever to discount, revenue would be $0.6 \times 800 + 0.4 \times 100 = 480 + 40 = 520$, a profit of only Rs. 20. The 25% discount is worth Rs. 150 in expectation.

Example 36 [Adobe · pattern]

A shopkeeper marks his goods up by $m\%$ and then gives a discount of $m\%$ on the marked price. Show that he always loses, and find m if his loss is 4%.

SOLUTION

Insight: the markup is taken on CP and the discount on the bigger MP, so they cannot cancel. Algebraically it is a difference of squares.

Take CP = 100.

$$MP = 100\left(1 + \frac{m}{100}\right).$$

$$SP = MP \times \left(1 - \frac{m}{100}\right) = 100\left(1 + \frac{m}{100}\right)\left(1 - \frac{m}{100}\right).$$

Use $(1 + a)(1 - a) = 1 - a^2$ with $a = \frac{m}{100}$:

$$SP = 100\left(1 - \frac{m^2}{10000}\right) = 100 - \frac{m^2}{100}$$

Since $\frac{m^2}{100} > 0$ for any $m \neq 0$, SP is always below 100. He always loses.

Loss = $\frac{m^2}{100}$ on a CP of 100, so

$$\text{Loss}\% = \frac{m^2}{100}$$

Set $\frac{m^2}{100} = 4$, giving $m^2 = 400$ and $m = 20$.

Check with $m = 20$: CP 100, MP 120, SP = $120 \times 0.8 = 96$. Loss 4 on 100 = 4%. Correct.

Answer: Always a loss of $\frac{m^2}{100}\%$; a 4% loss means $m = 20$

Example 37 [Uber · pattern]

A dealer sells at cost price but uses a faulty balance. On a nominal 1 kg sale his gain is 25%. He now sells in nominal 5 kg lots with the same balance, but wants his gain on such a lot to be exactly 10%. What percentage discount must he give on his quoted price?

SOLUTION

Insight: the faulty balance is a fixed multiplier on revenue. Find that multiplier once, then choose the discount that pulls the product down to 1.10.

Step 1 — how heavy is his “1 kg”?

Let the weight he actually gives be w grams.

$$\text{Gain}\% = \frac{1000 - w}{w} \times 100 = 25$$

$$1000 - w = 0.25w$$

$$1000 = 1.25w$$

$$w = \frac{1000}{1.25} = 800 \text{ g}$$

Step 2 — the 5 kg lot.

Nominal 5 kg means he delivers $5 \times 800 = 4000$ g.

Let the honest cost be c per gram.

His cost = $4000c$.

His quoted price (cost price of 5000 g) = $5000c$.

Step 3 — the discount.

Let the discount be $d\%$. Revenue = $5000c(1 - \frac{d}{100})$.

For a 10% gain: revenue = $1.10 \times 4000c = 4400c$.

$$5000c \left(1 - \frac{d}{100}\right) = 4400c$$

Cancel c :

$$1 - \frac{d}{100} = \frac{4400}{5000} = 0.88$$

$$\frac{d}{100} = 0.12$$

$$d = 12$$

Check: $5000 \times 0.88 = 4400$ against a cost of 4000, a gain of $\frac{400}{4000} = 10\%$. Correct.

Answer: A 12% discount

Example 38 [Goldman Sachs · pattern]

A trader has exactly Rs. 9,000 and must spend all of it. Item A costs Rs. 300 and sells at a 20% profit. Item B costs Rs. 200 and sells at a 35% profit. He must buy at least 10 of each, and only whole items. What mix gives the largest total profit, and how much is it?

SOLUTION

Insight: with a fixed budget, rank the items by profit per rupee spent, not by profit per item. Item A earns more per item but less per rupee.

Profit per item:

$$\text{A: } 300 \times 0.20 = 60.$$

$$\text{B: } 200 \times 0.35 = 70.$$

Profit per rupee spent:

$$\text{A: } \frac{60}{300} = 0.20.$$

$$\text{B: } \frac{70}{200} = 0.35.$$

B is better per rupee, so push money into B and cut A to its floor of 10.

Budget equation. Let a be the number of A and b the number of B.

$$300a + 200b = 9000$$

Divide by 100:

$$3a + 2b = 90$$

For b to be a whole number, $90 - 3a$ must be even, so $3a$ must be even, so a must be even.

Smallest even a with $a \geq 10$ is $a = 10$.

$$2b = 90 - 30 = 60 \quad b = 30$$

Check $b = 30 \geq 10$. Valid.

$$\text{Profit} = 10 \times 60 + 30 \times 70 = 600 + 2100 = 2700.$$

Compare the next options to be sure.

$$a = 12: 2b = 90 - 36 = 54, b = 27. \text{ Profit} = 720 + 1890 = 2610.$$

$$a = 14: 2b = 90 - 42 = 48, b = 24. \text{ Profit} = 840 + 1680 = 2520.$$

Profit falls as a rises, exactly as the per-rupee ranking predicted.

$$\text{Overall profit\%} = \frac{2700}{9000} \times 100 = 30\%.$$

Answer: 10 of A and 30 of B; maximum profit Rs. 2,700 (30% on the Rs. 9,000)

PRACTICE · TIER 3 target: 4 min/Q

- Two cars are sold at Rs. 4,80,000 each, one at a 25% profit and one at a 25% loss. Find the net loss% and the net loss in rupees.
- A dealer's buying weight gives him 25% more than he pays for, and his selling weight gives the buyer 20% less than charged for. He quotes cost price. Find his overall gain%.
- A stall sells 120 plates a day at THB 60. Each plate costs THB 35. Every THB 1 cut in price adds 8 plates a day. Find the price that maximises daily profit and that profit.
- A shopkeeper marks up by $m\%$ and then discounts by $m\%$, ending with a 9% loss. Find m .
- A seller buys an item for Rs. 2,000. It sells at Rs. 3,000 with probability 0.5. If not, he cuts the price to Rs. 2,400 and it then sells with probability 0.8; otherwise it is scrapped for Rs. 1,200. Find the expected profit and expected profit%.
- A trader must spend exactly Rs. 12,000. Item P costs Rs. 400 and sells at a 25% profit; item Q costs Rs. 250 and sells at a 30% profit. He must buy at least 5 of each, whole items only. Find the maximum total profit.
- A subscription box service has a fixed cost of S\$9,000 a month and a variable cost of S\$14 per box. It sells 500 boxes a month at S\$40. If a 10% price cut raises volume by 25%, does monthly profit rise or fall, and by how much?
- A dealer sells at cost price, but his "1 kg" weight is really 875 g. Find his gain%. What discount on his quoted price would bring his gain down to exactly 5%?

ANSWER KEY

- 6.25% loss, Rs. 64,000. By the rule, $\text{loss\%} = \frac{25^2}{100} = 6.25$. In money: $\text{CP}_1 = \frac{480000}{1.25} = 384000$ and $\text{CP}_2 = \frac{480000}{0.75} = 640000$, total CP = 10, 24, 000 against SP = 9, 60, 000.
- 56.25%. Buying factor = $\frac{1250}{1000} = 1.25$. Selling factor = $\frac{1000}{800} = 1.25$. Together $1.25 \times 1.25 = 1.5625$. The two cheats multiply; adding $25 + 25 = 50$ is wrong.
- THB 55, profit THB 3,200. With n baht cut: profit = $(25 - n)(120 + 8n) = 3000 + 80n - 8n^2$, peaking at $n = \frac{80}{16} = 5$. Then price 55, quantity 160, margin 20. Check $n = 4$ gives $21 \times 152 = 3192$ and $n = 6$ gives $19 \times 168 = 3192$, both lower. Midpoint rule: choke = $60 + \frac{120}{8} = 75$, so best price = $\frac{35+75}{2} = 55$.
- $m = 30$. Loss% = $\frac{m^2}{100} = 9$, so $m^2 = 900$. A markup and discount of the same size never cancel because they sit on different bases.
- Rs. 580, i.e. 29%. $E = 0.5 \times 3000 + 0.5 \times (0.8 \times 2400 + 0.2 \times 1200) = 1500 + 0.5 \times 2160 = 2580$. Profit = $2580 - 2000$.
- Rs. 3,500. Profit per rupee: P gives 0.25, Q gives 0.30, so minimise P. $400p + 250q = 12000$ becomes $8p + 5q = 240$, which forces p to be a multiple of 5, so $p = 5$ and $q = 40$. Profit = $5 \times 100 + 40 \times 75 = 500 + 3000$.
- It rises by S\$750. Now: contribution = 26, so $500 \times 26 - 9000 = 13000 - 9000 = 4000$. After: price 36, contribution 22, volume 625, so $625 \times 22 - 9000 = 13750 - 9000 = 4750$. The 25% volume gain beats the 15.4% contribution loss.

8. Gain = $\frac{1}{7} = 14\frac{2}{7}\%$ (about 14.29%), and the discount needed is 8.125%. Gain% = $\frac{1000-875}{875} \times 100 = \frac{125}{875} \times 100$. For a 5% gain: $1000c(1 - \frac{d}{100}) = 1.05 \times 875c = 918.75c$, so $1 - \frac{d}{100} = 0.91875$ and $d = 8.125$.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 30 minutes for all 20

1. CP is Rs. 850 and SP is Rs. 1,020. Find the profit%.
2. Find the single discount equivalent to 40% followed by 25%.
3. An article sold for Rs. 2,070 gives a 15% profit. Find the CP.
4. A trader marks up 25% and then gives a 20% discount. Find his profit or loss%.
5. A shopkeeper sells at cost price but uses a 750 g weight for 1 kg. Find his gain%.
6. Two pens are sold at Rs. 480 each, one at a 20% gain and one at a 20% loss. Find the net loss in rupees.
7. The cost price of 25 items equals the selling price of 20 items. Find the profit%.
8. A unit has a fixed cost of Rs. 54,000, a variable cost of Rs. 45 per piece and a selling price of Rs. 90 per piece. Find the break-even number of pieces.
9. An item marked S\$650 is sold for S\$520. Find the discount%.
10. A seller wants a 25% profit after giving a 20% discount. By what percent above cost must he mark the goods?
11. A trader buys 250 kg of tomatoes at THB 18 per kg. 10% spoil. He sells the rest at THB 25 per kg. Find his profit%.
12. A producer sells to a wholesaler at a 15% profit, the wholesaler to a retailer at a 20% profit, and the retailer to a customer at a 25% profit. The producer's cost is Rs. 2,000. What does the customer pay?
13. If the profit is 25% of the cost price, what percentage of the selling price is it?
14. A dealer marks his goods 40% above cost, uses an 800 g weight for 1 kg, and sells at the marked price. Find his gain%.
15. A markup of $m\%$ followed by a discount of $m\%$ leaves a loss of 2.25%. Find m .
16. A firm sells 80 units at Rs. 200 each; each unit costs Rs. 140. Every Rs. 5 cut in price adds 10 units of sales. Find the price that maximises profit.
17. A seller buys at Rs. 600. The item sells at Rs. 900 with probability 0.7; otherwise it is cleared at Rs. 700. Find the expected profit%.
18. A phone marked S\$900 is sold for S\$648 after a 20% discount followed by a second discount. Find the second discount%.
19. A trader's selling price does not change but his cost price falls by 20%. His profit was 20%. Find his new profit%.
20. A batch of 200 items costs Rs. 50 each plus a fixed cost of Rs. 5,000. All are sold at Rs. 90 each. Find the profit%.

ANSWER KEY

1. 20%. $1020 - 850 = 170$; $\frac{170}{850} = 0.20$.
2. 55%. $0.60 \times 0.75 = 0.45$, so 55% is taken off. (Adding to 65% is the trap.)
3. Rs. 1,800. $\frac{2070}{1.15} = 1800$.
4. Neither — exactly no profit, no loss. $1.25 \times 0.80 = 1.00$. Or $25 - 20 - \frac{25 \times 20}{100} = 5 - 5 = 0$.
5. $33\frac{1}{3}\%$. $\frac{1000-750}{750} \times 100 = \frac{250}{750} \times 100$.
6. Rs. 40 (a 4% loss). $CP_1 = \frac{480}{1.2} = 400$, $CP_2 = \frac{480}{0.8} = 600$; total CP 1000, total SP 960.
7. 25%. $\frac{25-20}{20} \times 100 = \frac{5}{20} \times 100$.
8. 1,200 pieces. Contribution = $90 - 45 = 45$; $\frac{54000}{45} = 1200$.
9. 20%. $650 - 520 = 130$; $\frac{130}{650} = 0.20$.
10. 56.25%. $\frac{1.25}{0.80} = 1.5625$.

11. 25%. $CP = 250 \times 18 = 4500$; left 225 kg; $SP = 225 \times 25 = 5625$; profit 1125; $\frac{1125}{4500} = 0.25$.
12. Rs. 3,450. $2000 \times 1.15 = 2300$; $\times 1.20 = 2760$; $\times 1.25 = 3450$.
13. 20%. Take CP 100, SP 125, profit 25; $\frac{25}{125} = 0.20$.
14. 75%. $1.40 \times \frac{1000}{800} = 1.40 \times 1.25 = 1.75$. The markup and the false weight multiply.
15. $m = 15$. $\frac{m^2}{100} = 2.25 \rightarrow m^2 = 225$.
16. Rs. 190 (profit Rs. 5,000). With n cuts of Rs. 5: profit $= (60 - 5n)(80 + 10n) = 4800 + 200n - 50n^2$, peak at $n = \frac{200}{100} = 2$. Price = 190, quantity 100, margin 50. Check $n = 1$ gives $55 \times 90 = 4950$ and $n = 3$ gives $45 \times 110 = 4950$. Midpoint rule: Rs. 5 buys 10 units so $k = 2$; choke $= 200 + \frac{80}{2} = 240$, best price $= \frac{140+240}{2} = 190$.
17. 40%. $E = 0.7 \times 900 + 0.3 \times 700 = 630 + 210 = 840$; profit 240; $\frac{240}{600} = 0.40$.
18. 10%. $900 \times 0.8 = 720$; $\frac{648}{720} = 0.90$.
19. 50%. CP 100 $\rightarrow$ SP 120. New CP 80, profit 40, $\frac{40}{80} = 0.50$.
20. 20%. Total cost $= 200 \times 50 + 5000 = 10000 + 5000 = 15000$; $SP = 200 \times 90 = 18000$; profit 3000; $\frac{3000}{15000} = 0.20$. (Forgetting the fixed cost gives 80% — the trap.)

ONE-PAGE REVISION CARD

The bases — get these right and half the traps die

- Profit% and loss% are on **CP**.
- Discount% is on **MP**.
- Commission and platform fees behave like discounts: they cut SP.

Core formulas

$$\text{Profit\%} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100 \quad \text{SP} = \text{CP} \frac{100 + p}{100} \quad \text{CP} = \frac{100 \text{ SP}}{100 + p}$$

$$\text{SP} = \text{MP} \frac{100 - d}{100} \quad \text{MP}(100 - d) = \text{CP}(100 + p)$$

Shortcuts worth memorising

- Markup $m\%$, discount $d\%$: Profit% = $m - d - \frac{md}{100}$.
- Two discounts: single = $d_1 + d_2 - \frac{d_1 d_2}{100}$.
- Three or more discounts: multiply the remaining factors, e.g. $0.75 \times 0.80 \times 0.90 = 0.54$, so 46% off.
- Same SP, $+x\%$ and $-x\%$: always a **loss** of $\frac{x^2}{100}\%$.
- Markup $m\%$ then discount $m\%$: always a **loss** of $\frac{m^2}{100}\%$.
- CP of a articles = SP of b articles: Profit% = $\frac{a-b}{b} \times 100$.
- False weight, selling at CP: Gain% = $\frac{\text{true} - \text{given}}{\text{given}} \times 100$.
- False weight **and** markup: multiply the factors, $\text{SP}/\text{CP} = ((100 + m)/100) \times (\text{true}/\text{given})$.
- Markup needed for profit $p\%$ after discount $d\%$: $\text{MP}/\text{CP} = (100 + p)/(100 - d)$.
- Profit is $p\%$ of CP $\rightarrow$ it is $\frac{p}{100+p} \times 100\%$ of SP.
- Break-even units = $\frac{\text{fixed cost}}{\text{SP} - \text{variable cost}}$.
- Linear demand: best price = (cost + choke price)/2, where choke price = current price + current quantity / k and k is the extra units sold per 1 unit of price cut. (Or expand the quadratic and take its vertex.)

The top 5 traps

1. **Adding successive discounts.** 30% then 10% is 37% off, not 40%.
2. **Using the wrong base.** A 10% loss then a 20% gain are both on CP — return to CP before moving on. Never build a new percentage on an old SP.
3. **False weight base.** Gain is on the weight **given**, not on 1000 g. 900 g for a kilo is $11\frac{1}{9}\%$ gain, not 10%.
4. **Forgetting fixed costs.** A batch's CP is item cost $\times$ quantity **plus** fixed cost. Unsold stock is still paid for; it lowers SP, it does not lower CP.
5. **Assuming equal-and-opposite cancels.** $+x\%$ and $-x\%$ on equal SPs is a loss. Markup $m\%$ then discount $m\%$ is a loss. Different bases never cancel.

Fast fraction table for this chapter

%	12.5	16⅔	20	25	33⅓
fraction	1/8	1/6	1/5	1/4	1/3

Simple & Compound Interest

Simple interest adds. Compound interest multiplies.

What you need to know

WHAT YOU NEED TO KNOW

Symbols. P = principal, R = rate per cent per year, T or n = time in years, A = amount (principal + interest).

Simple interest — interest only on the original principal

$$SI = \frac{PRT}{100} \quad A = P + SI = P \left(1 + \frac{RT}{100} \right)$$

$$P = \frac{100 SI}{RT} \quad R = \frac{100 SI}{PT} \quad T = \frac{100 SI}{PR}$$

Compound interest — interest on interest

$$A = P \left(1 + \frac{R}{100} \right)^n \quad CI = A - P = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right]$$

Compounding more than once a year (nominal rate $R\%$ per year) — divide the rate, multiply the periods:

Compounded	yearly	half-yearly	quarterly
Rate, periods in n yr	$R\%, n$	$R/2\%, 2n$	$R/4\%, 4n$
Amount	$P(1 + R/100)^n$	$P(1 + R/200)^{2n}$	$P(1 + R/400)^{4n}$

Monthly: rate $R/12\%$, periods $12n$, amount $P(1 + R/1200)^{12n}$.

Effective annual rate with k compoundings a year: $EAR = [(1 + R/(100k))^k - 1] \times 100\%$.

Difference between CI and SI (same P and R), and the 2-year ratio:

$$2 \text{ yr: } D = P \left(\frac{R}{100} \right)^2 \quad 3 \text{ yr: } D = P \frac{R^2(300 + R)}{10^6} \quad \frac{CI}{SI} = 1 + \frac{R}{200}$$

Doubling and multiplying

- SI: doubles when $RT = 100$. Doubles in $T \rightarrow$ triples in $2T \rightarrow m$ times in $(m - 1)T$.
- CI: m times in $T \rightarrow m^2$ times in $2T \rightarrow m^3$ times in $3T$.
- **Rule of 72:** at CI, doubling time $\approx 72/R$ years.

Growth, decay, depreciation (compound interest in disguise). Value after n years $= P(1 \pm R/100)^n$ — use $+$ for growth (population, revenue, prices) and $-$ for decay.

Equal annual instalments of x on a loan P . At CI, discount every payment back to today, writing $v = 1 + R/100$: $P = x/v + x/v^2 + \dots + x/v^n$. At SI, instead compare values at the end of year n — each instalment earns simple interest for the years still left.

EMI (reducing balance), with $i = R/1200$ per month and n months: $EMI = Pi(1 + i)^n / ((1 + i)^n - 1)$.

Flat-rate loan. Total interest $= (PRT)/100$ and $EMI = (P + \text{total interest})/(\text{number of months})$. The true reducing-balance rate is roughly **twice** the quoted flat rate.

Real return. Nominal $R\%$, inflation $f\%$: real factor per year $= (1 + R/100)/(1 + f/100)$. Divide, never subtract.

Warm-up

WARM-UP

build the reflex

Example 1

Find the simple interest on Rs. 8,000 at 9% per year for 3 years.

SOLUTION

$$SI = \frac{8000 \times 9 \times 3}{100}.$$

$$8000 \times 9 = 72000. \text{ Then } 72000 \times 3 = 216000.$$

$$\frac{216000}{100} = 2160.$$

Answer: Rs. 2,160

Example 2

The simple interest on Rs. 6,000 for 4 years is Rs. 1,440. Find the rate.

SOLUTION

$$R = \frac{100 \times SI}{PT} = \frac{100 \times 1440}{6000 \times 4}.$$

$$100 \times 1440 = 144000. \text{ And } 6000 \times 4 = 24000.$$

$$\frac{144000}{24000} = 6.$$

Answer: 6% per year

Example 3

Find the amount on Rs. 12,500 at 8% simple interest for 5 years.

SOLUTION

$$SI = \frac{12500 \times 8 \times 5}{100}.$$

$$12500 \times 8 = 100000. \text{ Then } 100000 \times 5 = 500000.$$

$$\frac{500000}{100} = 5000.$$

$$A = 12500 + 5000 = 17500.$$

Answer: Rs. 17,500

Example 4

Find the compound interest on Rs. 10,000 at 10% per year for 2 years, compounded yearly.

SOLUTION

$$\text{Year 1: } 10000 \times 1.10 = 11000.$$

$$\text{Year 2: } 11000 \times 1.10 = 12100.$$

$$CI = 12100 - 10000 = 2100.$$

Answer: Rs. 2,100

Example 5

Find the compound interest on Rs. 5,000 at 20% per year for 2 years.

SOLUTION

$$\text{Year 1: } 5000 \times 1.20 = 6000.$$

$$\text{Year 2: } 6000 \times 1.20 = 7200.$$

$$CI = 7200 - 5000 = 2200.$$

Answer: Rs. 2,200

Example 6

Find the difference between CI and SI on Rs. 20,000 at 5% per year for 2 years.

SOLUTION

$$D = P\left(\frac{R}{100}\right)^2 = 20000 \times \left(\frac{5}{100}\right)^2 = 20000 \times (0.05)^2.$$

$$(0.05)^2 = 0.0025.$$

$$20000 \times 0.0025 = 50.$$

Answer: Rs. 50

Example 7

Find the compound interest on Rs. 16,000 at 10% per year for 1 year, compounded half-yearly.

SOLUTION

Half-yearly rate = $\frac{10}{2} = 5\%$. Number of periods = 2.

$$\text{Period 1: } 16000 \times 1.05 = 16800.$$

$$\text{Period 2: } 16800 \times 1.05 = 17640.$$

$$CI = 17640 - 16000 = 1640.$$

Answer: Rs. 1,640

Example 8

In how many years will Rs. 4,000 double itself at 12.5% simple interest?

SOLUTION

To double, the interest earned must equal the principal: $SI = 4000$.

$$T = \frac{100 \times SI}{PR} = \frac{100 \times 4000}{4000 \times 12.5} = \frac{100}{12.5} = 8.$$

Answer: 8 years

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Find the simple interest on Rs. 24,000 at 7.5% per year for 2 years and 8 months.

SOLUTION

First turn the time into years.

$$8 \text{ months} = \frac{8}{12} = \frac{2}{3} \text{ of a year.}$$

$$T = 2 + \frac{2}{3} = \frac{8}{3} \text{ years.}$$

$$\text{Interest for one year} = \frac{24000 \times 7.5}{100}.$$

$$24000 \times 7.5 = 180000. \text{ Then } \frac{180000}{100} = 1800.$$

$$\text{Interest for } \frac{8}{3} \text{ years} = 1800 \times \frac{8}{3}.$$

$$\frac{1800}{3} = 600. \text{ Then } 600 \times 8 = 4800.$$

Answer: Rs. 4,800

TRAP

Never plug “2 years 8 months” in as $T = 2.8$. Months must be divided by 12, not by 10. 2.8 would give Rs. 5,040 — wrong by Rs. 240.

Example 10 [Accenture pattern]

A sum of money at simple interest amounts to Rs. 9,560 in 3 years and to Rs. 10,600 in 5 years. Find the sum and the rate.

SOLUTION

Key idea: with SI, every year adds exactly the same amount.

Growth over 2 years (from year 3 to year 5) = $10600 - 9560 = 1040$.

Interest for 1 year = $\frac{1040}{2} = 520$.

Interest for 3 years = $520 \times 3 = 1560$.

$P = 9560 - 1560 = 8000$.

$R = \frac{100 \times 520}{8000 \times 1} = \frac{52000}{8000} = 6.5$.

Answer: Sum Rs. 8,000, rate 6.5% per year

Example 11 [Infosys pattern]

Find the compound interest on Rs. 12,000 at 10% per year for 3 years.

SOLUTION

Year 1: $12000 \times 1.10 = 13200$.

Year 2: $13200 \times 1.10 = 14520$.

Year 3: $14520 \times 1.10 = 15972$.

CI = $15972 - 12000 = 3972$.

Answer: Rs. 3,972

SHORTCUT

Learn the 10% powers by heart — they appear constantly:

$$1.1^2 = 1.21 \quad 1.1^3 = 1.331 \quad 1.1^4 = 1.4641 \quad 1.1^5 = 1.61051$$

Also $1.05^2 = 1.1025$, $1.05^3 = 1.157625$, $1.2^2 = 1.44$, $1.2^3 = 1.728$, $1.25^3 = 1.953125$.

Example 12 [Wipro pattern]

The difference between the compound interest and the simple interest on a sum for 2 years at 10% per year is Rs. 360. Find the sum.

SOLUTION

$$D = P \left(\frac{R}{100} \right)^2.$$

$$\left(\frac{10}{100} \right)^2 = (0.1)^2 = 0.01.$$

$$\text{So } 0.01P = 360.$$

$$P = \frac{360}{0.01} = 36000.$$

Check. SI = $\frac{36000 \times 10 \times 2}{100} = 7200$. CI: $36000 \times 1.21 = 43560$, so CI = 7560. Difference = $7560 - 7200 = 360$. Correct.

Answer: Rs. 36,000

SHORTCUT

Why the 2-year difference is $P\left(\frac{R}{100}\right)^2$. SI ignores the interest earned in year 1. CI pays interest on it. So the gap is simply “ $R\%$ of the first year’s interest”:

$$\frac{R}{100} \times \frac{PR}{100} = \frac{PR^2}{10000} = P\left(\frac{R}{100}\right)^2$$

Example 13 [Capgemini pattern]

Find the compound interest on Rs. 50,000 at 8% per year for 6 months, compounded quarterly.

SOLUTION

Quarterly rate = $\frac{8}{4} = 2\%$.

6 months = 2 quarters, so 2 periods.

Period 1: $50000 \times 1.02 = 51000$.

Period 2: $51000 \times 1.02 = 52020$.

CI = $52020 - 50000 = 2020$.

Answer: Rs. 2,020

TRAP

“Compounded quarterly” changes **two** things at once: the rate is divided by 4 **and** the number of periods is multiplied by 4. Students who divide the rate but forget to multiply the periods (or the reverse) lose the mark every time.

Example 14 [Cognizant pattern]

A sum of money doubles itself in 8 years at simple interest. In how many years will it triple?

SOLUTION

Doubling means interest earned = P .

$P = \frac{P \times R \times 8}{100}$, so $R \times 8 = 100$ and $R = 12.5\%$.

Tripling means interest earned = $2P$.

$2P = \frac{P \times 12.5 \times T}{100}$

$2 = \frac{12.5T}{100}$

$12.5T = 200$

$T = \frac{200}{12.5} = 16$.

Answer: 16 years

SHORTCUT

At **simple** interest the interest earned is a straight line, so the shortcut is just counting:

doubling in $T \rightarrow$ tripling in $2T \rightarrow m$ times in $(m - 1)T$ years.

At **compound** interest it is completely different — see Example 16.

Example 15 [TCS NQT pattern]

A sum amounts to Rs. 8,820 in 2 years at 5% per year compound interest. Find the sum.

SOLUTION

$A = P(1.05)^2$.

$$(1.05)^2 = 1.1025.$$

$$P = \frac{8820}{1.1025}.$$

$$\text{Multiply top and bottom by 10000: } P = \frac{88200000}{11025}.$$

$$11025 \times 8000 = 88200000, \text{ so } P = 8000.$$

Check. $8000 \times 1.05 = 8400$; $8400 \times 1.05 = 8820$. Correct.

Answer: Rs. 8,000

Example 16 [Accenture pattern]

A sum of money at compound interest becomes 3 times itself in 5 years. In how many years will it become 27 times itself?

SOLUTION

Key idea: at CI the multiplier, not the interest, is what repeats.

In 5 years the money is multiplied by 3.

In another 5 years it is multiplied by 3 again: $3 \times 3 = 9$ times, at 10 years.

In another 5 years: $9 \times 3 = 27$ times, at 15 years.

Since $27 = 3^3$, we need $3 \times 5 = 15$ years.

Answer: 15 years

TRAP

Do not answer $\frac{27}{3} \times 5 = 45$ years. That is the **simple interest** style of thinking. At CI you ask “how many times do I multiply by 3?” — the answer is 3 times, so 15 years.

Example 17 [Infosys pattern]

A machine bought for Rs. 64,000 loses 25% of its value every year. Find its value after 3 years. After how many full years does its value first drop below Rs. 20,000?

SOLUTION

Each year the machine keeps $100 - 25 = 75\%$ of its value, so multiply by 0.75.

$$\text{Year 1: } 64000 \times 0.75 = 48000.$$

$$\text{Year 2: } 48000 \times 0.75 = 36000.$$

$$\text{Year 3: } 36000 \times 0.75 = 27000.$$

$$\text{Year 4: } 27000 \times 0.75 = 20250. \text{ Still above 20,000.}$$

$$\text{Year 5: } 20250 \times 0.75 = 15187.50. \text{ Now below 20,000.}$$

Answer: Rs. 27,000 after 3 years; it first drops below Rs. 20,000 after 5 years

Example 18 [Wipro pattern]

The population of a town is 1,25,000 and grows at 6% per year. Find the population after 2 years and the increase over those 2 years.

SOLUTION

$$\text{Year 1: } 125000 \times 1.06.$$

$$125000 \times 0.06 = 7500, \text{ so the population is } 125000 + 7500 = 132500.$$

$$\text{Year 2: } 132500 \times 1.06.$$

$$132500 \times 0.06 = 7950, \text{ so the population is } 132500 + 7950 = 140450.$$

$$\text{Increase} = 140450 - 125000 = 15450.$$

Answer: 1,40,450; an increase of 15,450

Example 19 [Capgemini pattern]

A man borrows Rs. 21,000 at 10% simple interest and repays it in 2 equal annual instalments. Find each instalment.

SOLUTION

Compare everything at the same date — the end of year 2.

What he owes at the end of year 2:

$$SI = \frac{21000 \times 10 \times 2}{100} = 4200.$$

$$\text{Amount owed} = 21000 + 4200 = 25200.$$

What he pays. Let each instalment be x .

The first instalment is paid at the end of year 1, so it sits with the lender for 1 more year and earns 10% simple interest:

$$\text{value at end of year 2} = x + \frac{x \times 10 \times 1}{100} = 1.1x.$$

The second instalment is paid at the end of year 2, so its value is just x .

$$\text{Total paid (valued at end of year 2)} = 1.1x + x = 2.1x.$$

Set the two equal:

$$2.1x = 25200$$

$$x = \frac{25200}{2.1} = \frac{252000}{21} = 12000.$$

Answer: Rs. 12,000 per instalment

Example 20 [Cognizant pattern]

A loan of Rs. 10,500 at 10% per year compound interest is cleared in 2 equal annual instalments. Find each instalment.

SOLUTION

Let each instalment be x .

At CI we discount each payment back to today:

$$10500 = \frac{x}{1.1} + \frac{x}{1.1^2} = \frac{x}{1.1} + \frac{x}{1.21}$$

Multiply every term by 1.21 to clear the fractions:

$$10500 \times 1.21 = 1.1x + x$$

$$10500 \times 1.21 = 10500 + 10500 \times 0.21 = 10500 + 2205 = 12705.$$

$$12705 = 2.1x$$

$$x = \frac{12705}{2.1} = \frac{127050}{21} = 6050$$

Check. $\frac{6050}{1.1} = 5500$ and $\frac{6050}{1.21} = 5000$. Sum = 10500. Correct.

Answer: Rs. 6,050 per instalment

PRACTICE · TIER 1 target: 60 s/Q

1. Find the simple interest on Rs. 15,000 at 8% per year for 4 years.
2. Rs. 9,000 gives a simple interest of Rs. 2,700 in 3 years. Find the rate.
3. In how many years does Rs. 7,200 at 6% simple interest give Rs. 1,296 of interest?
4. Find the compound interest on Rs. 16,000 at 15% per year for 2 years.
5. Find the difference between CI and SI on Rs. 45,000 at 4% per year for 2 years.

6. Find the amount on Rs. 20,000 at 10% per year for 1.5 years, compounded half-yearly.
7. A sum doubles itself in 10 years at simple interest. Find the rate.
8. A sum doubles itself in 4 years at compound interest. In how many years will it become 16 times?
9. A sum amounts to Rs. 13,310 in 3 years at 10% compound interest. Find the sum.
10. A machine worth Rs. 50,000 depreciates 20% a year. Find its value after 3 years.
11. A sum at simple interest amounts to Rs. 12,400 in 4 years and Rs. 14,200 in 7 years. Find the sum and the rate.
12. A loan of Rs. 8,400 at 10% simple interest is repaid in 2 equal annual instalments. Find each instalment.
13. Rs. 6,000 is invested for 2 years, once at 10% compound interest and once at 12% simple interest. Which gives more interest, and by how much?
14. A town's population of 80,000 grows 5% a year. Find the population after 2 years.
15. Find the effective annual rate for a nominal 12% per year compounded quarterly.

ANSWER KEY

1. Rs. 4,800. $\frac{15000 \times 8 \times 4}{100}$.
2. 10%. $R = \frac{100 \times 2700}{9000 \times 3} = \frac{270000}{27000}$.
3. 3 years. $T = \frac{100 \times 1296}{7200 \times 6} = \frac{129600}{43200}$.
4. Rs. 5,160. $16000 \times 1.15 = 18400$; $18400 \times 1.15 = 21160$; $21160 - 16000$.
5. Rs. 72. $45000 \times (0.04)^2 = 45000 \times 0.0016$.
6. Rs. 23,152.50. Rate 5%, 3 periods: 21000, 22050, 23152.50.
7. 10%. Doubling needs $RT = 100$ with $T = 10$.
8. 16 years. $16 = 2^4$, so 4 doubling periods of 4 years each.
9. Rs. 10,000. $\frac{13310}{1.331} = 10000$.
10. Rs. 25,600. $50000 \times 0.8 = 40000$; $\times 0.8 = 32000$; $\times 0.8 = 25600$.
11. Sum Rs. 10,000, rate 6%. Growth over 3 years = 1800, so 1 year = 600; 4-year interest = 2400; $P = 12400 - 2400$; $R = \frac{600}{10000} \times 100$.
12. Rs. 4,800. Owed at end of year 2 = $8400 + 1680 = 10080$; $2.1x = 10080$.
13. Simple interest gives more, by Rs. 180. CI = $6000 \times 1.21 - 6000 = 1260$; SI = $\frac{6000 \times 12 \times 2}{100} = 1440$. At short times a higher simple rate can beat a lower compound rate.
14. 88,200. $80000 \times 1.05 = 84000$; $84000 \times 1.05 = 88200$.
15. About 12.55%. $(1.03)^4 = 1.12550881$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [DBS · pattern]

A fixed deposit of S\$40,000 pays 6% per year compounded half-yearly. Find the amount after 2 years and the effective annual rate.

SOLUTION

Half-yearly rate = $\frac{6}{2} = 3\%$. Periods = $2 \times 2 = 4$.

Period 1: $40000 \times 1.03 = 41200$.

Period 2: $41200 \times 1.03 = 42436$. (Working: $41200 \times 0.03 = 1236$; $41200 + 1236 = 42436$.)

Period 3: $42436 \times 1.03 = 43709.08$. (Working: $42436 \times 0.03 = 1273.08$; $42436 + 1273.08 = 43709.08$.)

Period 4: $43709.08 \times 1.03 = 45020.35$. (Working: $43709.08 \times 0.03 = 1311.27$; $43709.08 + 1311.27 = 45020.35$.)

Effective annual rate. One year is two 3% periods:

$(1.03)^2 = 1.0609$, so the EAR is 6.09%.

Answer: S\$45,020.35; effective annual rate 6.09%

SHORTCUT

Once you have the EAR you can jump straight to any whole number of years: $40000 \times (1.0609)^2 = 40000 \times 1.12550881 = 45020.35$. Same answer, two multiplications instead of four.

Example 22 [Grab · pattern]

A merchant needs THB 2,00,000 for 3 years. Lender A charges 12% per year simple interest. Lender B charges 10% per year compounded annually. Which loan is cheaper, and by how much?

SOLUTION

Lender A (simple).

$$SI = \frac{200000 \times 12 \times 3}{100}$$

$200000 \times 12 = 2400000$. Then $2400000 \times 3 = 7200000$.

$$\frac{7200000}{100} = 72000.$$

Lender B (compound).

Year 1: $200000 \times 1.1 = 220000$.

Year 2: $220000 \times 1.1 = 242000$.

Year 3: $242000 \times 1.1 = 266200$.

Interest = $266200 - 200000 = 66200$.

Compare. $72000 - 66200 = 5800$.

Answer: Lender B is cheaper by THB 5,800

A lower compound rate can still beat a higher simple rate over a short period. Over a long enough period compounding always wins — but “long enough” here would be well past 3 years.

Example 23 [Shopee · pattern]

The difference between the compound interest and the simple interest on a sum for 3 years at 10% per year is Rs. 3,100. Find the sum.

SOLUTION

$$D = P \frac{R^2(300 + R)}{10^6}$$

Put $R = 10$:

$R^2 = 100$ and $300 + R = 310$.

$$D = P \frac{100 \times 310}{1000000} = P \times \frac{31000}{1000000} = 0.031P$$

$0.031P = 3100$

$$P = \frac{3100}{0.031} = 100000.$$

Check. SI = $\frac{100000 \times 10 \times 3}{100} = 30000$. CI: $100000 \times 1.331 = 133100$, so CI = 33100. Difference = $33100 - 30000 = 3100$. Correct.

Answer: Rs. 1,00,000

Example 24 [GIC · pattern]

\$25,000 is invested at 8% per year compounded annually. After 2 years the rate falls to 6% for one more year. Find the amount at the end of year 3.

SOLUTION

Year 1: $25000 \times 1.08 = 27000$.

Year 2: $27000 \times 1.08 = 29160$. (Working: $27000 \times 0.08 = 2160$.)

Year 3 at the new rate: 29160×1.06 .

$29160 \times 0.06 = 1749.60$.

$29160 + 1749.60 = 30909.60$.

Answer: S\$30,909.60

TRAP

Do not average the rates. $\frac{8+8+6}{3} = 7.33\%$ would give $25000 \times (1.0733)^3 = 30910.31$, which is Rs. 0.71 too high. It looks close only because the rates here are close. Rates that change must be applied in order, one factor at a time.

Example 25 [SCB · pattern]

A loan of Rs. 33,100 at 10% per year compound interest is cleared in 3 equal annual instalments. Find each instalment.

SOLUTION

Let each instalment be x . Discount all three back to today:

$$33100 = \frac{x}{1.1} + \frac{x}{1.21} + \frac{x}{1.331}$$

Multiply every term by 1.331:

$$33100 \times 1.331 = 1.21x + 1.1x + x$$

Left side: $33100 \times 1.331 = 33100 + 33100 \times 0.331$.

$33100 \times 0.331 = 10956.1$.

So the left side = $33100 + 10956.1 = 44056.1$.

Right side = $3.31x$.

$$x = \frac{44056.1}{3.31} = 13310$$

Check. $\frac{13310}{1.1} = 12100$; $\frac{13310}{1.21} = 11000$; $\frac{13310}{1.331} = 10000$. Sum = $12100 + 11000 + 10000 = 33100$. Correct.

Answer: Rs. 13,310 per instalment

SHORTCUT

For n instalments at 10%, multiply the loan by 1.1^n and divide by $(1.1^{n-1} + \dots + 1.1 + 1)$.

$n = 2$: divide by 2.1. $n = 3$: divide by 3.31. $n = 4$: divide by 4.641.

Those divisors are just 1.1^n figures with the decimal shifted — easy to recall.

Example 26 [Agoda · pattern]

A person takes a “flat rate” loan of Rs. 1,20,000 for 2 years at 12% per year. Under this scheme the total interest is worked out as simple interest on the whole amount, and principal plus interest is split into 24 equal monthly payments. Find the EMI, and estimate the true annual rate the borrower is really paying.

SOLUTION

Step 1 — total interest (flat).

$$SI = \frac{120000 \times 12 \times 2}{100}.$$

$$120000 \times 12 = 1440000. \text{ Then } 1440000 \times 2 = 2880000.$$

$$\frac{2880000}{100} = 28800.$$

Step 2 — EMI.

$$\text{Total to repay} = 120000 + 28800 = 148800.$$

$$EMI = \frac{148800}{24} = 6200.$$

Step 3 — the true rate.

The borrower does not owe Rs. 1,20,000 for the whole 2 years. The balance falls in 24 steps to zero, so the average amount outstanding is

$$120000 \times \frac{25}{48} = 62500$$

(The factor is $(n+1)/(2n)$ with $n = 24$ payments, because the balance is still the full 1,20,000 at the start of month 1.)

He pays Rs. 28,800 of interest over 2 years on an average balance of Rs. 62,500:

$$\text{true rate} \approx \frac{28800}{62500 \times 2} \times 100 = \frac{28800}{125000} \times 100 = 23.04\%$$

This shortcut always runs a little high; the exact reducing-balance rate here is about 21.6% per year. Either way the message is the same — the loan costs far more than 12%.

Answer: EMI Rs. 6,200; true rate roughly 23% per year by the quick estimate (about 21.6% exactly) — near enough double the quoted 12%

TRAP

A “12% flat” loan is **not** a 12% loan. Flat rate charges you interest on the full principal even in the last month, when you owe almost nothing. Always double the flat rate for a rough comparison with a bank’s reducing-balance rate.

Example 27 [LINE MAN · pattern]

An investment grows from THB 50,000 to THB 72,000 in 2 years at compound interest. Find the annual rate. At the same rate, how long from the start until it reaches THB 1,24,416?

SOLUTION**Rate.**

$$50000 \left(1 + \frac{R}{100}\right)^2 = 72000$$

$$\left(1 + \frac{R}{100}\right)^2 = \frac{72000}{50000} = 1.44$$

$$1 + \frac{R}{100} = \sqrt{1.44} = 1.2$$

$$R = 20\%.$$

Time to reach 1,24,416. Keep multiplying by 1.2.

$$\text{End of year 1: } 50000 \times 1.2 = 60000.$$

$$\text{End of year 2: } 60000 \times 1.2 = 72000.$$

$$\text{End of year 3: } 72000 \times 1.2 = 86400.$$

$$\text{End of year 4: } 86400 \times 1.2 = 103680.$$

$$\text{End of year 5: } 103680 \times 1.2 = 124416.$$

Answer: 20% per year; 5 years from the start

Example 28 [Razer · pattern]

Rs. 60,000 is split into two parts. One part is lent at 8% simple interest and the other at 10% simple interest, both for 3 years. The total interest is Rs. 15,600. Find the two parts.

SOLUTION

Let the part at 8% be x . The other part is $60000 - x$.

Interest over 3 years at 8% = $\frac{x \times 8 \times 3}{100} = 0.24x$.

Interest over 3 years at 10% = $\frac{(60000 - x) \times 10 \times 3}{100} = 0.30(60000 - x)$.

$0.30 \times 60000 = 18000$, so this is $18000 - 0.30x$.

Total:

$$0.24x + 18000 - 0.30x = 15600$$

$$18000 - 0.06x = 15600$$

$$0.06x = 2400$$

$$x = \frac{2400}{0.06} = 40000$$

Other part = $60000 - 40000 = 20000$.

Check. $40000 \times 0.24 = 9600$ and $20000 \times 0.30 = 6000$. Sum = 15600. Correct.

Answer: Rs. 40,000 at 8% and Rs. 20,000 at 10%

Example 29 [Sea · pattern]

A seller's monthly revenue is S\$40,000. It rises 25% in month 1, rises another 20% in month 2, then falls 20% in month 3. Find the revenue after month 3 and the overall percentage change.

SOLUTION

Treat every change as a multiplying factor.

Month 1: $40000 \times 1.25 = 50000$.

Month 2: $50000 \times 1.20 = 60000$.

Month 3: $60000 \times 0.80 = 48000$.

Overall change = $48000 - 40000 = 8000$.

Percentage change = $\frac{8000}{40000} \times 100 = 0.2 \times 100 = 20\%$.

Answer: S\$48,000; an overall rise of 20%

TRAP

+25, +20, -20 does **not** add up to +25%. The 20% fall is taken on a much bigger base than the 20% rise was, so it wipes out more. Always multiply: $1.25 \times 1.20 \times 0.80 = 1.20$.

Example 30 [DBS · pattern]

Rs. 1,00,000 is to be invested for 1 year. Scheme A pays 10% compounded annually. Scheme B pays 9.6% compounded quarterly. Which pays more, and by how much?

SOLUTION

Scheme A.

$100000 \times 1.10 = 110000$.

Scheme B.

Quarterly rate = $\frac{9.6}{4} = 2.4\%$, so the factor is 1.024 and there are 4 periods.

$(1.024)^2 = 1.048576$. (Working: $1.024 \times 1.024 = 1.024 + 1.024 \times 0.024 = 1.024 + 0.024576$.)

$$(1.024)^4 = (1.048576)^2 = 1.09951163. \quad (\text{Working: } 1.048576 + 1.048576 \times 0.048576 = 1.048576 + 0.050936 = 1.099512.)$$

$$\text{Amount} = 100000 \times 1.09951163 = 109951.16.$$

$$\text{Compare. } 110000 - 109951.16 = 48.84.$$

Answer: Scheme A pays more, by about Rs. 48.84

Quarterly compounding lifts 9.6% to an effective 9.951%, still just short of a plain 10%. Frequent compounding helps, but it cannot rescue a rate that is 0.4 points too low.

PRACTICE · TIER 2 target: 100 s/Q

- Find the amount on S\$60,000 at 8% per year for 1.5 years, compounded half-yearly.
- Find the difference between CI and SI on Rs. 80,000 at 5% per year for 3 years.
- A sum amounts to Rs. 17,640 in 2 years at 5% compound interest. Find the sum.
- A loan of Rs. 21,000 at 10% per year compound interest is cleared in 2 equal annual instalments. Find each instalment.
- A flat-rate loan of Rs. 90,000 is taken for 3 years at 10% flat, repaid in 36 EMIs. Find the EMI.
- An investment grows from S\$32,000 to S\$50,000 in 2 years at compound interest. Find the annual rate.
- Rs. 50,000 is invested for 2 years. Option 1: 12% simple interest. Option 2: 11% compound interest. Which gives more interest, and by how much?
- Rs. 90,000 is split, one part at 6% simple interest and the rest at 11% simple interest, both for 2 years. The total interest is Rs. 15,000. Find the two parts.
- A sum becomes 8 times itself in 12 years at compound interest. In how many years does it double?
- A shop's revenue of THB 2,50,000 rises 20%, rises 20% again, then falls 25%. Find the final revenue and the overall percentage change.
- Find the effective annual rate for a nominal 16% per year compounded quarterly.
- A machine costing S\$80,000 depreciates 15% a year. Find its value after 2 years and the total depreciation.

ANSWER KEY

- S\$67,491.84. Rate 4%, 3 periods: 62400, 64896, 67491.84.
- Rs. 610. $D = 80000 \times \frac{25 \times 305}{10^6} = 80000 \times 0.007625$. (Check: SI = 12000, CI = $80000 \times 1.157625 - 80000 = 12610$.)
- Rs. 16,000. $\frac{17640}{1.1025} = 16000$.
- Rs. 12,100. $21000 \times 1.21 = 25410 = 2.1x$. (Check: $\frac{12100}{1.1} = 11000$, $\frac{12100}{1.21} = 10000$, sum 21,000.)
- Rs. 3,250. Interest = $\frac{90000 \times 10 \times 3}{100} = 27000$; total = 117000; $\frac{117000}{36}$.
- 25%. $\frac{50000}{32000} = 1.5625$ and $\sqrt{1.5625} = 1.25$.
- Simple interest gives more, by Rs. 395. SI = 12000; CI = $50000 \times 1.2321 - 50000 = 11605$.
- Rs. 48,000 at 6% and Rs. 42,000 at 11%. $0.12x + 0.22(90000 - x) = 15000$ gives $19800 - 0.10x = 15000$.
- 4 years. $8 = 2^3$, so 12 years holds three doubling periods.
- THB 2,70,000, a rise of 8%. $1.2 \times 1.2 \times 0.75 = 1.08$. (Not $20 + 20 - 25 = 15\%$.)
- About 16.99%. $(1.04)^4 = 1.16985856$.
- S\$57,800; depreciation S\$22,200. $80000 \times 0.85 = 68000$; $68000 \times 0.85 = 57800$.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Goldman Sachs · pattern]

Prove that for the same principal, rate and a period of 2 years, the difference between compound and simple interest is exactly $P\left(\frac{R}{100}\right)^2$. Then use it: the difference is Rs. 96 at 8% for 2 years. Find the principal.

SOLUTION

Insight: the whole gap is one thing — the interest that CI pays on year 1's interest, and SI does not.

Write $r = \frac{R}{100}$.

Simple interest over 2 years:

$$SI = 2Pr$$

Compound interest over 2 years:

$$CI = P(1+r)^2 - P = P(1+2r+r^2) - P = 2Pr + Pr^2$$

Difference:

$$D = (CI) - (SI) = (2Pr + Pr^2) - 2Pr = Pr^2 = P\left(\frac{R}{100}\right)^2$$

Read the meaning. Year 1's interest is Pr . Under CI that amount itself earns interest for year 2, giving $r \times Pr = Pr^2$. Under SI it earns nothing. So the gap is exactly Pr^2 — nothing else differs.

Now the number. $R = 8$, so $r = 0.08$ and $r^2 = 0.0064$.

$$0.0064P = 96$$

$$P = \frac{96}{0.0064}$$

Multiply top and bottom by 10000: $P = \frac{960000}{64} = 15000$.

Check. $SI = \frac{15000 \times 8 \times 2}{100} = 2400$. $CI: 15000 \times (1.08)^2 = 15000 \times 1.1664 = 17496$, so $CI - SI = 2496 - 2400 = 96$. Correct.

Answer: $D = P\left(\frac{R}{100}\right)^2$; the principal is Rs. 15,000

Example 32 [D. E. Shaw · pattern]

At 9% per year compound interest, how long does money take to double? Compare your exact answer with the “Rule of 72”.

SOLUTION

Insight: at compound interest the doubling time depends only on the rate, not on how much money you start with — because the principal cancels.

We need the smallest n with $(1.09)^n \geq 2$.

Build the powers by repeated squaring — far faster than multiplying nine times.

$$(1.09)^2 = 1.09 \times 1.09 = 1.09 + 1.09 \times 0.09 = 1.09 + 0.0981 = 1.1881.$$

$$(1.09)^4 = (1.1881)^2 = 1.1881 + 1.1881 \times 0.1881.$$

$$1.1881 \times 0.1881 = 0.22348.$$

$$\text{So } (1.09)^4 = 1.41158.$$

$$(1.09)^8 = (1.41158)^2 = 1.41158 + 1.41158 \times 0.41158.$$

$$1.41158 \times 0.41158 = 0.58098.$$

So $(1.09)^8 = 1.99256$. That is just **under 2** — eight years is not quite enough.

$$(1.09)^9 = 1.99256 \times 1.09 = 1.99256 + 1.99256 \times 0.09 = 1.99256 + 0.17933 = 2.17189.$$

So money doubles a little after 8 years; the first **whole** year by which it has doubled is 9.

Rule of 72: doubling time $\approx 72/R = \frac{72}{9} = 8$ years. The rule gives 8, and the true figure is about 8.04 years — an error of roughly 16 days.

Answer: Just over 8 years (8 full years leaves you at 1.9926 times; 9 full years gives 2.1719 times). The Rule of 72 estimate of 8 years is accurate here.

SHORTCUT

Rule of 72: doubling time $\approx \frac{72}{R}$ years at compound interest. It is very good for rates between about 5% and 12%, and drifts at extreme rates. At 1%: the rule says 72, the truth is 69.7. At 30%: the rule says 2.4, the truth is 2.64.

Example 33 [Amazon · pattern]

A laptop can be bought two ways. Plan A: pay Rs. 60,000 today. Plan B: pay Rs. 22,000 today, Rs. 22,000 at the end of year 1, and Rs. 22,000 at the end of year 2. Money earns 10% per year compound interest. Which plan costs less, and by how much in today's rupees?

SOLUTION

Insight: Rs. 22,000 paid a year from now costs you less than Rs. 22,000 today, because you could have kept that money earning 10%. Pull every payment back to today before adding.

Plan A. Cost today = 60000.

Plan B. Discount each payment.

Payment today: 22000.

Payment after 1 year: $\frac{22000}{1.1}$.

$$\frac{22000}{1.1} = \frac{220000}{11} = 20000.$$

Payment after 2 years: $\frac{22000}{1.21}$.

$$\frac{22000}{1.21} = \frac{2200000}{121} = 18181.82. \text{ (Working: } 121 \times 18181 = 2199901; \text{ the remainder 99 gives 0.818.)}$$

$$\text{Total today} = 22000 + 20000 + 18181.82 = 60181.82.$$

Compare. $60181.82 - 60000 = 181.82$.

Plan A is cheaper by Rs. 181.82 in today's money.

Answer: Plan A, by about Rs. 181.82 in present-day value

Note how close it is. At a rate of about 10.36% the two plans would cost the same. The instalment plan looks free, but it is only “free” if the interest rate is high enough — spreading payments is a hidden loan running the other way.

Example 34 [Google · pattern]

Rs. 1,00,000 is invested for 1 year at a nominal 12% per year. Find the amount when it is compounded yearly, half-yearly, quarterly and monthly. What happens as the compounding gets ever more frequent?

SOLUTION

Insight: more frequent compounding always helps, but each extra split helps less than the one before. The amount converges to a ceiling, it does not grow without limit.

Yearly. $100000 \times 1.12 = 112000$.

Half-yearly. Rate 6%, 2 periods.

$$(1.06)^2 = 1.06 + 1.06 \times 0.06 = 1.06 + 0.0636 = 1.1236.$$

$$\text{Amount} = 112360.$$

Quarterly. Rate 3%, 4 periods.

$$(1.03)^2 = 1.0609.$$

$$(1.03)^4 = (1.0609)^2 = 1.0609 + 1.0609 \times 0.0609 = 1.0609 + 0.06461 = 1.12551.$$

$$\text{Amount} = 112550.88.$$

Monthly. Rate 1%, 12 periods.

$$(1.01)^2 = 1.0201.$$

$$(1.01)^4 = (1.0201)^2 = 1.0201 + 1.0201 \times 0.0201 = 1.0201 + 0.020504 = 1.040604.$$

$$(1.01)^8 = (1.040604)^2 = 1.040604 + 1.040604 \times 0.040604 = 1.040604 + 0.042253 = 1.082857.$$

$$(1.01)^{12} = (1.01)^8 \times (1.01)^4 = 1.082857 \times 1.040604 = 1.082857 + 1.082857 \times 0.040604 = 1.082857 + 0.043968 = 1.126825.$$

$$\text{Amount} = 112682.50.$$

The pattern.

Compounding	Amount (Rs.)	Gain over the line above
yearly	1,12,000.00	–
half-yearly	1,12,360.00	360.00
quarterly	1,12,550.88	190.88
monthly	1,12,682.50	131.62

Each step gains less than the last: 360, then 191, then 132. The gains are shrinking towards zero.

The ceiling. As the number of splits k grows without bound,

$$\left(1 + \frac{0.12}{k}\right)^k \rightarrow e^{0.12} = 1.127497$$

so the amount can never exceed about Rs. 1,12,749.69 — daily or even by-the-second compounding at 12% buys you less than Rs. 70 more than monthly.

Answer: Rs. 1,12,000 / Rs. 1,12,360 / Rs. 1,12,550.88 / Rs. 1,12,682.50; the amount rises with frequency but is capped at about Rs. 1,12,749.69

Example 35 [Microsoft · pattern]

Derive the EMI. A loan of Rs. 1,00,000 is taken at 12% per year, charged at 1% per month on the outstanding balance, and repaid in 3 equal monthly instalments. Find the EMI and show the month-by-month balance.

SOLUTION

Insight: an EMI is simply the payment whose present value, discounted at the loan's own rate, equals the loan. Set that equation and solve.

Let the EMI be x and the monthly factor be 1.01.

$$100000 = \frac{x}{1.01} + \frac{x}{(1.01)^2} + \frac{x}{(1.01)^3}$$

$$(1.01)^2 = 1.0201 \text{ and } (1.01)^3 = 1.0201 \times 1.01 = 1.030301.$$

Multiply every term by 1.030301:

$$100000 \times 1.030301 = x(1.0201 + 1.01 + 1)$$

$$\text{Left side} = 103030.10.$$

$$\text{Right side bracket} = 3.0301.$$

$$x = \frac{103030.10}{3.0301}$$

$$3.0301 \times 34000 = 103023.40.$$

$$\text{Remainder} = 103030.10 - 103023.40 = 6.70.$$

$$\frac{6.70}{3.0301} = 2.21.$$

$$x = 34002.21$$

Same answer from the standard formula with $i = 0.01$ and $n = 3$:

$$\text{EMI} = \frac{Pi(1+i)^n}{(1+i)^n - 1} = \frac{100000 \times 0.01 \times 1.030301}{0.030301} = \frac{1030.301}{0.030301} = 34002.21$$

Month-by-month check.

Month	Opening	Interest 1%	Principal repaid	Closing
1	1,00,000.00	1,000.00	33,002.21	66,997.79
2	66,997.79	669.98	33,332.23	33,665.56
3	33,665.56	336.66	33,665.55	0.01

The balance lands on zero (the 1 paisa is rounding). The EMI is correct.

Notice the shape: in month 1 only 97% of the payment goes to principal; by month 3 it is 99%. On a long loan the early EMIs are almost all interest.

Answer: EMI Rs. 34,002.21

Example 36 [Adobe · pattern]

For a certain sum at a certain rate, the compound interest over 2 years is Rs. 2,652 and the simple interest over the same 2 years is Rs. 2,600. Find the rate and the sum.

SOLUTION

Insight: divide the two, not subtract them. The ratio kills the principal and leaves the rate alone.

Write $r = \frac{R}{100}$.

$$\text{SI} = 2Pr \quad \text{CI} = 2Pr + Pr^2$$

$$\frac{\text{CI}}{\text{SI}} = \frac{2Pr + Pr^2}{2Pr} = \frac{Pr(2+r)}{2Pr} = \frac{2+r}{2} = 1 + \frac{r}{2}$$

Since $r = \frac{R}{100}$, this is $1 + \frac{R}{200}$.

Use the numbers.

$$\frac{2652}{2600} = 1 + \frac{R}{200}$$

$$\frac{2652}{2600} = 1.02.$$

$$1.02 = 1 + \frac{R}{200}$$

$$\frac{R}{200} = 0.02$$

$$R = 4\%$$

Now the sum. Use SI.

$$2600 = \frac{P \times 4 \times 2}{100} = \frac{8P}{100} = 0.08P$$

$$P = \frac{2600}{0.08} = 32500$$

Check. $32500 \times 1.04 = 33800$; $33800 \times 1.04 = 35152$. $\text{CI} = 35152 - 32500 = 2652$. Correct. $\text{SI} = 32500 \times 0.08 = 2600$. Correct.

Answer: Rate 4% per year; sum Rs. 32,500

SHORTCUT

Memorise the one-liner for 2 years:

$$\frac{CI}{SI} = 1 + \frac{R}{200}$$

So at 10%, CI is exactly 1.05 times SI. At 8%, exactly 1.04 times. It turns a two-unknown problem into a single division.

Example 37 [Uber · pattern]

A driver takes a flat-rate loan of Rs. 60,000 for 1 year at 10% flat, repaid in 12 EMIs. Find the EMI. Then show that the true monthly rate he is paying is close to 1.5%, and state the true annual rate.

SOLUTION

Insight: a flat loan charges interest on the full Rs. 60,000 for all 12 months, but the borrower's average balance over the year is only about half of that. Halving the balance roughly doubles the effective rate.

Step 1 — EMI.

$$\text{Flat interest} = \frac{60000 \times 10 \times 1}{100} = 6000.$$

$$\text{Total repaid} = 60000 + 6000 = 66000.$$

$$\text{EMI} = \frac{66000}{12} = 5500.$$

Step 2 — a quick estimate of the true rate.

The balance falls roughly in a straight line from 60,000 to 0, so the average outstanding balance is about

$$60000 \times \frac{13}{24} = 32500$$

(The factor $\frac{13}{24}$, not $\frac{1}{2}$, because there are 12 payments and the balance is still 60,000 at the start of month 1.)

$$\text{true annual rate} \approx \frac{6000}{32500} \times 100 = 18.46\%$$

Step 3 — check it properly. The true monthly rate i satisfies

$$60000 = 5500 \times \frac{1 - (1 + i)^{-12}}{i}$$

Try $i = 0.015$ (1.5% per month).

$$(1.015)^2 = 1.030225.$$

$$(1.015)^4 = (1.030225)^2 = 1.030225 + 1.030225 \times 0.030225 = 1.030225 + 0.031139 = 1.061364.$$

$$(1.015)^8 = (1.061364)^2 = 1.061364 + 1.061364 \times 0.061364 = 1.061364 + 0.065130 = 1.126494.$$

$$(1.015)^{12} = 1.126494 \times 1.061364 = 1.126494 + 1.126494 \times 0.061364 = 1.126494 + 0.069125 = 1.195619.$$

$$\frac{1}{1.195619} = 0.836388.$$

$$1 - 0.836388 = 0.163612.$$

$$\frac{0.163612}{0.015} = 10.9075.$$

$$5500 \times 10.9075 = 59991.$$

That is Rs. 9 short of Rs. 60,000, so the true rate is a hair **below** 1.5% per month — about 1.4977%.

Annual figures.

$$\text{Nominal annual rate} = 12 \times 1.4977 \approx 17.97\%, \text{ call it } 18\%.$$

$$\text{Effective annual rate} = (1.015)^{12} - 1 \approx 0.1956, \text{ i.e. about } 19.6\%.$$

Answer: EMI Rs. 5,500; true rate about 1.5% per month, i.e. about 18% per year nominal — roughly 1.8 times the quoted 10% flat

TRAP

Flat rate, “add-on” rate and “reducing balance” rate are three different things. Comparing a 10% flat loan with a 14% bank loan and picking the flat one is a classic and expensive error: the flat loan really costs about 18%.

Example 38 [Goldman Sachs · pattern]

An investment returns 12% a year for 3 years. Inflation runs at 5% a year over the same period. Find the real growth of the money over the 3 years, and the real rate per year.

SOLUTION

Insight: a nominal return and inflation are both multipliers acting on the same money. The real result is their ratio, never their difference.

Nominal factor over 3 years.

$$(1.12)^2 = 1.12 + 1.12 \times 0.12 = 1.12 + 0.1344 = 1.2544.$$

$$(1.12)^3 = 1.2544 \times 1.12 = 1.2544 + 1.2544 \times 0.12 = 1.2544 + 0.150528 = 1.404928.$$

Inflation factor over 3 years.

$$(1.05)^2 = 1.1025.$$

$$(1.05)^3 = 1.1025 \times 1.05 = 1.1025 + 0.055125 = 1.157625.$$

Real factor.

$$\text{real factor} = \frac{1.404928}{1.157625}$$

$$1.157625 \times 1.21 = 1.157625 + 1.157625 \times 0.21 = 1.157625 + 0.243101 = 1.400726.$$

$$\text{Remainder} = 1.404928 - 1.400726 = 0.004202.$$

$$\frac{0.004202}{1.157625} = 0.00363.$$

$$\text{real factor} = 1.21363$$

So the real growth over 3 years is about **21.36%**.

Real rate per year.

$$\frac{1.12}{1.05} = 1.0666667$$

so the real rate is 6.6667%, i.e. $6\frac{2}{3}\%$ per year.

Check: $(1.0666667)^2 = 1.137778$ and $1.137778 \times 1.0666667 = 1.213630$. Matches.

The wrong method, for contrast. Subtracting gives $12 - 5 = 7\%$ a year, and $(1.07)^3 = 1.225043$, i.e. 22.5% — an overstatement of more than a full percentage point over 3 years.

Answer: Real growth about 21.36% over 3 years; real rate $6\frac{2}{3}\%$ per year

PRACTICE · TIER 3 target: 4 min/Q

1. The difference between CI and SI on a sum for 2 years at 6% per year is Rs. 180. Find the sum.
2. Use the Rule of 72 to estimate the doubling time at 6% compound interest, then check whether 12 years is actually enough by computing $(1.06)^{12}$.
3. Plan A: pay Rs. 45,000 today. Plan B: pay Rs. 16,000 today and Rs. 16,000 at the end of each of the next 2 years. Money earns 10% per year. Which plan is cheaper in today's money, and by how much?
4. Rs. 2,00,000 is invested for 1 year at a nominal 8% per year. Find the amount when compounded yearly, half-yearly and quarterly.
5. A loan of Rs. 50,000 at 1% per month on the reducing balance is repaid in 3 equal monthly instalments. Find the EMI.
6. For a certain sum and rate, the CI for 2 years is Rs. 1,640 and the SI for 2 years is Rs. 1,600. Find the rate and the sum.

7. A flat-rate loan of Rs. 1,20,000 is taken for 2 years at 9% flat, repaid in 24 EMIs. Find the EMI and estimate the true annual rate.
8. An investment returns 15% a year for 2 years while inflation runs at 6% a year. Find the real growth over the 2 years and the real rate per year.

ANSWER KEY

1. Rs. 50,000. $D = P(0.06)^2 = 0.0036P = 180$. The gap is just the interest earned on year 1's interest, so it is tiny compared with the interest itself (SI here is Rs. 6,000).
2. The rule says $\frac{72}{6} = 12$ years, and 12 years is indeed enough. Squaring up: $(1.06)^2 = 1.1236$; $(1.06)^4 = 1.26247696$; $(1.06)^8 = 1.593848$; $(1.06)^{12} = 1.593848 \times 1.26247696 = 2.012196$. Since that exceeds 2, the money has just passed double at 12 years.
3. Plan B is cheaper by about Rs. 1,231. $PV(B) = 16000 + \frac{16000}{1.1} + \frac{16000}{1.21} = 16000 + 14545.45 + 13223.14 = 43768.59$, against Plan A's 45,000. Unlike Example 33 the down payment here is a much smaller share of the total, so more of the cost is pushed into the future and discounting bites harder.
4. Yearly Rs. 2,16,000; half-yearly Rs. 2,16,320 (rate 4%, $(1.04)^2 = 1.0816$); quarterly Rs. 2,16,486.43 (rate 2%, $(1.02)^4 = 1.08243216$). The steps shrink: +320, then +166.43.
5. Rs. 17,001.11. $50000 \times 1.030301 = 51515.05$, and dividing by $(1.0201 + 1.01 + 1) = 3.0301$ gives 17,001.11 — exactly half the answer in Example 35, because the EMI is proportional to the loan.
6. Rate 5%, sum Rs. 16,000. $\frac{CI}{SI} = \frac{1640}{1600} = 1.025 = 1 + \frac{R}{200}$, so $R = 5$. Then $SI = 0.10P = 1600$. (Check: $16000 \times 1.1025 = 17640$, so $CI = 1640$.)
7. EMI Rs. 5,900; true rate about 17.3% per year. Flat interest $= \frac{120000 \times 9 \times 2}{100} = 21600$; total $= 141600$; $\frac{141600}{24} = 5900$. Average balance $\approx 120000 \times \frac{25}{48} = 62500$, so the true rate $\approx \frac{21600}{62500 \times 2} \times 100 = 17.28\%$ — close to double the quoted 9%.
8. Real growth about 17.70% over 2 years; real rate about 8.49% per year. Nominal $(1.15)^2 = 1.3225$; inflation $(1.06)^2 = 1.1236$; ratio $= \frac{1.3225}{1.1236} = 1.17702$. Per year $\frac{1.15}{1.06} = 1.084906$. Subtracting would have given 9% a year, which overstates it.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 30 minutes for all 20

1. Find the simple interest on Rs. 18,000 at 7% per year for 5 years.
2. Find the compound interest on Rs. 8,000 at 25% per year for 2 years.
3. A sum doubles in 6 years at simple interest. In how many years will it triple?
4. Find the difference between CI and SI on Rs. 25,000 at 8% per year for 2 years.
5. Find the compound interest on Rs. 30,000 at 20% per year for 1 year, compounded half-yearly.
6. A sum amounts to Rs. 9,261 in 3 years at 5% compound interest. Find the sum.
7. A loan of Rs. 16,800 at 10% compound interest is cleared in 2 equal annual instalments. Find each instalment.
8. A machine worth Rs. 1,00,000 depreciates 10% a year. Find its value after 3 years.
9. A sum at simple interest amounts to Rs. 11,500 in 3 years and Rs. 12,500 in 5 years. Find the sum and the rate.
10. A flat-rate loan of Rs. 72,000 for 2 years at 11% flat is repaid in 24 EMIs. Find the EMI.
11. A sum becomes 4 times itself in 6 years at compound interest. In how many years will it become 64 times?
12. A town's population of 40,000 falls 5% a year. Find the population after 2 years.
13. Rs. 50,000 is split, one part at 9% simple interest and the rest at 12% simple interest, both for 2 years. The total interest is Rs. 10,200. Find the two parts.
14. Find the effective annual rate for a nominal 10% per year compounded half-yearly.
15. Find the difference between CI and SI on Rs. 10,000 at 20% per year for 3 years.
16. Which costs less today: paying Rs. 33,000 now, or paying Rs. 12,000 now and Rs. 12,000 at the end of each of the next 2 years, when money earns 10% per year?
17. An investment returns 10% in a year while inflation is 4%. Find the real rate of return for that year.

18. Find the compound interest on Rs. 64,000 at 12.5% per year for 3 years.
19. For how long must Rs. 15,000 be lent at 8% simple interest to earn Rs. 4,800?
20. Rs. 1,00,000 is invested for 2 years, once at 10% compound interest and once at 10.5% simple interest. Compare the interest earned.

ANSWER KEY

1. Rs. 6,300. $\frac{18000 \times 7 \times 5}{100}$.
2. Rs. 4,500. $8000 \times 1.25 = 10000$; $\times 1.25 = 12500$; minus 8,000.
3. 12 years. At SI, doubling in T means tripling in $2T$. (Rate = $\frac{100}{6} = 16\frac{2}{3}\%$.)
4. Rs. 160. $25000 \times (0.08)^2 = 25000 \times 0.0064$.
5. Rs. 6,300. Half-yearly rate 10%, 2 periods: 33000, then 36300; $36300 - 30000$.
6. Rs. 8,000. $\frac{9261}{1.157625} = 8000$. (Check: $8000 \rightarrow 8400 \rightarrow 8820 \rightarrow 9261$.)
7. Rs. 9,680. $16800 \times 1.21 = 20328$ and $\frac{20328}{2.1} = 9680$. (Check: $\frac{9680}{1.1} = 8800$, $\frac{9680}{1.21} = 8000$, sum 16,800.)
8. Rs. 72,900. $100000 \times 0.9 = 90000$; $\times 0.9 = 81000$; $\times 0.9 = 72900$.
9. Sum Rs. 10,000, rate 5%. Two years of growth = 1000, so 1 year = 500; 3-year interest = 1500; $P = 11500 - 1500$.
10. Rs. 3,660. Interest = $\frac{72000 \times 11 \times 2}{100} = 15840$; total = 87840; $\frac{87840}{24}$.
11. 18 years. $64 = 4^3$, so three blocks of 6 years. (Answering $\frac{64}{4} \times 6 = 96$ is the SI-style trap.)
12. 36,100. $40000 \times 0.95 = 38000$; $38000 \times 0.95 = 36100$.
13. Rs. 30,000 at 9% and Rs. 20,000 at 12%. $0.18x + 0.24(50000 - x) = 10200$ gives $12000 - 0.06x = 10200$.
14. 10.25%. $(1.05)^2 = 1.1025$.
15. Rs. 1,280. $D = P \frac{R^2(300+R)}{10^6} = 10000 \times \frac{400 \times 320}{10^6} = 10000 \times 0.128$. (Check: SI = 6000; CI = $10000 \times 1.728 - 10000 = 7280$.)
16. The instalment plan is cheaper, by about Rs. 173.55. $PV = 12000 + \frac{12000}{1.1} + \frac{12000}{1.21} = 12000 + 10909.09 + 9917.36 = 32826.45$.
17. About 5.77%. $\frac{1.10}{1.04} = 1.057692$. Subtracting would give 6%, which is too high.
18. Rs. 27,125. Factor 1.125: $64000 \rightarrow 72000 \rightarrow 81000 \rightarrow 91125$; $91125 - 64000$.
19. 4 years. $T = \frac{100 \times 4800}{15000 \times 8} = \frac{480000}{120000}$.
20. They are exactly equal — Rs. 21,000 each. CI = $100000 \times 1.21 - 100000 = 21000$; SI = $\frac{100000 \times 10.5 \times 2}{100} = 21000$. This is the rule $\frac{CI}{SI} = 1 + \frac{R}{200}$ in action: at 10% over 2 years, CI equals SI at 10.5%.

ONE-PAGE REVISION CARD

The two engines

- **Simple interest adds the same slab every year.** Interest never earns interest.
- **Compound interest multiplies by the same factor every year.** Interest earns interest.

Core formulas

$$SI = \frac{PRT}{100} \quad A = P \left(1 + \frac{RT}{100} \right)$$

$$A = P \left(1 + \frac{R}{100} \right)^n \quad CI = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right]$$

Compounding frequency

half-yearly: rate $R/2$, periods $2n$ quarterly: rate $R/4$, periods $4n$ monthly: rate $R/12$, periods $12n$
 k compoundings a year: $EAR = [(1 + R/(100k))^k - 1] \times 100$ per cent.

Shortcuts worth memorising

$$2 \text{ yr gap: } D = P \left(\frac{R}{100} \right)^2 \quad 3 \text{ yr gap: } D = P \frac{R^2(300 + R)}{10^6} \quad 2 \text{ yr ratio: } \frac{CI}{SI} = 1 + \frac{R}{200}$$

- SI doubling: $RT = 100$. Double in $T \rightarrow$ triple in $2T \rightarrow m$ times in $(m-1)T$.
- CI multiplying: m times in $T \rightarrow m^2$ times in $2T \rightarrow m^3$ times in $3T$.
- Rule of 72: doubling time $\approx \frac{72}{R}$ years.
- n instalments at 10%: multiply the loan by 1.1^n , then divide by 2.1, 3.31 or 4.641 for $n = 2, 3, 4$.
- $EMI = \frac{Pi(1+i)^n}{(1+i)^n - 1}$, with $i = \frac{R}{1200}$.
- Flat rate $\rightarrow$ true reducing-balance rate is roughly **double**.
- Real return: divide $(1 + \frac{R}{100})$ by $(1 + \frac{f}{100})$; never subtract.

Powers to know cold

$1.05^2 = 1.1025$	$1.05^3 = 1.157625$	$1.1^2 = 1.21$	$1.1^3 = 1.331$	$1.1^4 = 1.4641$
$1.2^2 = 1.44$	$1.2^3 = 1.728$	$1.25^2 = 1.5625$	$1.25^3 = 1.953125$	$1.15^2 = 1.3225$

The top 5 traps

1. **Months divided by 10.** “2 years 8 months” is $\frac{8}{3}$ years, not 2.8 years.
2. **Half-yearly done halfway.** Halve the rate **and** double the periods. Doing only one of the two is the single most common slip in this chapter.
3. **SI thinking applied to CI multiples.** At CI, “3 times in 5 years” gives 27 times in 15 years — not 45. Count the multiplications, do not scale the time.
4. **Flat rate read as a real rate.** A 10% flat loan really costs about 18%. Always double it before comparing.
5. **Subtracting inflation.** A 12% return with 5% inflation is $1.12/1.05 = 6\frac{2}{3}\%$ real, not 7%. Ratios, not differences.

Time, Speed & Distance

One relation, ten disguises.

What you need to know

WHAT YOU NEED TO KNOW

The one relation

$$\text{speed} = \frac{\text{distance}}{\text{time}} \quad \text{distance} = \text{speed} \times \text{time} \quad \text{time} = \frac{\text{distance}}{\text{speed}}$$

Unit conversion

$$1 \text{ km/h} = \frac{5}{18} \text{ m/s} \quad 1 \text{ m/s} = \frac{18}{5} \text{ km/h}$$

Memorise: 18 km/h = 5 m/s, 36 = 10, 54 = 15, 72 = 20, 90 = 25, 108 = 30.

Fixed distance Speed and time are inversely proportional. If speeds are in ratio $a : b$, then times are in ratio $b : a$.

Average speed

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

Two equal distances at u and v : $2uv/(u+v)$ (harmonic mean). Three equal distances at u, v, w : $3uvw/(uv+vw+wu)$. Equal **times** at u and v : $(u+v)/2$.

Late and early Same distance, speeds $s_1 < s_2$, time difference Δt (in hours):

$$d = \frac{s_1 s_2}{s_2 - s_1} \times \Delta t$$

Relative speed Opposite directions: $u + v$. Same direction: $|u - v|$.

Trains (train length L , platform length P)

- Crossing a pole or a standing man: time = L/s
- Crossing a platform or bridge: time = $(L + P)/s$
- Crossing a moving person: use relative speed, distance = L
- Two trains crossing: distance = $L_1 + L_2$, use relative speed

Boats and streams (boat b , current c)

$$\text{downstream} = b + c \quad \text{upstream} = b - c$$

$$b = \frac{\text{down} + \text{up}}{2} \quad c = \frac{\text{down} - \text{up}}{2}$$

Races “A beats B by x m” $\Rightarrow$ when A finishes the race, B has run (race $-x$) m. “A beats B by t s” $\Rightarrow$ B needs t more seconds to finish. “A gives B a start of x m” $\Rightarrow$ A runs the full race, B runs (race $-x$) m.

Circular tracks (track length C , speeds $u > v$)

- First meeting, same direction: $C/(u - v)$
- First meeting, opposite directions: $C/(u + v)$
- Both back at the start together: LCM of the two lap times
- Distinct meeting points (speed ratio $a : b$ in lowest terms): $a + b$ if opposite, $a - b$ if same direction

Meeting then continuing Two bodies start at the same time towards each other and meet. If afterwards they need t_1 and t_2 more hours:

$$\text{time to meet} = \sqrt{t_1 t_2} \quad \text{speed}_1 : \text{speed}_2 = \sqrt{t_2} : \sqrt{t_1}$$

Warm-up

WARM-UP

build the reflex

Example 1

Convert 72 km/h into m/s.

SOLUTION

$$72 \times \frac{5}{18} = \frac{72 \times 5}{18} = \frac{360}{18} = 20 \text{ m/s.}$$

Answer: 20 m/s

Example 2

Convert 25 m/s into km/h.

SOLUTION

$$25 \times \frac{18}{5} = \frac{25 \times 18}{5} = \frac{450}{5} = 90 \text{ km/h.}$$

Answer: 90 km/h

Example 3

A car covers 240 km in 4 hours. Find its speed.

SOLUTION

$$\text{speed} = \frac{240}{4} = 60 \text{ km/h.}$$

Answer: 60 km/h

Example 4

A bus runs at 45 km/h for 2.5 hours. How far does it go?

SOLUTION

$$\text{distance} = 45 \times 2.5 = 112.5 \text{ km.}$$

Answer: 112.5 km

Example 5

How long does a scooter at 60 km/h take to cover 150 km?

SOLUTION

$$\text{time} = \frac{150}{60} = 2.5 \text{ hours} = 2 \text{ hours } 30 \text{ minutes.}$$

Answer: 2 hours 30 minutes

Example 6

A train 180 m long runs at 54 km/h. How long does it take to pass a pole?

SOLUTION

Speed = $54 \times \frac{5}{18} = 15$ m/s.
 Distance to cover = its own length = 180 m.
 Time = $\frac{180}{15} = 12$ s.

Answer: 12 seconds

Example 7

A boat rows at 12 km/h in still water. The stream flows at 3 km/h. Find the downstream and upstream speeds.

SOLUTION

Downstream = $12 + 3 = 15$ km/h.
 Upstream = $12 - 3 = 9$ km/h.

Answer: 15 km/h and 9 km/h

Example 8

Two cyclists ride towards each other at 20 km/h and 30 km/h. What is their relative speed?

SOLUTION

Opposite directions, so add: $20 + 30 = 50$ km/h.

Answer: 50 km/h

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

A train covers 480 km at a steady speed. If its speed had been 8 km/h more, the trip would have taken 3 hours less. Find the original speed.

SOLUTION

Let the original speed be s km/h.

Original time = $480/s$. New time = $480/(s + 8)$.

$$\frac{480}{s} - \frac{480}{s + 8} = 3$$

Multiply both sides by $s(s + 8)$:

$$480(s + 8) - 480s = 3s(s + 8)$$

$$480 \times 8 = 3s^2 + 24s$$

$$3840 = 3s^2 + 24s$$

Divide by 3:

$$1280 = s^2 + 8s \Rightarrow s^2 + 8s - 1280 = 0$$

$$s = \frac{-8 + \sqrt{64 + 5120}}{2} = \frac{-8 + \sqrt{5184}}{2} = \frac{-8 + 72}{2} = \frac{64}{2} = 32$$

Check: $480/32 = 15$ h and $480/40 = 12$ h. Difference 3 h. Correct.

Answer: 32 km/h

Example 10 [Infosys pattern]

A man drives 60 km at 30 km/h and the next 60 km at 60 km/h. Find his average speed for the whole 120 km.

SOLUTION

Time for first part = $\frac{60}{30} = 2$ h.

Time for second part = $\frac{60}{60} = 1$ h.

Total time = 3 h. Total distance = 120 km.

$$\text{average speed} = \frac{120}{3} = 40 \text{ km/h}$$

Harmonic-mean check: $\frac{2 \times 30 \times 60}{30 + 60} = \frac{3600}{90} = 40$ km/h.

Answer: 40 km/h

TRAP

Average speed for two equal distances is **not** $(30 + 60)/2 = 45$ km/h. The slower leg eats more time, so the average always sits closer to the slower speed. Use $2uv/(u + v)$ only when the two **distances** are equal.

Example 11 [Wipro pattern]

Walking at $4/5$ of his usual speed, a student reaches school 10 minutes late. Find his usual time.

SOLUTION

Speed ratio (new : usual) = 4 : 5.

For a fixed distance, time ratio is the flip: (new : usual) = 5 : 4.

Let usual time = $4k$ minutes. Then new time = $5k$ minutes.

Extra time = $5k - 4k = k = 10$ minutes.

Usual time = $4k = 4 \times 10 = 40$ minutes.

Answer: 40 minutes

SHORTCUT

“Speed becomes a/b of usual” $\Rightarrow$ “time becomes b/a of usual”. Extra time = $(b/a - 1)$ of the usual time. Here $5/4 - 1 = 1/4$, and $1/4$ of the usual time is 10 min, so usual time = 40 min. One line, no equations.

Example 12 [Capgemini pattern]

A train 240 m long passes a platform 360 m long in 30 seconds. Find its speed in km/h.

SOLUTION

Distance covered = $240 + 360 = 600$ m.

Speed = $\frac{600}{30} = 20$ m/s.

In km/h: $20 \times \frac{18}{5} = \frac{360}{5} = 72$ km/h.

Answer: 72 km/h

TRAP

A train crossing a **pole** covers only its own length. A train crossing a **platform** covers its length plus the platform’s length. Students lose easy marks by using the wrong one.

Example 13 [TCS NQT pattern]

Two trains, 150 m and 200 m long, run on parallel tracks in opposite directions at 60 km/h and 40 km/h. How long do they take to cross each other completely?

SOLUTION

Opposite directions, so relative speed = $60 + 40 = 100$ km/h.

In m/s: $100 \times \frac{5}{18} = \frac{500}{18} = \frac{250}{9}$ m/s.

Distance to cover = $150 + 200 = 350$ m.

$$\text{time} = \frac{350}{250/9} = 350 \times \frac{9}{250} = \frac{3150}{250} = 12.6 \text{ s}$$

Answer: 12.6 seconds

Example 14 [Accenture pattern]

Train A is 180 m long and runs at 72 km/h. Train B is 120 m long and runs at 54 km/h on a parallel track in the same direction. How long does A take to overtake B completely?

SOLUTION

Same direction, so relative speed = $72 - 54 = 18$ km/h.

In m/s: $18 \times \frac{5}{18} = 5$ m/s.

Distance to cover = $180 + 120 = 300$ m.

$$\text{time} = \frac{300}{5} = 60 \text{ s}$$

Answer: 60 seconds

Example 15 [Cognizant pattern]

A boat covers 36 km downstream in 2 hours and the same 36 km upstream in 3 hours. Find the speed of the boat in still water and the speed of the stream.

SOLUTION

Downstream speed = $\frac{36}{2} = 18$ km/h.

Upstream speed = $\frac{36}{3} = 12$ km/h.

$$b = \frac{18 + 12}{2} = \frac{30}{2} = 15 \text{ km/h}$$

$$c = \frac{18 - 12}{2} = \frac{6}{2} = 3 \text{ km/h}$$

Answer: Boat 15 km/h, stream 3 km/h

Example 16 [Infosys pattern]

A boat's speed in still water is 10 km/h and the stream flows at 2 km/h. How long does the boat take to go 48 km downstream and come back?

SOLUTION

Downstream speed = $10 + 2 = 12$ km/h $\Rightarrow$ time = $\frac{48}{12} = 4$ h.

Upstream speed = $10 - 2 = 8$ km/h $\Rightarrow$ time = $\frac{48}{8} = 6$ h.

Total = $4 + 6 = 10$ h.

Answer: 10 hours

Example 17 [Wipro pattern]

In a 1000 m race A beats B by 100 m. In a 1000 m race B beats C by 50 m. By how much does A beat C in a 1000 m race?

SOLUTION

When A runs 1000 m, B runs 900 m.

When B runs 1000 m, C runs 950 m.

So when B runs 900 m, C runs

$$900 \times \frac{950}{1000} = 900 \times 0.95 = 855 \text{ m}$$

A finishes 1000 m at the moment C has run 855 m.

A beats C by $1000 - 855 = 145$ m.

Answer: 145 m

Example 18 [Accenture pattern]

In a 400 m race A beats B by 16 m or, equivalently, by 4 seconds. Find the time each runner takes for 400 m.

SOLUTION

“Beats by 16 m or 4 s” means B needs 4 more seconds to cover the last 16 m.

B’s speed = $\frac{16}{4} = 4$ m/s.

B’s time for 400 m = $\frac{400}{4} = 100$ s.

A’s time = $100 - 4 = 96$ s.

Answer: A: 96 s, B: 100 s

TRAP

“A beats B by 16 m” and “A beats B by 4 s” are two different measurements of the same gap. Do not add them. Use them together to get B’s speed: metres $\div$ seconds.

Example 19 [TCS NQT pattern]

Two cities are 300 km apart. A car leaves city A at 60 km/h towards B. At the same moment another car leaves B at 40 km/h towards A. When do they meet and how far from A?

SOLUTION

They approach each other, so the gap closes at $60 + 40 = 100$ km/h.

$$\text{time to meet} = \frac{300}{100} = 3 \text{ hours}$$

Distance from A = $60 \times 3 = 180$ km.

Check: from B, $40 \times 3 = 120$ km, and $180 + 120 = 300$ km. Correct.

Answer: After 3 hours, 180 km from A

Example 20 [Capgemini pattern]

Two runners start together from the same point on a 400 m circular track. Their speeds are 5 m/s and 3 m/s. After how long do they first meet again if (a) they run in the same direction, (b) they run in opposite directions?

SOLUTION

(a) Same direction. The faster must gain one full lap on the slower.

Relative speed = $5 - 3 = 2$ m/s.

$$\text{time} = \frac{400}{2} = 200 \text{ s}$$

(b) Opposite directions. Together they must cover one full lap.

Relative speed = $5 + 3 = 8$ m/s.

$$\text{time} = \frac{400}{8} = 50 \text{ s}$$

Answer: (a) 200 s (b) 50 s

SHORTCUT

On a circular track the **first meeting** question is always “track length $\div$ relative speed”. Same direction subtract, opposite direction add. That is the whole idea.

PRACTICE · TIER 1 target: 60 s/Q

1. Convert 108 km/h into m/s.
2. A cyclist covers 90 km in 4 hours 30 minutes. Find his speed.
3. A car travels from P to Q at 40 km/h and returns at 60 km/h. Find the average speed for the round trip.
4. A train 150 m long runs at 90 km/h. How long does it take to cross a pole?
5. A train 200 m long crosses a bridge 400 m long in 24 seconds. Find its speed in km/h.
6. Two trains 120 m and 180 m long run in opposite directions at 42 km/h and 30 km/h. Find the time they take to cross each other.
7. A boat covers 24 km downstream in 2 hours and 24 km upstream in 3 hours. Find the still-water speed and the stream speed.
8. A man rows at 8 km/h in still water. The stream flows at 2 km/h. How long does he take to row 30 km downstream?
9. Walking at $\frac{3}{4}$ of his usual speed, a man reaches office 15 minutes late. Find his usual time.
10. In a 400 m race A beats B by 40 m. Find the ratio of their speeds.
11. Two cars start at the same time from towns 420 km apart and move towards each other at 50 km/h and 70 km/h. After how long do they meet?
12. A train at 60 km/h crosses a man walking at 6 km/h in the same direction in 12 seconds. Find the length of the train.
13. A 90 km journey is covered with $\frac{1}{3}$ of the distance at 30 km/h and the rest at 45 km/h. Find the total time.
14. A runner takes 5 minutes for one lap of a 1000 m track. Find his speed in km/h.
15. Two runners start together from one point of a 300 m circular track and run in opposite directions at 4 m/s and 6 m/s. When do they first meet?

ANSWER KEY

1. **30 m/s** — $108 \times \frac{5}{18} = 30$.
2. **20 km/h** — 4 h 30 min = 4.5 h; $90/4.5 = 20$.
3. **48 km/h** — $2 \times 40 \times 60/100 = 4800/100 = 48$. Not 50.
4. **6 s** — 90 km/h = 25 m/s; $150/25 = 6$.
5. **90 km/h** — $(200 + 400)/24 = 25 \text{ m/s} = 25 \times 18/5 = 90$.
6. **15 s** — relative 72 km/h = 20 m/s; $(120 + 180)/20 = 15$.
7. **Boat 10 km/h, stream 2 km/h** — down 12, up 8; $(12 + 8)/2 = 10$, $(12 - 8)/2 = 2$.
8. **3 hours** — downstream speed $8 + 2 = 10$; $30/10 = 3$.
9. **45 minutes** — time ratio 4 : 3 flipped; extra $\frac{1}{3}$ of usual = 15 min, so usual = 45 min.
10. **10 : 9** — same time, so speeds are as distances: $400 : 360 = 10 : 9$.
11. **3.5 hours** — closing speed 120 km/h; $420/120 = 3.5$.
12. **180 m** — relative 54 km/h = 15 m/s; $15 \times 12 = 180$.
13. **2 hours 20 minutes** — $30/30 = 1 \text{ h}$ and $60/45 = 4/3 \text{ h}$; total $7/3 \text{ h}$.
14. **12 km/h** — 1000 m in 300 s = $10/3 \text{ m/s} = (10/3) \times 18/5 = 12$.
15. **30 s** — relative $4 + 6 = 10 \text{ m/s}$; $300/10 = 30$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A driver takes a 24 km fare. The first 8 km are in heavy traffic at 16 km/h. The remaining 16 km are on the expressway at 48 km/h. Find the average speed for the whole trip.

SOLUTION

$$\text{Time in traffic} = \frac{8}{16} = 0.5 \text{ h.}$$

$$\text{Time on expressway} = \frac{16}{48} = \frac{1}{3} \text{ h.}$$

$$\text{Total time} = \frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6} \text{ h.}$$

$$\text{average speed} = \frac{24}{5/6} = 24 \times \frac{6}{5} = \frac{144}{5} = 28.8 \text{ km/h}$$

Answer: 28.8 km/h

Example 22 [Shopee · pattern]

A rider leaves the hub at 09:00 and rides at 30 km/h along a fixed route. A second rider leaves the same hub at 09:30 along the same route at 45 km/h. At what time does the second rider catch the first, and how far from the hub?

SOLUTION

$$\text{Head start of the first rider in 30 minutes} = 30 \times 0.5 = 15 \text{ km.}$$

$$\text{The gap closes at } 45 - 30 = 15 \text{ km/h.}$$

$$\text{catch-up time} = \frac{15}{15} = 1 \text{ hour after 09:30} = 10 : 30$$

$$\text{Distance from hub} = 45 \times 1 = 45 \text{ km.}$$

Check: the first rider has ridden from 09:00 to 10:30, that is 1.5 h at 30 km/h = 45 km. Correct.

Answer: 10:30, at 45 km from the hub

Example 23 [GIC · pattern]

A train crosses a man standing on a platform in 9 seconds. It crosses the whole 240 m platform in 24 seconds. Find the length and the speed of the train.

SOLUTION

Let the length be L m and the speed be s m/s.

$$\text{Crossing the man: } L = 9s.$$

$$\text{Crossing the platform: } L + 240 = 24s.$$

Subtract the first from the second:

$$240 = 24s - 9s = 15s \Rightarrow s = \frac{240}{15} = 16 \text{ m/s}$$

$$\text{Then } L = 9 \times 16 = 144 \text{ m.}$$

$$\text{Speed in km/h} = 16 \times \frac{18}{5} = \frac{288}{5} = 57.6 \text{ km/h.}$$

Answer: Length 144 m, speed 16 m/s = 57.6 km/h

SHORTCUT

Two crossing times, one train: **subtract the two equations**. The train length cancels and you are left with (platform length) = (difference of times) $\times$ speed.

Example 24 [Agoda · pattern]

Two express trains start at the same moment from two cities and travel towards each other. After they pass each other, the first needs 9 more hours to finish its trip and the second needs 4 more hours. The first train runs at 40 km/h. Find the second train's speed and the distance between the cities.

SOLUTION

Let the speeds be $a = 40$ and b , and let t be the time until they meet.

After meeting, train 1 must still cover the distance train 2 already covered, that is bt :

$$\frac{bt}{a} = 9 \Rightarrow bt = 9a$$

After meeting, train 2 must still cover at :

$$\frac{at}{b} = 4 \Rightarrow at = 4b$$

Multiply the two original relations:

$$\frac{bt}{a} \times \frac{at}{b} = 9 \times 4 \Rightarrow t^2 = 36 \Rightarrow t = 6 \text{ h}$$

From $at = 4b$: $40 \times 6 = 4b \Rightarrow b = \frac{240}{4} = 60 \text{ km/h}$.

Distance = $(a + b) \times t = (40 + 60) \times 6 = 600 \text{ km}$.

Check: train 1 still has $60 \times 6 = 360 \text{ km}$, and $360/40 = 9 \text{ h}$.

Train 2 still has $40 \times 6 = 240 \text{ km}$, and $240/60 = 4 \text{ h}$. Both correct.

Answer: Second train 60 km/h; cities 600 km apart

SHORTCUT

Meeting problems of this type collapse to two facts:

$$t = \sqrt{t_1 t_2} \quad \text{speed}_1 : \text{speed}_2 = \sqrt{t_2} : \sqrt{t_1}$$

Here $t = \sqrt{9 \times 4} = 6 \text{ h}$ and $40 : b = \sqrt{4} : \sqrt{9} = 2 : 3$, so $b = 60$.

Example 25 [DBS · pattern]

A river ferry travels 45 km downstream and then 45 km back upstream, taking 8 hours in all. The current runs at 3 km/h. Find the ferry's speed in still water.

SOLUTION

Let the still-water speed be $b \text{ km/h}$.

$$\frac{45}{b+3} + \frac{45}{b-3} = 8$$

Combine the left side over $(b+3)(b-3) = b^2 - 9$:

$$\frac{45(b-3) + 45(b+3)}{b^2 - 9} = 8$$

$$\frac{45b - 135 + 45b + 135}{b^2 - 9} = 8 \Rightarrow \frac{90b}{b^2 - 9} = 8$$

$$90b = 8b^2 - 72 \Rightarrow 8b^2 - 90b - 72 = 0$$

Divide by 2:

$$4b^2 - 45b - 36 = 0$$

$$b = \frac{45 + \sqrt{2025 + 576}}{8} = \frac{45 + \sqrt{2601}}{8} = \frac{45 + 51}{8} = \frac{96}{8} = 12$$

Check: downstream $45/15 = 3 \text{ h}$, upstream $45/9 = 5 \text{ h}$, total 8 h. Correct.

Answer: 12 km/h

Example 26 [Sea Group · pattern]

A courier van leaves the depot at 07:30. It drives 200 km at 50 km/h, takes a 1 hour break, then drives another 120 km at 40 km/h. Find the arrival time and the average speed for the whole run, counting the break.

SOLUTION

$$\text{Leg 1 time} = \frac{200}{50} = 4 \text{ h.}$$

$$\text{Break} = 1 \text{ h.}$$

$$\text{Leg 2 time} = \frac{120}{40} = 3 \text{ h.}$$

$$\text{Total elapsed time} = 4 + 1 + 3 = 8 \text{ h.}$$

$$\text{Arrival} = 07 : 30 + 8 \text{ h} = 15 : 30.$$

$$\text{Total distance} = 200 + 120 = 320 \text{ km.}$$

$$\text{average speed} = \frac{320}{8} = 40 \text{ km/h}$$

Answer: Arrives 15:30; average 40 km/h

TRAP

A rest break adds to **time** but not to **distance**. If the question says “including stoppages”, the break must go into the denominator. Excluding the break here the average would be $320/7 \approx 45.7$ km/h, a different answer.

Example 27 [SCB · pattern]

A train 180 m long runs at 63 km/h. A jogger runs at 9 km/h on a path beside the track. How long does the train take to pass him if (a) he runs in the same direction, (b) he runs in the opposite direction?

SOLUTION

The train must cover only its own length, 180 m, relative to the jogger.

(a) Same direction. Relative speed = $63 - 9 = 54 \text{ km/h} = 54 \times \frac{5}{18} = 15 \text{ m/s.}$

$$\text{time} = \frac{180}{15} = 12 \text{ s}$$

(b) Opposite direction. Relative speed = $63 + 9 = 72 \text{ km/h} = 72 \times \frac{5}{18} = 20 \text{ m/s.}$

$$\text{time} = \frac{180}{20} = 9 \text{ s}$$

Answer: (a) 12 s (b) 9 s

Example 28 [LINE MAN · pattern]

A cyclist rides up a hill road at 12 km/h and comes down the same road at 20 km/h. The whole ride takes 4 hours. Find the average speed and the one-way length of the road.

SOLUTION

The two distances are equal, so use the harmonic mean:

$$\text{average speed} = \frac{2 \times 12 \times 20}{12 + 20} = \frac{480}{32} = 15 \text{ km/h}$$

$$\text{Total distance} = \text{average speed} \times \text{total time} = 15 \times 4 = 60 \text{ km.}$$

$$\text{One-way length} = \frac{60}{2} = 30 \text{ km.}$$

$$\text{Check: up } 30/12 = 2.5 \text{ h, down } 30/20 = 1.5 \text{ h, total 4 h. Correct.}$$

Answer: 15 km/h; road is 30 km one way

Example 29 [Razer · pattern]

Two testers ride on a 600 m circular test track, starting together from the same point and going in the same direction at 8 m/s and 5 m/s. (a) When do they first meet again? (b) When are they next together at the starting point? (c) How many distinct points of the track will they ever meet at?

SOLUTION

(a) Relative speed = $8 - 5 = 3$ m/s. The faster needs to gain one lap.

$$\text{time} = \frac{600}{3} = 200 \text{ s}$$

(b) Lap time of the faster = $\frac{600}{8} = 75$ s. Lap time of the slower = $\frac{600}{5} = 120$ s.

$$75 = 3 \times 5^2, \quad 120 = 2^3 \times 3 \times 5.$$

$$\text{LCM} = 2^3 \times 3 \times 5^2 = 600 \text{ s}.$$

(c) Speeds in lowest terms are $8 : 5$. Same direction, so the number of distinct meeting points is $8 - 5 = 3$.

Check: meetings happen at $t = 200, 400, 600$ s.

At 200 s the faster is at $8 \times 200 = 1600$ m, and $1600 - 2 \times 600 = 400$ m.

At 400 s: $3200 - 5 \times 600 = 200$ m.

At 600 s: $4800 - 8 \times 600 = 0$ m.

The three points are the 400 m mark, the 200 m mark and the start.

Answer: (a) 200 s (b) 600 s (c) 3 points

Example 30 [Grab · pattern]

A commuter leaves home at the same time every day. At 40 km/h she reaches the office 8 minutes late. At 50 km/h she reaches 4 minutes early. Find the distance to the office.

SOLUTION

The two trips differ by $8 + 4 = 12$ minutes = $\frac{12}{60} = \frac{1}{5}$ h.

Let the distance be d km.

$$\frac{d}{40} - \frac{d}{50} = \frac{1}{5}$$

LCM of 40 and 50 is 200:

$$\frac{5d}{200} - \frac{4d}{200} = \frac{1}{5} \Rightarrow \frac{d}{200} = \frac{1}{5}$$

$$d = \frac{200}{5} = 40 \text{ km}$$

Check: $40/40 = 1$ h = 60 min; $40/50 = 0.8$ h = 48 min. Difference 12 min. Correct.

Answer: 40 km

SHORTCUT

Late-and-early questions always reduce to

$$d = \frac{s_1 s_2}{s_2 - s_1} \times \Delta t$$

with Δt in hours. Here $d = (40 \times 50)/10 \times 1/5 = 200 \times 1/5 = 40$ km.

PRACTICE · TIER 2 target: 100 s/Q

1. A driver covers 30 km at 45 km/h, waits 20 minutes, then covers 20 km at 40 km/h. Find the total time and the average speed for the 50 km including the wait.

2. A ferry takes 5 hours downstream and 7 hours upstream between two ports 210 km apart. Find the ferry's still-water speed and the current's speed.
3. Two trains leave two cities 350 km apart at 08:00 and travel towards each other at 80 km/h and 60 km/h. At what time do they meet, and how far from the first city?
4. A train crosses a standing man in 8 seconds and a 264 m platform in 20 seconds. Find its length and its speed in km/h.
5. A bus averages 36 km/h including stops and 45 km/h excluding stops. How many minutes per hour does it stop?
6. A cyclist rides out at 15 km/h and returns along the same road at 10 km/h. The whole ride takes 5 hours. Find the one-way distance.
7. Two riders start together on a 1.2 km circular track in the same direction at 18 km/h and 12 km/h. When do they first meet again, and how many laps has the faster rider done by then?
8. A van leaves a depot at 09:00 at 40 km/h. A motorbike leaves the same depot at 10:00 at 60 km/h on the same road. When and where does the bike catch the van?
9. In a 1200 m race A beats B by 120 m. In a 1200 m race B beats C by 100 m. By how much does A beat C over 1200 m?
10. A man rows at 9 km/h in still water. The river flows at 3 km/h. He rows to a point downstream and returns, taking 5 hours in all. How far is the point?
11. A train running at 54 km/h takes 60 seconds to pass a 150 m train moving at 36 km/h in the same direction. Find the length of the first train.
12. A car covers a certain distance at 60 km/h and returns at 40 km/h. The return trip takes 1 hour longer. Find the one-way distance.

ANSWER KEY

1. **1.5 h; 331/3 km/h** — $40 + 20 + 30 = 90$ min; $50/1.5 = 100/3$.
2. **36 km/h and 6 km/h** — down $210/5 = 42$, up $210/7 = 30$; half-sum 36, half-difference 6.
3. **10:30, 200 km** — closing speed 140; $350/140 = 2.5$ h; $80 \times 2.5 = 200$.
4. **176 m; 79.2 km/h** — $264 = (20 - 8)s \Rightarrow s = 22$ m/s; $L = 8 \times 22 = 176$; $22 \times 18/5 = 79.2$.
5. **12 minutes** — in one hour it does 36 km, which needs $36/45$ h = 48 min of motion.
6. **30 km** — average = $2 \times 15 \times 10/25 = 12$; total $12 \times 5 = 60$; half is 30.
7. **12 minutes; 3 laps** — relative 6 km/h = $5/3$ m/s; $1200/(5/3) = 720$ s; faster covers $5 \times 720 = 3600$ m = 3 laps.
8. **12:00, 120 km from the depot** — head start 40 km, closing 20 km/h, so 2 h after 10:00.
9. **210 m** — B runs 1080 when A runs 1200; C runs $1080 \times 1100/1200 = 990$; $1200 - 990 = 210$.
10. **20 km** — $d/12 + d/6 = 5 \Rightarrow 3d = 60$.
11. **150 m** — relative 18 km/h = 5 m/s; total length $5 \times 60 = 300$; $300 - 150 = 150$.
12. **120 km** — $d/40 - d/60 = 1 \Rightarrow d/120 = 1$.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Google · pattern]

Two ships are 180 km apart and sail straight towards each other at 30 km/h and 15 km/h. A survey drone starts from the first ship at the same moment, flies at 60 km/h to the second ship, turns instantly, flies back to the first, and keeps shuttling until the ships meet. What total distance does the drone fly?

SOLUTION

Insight: do not chase the shuttle legs. The drone flies for exactly as long as the ships take to meet, so only the total time matters.

The gap closes at $30 + 15 = 45$ km/h.

$$\text{time until the ships meet} = \frac{180}{45} = 4 \text{ hours}$$

The drone flies for the whole of those 4 hours without stopping.

$$\text{distance} = 60 \times 4 = 240 \text{ km}$$

Answer: 240 km

The individual legs form an infinite geometric series. Summing it gives the same 240 km, but the time argument gets there in one line.

Example 32 [Amazon · pattern]

A delivery van must average 60 km/h over a 120 km route. It covers the first 60 km at 30 km/h. What speed must it hold over the remaining 60 km to hit the target average?

SOLUTION

Insight: an average speed is a constraint on total time, and the time budget can be spent before the trip is over.

Time budget for the whole route:

$$\frac{120}{60} = 2 \text{ hours}$$

Time already spent on the first half:

$$\frac{60}{30} = 2 \text{ hours}$$

The full budget is gone and 60 km still remain. Even at infinite speed the second half takes more than zero time, so the total time is strictly greater than 2 hours.

Algebra says the same thing. Let the needed speed be v :

$$\frac{2 \times 30 \times v}{30 + v} = 60 \Rightarrow 60v = 60(30 + v) \Rightarrow 60v = 1800 + 60v$$
$$0 = 1800$$

A false statement, so no such v exists.

Answer: Impossible — the target cannot be reached

TRAP

If the first half of a journey is done at half the required average speed, the trip is already lost. More generally, with equal halves at u and v , the average $2uv/(u + v)$ is always less than $2u$. So the average can never be doubled by the second half alone.

Example 33 [Goldman Sachs · pattern]

A rower is going upstream. As he passes under a bridge his sealed water bottle falls into the river and floats away. He keeps rowing upstream for 1 hour, then turns round and rows downstream at the same effort. He picks up the bottle 6 km downstream of the bridge. Find the speed of the current.

SOLUTION

Insight: measure everything relative to the water. Relative to the water the bottle never moves, and the rower's speed is the same both ways.

Let the rower's speed relative to the water be r and the current be c .

Relative to the water:

- Going up, he moves away from the bottle at r for 1 hour.

- Coming back, he closes the same relative gap at r , so it also takes 1 hour.

Total time from the drop to the pick-up = $1 + 1 = 2$ hours.

The bottle itself just drifts with the current. In those 2 hours it travelled 6 km.

$$c = \frac{6}{2} = 3 \text{ km/h}$$

Ground check with $r = 7$ km/h, $c = 3$ km/h. Upstream speed = 4 km/h, so in 1 h he is 4 km above the bridge and the bottle is 3 km below it: gap 7 km. Downstream he does 10 km/h, the bottle 3 km/h, so the gap closes at 7 km/h, taking 1 h. At that moment the bottle has drifted $3 \times 2 = 6$ km. Correct, and r never entered the answer.

Answer: 3 km/h

Example 34 [Microsoft · pattern]

Two athletes run round a circular track starting together from the same point. Their speeds are in the ratio 5 : 3. How many distinct points of the track do they ever meet at, (a) running in opposite directions, (b) running in the same direction? Explain why.

SOLUTION

Insight: meetings are equally spaced in time, so the meeting points are equally spaced round the track. Count how many gaps fit into one lap.

Let the track length be C and the speeds be $5k$ and $3k$.

(a) Opposite directions. They meet every $C/(5k + 3k) = C/(8k)$ seconds.

Between two meetings the faster athlete advances

$$5k \times \frac{C}{8k} = \frac{5C}{8}$$

So each new meeting point sits $5C/8$ further round the track than the previous one.

The points are $0, 5C/8, 10C/8, 15C/8, \dots$, that is $0, 5/8, 2/8, 7/8, 4/8, 1/8, 6/8, 3/8$ of a lap, then back to 0.

Since $\gcd(5, 8) = 1$, the multiples of $5/8$ run through all 8 eighths before repeating. **8 distinct points.**

(b) Same direction. They meet every $C/(5k - 3k) = C/(2k)$ seconds, and the faster advances

$$5k \times \frac{C}{2k} = \frac{5C}{2} = 2C + \frac{C}{2}$$

which is half a lap past the previous point. The points are $0, C/2, 0, C/2, \dots$ **2 distinct points.**

General rule, with the speed ratio written in lowest terms as $a : b$:

opposite directions give $a + b$ points, same direction gives $a - b$ points.

Answer: (a) 8 points (b) 2 points

SHORTCUT

Always reduce the speed ratio first. Speeds 6 : 4 behave like 3 : 2, giving 5 meeting points in opposite directions and 1 in the same direction, not 10 and 2.

Example 35 [D. E. Shaw · pattern]

Two friends must travel 60 km together and arrive at the same time. They have one motorbike, which carries only one person, at 30 km/h. Walking speed is 6 km/h. The bike may be parked anywhere on the road and picked up later. What is the least time in which both can arrive?

SOLUTION

Insight: the bike must never sit idle any longer than necessary, and by symmetry the best plan gives each friend the same riding distance.

Plan: A rides the first x km, parks the bike and walks the rest. B walks the first x km, finds the bike and rides the remaining $60 - x$ km.

$$\begin{aligned}\text{A's time: } & \frac{x}{30} + \frac{60-x}{6} \\ \text{B's time: } & \frac{x}{6} + \frac{60-x}{30}\end{aligned}$$

They must arrive together, so set the two equal:

$$\frac{x}{30} + \frac{60-x}{6} = \frac{x}{6} + \frac{60-x}{30}$$

Multiply everything by 30:

$$\begin{aligned}x + 5(60-x) &= 5x + (60-x) \\ x + 300 - 5x &= 5x + 60 - x \\ 300 - 4x &= 4x + 60 \\ 240 = 8x &\Rightarrow x = 30 \text{ km}\end{aligned}$$

Common arrival time:

$$\frac{30}{30} + \frac{30}{6} = 1 + 5 = 6 \text{ hours}$$

Feasibility check: A parks the bike at the 30 km mark after 1 hour. B walks 30 km at 6 km/h and reaches that mark at hour 5, when the bike is waiting. B then rides 30 km in 1 hour and arrives at hour 6, together with A. Valid.

Compare: both walking the whole way would take $60/6 = 10$ hours. The shared bike saves 4 hours.

Answer: 6 hours

Example 36 [Uber · pattern]

A man walks along a straight bus route. Buses run in both directions at the same constant speed and at equal time intervals. A bus overtakes him every 10 minutes, and a bus meets him head-on every 6 minutes. Find the interval between consecutive buses and the ratio of bus speed to walking speed.

SOLUTION

Insight: the distance between two consecutive buses is fixed. Each encounter means the man and one bus together closed exactly that fixed gap.

Let the bus speed be v , the man's speed be u , and the interval between buses be T minutes.

Consecutive buses on the same side are a distance vT apart.

Buses coming from behind close the gap at $v - u$:

$$(v - u) \times 10 = vT$$

Buses coming towards him close the gap at $v + u$:

$$(v + u) \times 6 = vT$$

The right sides are equal, so:

$$\begin{aligned}10v - 10u &= 6v + 6u \\ 4v &= 16u \Rightarrow v = 4u\end{aligned}$$

Substitute into the first relation:

$$(4u - u) \times 10 = 4uT \Rightarrow 30u = 4uT \Rightarrow T = \frac{30}{4} = 7.5 \text{ minutes}$$

Check with $u = 5$ km/h, $v = 20$ km/h. Gap = $20 \times 7.5/60 = 2.5$ km.

Overtaking: $2.5/15$ h = 10 min. Head-on: $2.5/25$ h = 6 min. Both correct.

Answer: Interval 7.5 minutes; bus speed is 4 times walking speed

SHORTCUT

For this family, with overtaking gap p and head-on gap q :

$$T = \frac{2pq}{p+q} \quad \frac{v}{u} = \frac{p+q}{p-q}$$

Here $T = (2 \times 10 \times 6)/16 = 120/16 = 7.5$ min and $v/u = 16/4 = 4$.

Example 37 [Adobe · pattern]

An airport walkway is 120 m long. A traveller who walks along it while it moves takes 40 seconds end to end. Standing still on it takes 60 seconds. How long would the same traveller take to walk 120 m on the fixed floor beside it?

SOLUTION

Insight: speeds add, times do not. Convert every statement into a speed first.

Walkway speed (standing still):

$$\frac{120}{60} = 2 \text{ m/s}$$

Walking while it moves (speeds add):

$$\frac{120}{40} = 3 \text{ m/s}$$

So the traveller's own walking speed is

$$3 - 2 = 1 \text{ m/s}$$

On the fixed floor:

$$\text{time} = \frac{120}{1} = 120 \text{ s}$$

Answer: 120 seconds

TRAP

Do not subtract the times: $60 - 40 = 20$ s is wrong. Time is not additive, speed is. Convert to m/s, subtract, convert back.

Example 38 [Google · pattern]

Three runners start together from the same point of a 600 m circular track and run in the same direction at 8 m/s, 5 m/s and 2 m/s. (a) After how long are all three together again for the first time? (b) After how long are all three back at the starting point together?

SOLUTION

Insight: three runners are together exactly when every pair is together. So find each pair's meeting period, then take the LCM. Being together anywhere is a weaker condition than being together at the start.

(a) Together anywhere. For each pair, the meeting period is $600 \div (\text{speed difference})$.

$$(8, 5) : \frac{600}{3} = 200 \text{ s} \quad (8, 2) : \frac{600}{6} = 100 \text{ s} \quad (5, 2) : \frac{600}{3} = 200 \text{ s}$$

All three coincide at the common multiples of 200, 100 and 200.

$\text{LCM}(200, 100, 200) = 200$ s.

Check at $t = 200$ s:

Runner 1: $8 \times 200 = 1600$ m; $1600 - 2 \times 600 = 400$ m.

Runner 2: $5 \times 200 = 1000$ m; $1000 - 600 = 400$ m.

Runner 3: $2 \times 200 = 400$ m.

All three stand at the 400 m mark. Correct.

(b) Together at the start. Lap times are

$$\frac{600}{8} = 75 \text{ s}, \quad \frac{600}{5} = 120 \text{ s}, \quad \frac{600}{2} = 300 \text{ s}$$

$$75 = 3 \times 5^2, \quad 120 = 2^3 \times 3 \times 5, \quad 300 = 2^2 \times 3 \times 5^2$$

$$\text{LCM} = 2^3 \times 3 \times 5^2 = 600 \text{ s}$$

Answer: (a) 200 s (b) 600 s

PRACTICE · TIER 3 target: 4 min/Q

- Two trains start at the same time from A and B and move towards each other. After they cross, the first takes 16 hours to reach B and the second takes 25 hours to reach A. Find the ratio of their speeds.
- A boat's still-water speed is 3 times the speed of the stream. Going 96 km upstream takes 8 hours longer than going 96 km downstream. Find the speed of the stream.
- A man walks along a bus route. Buses overtake him every 12 minutes and meet him every 4 minutes. All buses have the same speed and run at equal intervals. Find the interval between buses and the ratio of bus speed to walking speed.
- Two runners start together on a 480 m circular track in the same direction at 12 m/s and 8 m/s. (a) When do they first meet? (b) How many distinct meeting points are there? (c) When are they both back at the start together?
- A walker and a cyclist start together from A towards B, 36 km away. The cyclist rides at 18 km/h, reaches B, turns round at once and rides back. The walker walks at 6 km/h. How far from A do they meet?
- A driver covers the first 40 km of an 80 km route at 40 km/h. What speed must he hold over the remaining 40 km so that his average for the whole route is 80 km/h?
- Three runners start together from the same point of a 360 m circular track and run in the same direction at 9 m/s, 6 m/s and 4 m/s. After how long are all three together again?
- A train passes a 210 m platform in 25 seconds and a 330 m platform in 31 seconds. Find its speed and its length.

ANSWER KEY

- 5 : 4** — with $t_1 = 16$ and $t_2 = 25$, the ratio is $\text{speed}_1 : \text{speed}_2 = \sqrt{t_2} : \sqrt{t_1} = \sqrt{25} : \sqrt{16} = 5 : 4$. The train that needs **longer** after crossing is the slower one. Here the second train needs 25 h against 16 h, so the second train is the slower, and the first is faster in the ratio 5 : 4.
- 3 km/h** — let the stream be c , so the boat is $3c$, downstream $4c$, upstream $2c$. Then $96/(2c) - 96/(4c) = 8$, that is $48/c - 24/c = 8$, so $24/c = 8$ and $c = 3$. Boat 9 km/h: upstream 16 h, downstream 8 h, difference 8 h.
- 6 minutes; ratio 2 : 1** — $(v - u)12 = vT$ and $(v + u)4 = vT$ give $12v - 12u = 4v + 4u$, so $8v = 16u$ and $v = 2u$. Then $vT = (2u - u)12 = 12u$, so $T = 6$ min.
- (a) 120 s (b) 1 point (c) 120 s** — relative speed 4 m/s gives $480/4 = 120$ s. Ratio $12 : 8 = 3 : 2$, and $3 - 2 = 1$, so they only ever meet at one point. Lap times 40 s and 60 s have LCM 120 s, the same instant, which confirms the single point is the start.
- 18 km from A** — let the meeting be at time t . The walker is at $6t$ from A. The cyclist has ridden $36 + (36 - 6t) = 72 - 6t$. So $18t = 72 - 6t$, giving $24t = 72$ and $t = 3$ h. The walker is at $6 \times 3 = 18$ km.
- Impossible** — the whole route at 80 km/h allows exactly 1 hour, and the first 40 km at 40 km/h already used 1 hour. No finite speed can fix a time budget that is already spent.
- 360 s (6 minutes)** — pair periods are $360/3 = 120$ s, $360/5 = 72$ s and $360/2 = 180$ s. $120 = 2^3 \cdot 3 \cdot 5$, $72 = 2^3 \cdot 3^2$, $180 = 2^2 \cdot 3^2 \cdot 5$, so the LCM is $2^3 \cdot 3^2 \cdot 5 = 360$ s. At 360 s all three have run a whole number of laps, so they meet at the start.
- 20 m/s (72 km/h); 290 m** — subtracting the two crossing equations kills the length: $330 - 210 = (31 - 25)s$, so $120 = 6s$ and $s = 20$ m/s. Then $L + 210 = 25 \times 20 = 500$, so $L = 290$ m. Check: $(290 + 330)/20 = 31$ s.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 25 questions in 30 min pace · 20 questions here

1. A car covers 260 km in 4 hours. Find its speed.
2. Convert 12.5 m/s into km/h.
3. A train 210 m long crosses a pole in 14 seconds. Find its speed in km/h.
4. Walking at 5 km/h a man reaches 6 minutes late; at 6 km/h he reaches 4 minutes early. Find the distance.
5. A journey is split into three equal distances covered at 20, 30 and 60 km/h. Find the average speed.
6. A boat rows at 15 km/h in still water; the stream flows at 5 km/h. How long does 60 km downstream take?
7. Two trains 100 m and 150 m long run in the same direction at 54 km/h and 36 km/h. How long does the faster take to pass the slower?
8. In a 300 m race A beats B by 30 m. B runs at 4.5 m/s. Find A's speed.
9. A cyclist covers a distance in 3 hours at 24 km/h. At what speed must he ride to cover it in 2 hours?
10. Two men start from the same point in opposite directions at 4 km/h and 5 km/h. How far apart are they after 3 hours?
11. A train at 72 km/h crosses a 300 m platform in 30 seconds. Find the train's length.
12. A boat takes 4 hours for 24 km downstream and 6 hours for 24 km upstream. Find its still-water speed.
13. A runner runs at 10 km/h for 2 hours and then at 15 km/h for 1 hour. Find his average speed.
14. Two cars 480 km apart move towards each other at 65 km/h and 55 km/h. After how long do they meet?
15. A car's speed is increased by 25 percent. By what percent does the time for a fixed distance fall?
16. Two cyclists start together from one point of an 800 m circular track and ride in opposite directions at 20 m/s and 12 m/s. When do they first meet?
17. A train at 45 km/h crosses another train 200 m long coming the other way at 36 km/h in 20 seconds. Find the length of the first train.
18. A person travels from X to Y at 30 km/h and returns at 45 km/h, taking 5 hours in all. Find the distance XY.
19. A boat's downstream speed is 3 times its upstream speed. Find the ratio of the boat's still-water speed to the stream's speed.
20. A 240 km journey is done half at 40 km/h and half at 60 km/h. Find the total time.

ANSWER KEY

1. **65 km/h** — $260/4$.
2. **45 km/h** — $12.5 \times 18/5$.
3. **54 km/h** — $210/14 = 15$ m/s, then $\times 18/5$.
4. **5 km** — difference 10 min = $1/6$ h; $d/5 - d/6 = 1/6 \Rightarrow d/30 = 1/6$.
5. **30 km/h** — $3uvw/(uv + vw + wu) = (3 \times 36000)/(600 + 1800 + 1200) = 108000/3600$.
6. **3 hours** — downstream 20 km/h; $60/20$.
7. **50 s** — relative 18 km/h = 5 m/s; $(100 + 150)/5$.
8. **5 m/s** — B covers 270 m in $270/4.5 = 60$ s; A does 300 m in the same 60 s.
9. **36 km/h** — distance $24 \times 3 = 72$ km; $72/2$.
10. **27 km** — separating at 9 km/h for 3 h.
11. **300 m** — 72 km/h = 20 m/s; $20 \times 30 = 600$; $600 - 300$.
12. **5 km/h** — down 6, up 4; $(6 + 4)/2$.
13. **$35/3 \approx 11.67$ km/h** — 35 km in 3 hours. Not $(10 + 15)/2$.
14. **4 hours** — $480/120$.
15. **20 percent** — speed $\times 5/4$ means time $\times 4/5$, a fall of $1/5$.
16. **25 s** — relative 32 m/s; $800/32$.
17. **250 m** — relative 81 km/h = 22.5 m/s; $22.5 \times 20 = 450$ m total; $450 - 200$.
18. **90 km** — average = $2 \times 30 \times 45/75 = 36$ km/h; total distance $36 \times 5 = 180$ km; one way 90 km.
19. **2 : 1** — $b + c = 3(b - c) \Rightarrow 4c = 2b$.
20. **5 hours** — $120/40 = 3$ h and $120/60 = 2$ h.

ONE-PAGE REVISION CARD

The one relation

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

1 km/h = 5/18 m/s. 1 m/s = 18/5 km/h. 18, 36, 54, 72, 90, 108 km/h = 5, 10, 15, 20, 25, 30 m/s.

Fixed distance $\Rightarrow$ speed ratio $a : b$ means time ratio $b : a$. Speed $\times k$ means time $\times 1/k$.

Average speed = total distance / total time, always. Equal distances at u, v : $2uv/(u + v)$. Three equal distances: $3uvw/(uv + vw + wu)$. Equal times at u, v : $(u + v)/2$.

Late and early: $d = (s_1 s_2)/(s_2 - s_1) \times \Delta t$, with Δt in hours.

Relative speed: opposite $u + v$; same direction $|u - v|$.

Trains: pole L/s ; platform $(L + P)/s$; two trains $(L_1 + L_2)/\text{relative speed}$. Two crossing times, one train: subtract the equations, the length cancels.

Boats: down $= b + c$, up $= b - c$, and $b = (\text{down} + \text{up})/2$, $c = (\text{down} - \text{up})/2$.

Races: “beats by x m” fixes distances at one instant; “beats by t s” fixes times. Chain them through the middle runner, never by adding gaps.

Circular track: first meeting $= C/(u \pm v)$. Back at start together = LCM of lap times. Distinct meeting points from ratio $a : b$ in lowest terms: $a + b$ opposite, $a - b$ same.

Meet then continue: $t = \sqrt{t_1 t_2}$ and $s_1 : s_2 = \sqrt{t_2} : \sqrt{t_1}$.

Bus interval problem: overtake gap p , head-on gap $q \Rightarrow T = 2pq/(p + q)$, $v/u = (p + q)/(p - q)$.

Shuttling drone or bird: only the meeting time matters. Distance = its speed $\times$ that time.

Boat and floating object: work relative to the water. Time away = time back. The object’s drift gives the current directly.

Top five traps

1. Averaging two speeds for equal distances. Use the harmonic mean, never $(u + v)/2$.
2. Forgetting the train’s own length when it crosses a platform, bridge or another train.
3. Mixing units: metres with km/h. Convert first, every time.
4. Treating “beats by 16 m” and “beats by 4 s” as two gaps to be added. They describe the same gap.
5. Counting rest stops as distance, or leaving them out of the total time when the question says “including stoppages”.

Time & Work

Turn days into rates, then just add.

What you need to know

WHAT YOU NEED TO KNOW

The one idea: work per day If a worker finishes a job in a days, then in one day that worker does $1/a$ of the job.

$$\text{work done} = \text{rate} \times \text{time}$$

Combining workers (add the rates, never the days)

$$\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} \Rightarrow \text{together they take } \frac{ab}{a+b} \text{ days}$$

Three workers: $\frac{abc}{ab+bc+ca}$ days.

The LCM method (faster in an exam) Let total work = LCM of the given days. Then each worker's daily output is a whole number. Example: $a = 12$, $b = 18$. Total = 36 units. A does 3 units/day, B does 2 units/day.

Efficiency Efficiency is proportional to $1/\text{time}$. Times in ratio $a : b \Rightarrow$ efficiencies in ratio $b : a$. "A is p percent more efficient than B" $\Rightarrow$ A's time = B's time $\div (1 + p/100)$.

The work equation

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

M men, D days, H hours per day, W amount of work.

Wages Pay is split in the ratio of **work actually done**, which for equal working time is the ratio of efficiencies.

Pipes and cisterns Filling pipe: rate is positive. Emptying pipe or leak: rate is negative.

$$\text{net rate} = \sum \text{filling rates} - \sum \text{emptying rates}$$

Tank fills in $1 \div (\text{net rate})$. A negative net rate means the tank never fills.

Leak from two times Pipe alone fills in a h; with a leak it takes b h ($b > a$):

$$\text{leak rate} = \frac{1}{a} - \frac{1}{b} \Rightarrow \text{leak alone empties in } \frac{ab}{b-a} \text{ hours}$$

Alternate days Work out the amount finished in one full 2-day block, see how many whole blocks fit, then finish the remainder day by day. The last day is usually a fraction.

Three pair-totals If the pairs take p , q and r days (A and B in p , B and C in q , C and A in r):

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 2 \times (\text{one day's work of } A + B + C)$$

So the rate of all three together is half that sum, and each single rate is one subtraction away.

Warm-up

WARM-UPbuild the reflex

Example 1

A can finish a job in 12 days. How much does he finish in one day?

SOLUTION

One day's work = $\frac{1}{12}$ of the job.

Answer: 1/12 of the job**Example 2**

A does a job in 10 days and B in 15 days. How long do they take together?

SOLUTION

Together in one day: $\frac{1}{10} + \frac{1}{15}$. LCM of 10 and 15 is 30.

$$\frac{3}{30} + \frac{2}{30} = \frac{5}{30} = \frac{1}{6}$$

So they finish in 6 days.

Answer: 6 days**Example 3**

A takes 20 days and B takes 30 days. Find the time together.

SOLUTION

$$\frac{1}{20} + \frac{1}{30} = \frac{3}{60} + \frac{2}{60} = \frac{5}{60} = \frac{1}{12}$$

Time together = 12 days.

Answer: 12 days**Example 4**

A completes $\frac{1}{5}$ of a job in 4 days. How long does the whole job take him?

SOLUTION

$\frac{1}{5}$ of the job takes 4 days, so the whole job takes $4 \times 5 = 20$ days.

Answer: 20 days**Example 5**

Pipe P fills a tank in 8 hours and pipe Q fills it in 12 hours. How long do both together take?

SOLUTION

$$\frac{1}{8} + \frac{1}{12} = \frac{3}{24} + \frac{2}{24} = \frac{5}{24}$$

Time = $\frac{24}{5} = 4.8$ hours.

Answer: 4.8 hours**Example 6**

A tap fills a tank in 6 hours. A leak can empty the full tank in 18 hours. With both working, how long does the tank take to fill?

SOLUTION

Net rate = $\frac{1}{6} - \frac{1}{18} = \frac{3}{18} - \frac{1}{18} = \frac{2}{18} = \frac{1}{9}$ per hour.
Time = 9 hours.

Answer: 9 hours

Example 7

A is twice as efficient as B. B alone takes 18 days. How long does A take?

SOLUTION

Twice the efficiency means half the time: $\frac{18}{2} = 9$ days.

Answer: 9 days

Example 8

Ten men build a wall in 12 days. How long would one man take?

SOLUTION

Total work = $10 \times 12 = 120$ man-days.

One man needs $\frac{120}{1} = 120$ days.

Answer: 120 days

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

A and B together finish a job in 12 days. A alone takes 20 days. How long does B alone take?

SOLUTION

B's one-day work = (together) – (A alone).

$$\frac{1}{12} - \frac{1}{20}$$

LCM of 12 and 20 is 60.

$$\frac{5}{60} - \frac{3}{60} = \frac{2}{60} = \frac{1}{30}$$

So B alone takes 30 days.

Answer: 30 days

TRAP

Never write “together they take 12, A takes 20, so B takes $20 - 12 = 8$ days”. Days do not subtract. **Rates** subtract. Convert to 1/days first, every single time.

Example 10 [Infosys pattern]

A can do a job in 12 days and B in 18 days. They work together for 4 days, then A leaves and B finishes the rest. Find the total number of days taken.

SOLUTION

Use the LCM method. LCM of 12 and 18 is 36, so let the job be 36 units.

A does $\frac{36}{12} = 3$ units per day.

B does $\frac{36}{18} = 2$ units per day.

Together they do $3 + 2 = 5$ units per day.

In 4 days together: $5 \times 4 = 20$ units.

Remaining work = $36 - 20 = 16$ units.

B alone at 2 units per day needs $\frac{16}{2} = 8$ days.

Total = $4 + 8 = 12$ days.

Answer: 12 days

SHORTCUT

The LCM method kills fractions. Set total work = LCM of the given day counts, and every rate becomes a whole number. You will finish these questions in half the time.

Example 11 [Wipro pattern]

A, B and C can do a job alone in 10, 12 and 15 days. How long do all three together take?

SOLUTION

LCM of 10, 12 and 15 is 60. Let the job be 60 units.

A: $\frac{60}{10} = 6$ units per day.

B: $\frac{60}{12} = 5$ units per day.

C: $\frac{60}{15} = 4$ units per day.

Together: $6 + 5 + 4 = 15$ units per day.

$$\text{time} = \frac{60}{15} = 4 \text{ days}$$

Answer: 4 days

Example 12 [Accenture pattern]

A and B together take 12 days, B and C together take 15 days, and A and C together take 20 days. How long would each take alone, and how long would all three take together?

SOLUTION

Add the three one-day works:

$$\frac{1}{12} + \frac{1}{15} + \frac{1}{20}$$

LCM of 12, 15 and 20 is 60.

$$\frac{5}{60} + \frac{4}{60} + \frac{3}{60} = \frac{12}{60} = \frac{1}{5}$$

But each of A, B and C has been counted twice in that sum, so

$$2 \times (A + B + C) = \frac{1}{5} \Rightarrow A + B + C = \frac{1}{10}$$

All three together: **10 days.**

A alone = $(A + B + C) - (B + C) = \frac{1}{10} - \frac{1}{15} = \frac{3}{30} - \frac{2}{30} = \frac{1}{30} \Rightarrow 30$ days.

B alone = $(A + B + C) - (A + C) = \frac{1}{10} - \frac{1}{20} = \frac{2}{20} - \frac{1}{20} = \frac{1}{20} \Rightarrow 20$ days.

C alone = $(A + B + C) - (A + B) = \frac{1}{10} - \frac{1}{12} = \frac{6}{60} - \frac{5}{60} = \frac{1}{60} \Rightarrow 60$ days.

Check: $\frac{1}{30} + \frac{1}{20} = \frac{2}{60} + \frac{3}{60} = \frac{5}{60} = \frac{1}{12}$. Matches the given A and B figure.

Answer: All three 10 days; A 30, B 20, C 60 days

SHORTCUT

Given the three pair-totals, **add them and halve**. That single move gives the rate of all three together, and every individual rate is then one subtraction away.

Example 13 [Capgemini pattern]

A is 50 percent more efficient than B. A alone finishes a job in 12 days. How long do A and B together take?

SOLUTION

A's efficiency = $1.5 \times$ B's efficiency, so A's time = B's time $\div 1.5$.

$$12 = \frac{\text{B's time}}{1.5} \Rightarrow \text{B's time} = 12 \times 1.5 = 18 \text{ days}$$

Together:

$$\frac{1}{12} + \frac{1}{18} = \frac{3}{36} + \frac{2}{36} = \frac{5}{36}$$
$$\text{time} = \frac{36}{5} = 7.2 \text{ days}$$

Answer: 7.2 days

Example 14 [Cognizant pattern]

A can do a job in 12 days and B in 15 days. They work together and are paid Rs. 5,400 in all. How should the money be split?

SOLUTION

Pay follows work done, and both work the same number of days, so pay follows the rates.

Rate ratio = $\frac{1}{12} : \frac{1}{15}$. Multiply both by 60:

$$\frac{60}{12} : \frac{60}{15} = 5 : 4$$

Total parts = $5 + 4 = 9$.

One part = $\frac{5400}{9} = 600$.

A gets $5 \times 600 = \text{Rs. } 3,000$.

B gets $4 \times 600 = \text{Rs. } 2,400$.

Check: $3000 + 2400 = 5400$. Correct.

Answer: A: Rs. 3,000, B: Rs. 2,400

TRAP

Do not split wages as $12 : 15$. The faster worker must earn **more**, not less. Wages go in the ratio of one-day work, which is the **inverse** of the day counts.

Example 15 [TCS NQT pattern]

Twelve workers, working 8 hours a day, build a wall in 15 days. How many days would 18 workers need, working 10 hours a day?

SOLUTION

Total work in worker-hours:

$$12 \times 8 \times 15 = 96 \times 15 = 1440 \text{ worker-hours}$$

New team does $18 \times 10 = 180$ worker-hours per day.

$$\text{days} = \frac{1440}{180} = 8$$

Answer: 8 days

Example 16 [Wipro pattern]

A pipe fills a tank in 10 hours. Because of a leak at the bottom it actually takes 15 hours. How long would the leak alone take to empty the full tank?

SOLUTION

Let the leak empty the full tank in L hours.

$$\begin{aligned}\frac{1}{10} - \frac{1}{L} &= \frac{1}{15} \\ \frac{1}{L} &= \frac{1}{10} - \frac{1}{15} = \frac{3}{30} - \frac{2}{30} = \frac{1}{30} \\ L &= 30 \text{ hours}\end{aligned}$$

Answer: 30 hours

Example 17 [Accenture pattern]

Pipes A and B fill a tank in 12 and 16 hours. Outlet pipe C empties the full tank in 24 hours. If all three are open, how long does the tank take to fill?

SOLUTION

LCM of 12, 16 and 24 is 48. Let the tank be 48 units.

A: $\frac{48}{12} = 4$ units per hour (in).

B: $\frac{48}{16} = 3$ units per hour (in).

C: $\frac{48}{24} = 2$ units per hour (out).

Net = $4 + 3 - 2 = 5$ units per hour.

$$\text{time} = \frac{48}{5} = 9.6 \text{ hours}$$

Answer: 9.6 hours

TRAP

The outlet pipe gets a **minus** sign. Adding all three rates gives $48/9 \approx 5.3$ hours, a classic wrong option planted in these papers.

Example 18 [Infosys pattern]

A can do a job in 9 days and B in 18 days. They work on alternate days with A starting. In how many days is the job finished?

SOLUTION

LCM of 9 and 18 is 18. Let the job be 18 units.

A does $\frac{18}{9} = 2$ units per day. B does $\frac{18}{18} = 1$ unit per day.

One 2-day block (A then B) = $2 + 1 = 3$ units.

Five blocks = 10 days and $5 \times 3 = 15$ units.

Remaining = $18 - 15 = 3$ units.

Day 11 is A's turn: he does 2 units, total 17 units, 1 unit left.

Day 12 is B's turn: he does exactly 1 unit, finishing the job.

Total = 12 days.

Answer: 12 days

Example 19 [Capgemini pattern]

Eighteen men can finish a job in 20 days. After 5 days, 3 men leave. How many more days does the job take?

SOLUTION

Total work = $18 \times 20 = 360$ man-days.

Done in 5 days = $18 \times 5 = 90$ man-days.

Remaining = $360 - 90 = 270$ man-days.

Men left = $18 - 3 = 15$.

$$\text{days needed} = \frac{270}{15} = 18 \text{ days}$$

Total elapsed = $5 + 18 = 23$ days.

Answer: 18 more days (23 days in all)

Example 20 [TCS NQT pattern]

A contractor agrees to finish a road in 40 days and puts 30 men on it. After 25 days only half the road is done. How many extra men must he hire to finish on time?

SOLUTION

Work done in 25 days = $30 \times 25 = 750$ man-days, and that is half the road.

So the whole road = $2 \times 750 = 1500$ man-days.

Remaining work = 750 man-days.

Days left = $40 - 25 = 15$.

$$\text{men needed} = \frac{750}{15} = 50$$

He already has 30, so he must hire $50 - 30 = 20$ more.

Answer: 20 extra men

PRACTICE · TIER 1 target: 60 s/Q

1. A does a job in 16 days and B in 24 days. How long together?
2. A and B together take 8 days. A alone takes 12 days. Find B's time alone.
3. A, B and C alone take 12, 15 and 20 days. Find the time for all three together.
4. Fifteen men complete a job in 20 days. How long would 25 men take?
5. A can do $\frac{1}{4}$ of a job in 5 days. How long does the whole job take him?
6. A is three times as efficient as B. Together they take 9 days. Find each one's time alone.
7. Pipe A fills a tank in 20 minutes and pipe B in 30 minutes. How long do both take together?
8. A tap fills a tank in 12 hours. A leak can empty the full tank in 20 hours. With both open, how long does the tank take to fill?
9. A and B finish a job in 10 days and are paid Rs. 9,000. A alone would take 15 days. Find each one's share.
10. Eight men working 9 hours a day finish a job in 20 days. How many days will 12 men working 6 hours a day take?
11. A alone takes 18 days and B alone takes 9 days. They work together for 3 days. What fraction of the job is left?
12. A takes 24 days and B takes 36 days. They work together for 6 days, then A leaves. How many more days does B need?
13. Two pipes fill a tank in 15 and 20 minutes. Both are opened; after 5 minutes the first is closed. Find the total filling time.

14. Twenty workers build a wall in 30 days. How many workers are needed to build it in 12 days?
 15. A is 20 percent more efficient than B. B alone takes 30 days. Find A's time alone.

ANSWER KEY

1. **9.6 days** — $1/16 + 1/24 = 5/48$, so $48/5$.
2. **24 days** — $1/8 - 1/12 = 1/24$.
3. **5 days** — $5/60 + 4/60 + 3/60 = 12/60 = 1/5$.
4. **12 days** — $15 \times 20 = 300$ man-days; $300/25$.
5. **20 days** — 5×4 .
6. **A 12 days, B 36 days** — if B's rate is r , A's is $3r$ and $4r = 1/9$, so $r = 1/36$.
7. **12 minutes** — $1/20 + 1/30 = 5/60 = 1/12$.
8. **30 hours** — $1/12 - 1/20 = 2/60 = 1/30$.
9. **A Rs. 6,000, B Rs. 3,000** — B's rate = $1/10 - 1/15 = 1/30$, so B alone takes 30 days; ratio $1/15 : 1/30 = 2 : 1$.
10. **20 days** — $8 \times 9 \times 20 = 1440$ worker-hours; $12 \times 6 = 72$ per day.
11. **$1/2$ left** — combined rate $1/18 + 1/9 = 1/6$; in 3 days they do $1/2$.
12. **21 days** — combined $5/72$; 6 days gives $30/72 = 5/12$; remaining $7/12$ at $1/36$ per day.
13. **$40/3 \approx 13.33$ minutes** — 5 minutes together gives $35/60 = 7/12$; remaining $5/12$ at $1/20$ per minute takes $25/3$ minutes.
14. **50 workers** — $20 \times 30 = 600$ worker-days; $600/12$.
15. **25 days** — $30 \div 1.2$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A support team of 6 agents clears 480 tickets in 4 hours. All agents work at the same rate. How many agents are needed to clear 900 tickets in 5 hours?

SOLUTION

Rate per agent per hour:

$$\frac{480}{6 \times 4} = \frac{480}{24} = 20 \text{ tickets}$$

Required throughput:

$$\frac{900}{5} = 180 \text{ tickets per hour}$$

Agents needed:

$$\frac{180}{20} = 9$$

Answer: 9 agents

Example 22 [Shopee · pattern]

A team of 4 senior packers can pack one shipment in 9 hours. A team of 6 junior packers can pack the same shipment in 18 hours. How long will 2 seniors and 3 juniors take, working together?

SOLUTION

Take the shipment as 1 unit of work.

4 seniors for 9 hours do the whole shipment:

$$4s \times 9 = 1 \Rightarrow s = \frac{1}{36} \text{ shipment per hour per senior}$$

6 juniors for 18 hours do the whole shipment:

$$6j \times 18 = 1 \Rightarrow j = \frac{1}{108} \text{ shipment per hour per junior}$$

Mixed team rate:

$$2s + 3j = \frac{2}{36} + \frac{3}{108} = \frac{6}{108} + \frac{3}{108} = \frac{9}{108} = \frac{1}{12}$$

Time = 12 hours.

Answer: 12 hours

Example 23 [DBS · pattern]

A migration project is estimated at 1,440 person-hours. A team of 8 people works 6 hours a day. After 10 days, 2 more people join the team on the same 6-hour day. How many days does the project take in all?

SOLUTION

Team output before the joiners: $8 \times 6 = 48$ person-hours per day.

Done in 10 days: $48 \times 10 = 480$ person-hours.

Remaining: $1440 - 480 = 960$ person-hours.

New team output: $10 \times 6 = 60$ person-hours per day.

$$\text{days needed} = \frac{960}{60} = 16$$

Total = $10 + 16 = 26$ days.

Answer: 26 days

Example 24 [GIC · pattern]

A 760 litre tank is empty. Inlet A is opened at the start and delivers 40 litres per minute. Inlet B, at 30 litres per minute, is opened 4 minutes later. A drain valve removing 25 litres per minute is opened 6 minutes after B. When is the tank full?

SOLUTION

Minute 0 to 4 — only A.

$$40 \times 4 = 160 \text{ litres}$$

Minute 4 to 10 — A and B, that is $40 + 30 = 70$ litres per minute for 6 minutes.

$$70 \times 6 = 420 \text{ litres}$$

Running total = $160 + 420 = 580$ litres.

From minute 10 — A and B in, drain out: $40 + 30 - 25 = 45$ litres per minute.

Still needed = $760 - 580 = 180$ litres.

$$\frac{180}{45} = 4 \text{ minutes}$$

Tank is full at minute $10 + 4 = 14$.

Answer: 14 minutes after A was opened

Example 25 [Agoda · pattern]

A and B can complete a data migration alone in 20 and 30 days. They start together, but A is pulled onto another project 5 days before the migration is finished. How many days does the whole migration take?

SOLUTION

Let the total time be T days.

B works for all T days. A works for $T - 5$ days.

$$\frac{T-5}{20} + \frac{T}{30} = 1$$

Multiply everything by 60:

$$3(T-5) + 2T = 60$$

$$3T - 15 + 2T = 60$$

$$5T = 75 \Rightarrow T = 15 \text{ days}$$

Check: A works 10 days $\Rightarrow 10/20 = 1/2$. B works 15 days $\Rightarrow 15/30 = 1/2$. Sum = 1. Correct.

Answer: 15 days

TRAP

“A leaves 5 days before the end” does **not** mean A worked 5 days. It means A worked $T - 5$ days, where T is the answer you are still looking for. Set up the equation in T and let the algebra find it.

Example 26 [SCB · pattern]

A, B and C together complete an audit in 10 days and are paid S\$18,000. A alone would take 24 days and B alone 30 days. Find each person's share.

SOLUTION

C's daily rate:

$$\frac{1}{10} - \frac{1}{24} - \frac{1}{30}$$

LCM of 10, 24 and 30 is 120.

$$\frac{12}{120} - \frac{5}{120} - \frac{4}{120} = \frac{3}{120} = \frac{1}{40}$$

So C alone would take 40 days.

All three work the same 10 days, so pay goes in the ratio of rates, that is

$$5 : 4 : 3 \quad (\text{the numerators over 120})$$

Total parts = 12, so one part = $\frac{18000}{12} = 1500$.

A: $5 \times 1500 = \text{S\$}7,500$.

B: $4 \times 1500 = \text{S\$}6,000$.

C: $3 \times 1500 = \text{S\$}4,500$.

Check: $7500 + 6000 + 4500 = 18000$. Correct.

Answer: A S\$7,500, B S\$6,000, C S\$4,500

SHORTCUT

When you put all the rates over a **common** denominator, the numerators are already the wage ratio. No extra step is needed.

Example 27 [LINE MAN · pattern]

A fleet of 15 riders was scheduled to deliver 1,800 orders in 6 days, every rider working at the same rate. At the end of day 3 the order book grew to 2,400 orders and 3 riders resigned. How many extra riders must be hired from day 4 so that all 2,400 orders still go out by the end of day 6?

SOLUTION

Rate per rider per day:

$$\frac{1800}{15 \times 6} = \frac{1800}{90} = 20 \text{ orders}$$

Delivered in the first 3 days:

$$15 \times 3 \times 20 = 900 \text{ orders}$$

Still to deliver: $2400 - 900 = 1500$ orders in 3 days.

$$\text{orders per day needed} = \frac{1500}{3} = 500$$

$$\text{riders needed} = \frac{500}{20} = 25$$

Riders in hand from day 4 = $15 - 3 = 12$.

Extra riders = $25 - 12 = 13$.

Answer: 13 extra riders

Example 28 [Razer · pattern]

Pipes P and Q can fill a tank in 6 hours and 8 hours. Both are opened together, but a leak at the bottom means the tank fills in 4 hours. How long would the leak alone take to empty the full tank?

SOLUTION

Rate of P and Q together:

$$\frac{1}{6} + \frac{1}{8} = \frac{4}{24} + \frac{3}{24} = \frac{7}{24}$$

With the leak, the observed rate is $\frac{1}{4} = \frac{6}{24}$.

$$\text{leak rate} = \frac{7}{24} - \frac{6}{24} = \frac{1}{24}$$

The leak alone empties the full tank in 24 hours.

Answer: 24 hours

Example 29 [Sea Group · pattern]

A build task needs 24 working days. The team works 6 days a week with Sunday off, and starts on a Monday. On which calendar day of the project is the task finished?

SOLUTION

Each calendar week gives 6 working days.

$24 \div 6 = 4$ full weeks of working days, with nothing left over.

Week 1: working days 1 to 6 fall on calendar days 1 to 6 (Mon to Sat). Calendar day 7 is Sunday, no work.

Week 2: working days 7 to 12 on calendar days 8 to 13.

Week 3: working days 13 to 18 on calendar days 15 to 20.

Week 4: working days 19 to 24 on calendar days 22 to 27.

So working day 24 is calendar day 27, a Saturday.

Answer: Calendar day 27 (a Saturday)

TRAP

Do not answer 28 calendar days. Four weeks contain four Sundays, but the last Sunday comes **after** the job is already done, so it is not counted.

Example 30 [Grab · pattern]

Engineer A finishes $\frac{3}{5}$ of a task in 9 days working alone. Engineer B then joins, and the two of them finish the remaining part in 4 days. How long would B alone have taken for the whole task?

SOLUTION

A's daily rate:

$$\frac{3/5}{9} = \frac{3}{45} = \frac{1}{15}$$

Remaining work = $1 - \frac{3}{5} = \frac{2}{5}$, done by both in 4 days, so the combined rate is

$$\frac{2/5}{4} = \frac{2}{20} = \frac{1}{10}$$

B's rate:

$$\frac{1}{10} - \frac{1}{15} = \frac{3}{30} - \frac{2}{30} = \frac{1}{30}$$

B alone would take 30 days.

Answer: 30 days

PRACTICE · TIER 2 target: 100 s/Q

- Five support agents clear 600 tickets in 8 hours. How many agents are needed to clear 1,080 tickets in 6 hours?
- A and B alone take 12 and 24 days. They start together, but A leaves 3 days before the job is finished. How many days does the job take in all?
- Pipe A fills a tank in 6 hours, pipe B in 12 hours and pipe C in 4 hours. A and B are opened first; C joins 2 hours later. Find the total filling time.
- A full 1,800 litre tank has a drain that empties 60 litres per minute and an inlet that fills 45 litres per minute. Both are opened. How long until the tank is empty?
- An order of 2,400 units must be produced. Ten machines each make 8 units per hour and run 6 hours a day. How many days does the order take?
- Four men and 6 women finish a task in 8 days. Three men and 7 women finish the same task in 10 days. How long would 10 women take?
- A project is estimated at 960 person-days and a team of 16 starts work. After 20 days the client doubles the scope. How many extra people are needed so the project still ends 60 days after the start?
- Pipes P and Q fill a tank in 20 and 30 minutes. Both are opened, but Q is shut 5 minutes before the tank is full. Find the total filling time.
- A and B together take 10 days, B and C take 12 days, A and C take 15 days. How long would A alone take?
- A contractor hires 40 workers to finish a job in 50 days. After 30 days only 40 percent of the job is done. How many extra workers must he hire to finish on time?
- A tap fills a tank in 6 hours. When the tank is exactly half full, three more taps of the same rate are opened. Find the total filling time.
- A, B and C together finish a job in 8 days and are paid THB 14,400. A alone would take 24 days and B alone 32 days. Find each share.

ANSWER KEY

- 12 agents** — per agent per hour = $600/40 = 15$; need $1080/6 = 180$ per hour.
- 10 days** — $(T - 3)/12 + T/24 = 1$ gives $2T - 6 + T = 24$, so $T = 10$. Check: A 7 days = $7/12$, B 10 days = $5/12$.
- 3 hours** — A and B give $1/6 + 1/12 = 1/4$ per hour, so 2 hours gives half the tank. With C the rate is $1/6 + 1/12 + 1/4 = 1/2$ per hour, so the other half takes 1 hour.

4. **2 hours** — net outflow $60 - 45 = 15$ litres per minute; $1800/15 = 120$ minutes.
5. **5 days** — output per day $= 10 \times 8 \times 6 = 480$ units; $2400/480$.
6. **40 days** — $4m + 6w = 1/8$ and $3m + 7w = 1/10$. Multiply by 3 and 4: $12m + 18w = 3/8$ and $12m + 28w = 2/5$. Subtract: $10w = 1/40$, so 10 women take 40 days.
7. **24 extra people** — done $= 16 \times 20 = 320$ person-days; new total 1,920; remaining 1,600 in 40 days needs 40 people.
8. **14 minutes** — $T/20 + (T - 5)/30 = 1$ gives $3T + 2T - 10 = 60$, so $T = 14$.
9. **24 days** — the three pair-rates sum to $1/10 + 1/12 + 1/15 = 6/60 + 5/60 + 4/60 = 1/4$, so all three together have rate $1/8$; $A = 1/8 - 1/12 = 3/24 - 2/24 = 1/24$.
10. **50 extra workers** — $40 \times 30 = 1200$ person-days is 40 percent, so the job is 3,000 person-days; remaining 1,800 in 20 days needs 90 workers.
11. **3 hours 45 minutes** — first half takes 3 hours; four taps give $4/6 = 2/3$ of a tank per hour, so the second half takes $(1/2) \div (2/3) = 3/4$ hour.
12. **A THB 4,800, B THB 3,600, C THB 6,000** — over 96 the rates are 12, 4, 3 for all three, A and B, so C is $12 - 4 - 3 = 5$; ratio 4 : 3 : 5 and one part is 1,200.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Google · pattern]

A can finish a job in 6 days and B in 9 days. They work on alternate days. Show that it matters who starts, and find the total time in each case.

SOLUTION

Insight: full 2-day blocks are identical whoever starts. Only the ragged end differs, and the worker who starts gets one extra turn in that ragged end.

LCM of 6 and 9 is 18, so let the job be 18 units.

A does $\frac{18}{6} = 3$ units per day. B does $\frac{18}{9} = 2$ units per day.

One 2-day block $= 3 + 2 = 5$ units.

Case 1: A starts.

Three blocks $= 6$ days and $3 \times 5 = 15$ units. Remaining $= 3$ units.

Day 7 is A's turn and A does exactly 3 units. Job done.

$$\text{total} = 7 \text{ days}$$

Case 2: B starts.

Three blocks $= 6$ days and 15 units, exactly as before. Remaining $= 3$ units.

Day 7 is B's turn: B does 2 units, leaving 1 unit.

Day 8 is A's turn: A works at 3 units per day and needs only 1 unit, taking $\frac{1}{3}$ of a day.

$$\text{total} = 7 + \frac{1}{3} = \frac{22}{3} \approx 7.33 \text{ days}$$

So starting with the faster worker saves $1/3$ of a day.

Answer: A first: 7 days. B first: $22/3 \approx 7.33$ days

SHORTCUT

For alternate-day work, always put the **faster** worker first when you are asked for the minimum time. The blocks are identical; the faster worker simply eats more of the leftover.

Example 32 [Goldman Sachs · pattern]

A ticket is assigned at random, with equal probability, to one of three engineers. Working alone they would need 6, 9 and 18 days. (a) What is the expected number of days to close the ticket? (b) If instead all three worked on it together, how long would it take? (c) Explain the gap between the two answers.

SOLUTION

Insight: expectation averages the times, but teamwork averages the rates. Averaging rates and then inverting gives a much smaller number.

(a) Each engineer is equally likely, so each has probability $\frac{1}{3}$.

$$E = \frac{1}{3} \times 6 + \frac{1}{3} \times 9 + \frac{1}{3} \times 18 = \frac{6 + 9 + 18}{3} = \frac{33}{3} = 11 \text{ days}$$

(b) Add the rates. LCM of 6, 9 and 18 is 18.

$$\frac{1}{6} + \frac{1}{9} + \frac{1}{18} = \frac{3}{18} + \frac{2}{18} + \frac{1}{18} = \frac{6}{18} = \frac{1}{3}$$

time = 3 days

(c) Part (a) is the arithmetic mean of the three times, 11 days.

Part (b) is one third of the harmonic mean of the three times:

$$\text{harmonic mean} = \frac{3}{1/6 + 1/9 + 1/18} = \frac{3}{1/3} = 9 \text{ days}$$

and three workers at that average rate finish in $9/3 = 3$ days.

The harmonic mean is never larger than the arithmetic mean, and the slow engineer drags the arithmetic mean up while barely affecting the combined rate. That is the whole gap.

Answer: (a) 11 days (b) 3 days

Example 33 [D. E. Shaw · pattern]

Thirty workers can dig a canal in 10 days. To cut costs the contractor sends one worker home at the end of every day, starting with the end of day 1. In how many days is the canal finished?

SOLUTION

Insight: daily output is no longer constant, it is an arithmetic progression. Sum the progression and compare with the total work.

Total work = $30 \times 10 = 300$ worker-days.

Workers present on day $i = 30 - (i - 1) = 31 - i$.

Day 1: 30, day 2: 29, day 3: 28, and so on.

Work finished after k days:

$$S_k = \sum_{i=1}^k (31 - i) = 31k - \frac{k(k+1)}{2}$$

Try $k = 12$:

$$S_{12} = 31 \times 12 - \frac{12 \times 13}{2} = 372 - 78 = 294$$

That is short of 300, so the canal is not done on day 12.

Try $k = 13$:

$$S_{13} = 31 \times 13 - \frac{13 \times 14}{2} = 403 - 91 = 312$$

That overshoots 300, so the canal is finished during day 13.

Work left at the start of day 13 = $300 - 294 = 6$ worker-days.

Workers on day 13 = $31 - 13 = 18$.

$$\text{fraction of day 13 needed} = \frac{6}{18} = \frac{1}{3}$$

$$\text{Total} = 12 + \frac{1}{3} = \frac{37}{3} \approx 12.33 \text{ days.}$$

Answer: 12 $\frac{1}{3}$ days — it finishes one third of the way into day 13

TRAP

A shrinking crew does not always finish. The most a crew of n shrinking by one per day can ever deliver is $n(n+1)/2$ worker-days. Check that this exceeds the total work before you start summing.

Example 34 [Microsoft · pattern]

Pipe A alone fills a tank in 12 minutes and pipe B alone in 18 minutes. Both are opened at the start, but A is shut off some minutes before the tank is full. The tank is full exactly 12 minutes after the start. For how many minutes before the end was A shut off?

SOLUTION

Insight: B runs the whole time, so compute B's contribution first. Whatever is left is A's, and A's running time follows at once.

B runs for the full 12 minutes:

$$12 \times \frac{1}{18} = \frac{12}{18} = \frac{2}{3} \text{ of the tank}$$

So A must supply

$$1 - \frac{2}{3} = \frac{1}{3} \text{ of the tank}$$

A fills at $\frac{1}{12}$ of the tank per minute, so A's running time is

$$(1/3) \div (1/12) = \frac{1}{3} \times 12 = 4 \text{ minutes}$$

A ran for the first 4 minutes of the 12, so it was shut off

$$12 - 4 = 8 \text{ minutes before the end}$$

Check: $4 \times 1/12 = 1/3$ and $12 \times 1/18 = 2/3$, and $1/3 + 2/3 = 1$. Correct.

Answer: 8 minutes before the end

Example 35 [Uber · pattern]

Six men or 9 women can finish a survey in 20 days. (a) How long will 4 men and 12 women take? (b) If the team in (a) is paid Rs. 90,000 in total, and pay follows work done, what does each man and each woman receive?

SOLUTION

Insight: convert everything into one currency of labour. Here, express every woman as a fraction of a man.

(a) Six men and 9 women do the same job in the same time, so

$$6 \text{ men} = 9 \text{ women} \Rightarrow 1 \text{ man} = \frac{9}{6} = \frac{3}{2} \text{ women} \Rightarrow 1 \text{ woman} = \frac{2}{3} \text{ man}$$

Total work = 6 men $\times$ 20 days = 120 man-days.

The new team, in man-equivalents:

$$4 + 12 \times \frac{2}{3} = 4 + 8 = 12 \text{ men}$$

$$\text{days} = \frac{120}{12} = 10 \text{ days}$$

(b) Over those 10 days:

Men contribute $4 \times 10 = 40$ man-days.

Women contribute $8 \times 10 = 80$ man-days (using their man-equivalent of 8).

Total = 120 man-days, as expected.

Men's share = $\frac{40}{120} \times 90000 = \frac{1}{3} \times 90000 = \text{Rs. } 30,000$, split between 4 men:

$$\frac{30000}{4} = 7500$$

so each man gets Rs. 7,500.

Women's share = $\frac{80}{120} \times 90000 = \frac{2}{3} \times 90000 = \text{Rs. } 60,000$, split between 12 women:

$$\frac{60000}{12} = 5000$$

so each woman gets Rs. 5,000.

Check: $30000 + 60000 = 90000$, and a man earns $7500/5000 = 3/2$ times a woman, matching their efficiency ratio.

Answer: (a) 10 days (b) Rs. 7,500 per man, Rs. 5,000 per woman

Example 36 [Adobe · pattern]

Two men and 3 boys finish a job in 10 days. Three men and 2 boys finish the same job in 8 days. In how many days will 1 man and 1 boy finish it?

SOLUTION

Insight: the question asks for $m + b$, not for m and b separately. Adding the two equations gives $5m + 5b$ in one step, so never solve the system.

Let m and b be the daily work of one man and one boy.

$$10(2m + 3b) = 1 \Rightarrow 2m + 3b = \frac{1}{10}$$

$$8(3m + 2b) = 1 \Rightarrow 3m + 2b = \frac{1}{8}$$

Add them:

$$5m + 5b = \frac{1}{10} + \frac{1}{8} = \frac{4}{40} + \frac{5}{40} = \frac{9}{40}$$

$$m + b = \frac{9}{200}$$

$$\text{days} = \frac{200}{9} \approx 22.22 \text{ days}$$

Verification by solving fully. From $2m + 3b = 0.1$ and $3m + 2b = 0.125$:

multiply the first by 3 and the second by 2 to get $6m + 9b = 0.3$ and $6m + 4b = 0.25$;

subtract: $5b = 0.05$, so $b = 0.01$; then $2m = 0.1 - 0.03 = 0.07$, so $m = 0.035$.

$m + b = 0.045 = 9/200$. Same answer.

Answer: $200/9 \approx 22.22$ days

Example 37 [Amazon · pattern]

A can finish a job in 12 days and B in 20 days. A works alone for the first k days and then hands over to B, who finishes it. The whole job must be delivered within 16 days. Find the smallest whole number of days k that A must work.

SOLUTION

Insight: write the total elapsed time as a single linear function of k . Every extra day A works removes more than one day from the schedule, because A is faster.

After k days A has done $\frac{k}{12}$ of the job.

B must do the rest, $1 - \frac{k}{12}$, at $\frac{1}{20}$ per day, so B works

$$20\left(1 - \frac{k}{12}\right) = 20 - \frac{20k}{12} = 20 - \frac{5k}{3} \text{ days}$$

Total elapsed time:

$$T = k + 20 - \frac{5k}{3} = 20 - \frac{2k}{3}$$

Each day A works shortens the schedule by $\frac{2}{3}$ of a day. Require $T \leq 16$:

$$20 - \frac{2k}{3} \leq 16 \Rightarrow \frac{2k}{3} \geq 4 \Rightarrow k \geq 6$$

Smallest whole number: $k = 6$.

Check: A works 6 days and does $6/12 = 1/2$. B must do $1/2$, which takes $20 \times 1/2 = 10$ days.

Total = $6 + 10 = 16$ days, exactly on the deadline.

Answer: $k = 6$ days

Example 38 [Google · pattern]

Three taps fill a tank in 12, 15 and 30 hours. They are opened one at a time for exactly one hour each, in rotation, starting with the fastest. (a) How long does the tank take to fill? (b) Show that the answer changes if the rotation starts with the slowest instead.

SOLUTION

Insight: a full 3-hour cycle always delivers the same amount whatever the order. Order only decides how the final partial cycle is filled, and the fastest tap clears the remainder quickest.

LCM of 12, 15 and 30 is 60, so let the tank be 60 units.

Tap A: $\frac{60}{12} = 5$ units per hour.

Tap B: $\frac{60}{15} = 4$ units per hour.

Tap C: $\frac{60}{30} = 2$ units per hour.

One cycle of 3 hours = $5 + 4 + 2 = 11$ units.

(a) Order A, B, C.

Five cycles = 15 hours and $5 \times 11 = 55$ units.

Remaining = $60 - 55 = 5$ units.

Hour 16 belongs to A, who delivers exactly 5 units. Tank full.

$$\text{time} = 16 \text{ hours}$$

(b) Order C, B, A.

Five cycles = 15 hours and 55 units, exactly as before. Remaining = 5 units.

Hour 16 belongs to C: 2 units, running total 57, remaining 3.

Hour 17 belongs to B, who fills 4 units per hour but needs only 3, taking

$$\frac{3}{4} = 0.75 \text{ of an hour}$$

$$\text{time} = 16 + 0.75 = 16.75 \text{ hours}$$

The gap is 45 minutes, created entirely inside the last partial cycle.

Answer: (a) 16 hours (b) 16.75 hours with the reversed order

PRACTICE · TIER 3 target: 4 min/Q

1. A and B together finish a job in 4 days. A alone takes 6 days more than B alone. Find how long each takes alone.

2. Thirty workers can finish a job in 12 days. One worker is sent home at the end of each day, beginning with the end of day 1. In how many days is the job finished?
3. Three taps fill a tank in 10, 15 and 20 hours. They are opened one hour each in rotation, in that order. How long does the tank take to fill?
4. An engineer finishes 60 percent of a task in 15 days working alone. A second engineer then joins and together they finish the remaining part in 5 days. How long would the second engineer alone have taken for the whole task?
5. Two pipes fill a tank in 30 and 45 minutes. Both are opened together. After how many minutes must the second pipe be closed so that the tank fills in exactly 25 minutes?
6. A can do a job in 15 days and B in 25 days. A works alone for the first k days and B finishes the rest. The job must be delivered within 21 days. Find the least whole number k .
7. A task is given at random, with equal probability, to one of three engineers who would take 4, 6 and 12 days alone. Find the expected time, and compare it with the time if all three worked together.
8. Eight men or 12 boys can do a piece of work in 25 days. In how many days will 6 men and 11 boys do it?

ANSWER KEY

1. **B 6 days, A 12 days** — let B take x , so A takes $x + 6$. Then $1/x + 1/(x + 6) = 1/4$ gives $4(2x + 6) = x^2 + 6x$, so $x^2 - 2x - 24 = 0$ and $x = 6$ (the root -4 is rejected). Check: $1/6 + 1/12 = 1/4$.
2. **Exactly 16 days** — total work = 360 worker-days. Day i has $31 - i$ workers, so after k days the total is $31k - k(k + 1)/2$. At $k = 16$ this is $496 - 136 = 360$, exactly the whole job. Equivalently $30 + 29 + \dots + 15 = (30 + 15) \times 16/2 = 360$.
3. **13.5 hours** — take the tank as 60 units, so the rates are 6, 4 and 3 units per hour and one 3-hour cycle gives 13 units. Four cycles give 52 units in 12 hours. Hour 13 (first tap) adds 6, reaching 58. Hour 14 (second tap) needs only 2 of its 4 units, taking half an hour.
4. **25 days** — the first engineer's rate is $0.6/15 = 1/25$. The pair's rate is $0.4/5 = 2/25$. So the second engineer's rate is $2/25 - 1/25 = 1/25$, meaning 25 days alone. The two are equally fast.
5. **After 7.5 minutes** — the first pipe runs all 25 minutes and fills $25/30 = 5/6$. The second must supply $1/6$ at $1/45$ per minute, that is $45/6 = 7.5$ minutes.
6. **$k = 6$** — B works $25(1 - k/15) = 25 - 5k/3$ days, so the total is $T = k + 25 - 5k/3 = 25 - 2k/3$. Setting $T \leq 21$ gives $2k/3 \geq 4$, so $k \geq 6$. At $k = 6$, A does $6/15 = 2/5$ and B needs $25 \times 3/5 = 15$ days, total exactly 21.
7. **Expected $22/3 \approx 7.33$ days; together 2 days** — the expectation is $(4 + 6 + 12)/3 = 22/3$. Together the rates add: $1/4 + 1/6 + 1/12 = 3/12 + 2/12 + 1/12 = 1/2$, so 2 days. Expectation averages times, teamwork averages rates, and that is why the gap is so wide.
8. **15 days** — 8 men = 12 boys, so 1 man = 1.5 boys. Total work = $8 \times 25 = 200$ man-days = 300 boy-days. The team of 6 men and 11 boys is $9 + 11 = 20$ boys, so $300/20 = 15$ days.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 20 questions · 30 minutes

1. A does a job in 25 days and B in 20 days. How long do they take together?
2. Twelve men finish a job in 18 days. How long would 9 men take?
3. A pipe fills a tank in 9 hours, but with a leak it takes 12 hours. How long would the leak alone take to empty the full tank?
4. A is twice as good a workman as B. Together they finish in 14 days. How long would B alone take?
5. A, B and C alone take 20, 30 and 60 days. How long do all three together take?
6. Two pipes fill a tank in 10 and 15 minutes. A waste pipe empties it in 30 minutes. All three are open. How long does the tank take to fill?
7. A and B together finish a job in 12 days and are paid Rs. 12,000. A alone would take 20 days. Find B's share.
8. A can do a job in 30 days. He works 10 days and leaves. B finishes the remaining work in 20 days. How long would B alone take for the whole job?

9. Five machines produce 250 items in 5 hours. How many items will 8 machines produce in 6 hours?
10. A alone takes 12 days and B alone 24 days. They work on alternate days with A starting. How long does the job take?
11. Twenty men working 8 hours a day finish a job in 15 days. How many days will 25 men working 6 hours a day take?
12. A tank has two inlets that fill it in 12 and 18 minutes and one outlet that empties it in 36 minutes. All three are open. How long does the tank take to fill?
13. A finishes $\frac{2}{5}$ of a job in 8 days. How many more days does he need to finish the rest?
14. Four men and 4 boys finish a job in 6 days. Two men and 8 boys finish the same job in 8 days. How long will 1 man and 1 boy take?
15. A leak can empty a full tank in 15 hours. An inlet filling at 4 litres per minute is opened and the full tank now empties in 20 hours. Find the capacity of the tank.
16. A contractor plans a job for 24 days with 45 men. After 15 days only half the job is done. How many extra men are needed to finish on time?
17. Thirty workers can complete a job in 24 days. After 6 days, 10 more workers join. How many days does the job take in all?
18. A is 25 percent less efficient than B. B alone takes 18 days. How long does A take alone?
19. Two typists together type a report in 6 hours. The first alone would take 10 hours. How long would the second alone take?
20. Three pipes fill a cistern in 5, 10 and 20 hours. How long do all three together take?

ANSWER KEY

1. **100/9 $\approx$ 11.11 days** — $\frac{1}{25} + \frac{1}{20} = \frac{9}{100}$.
2. **24 days** — $12 \times 18 = 216$ man-days; $216/9$.
3. **36 hours** — $\frac{1}{9} - \frac{1}{12} = \frac{1}{36}$.
4. **42 days** — if B's rate is r then A's is $2r$ and $3r = \frac{1}{14}$, so $r = \frac{1}{42}$.
5. **10 days** — $\frac{3}{60} + \frac{2}{60} + \frac{1}{60} = \frac{6}{60} = \frac{1}{10}$.
6. **7.5 minutes** — $\frac{3}{30} + \frac{2}{30} - \frac{1}{30} = \frac{4}{30} = \frac{2}{15}$.
7. **Rs. 4,800** — B's rate = $\frac{1}{12} - \frac{1}{20} = \frac{1}{30}$; ratio A : B = $\frac{1}{20} : \frac{1}{30} = 3 : 2$, so B takes $\frac{2}{5}$ of 12,000.
8. **30 days** — A did $\frac{10}{30} = \frac{1}{3}$, so B did $\frac{2}{3}$ in 20 days, giving 30 days for the whole job.
9. **480 items** — one machine makes $250/25 = 10$ items per hour; $8 \times 6 \times 10$.
10. **16 days** — take the job as 24 units, so A does 2 and B does 1 per day; each 2-day block gives 3 units, and 8 blocks give 24 units in 16 days.
11. **16 days** — $20 \times 8 \times 15 = 2400$ worker-hours; $25 \times 6 = 150$ per day.
12. **9 minutes** — take 36 units: $3 + 2 - 1 = 4$ units per minute.
13. **12 more days** — his rate is $(\frac{2}{5})/8 = \frac{1}{20}$, so the whole job takes 20 days and 12 remain.
14. **24 days** — the first statement alone gives $4(m + b) = \frac{1}{6}$, so $m + b = \frac{1}{24}$. The second statement is consistent but not needed.
15. **14,400 litres** — the inlet's rate as a fraction is $\frac{1}{15} - \frac{1}{20} = \frac{1}{60}$ of the tank per hour; 4 litres per minute is 240 litres per hour, so the tank is 240×60 .
16. **30 extra men** — $45 \times 15 = 675$ man-days is half the job, so the job is 1,350 man-days; the remaining 675 in 9 days needs 75 men.
17. **19.5 days** — total $30 \times 24 = 720$ worker-days; 180 done in 6 days; the remaining 540 with 40 workers takes 13.5 days.
18. **24 days** — A's rate is 0.75 of B's, so A's time is $18 \div 0.75$.
19. **15 hours** — $\frac{1}{6} - \frac{1}{10} = \frac{2}{30} = \frac{1}{15}$.
20. **20/7 $\approx$ 2.86 hours** — $\frac{1}{5} + \frac{1}{10} + \frac{1}{20} = \frac{7}{20}$.

ONE-PAGE REVISION CARD

The one idea Finish in a days $\Rightarrow$ one day's work is $1/a$. Add rates, never days.

$$\text{together} = \frac{1}{a} + \frac{1}{b} \Rightarrow \text{time} = \frac{ab}{a+b}$$

LCM method: let total work = LCM of the day counts. Every rate becomes a whole number and all fractions disappear.

Efficiency: efficiency $\propto 1/\text{time}$. Times $a : b \Rightarrow$ efficiencies $b : a$. " p percent more efficient" $\Rightarrow$ time $\div (1 + p/100)$.

Work equation: $(M_1 D_1 H_1)/W_1 = (M_2 D_2 H_2)/W_2$. Man-days (or worker-hours) is the safest way to handle crews that change size.

Wages: split in the ratio of work actually done. Put all rates over one common denominator and read the numerators as the ratio.

Three pair-totals: if the pairs take p, q, r days, add $1/p + 1/q + 1/r$ and halve it to get the rate of all three; then subtract one pair rate to get the missing person.

Pipes: filling positive, emptying negative, net rate = sum. Fills in a , with a leak takes b ; leak alone empties in $ab/(b - a)$.

Alternate days: value one 2-day block, fit whole blocks, then finish day by day. Put the faster worker first to minimise the total.

Someone leaves d days before the end: that person worked $T - d$ days, where T is the unknown total. Write one equation in T .

Shrinking crew: daily output is an AP. Work after k days = $\sum$ of the AP. A crew of n losing one per day can never deliver more than $n(n+1)/2$ worker-days.

Expectation vs teamwork: picking one worker at random averages the **times**; putting them all on the job averages the **rates**. The second is always far faster.

Top five traps

1. Subtracting days instead of rates. B is not $(20 - 12)$ days; B's rate is $1/12 - 1/20$.
2. Giving the outlet pipe a plus sign, or forgetting the leak entirely.
3. Splitting wages in the ratio of the day counts. It must be the inverse ratio.
4. Reading "leaves 5 days before the end" as "worked 5 days".
5. Stopping an alternate-day or rotation problem at a whole day. The last turn is usually a fraction of a day, and the final answer is rarely an integer.

Permutations & Combinations

Count without listing. Order first, then choose.

What you need to know

WHAT YOU NEED TO KNOW

Notation. nP_r = number of **arrangements** of r things out of n (order matters). ${}^nC_r = \binom{n}{r}$ = number of **selections** of r things out of n (order does not matter).

The one question to ask first: does changing the order make a new answer? Yes $\rightarrow$ permutation. No $\rightarrow$ combination.

Counting principle.

- AND $\rightarrow$ multiply. (Do this, then that.)
- OR $\rightarrow$ add. (Do this, or instead that. Cases must not overlap.)

Factorials. $n! = n(n-1)(n-2)\cdots 2 \cdot 1$, and $0! = 1$. $1! = 1$, $2! = 2$, $3! = 6$, $4! = 24$, $5! = 120$, $6! = 720$, $7! = 5040$, $8! = 40320$, $9! = 362880$, $10! = 3628800$.

Core formulas.

$${}^nP_r = \frac{n!}{(n-r)!} = \underbrace{n(n-1)\cdots}_{r \text{ factors}} \quad {}^nC_r = \frac{n!}{r!(n-r)!} = \frac{{}^nP_r}{r!}$$

- ${}^nC_r = {}^nC_{n-r}$ ${}^nC_0 = {}^nC_n = 1$ ${}^nC_1 = n$
- ${}^nC_2 = \frac{n(n-1)}{2}$ ${}^nC_3 = \frac{n(n-1)(n-2)}{6}$
- ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$ (Pascal)
- ${}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n$ so: subsets of an n -set $= 2^n$; at least one item $= 2^n - 1$

Repetition allowed. r slots, n choices each $\rightarrow n^r$ (codes, PINs, passwords).

Alike objects in a row. n objects with p alike of one kind, q alike of another:

$$\text{arrangements} = \frac{n!}{p!q!r!\cdots}$$

Circular.

- n people round a table: $(n-1)!$
- Necklace / garland (flip allowed, clockwise = anticlockwise): $\frac{(n-1)!}{2}$

Two standard moves.

- **Together** $\rightarrow$ glue the group into one unit, then multiply by the internal arrangements.
- **No two together** $\rightarrow$ arrange the others first, then drop the special ones into the **gaps**.
- r chosen from n in a row with no two adjacent: ${}^{n-r+1}C_r$.

Identical objects into distinct groups ($x_1 + x_2 + \cdots + x_r = n$):

- each group ≥ 1 : ${}^{n-1}C_{r-1}$
- each group ≥ 0 : ${}^{n+r-1}C_{r-1}$

Distinct objects into distinct boxes.

- any number per box: r^n
- into exactly 3 boxes, none empty: $3^n - 3 \cdot 2^n + 3$

Dividing into groups. mn distinct objects into m groups of n each: labelled groups $\rightarrow \frac{(mn)!}{(n!)^m}$; unlabelled groups $\rightarrow \frac{(mn)!}{(n!)^m \cdot m!}$.

Derangements (nothing in its own place):

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots \pm \frac{1}{n!} \right)$$

$D_1 = 0, D_2 = 1, D_3 = 2, D_4 = 9, D_5 = 44, D_6 = 265, D_7 = 1854$. Exactly k items in the right place out of n : ${}^nC_k \cdot D_{n-k}$.

Geometry counts. From n points, no 3 collinear: lines = nC_2 , triangles = nC_3 . Diagonals of an n -gon = $\frac{n(n-3)}{2}$. Rectangles in a grid of $a \times b$ cells = ${}^{a+1}C_2 \times {}^{b+1}C_2$.

Grid paths. m steps east and n steps north: $\frac{(m+n)!}{m!n!}$.

Sum of all numbers formed by n distinct non-zero digits, all n used:

$$\text{Sum} = (n-1)! \times (\text{sum of digits}) \times \underbrace{111\cdots 1}_{n \text{ ones}}$$

Warm-up

WARM-UP

build the reflex

Example 1

Find $6!$, $0!$ and $\frac{5!}{3!}$.

SOLUTION

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720.$$

$$0! = 1 \text{ (by definition).}$$

$$\frac{5!}{3!} = \frac{5 \times 4 \times 3!}{3!} = 5 \times 4 = 20.$$

Answer: 720, 1, 20

Example 2

Find 8P_3 .

SOLUTION

8P_3 means 3 factors starting from 8.

$$= 8 \times 7 \times 6 = 56 \times 6 = 336.$$

Answer: 336

Example 3

Find ${}^{10}C_3$.

SOLUTION

$${}^{10}C_3 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = \frac{720}{6} = 120.$$

Answer: 120

Example 4

Find ${}^{12}C_{10}$.

SOLUTION

Use ${}^nC_r = {}^nC_{n-r}$. So ${}^{12}C_{10} = {}^{12}C_2$.

$${}^{12}C_2 = \frac{12 \times 11}{2} = \frac{132}{2} = 66.$$

Answer: 66

SHORTCUT

Never compute nC_r with r bigger than $\frac{n}{2}$. Flip it first: ${}^{20}C_{17} = {}^{20}C_3 = \frac{20 \times 19 \times 18}{6} = 1140$. Three multiplications instead of seventeen.

Example 5

From 5 different letters, how many 3-letter codes are possible (a) with no letter repeated, (b) with repetition allowed?

SOLUTION

(a) Slot 1: 5 ways. Slot 2: 4 ways left. Slot 3: 3 ways left. $5 \times 4 \times 3 = 60$.

(b) Every slot has all 5 letters available. $5 \times 5 \times 5 = 125$.

Answer: (a) 60 (b) 125

Example 6

How many arrangements of the letters of **LEVEL**?

SOLUTION

Letters: L, E, V, E, L $\rightarrow$ 5 letters. L appears 2 times, E appears 2 times.

$$\text{Arrangements} = \frac{5!}{2! \cdot 2!} = \frac{120}{2 \times 2} = \frac{120}{4} = 30.$$

Answer: 30

Example 7

In how many ways can 7 friends sit around a circular table?

SOLUTION

Circular: fix one person, arrange the rest.

$$(7 - 1)! = 6! = 720.$$

Answer: 720

Example 8

A set has 5 different elements. How many subsets does it have? How many are non-empty?

SOLUTION

Each element is either in or out: 2 choices each.

$$\text{Total subsets} = 2^5 = 32.$$

Remove the empty subset: $32 - 1 = 31$.

Answer: 32 subsets; 31 non-empty

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

How many 4-digit numbers can be formed using the digits 1, 2, 3, 4, 5, 6, 7 if no digit repeats?

SOLUTION

Order matters (1234 and 4321 are different numbers). No zero to worry about.

$${}^7P_4 = 7 \times 6 \times 5 \times 4.$$

$$7 \times 6 = 42. 42 \times 5 = 210. 210 \times 4 = 840.$$

Answer: 840

Example 10 [Infosys pattern]

How many 4-digit numbers can be formed using 0, 1, 2, 3, 4, 5 with no digit repeated?

SOLUTION

The thousands place cannot be 0.

Thousands place: 5 choices (1, 2, 3, 4, 5).

Now 5 digits are left (including 0) for the other 3 places: $5 \times 4 \times 3 = 60$.

$$\text{Total} = 5 \times 60 = 300.$$

Answer: 300

TRAP

The most common loss of marks in this chapter: forgetting that a number cannot start with 0. Here the careless answer is ${}^6P_4 = 360$. The 60 extra “numbers” all start with 0, so they are only 3-digit numbers.

Example 11 [Accenture pattern]

A committee of 3 men and 2 women must be formed from 5 men and 4 women. In how many ways?

SOLUTION

Choosing a committee: order does not matter $\rightarrow$ combinations.

$$\text{Men: } {}^5C_3 = {}^5C_2 = \frac{5 \times 4}{2} = 10.$$

$$\text{Women: } {}^4C_2 = \frac{4 \times 3}{2} = 6.$$

Men AND women $\rightarrow$ multiply: $10 \times 6 = 60$.

Answer: 60

Example 12 [Wipro pattern]

From 6 men and 5 women, a committee of 4 is formed with **at least 3 women**. In how many ways?

SOLUTION

“At least 3 women” out of 4 members means 3 women or 4 women. These cases do not overlap, so add.

$$\text{Case A — 3 women, 1 man: } {}^5C_3 \times {}^6C_1 = 10 \times 6 = 60. ({}^5C_3 = {}^5C_2 = \frac{5 \times 4}{2} = 10.)$$

$$\text{Case B — 4 women, 0 men: } {}^5C_4 \times {}^6C_0 = 5 \times 1 = 5.$$

$$\text{Total} = 60 + 5 = 65.$$

Answer: 65

TRAP

“At least 3” is **not** ${}^5C_3 \times {}^8C_1$. That kind of shortcut counts the same committee more than once. Always split “at least” into clean, separate cases and add.

Example 13 [TCS NQT pattern]

How many arrangements are there of the letters of the word **PLACEMENT**?

SOLUTION

Write the letters out: P, L, A, C, E, M, E, N, T → 9 letters.

Repeats: E appears 2 times. All others once.

$$\text{Arrangements} = \frac{9!}{2!} = \frac{362880}{2} = 181440.$$

Answer: 181440

Example 14 [Capgemini pattern]

For the word **BANANA**: (a) how many arrangements? (b) how many start with B?

SOLUTION

(a) Letters: B, A, N, A, N, A → 6 letters. A appears 3 times, N appears 2 times.

$$\frac{6!}{3! \cdot 2!} = \frac{720}{6 \times 2} = \frac{720}{12} = 60.$$

(b) Fix B in the first place. The remaining 5 letters are A, N, A, N, A: A three times, N twice.

$$\frac{5!}{3! \cdot 2!} = \frac{120}{12} = 10.$$

Answer: (a) 60 (b) 10

Example 15 [Cognizant pattern]

In how many arrangements of **COMPUTER** do all the vowels stay together?

SOLUTION

C, O, M, P, U, T, E, R → 8 distinct letters.

Vowels: O, U, E (3 of them). Consonants: C, M, P, T, R (5 of them).

Glue the 3 vowels into one block. Now count units: 5 consonants + 1 block = 6 units.

Arrange the 6 units: $6! = 720$.

Arrange the vowels inside the block: $3! = 6$.

Total = $720 \times 6 = 4320$.

Answer: 4320

SHORTCUT

Together = glue. Tie the group with a rubber band, count the units, then untie it and multiply by the arrangements inside. Two multiplications, always.

Example 16 [TCS NQT pattern]

In how many arrangements of **LEADING** do no two vowels sit next to each other?

SOLUTION

L, E, A, D, I, N, G → 7 distinct letters.

Consonants: L, D, N, G (4). Vowels: E, A, I (3).

Step 1 — seat the consonants first. $4! = 24$ ways.

Step 2 — find the gaps. With 4 consonants in a row there are 5 gaps: $_ C _ C _ C _ C _$

Step 3 — put the 3 vowels into 3 different gaps. Order matters, so ${}^5P_3 = 5 \times 4 \times 3 = 60$.

Total = $24 \times 60 = 1440$.

Answer: 1440

SHORTCUT

No two together = gap method. Seat the “ordinary” items first, count the gaps (always one more than the number of ordinary items), then place the “special” items into separate gaps. This is faster and safer than “total minus together” whenever there are 3 or more special items.

Example 17 [Accenture pattern]

At a meeting 12 people are present. (a) Each shakes hands with every other exactly once. How many handshakes? (b) Instead, each person gives a business card to every other person. How many cards are given?

SOLUTION

(a) A handshake between P and Q is the same as between Q and P $\rightarrow$ order does not matter.

$${}^{12}C_2 = \frac{12 \times 11}{2} = \frac{132}{2} = 66.$$

(b) A card from P to Q is **not** the same as a card from Q to P $\rightarrow$ order matters.

$${}^{12}P_2 = 12 \times 11 = 132.$$

Answer: (a) 66 (b) 132

TRAP

Handshakes, matches between teams, lines through points $\rightarrow$ combinations. Gifts, letters, one-way flights between cities, ordered pairs $\rightarrow$ permutations. Read the verb carefully. The two answers always differ by a factor of 2 in the pair case.

Example 18 [Infosys pattern]

How many diagonals does a 12-sided polygon have?

SOLUTION

Join any 2 of the 12 vertices: ${}^{12}C_2 = \frac{12 \times 11}{2} = 66$ line segments.

Of these, 12 are the sides of the polygon, not diagonals.

$$\text{Diagonals} = 66 - 12 = 54.$$

(Formula check: $\frac{n(n-3)}{2} = \frac{12 \times 9}{2} = \frac{108}{2} = 54$. Matches.)

Answer: 54

Example 19 [Wipro pattern]

5 boys and 3 girls stand in a row. In how many ways can they stand if all 3 girls stay together?

SOLUTION

Glue the 3 girls into one block.

$$\text{Units} = 5 \text{ boys} + 1 \text{ block} = 6 \text{ units.}$$

$$\text{Arrange units: } 6! = 720.$$

$$\text{Arrange the girls inside the block: } 3! = 6.$$

$$\text{Total} = 720 \times 6 = 4320.$$

Answer: 4320

Example 20 [TCS NQT pattern]

There are 10 points in a plane. Exactly 4 of them lie on one straight line; no other 3 points are collinear. (a) How many triangles can be formed? (b) How many straight lines?

SOLUTION

(a) Any 3 points give a triangle, **unless** all 3 are on the same line.

All choices: ${}^{10}C_3 = \frac{10 \times 9 \times 8}{6} = \frac{720}{6} = 120$.

Bad choices (all 3 from the 4 collinear points): ${}^4C_3 = 4$.

Triangles = $120 - 4 = 116$.

(b) Any 2 points give a line.

All choices: ${}^{10}C_2 = \frac{10 \times 9}{2} = 45$.

The 4 collinear points give ${}^4C_2 = 6$ pairs, but all 6 pairs give the **same** single line.

Lines = $45 - 6 + 1 = 40$.

Answer: (a) 116 (b) 40

PRACTICE · TIER 1 target: 60 s/Q

1. Find 7P_3 .
2. Find ${}^{15}C_{13}$.
3. How many 3-digit numbers with no repeated digit can be made from the digits 1 to 9?
4. How many 4-character codes can be made from the 26 English capital letters if repetition is allowed?
5. How many arrangements are there of the letters of **SUCCESS**?
6. How many arrangements are there of the letters of **MISSISSIPPI**?
7. In how many ways can 4 books be chosen from 9 different books?
8. A committee of 5 is chosen from 7 men and 4 women, with exactly 2 women. In how many ways?
9. In how many ways can 6 people be seated around a circular table?
10. At a party every person shook hands with every other person exactly once. There were 45 handshakes. How many people were there?
11. How many diagonals does a regular octagon have?
12. How many arrangements of the letters of **ORANGE** begin with a vowel?
13. 4 different Maths books and 3 different Physics books are placed on a shelf so that all the Physics books are together. In how many ways?
14. How many non-empty subsets does a set of 6 elements have?
15. With 5 different flags, how many signals can be made by hoisting one or more flags one above the other?

ANSWER KEY

1. $210 - 7 \times 6 \times 5$.
2. 105 — flip: ${}^{15}C_{13} = {}^{15}C_2 = \frac{15 \times 14}{2}$.
3. 504 — $9 \times 8 \times 7$; no zero is available so no first-digit problem.
4. 456976 — repetition allowed: 26^4 .
5. 420 — 7 letters, S three times, C twice: $\frac{5040}{6 \times 2}$.
6. 34650 — 11 letters: I four times, S four times, P twice: $\frac{39916800}{24 \times 24 \times 2} = \frac{39916800}{1152}$.
7. $126 - {}^9C_4 = \frac{9 \times 8 \times 7 \times 6}{24} = \frac{3024}{24}$.
8. $210 - {}^4C_2 \times {}^7C_3 = 6 \times 35$. “Exactly 2 women” fixes the men count at 3.
9. $120 - (6 - 1)! = 5! = 120$. Answer 720 means you forgot it is circular.
10. $10 - \frac{n(n-1)}{2} = 45 \rightarrow n(n-1) = 90 \rightarrow n = 10$.
11. $20 - \frac{8 \times 5}{2}$.
12. 360 — vowels O, A, E: 3 choices for the first letter, then $5! = 120$ for the rest.
13. 720 — glue: 5 units $\rightarrow 5! = 120$, times $3! = 6$ inside.
14. $63 - 2^6 - 1 = 64 - 1$.
15. $325 - {}^5P_1 + {}^5P_2 + {}^5P_3 + {}^5P_4 + {}^5P_5 = 5 + 20 + 60 + 120 + 120$. Order up the pole matters.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A rider app lists 7 open pickup points. A rider will visit exactly 4 of them, one after the other. (a) In how many different routes can he do this? (b) In how many of these is a particular point (the central kitchen) visited first?

SOLUTION

(a) The order of visits matters, so this is an arrangement.

$${}^7P_4 = 7 \times 6 \times 5 \times 4 = 840.$$

(b) Fix the central kitchen in position 1. Now 3 more stops must be chosen and ordered from the remaining 6 points.

$${}^6P_3 = 6 \times 5 \times 4 = 120.$$

Answer: (a) 840 (b) 120

Example 22 [Shopee · pattern]

A seller has 10 listed products. She builds a “bundle” of any 4 of them. But the phone case and the screen guard must never be in the same bundle (they clash in the promo rules). How many bundles are possible?

SOLUTION

Step 1 — all bundles, ignoring the rule. ${}^{10}C_4 = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = \frac{5040}{24} = 210.$

Step 2 — bad bundles that contain both clashing items. Both are already in. Choose 2 more from the remaining 8 products. ${}^8C_2 = \frac{8 \times 7}{2} = 28.$

Step 3 — subtract. $210 - 28 = 182.$

Answer: 182

SHORTCUT

When a rule says “these two must NOT both appear”, it is almost always faster to count **all** cases and subtract the cases where both **do** appear. Forcing both in is a single, easy count.

Example 23 [DBS · pattern]

8 managers sit at a round table. Two of them, Arun and Bala, must not sit next to each other. In how many ways can they be seated?

SOLUTION

Step 1 — all circular seatings. $(8 - 1)! = 7! = 5040.$

Step 2 — seatings where Arun and Bala ARE together. Glue them into one unit. Units = 6 other managers + 1 block = 7 units around a circle.

Circular arrangements of 7 units = $(7 - 1)! = 6! = 720.$

Inside the block Arun–Bala or Bala–Arun: $2! = 2.$

Together = $720 \times 2 = 1440.$

Step 3 — subtract. $5040 - 1440 = 3600.$

Answer: 3600

TRAP

In a circular “together” problem you glue first and **then** apply $(k - 1)!$ to the new number of units. Writing $(8 - 1)!$ and then gluing gives $7! \times 2 = 10080$, which is larger than the total. If a sub-count comes out bigger than the total, you glued in the wrong order.

Example 24 [SCB · pattern]

12 identical laptops are sent to 4 branch offices. Every office must get at least one. In how many ways can they be split?

SOLUTION

The laptops are identical, so only the **numbers** matter. We need $x_1 + x_2 + x_3 + x_4 = 12$, each $x_i \geq 1$.

Picture 12 laptops in a row with 11 gaps between them: $L | L | L \cdots L$

Put 3 dividers into 3 of the 11 gaps. That cuts the row into 4 non-empty parts.

$${}^{11}C_3 = \frac{11 \times 10 \times 9}{3 \times 2 \times 1} = \frac{990}{6} = 165.$$

Answer: 165

Example 25 [LINE MAN · pattern]

10 identical THB 100 vouchers are given to 4 riders. (a) Any rider may get none. In how many ways? (b) Every rider must get at least 2. In how many ways?

SOLUTION

(a) $x_1 + x_2 + x_3 + x_4 = 10$, each $x_i \geq 0$. Use ${}^{n+r-1}C_{r-1}$ with $n = 10$, $r = 4$.

$${}^{13}C_3 = \frac{13 \times 12 \times 11}{6} = \frac{1716}{6} = 286.$$

(b) First hand out the compulsory 2 vouchers to each rider: $4 \times 2 = 8$ used.

Vouchers left = $10 - 8 = 2$, now free to give to anyone.

$$y_1 + y_2 + y_3 + y_4 = 2 \text{ with each } y_i \geq 0: {}^{2+4-1}C_3 = {}^5C_3 = {}^5C_2 = \frac{5 \times 4}{2} = 10.$$

Answer: (a) 286 (b) 10

SHORTCUT

“Each must get at least k ” $\rightarrow$ pay the minimum first, then the problem becomes the plain “at least 0” case. Only the leftover count changes.

Example 26 [GIC · pattern]

12 different analysts are split into 3 groups of 4. (a) The groups are named Equity, Credit and Macro. In how many ways? (b) The groups have no names — only the split matters. In how many ways?

SOLUTION

(a) Pick Equity’s 4 from 12, then Credit’s 4 from the remaining 8, then Macro gets the last 4.

$${}^{12}C_4 = \frac{12 \times 11 \times 10 \times 9}{24} = \frac{11880}{24} = 495.$$

$${}^8C_4 = \frac{8 \times 7 \times 6 \times 5}{24} = \frac{1680}{24} = 70.$$

$${}^4C_4 = 1.$$

$$\text{Total} = 495 \times 70 \times 1 = 34650.$$

(b) Without names, the same 3 sets can be labelled in $3! = 6$ orders, and all 6 were counted separately above.

$$\frac{34650}{6} = 5775.$$

Answer: (a) 34650 (b) 5775

TRAP

Divide by $m!$ **only** when the groups are of equal size **and** unlabelled. If the groups have different sizes (say 5, 4, 3), do not divide — the sizes themselves already tell the groups apart.

Example 27 [Agoda · pattern]

A traveller walks on a city grid. Her hotel is 5 blocks east and 3 blocks north of the station. She only walks east or north. (a) How many shortest routes are there? (b) How many of them pass through a coffee shop that is 2 blocks east and 1 block north of the station?

SOLUTION

(a) Every shortest route is a word of 5 E's and 3 N's, in some order. Total letters = 8.

$$\frac{8!}{5! \cdot 3!} = \frac{40320}{120 \times 6} = \frac{40320}{720} = 56.$$

(b) Split the journey at the coffee shop.

Station → coffee shop: 2 E and 1 N → $\frac{3!}{2! \cdot 1!} = \frac{6}{2} = 3$ ways.

Coffee shop → hotel: still needs $5 - 2 = 3$ E and $3 - 1 = 2$ N → $\frac{5!}{3! \cdot 2!} = \frac{120}{12} = 10$ ways.

Multiply: $3 \times 10 = 30$.

Answer: (a) 56 (b) 30

Example 28 [Razer · pattern]

A 6-digit account PIN must not start with 0 and must have no repeated digit. How many such PINs exist?

SOLUTION

Digits available: 0 to 9, that is 10 digits.

Position 1: cannot be 0 → 9 choices.

Positions 2 to 6: 9 digits remain (0 is now allowed), and no repeats: $9 \times 8 \times 7 \times 6 \times 5$.

$$9 \times 8 = 72. 72 \times 7 = 504. 504 \times 6 = 3024. 3024 \times 5 = 15120.$$

$$\text{Total} = 9 \times 15120 = 136080.$$

Answer: 136080

Example 29 [Sea/Shopee · pattern]

A project team of 5 is chosen from 6 staff in Singapore and 5 staff in Thailand. The team must contain at least 2 from each country. In how many ways?

SOLUTION

Team size is 5. “At least 2 from each” allows only these splits: (2 SG, 3 TH) and (3 SG, 2 TH). A (4,1) split fails the Thailand condition, (1,4) fails Singapore.

$$\text{Case A — 2 SG, 3 TH: } {}^6C_2 = \frac{6 \times 5}{2} = 15. {}^5C_3 = {}^5C_2 = \frac{5 \times 4}{2} = 10. 15 \times 10 = 150.$$

$$\text{Case B — 3 SG, 2 TH: } {}^6C_3 = \frac{6 \times 5 \times 4}{6} = \frac{120}{6} = 20. {}^5C_2 = 10. 20 \times 10 = 200.$$

$$\text{Total} = 150 + 200 = 350.$$

Answer: 350

Example 30 [Grab · pattern]

A waiting bench has 9 seats in a row. 4 riders must sit so that no two of them are next to each other. In how many ways?

SOLUTION

Step 1 — the empty seats first. $9 - 4 = 5$ seats stay empty.

Lay out the 5 empty seats. They create $5 + 1 = 6$ gaps (including the two ends): _ e _ e _ e _ e _

Step 2 — choose 4 of the 6 gaps (one rider per gap keeps them apart): ${}^6C_4 = {}^6C_2 = \frac{6 \times 5}{2} = 15$.

Step 3 — the riders are different people, so order them: $4! = 24$.

Total = $15 \times 24 = 360$.

(Formula check: ${}^{n-r+1}C_r \times r! = {}^6C_4 \times 4! = 360$. Matches.)

Answer: 360

PRACTICE · TIER 2 target: 100 s/Q

1. 8 people sit around a round table. 3 particular people must sit together. In how many ways?
2. 15 identical cartons are sent to 5 stores, each store getting at least one. In how many ways?
3. 7 identical discount coupons are given to 3 users; a user may get none. In how many ways?
4. A courier must go 4 blocks east and 4 blocks north. How many shortest routes?
5. From 8 drivers, 3 are chosen for the morning shift and 3 different ones for the evening shift. In how many ways?
6. A bundle of 5 items is made from 12 items, and 2 particular items must both be included. How many bundles?
7. A playlist plays 4 songs in order, chosen from 10 different songs. How many playlists?
8. 10 different interns are split into 2 unnamed groups of 5. In how many ways?
9. How many strings of length 5 can be made from the letters A, B, C with repetition allowed?
10. 3 people sit on a row of 7 seats so that no two are adjacent. In how many ways?
11. 5 people sit at a round table; 2 particular people must sit together. In how many ways?
12. How many 4-digit even numbers with no repeated digit can be made from the digits 1, 2, 3, 4, 5?

ANSWER KEY

1. 720 — glue the 3: units = $5 + 1 = 6$, circular $\rightarrow 5! = 120$, times $3! = 6$.
2. 1001 — ${}^{14}C_4 = \frac{14 \times 13 \times 12 \times 11}{24} = \frac{24024}{24}$.
3. 36 — zero allowed: ${}^{7+3-1}C_2 = {}^9C_2 = 36$.
4. 70 — $\frac{8!}{4!4!} = \frac{40320}{576}$.
5. 560 — ${}^8C_3 \times {}^5C_3 = 56 \times 10$. The two shifts are different, so do not divide by $2!$.
6. 120 — the 2 are fixed, pick 3 more from 10: ${}^{10}C_3$.
7. 5040 — order matters: ${}^{10}P_4 = 10 \times 9 \times 8 \times 7$.
8. 126 — ${}^{10}C_5 = 252$, then divide by $2!$ because the groups are unnamed.
9. 243 — 3^5 .
10. 60 — 4 empty seats give 5 gaps: ${}^5C_3 \times 3! = 10 \times 6$.
11. 12 — glue: 4 units circular $\rightarrow 3! = 6$, times $2! = 2$.
12. 48 — last digit must be 2 or 4 (2 ways), then $4 \times 3 \times 2 = 24$ for the first three places: 2×24 .

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Goldman Sachs · pattern]

5 signed letters and their 5 addressed envelopes are mixed up. A clerk puts one letter into each envelope at random. In how many ways can **no** letter end up in its own envelope?

SOLUTION

Insight: count all arrangements, then throw out every arrangement in which at least one letter is correct — by inclusion–exclusion, alternating signs.

Let A_i = the set of arrangements where letter i IS in its own envelope.

Number with a chosen set of k letters fixed correctly = $(5 - k)!$, and there are 5C_k such sets.

$$D_5 = 5! - {}^5C_1 4! + {}^5C_2 3! - {}^5C_3 2! + {}^5C_4 1! - {}^5C_5 0!$$

Term by term:

- $5! = 120$
- ${}^5C_1 \times 4! = 5 \times 24 = 120$
- ${}^5C_2 \times 3! = 10 \times 6 = 60$
- ${}^5C_3 \times 2! = 10 \times 2 = 20$
- ${}^5C_4 \times 1! = 5 \times 1 = 5$
- ${}^5C_5 \times 0! = 1 \times 1 = 1$

$$D_5 = 120 - 120 + 60 - 20 + 5 - 1 = 44.$$

Answer: 44

Example 32 [D. E. Shaw · pattern]

7 letters are put into 7 addressed envelopes at random. In how many ways do **exactly 3** letters land in the correct envelope?

SOLUTION

Insight: “exactly 3 correct” splits cleanly into two independent jobs — choose which 3 are correct, then force the remaining 4 to be all wrong. The second job is a derangement, not a free arrangement.

Step 1 — choose the 3 correct letters: ${}^7C_3 = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = \frac{210}{6} = 35$.

Step 2 — the other 4 letters must all be wrong. That is D_4 .

$$D_4 = 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right).$$

Inside the bracket, take 24 as the common denominator: $1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} = \frac{24 - 24 + 12 - 4 + 1}{24} = \frac{9}{24}$.

So $D_4 = 24 \times \frac{9}{24} = 9$.

Step 3 — multiply: $35 \times 9 = 315$.

Answer: 315

TRAP

If Step 2 were “4!” you would be allowing some of those 4 to also be correct, and the count would include cases with 4, 5, 6 or 7 correct letters. “Exactly k ” always forces a derangement on the rest: ${}^nC_k \times D_{n-k}$.

Example 33 [Amazon · pattern]

5 different packages are put into 3 different lockers. Every locker must get at least one package. In how many ways?

SOLUTION

Insight: the packages are different and the lockers are different, so start from the free count 3^5 , then remove the arrangements that leave a locker empty — again with inclusion–exclusion.

All ways (lockers may be empty): each package picks one of 3 lockers. $3^5 = 243$.

Remove the ways that miss at least one locker.

Ways that use at most 2 chosen lockers: pick which locker is banned (${}^3C_1 = 3$ ways), then each package has 2 choices: $2^5 = 32$. Subtract $3 \times 32 = 96$.

But an arrangement using only 1 locker was subtracted twice (it misses two lockers), so add it back. Pick which single locker is used: ${}^3C_2 = 3$ ways of banning two lockers, then $1^5 = 1$. Add $3 \times 1 = 3$.

$$243 - 96 + 3 = 150.$$

Answer: 150

Example 34 [Google · pattern]

All possible 4-digit numbers are formed using the digits 2, 3, 5, 7, each digit used exactly once. Find the sum of all these numbers.

SOLUTION

Insight: do not add 24 numbers. Count how many times each digit sits in each place — by symmetry, every digit visits every place equally often.

Total numbers = $4! = 24$.

Fix digit 2 in the units place. The other 3 digits fill the other 3 places in $3! = 6$ ways. So each digit appears in the units place exactly 6 times. The same is true for tens, hundreds and thousands.

Sum of the digits = $2 + 3 + 5 + 7 = 17$.

Sum contributed by the units place = $6 \times 17 = 102$. Tens place = 102×10 . Hundreds = 102×100 . Thousands = 102×1000 .

$$\text{Total} = 102 \times (1 + 10 + 100 + 1000) = 102 \times 1111$$

$$102 \times 1111 = 102 \times 1000 + 102 \times 100 + 102 \times 10 + 102 = 102000 + 10200 + 1020 + 102 = 113322.$$

Check: average number = $\frac{113322}{24} = 4721.75$, and the average digit is $\frac{17}{4} = 4.25$, giving 4721.75. Matches.

Answer: 113322

SHORTCUT

General rule: with n distinct non-zero digits all used, each digit appears $(n - 1)!$ times in each place, so Sum = $(n - 1)! \times (\text{sum of digits}) \times \left(\underbrace{11 \dots 1}_n \right)$. For 4 digits the repunit is 1111; for 3 digits it is 111.

Example 35 [Microsoft · pattern]

An 8×8 chessboard is drawn. (a) How many rectangles are there in the figure? (b) How many of them are squares?

SOLUTION

Insight: a rectangle is not a shape you hunt for — it is fixed the moment you choose two vertical lines and two horizontal lines.

(a) An 8×8 board of cells is made of 9 vertical lines and 9 horizontal lines.

Choose 2 vertical lines: ${}^9C_2 = \frac{9 \times 8}{2} = 36$. Choose 2 horizontal lines: ${}^9C_2 = 36$.

Rectangles = $36 \times 36 = 1296$.

(b) A $k \times k$ square needs its top-left corner in a $(9 - k) \times (9 - k)$ block of positions, so there are $(9 - k)^2$ squares of side k .

$$k = 1 : 8^2 = 64, k = 2 : 7^2 = 49, k = 3 : 36, k = 4 : 25, k = 5 : 16, k = 6 : 9, k = 7 : 4, k = 8 : 1.$$

$$64 + 49 = 113. 113 + 36 = 149. 149 + 25 = 174. 174 + 16 = 190. 190 + 9 = 199. 199 + 4 = 203. 203 + 1 = 204.$$

Answer: (a) 1296 rectangles (b) 204 squares

Example 36 [Uber · pattern]

In how many ways can 3 numbers be chosen from 1, 2, 3, ..., 20 so that no two chosen numbers are consecutive?

SOLUTION

Insight: shrink the problem. Subtract a growing amount from each chosen number so that “no two consecutive” turns into “all different” — a plain combination.

Let the chosen numbers be $a < b < c$ with $b \geq a + 2$ and $c \geq b + 2$.

Define $a' = a$, $b' = b - 1$, $c' = c - 2$.

Then $b' - a' = b - a - 1 \geq 1$ and $c' - b' = c - b - 1 \geq 1$, so $a' < b' < c'$ with no gap condition at all.

Range: $a' \geq 1$ and $c' = c - 2 \leq 20 - 2 = 18$.

So a', b', c' are any 3 different numbers from 1 to 18. And the map is reversible, so the counts are equal.

$${}^{18}C_3 = \frac{18 \times 17 \times 16}{3 \times 2 \times 1} = \frac{4896}{6} = 816.$$

(Formula check: ${}^{n-r+1}C_r = {}^{20-3+1}C_3 = {}^{18}C_3$. Matches.)

Answer: 816

Example 37 [Adobe · pattern]

How many solutions in non-negative integers does $x + y + z = 20$ have, if $x \leq 8$?

SOLUTION

Insight: an upper bound is handled the same way as a “must not happen” rule — count everything, then subtract the cases that break the bound. And the broken cases are themselves a clean stars-and-bars count after a substitution.

Step 1 — all non-negative solutions. ${}^{20+3-1}C_{3-1} = {}^{22}C_2 = \frac{22 \times 21}{2} = \frac{462}{2} = 231$.

Step 2 — the bad ones, where $x \geq 9$. Put $x = 9 + x'$ with $x' \geq 0$. Then $9 + x' + y + z = 20 \rightarrow x' + y + z = 11$.

$${}^{11+3-1}C_{3-1} = {}^{13}C_2 = \frac{13 \times 12}{2} = \frac{156}{2} = 78.$$

Step 3 — subtract. $231 - 78 = 153$.

Answer: 153

Example 38 [Google · pattern]

In how many ways can 10 be written as an **ordered** sum of positive integers? (For example 10, $7 + 3$, $3 + 7$, $1 + 1 + 8$ all count as different.)

SOLUTION

Insight: an ordered sum is just a way of cutting a row of ten 1's into blocks. Every cut position is an independent yes/no choice.

Write 10 as ten 1's in a row: 1 1 1 1 1 1 1 1 1 1

There are $10 - 1 = 9$ gaps between neighbouring 1's.

In each gap, either place a “+” (start a new part) or join the two 1's into the same part. Two choices, independently, in each of the 9 gaps.

$$2^9 = 512.$$

Cross-check with the parts count: the number of ordered sums with exactly k parts is ${}^9C_{k-1}$, and ${}^9C_0 + {}^9C_1 + \dots + {}^9C_9 = 2^9 = 512$. Matches.

Answer: 512

PRACTICE · TIER 3 target: 4 min/Q

- 6 letters are placed at random into 6 addressed envelopes. In how many ways does no letter reach its correct envelope?
- 5 letters are placed at random into 5 addressed envelopes. In how many ways do exactly 2 reach the correct envelope?
- 6 different balls are put into 3 different boxes so that no box is empty. In how many ways?
- All possible 3-digit numbers are formed from the digits 1, 3, 5, 7 using each digit at most once. Find the sum of all of them.
- In how many ways can 4 numbers be chosen from 1 to 15 so that no two are consecutive?
- How many solutions in non-negative integers does $a + b + c + d = 12$ have if every variable is at most 6?
- How many squares are there in a 6×6 grid of cells?

8. How many 5-digit numbers have their digits in strictly increasing order?

ANSWER KEY

1. **265.** This is D_6 . Using $D_n = (n-1)(D_{n-1} + D_{n-2})$ with $D_4 = 9$ and $D_5 = 44$: $D_6 = 5(44 + 9) = 5 \times 53 = 265$. The recurrence is far faster than the full inclusion-exclusion sum, and worth memorising up to D_6 .
2. **20.** Choose which 2 are correct: ${}^5C_2 = 10$. The remaining 3 must all be wrong: $D_3 = 2$. So $10 \times 2 = 20$. Using $3!$ instead of D_3 gives 60, which wrongly includes cases with more than 2 correct.
3. **540.** Free count $3^6 = 729$. Subtract the ones missing a box: ${}^3C_1 \times 2^6 = 3 \times 64 = 192$. Add back the ones using a single box: ${}^3C_2 \times 1^6 = 3$. $729 - 192 + 3 = 540$.
4. **10656.** Total numbers $= {}^4P_3 = 24$. Each of the 4 digits appears $\frac{24}{4} = 6$ times in each of the 3 places. Sum of digits $= 1 + 3 + 5 + 7 = 16$, so each place contributes $6 \times 16 = 96$. Total $= 96 \times 111 = 10656$.
5. **495.** Shrink: subtract 0, 1, 2, 3 from the four chosen numbers in order, giving 4 distinct values in the range 1 to $15 - 3 = 12$. So ${}^{12}C_4 = \frac{12 \times 11 \times 10 \times 9}{24} = \frac{11880}{24} = 495$.
6. **231.** All solutions: ${}^{15}C_3 = 455$. Bad ones with a chosen variable ≥ 7 : substitute and get a total of 5, giving ${}^8C_3 = 56$ each, and there are 4 variables: $4 \times 56 = 224$. Two variables cannot both be ≥ 7 since $14 > 12$, so nothing is added back. $455 - 224 = 231$.
7. **91.** Squares of side k number $(7-k)^2$: $36 + 25 + 16 + 9 + 4 + 1 = 91$. (Rectangles would be ${}^7C_2 \times {}^7C_2 = 441$.)
8. **126.** Once you choose 5 different digits, exactly one increasing order exists — so this is a pure selection. The digit 0 can never be used, because in an increasing number it would have to come first. So choose 5 from 1..9: ${}^9C_5 = {}^9C_4 = \frac{9 \times 8 \times 7 \times 6}{24} = \frac{3024}{24} = 126$.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 90 s/Q · 30 min total

1. Find 9P_2 .
2. How many arrangements are there of the letters of **ACCOUNT**?
3. From 5 red and 4 blue pens, 2 red and 1 blue are chosen. In how many ways?
4. In how many ways can 4 friends sit around a circular table?
5. How many 3-digit numbers with no repeated digit can be formed from the digits 0 to 9?
6. 20 identical pens are given to 4 students, each getting at least one. In how many ways?
7. How many diagonals does a 15-sided polygon have?
8. A team of 4 is chosen from 5 men and 4 women with at least 1 woman. In how many ways?
9. 7 letters go into 7 addressed envelopes at random. In how many ways does exactly 1 reach the right envelope?
10. A delivery route goes 6 blocks east and 4 blocks north. How many shortest routes?
11. A 4-character password uses the 26 capital letters and the 10 digits, with repetition allowed. How many passwords?
12. How many arrangements are there of the letters of **ENGINEERING**?
13. In how many ways can 3 numbers be chosen from 1 to 30 so that no two are consecutive?
14. 5 different prizes are given to 3 students; a student may get any number of prizes. In how many ways?
15. From 4 different fruits, at least one must be chosen. In how many ways?
16. 6 people sit at a round table; 2 particular people must not sit together. In how many ways?
17. Find the sum of all 3-digit numbers formed from the digits 2, 4, 6 with no digit repeated.
18. 12 distinct books are split into 2 unnamed groups of 6. In how many ways?
19. How many 4-letter arrangements can be made from the letters of **DELTA** with no letter repeated?
20. In how many ways can 3 identical red balls and 2 identical blue balls be placed in a row?

ANSWER KEY

1. $72 - 9 \times 8$.
2. 2520 — 7 letters, C twice: $\frac{5040}{2}$.
3. $40 - {}^5C_2 \times {}^4C_1 = 10 \times 4$.
4. $6 - (4 - 1)! = 3!$.
5. 648 — first digit 9 ways (not 0), then $9 \times 8: 9 \times 9 \times 8$.
6. $969 - {}^{19}C_3 = \frac{19 \times 18 \times 17}{6} = \frac{5814}{6}$.
7. $90 - \frac{15 \times 12}{2}$.
8. 121 — all ${}^9C_4 = 126$, minus all-men ${}^5C_4 = 5$.
9. $1855 - {}^7C_1 \times D_6 = 7 \times 265$.
10. $210 - \frac{10!}{6!4!} = {}^{10}C_4$.
11. 1679616 — 36 symbols, 4 slots: 36^4 .
12. 277200 — 11 letters: E three times, N three times, G twice, I twice. $\frac{39916800}{6 \times 6 \times 2 \times 2} = \frac{39916800}{144}$.
13. $3276 - {}^{30-3+1}C_3 = {}^{28}C_3 = \frac{28 \times 27 \times 26}{6} = \frac{19656}{6}$.
14. 243 — each of the 5 distinct prizes picks a student: 3^5 .
15. $15 - 2^4 - 1$.
16. 72 — total $5! = 120$; together $= 4! \times 2 = 48$; $120 - 48$.
17. 2664 — 6 numbers; each digit appears $\frac{6}{3} = 2$ times per place; digit sum $= 12$; $2 \times 12 = 24$ per place; 24×111 .
18. $462 - {}^{12}C_6 = 924$, divided by $2!$ for unnamed groups.
19. $120 - {}^5P_4 = 5 \times 4 \times 3 \times 2$.
20. $10 - \frac{5!}{3!2!} = \frac{120}{12}$.

ONE-PAGE REVISION CARD

THE FIRST QUESTION, ALWAYS: does order matter? Order matters $\rightarrow {}^nP_r$. Order does not matter $\rightarrow {}^nC_r$.

CORE

- ${}^nP_r = \frac{n!}{(n-r)!} = r$ factors counting down from n
- ${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{n!}{r!(n-r)!}$, and ${}^nC_r = {}^nC_{n-r}$
- ${}^nC_2 = \frac{n(n-1)}{2}$, ${}^nC_3 = \frac{n(n-1)(n-2)}{6}$, $\sum_r {}^nC_r = 2^n$
- $0! = 1$; $5! = 120$, $6! = 720$, $7! = 5040$, $8! = 40320$, $9! = 362880$, $10! = 3628800$

THE PATTERNS

Repetition allowed, r slots	n^r
Letters with repeats	$\frac{n!}{p!q!\dots}$
Round table, n people	$(n-1)!$
Necklace	$\frac{(n-1)!}{2}$
Group together	glue $\rightarrow (k)! \times (\text{units arrangement})$
No two together (r of n in a row)	${}^{n-r+1}C_r$, then $\times r!$ if distinct
Identical into r groups, each ≥ 1	${}^{n-1}C_{r-1}$
Identical into r groups, each ≥ 0	${}^{n+r-1}C_{r-1}$
mn distinct into m labelled groups of n	$\frac{(mn)!}{(n!)^m}$
...into m unlabelled groups	$\frac{(mn)!}{(n!)^m m!}$
n distinct into 3 distinct boxes, none empty	$3^n - 3 \cdot 2^n + 3$
Grid, m east and n north	$\frac{(m+n)!}{m!n!}$
Diagonals of n -gon	$\frac{n(n-3)}{2}$
Rectangles in $a \times b$ cells	${}^{a+1}C_2 \times {}^{b+1}C_2$
Derangement	$D_n = (n-1)(D_{n-1} + D_{n-2})$
Exactly k correct out of n	${}^nC_k \times D_{n-k}$
Sum of all numbers, n distinct digits	$(n-1)! \times (\text{digit sum}) \times 11\dots 1$

$$D_1 = 0, D_2 = 1, D_3 = 2, D_4 = 9, D_5 = 44, D_6 = 265, D_7 = 1854.$$

THE FIVE SHORTCUTS

1. Flip nC_r to ${}^nC_{n-r}$ whenever $r > \frac{n}{2}$.
2. "Must not both appear" $\rightarrow$ total minus (both forced in).
3. "Each gets at least k " $\rightarrow$ pay the minimum first, then count the leftover freely.
4. "Together" $\rightarrow$ glue and multiply by the inside arrangements.
5. "No two together" $\rightarrow$ seat the others, then use the gaps (1 more gap than items).

THE TOP FIVE TRAPS

1. **Leading zero.** A number cannot begin with 0. Handle the first place separately.
2. **"At least" done in one shot.** Split into disjoint cases and add. A single clever product double-counts.
3. **Circular gluing order.** Glue first, count the new units, only then apply $(k-1)!$.
4. **Dividing by $m!$ when you should not.** Divide only for equal-sized **unlabelled** groups. Named groups (Team A, Team B) are already distinct.
5. **Using $(n-k)!$ instead of D_{n-k} for "exactly k correct".** The rest must be forced wrong, not left free.

Probability

Count the good cases. Divide. Then check the condition.

What you need to know

WHAT YOU NEED TO KNOW

Classical probability. For equally likely outcomes:

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

Always $0 \leq P(E) \leq 1$. $P(\text{certain}) = 1$, $P(\text{impossible}) = 0$.

Complement. $P(\text{not } E) = 1 - P(E)$. Use this for every “at least one” question.

Odds. Odds in favour $a : b \rightarrow P = \frac{a}{a+b}$. Odds against $a : b \rightarrow P = \frac{b}{a+b}$.

Addition rule.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually exclusive (cannot both happen) $\rightarrow P(A \cap B) = 0$.

Multiplication rule.

$$P(A \cap B) = P(A) \cdot P(B | A)$$

Independent (one does not affect the other) $\rightarrow P(A \cap B) = P(A) \cdot P(B)$.

Conditional probability.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

In counting language: shrink the sample space to B , then count inside it.

Total probability. If $B_1, B_2, \dots, B_n$ cover all cases and do not overlap:

$$P(E) = P(B_1)P(E | B_1) + P(B_2)P(E | B_2) + \dots$$

Bayes' theorem. (Cause given effect.)

$$P(B_i | E) = \frac{P(B_i)P(E | B_i)}{\sum_j P(B_j)P(E | B_j)}$$

Expected value. If X takes value x_i with probability p_i :

$$E[X] = \sum x_i p_i$$

Linearity: $E[X + Y] = E[X] + E[Y]$ — true even when X and Y are **not** independent. **Indicator trick:** to count “how many of these n things happen”, add the n separate probabilities.

Binomial. n independent trials, success probability p each, $q = 1 - p$:

$$P(\text{exactly } r \text{ successes}) = {}^n C_r p^r q^{n-r}$$

Mean = np . Variance = npq .

Geometric (waiting). Repeat until the first success, probability p each try:

$$P(\text{first success on try } k) = q^{k-1}p, \quad E[\text{number of tries}] = \frac{1}{p}$$

It is **memoryless**: past failures do not change what is left to wait.

Standard sample spaces.

- One die: 6 outcomes. Two dice: 36 outcomes.
- Sum on two dice — number of ways:

Sum	2	3	4	5	6	7	8	9	10	11 / 12
Ways	1	2	3	4	5	6	5	4	3	2 / 1

- n coins: 2^n outcomes; exactly r heads in nC_r ways.
- A deck: 52 cards, 26 red, 26 black, 4 suits of 13, 4 aces, 12 face cards (J, Q, K), 6 red face cards.
- Drawing r balls at once from a bag = combinations: total = nC_r .

Calendar. Ordinary year = 52 weeks + 1 odd day $\rightarrow P(53 \text{ of a given weekday}) = \frac{1}{7}$. Leap year = 52 weeks + 2 odd days $\rightarrow P = \frac{2}{7}$.

Warm-up

WARM-UP

build the reflex

Example 1

A fair die is rolled. Find (a) $P(\text{even})$, (b) $P(\text{more than 4})$.

SOLUTION

Total outcomes = 6.

(a) Even faces: 2, 4, 6 $\rightarrow$ 3 favourable. $P = \frac{3}{6} = \frac{1}{2}$.

(b) More than 4: 5, 6 $\rightarrow$ 2 favourable. $P = \frac{2}{6} = \frac{1}{3}$.

Answer: (a) $\frac{1}{2}$ (b) $\frac{1}{3}$

Example 2

One card is drawn from a well-shuffled deck of 52. Find (a) $P(\text{king})$, (b) $P(\text{spade})$.

SOLUTION

(a) 4 kings. $P = \frac{4}{52} = \frac{1}{13}$.

(b) 13 spades. $P = \frac{13}{52} = \frac{1}{4}$.

Answer: (a) $\frac{1}{13}$ (b) $\frac{1}{4}$

Example 3

Two fair coins are tossed. Find (a) $P(\text{exactly one head})$, (b) $P(\text{at least one head})$.

SOLUTION

Outcomes: HH, HT, TH, TT $\rightarrow$ 4 in all.

(a) Exactly one head: HT, TH $\rightarrow$ 2. $P = \frac{2}{4} = \frac{1}{2}$.

(b) At least one head: all except TT $\rightarrow$ 3. $P = \frac{3}{4}$.

Answer: (a) $\frac{1}{2}$ (b) $\frac{3}{4}$

Example 4

A bag has 5 red and 3 blue balls. One ball is drawn. Find $P(\text{blue})$.

SOLUTION

Total balls = $5 + 3 = 8$. Blue = 3.

$$P = \frac{3}{8}.$$

Answer: $\frac{3}{8}$

Example 5

Two fair dice are rolled. Find $P(\text{sum} = 7)$.

SOLUTION

Total outcomes = $6 \times 6 = 36$.

Sum 7: (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) $\rightarrow$ 6 ways.

$$P = \frac{6}{36} = \frac{1}{6}.$$

Answer: $\frac{1}{6}$

Example 6

(a) $P(A) = 0.35$. Find $P(\text{not } A)$. (b) The odds in favour of an event are 3 : 5. Find its probability.

SOLUTION

(a) $P(\text{not } A) = 1 - 0.35 = 0.65$.

(b) Odds in favour 3 : 5 means 3 favourable parts and 5 unfavourable parts. Total parts = $3 + 5 = 8$. $P = \frac{3}{8}$.

Answer: (a) 0.65 (b) $\frac{3}{8}$

Example 7

A and B are independent with $P(A) = 0.6$ and $P(B) = 0.5$. Find (a) $P(\text{both})$, (b) $P(\text{neither})$.

SOLUTION

(a) Independent $\rightarrow$ multiply. $P(A \cap B) = 0.6 \times 0.5 = 0.30$.

(b) $P(\text{not } A) = 1 - 0.6 = 0.4$ and $P(\text{not } B) = 1 - 0.5 = 0.5$. $P(\text{neither}) = 0.4 \times 0.5 = 0.20$.

Answer: (a) 0.30 (b) 0.20

Example 8

$P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{6}$. Find $P(A \cup B)$.

SOLUTION

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{6}.$$

Common denominator 6: $\frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$.

Answer: $\frac{2}{3}$

Tier 1 — Service companies**TIER 1**

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Two fair dice are rolled. Find the probability that the sum is at least 10.

SOLUTION

Total outcomes = 36.

Sum = 10: (4, 6), (5, 5), (6, 4) → 3 ways. Sum = 11: (5, 6), (6, 5) → 2 ways. Sum = 12: (6, 6) → 1 way.

Favourable = 3 + 2 + 1 = 6.

$$P = \frac{6}{36} = \frac{1}{6}.$$

Answer: $\frac{1}{6}$

Example 10 [Accenture pattern]

One card is drawn from a deck of 52. Find the probability that it is a face card **or** a red card.

SOLUTION

Face cards (J, Q, K in 4 suits) = 12. Red cards = 26. Cards that are both red and a face card = 6 (J, Q, K of hearts and of diamonds).

$$P(F \cup R) = P(F) + P(R) - P(F \cap R) = \frac{12}{52} + \frac{26}{52} - \frac{6}{52} = \frac{32}{52}.$$

$$\frac{32}{52} = \frac{8}{13}.$$

Answer: $\frac{8}{13}$

TRAP

Do not add $\frac{12}{52} + \frac{26}{52} = \frac{38}{52}$. The 6 red face cards would be counted twice. Whenever two events can happen together, the overlap must be removed exactly once.

Example 11 [Infosys pattern]

A bag has 4 red, 5 green and 3 blue balls. Two balls are drawn together (without replacement). Find (a) P (both red), (b) P (one red and one green).

SOLUTION

Total balls = 4 + 5 + 3 = 12.

Total ways to draw 2 = ${}^{12}C_2 = \frac{12 \times 11}{2} = 66$.

(a) Both red: ${}^4C_2 = \frac{4 \times 3}{2} = 6$. $P = \frac{6}{66} = \frac{1}{11}$.

(b) One red and one green: ${}^4C_1 \times {}^5C_1 = 4 \times 5 = 20$. $P = \frac{20}{66} = \frac{10}{33}$.

Answer: (a) $\frac{1}{11}$ (b) $\frac{10}{33}$

SHORTCUT

Two balls drawn together = draw them one by one. For “both red” you may also write $\frac{4}{12} \times \frac{3}{11} = \frac{12}{132} = \frac{1}{11}$ — the same answer, with no nC_r at all.

For a **mixed** pair like “one red and one green” you must allow both orders: $\frac{4}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{4}{11} = \frac{20}{132} + \frac{20}{132} = \frac{40}{132} = \frac{10}{33}$. Same answer again. Use whichever line you can write faster, but remember: same colour → one order, different colours → two orders.

Example 12 [Wipro pattern]

Three fair coins are tossed. Find (a) P (at least one head), (b) P (exactly two heads).

SOLUTION

Total outcomes = $2^3 = 8$.

(a) The only outcome with no head is TTT, which is 1 outcome. P (no head) = $\frac{1}{8}$, so P (at least one head) = $1 - \frac{1}{8} = \frac{7}{8}$.

(b) Exactly two heads: ${}^3C_2 = 3$ outcomes (HHT, HTH, THH). $P = \frac{3}{8}$.

Answer: (a) $\frac{7}{8}$ (b) $\frac{3}{8}$

SHORTCUT

“At least one” → **turn it around**. $P(\text{at least one}) = 1 - P(\text{none})$. “None” is a single easy product; “at least one” is a long sum of cases. This single move saves more time in this chapter than any other.

Example 13 [Capgemini pattern]

Anjali can solve a problem with probability $\frac{1}{3}$ and Bhavin with probability $\frac{1}{4}$, independently. Find (a) $P(\text{the problem gets solved})$, (b) $P(\text{exactly one of them solves it})$.

SOLUTION

$$P(\text{Anjali fails}) = 1 - \frac{1}{3} = \frac{2}{3}. P(\text{Bhavin fails}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

$$(a) P(\text{both fail}) = \frac{2}{3} \times \frac{3}{4} = \frac{6}{12} = \frac{1}{2}. P(\text{solved}) = 1 - \frac{1}{2} = \frac{1}{2}.$$

$$(b) \text{Anjali solves and Bhavin fails: } \frac{1}{3} \times \frac{3}{4} = \frac{3}{12}. \text{ Bhavin solves and Anjali fails: } \frac{2}{3} \times \frac{1}{4} = \frac{2}{12}.$$

$$\text{These two cases cannot happen together, so add: } \frac{3}{12} + \frac{2}{12} = \frac{5}{12}.$$

Answer: (a) $\frac{1}{2}$ (b) $\frac{5}{12}$

Example 14 [Cognizant pattern]

A number is picked at random from 1 to 50. Find the probability that it is divisible by 3 or by 5.

SOLUTION

Multiples of 3 up to 50: $50 \div 3 = 16$ (remainder 2) → 16 numbers. Multiples of 5 up to 50: $50 \div 5 = 10$ → 10 numbers. Multiples of 15 (both) up to 50: $50 \div 15 = 3$ (remainder 5) → 3 numbers.

$$\text{Favourable} = 16 + 10 - 3 = 23.$$

$$P = \frac{23}{50}.$$

Answer: $\frac{23}{50}$

Example 15 [TCS NQT pattern]

Two fair dice are rolled. Find the probability that the product of the numbers is even.

SOLUTION

The product is odd only when **both** numbers are odd.

Odd faces: 1, 3, 5 → 3 choices on each die. Both odd = $3 \times 3 = 9$ outcomes.

$$P(\text{product odd}) = \frac{9}{36} = \frac{1}{4}.$$

$$P(\text{product even}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Answer: $\frac{3}{4}$

Example 16 [Accenture pattern]

The letters of the word **TABLE** are arranged at random. Find the probability that the two vowels come together.

SOLUTION

T, A, B, L, E → 5 different letters.

$$\text{Total arrangements} = 5! = 120.$$

Vowels are A and E. Glue them into one block. Units = T, B, L + 1 block = 4 units → $4! = 24$. Inside the block: $2! = 2$. Favourable = $24 \times 2 = 48$.

$$P = \frac{48}{120} = \frac{2}{5}.$$

Answer: $\frac{2}{5}$

Example 17 [Infosys pattern]

A box holds 6 good bulbs and 4 defective bulbs. Three bulbs are drawn at random together. Find (a) $P(\text{all good})$, (b) $P(\text{exactly one defective})$.

SOLUTION

Total bulbs = 10. Total ways = ${}^{10}C_3 = \frac{10 \times 9 \times 8}{6} = \frac{720}{6} = 120$.

(a) All 3 good: ${}^6C_3 = \frac{6 \times 5 \times 4}{6} = \frac{120}{6} = 20$. $P = \frac{20}{120} = \frac{1}{6}$.

(b) 1 defective and 2 good: ${}^4C_1 \times {}^6C_2 = 4 \times 15 = 60$. $P = \frac{60}{120} = \frac{1}{2}$.

Answer: (a) $\frac{1}{6}$ (b) $\frac{1}{2}$

Example 18 [Wipro pattern]

Find the probability that (a) an ordinary year has 53 Sundays, (b) a leap year has 53 Sundays.

SOLUTION

(a) An ordinary year has 365 days. $365 = 52 \times 7 + 1$.

So there are 52 full weeks (giving 52 Sundays for sure) plus 1 extra day. That extra day is equally likely to be any of the 7 weekdays. A 53rd Sunday happens only if the extra day is a Sunday.

$$P = \frac{1}{7}.$$

(b) A leap year has 366 days. $366 = 52 \times 7 + 2$.

The 2 extra days are consecutive. The 7 equally likely pairs are: (Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat), (Sat, Sun).

Sunday appears in 2 of these 7 pairs.

$$P = \frac{2}{7}.$$

Answer: (a) $\frac{1}{7}$ (b) $\frac{2}{7}$

Example 19 [TCS NQT pattern]

Two cards are drawn one after the other from a deck of 52, without replacement. Find $P(\text{both are aces})$.

SOLUTION

First card an ace: $\frac{4}{52} = \frac{1}{13}$.

Now 51 cards are left and only 3 aces remain. Second card an ace, given the first was: $\frac{3}{51} = \frac{1}{17}$.

$$P = \frac{1}{13} \times \frac{1}{17} = \frac{1}{221}.$$

Answer: $\frac{1}{221}$

TRAP

With replacement or without? If the first card is put back, the second probability is again $\frac{4}{52}$ and the answer becomes $\frac{1}{169}$. Read the sentence again before you multiply. In “drawn together” or “drawn one after the other” problems, assume **without** replacement unless the words “with replacement” appear.

Example 20 [Capgemini pattern]

Ravi speaks the truth 75% of the time and Sita 80% of the time. They each describe the same event independently. Find the probability that they contradict each other.

SOLUTION

$$P(R \text{ truth}) = 0.75, P(R \text{ lie}) = 0.25. P(S \text{ truth}) = 0.80, P(S \text{ lie}) = 0.20.$$

They contradict when exactly one of them is lying.

$$\text{Ravi truth AND Sita lie: } 0.75 \times 0.20 = 0.15. \text{ Ravi lie AND Sita truth: } 0.25 \times 0.80 = 0.20.$$

These cases are separate, so add: $0.15 + 0.20 = 0.35$.

| $0.35 = 35\%$.

Answer: 0.35, that is 35%

PRACTICE · TIER 1 target: 60 s/Q

1. A fair die is rolled. Find P (the number is prime).
2. Two fair dice are rolled. Find P (sum = 9).
3. One card is drawn from a deck. Find P (neither a king nor a queen).
4. A bag has 6 white and 4 black balls. One is drawn. Find P (black).
5. Two fair coins are tossed. Find P (no head).
6. Four fair coins are tossed. Find P (exactly 3 heads).
7. A and B are independent, $P(A) = 0.4$, $P(B) = 0.3$. Find $P(A \cup B)$.
8. A number is picked from 1 to 20. Find P (it is divisible by 4).
9. A bag has 5 red and 7 blue balls. Two are drawn together. Find P (both blue).
10. Two fair dice are rolled. Find P (a doublet) — both dice show the same number.
11. A ticket is drawn from tickets numbered 1 to 15. Find P (multiple of 3 or of 5).
12. $P(A) = 0.5$, $P(B) = 0.6$, $P(A \cup B) = 0.8$. Find $P(A \cap B)$.
13. Three students solve a problem independently with probabilities $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. Find P (at least one solves it).
14. One card is drawn from a deck. Find P (a red face card).
15. The letters of the word **APPLE** are arranged at random. Find P (the two P's are together).

ANSWER KEY

1. $\frac{1}{2}$ — primes on a die are 2, 3, 5 $\rightarrow \frac{3}{6}$. (1 is not prime; answering $\frac{2}{3}$ means you counted it.)
2. $\frac{1}{9}$ — (3, 6), (4, 5), (5, 4), (6, 3) $\rightarrow \frac{4}{36}$.
3. $\frac{11}{13}$ — 4 kings + 4 queens = 8 excluded; $\frac{44}{52}$.
4. $\frac{2}{5}$ — $\frac{4}{10}$.
5. $\frac{1}{4}$ — only TT out of 4 outcomes.
6. $\frac{1}{4}$ — ${}^4C_3 = 4$ out of $2^4 = 16$.
7. 0.58 — $0.4 + 0.3 - (0.4 \times 0.3) = 0.7 - 0.12$. Answering 0.7 means you treated them as mutually exclusive.
8. $\frac{1}{4}$ — 4, 8, 12, 16, 20 $\rightarrow \frac{5}{20}$.
9. $\frac{7}{22}$ — ${}^7C_2 / {}^{12}C_2 = \frac{21}{66}$.
10. $\frac{1}{6}$ — 6 doublets out of 36.
11. $\frac{7}{15}$ — multiples of 3: 5; of 5: 3; of 15: 1. $5 + 3 - 1 = 7$.
12. 0.3 — $P(A \cap B) = 0.5 + 0.6 - 0.8$.
13. $\frac{3}{4}$ — $P(\text{none}) = \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{6}{24} = \frac{1}{4}$; answer = $1 - \frac{1}{4}$.
14. $\frac{3}{26}$ — 6 red face cards out of 52.
15. $\frac{2}{5}$ — total = $5!/2! = 60$; PP glued gives $4! = 24$ (do not multiply by $2!$, the two P's are identical); $\frac{24}{60}$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

60% of a city's rides are inside the city centre and 40% are suburban. A city-centre ride finishes within 20 minutes with probability 0.90; a suburban ride with probability 0.70. A ride is picked at random. Find the probability that it finished within 20 minutes.

| **SOLUTION**

Two possible “causes”: city centre (C) or suburban (S). They cover everything and do not overlap.
 $P(C) = 0.60$, $P(S) = 0.40$. $P(F | C) = 0.90$, $P(F | S) = 0.70$, where F = finished in 20 min.
 Total probability:

$$\begin{aligned} P(F) &= P(C)P(F | C) + P(S)P(F | S) \\ &= 0.60 \times 0.90 + 0.40 \times 0.70 \\ &= 0.54 + 0.28 = 0.82. \end{aligned}$$

Answer: 0.82, that is 82%

Example 22 [Grab · pattern]

Use the same data as Example 21. A ride is found to have finished within 20 minutes. Find the probability that it was a city-centre ride.

SOLUTION

Now the direction is reversed: we know the **effect** and ask about the **cause**. That is Bayes.

$$P(C | F) = \frac{P(C)P(F | C)}{P(F)}$$

Numerator = $0.60 \times 0.90 = 0.54$ (computed in Example 21). Denominator = 0.82.

$$P(C | F) = \frac{0.54}{0.82} = \frac{54}{82} = \frac{27}{41}.$$

As a decimal: $27 \div 41 = 0.6585... \approx 0.659$.

Answer: $\frac{27}{41} \approx 0.659$

SHORTCUT

Bayes is just a two-line table. Write down the product $P(\text{cause}) \times P(\text{effect} | \text{cause})$ for each cause, add the column to get $P(\text{effect})$, then divide the row you want by that total. You never need to recall the full formula.

Cause	Prior	$P(F \cdot)$	Product
City	0.60	0.90	0.54
Suburb	0.40	0.70	0.28
		Total	0.82

Example 23 [Shopee · pattern]

Warehouse A ships 70% of all orders and damages 2% of what it ships. Warehouse B ships the other 30% and damages 5%. Find the probability that a randomly chosen order arrives damaged.

SOLUTION

$$P(A) = 0.70, P(D | A) = 0.02 \rightarrow \text{product} = 0.70 \times 0.02 = 0.014.$$

$$P(B) = 0.30, P(D | B) = 0.05 \rightarrow \text{product} = 0.30 \times 0.05 = 0.015.$$

$$P(D) = 0.014 + 0.015 = 0.029.$$

$$0.029 = 2.9\%.$$

Answer: 0.029, that is 2.9%

Example 24 [Shopee · pattern]

Continuing Example 23: an order arrives damaged. Find the probability that it came from warehouse B.

SOLUTION

$$P(B | D) = \frac{P(B)P(D | B)}{P(D)} = \frac{0.015}{0.029}$$

Multiply top and bottom by 1000: $\frac{15}{29}$.

$15 \div 29 = 0.5172... \approx 0.517$.

Answer: $\frac{15}{29} \approx 0.517$

TRAP

Warehouse B ships less than half the orders, yet it is the more likely source of a damaged one. A small group with a high failure rate can dominate. Never answer a Bayes question with the prior (0.30) — the evidence has changed it.

Example 25 [DBS · pattern]

A bank runs a promo spin. A customer wins S\$50 with probability 0.02, wins S\$10 with probability 0.10, and wins nothing otherwise. 5,000 customers spin. (a) What is the expected payout per customer? (b) What total payout should the bank budget for?

SOLUTION

(a) List value and probability:

Prize	S\$50	S\$10
Probability	0.02	0.10

Nothing: probability = $1 - 0.02 - 0.10 = 0.88$, value 0.

$$E = 50 \times 0.02 + 10 \times 0.10 + 0 \times 0.88$$

$$= 1.00 + 1.00 + 0 = 2.00, \text{ that is S\$2.00 per customer.}$$

(b) Expected total = $5000 \times 2.00 = 10000$, that is S\$10,000.

Answer: (a) S\$2 per customer (b) S\$10,000

Example 26 [Agoda · pattern]

8 guests each hold a booking. Each guest cancels independently with probability 0.15. Find the probability that exactly 2 cancel.

SOLUTION

This is binomial with $n = 8$, $p = 0.15$, $q = 0.85$, $r = 2$.

$$P = {}^8C_2(0.15)^2(0.85)^6$$

Piece 1: ${}^8C_2 = \frac{8 \times 7}{2} = 28$.

Piece 2: $(0.15)^2 = 0.0225$.

Piece 3: $(0.85)^6$. $0.85^2 = 0.7225$. $0.85^3 = 0.7225 \times 0.85 = 0.614125$. $0.85^6 = (0.85^3)^2 = 0.614125 \times 0.614125 = 0.377150$ (to 6 places).

Multiply: $28 \times 0.0225 = 0.63$. $0.63 \times 0.377150 = 0.237604$.

Answer: ≈ 0.2376 , that is about 23.8%

Example 27 [SCB · pattern]

A credit officer reviews 3 independent loan applications. Each is approved with probability 0.6. Find the probability that at least 2 are approved.

SOLUTION

Binomial with $n = 3$, $p = 0.6$, $q = 0.4$.

“At least 2” means exactly 2 or exactly 3.

Exactly 2: ${}^3C_2(0.6)^2(0.4)^1 = 3 \times 0.36 \times 0.4. 3 \times 0.36 = 1.08. 1.08 \times 0.4 = 0.432.$

Exactly 3: ${}^3C_3(0.6)^3 = 1 \times 0.216 = 0.216. (0.6^2 = 0.36, 0.36 \times 0.6 = 0.216.)$

$$P = 0.432 + 0.216 = 0.648.$$

Answer: 0.648

Example 28 [GIC · pattern]

A portfolio holds 12 bonds, of which 4 are rated junk. An auditor picks 3 bonds at random. Find the probability that at least one junk bond is picked.

SOLUTION

Use the complement.

$$\text{Total ways to pick 3 of 12: } {}^{12}C_3 = \frac{12 \times 11 \times 10}{6} = \frac{1320}{6} = 220.$$

$$\text{No junk bond means all 3 come from the 8 good bonds: } {}^8C_3 = \frac{8 \times 7 \times 6}{6} = \frac{336}{6} = 56.$$

$$P(\text{no junk}) = \frac{56}{220} = \frac{14}{55}.$$

$$P(\text{at least one junk}) = 1 - \frac{14}{55} = \frac{41}{55}.$$

$$41 \div 55 = 0.7454... \approx 0.745.$$

Answer: $\frac{41}{55} \approx 0.745$

Example 29 [LINE MAN · pattern]

A rider has 4 deliveries in a shift. Each is late independently with probability 0.2. (a) Find the expected number of late deliveries. (b) Find $P(\text{at least one is late})$.

SOLUTION

$$(a) \text{ Binomial mean} = np = 4 \times 0.2 = 0.8 \text{ deliveries.}$$

$$(b) P(\text{one delivery not late}) = 1 - 0.2 = 0.8.$$

$$P(\text{none late}) = 0.8^4. 0.8^2 = 0.64. 0.8^4 = 0.64 \times 0.64 = 0.4096.$$

$$P(\text{at least one late}) = 1 - 0.4096 = 0.5904.$$

Answer: (a) 0.8 (b) 0.5904

Example 30 [Razer · pattern]

Each loot box contains a rare skin with probability $\frac{1}{8}$, independently of the others. A player keeps opening boxes until the first rare skin appears. (a) Find the expected number of boxes he must open. (b) Find $P(\text{the first rare skin is in the 3rd box})$.

SOLUTION

$$(a) \text{ Waiting for a first success with } p = \frac{1}{8}: E = \frac{1}{p} = 1 \div \frac{1}{8} = 8 \text{ boxes.}$$

(b) Box 1 fails, box 2 fails, box 3 succeeds. They are independent, so multiply:

$$\frac{7}{8} \times \frac{7}{8} \times \frac{1}{8} = \frac{49}{512}.$$

$$49 \div 512 = 0.0957.$$

Answer: (a) 8 boxes (b) $\frac{49}{512} \approx 0.0957$

PRACTICE · TIER 2 target: 100 s/Q

- Machine A makes 60% of the parts with a 3% defect rate; machine B makes 40% with a 6% defect rate. Find $P(\text{a random part is defective})$.
- Using the data of Q1, a part is found defective. Find $P(\text{it came from A})$.

- Two cards are drawn from a deck without replacement. Given that the first was a king, find $P(\text{the second is also a king})$.
- A roll of 5 coupons contains exactly 2 winning coupons. Two coupons are drawn together. Find $P(\text{both are winners})$.
- A game costs Rs. 12 to play. You roll a fair die and are paid Rs. 60 if a 6 appears, nothing otherwise. Find the expected profit per game.
- A fair coin is tossed 6 times. Find $P(\text{exactly 4 heads})$.
- $P(A) = 0.3$ and $P(B | A) = 0.5$. Find $P(A \cap B)$.
- A lot of 10 bulbs contains 3 defective ones. Two bulbs are drawn together. Find $P(\text{exactly one is defective})$.
- A rider makes 5 deliveries, each late independently with probability 0.1. Find $P(\text{at most one is late})$.
- Find the expected value of one roll of a fair die.
- A test has 4 multiple-choice questions with 4 options each. A student guesses every answer. Find $P(\text{exactly 2 correct})$.
- Box A holds 3 red and 2 white balls; box B holds 2 red and 4 white balls. A box is chosen at random, then one ball is drawn from it. Find $P(\text{the ball is red})$.

ANSWER KEY

- $0.042 - 0.6 \times 0.03 = 0.018$ and $0.4 \times 0.06 = 0.024$; add.
- $\frac{3}{7} \approx 0.4286 - 0.018/0.042$. Answering 0.6 means you reported the prior instead of the posterior.
- $\frac{1}{17} - 3 \text{ kings remain among } 51 \text{ cards: } \frac{3}{51}$.
- $\frac{1}{10} - {}^2C_2 / {}^5C_2 = \frac{1}{10}$.
- Rs. 2 — expected payout $= 60 \times \frac{1}{6} = 10$; profit $= 10 - 12 = -2$, so an expected loss of Rs. 2 per game.
- $\frac{15}{64} - {}^6C_4 / 2^6 = \frac{15}{64}$.
- $0.15 - P(A) \times P(B | A) = 0.3 \times 0.5$.
- $\frac{7}{15} - {}^3C_1 \times {}^{10}C_2 = \frac{21}{45}$.
- $0.91854 - \text{none late} = 0.9^5 = 0.59049$; exactly one $= 5 \times 0.1 \times 0.9^4 = 5 \times 0.1 \times 0.6561 = 0.32805$; add.
- $3.5 - (1 + 2 + 3 + 4 + 5 + 6)/6 = \frac{21}{6}$.
- $\frac{27}{128} \approx 0.211 - {}^4C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^2 = 6 \times \frac{1}{16} \times \frac{9}{16} = \frac{54}{256}$.
- $\frac{7}{15} - \frac{1}{2} \times \frac{3}{5} + \frac{1}{2} \times \frac{2}{6} = \frac{3}{10} + \frac{1}{6} = \frac{9}{30} + \frac{5}{30} = \frac{14}{30}$. The boxes hold different totals, so you cannot pool the balls.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Goldman Sachs · pattern]

A disease affects 0.5% of a population. A screening test detects the disease in 99% of people who have it, and wrongly reports positive for 4% of people who do not. A randomly chosen person tests positive. Find the probability that the person actually has the disease.

SOLUTION

Insight: with a rare disease, the false positives come from a huge healthy group, so they can easily outnumber the true positives. The base rate decides the answer, not the 99%.

Let D = has disease, T = tests positive.

$P(D) = 0.005$, so $P(\text{no } D) = 0.995$. $P(T | D) = 0.99$, $P(T | \text{no } D) = 0.04$.

True positives: $0.005 \times 0.99 = 0.00495$.

False positives: $0.995 \times 0.04 = 0.0398$.

All positives: $P(T) = 0.00495 + 0.0398 = 0.04475$.

$$P(D | T) = \frac{0.00495}{0.04475}$$

Multiply top and bottom by 100000: $\frac{495}{4475}$. Divide both by 5: $\frac{99}{895}$.

$99 \div 895 = 0.11061... \approx 0.1106$.

Sanity check with 100,000 people: 500 have the disease and 495 of them test positive; 99,500 are healthy and 3,980 of them test positive. Total positives = $495 + 3980 = 4475$, of which 495 are true. $\frac{495}{4475} = 0.1106$. Matches.

Answer: $\approx 11.1\%$

TRAP

The “99% accurate test” instinct says 99%. The correct answer is about 11%. Whenever a condition is rare, always run the 100,000-people table — it converts the whole problem into two integers and removes all doubt.

Example 32 [Google · pattern]

5 people are picked at random. Assume all 12 birth months are equally likely and independent. Find the probability that at least two of them share a birth month.

SOLUTION

Insight: “at least two share” has too many cases. Its opposite — “all five are different” — is a single falling product.

Total equally likely outcomes: each person has 12 possible months. 12^5 . $12^2 = 144$, $12^3 = 1728$, $12^4 = 20736$, $12^5 = 248832$.

All different: person 1 has 12 free choices, person 2 must avoid 1 month (11 left), and so on. $12 \times 11 \times 10 \times 9 \times 8$.

$12 \times 11 = 132$. $132 \times 10 = 1320$. $1320 \times 9 = 11880$. $11880 \times 8 = 95040$.

$$P(\text{all different}) = \frac{95040}{248832} = 0.381944...$$

$$P(\text{at least two share}) = 1 - 0.381944 = 0.618056$$

Answer: ≈ 0.618 , that is about **61.8%**

Example 33 [Amazon · pattern]

5 letters are placed at random into 5 addressed envelopes, one letter per envelope. Find the expected number of letters that land in the correct envelope.

SOLUTION

Insight: do not find the whole distribution. Define one indicator per letter and add the expectations — linearity of expectation does not care that the events are tangled together.

Let $X_i = 1$ if letter i is in its own envelope, and 0 otherwise. The total number of correct letters is $X = X_1 + X_2 + X_3 + X_4 + X_5$.

For a single letter i : by symmetry it is equally likely to land in any of the 5 envelopes.

$$P(X_i = 1) = \frac{1}{5} \rightarrow E[X_i] = 1 \times \frac{1}{5} + 0 \times \frac{4}{5} = \frac{1}{5}$$

By linearity:

$$E[X] = E[X_1] + \dots + E[X_5] = 5 \times \frac{1}{5} = 1$$

Note how little we needed: the X_i are strongly dependent (if 4 letters are correct the 5th must be too), and linearity still holds.

Check against the exact distribution. Out of $5! = 120$ arrangements, the number with exactly k correct is ${}^5C_k D_{5-k}$, using $D_0 = 1, D_1 = 0, D_2 = 1, D_3 = 2, D_4 = 9, D_5 = 44$:

k correct	0	1	2	3	4	5
Arrangements	44	45	20	10	0	1

$$\frac{0 \times 44 + 1 \times 45 + 2 \times 20 + 3 \times 10 + 4 \times 0 + 5 \times 1}{120} = \frac{45 + 40 + 30 + 0 + 5}{120} = \frac{120}{120} = 1. \text{ Matches.}$$

Answer: 1 letter, on average

SHORTCUT

Indicator trick. To find “the expected number of things that happen”, write the count as a sum of 0/1 variables and add the individual probabilities. Independence is never needed. This one idea answers most “expected number of ...” interview questions in two lines.

Example 34 [D. E. Shaw · pattern]

A fair die is rolled repeatedly until a 6 appears. (a) Find the expected number of rolls. (b) You have already rolled 3 times with no 6. Find the expected number of **further** rolls.

SOLUTION

Insight: set up an equation in the answer itself. After one failed roll you are standing exactly where you started — the process has no memory.

(a) Let E be the expected number of rolls.

Roll once — that costs 1 roll, always. With probability $\frac{1}{6}$ you get a 6 and stop. With probability $\frac{5}{6}$ you do not, and you are back at the start, expecting E more rolls.

$$E = 1 + \frac{5}{6}E$$

$$E - \frac{5}{6}E = 1 \rightarrow \frac{1}{6}E = 1 \rightarrow E = 6$$

(b) The die does not remember the 3 failures. The expected number of further rolls is again 6.

(Total expected rolls for such a player is $3 + 6 = 9$, but the **remaining** wait is unchanged at 6.)

Answer: (a) 6 rolls (b) 6 more rolls

Example 35 [Microsoft · pattern]

Each cereal packet contains one of 3 different toys, equally likely and independent. Find the expected number of packets needed to collect all 3 toys.

SOLUTION

Insight: split the wait into stages. In each stage you are simply waiting for a first success, and Example 34 says that wait is $\frac{1}{p}$.

Stage 1 — get any new toy. The first packet is always new. $p = \frac{3}{3} = 1$, expected packets = 1.

Stage 2 — get a toy different from the one you have. 2 of the 3 toys are new. $p = \frac{2}{3}$, expected packets = $1 \div \frac{2}{3} = \frac{3}{2}$.

Stage 3 — get the last missing toy. 1 of the 3 toys is new. $p = \frac{1}{3}$, expected packets = $1 \div \frac{1}{3} = 3$.

By linearity the total expectation is the sum of the stage expectations:

$$E = 1 + \frac{3}{2} + 3 = \frac{2}{2} + \frac{3}{2} + \frac{6}{2} = \frac{11}{2} = 5.5$$

(General form: $E = n(1 + \frac{1}{2} + \dots + \frac{1}{n})$. Here $3(1 + \frac{1}{2} + \frac{1}{3}) = 3 \times \frac{11}{6} = 5.5$.)

Answer: 5.5 packets

Example 36 [Uber · pattern]

Two fair dice are rolled. You are told that at least one of them shows a 4. Find the probability that the sum is 7.

SOLUTION

Insight: conditioning is not multiplying by anything. It means throwing away every outcome that does not match the information, and recounting inside what is left.

New sample space — outcomes with at least one 4. First die is 4: (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6) → 6 outcomes. Second die is 4: (1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 4) → 6 outcomes. (4, 4) was counted twice.

Total = $6 + 6 - 1 = 11$ outcomes.

Favourable — sum 7 among these 11. (4, 3) and (3, 4). That is 2 outcomes.

$$P = \frac{2}{11}$$

Answer: $\frac{2}{11}$

TRAP

The answer is **not** $\frac{1}{6}$. Knowing a 4 is present throws away every outcome with no 4 at all — that is $5 \times 5 = 25$ outcomes — leaving $36 - 25 = 11$. The space is reshaped, and 11 is not a multiple of 6. Also note the wording: “at least one shows a 4” gives 11 outcomes, while “the first die shows a 4” gives 6 outcomes and the answer $\frac{1}{6}$. One word changes the denominator.

Example 37 [Adobe · pattern]

A company prints 1,000 scratch cards. One card wins Rs. 20,000, five cards win Rs. 2,000 each, and fifty cards win Rs. 100 each. The rest win nothing. Each card sells for Rs. 60. (a) Find the expected prize per card. (b) Find the company’s expected profit on the whole print run.

SOLUTION

Insight: the expected prize per card is just the total prize money spread over all cards. No probability table is needed once you see that.

(a) Total prize money: $1 \times 20000 = 20000$ $5 \times 2000 = 10000$ $50 \times 100 = 5000$

Total = $20000 + 10000 + 5000 = 35000$, that is Rs. 35,000.

Expected prize per card = $\frac{35000}{1000} = 35$, that is Rs. 35.

(b) Revenue per card is Rs. 60. Expected prize cost per card is Rs. 35.

Expected profit per card = $60 - 35 = 25$, that is Rs. 25.

Whole run: $1000 \times 25 = 25000$, that is Rs. 25,000.

(Check by totals: revenue = $1000 \times 60 = 60000$; prizes = 35000; profit = $60000 - 35000 = 25000$. Matches.)

Answer: (a) Rs. 35 (b) Rs. 25,000

Example 38 [Google · pattern]

Priya and Kwan toss a fair coin one after the other, Priya first. The first person to toss a head wins. Find the probability that Priya wins.

SOLUTION

Insight: after two failed tosses the game is identical to the start, but it is Priya’s turn again. So write her winning probability in terms of itself.

Let $p = P(\text{Priya wins})$.

Toss 1 (Priya). Head with probability $\frac{1}{2} \rightarrow$ she wins immediately.

Toss 2 (Kwan). Reached only if Priya missed (probability $\frac{1}{2}$). If Kwan gets a head (probability $\frac{1}{2}$) Priya loses. If both miss — probability $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ — the position is exactly the starting position, with Priya to toss. Her chance from there is again p .

$$p = \frac{1}{2} + \frac{1}{4}p$$

$$p - \frac{1}{4}p = \frac{1}{2} \rightarrow \frac{3}{4}p = \frac{1}{2} \rightarrow p = \frac{1}{2} \times \frac{4}{3} = \frac{2}{3}$$

Check by series: $p = \frac{1}{2} + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^5 + \cdots = \frac{\frac{1}{2}}{1 - \frac{1}{4}} = \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{2}{3}$. Matches.

Answer: $\frac{2}{3}$

PRACTICE · TIER 3 target: 4 min/Q

1. A disease affects 1% of a population. A test is positive for 95% of people who have it and for 10% of people who do not. A person tests positive. Find the probability that the person has the disease.
2. 4 people are picked at random. Assume 12 equally likely birth months. Find the probability that at least two share a birth month.
3. A fair coin is tossed 10 times. Find the expected number of positions i (with $1 \leq i \leq 9$) at which toss i and toss $i + 1$ are both heads.
4. A fair die is rolled until a number greater than 4 appears. Find the expected number of rolls.
5. Each packet holds one of 4 equally likely toys. Find the expected number of packets needed to collect all 4.
6. Two fair dice are rolled and you are told at least one shows a 3. Find $P(\text{the sum is } 8)$.
7. Aarav and Bao roll a fair die alternately, Aarav first. The first to roll a 6 wins. Find $P(\text{Aarav wins})$.
8. 3 cards are drawn from a deck of 52 without replacement. Find the expected number of aces drawn.

ANSWER KEY

1. $\approx 8.76\%$. True positives = $0.01 \times 0.95 = 0.0095$. False positives = $0.99 \times 0.10 = 0.099$. All positives = 0.1085 . Answer = $0.0095/0.1085 = 0.08756$. In 100,000 people: 950 true positives against 9,900 false ones — the healthy group is 99 times larger, so its 10% error rate swamps the test's 95% hit rate.
2. ≈ 0.4271 . All different = $(12 \times 11 \times 10 \times 9)/12^4 = \frac{11880}{20736} = 0.572917$. Answer = $1 - 0.572917 = 0.427083$. Note how fast this grows: 5 people already gave 0.618 in Example 32.
3. 2.25. Use indicators. There are 9 adjacent pairs. For each pair, $P(\text{both heads}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. By linearity $E = 9 \times \frac{1}{4} = \frac{9}{4} = 2.25$. The pairs overlap and are dependent, which does not matter for expectation.
4. 3 rolls. "Greater than 4" means 5 or 6, so $p = \frac{2}{6} = \frac{1}{3}$. Waiting for a first success gives $E = 1/p = 3$.
5. $\frac{25}{3} \approx 8.33$ packets. Stage waits are $1, \frac{4}{3}, \frac{4}{2} = 2$ and $\frac{4}{1} = 4$. Sum = $1 + \frac{4}{3} + 2 + 4 = 7 + \frac{4}{3} = \frac{25}{3}$. Equivalently $4(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}) = 4 \times \frac{25}{12}$.
6. $\frac{2}{11}$. Outcomes containing at least one 3: $6 + 6 - 1 = 11$. Those summing to 8: (3, 5) and (5, 3), so 2. Answer $\frac{2}{11}$, not the unconditional $\frac{5}{36}$.
7. $\frac{6}{11}$. Let p be Aarav's chance. He wins at once with probability $\frac{1}{6}$; with probability $\frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$ both miss and the game restarts with him to roll. So $p = \frac{1}{6} + \left(\frac{25}{36}\right)p \rightarrow \left(\frac{11}{36}\right)p = \frac{1}{6} \rightarrow p = \frac{36}{66} = \frac{6}{11}$. Moving first is worth more than half.
8. $\frac{3}{13} \approx 0.2308$. Indicators again: let $X_i = 1$ if the i -th card drawn is an ace. By symmetry each drawn card is an ace with probability $\frac{4}{52} = \frac{1}{13}$, whatever its position. $E = 3 \times \frac{1}{13} = \frac{3}{13}$. "Without replacement" changes the distribution but not this expectation.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 90 s/Q · 30 min total

1. A fair die is rolled. Find $P(\text{the number is odd})$.
2. Two fair dice are rolled. Find $P(\text{sum} = 5)$.

3. One card is drawn from a deck. Find $P(\text{an ace or a spade})$.
4. A bag has 3 red and 5 blue balls. Two are drawn together. Find $P(\text{both are the same colour})$.
5. Five fair coins are tossed. Find $P(\text{at least one tail})$.
6. A and B are independent with $P(A) = 0.7$ and $P(B) = 0.4$. Find $P(\text{exactly one of them happens})$.
7. Machine A makes 70% of the output with a 2% defect rate; machine B makes 30% with a 5% defect rate. A defective item is found. Find $P(\text{it came from B})$.
8. A fair die is rolled and you are paid twice the number shown, in rupees. Find the expected payment.
9. 5 independent trials each succeed with probability 0.2. Find $P(\text{exactly one success})$.
10. Find the probability that a leap year contains 53 Mondays.
11. The letters of **LEVEL** are arranged at random. Find $P(\text{the arrangement starts with L})$.
12. A lot of 12 bulbs contains 4 defective ones. Three are drawn together. Find $P(\text{none is defective})$.
13. A fair die is rolled 30 times. Find the expected number of sixes.
14. $P(A) = 0.6$ and $P(B | A) = 0.25$. Find $P(A \cap B)$.
15. Two fair dice are rolled. Given that the sum is even, find $P(\text{both dice show odd numbers})$.
16. A fair coin is tossed until the first head appears. Find $P(\text{exactly 4 tosses are needed})$.
17. Three friends each choose a day of the week at random. Find $P(\text{all three choose different days})$.
18. A bag has 4 green and 6 yellow balls. One ball is drawn, its colour noted, and it is put back; then another is drawn. Find $P(\text{both are green})$.
19. The odds against an event are 4 : 7. Find the probability that the event happens.
20. A game costs Rs. 20 to enter. You win Rs. 100 with probability 0.15, and nothing otherwise. Find the expected profit per game.

ANSWER KEY

1. $\frac{1}{2}$ — 1, 3, 5 out of 6.
2. $\frac{1}{9}$ — (1, 4), (2, 3), (3, 2), (4, 1) $\rightarrow \frac{4}{36}$.
3. $\frac{4}{13}$ — $4 + 13 - 1 = 16$ (the ace of spades is in both); $\frac{16}{52}$.
4. $\frac{13}{28}$ — $[{}^3C_2 + {}^5C_2] / {}^8C_2 = \frac{3+10}{28}$.
5. $\frac{31}{32}$ — $1 - (\frac{1}{2})^5$.
6. 0.54 — $0.7 \times 0.6 + 0.3 \times 0.4 = 0.42 + 0.12$.
7. $\frac{15}{29} \approx 0.517$ — products $0.7 \times 0.02 = 0.014$ and $0.3 \times 0.05 = 0.015$; total 0.029; take $0.015/0.029$.
8. Rs. 7 — $E[2X] = 2E[X] = 2 \times 3.5$.
9. 0.4096 — ${}^5C_1 (0.2)(0.8)^4 = 5 \times 0.2 \times 0.4096 = 1 \times 0.4096$.
10. $\frac{2}{7}$ — 366 days give 2 odd days; Monday appears in 2 of the 7 consecutive-day pairs.
11. $\frac{2}{5}$ — total = $5!/(2!2!) = 30$; starting with L leaves E, V, E, L $\rightarrow 4!/2! = 12$; $\frac{12}{30}$.
12. $\frac{14}{55}$ — ${}^8C_3 / {}^{12}C_3 = \frac{56}{220}$.
13. 5 — $np = 30 \times \frac{1}{6}$.
14. 0.15 — 0.6×0.25 .
15. $\frac{1}{2}$ — sum even means both odd (9 ways) or both even (9 ways), so 18 outcomes; both odd is 9 of them.
16. $\frac{1}{16}$ — three tails then a head: $(\frac{1}{2})^3 \times \frac{1}{2}$.
17. $\frac{30}{49}$ — $(7 \times 6 \times 5) / 7^3 = \frac{210}{343}$.
18. $\frac{4}{25}$ — with replacement the draws are independent: $(\frac{4}{10})^2 = 0.16$.
19. $\frac{7}{11}$ — odds **against** 4 : 7 means 4 parts against and 7 parts for, so $P = \frac{7}{11}$. Answering $\frac{4}{11}$ means you read “against” as “in favour”.
20. —Rs. 5 — expected winnings = $100 \times 0.15 = 15$; profit = $15 - 20 = -5$, an expected loss of Rs. 5.

ONE-PAGE REVISION CARD

THE CORE

$$P(E) = \frac{\text{favourable}}{\text{total}}, \quad P(\text{not } E) = 1 - P(E)$$

Odds in favour $a : b \rightarrow \frac{a}{a+b}$. Odds against $a : b \rightarrow \frac{b}{a+b}$.

THE FOUR RULES

Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Multiplication	$P(A \cap B) = P(A) \cdot P(B A)$
Independent	$P(A \cap B) = P(A) \cdot P(B)$
Conditional	$P(A B) = P(A \cap B) / P(B)$

TOTAL PROBABILITY & BAYES — build the table, never the formula:

Cause	Prior	$P(E \text{cause})$	Product
B_1	$P(B_1)$	$P(E B_1)$	multiply
B_2	$P(B_2)$	$P(E B_2)$	multiply
		Total = $P(E)$	add the column

Then $P(B_i | E) = \frac{\text{its product}}{\text{the column total}}$.

EXPECTATION

- $E[X] = \sum x_i p_i$; $E[aX] = aE[X]$
- Linearity:** $E[X + Y] = E[X] + E[Y]$, dependent or not
- Indicator trick:** expected count = sum of the individual probabilities
- Waiting for a first success: $E = 1/p$; memoryless
- Collect all n coupons: $E = n(1 + \frac{1}{2} + \dots + \frac{1}{n})$

BINOMIAL — n independent trials, success probability p :

$$P(r \text{ successes}) = {}^n C_r p^r q^{n-r}, \quad \text{mean} = np, \quad \text{variance} = npq$$

STANDARD NUMBERS

- Two dice: 36 outcomes. Sum ways: 2 and 12 $\rightarrow$ 1; 3 and 11 $\rightarrow$ 2; 4 and 10 $\rightarrow$ 3; 5 and 9 $\rightarrow$ 4; 6 and 8 $\rightarrow$ 5; 7 $\rightarrow$ 6. Doublets: 6.
- Deck: 52 cards, 13 per suit, 4 aces, 12 face cards, 6 red face cards.
- n coins: 2^n outcomes, exactly r heads in ${}^n C_r$ ways.
- 53 of a given weekday: ordinary year $\frac{1}{7}$, leap year $\frac{2}{7}$.

THE FIVE SHORTCUTS

- “At least one” $\rightarrow 1 - P(\text{none})$.
- Drawing several balls at once $\rightarrow$ use ${}^n C_r$ on top and bottom; the ordering cancels.
- Bayes $\rightarrow$ two-line product table, then divide by the column total.
- “Expected number of ...” $\rightarrow$ indicators plus linearity, in two lines.
- Repeating games $\rightarrow$ write p in terms of itself and solve one equation.

THE TOP FIVE TRAPS

- Adding overlapping events.** $P(A) + P(B)$ is wrong unless A and B cannot both happen. Subtract $P(A \cap B)$.
- With vs without replacement.** Without replacement the second denominator drops by one — and the numerator may drop too.

3. **Answering a Bayes question with the prior.** Evidence changes the probability; a small high-risk group can dominate the positives.
4. **Conditioning by multiplying.** $P(A \mid B)$ means shrinking the sample space to B and recounting inside it. “At least one die shows a 4” gives 11 outcomes, not 6, not 12.
5. **Mixing up odds and probability.** Odds $a : b$ means a parts to b parts, so the probability has denominator $a + b$, not b .

Algebra Essentials

Equations, progressions, powers, logs, inequalities

What you need to know

WHAT YOU NEED TO KNOW

Identities

- $(a + b)^2 = a^2 + 2ab + b^2$ $(a - b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a - b)(a + b)$
- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
- If $x + \frac{1}{x} = k$: $x^2 + \frac{1}{x^2} = k^2 - 2$, $x^3 + \frac{1}{x^3} = k^3 - 3k$
- If $x - \frac{1}{x} = k$: $x^2 + \frac{1}{x^2} = k^2 + 2$, $x^3 - \frac{1}{x^3} = k^3 + 3k$

Linear equations

- $a_1x + b_1y = c_1$, $a_2x + b_2y = c_2$.
- Unique solution if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. No solution if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. Infinite if all three equal.

Quadratic $ax^2 + bx + c = 0$, $a \neq 0$

- $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, discriminant $D = b^2 - 4ac$
- $D > 0$: two real distinct. $D = 0$: two equal. $D < 0$: no real root.
- Sum of roots $\alpha + \beta = -\frac{b}{a}$. Product $\alpha\beta = \frac{c}{a}$.
- Equation from roots: $x^2 - (\text{sum})x + (\text{product}) = 0$
- $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$; $|\alpha - \beta| = \frac{\sqrt{D}}{|a|}$
- Minimum / maximum of $ax^2 + bx + c$ is at $x = -\frac{b}{2a}$; value $= c - \frac{b^2}{4a}$.

Progressions

- AP: $t_n = a + (n - 1)d$; $S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$
- GP: $t_n = ar^{n-1}$; $S_n = \frac{a(r^n - 1)}{r - 1}$ for $r \neq 1$
- Infinite GP ($|r| < 1$): $S_\infty = \frac{a}{1 - r}$
- HP: reciprocals form an AP. n -th term of HP $= \frac{1}{a + (n - 1)d}$.
- $1 + 2 + \dots + n = \frac{n(n+1)}{2}$; $1^2 + \dots + n^2 = n(n+1)\frac{2n+1}{6}$; $1^3 + \dots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$
- For two positives: $AM = \frac{a+b}{2}$, $GM = \sqrt{ab}$, $HM = \frac{2ab}{a+b}$; $GM^2 = AM \times HM$; $AM \geq GM \geq HM$

Indices

- $a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$
- $a^0 = 1$; $a^{-n} = \frac{1}{a^n}$; $a^{\frac{m}{n}} = \sqrt[n]{a^m}$

Surds

- $\frac{1}{\sqrt{a} - \sqrt{b}} = \frac{\sqrt{a} + \sqrt{b}}{a - b}$ (multiply by the conjugate)
- $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$

Logarithms (base $b > 0$, $b \neq 1$)

- $\log_b x = y \leftrightarrow b^y = x$
- $\log(mn) = \log m + \log n$; $\log\left(\frac{m}{n}\right) = \log m - \log n$; $\log(m^p) = p \log m$
- $\log_b b = 1$; $\log_b 1 = 0$; change of base: $\log_b a = \frac{\log_c a}{\log_c b} = \frac{1}{\log_a b}$
- Digits in N (base 10) $= \lfloor \log_{10} N \rfloor + 1$
- Useful: $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$, $\log_{10} 7 = 0.8451$

Inequalities

- Multiplying or dividing by a negative number flips the sign.
- $|x| < a \leftrightarrow -a < x < a$; $|x| > a \leftrightarrow x < -a$ or $x > a$
- For $x > 0$: $x + \frac{k}{x} \geq 2\sqrt{k}$, equality at $x = \sqrt{k}$.

Functions

- $(f \circ g)(x) = f(g(x))$ — inner function first.
- Inverse: swap x and y in $y = f(x)$, then solve for y .
- Domain rule: denominator $\neq 0$, and the inside of a square root ≥ 0 .
- Remainder when $f(x)$ is divided by $(x - a)$ is $f(a)$.

Warm-up

WARM-UP

build the reflex

Example 1

Solve $5x - 7 = 3x + 9$.

SOLUTION

Bring $3x$ to the left: $5x - 3x - 7 = 9$, so $2x - 7 = 9$.

Add 7: $2x = 16$.

Divide by 2: $x = 8$.

Answer: $x = 8$

Example 2

Simplify $81^{3/4} \div 27^{2/3}$.

SOLUTION

$81 = 3^4$, so $81^{3/4} = 3^{4 \times 3/4} = 3^3 = 27$.

$27 = 3^3$, so $27^{2/3} = 3^{3 \times 2/3} = 3^2 = 9$.

$27 \div 9 = 3$.

Answer: 3

Example 3

Rationalise $\frac{1}{\sqrt{5}-2}$.

SOLUTION

Multiply top and bottom by the conjugate $\sqrt{5} + 2$.

Bottom: $(\sqrt{5} - 2)(\sqrt{5} + 2) = 5 - 4 = 1$.

Top: $\sqrt{5} + 2$.

So the value is $\frac{\sqrt{5}+2}{1} = \sqrt{5} + 2$.

Answer: $\sqrt{5} + 2$

Example 4

Find the 12th term of the AP 5, 9, 13, ...

SOLUTION

$a = 5$, $d = 9 - 5 = 4$.

$t_{12} = a + 11d = 5 + 11 \times 4 = 5 + 44 = 49$.

Answer: 49

Example 5

Find the sum of the first 20 terms of 3, 7, 11, ...

SOLUTION

$$a = 3, d = 4, n = 20.$$

$$S_{20} = \frac{20}{2} [2 \times 3 + 19 \times 4] = 10[6 + 76] = 10 \times 82 = 820.$$

Answer: 820

Example 6

Find the 6th term of the GP 2, 6, 18, ..., and the sum of $8 + 4 + 2 + \dots$ to infinity.

SOLUTION

$$\text{GP: } a = 2, r = 6 \div 2 = 3. t_6 = ar^5 = 2 \times 3^5 = 2 \times 243 = 486.$$

$$\text{Infinite GP: } a = 8, r = 4 \div 8 = \frac{1}{2}.$$

$$S_{\infty} = \frac{8}{1 - \frac{1}{2}} = \frac{8}{\frac{1}{2}} = 16.$$

Answer: $t_6 = 486$; sum = 16

Example 7

Evaluate $\log_2 64 - \log_3 27$.

SOLUTION

$$2^6 = 64, \text{ so } \log_2 64 = 6.$$

$$3^3 = 27, \text{ so } \log_3 27 = 3.$$

$$6 - 3 = 3.$$

Answer: 3

Example 8

Solve $|x - 3| = 7$.

SOLUTION

Two cases.

$$\text{Case 1: } x - 3 = 7 \rightarrow x = 10.$$

$$\text{Case 2: } x - 3 = -7 \rightarrow x = -4.$$

Answer: $x = 10$ or $x = -4$

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Three pens and four notebooks cost Rs. 250. Five pens and two notebooks cost Rs. 230. Find the price of one pen and one notebook.

SOLUTION

Let one pen cost p and one notebook cost n .

$$3p + 4n = 250 \dots (i)$$

$$5p + 2n = 230 \dots (ii)$$

Multiply (ii) by 2 so the n terms match: $10p + 4n = 460 \dots (iii)$

Subtract (i) from (iii): $(10p - 3p) + (4n - 4n) = 460 - 250$

$$7p = 210, \text{ so } p = 30.$$

$$\text{Put } p = 30 \text{ in (i): } 3 \times 30 + 4n = 250 \rightarrow 90 + 4n = 250 \rightarrow 4n = 160 \rightarrow n = 40.$$

Check in (ii): $5 \times 30 + 2 \times 40 = 150 + 80 = 230$. Correct.

Answer: Pen Rs. 30, notebook Rs. 40

SHORTCUT

To kill one variable, make its coefficients equal, not “nice”. Here $4n$ and $2n$: doubling the second equation is one step. Do not cross-multiply the whole system — that creates big numbers you will misadd.

Example 10 [Infosys pattern]

One root of $x^2 - 7x + k = 0$ is 3. Find k and the other root.

SOLUTION

A root satisfies the equation. Put $x = 3$:

$$3^2 - 7 \times 3 + k = 0$$

$$9 - 21 + k = 0$$

$$-12 + k = 0, \text{ so } k = 12.$$

Now the equation is $x^2 - 7x + 12 = 0$.

Sum of roots = 7, so the other root = $7 - 3 = 4$.

Check: $4^2 - 7 \times 4 + 12 = 16 - 28 + 12 = 0$. Correct.

Answer: $k = 12$, other root = 4

Example 11 [Capgemini pattern]

If α and β are the roots of $x^2 - 5x + 3 = 0$, find (a) $\alpha^2 + \beta^2$ and (b) $\frac{1}{\alpha} + \frac{1}{\beta}$.

SOLUTION

Here $a = 1$, $b = -5$, $c = 3$.

Sum $\alpha + \beta = -\frac{b}{a} = 5$. Product $\alpha\beta = \frac{c}{a} = 3$.

$$(a) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 5^2 - 2 \times 3 = 25 - 6 = 19.$$

$$(b) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta} = \frac{5}{3}.$$

Answer: (a) 19 (b) $\frac{5}{3}$

TRAP

Sum of roots is $-\frac{b}{a}$, not $\frac{b}{a}$. In $x^2 - 5x + 3$ the value of b is -5 , so the sum is $-\frac{-5}{1} = +5$. Writing -5 here is the single most common slip in this topic.

Example 12 [Wipro pattern]

For which values of k does $4x^2 + kx + 9 = 0$ have equal roots?

SOLUTION

Equal roots means $D = 0$, where $D = b^2 - 4ac$.

$$a = 4, b = k, c = 9.$$

$$D = k^2 - 4 \times 4 \times 9 = k^2 - 144.$$

$$\text{Set } D = 0: k^2 = 144, \text{ so } k = 12 \text{ or } k = -12.$$

Answer: $k = \pm 12$

Example 13 [TCS NQT pattern]

Meera joins at a monthly salary of Rs. 24,000. Her monthly salary rises by Rs. 1,500 at the start of every year. What is her monthly salary in the 8th year, and how much does she earn in total over 8 years?

SOLUTION

Monthly salaries form an AP: $a = 24000$, $d = 1500$.

Year 8 monthly salary: $t_8 = 24000 + 7 \times 1500 = 24000 + 10500 = 34500$.

Sum of the 8 monthly figures: $S_8 = \frac{8}{2}(a + t_8) = 4(24000 + 34500) = 4 \times 58500 = 234000$.

That is the total of the 8 monthly rates. Each rate is paid for 12 months, so over 8 full years she earns $12 \times 234000 = 2808000$, that is Rs. 28,08,000.

Answer: Year-8 salary Rs. 34,500; 8-year total Rs. 28,08,000

Example 14 [Accenture pattern]

Which term of the AP 7, 11, 15, ... is 143?

SOLUTION

$a = 7$, $d = 4$.

$t_n = a + (n - 1)d = 143$

$7 + (n - 1) \times 4 = 143$

$(n - 1) \times 4 = 136$

$n - 1 = 34$

$n = 35$.

Answer: 35th term

Example 15 [Cognizant pattern]

A machine bought for Rs. 80,000 loses 25% of its value every year. What is its value after 4 years?

SOLUTION

Each year the value becomes $100\% - 25\% = 75\%$ of the year before.

So the values form a GP with $a = 80000$ and $r = 0.75$.

Value after 4 years $= 80000 \times (0.75)^4$.

$(0.75)^2 = 0.5625$.

$(0.75)^4 = 0.5625 \times 0.5625 = 0.31640625$.

$80000 \times 0.31640625 = 25312.5$.

Answer: Rs. 25,312.50

SHORTCUT

For “falls by 25% a year”, never subtract 25% four separate times. Multiply by 0.75 four times — one power, no chain of subtractions. Same for growth: +12% for n years is $\times (1.12)^n$.

Example 16 [Infosys pattern]

Find the sum of the first 10 terms of 3, 6, 12, 24, ...

SOLUTION

$a = 3$, $r = 6 \div 3 = 2$, $n = 10$.

$S_n = \frac{a(r^n - 1)}{r - 1} = 3 \frac{2^{10} - 1}{2 - 1}$.

$2^{10} = 1024$, so $2^{10} - 1 = 1023$.

$S_{10} = 3 \times 1023 \div 1 = 3069$.

Answer: 3069

Example 17 [TCS NQT pattern]Solve $9^{x+1} = 27^{x-1}$.**SOLUTION**Write both sides with base 3. $9 = 3^2$ and $27 = 3^3$.Left: $(3^2)^{x+1} = 3^{2x+2}$.Right: $(3^3)^{x-1} = 3^{3x-3}$.Equal bases means equal powers: $2x + 2 = 3x - 3$.

$$2 + 3 = 3x - 2x$$

$$x = 5.$$

Check: $9^6 = 3^{12}$ and $27^4 = 3^{12}$. Correct.**Answer:** $x = 5$ **Example 18** [Capgemini pattern]Solve $\log_2 x + \log_2(x - 2) = 3$.**SOLUTION**Add the logs: $\log_2[x(x - 2)] = 3$.Rewrite in power form: $x(x - 2) = 2^3 = 8$.

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0, \text{ so } x = 4 \text{ or } x = -2.$$

A log needs a positive input. $x = -2$ makes $\log_2 x$ undefined, so reject it.Check $x = 4$: $\log_2 4 + \log_2 2 = 2 + 1 = 3$. Correct.**Answer:** $x = 4$ **TRAP**

Every log equation must end with a check. Solving the algebra gives candidates, not answers. Any candidate that makes some log take a zero or negative input is thrown away.

Example 19 [Accenture pattern]Solve $3x - 7 \leq 5x + 3$ and state the least integer value of x .**SOLUTION**

$$3x - 7 \leq 5x + 3$$

Take $5x$ to the left and -7 to the right: $3x - 5x \leq 3 + 7$

$$-2x \leq 10$$

Divide by -2 and flip the sign: $x \geq -5$.The least integer that is ≥ -5 is -5 .**Answer:** $x \geq -5$; least integer = -5 **Example 20** [Wipro pattern]Given $f(x) = 3x - 4$ and $g(x) = x^2 + 1$, find $f(g(2))$ and $g(f(2))$.**SOLUTION**

Inner function first.

$$g(2) = 2^2 + 1 = 4 + 1 = 5. \text{ Then } f(5) = 3 \times 5 - 4 = 15 - 4 = 11.$$

$$f(2) = 3 \times 2 - 4 = 6 - 4 = 2. \text{ Then } g(2) = 2^2 + 1 = 5.$$

So $f(g(2)) = 11$ and $g(f(2)) = 5$.**Answer:** $f(g(2)) = 11, g(f(2)) = 5$

TRAP

$f(g(x))$ and $g(f(x))$ are different animals. The answers above (11 and 5) are not equal. Read the brackets from the inside out, every time.

PRACTICE · TIER 1 target: 60 s/Q

- Solve $7x + 5 = 4x + 26$.
- Solve the pair $2x + 3y = 31$, $5x - y = 18$.
- Find the roots of $x^2 - 11x + 30 = 0$.
(a) 5, 6 (b) -5, -6 (c) 3, 10 (d) 2, 15
- If the roots of $x^2 + px + 12 = 0$ are 3 and 4, find p .
- If α, β are the roots of $2x^2 - 9x + 4 = 0$, find $\alpha + \beta$ and $\alpha\beta$.
- For what value of m does $x^2 + 6x + m = 0$ have equal roots?
(a) 9 (b) 6 (c) 36 (d) 3
- Find the 25th term of the AP 8, 15, 22, ...
- How many terms are there in the AP 12, 18, 24, ..., 150?
- Find the sum of all multiples of 7 between 1 and 200.
- The 4th term of a GP is 54 and the first term is 2. Find the common ratio.
- Find the sum to infinity of $27 + 9 + 3 + \dots$
- Simplify $(32)^{2/5} \times (125)^{1/3}$.
- If $\log_{10} 2 = 0.3010$, find $\log_{10} 32$.
- Solve $2 \log_5 x = \log_5 36$, with $x > 0$.
- Solve the inequality $4 - 3x > 13$.

ANSWER KEY

- $x = 7$. $7x - 4x = 26 - 5 \rightarrow 3x = 21$.
- $x = 5, y = 7$. Multiply the second by 3: $15x - 3y = 54$; add to the first: $17x = 85 \rightarrow x = 5$, then $y = 5(5) - 18 = 7$.
- (a) 5, 6 — they add to 11 and multiply to 30. (b) is the sign-flip error, (c) and (d) multiply to 30 but do not add to 11.
- $p = -7$. Sum of roots $= 7 = -p$.
- $\alpha + \beta = \frac{9}{2}$, $\alpha\beta = \frac{4}{2} = 2$.
- (a) 9. $D = 36 - 4m = 0 \rightarrow m = 9$. (b) is b itself, (c) is b^2 , (d) is $b/2$.
176. $a = 8, d = 7$; $8 + 24 \times 7 = 8 + 168$.
24. $12 + (n - 1)6 = 150 \rightarrow n - 1 = 23$.
2842. Terms 7 to 196: $n = 28$; $S = \frac{28}{2}(7 + 196) = 14 \times 203$.
- $r = 3$. $2r^3 = 54 \rightarrow r^3 = 27$.
- 40.5. $a = 27, r = \frac{1}{3}$; $27 \div (\frac{2}{3}) = \frac{81}{2}$.
20. $32^{2/5} = 2^2 = 4$ and $125^{1/3} = 5$; $4 \times 5 = 20$.
- 1.5050. $\log 32 = 5 \log 2 = 5 \times 0.3010$.
- $x = 6$. $\log_5 x^2 = \log_5 36 \rightarrow x^2 = 36$; reject -6 .
- $x < -3$. $-3x > 9$, divide by -3 and flip.

Tier 2 — Singapore & Thailand**TIER 2**

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [LINE MAN · pattern]

A rider in Bangkok earns THB 420 on his first day on a new incentive plan. Every following day his earning rises by THB 35. On which day does his cumulative earning first reach THB 10,000?

SOLUTION

Daily earnings form an AP: $a = 420$, $d = 35$.

Cumulative earning after n days:

$$S_n = \frac{n}{2}[2 \times 420 + (n-1) \times 35] = \frac{n}{2}[840 + 35n - 35] = \frac{n}{2}[805 + 35n].$$

$$\text{Try } n = 15: S_{15} = \frac{15}{2}[805 + 525] = \frac{15}{2} \times 1330 = 15 \times 665 = 9975.$$

That is below 10,000.

$$\text{Try } n = 16: S_{16} = \frac{16}{2}[805 + 560] = 8 \times 1365 = 10920.$$

So the total crosses 10,000 on day 16.

Answer: Day 16

SHORTCUT

For “when does the running total first cross T ”, do not solve the quadratic. Estimate $n \approx T \div$ average daily amount, then test that n and $n \pm 1$. Two multiplications beat the quadratic formula every time.

Example 22 [Shopee · pattern]

A seller records 2,500 orders on day 1 of a campaign. Orders grow by 20% each day. Find the orders on day 5 and the total orders over the 5 days.

SOLUTION

GP with $a = 2500$ and $r = 1.2$.

Day 5 is the 5th term: $t_5 = 2500 \times (1.2)^4$.

$$(1.2)^2 = 1.44.$$

$$(1.2)^4 = 1.44 \times 1.44 = 2.0736.$$

$$t_5 = 2500 \times 2.0736 = 5184.$$

$$\text{Total of 5 days: } S_5 = \frac{a(r^5 - 1)}{r - 1}.$$

$$(1.2)^5 = 2.0736 \times 1.2 = 2.48832.$$

$$S_5 = 2500 \times \frac{2.48832 - 1}{0.2} = 2500 \times \frac{1.48832}{0.2} = 2500 \times 7.4416 = 18604.$$

Answer: Day 5: 5,184 orders. Five-day total: 18,604 orders

Example 23 [Grab · pattern]

A subscription team models its daily profit in Singapore dollars as $P(x) = -2x^2 + 240x - 4000$, where x is the number of premium sign-ups in a day. Find (a) the sign-ups that maximise profit and the maximum profit, and (b) the two break-even values of x .

SOLUTION

(a) For $ax^2 + bx + c$ with $a < 0$, the maximum is at $x = -\frac{b}{2a}$.

$$a = -2, b = 240.$$

$$x = -\frac{240}{2 \times (-2)} = -\frac{240}{-4} = 60.$$

$$P(60) = -2 \times 60^2 + 240 \times 60 - 4000$$

$$= -2 \times 3600 + 14400 - 4000$$

$$= -7200 + 14400 - 4000 = 3200.$$

(b) Break-even means $P(x) = 0$:

$$-2x^2 + 240x - 4000 = 0. \text{ Divide by } -2:$$

$$x^2 - 120x + 2000 = 0$$

$$D = 120^2 - 4 \times 1 \times 2000 = 14400 - 8000 = 6400; \sqrt{D} = 80.$$

$$x = \frac{120 \pm 80}{2} \rightarrow x = \frac{200}{2} = 100 \text{ or } x = \frac{40}{2} = 20.$$

Answer: (a) 60 sign-ups, profit S\$3,200 (b) break-even at $x = 20$ and $x = 100$

Notice the shape: profit is zero at 20 and at 100, and the peak sits exactly midway at 60. The vertex of a parabola is always the midpoint of its two roots. That is a free check.

Example 24 [DBS · pattern]

A fund grows 8% a year. How many **full** years must an investment stay in before it is worth at least double its starting value? Use $\log_{10} 2 = 0.3010$ and $\log_{10} 1.08 = 0.0334$.

SOLUTION

Let the start value be A . After n years it is $A(1.08)^n$.

Doubling: $A(1.08)^n = 2A \rightarrow (1.08)^n = 2$.

Take $\log_{10}$ of both sides: $n \log_{10} 1.08 = \log_{10} 2$.

$n = 0.3010 \div 0.0334 = 9.01$ years.

Exact doubling happens at $n = 9.01$, which is **more** than 9.

So after 9 full years the fund is still a hair short of double.

The next full year carries it past double.

So 10 full years are needed.

Answer: 10 full years (exact doubling time ≈ 9.01 years)

Example 25 [GIC · pattern]

Three numbers are in AP. Their sum is 27 and their product is 585. Find the numbers.

SOLUTION

Take the three terms as $a - d$, a , $a + d$. This makes the sum easy.

Sum: $(a - d) + a + (a + d) = 3a = 27 \rightarrow a = 9$.

Product: $(a - d) \times a \times (a + d) = a(a^2 - d^2) = 585$.

$9(81 - d^2) = 585$

$81 - d^2 = \frac{585}{9} = 65$

$d^2 = 81 - 65 = 16$, so $d = 4$ or $d = -4$.

With $d = 4$: the numbers are 5, 9, 13.

Check: $5 + 9 + 13 = 27$ and $5 \times 9 \times 13 = 585$. Correct.

Answer: 5, 9, 13

SHORTCUT

Three terms in AP: use $a - d$, a , $a + d$. Four terms: use $a - 3d$, $a - d$, $a + d$, $a + 3d$ (common difference $2d$).

Three terms in GP: use $\frac{a}{r}$, a , ar . The middle-symmetry makes the sum or product collapse in one line.

Example 26 [Agoda · pattern]

A shuttle covers the same stretch three times: at 30 km/h, then 40 km/h, then 60 km/h. Find its average speed for the whole trip.

SOLUTION

Equal distances, different speeds $\rightarrow$ use the harmonic mean, not the plain average.

Let each stretch be d km. Total distance = $3d$.

Time = $\frac{d}{30} + \frac{d}{40} + \frac{d}{60}$.

LCM of 30, 40, 60 is 120.

$\frac{d}{30} = 4\frac{d}{120}$, $\frac{d}{40} = 3\frac{d}{120}$, $\frac{d}{60} = 2\frac{d}{120}$.

Total time = $\frac{4d+3d+2d}{120} = 9\frac{d}{120}$.

Average speed = $\frac{3d}{9d/120} = 3d \times \frac{120}{9d} = \frac{360}{9} = 40$ km/h.

Answer: 40 km/h

TRAP

The plain average of 30, 40 and 60 is $\frac{130}{3} = 43.33$ km/h. That is wrong. Equal distances need the harmonic mean; equal times need the plain mean. Ask which one is equal before you average.

Example 27 [Sea · pattern]

Given $5^x = 3^y = 15^z$ with x, y, z all non-zero, show that $\frac{1}{z} = \frac{1}{x} + \frac{1}{y}$. Then find z when $x = 10$ and $y = 15$.

SOLUTION

Let the common value be k , so $5^x = k$, $3^y = k$ and $15^z = k$.

Raise each to the reciprocal power to free the base:

$$5 = k^{1/x}, \quad 3 = k^{1/y}, \quad 15 = k^{1/z}.$$

But $15 = 5 \times 3$, so

$$k^{1/z} = k^{1/x} \times k^{1/y} = k^{1/x+1/y}.$$

Same base, so the powers match:

$$\frac{1}{z} = \frac{1}{x} + \frac{1}{y}$$

Now put $x = 10$ and $y = 15$:

$$\frac{1}{z} = \frac{1}{10} + \frac{1}{15}.$$

LCM of 10 and 15 is 30: $\frac{1}{10} = \frac{3}{30}$ and $\frac{1}{15} = \frac{2}{30}$.

$$\frac{1}{z} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}, \text{ so } z = 6.$$

Answer: $\frac{1}{z} = \frac{1}{x} + \frac{1}{y}; \quad z = 6$

Example 28 [SCB · pattern]

A Bangkok food stall makes THB 38 profit per plate and pays THB 4,560 rent a day. The kitchen can serve at most 260 plates a day. For what numbers of plates is the daily net profit at least THB 1,900? What is the best possible daily net profit?

SOLUTION

Let n be the plates sold in a day.

$$\text{Net profit} = 38n - 4560.$$

$$\text{Condition: } 38n - 4560 \geq 1900$$

$$38n \geq 1900 + 4560 = 6460$$

$$n \geq \frac{6460}{38} = 170.$$

$$\text{Capacity gives } n \leq 260.$$

$$\text{So } 170 \leq n \leq 260.$$

$$\text{Best case } n = 260: \text{ net profit} = 38 \times 260 - 4560 = 9880 - 4560 = 5320.$$

Answer: $170 \leq n \leq 260$ plates; maximum net profit THB 5,320

Example 29 [Razer · pattern]

For $f(x) = \frac{2x+3}{x-1}$, find $f^{-1}(x)$ and verify it using $x = 4$.

SOLUTION

$$\text{Write } y = \frac{2x+3}{x-1}.$$

$$\text{Cross-multiply: } y(x-1) = 2x+3$$

$$yx - y = 2x + 3$$

$$\text{Collect } x \text{ terms on one side: } yx - 2x = y + 3$$

$$x(y-2) = y+3$$

$$x = \frac{y+3}{y-2}.$$

$$\text{Swap the letters: } f^{-1}(x) = \frac{x+3}{x-2}.$$

$$\text{Verify: } f(4) = \frac{2 \times 4 + 3}{4 - 1} = \frac{11}{3}.$$

$$f^{-1}\left(\frac{11}{3}\right) = \frac{\frac{11}{3}+3}{\frac{11}{3}-2} = \frac{\frac{11}{3}+\frac{9}{3}}{\frac{11}{3}-\frac{6}{3}} = \frac{\frac{20}{3}}{\frac{5}{3}} = \frac{20}{5} = 4.$$

It returns 4, so the inverse is right.

Answer: $f^{-1}(x) = \frac{x+3}{x-2}$

Example 30 [Grab · pattern]

Three **positive** numbers are in AP with sum 15. If 1, 4 and 19 are added to them in order, the results are in GP. Find the original numbers.

SOLUTION

Let the AP be $a - d, a, a + d$.

Sum: $3a = 15 \rightarrow a = 5$. So the numbers are $5 - d, 5, 5 + d$.

After adding: $(5 - d + 1), (5 + 4), (5 + d + 19)$, i.e. $6 - d, 9, 24 + d$.

GP condition: the middle term squared equals the product of the outer two.

$$9^2 = (6 - d)(24 + d)$$

$$81 = 144 + 6d - 24d - d^2$$

$$81 = 144 - 18d - d^2$$

$$d^2 + 18d - 63 = 0$$

$$D = 18^2 + 4 \times 63 = 324 + 252 = 576; \sqrt{D} = 24.$$

$$d = \frac{-18 \pm 24}{2} \rightarrow d = 3 \text{ or } d = -21.$$

Test $d = -21$: the numbers would be $5 - (-21) = 26, 5, 5 + (-21) = -16$.

The third one is negative, and the question says all three are positive. Reject $d = -21$.

(It does pass the GP test — adding gives 27, 9, 3, ratio $\frac{1}{3}$ — which is exactly why the word **positive** is in the question.)

Take $d = 3$: the numbers are 2, 5, 8.

Check: adding gives 3, 9, 27, which is a GP with ratio 3, and all three originals are positive. Correct.

Answer: 2, 5, 8

PRACTICE · TIER 2 target: 100 s/Q

1. A courier hub handles 180 parcels on Monday and 24 more each following day. On which day does the running total first pass 1,500 parcels?
2. A start-up's monthly users grow by 25% a month. If January has 4,096 users, find the users in May and the total across January to May.
3. A vendor's daily profit is $P(x) = -5x^2 + 300x - 2500$ for x combo meals. Find the profit-maximising x , the maximum profit, and both break-even points.
4. Three numbers in GP have product 216 and sum 26. Find them.
5. A boat covers 60 km downstream at 20 km/h and returns at 12 km/h. Find the average speed for the round trip.
6. A deposit of S\$5,000 grows at 6% a year. Using $\log_{10} 1.06 = 0.0253$ and $\log_{10} 1.5 = 0.1761$, find how many full years until it first exceeds S\$7,500.
7. If $f(x) = \frac{3x-1}{x+2}$, find $f^{-1}(5)$.
8. A river-cruise operator charges THB 1,200 per seat when 40 seats are booked. For every extra seat booked above 40, the price **every** passenger pays drops by THB 10. If n extra seats are booked, revenue is $(40 + n)(1200 - 10n)$. Find the n that maximises revenue, and that maximum revenue.
9. The sum of the first n terms of an AP is $3n^2 + 5n$. Find the 10th term.
10. An office lamp loses 15% of its brightness every 1,000 hours. After how many full 1,000-hour blocks does brightness fall below half? Use $\log_{10} 0.85 = -0.0706$ and $\log_{10} 0.5 = -0.3010$.
11. Solve for x : $2^{x+2} + 2^x = 80$.
12. A warehouse has 4 more shelves than aisles. Total storage slots equal 96, where slots = aisles $\times$ shelves. Find the number of aisles.

ANSWER KEY

- Day 7. $S_n = \frac{n}{2}[360 + 24(n-1)] = n(168 + 12n)$; $S_6 = 6(240) = 1440 < 1500$ and $S_7 = 7(252) = 1764 \geq 1500$.
- May = $4096(1.25)^4 = 4096 \times 2.44140625 = 10000$ users. Total = $4096 \frac{1.25^5 - 1}{0.25} = 4096 \times 8.20703125 = 33616$ users.
- $x = 30$; max = $-5(900) + 9000 - 2500 = 2000$. Break-even: $x^2 - 60x + 500 = 0 \rightarrow x = 10$ or 50 .
- 2, 6, 18. Take $\frac{a}{r}$, a , ar : product $a^3 = 216 \rightarrow a = 6$; $\frac{6}{r} + 6 + 6r = 26 \rightarrow 3r^2 - 10r + 3 = 0 \rightarrow r = 3$.
- 15 km/h. Equal distances: $HM = \frac{2 \times 20 \times 12}{20 + 12} = \frac{480}{32}$.
- 7 years. Need $(1.06)^n > 1.5 \rightarrow n > \frac{\ln 1.5}{\ln 1.06} = 6.96$.
- $f^{-1}(5) = -\frac{11}{2}$. Solve $\frac{3x-1}{x+2} = 5$: $3x-1 = 5x+10 \rightarrow -11 = 2x \rightarrow x = -\frac{11}{2}$. Check: $f(-\frac{11}{2}) = \frac{-\frac{33}{2}-1}{-\frac{11}{2}+2} = \frac{-\frac{35}{2}}{-\frac{7}{2}} = 5$.
- $n = 40$. Revenue = $48000 + 800n - 10n^2$; peak at $n = \frac{800}{20} = 40$. Then 80 seats at THB 800 each: $80 \times 800 = 64000$, i.e. THB 64,000.
62. $t_{10} = S_{10} - S_9 = (300 + 50) - (243 + 45) = 350 - 288$.
- 5 blocks. Need $n \times (-0.0706) < -0.3010 \rightarrow n > 4.26$.
- $x = 4$. $2^x(4+1) = 80 \rightarrow 2^x = 16$.
- 8 aisles. $a(a+4) = 96 \rightarrow a^2 + 4a - 96 = 0 \rightarrow (a+12)(a-8) = 0$.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Goldman Sachs · pattern]

Given $x + \frac{1}{x} = 5$, find $x^5 + \frac{1}{x^5}$ without finding x .

SOLUTION

Insight: power sums obey a recursion. $(x^a + x^{-a})(x^b + x^{-b}) = (x^{a+b} + x^{-(a+b)}) + (x^{a-b} + x^{-(a-b)})$.

Write $p_n = x^n + \frac{1}{x^n}$. We are given $p_1 = 5$.

Step 1. Square p_1 :

$$(x + \frac{1}{x})^2 = x^2 + 2 + \frac{1}{x^2} = 25, \text{ so } p_2 = 25 - 2 = 23.$$

Step 2. Cube p_1 :

$$(x + \frac{1}{x})^3 = x^3 + \frac{1}{x^3} + 3(x + \frac{1}{x}), \text{ so } 125 = p_3 + 3 \times 5.$$

$$p_3 = 125 - 15 = 110.$$

Step 3. Multiply p_2 and p_3 :

$$p_2 \times p_3 = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) = x^5 + \frac{1}{x} + x + \frac{1}{x^5} = p_5 + p_1.$$

$$23 \times 110 = 2530.$$

$$p_5 = 2530 - p_1 = 2530 - 5 = 2525.$$

Answer: $x^5 + \frac{1}{x^5} = 2525$

The same trick chains upward for free: $p_4 = p_2^2 - 2 = 529 - 2 = 527$, and $p_6 = p_3^2 - 2 = 12100 - 2 = 12098$. You never need the value of x (which here is irrational).

Example 32 [Google · pattern]

Find $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{99 \times 100}$.

SOLUTION

Insight: split each term into a difference so that neighbours cancel.

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Check on one term: $\frac{1}{2} - \frac{1}{3} = \frac{3-2}{6} = \frac{1}{6} = \frac{1}{2 \times 3}$. Correct.

So the sum becomes

$$\left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{99} - \frac{1}{100}\right).$$

Every middle piece appears once with a + and once with a -. They all cancel.

What survives is the very first and the very last:

$$1 - \frac{1}{100} = \frac{99}{100} = 0.99.$$

Answer: $\frac{99}{100}$

Example 33 [Amazon · pattern]

A fulfilment centre's daily cost in thousands of rupees is $C(x) = 3x + \frac{300}{x}$, where $x > 0$ is the number of loading bays kept open. Find the x that minimises cost and the minimum cost.

SOLUTION

Insight: a sum of the form “term + constant/term” is smallest when the two terms are equal. That is AM–GM.

For positive u, v : $\frac{u+v}{2} \geq \sqrt{uv}$, so $u + v \geq 2\sqrt{uv}$, with equality only when $u = v$.

Take $u = 3x$ and $v = \frac{300}{x}$.

$uv = 3x \times \frac{300}{x} = 900$, a constant — this is what makes the trick work.

So $C(x) \geq 2\sqrt{900} = 2 \times 30 = 60$.

Equality when $3x = \frac{300}{x} \rightarrow 3x^2 = 300 \rightarrow x^2 = 100 \rightarrow x = 10$ (take the positive root).

Check: $C(10) = 3 \times 10 + \frac{300}{10} = 30 + 30 = 60$.

Check a neighbour: $C(9) = 27 + 33.33 = 60.33$ and $C(11) = 33 + 27.27 = 60.27$. Both above 60.

Answer: $x = 10$ bays; minimum cost Rs. 60,000

SHORTCUT

AM–GM works only when the product of the parts is a constant. If it is not, force it: split a term into equal pieces. To minimise $2x + \frac{50}{x^2}$, write it as $x + x + \frac{50}{x^2}$ — now the product $x \times x \times \frac{50}{x^2} = 50$ is constant, and the bound is $3\sqrt[3]{50}$.

Example 34 [D. E. Shaw · pattern]

If α and β are the roots of $x^2 - 6x + 7 = 0$, find $\alpha^4 + \beta^4$ without solving the equation.

SOLUTION

Insight: build the answer from the sum and product alone, one power at a time.

$$\alpha + \beta = 6, \quad \alpha\beta = 7.$$

Step 1. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 36 - 14 = 22$.

Step 2. Square that result:

$$(\alpha^2 + \beta^2)^2 = \alpha^4 + 2\alpha^2\beta^2 + \beta^4$$

$$22^2 = \alpha^4 + \beta^4 + 2(\alpha\beta)^2$$

$$484 = \alpha^4 + \beta^4 + 2 \times 49$$

$$484 = \alpha^4 + \beta^4 + 98$$

$$\alpha^4 + \beta^4 = 386.$$

Sanity check: the roots are $3 \pm \sqrt{2}$, so $\alpha \approx 4.414$ and $\beta \approx 1.586$.

$4.414^4 \approx 379.6$ and $1.586^4 \approx 6.3$; the total is about 386. Correct.

Answer: 386

Example 35 [Microsoft · pattern]

Find the value of $x = \sqrt{20 + \sqrt{20 + \sqrt{20 + \dots}}}$, continuing for ever.

SOLUTION

Insight: the tail of the expression is a copy of the whole expression. Name it and the infinity disappears.

Inside the first square root sits $20 +$ (the same infinite expression), which is $20 + x$.

So $x = \sqrt{20 + x}$.

Square both sides: $x^2 = 20 + x$

$$x^2 - x - 20 = 0$$

$(x - 5)(x + 4) = 0$, so $x = 5$ or $x = -4$.

A square root is never negative, so reject -4 .

$x = 5$.

Feel the convergence: $\sqrt{20} = 4.472$, then $\sqrt{24.472} = 4.947$, then $\sqrt{24.947} = 4.995$. It is climbing to 5.

Answer: $x = 5$

Example 36 [Adobe · pattern]

Evaluate $\log_2 3 \times \log_3 4 \times \log_4 5 \times \dots \times \log_{63} 64$.

SOLUTION

Insight: change every log to one common base and the product telescopes.

Change of base: $\log_b a = \frac{\log a}{\log b}$, using any fixed base (say base 10).

The product becomes

$$\frac{\log 3}{\log 2} \times \frac{\log 4}{\log 3} \times \frac{\log 5}{\log 4} \times \dots \times \frac{\log 64}{\log 63}$$

Each numerator cancels the next denominator: $\log 3$ cancels, $\log 4$ cancels, and so on up to $\log 63$.

What is left is $\frac{\log 64}{\log 2} = \log_2 64$.

$2^6 = 64$, so the value is 6.

Answer: 6

Example 37 [Uber · pattern]

Two positive numbers a and b add to 18. What is the largest possible value of a^2b ?

SOLUTION

Insight: to use AM–GM on a^2b , split a into two equal halves so that all three parts can be made equal.

Write $a + b = \frac{a}{2} + \frac{a}{2} + b = 18$.

AM–GM on the three positive parts:

$$\frac{\frac{a}{2} + \frac{a}{2} + b}{3} \geq \sqrt[3]{\frac{a}{2} \times \frac{a}{2} \times b}$$

$$\frac{18}{3} \geq \sqrt[3]{\frac{a^2b}{4}}$$

$$6 \geq \sqrt[3]{\frac{a^2b}{4}}$$

Cube both sides: $216 \geq \frac{a^2b}{4}$, so $a^2b \leq 864$.

Equality needs all three parts equal: $\frac{a}{2} = b$.

With $a + b = 18$ and $a = 2b$: $2b + b = 18 \rightarrow b = 6, a = 12$.

Check: $a^2b = 144 \times 6 = 864$.

Check a neighbour: $a = 13, b = 5$ gives $169 \times 5 = 845 < 864$. And $a = 11, b = 7$ gives $121 \times 7 = 847 < 864$.

Answer: 864, at $a = 12$ and $b = 6$

Example 38 [Goldman Sachs · pattern]

Find $S = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$ continuing for ever.

SOLUTION

Insight: this is an AP (1, 2, 3, ...) times a GP ($\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$). Subtract a shifted copy of the sum and it collapses into a plain GP.

$$S = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$$

Multiply by $\frac{1}{2}$:

$$\frac{S}{2} = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \dots$$

Subtract the second line from the first, matching denominators:

$$S - \frac{S}{2} = \frac{1}{2} + \left(\frac{2}{4} - \frac{1}{4}\right) + \left(\frac{3}{8} - \frac{2}{8}\right) + \left(\frac{4}{16} - \frac{3}{16}\right) + \dots$$

$$\frac{S}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

The right side is an infinite GP with $a = \frac{1}{2}$ and $r = \frac{1}{2}$:

$$\frac{S}{2} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$$

So $S = 2$.

Partial check: $0.5 + 0.5 + 0.375 + 0.25 + 0.15625 = 1.78125$, and later terms keep adding smaller amounts towards 2.

Answer: $S = 2$

PRACTICE · TIER 3 target: 4 min/Q

1. If $x - \frac{1}{x} = 3$, find $x^3 - \frac{1}{x^3}$ and $x^4 + \frac{1}{x^4}$.
2. Find $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{99 \times 101}$.
3. For $x > 0$, find the least value of $4x + \frac{9}{x}$, and the x where it occurs.
4. If α, β are the roots of $x^2 - 4x + 1 = 0$, find $\alpha^3 + \beta^3$ and $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$.
5. Find the value of $x = \sqrt[3]{6 + \sqrt[3]{6 + \sqrt[3]{6 + \dots}}}$ — hint: set up the cubic and look for an integer root.
6. Evaluate $\log_3 5 \times \log_5 7 \times \log_7 9 \times \log_9 81$.
7. Two positive numbers have sum 30. Maximise ab^2 .
8. Find $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

ANSWER KEY

1. $x^3 - \frac{1}{x^3} = 36$ and $x^4 + \frac{1}{x^4} = 119$. From $x - \frac{1}{x} = 3$: cube it, $x^3 - \frac{1}{x^3} - 3(x - \frac{1}{x}) = 27$, so $x^3 - \frac{1}{x^3} = 27 + 9 = 36$. Also $x^2 + \frac{1}{x^2} = 3^2 + 2 = 11$, so $x^4 + \frac{1}{x^4} = 11^2 - 2 = 119$.
2. $\frac{50}{101}$. Here $\frac{1}{(2k-1)(2k+1)} = \frac{1}{2} \left[\frac{1}{2k-1} - \frac{1}{2k+1} \right]$. The sum telescopes to $\frac{1}{2} \left(1 - \frac{1}{101} \right) = \frac{1}{2} \times \frac{100}{101}$. Note the extra factor of $\frac{1}{2}$ — it is the gap between the two factors.
3. Least value 12, at $x = \frac{3}{2}$. AM-GM: $4x + \frac{9}{x} \geq 2\sqrt{4x \times \frac{9}{x}} = 2\sqrt{36} = 12$; equality when $4x = \frac{9}{x} \rightarrow x^2 = \frac{9}{4}$.
4. $\alpha^3 + \beta^3 = 52$ and $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 14$. Sum = 4, product = 1. $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = 64 - 12 = 52$. And $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{16-2}{1} = 14$.
5. $x = 2$. The tail repeats, so $x = \sqrt[3]{6 + x} \rightarrow x^3 = x + 6 \rightarrow x^3 - x - 6 = 0$. Test $x = 2$: $8 - 2 - 6 = 0$. Factor: $(x - 2)(x^2 + 2x + 3) = 0$; the quadratic has $D = 4 - 12 < 0$, so 2 is the only real root.
6. 4. Chain to base 3: the product telescopes to $\log_3 81 = 4$, since $3^4 = 81$. Each log's argument becomes the next log's base, so all middle logs cancel.
7. 4000, at $a = 10, b = 20$. Split b : $a + \frac{b}{2} + \frac{b}{2} = 30$, so $10 \geq \sqrt[3]{ab^2/4}$, giving $ab^2 \leq 4000$; equality at $a = \frac{b}{2}$.
8. $\frac{3}{4}$. Let $S = \sum_{n=1}^{\infty} \frac{n}{3^n}$. Then $\frac{S}{3} = \sum_{n=1}^{\infty} \frac{n}{3^{n+1}}$; subtracting gives $\frac{2S}{3} = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots = \frac{1/3}{1-1/3} = \frac{1}{2}$, so $S = \frac{3}{4}$.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 90 s/Q · 18 questions · 27 min

1. Solve $6x - 11 = 2x + 21$.
2. If α, β are the roots of $3x^2 - 12x + 5 = 0$, find $\alpha + \beta$.
3. Find the 18th term of the AP 4, 11, 18, ...
4. Simplify $(64)^{2/3} \div (16)^{1/2}$.
5. Solve $\log_3(x + 6) = 2$.
6. A car's resale value drops 20% a year from Rs. 9,00,000. Find its value after 3 years.
7. Find the sum $2 + 4 + 6 + \dots + 100$.
8. Solve $5 - 2x \geq 11$.
9. If $f(x) = x^2 - 3x$, find $f(f(2))$.
10. The sum to infinity of a GP is 45 and its first term is 30. Find the common ratio.
11. Find k so that $x^2 - kx + 25 = 0$ has equal roots.
12. If $x + \frac{1}{x} = 4$, find $x^3 + \frac{1}{x^3}$.
13. A phone plan costs THB 350 a month plus THB 2 per GB. For how many GB is the bill at most THB 500?
14. Find the remainder when $f(x) = x^4 - 5x^2 + 6x - 9$ is divided by $(x - 2)$.
15. Three numbers in AP have sum 45 and the largest is twice the smallest. Find them.
16. Find $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{20 \times 21}$.
17. For $x > 0$, find the minimum of $x + \frac{49}{x}$.
18. Solve $3^{2x} - 10 \times 3^x + 9 = 0$.

ANSWER KEY

1. $x = 8$. $4x = 32$.
2. Sum $= -\frac{-12}{3} = 4$.
3. 123 . $4 + 17 \times 7 = 4 + 119$.
4. 4 . $64^{2/3} = 4^2 = 16$; $16^{1/2} = 4$; $16 \div 4 = 4$.
5. $x = 3$. $x + 6 = 3^2 = 9$.
6. Rs. 4,60,800. $900000 \times (0.8)^3 = 900000 \times 0.512$.
7. 2550. 50 terms; $\frac{50}{2}(2 + 100) = 25 \times 102$.
8. $x \leq -3$. $-2x \geq 6$, divide by -2 and flip.
9. 10. $f(2) = 4 - 6 = -2$; then $f(-2) = (-2)^2 - 3(-2) = 4 + 6 = 10$.
10. $r = \frac{1}{3}$. $\frac{30}{1-r} = 45 \rightarrow 1 - r = \frac{2}{3}$.
11. $k = \pm 10$. $k^2 = 100$.
12. 52. $4^3 - 3 \times 4 = 64 - 12$.
13. At most 75 GB. $350 + 2g \leq 500 \rightarrow 2g \leq 150$.
14. -1 . Remainder $= f(2) = 16 - 20 + 12 - 9 = -1$.
15. 10, 15, 20. Take $a - d, a, a + d$: $3a = 45 \rightarrow a = 15$. Largest is twice smallest: $15 + d = 2(15 - d) \rightarrow 3d = 15 \rightarrow d = 5$.
16. $\frac{19}{42}$. Telescoping: $\frac{1}{2} - \frac{1}{21} = \frac{21-2}{42}$.
17. 14, at $x = 7$. AM-GM: $2\sqrt{49} = 14$.
18. $x = 0$ or $x = 2$. Let $t = 3^x$: $t^2 - 10t + 9 = 0 \rightarrow t = 1$ or 9 , so $3^x = 3^0$ or 3^2 .

ONE-PAGE REVISION CARD

Quadratics — $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Sum = $-\frac{b}{a}$, product = $\frac{c}{a}$. $D > 0$ distinct, $D = 0$ equal, $D < 0$ none. Vertex at $x = -\frac{b}{2a}$ — always the midpoint of the two roots.

AP — $t_n = a + (n-1)d$; $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$. Three terms: $a-d, a, a+d$.

GP — $t_n = ar^{n-1}$; $S_n = \frac{a(r^n-1)}{r-1}$; $S_\infty = \frac{a}{1-r}$ for $|r| < 1$. Three terms: $\frac{a}{r}, a, ar$.

Means — AM = $\frac{a+b}{2}$, GM = $\sqrt{ab}$, HM = $\frac{2ab}{a+b}$, and $GM^2 = AM \times HM$, with $AM \geq GM \geq HM$.

Power sums — $x + \frac{1}{x} = k \rightarrow x^2 + \frac{1}{x^2} = k^2 - 2$, $x^3 + \frac{1}{x^3} = k^3 - 3k$, and $p_{m+n} = p_m p_n - p_{m-n}$.

Logs — $\log(mn) = \log m + \log n$; $\log(m^p) = p \log m$; $\log_b a = \frac{1}{\log_a b}$; digits of $N = \lfloor \log_{10} N \rfloor + 1$.

Indices — $a^{\frac{m}{n}} = \sqrt[n]{a^m}$; $a^{-n} = \frac{1}{a^n}$; equal bases $\rightarrow$ equal powers.

Inequalities — flip on multiply/divide by a negative. For $x > 0$: $x + \frac{k}{x} \geq 2\sqrt{k}$ at $x = \sqrt{k}$.

Shortcuts worth memorising

- Growth/decay of $p\%$ for n periods: multiply by $(1 \pm \frac{p}{100})^n$. Never repeat-subtract.
- “Running total first crosses T ”: estimate n , then test n and $n \pm 1$. Skip the quadratic.
- $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ — telescopes. With a gap of 2, an extra $\frac{1}{2}$ appears in front.
- AM–GM only works when the product of the parts is constant. Split a term into equal pieces to force it.
- Nested radical or continued fraction: name the whole thing x , spot the copy inside, solve.

Top 5 traps

1. Sum of roots is $-\frac{b}{a}$. In $x^2 - 5x + 3$ it is $+5$, not -5 .
2. Dividing an inequality by a negative flips the sign. $-2x \leq 10$ gives $x \geq -5$.
3. Log solutions must be re-checked: any root making a log’s input zero or negative is rejected.
4. $f(g(x)) \neq g(f(x))$ in general. Work from the innermost bracket outward.
5. Equal distances at different speeds need the harmonic mean, not the plain average.

Geometry & Mensuration

Angles, shapes, areas, solids, coordinates, heights

What you need to know

WHAT YOU NEED TO KNOW

Lines and angles

- Angles on a straight line add to 180° . Angles round a point add to 360° .
- Vertically opposite angles are equal.
- Parallel lines cut by a transversal: corresponding angles equal, alternate angles equal, co-interior (same-side) angles add to 180° .

Triangles

- Angle sum = 180° . Exterior angle = sum of the two opposite interior angles.
- Sides a, b, c form a triangle only if $a + b > c$ for every pair.
- Pythagoras: $\text{hyp}^2 = \text{leg}_1^2 + \text{leg}_2^2$. Triples: $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, $(9, 40, 41)$, $(20, 21, 29)$ and their multiples.
- Area = $\frac{1}{2} \times \text{base} \times \text{height}$.
- Heron: $s = \frac{a+b+c}{2}$, Area = $\sqrt{s(s-a)(s-b)(s-c)}$.
- Equilateral side a : Area = $\frac{\sqrt{3}}{4}a^2$, height = $\frac{\sqrt{3}}{2}a$.
- Similar triangles: sides in ratio $k \rightarrow$ perimeters in ratio k , areas in ratio k^2 .
- Median splits a triangle into two equal areas. The three medians meet at the centroid, which divides each median 2 : 1 and cuts the triangle into 6 equal areas.
- Midpoint triangle (joining the three midpoints) has $\frac{1}{4}$ of the area.
- In a right triangle: inradius $r = \frac{a+b-c}{2}$, circumradius $R = \frac{c}{2}$ (c = hypotenuse).

Circles (radius r , diameter $d = 2r$)

- Circumference = $2\pi r$; Area = πr^2 . Use $\pi = \frac{22}{7}$ when r is a multiple of 7.
- Chord of length L at distance h from the centre: $r^2 = h^2 + \left(\frac{L}{2}\right)^2$.
- Sector of angle θ : arc = $\frac{\theta}{360} \times 2\pi r$, area = $\frac{\theta}{360} \times \pi r^2$.
- Angle at the centre = $2 \times$ angle at the circumference on the same arc. Angle in a semicircle = 90° .
- Tangent is perpendicular to the radius at the point of contact. Two tangents from an outside point are equal.
- Centres d apart, radii r_1, r_2 : direct common tangent = $\sqrt{d^2 - (r_1 - r_2)^2}$, transverse = $\sqrt{d^2 - (r_1 + r_2)^2}$.

Polygons (n sides, regular)

- Interior angle sum = $(n - 2) \times 180^\circ$. Each interior angle = $\frac{(n-2) \times 180}{n}$.
- Each exterior angle = $\frac{360}{n}$; the exterior angles always total 360° .
- Number of diagonals = $\frac{n(n-3)}{2}$.

2D areas

- Square side a : area a^2 , perimeter $4a$, diagonal $a\sqrt{2}$.
- Rectangle $l \times b$: area lb , perimeter $2(l + b)$, diagonal $\sqrt{l^2 + b^2}$.
- Parallelogram: base $\times$ height. Rhombus: $\frac{1}{2}d_1d_2$.
- Trapezium: $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$.

3D solids

Solid	Volume	Curved / lateral SA	Total SA
Cube, edge a	a^3	$4a^2$	$6a^2$

Cuboid l, b, h	lbh	$2h(l + b)$	$2(lb + bh + hl)$
Cylinder r, h	$\pi r^2 h$	$2\pi r h$	$2\pi r(r + h)$
Cone r, h, l	$\frac{1}{3}\pi r^2 h$	$\pi r l$	$\pi r(r + l)$
Sphere r	$\frac{4}{3}\pi r^3$	–	$4\pi r^2$
Hemisphere r	$\frac{2}{3}\pi r^3$	$2\pi r^2$	$3\pi r^2$

- Cone slant: $l = \sqrt{r^2 + h^2}$. Cuboid space diagonal = $\sqrt{l^2 + b^2 + h^2}$.
- Frustum (R, r, h): $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$.
- Melting or recasting keeps the **volume** the same, never the surface area.

Coordinate geometry

- Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Midpoint = $(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$.
- Section formula (ratio $m : n$): $(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n})$.
- Slope = $\frac{y_2 - y_1}{x_2 - x_1}$. Parallel: equal slopes. Perpendicular: $m_1 m_2 = -1$.
- Line: $y = mx + c$; intercept form $\frac{x}{a} + \frac{y}{b} = 1$.
- Area of triangle = $\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
- Distance from (x_0, y_0) to $ax + by + c = 0$ is $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

Trigonometry

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	–

- $\sin^2 \theta + \cos^2 \theta = 1$; $\tan \theta = \frac{\sin \theta}{\cos \theta}$.
- Heights and distances: $\tan(\text{angle of elevation}) = \frac{\text{height}}{\text{horizontal distance}}$.
- Angle of depression from a top point equals the angle of elevation from the bottom point.
- Useful decimals: $\sqrt{2} = 1.414$, $\sqrt{3} = 1.732$, $\frac{1}{\sqrt{3}} = 0.577$.

Warm-up

WARM-UP

build the reflex

Example 1

Three angles on one side of a straight line are $3x$, $2x$ and 40° . Find x .

SOLUTION

Angles on a straight line add to 180° .

$$3x + 2x + 40 = 180$$

$$5x = 180 - 40 = 140$$

$$x = \frac{140}{5} = 28^\circ.$$

Answer: $x = 28^\circ$

Example 2

The angles of a triangle are in the ratio 2 : 3 : 4. Find them.

SOLUTION

Let the angles be $2k, 3k, 4k$.

$$2k + 3k + 4k = 180 \rightarrow 9k = 180 \rightarrow k = 20.$$

$$\text{Angles: } 2 \times 20 = 40^\circ, 3 \times 20 = 60^\circ, 4 \times 20 = 80^\circ.$$

Check: $40 + 60 + 80 = 180$. Correct.

Answer: $40^\circ, 60^\circ, 80^\circ$

Example 3

A right triangle has legs 9 cm and 12 cm. Find the hypotenuse.

SOLUTION

$$\text{hyp}^2 = 9^2 + 12^2 = 81 + 144 = 225.$$

$$\text{hyp} = \sqrt{225} = 15 \text{ cm.}$$

(This is the (3, 4, 5) triple times 3.)

Answer: 15 cm

Example 4

Find the area of a triangle of base 14 cm and height 9 cm.

SOLUTION

$$\text{Area} = \frac{1}{2} \times 14 \times 9 = 7 \times 9 = 63 \text{ cm}^2.$$

Answer: 63 cm^2

Example 5

A circle has radius 7 cm. Find its area and circumference. Take $\pi = \frac{22}{7}$.

SOLUTION

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 22 \times 7 = 154 \text{ cm}^2.$$

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm.}$$

Answer: Area 154 cm^2 ; circumference 44 cm

Example 6

Find the volume and total surface area of a cuboid $12 \times 8 \times 5$ cm.

SOLUTION

$$\text{Volume} = lbh = 12 \times 8 \times 5 = 480 \text{ cm}^3.$$

$$\text{Surface} = 2(lb + bh + hl) = 2(12 \times 8 + 8 \times 5 + 5 \times 12)$$

$$= 2(96 + 40 + 60) = 2 \times 196 = 392 \text{ cm}^2.$$

Answer: 480 cm^3 ; 392 cm^2

Example 7

Find the distance between (2, 3) and (7, 15).

SOLUTION

$$d = \sqrt{(7-2)^2 + (15-3)^2} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

Answer: 13 units

Example 8

Evaluate $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$.

SOLUTION

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}, \sin 30^\circ = \frac{1}{2}, \cos 60^\circ = \frac{1}{2}.$$

$$\text{First part: } \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} = \frac{3}{4}.$$

$$\text{Second part: } \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

$$\text{Total} = \frac{3}{4} + \frac{1}{4} = 1.$$

Answer: 1

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Two parallel lines are cut by a transversal. A pair of co-interior angles measure $(3x + 10)^\circ$ and $(2x + 20)^\circ$. Find both angles.

SOLUTION

Co-interior angles (same side of the transversal, between the parallels) add to 180° .

$$(3x + 10) + (2x + 20) = 180$$

$$5x + 30 = 180$$

$$5x = 150$$

$$x = 30.$$

$$\text{First angle} = 3(30) + 10 = 90 + 10 = 100^\circ.$$

$$\text{Second angle} = 2(30) + 20 = 60 + 20 = 80^\circ.$$

$$\text{Check: } 100 + 80 = 180. \text{ Correct.}$$

Answer: 100° and 80°

Example 10 [Infosys pattern]

A man 1.8 m tall casts a shadow 2.4 m long. At the same moment a telecom tower casts a shadow 30 m long. Find the height of the tower.

SOLUTION

The sun's rays hit both at the same angle, so the two triangles are similar.

$\frac{\text{height}}{\text{shadow}}$ is the same for both.

$$\frac{1.8}{2.4} = \frac{h}{30}$$

$$h = 30 \times \frac{1.8}{2.4}$$

$$\frac{1.8}{2.4} = 0.75.$$

$$h = 30 \times 0.75 = 22.5 \text{ m.}$$

Answer: 22.5 m

SHORTCUT

In “same time, two shadows” problems you never need the angle. Just set $\frac{\text{height}}{\text{shadow}}$ equal for the two objects. Keep the ratio as a decimal ($1.8 \div 2.4 = 0.75$) and one multiplication finishes it.

Example 11 [Accenture pattern]

Find the area of a triangle with sides 13 cm, 14 cm and 15 cm.

SOLUTION

No height is given, so use Heron's formula.

$$s = \frac{13+14+15}{2} = \frac{42}{2} = 21.$$

$$\begin{aligned}
 s - a &= 21 - 13 = 8 \\
 s - b &= 21 - 14 = 7 \\
 s - c &= 21 - 15 = 6 \\
 \text{Area} &= \sqrt{21 \times 8 \times 7 \times 6}. \\
 21 \times 8 &= 168; \quad 7 \times 6 = 42; \quad 168 \times 42 = 7056. \\
 \sqrt{7056} &= 84 \text{ (since } 84^2 = 7056\text{)}. \\
 \text{Area} &= 84 \text{ cm}^2.
 \end{aligned}$$

Answer: 84 cm²

Example 12 [Wipro pattern]

A chord of a circle of radius 13 cm is 5 cm from the centre. Find the length of the chord.

SOLUTION

Drop a perpendicular from the centre to the chord. It bisects the chord.

This makes a right triangle: radius is the hypotenuse, the 5 cm is one leg, half the chord is the other.

$$13^2 = 5^2 + (\text{half chord})^2$$

$$169 = 25 + (\text{half chord})^2$$

$$(\text{half chord})^2 = 144$$

$$\text{half chord} = 12.$$

$$\text{Chord} = 2 \times 12 = 24 \text{ cm.}$$

Answer: 24 cm

TRAP

The distance from the centre is to the **chord**, and the perpendicular cuts the chord in half. Students often set $13^2 = 5^2 + \text{chord}^2$ and get 12 as the whole chord. Always double at the end.

Example 13 [Capgemini pattern]

A sector has radius 21 cm and angle 120° . Find its arc length, area and perimeter. Take $\pi = \frac{22}{7}$.

SOLUTION

$\frac{120}{360} = \frac{1}{3}$, so the sector is one third of the circle.

$$\text{Full circumference} = 2 \times \frac{22}{7} \times 21 = 2 \times 22 \times 3 = 132 \text{ cm.}$$

$$\text{Arc} = \frac{1}{3} \times 132 = 44 \text{ cm.}$$

$$\text{Full area} = \frac{22}{7} \times 21 \times 21 = 22 \times 3 \times 21 = 1386 \text{ cm}^2.$$

$$\text{Sector area} = \frac{1}{3} \times 1386 = 462 \text{ cm}^2.$$

$$\text{Perimeter of a sector} = \text{arc} + \text{two radii} = 44 + 21 + 21 = 86 \text{ cm.}$$

Answer: Arc 44 cm; area 462 cm²; perimeter 86 cm

TRAP

The perimeter of a sector is not just the arc. It is arc + $2r$, because the two straight edges are part of the boundary. Forgetting the $2r$ costs 42 cm here.

Example 14 [Cognizant pattern]

Each interior angle of a regular polygon is 162° . Find the number of sides and the number of diagonals.

SOLUTION

Interior + exterior = 180° , so each exterior angle = $180 - 162 = 18^\circ$.

Exterior angles always total 360° .

$$n = \frac{360}{18} = 20 \text{ sides.}$$

$$\text{Diagonals} = \frac{n(n-3)}{2} = \frac{20 \times 17}{2} = \frac{340}{2} = 170.$$

Answer: 20 sides; 170 diagonals

SHORTCUT

Always route interior-angle questions through the **exterior** angle. $n = 360 \div$ exterior is one division. Working from $\frac{(n-2)180}{n} = 162$ needs cross-multiplying and rearranging — three times the work, three times the risk.

Example 15 [TCS NQT pattern]

A rectangular garden is 40 m by 25 m. A path 2.5 m wide runs all round it on the outside. Find the area of the path.

SOLUTION

Outer rectangle: the path adds 2.5 m on **both** sides of each dimension.

$$\text{Outer length} = 40 + 2 \times 2.5 = 40 + 5 = 45 \text{ m.}$$

$$\text{Outer width} = 25 + 2 \times 2.5 = 25 + 5 = 30 \text{ m.}$$

$$\text{Outer area} = 45 \times 30 = 1350 \text{ m}^2.$$

$$\text{Garden area} = 40 \times 25 = 1000 \text{ m}^2.$$

$$\text{Path area} = 1350 - 1000 = 350 \text{ m}^2.$$

Answer: 350 m²

TRAP

A path “all round” adds its width **twice** to each dimension, not once. Using 42.5×27.5 instead of 45×30 is the standard wrong answer here.

Example 16 [Infosys pattern]

A cylinder has radius 7 cm and height 20 cm. Find its volume, curved surface area and total surface area. Take $\pi = \frac{22}{7}$.

SOLUTION

$$\text{Volume} = \pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 20 = 22 \times 7 \times 20 = 3080 \text{ cm}^3.$$

$$\text{Curved SA} = 2\pi r h = 2 \times \frac{22}{7} \times 7 \times 20 = 2 \times 22 \times 20 = 880 \text{ cm}^2.$$

$$\text{Area of one circular end} = \pi r^2 = \frac{22}{7} \times 49 = 154 \text{ cm}^2.$$

$$\text{Total SA} = 880 + 2 \times 154 = 880 + 308 = 1188 \text{ cm}^2.$$

Answer: 3080 cm³; CSA 880 cm²; TSA 1188 cm²

Example 17 [Accenture pattern]

A cone has radius 7 cm and height 24 cm. Find its slant height, curved surface area and volume. Take $\pi = \frac{22}{7}$.

SOLUTION

$$\text{Slant: } l = \sqrt{r^2 + h^2} = \sqrt{49 + 576} = \sqrt{625} = 25 \text{ cm.}$$

$$\text{Curved SA} = \pi r l = \frac{22}{7} \times 7 \times 25 = 22 \times 25 = 550 \text{ cm}^2.$$

$$\text{Volume} = \frac{1}{3} \pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 49 \times 24.$$

$$\frac{1}{3} \times 24 = 8, \text{ so this is } \frac{22}{7} \times 49 \times 8 = 22 \times 7 \times 8 = 1232 \text{ cm}^3.$$

Answer: $l = 25$ cm; CSA 550 cm²; volume 1232 cm³

Example 18 [Wipro pattern]

A metal sphere of radius 9 cm is melted and recast into small spheres of radius 3 cm. How many small spheres are formed?

SOLUTION

Melting keeps the volume the same.

$$\text{Big sphere volume} = \frac{4}{3} \pi (9)^3 = \frac{4}{3} \pi \times 729.$$

$$\text{Small sphere volume} = \frac{4}{3}\pi(3)^3 = \frac{4}{3}\pi \times 27.$$

$$\text{Number} = \frac{\frac{4}{3}\pi \times 729}{\frac{4}{3}\pi \times 27} = \frac{729}{27} = 27.$$

The $\frac{4}{3}\pi$ cancels — only the cube of the radii matters.

Answer: 27 spheres

SHORTCUT

For melting one solid into copies of another, cancel every common factor **before** multiplying. Here the whole answer is $(\frac{9}{3})^3 = 3^3 = 27$. Never compute $\frac{4}{3}\pi \times 729$ as a decimal.

Example 19 [Capgemini pattern]

Find the area of the triangle with vertices (1, 2), (4, 8) and (7, 3).

SOLUTION

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|.$$

$$x_1 = 1, y_1 = 2; \quad x_2 = 4, y_2 = 8; \quad x_3 = 7, y_3 = 3.$$

$$x_1(y_2 - y_3) = 1 \times (8 - 3) = 5$$

$$x_2(y_3 - y_1) = 4 \times (3 - 2) = 4$$

$$x_3(y_1 - y_2) = 7 \times (2 - 8) = 7 \times (-6) = -42$$

$$\text{Sum} = 5 + 4 - 42 = -33.$$

$$\text{Area} = \frac{1}{2} \times |-33| = \frac{33}{2} = 16.5 \text{ square units.}$$

Answer: 16.5 square units

Example 20 [TCS NQT pattern]

From a point 60 m from the foot of a tower, the angle of elevation of its top is 30° . Find the height of the tower. Take $\sqrt{3} = 1.732$.

SOLUTION

$$\tan(\text{elevation}) = \frac{\text{height}}{\text{distance}}.$$

$$\tan 30^\circ = \frac{h}{60}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{60}$$

$$h = \frac{60}{\sqrt{3}}.$$

$$\text{Rationalise: } h = \frac{60\sqrt{3}}{3} = 20\sqrt{3}.$$

$$h = 20 \times 1.732 = 34.64 \text{ m.}$$

Answer: $20\sqrt{3} \approx 34.64 \text{ m}$

PRACTICE · TIER 1 target: 60 s/Q

- Two angles on a straight line are $(2x + 15)^\circ$ and $(3x + 15)^\circ$. Find both angles.
- The angles of a triangle are in the ratio 3 : 4 : 5. Find the largest angle.
 - 75°
 - 60°
 - 90°
 - 80°
- A right triangle has legs 8 cm and 15 cm. Find the hypotenuse and the area.
- A circle has circumference 88 cm. Find its radius and area ($\pi = \frac{22}{7}$).
- A chord of length 16 cm lies in a circle of radius 17 cm. Find its distance from the centre.
 - 15 cm
 - 8 cm
 - 9 cm
 - $\sqrt{33}$ cm
- Each exterior angle of a regular polygon is 24° . Find the number of sides and the interior angle sum.
- A square has area 144 cm^2 . Find its perimeter and the length of its diagonal.
- A trapezium has parallel sides 15 cm and 25 cm, and height 12 cm. Find its area.
- A rhombus has diagonals 16 cm and 30 cm. Find its area and its side length.

10. A cube has edge 9 cm. Find its volume and total surface area.
11. A cylinder has radius 10.5 cm and height 8 cm. Find its volume ($\pi = \frac{22}{7}$).
12. A sphere has radius 21 cm. Find its surface area and volume ($\pi = \frac{22}{7}$).
13. Find the distance between $(-3, 4)$ and $(5, -2)$, and their midpoint.
14. A ladder leans against a wall at 45° and its foot is 5 m from the wall. How high up the wall does it reach?
15. The area of an equilateral triangle is $36\sqrt{3} \text{ cm}^2$. Find its side.

ANSWER KEY

1. 75° and 105° . $5x + 30 = 180 \rightarrow x = 30$.
2. (a) 75° . $3k + 4k + 5k = 180 \rightarrow k = 15$; largest = 5×15 . (b) is the middle angle, (c) assumes a right triangle, (d) comes from a $2 : 3 : 4$ split.
3. Hypotenuse 17 cm (the 8, 15, 17 triple); area = $\frac{1}{2} \times 8 \times 15 = 60 \text{ cm}^2$.
4. $r = 14 \text{ cm}$ ($2 \times \frac{22}{7} \times r = 88$); area = $\frac{22}{7} \times 196 = 616 \text{ cm}^2$.
5. (a) 15 cm. Half chord = 8; $17^2 - 8^2 = 289 - 64 = 225$, so the distance is 15. (b) reports the half chord itself, (c) is $17 - 8$ (subtracting instead of using Pythagoras), (d) is $\sqrt{17^2 - 16^2}$ — forgetting to halve the chord.
6. $n = \frac{360}{24} = 15$ sides; interior sum = $(15 - 2) \times 180 = 2340^\circ$.
7. Side = 12 cm; perimeter = 48 cm; diagonal = $12\sqrt{2} \approx 16.97 \text{ cm}$.
8. 240 cm^2 . $\frac{1}{2}(15 + 25) \times 12 = 20 \times 12$.
9. Area = $\frac{1}{2} \times 16 \times 30 = 240 \text{ cm}^2$; side = $\sqrt{8^2 + 15^2} = 17 \text{ cm}$ (half-diagonals are the legs).
10. 729 cm^3 ; $6 \times 81 = 486 \text{ cm}^2$.
11. 2772 cm^3 . $\frac{22}{7} \times 110.25 = 346.5$; 346.5×8 .
12. SA = $4 \times \frac{22}{7} \times 441 = 5544 \text{ cm}^2$; volume = $\frac{4}{3} \times \frac{22}{7} \times 9261 = 38808 \text{ cm}^3$.
13. Distance = $\sqrt{64 + 36} = 10$; midpoint = $(1, 1)$.
14. 5 m. At 45° the height equals the base distance, since $\tan 45^\circ = 1$.
15. 12 cm. $\frac{\sqrt{3}}{4}a^2 = 36\sqrt{3} \rightarrow a^2 = 144$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A Bangkok hub floor measuring 15 m by 6.4 m must be tiled with tiles of 40 cm by 30 cm. Tiles cost THB 32 each and the contractor orders 4% extra for breakage. Find the number of tiles ordered and the total cost.

SOLUTION

Work in centimetres so the units match.

Floor: 15 m = 1500 cm, 6.4 m = 640 cm.

Floor area = $1500 \times 640 = 960000 \text{ cm}^2$.

One tile = $40 \times 30 = 1200 \text{ cm}^2$.

Tiles needed = $\frac{960000}{1200} = 800$.

With 4% extra: $800 \times 1.04 = 832$ tiles.

Cost = 832×32 .

$832 \times 32 = 832 \times 30 + 832 \times 2 = 24960 + 1664 = 26624$.

Answer: 832 tiles; THB 26,624

SHORTCUT

Convert to one unit **before** you do anything else. Mixing metres and centimetres is the single biggest source of wrong tiling answers. An area ratio in cm^2 gives the tile count directly; no decimals appear anywhere.

Example 22 [SCB · pattern]

Sand is heaped in a cone of base radius 10.5 m and height 3 m at a site in Bangkok. Sand costs THB 250 per cubic metre. Find the value of the heap. Take $\pi = \frac{22}{7}$.

SOLUTION

Volume of a cone $= \frac{1}{3}\pi r^2 h$.

$$r^2 = 10.5 \times 10.5 = 110.25.$$

$\frac{1}{3} \times 3 = 1$, so the height and the $\frac{1}{3}$ cancel.

$$\text{Volume} = \frac{22}{7} \times 110.25 = 22 \times 15.75 = 346.5 \text{ m}^3.$$

(Here $110.25 \div 7 = 15.75$.)

$$\text{Value} = 346.5 \times 250.$$

$$346.5 \times 250 = 346.5 \times 1000 \div 4 = \frac{346500}{4} = 86625.$$

Answer: THB 86,625

Example 23 [Agoda · pattern]

A rectangular water tank is 2.4 m by 1.5 m by 1.2 m. It is filled by a pump delivering 300 litres per minute. How long does filling take? ($1 \text{ m}^3 = 1000 \text{ litres}$.)

SOLUTION

$$\text{Volume} = 2.4 \times 1.5 \times 1.2.$$

$$2.4 \times 1.5 = 3.6.$$

$$3.6 \times 1.2 = 4.32 \text{ m}^3.$$

$$\text{In litres: } 4.32 \times 1000 = 4320 \text{ litres.}$$

$$\text{Time} = \frac{4320}{300} = 14.4 \text{ minutes.}$$

$$0.4 \text{ minute} = 0.4 \times 60 = 24 \text{ seconds.}$$

Answer: 14.4 minutes, i.e. 14 min 24 s

Example 24 [LINE MAN · pattern]

A road roller is a cylinder of diameter 84 cm and length 1 m. How much road area does it level in 500 complete revolutions? Take $\pi = \frac{22}{7}$.

SOLUTION

One revolution lays down a strip whose area equals the roller's curved surface area.

$$\text{Diameter } 84 \text{ cm} \rightarrow \text{radius} = 42 \text{ cm} = 0.42 \text{ m.}$$

$$\text{Curved SA} = 2\pi rh = 2 \times \frac{22}{7} \times 0.42 \times 1.$$

$$0.42 \div 7 = 0.06.$$

$$= 2 \times 22 \times 0.06 = 2.64 \text{ m}^2 \text{ per revolution.}$$

$$\text{In 500 revolutions: } 2.64 \times 500 = 1320 \text{ m}^2.$$

Answer: 1320 m²

This is why the **length** of the roller, not its area of cross-section, drives the answer. A roller only paints the curved band; its two flat ends never touch the road.

Example 25 [Shopee · pattern]

A packing wire is bent into a circle of radius 28 cm. The same wire is straightened and bent into a square. By how much does the enclosed area drop? Take $\pi = \frac{22}{7}$.

SOLUTION

The wire length stays the same, so circumference = square perimeter.

$$\text{Circumference} = 2 \times \frac{22}{7} \times 28 = 2 \times 22 \times 4 = 176 \text{ cm.}$$

$$\text{Circle area} = \frac{22}{7} \times 28 \times 28 = 22 \times 4 \times 28 = 2464 \text{ cm}^2.$$

$$\text{Square side} = \frac{176}{4} = 44 \text{ cm.}$$

$$\text{Square area} = 44 \times 44 = 1936 \text{ cm}^2.$$

$$\text{Drop} = 2464 - 1936 = 528 \text{ cm}^2.$$

Answer: The area drops by 528 cm^2

TRAP

Equal perimeters never mean equal areas. For a fixed perimeter the circle always encloses the most area, and among rectangles the square encloses the most. Expect the shape change to **lose** area whenever you move away from a circle.

Example 26 [DBS · pattern]

A well of diameter 4 m is dug 14 m deep. The soil taken out is spread evenly to form a ring-shaped embankment 3 m wide around the well. Find the height of the embankment. Take $\pi = \frac{22}{7}$.

SOLUTION

Soil dug out = volume of the cylindrical well.

$$\text{Radius of the well} = \frac{4}{2} = 2 \text{ m.}$$

$$\text{Volume} = \pi r^2 h = \frac{22}{7} \times 4 \times 14 = 22 \times 4 \times 2 = 176 \text{ m}^3.$$

The embankment is a ring: inner radius 2 m, outer radius $2 + 3 = 5$ m.

$$\text{Ring area} = \pi(R^2 - r^2) = \frac{22}{7} \times (25 - 4) = \frac{22}{7} \times 21 = 66 \text{ m}^2.$$

$$\text{Height} = \frac{\text{volume}}{\text{area}} = \frac{176}{66} = \frac{8}{3} = 2.67 \text{ m.}$$

Answer: $\frac{8}{3} \approx 2.67 \text{ m}$

Example 27 [GIC · pattern]

An office window is 15 m above the ground. From it, the angle of elevation of the top of a building opposite is 30° and the angle of depression of that building's base is 45° . Find the height of the building. Take $\sqrt{3} = 1.732$.

SOLUTION

Let the horizontal distance between the two buildings be d .

Below the window (depression 45°):

$$\tan 45^\circ = \frac{15}{d} \rightarrow 1 = \frac{15}{d} \rightarrow d = 15 \text{ m.}$$

Above the window (elevation 30°), call that extra height x :

$$\tan 30^\circ = \frac{x}{d}$$

$$\frac{1}{\sqrt{3}} = \frac{x}{15}$$

$$x = \frac{15}{\sqrt{3}} = \frac{15\sqrt{3}}{3} = 5\sqrt{3} = 5 \times 1.732 = 8.66 \text{ m.}$$

$$\text{Total height} = 15 + 8.66 = 23.66 \text{ m.}$$

Answer: $15 + 5\sqrt{3} \approx 23.66 \text{ m}$

SHORTCUT

Always solve the depression part first. It usually gives the horizontal distance in one step (and at 45° that distance simply equals the known height). Feed that distance into the elevation part.

Example 28 [Razer · pattern]

A solid metal cylinder of radius 6 cm and height 15 cm is melted and cast into small cones of radius 3 cm and height 5 cm. How many cones are made?

SOLUTION

Volume is conserved.

$$\text{Cylinder} = \pi r^2 h = \pi \times 36 \times 15 = 540\pi \text{ cm}^3.$$

$$\text{One cone} = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 9 \times 5 = 15\pi \text{ cm}^3.$$

$$\text{Number} = \frac{540\pi}{15\pi} = \frac{540}{15} = 36.$$

The π cancels, so no decimals are needed anywhere.

Answer: 36 cones

Example 29 [Sea · pattern]

A bucket is a frustum with top radius 15 cm, bottom radius 10 cm and height 24 cm. Find its capacity in litres. Take $\pi = \frac{22}{7}$ and 1 litre = 1000 cm³.

SOLUTION

$$\text{Frustum volume} = \frac{1}{3}\pi h(R^2 + Rr + r^2).$$

$$R = 15, r = 10, h = 24.$$

$$R^2 = 225; Rr = 150; r^2 = 100.$$

$$\text{Sum} = 225 + 150 + 100 = 475.$$

$$\frac{1}{3} \times 24 = 8.$$

$$\text{Volume} = 8 \times \frac{22}{7} \times 475 = \frac{8 \times 22 \times 475}{7} = \frac{83600}{7} = 11942.86 \text{ cm}^3.$$

$$\text{In litres: } 11942.86 \div 1000 = 11.94 \text{ litres.}$$

Answer: About 11.94 litres

Example 30 [Grab · pattern]

A cylindrical tub of radius 10 cm holds water. A solid metal ball is dropped in and fully sinks; the water level rises by 2.88 cm. Find the radius of the ball.

SOLUTION

The water pushed up equals the ball's volume.

Water displaced = cylinder of radius 10 and height 2.88

$$= \pi \times 10^2 \times 2.88 = 288\pi \text{ cm}^3.$$

$$\text{Ball volume} = \frac{4}{3}\pi r^3.$$

$$\frac{4}{3}\pi r^3 = 288\pi$$

$$\text{Cancel } \pi: \frac{4}{3}r^3 = 288$$

$$r^3 = 288 \times \frac{3}{4} = 216$$

$$r = \sqrt[3]{216} = 6 \text{ cm.}$$

$$\text{Check: } \frac{4}{3}\pi \times 216 = 288\pi. \text{ Correct.}$$

Answer: 6 cm

PRACTICE · TIER 2 target: 100 s/Q

1. A room floor 12.5 m by 8 m is tiled with tiles of 50 cm by 40 cm costing Rs. 45 each. Find the number of tiles and the cost, with no wastage.
2. Two triangles are similar. Their areas are in the ratio 9 : 16 and the smaller one has perimeter 45 cm. Find the larger perimeter.
3. A 400 m running track has two straight sides and two semicircular ends of diameter 70 m. Find the length of each straight side and the area enclosed ($\pi = \frac{22}{7}$).
4. A cylindrical tank of radius 1.4 m and height 2.5 m is filled at 7 litres per second. How long does it take ($\pi = \frac{22}{7}$)?
5. A circular park has area 1386 m². Fencing costs Rs. 55 per metre. Find the total fencing cost ($\pi = \frac{22}{7}$).
6. A right triangle has vertices (0, 0), (6, 0) and (0, 8). Find its area, its hypotenuse, its inradius and its circumradius.
7. A cylindrical container of radius 6 cm and height 15 cm is filled with ice cream. It is shared into cones of radius 3 cm and height 12 cm, each topped with a hemisphere of radius 3 cm. How many portions are made?

8. A trapezoidal plot has parallel sides 48 m and 32 m, and the distance between them is 25 m. Land sells at Rs. 8,500 per m². Find the value of the plot.
9. A cube of edge 12 cm is cut into cubes of edge 3 cm. Find how many small cubes there are, and the ratio of the original surface area to the total surface area of all the pieces.
10. A hemispherical bowl of internal radius 9 cm is emptied into cylindrical bottles of radius 3 cm and height 3 cm. How many bottles are filled?
11. A vertical pole casts a 9 m shadow when the sun's elevation is 60°. Find the pole's height, and the shadow length when the elevation drops to 30°. Take $\sqrt{3} = 1.732$.
12. The line $3x + 4y = 24$ meets the axes at A and B . Find the area of triangle OAB and the perpendicular distance from the origin to the line.

ANSWER KEY

1. 500 tiles; Rs. 22,500. Floor = 1250 cm by 800 cm; one tile = $50 \times 40 = 2000 \text{ cm}^2$; tiles = $\frac{1250 \times 800}{2000} = 500$; cost = 500×45 .
2. 60 cm. Area ratio 9 : 16 $\rightarrow$ side (and perimeter) ratio 3 : 4; $45 \times \frac{4}{3}$.
3. The two semicircles make one full circle: $\frac{22}{7} \times 70 = 220$ m. Straights = $400 - 220 = 180$, so each is 90 m. Area = $90 \times 70 + \frac{22}{7} \times 35^2 = 6300 + 3850 = 10150 \text{ m}^2$.
4. 36 min 40 s. Volume = $\frac{22}{7} \times 1.96 \times 2.5 = 15.4 \text{ m}^3 = 15400 \text{ L}$; $\frac{15400}{7} = 2200 \text{ s} = 36 \text{ min } 40 \text{ s}$.
5. Rs. 7,260. $r^2 = 1386 \times \frac{7}{22} = 441 \rightarrow r = 21$; circumference = 132 m ; 132×55 .
6. Area = 24; hypotenuse = 10; inradius = $\frac{6+8-10}{2} = 2$; circumradius = $\frac{10}{2} = 5$.
7. 10 portions. Cylinder = 540π ; each portion = $36\pi + 18\pi = 54\pi$; $540 \div 54$.
8. Rs. 85,00,000. Area = $\frac{1}{2}(48 + 32) \times 25 = 1000 \text{ m}^2$; 1000×8500 .
9. 64 cubes. Original SA = $6 \times 144 = 864$; total after = $64 \times 6 \times 9 = 3456$; ratio 1 : 4.
10. 18 bottles. Bowl = $\frac{2}{3}\pi \times 729 = 486\pi$; bottle = $\pi \times 9 \times 3 = 27\pi$; $486 \div 27 = 18$. The π cancels.
11. Height = $9\sqrt{3} = 15.59 \text{ m}$. At 30°: shadow = $\frac{h}{\tan} 30^\circ = 9\sqrt{3} \times \sqrt{3} = 27 \text{ m}$.
12. $A(8, 0)$, $B(0, 6)$; area = $\frac{1}{2} \times 8 \times 6 = 24$; distance = $\frac{|-24|}{\sqrt{9+16}} = \frac{24}{5} = 4.8$.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Google · pattern]

Triangle ABC has area 96 cm^2 . The midpoints of its sides are joined to form triangle PQR . The midpoints of PQR are joined to form triangle XYZ . Find the area of XYZ .

SOLUTION

Insight: joining midpoints makes a triangle similar to the original with every side exactly half as long — and halving the sides quarters the area.

Why each side halves: PQ joins the midpoints of two sides of ABC , so by the midpoint theorem PQ is parallel to the third side and half its length. The same holds for the other two sides.

So PQR is similar to ABC with side ratio $k = \frac{1}{2}$.

Area ratio = $k^2 = \frac{1}{4}$.

Area of $PQR = 96 \times \frac{1}{4} = 24 \text{ cm}^2$.

Repeat the same step on PQR :

Area of $XYZ = 24 \times \frac{1}{4} = 6 \text{ cm}^2$.

General rule: after n midpoint steps the area is $\frac{96}{4^n}$. Here $n = 2$, so $\frac{96}{16} = 6$.

Answer: 6 cm^2

Example 32 [Amazon · pattern]

An ant walks on the surface of a closed box measuring $12 \times 4 \times 3$ cm, from one corner to the corner diagonally opposite. Find the shortest possible path length.

SOLUTION

Insight: unfold the box flat. On a flat sheet the shortest path is a straight line, so the answer is the smallest straight-line distance across an unfolding — not the space diagonal, which cuts through the air.

Each unfolding lays one edge alone against the **sum** of the other two. There are three such unfoldings, so test all three.

Unfolding 1: a rectangle 12 by $(4 + 3) = 7$.

Length $= \sqrt{12^2 + 7^2} = \sqrt{144 + 49} = \sqrt{193} = 13.89$ cm.

Unfolding 2: a rectangle $(12 + 3) = 15$ by 4.

Length $= \sqrt{15^2 + 4^2} = \sqrt{225 + 16} = \sqrt{241} = 15.52$ cm.

Unfolding 3: a rectangle $(12 + 4) = 16$ by 3.

Length $= \sqrt{16^2 + 3^2} = \sqrt{256 + 9} = \sqrt{265} = 16.28$ cm.

The smallest is $\sqrt{193} \approx 13.89$ cm.

Note the pattern: the best unfolding pairs the **largest** dimension alone against the sum of the other two.

Answer: $\sqrt{193} \approx 13.89$ cm

TRAP

The space diagonal $\sqrt{144 + 16 + 9} = \sqrt{169} = 13$ cm is shorter, but the ant cannot fly. It must stay on the surface, so 13 cm is not available. Read the word “surface” before you reach for the 3D diagonal.

Example 33 [Microsoft · pattern]

A sphere is packed snugly inside a cube of edge 12 cm. What fraction of the box is empty? Take $\pi = 3.1416$.

SOLUTION

Insight: the sphere touches all six faces, so its diameter equals the cube’s edge. The wasted fraction is then independent of the size of the cube.

Sphere diameter = 12, so radius $r = 6$ cm.

Cube volume $= 12^3 = 1728$ cm³.

Sphere volume $= \frac{4}{3}\pi r^3 = \frac{4}{3} \times 3.1416 \times 216$.

$\frac{4}{3} \times 216 = 288$.

$= 288 \times 3.1416 = 904.78$ cm³.

Empty volume $= 1728 - 904.78 = 823.22$ cm³.

Fraction empty $= \frac{823.22}{1728} = 0.4764$, i.e. about 47.6%.

General form: with edge a and radius $\frac{a}{2}$,

$\frac{\text{sphere}}{\text{cube}} = \frac{\frac{4}{3}\pi(\frac{a}{2})^3}{a^3} = \frac{4\pi a^3}{3 \times 8a^3} = \frac{\pi}{6} \approx 0.5236$.

So a sphere always fills $\frac{\pi}{6} \approx 52.4\%$ of its snug box, whatever the size, and 47.6% is always wasted.

Answer: About 47.6% empty (the sphere fills $\frac{\pi}{6} \approx 52.4\%$)

Example 34 [Goldman Sachs · pattern]

A delivery van must start at $A(1, 2)$, touch the x -axis at some point P , and end at $B(7, 4)$. Find the point P that makes the total trip $AP + PB$ shortest, and that shortest length.

SOLUTION

Insight: reflect one endpoint across the line. A path that bounces off a line has the same length as the straight path to the mirror image — and a straight line is the shortest route.

Reflect $A(1, 2)$ in the x -axis to get $A'(1, -2)$.

For any P on the x -axis, $AP = A'P$ (mirror symmetry).

So $AP + PB = A'P + PB$, which is smallest when A', P, B lie on one straight line.

Line $A'B$: slope $= \frac{4 - (-2)}{7 - 1} = \frac{6}{6} = 1$.

Equation through $A'(1, -2)$: $y - (-2) = 1(x - 1) \rightarrow y + 2 = x - 1 \rightarrow y = x - 3$.

It meets the x -axis where $y = 0$: $0 = x - 3 \rightarrow x = 3$.

So $P = (3, 0)$.

Shortest length $= A'B = \sqrt{(7-1)^2 + (4-(-2))^2} = \sqrt{36 + 36} = \sqrt{72} = 6\sqrt{2} = 8.49$.

Check directly: $AP = \sqrt{4 + 4} = 2\sqrt{2}$ and $PB = \sqrt{16 + 16} = 4\sqrt{2}$; total $= 6\sqrt{2}$. Correct.

Answer: $P = (3, 0)$; **shortest distance** $= 6\sqrt{2} \approx 8.49$

SHORTCUT

Reflection solves every “touch the line and come back” problem: shortest walk to a river and home, light bouncing off a mirror, a ball off a cushion. Reflect **one** point, draw the straight line, read off where it crosses.

Example 35 [D. E. Shaw · pattern]

Two circles of radii 9 cm and 4 cm touch each other externally. Find the length of their direct common tangent, and show it equals $2\sqrt{r_1 r_2}$.

SOLUTION

Insight: drop the smaller radius onto the larger one. The tangent, the line of centres, and the difference of radii form a right triangle.

Touching externally means the centres are $d = 9 + 4 = 13$ cm apart.

Both radii are perpendicular to the tangent, so they are parallel to each other.

Shift the short radius across: the right triangle has

- hypotenuse $= d = 13$,
- one leg $= r_1 - r_2 = 9 - 4 = 5$,
- other leg $=$ the tangent length t .

$$t = \sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \text{ cm.}$$

Now the general form. For externally touching circles $d = r_1 + r_2$, so

$$t^2 = (r_1 + r_2)^2 - (r_1 - r_2)^2.$$

$$\text{Expand: } (r_1^2 + 2r_1 r_2 + r_2^2) - (r_1^2 - 2r_1 r_2 + r_2^2) = 4r_1 r_2.$$

$$t = 2\sqrt{r_1 r_2}.$$

Check: $2\sqrt{9 \times 4} = 2 \times 6 = 12$. Matches.

Answer: $t = 12$ cm, and in general $t = 2\sqrt{r_1 r_2}$

Example 36 [Adobe · pattern]

A cone is cut by a plane parallel to its base, exactly halfway up. Find the ratio of the volume of the small top cone to the volume of the frustum below it.

SOLUTION

Insight: the top piece is a scaled copy of the whole cone. Scale the lengths by k and the volume scales by k^3 — so compute the whole and the top, then subtract.

Let the full cone have radius R and height H ; volume $V = \frac{1}{3}\pi R^2 H$.

Cutting halfway up means the small cone has height $\frac{H}{2}$.

By similar triangles its radius is $\frac{R}{2}$ (every length scales by $k = \frac{1}{2}$).

$$\text{Small cone volume } v = \frac{1}{3}\pi \left(\frac{R}{2}\right)^2 \left(\frac{H}{2}\right) = \frac{1}{3}\pi \left(\frac{R^2}{4}\right) \left(\frac{H}{2}\right) = \frac{1}{8} \times \frac{1}{3}\pi R^2 H = \frac{V}{8}.$$

$$\text{Frustum} = V - v = V - \frac{V}{8} = \frac{7V}{8}.$$

$$\text{Ratio (small cone : frustum)} = \frac{V}{8} : \frac{7V}{8} = 1 : 7.$$

Extend it: cutting at $\frac{1}{3}$ of the height from the apex gives $k = \frac{1}{3}$, so the top cone is $\frac{V}{27}$ and the frustum is $\frac{26V}{27}$ — ratio 1 : 26.

Answer: 1 : 7

Example 37 [Uber · pattern]

From a fixed point on level ground, the angle of elevation of a hovering drone is 30° . The drone then rises 40 m straight up, and the elevation becomes 60° . Find the drone's original height and its horizontal distance from the observer.

SOLUTION

Insight: the horizontal distance d never changes. Write the same d from both angles and set them equal — one unknown drops out.

Let the first height be h and the horizontal distance be d .

From the first sighting: $\tan 30^\circ = \frac{h}{d} \rightarrow \frac{1}{\sqrt{3}} = \frac{h}{d} \rightarrow d = h\sqrt{3}$.

From the second sighting: $\tan 60^\circ = \frac{h+40}{d} \rightarrow \sqrt{3} = \frac{h+40}{d} \rightarrow d = \frac{h+40}{\sqrt{3}}$.

Both expressions are the same d :

$$h\sqrt{3} = \frac{h+40}{\sqrt{3}}$$

Multiply both sides by $\sqrt{3}$:

$$3h = h + 40$$

$$2h = 40$$

$$h = 20 \text{ m.}$$

$$\text{Then } d = h\sqrt{3} = 20\sqrt{3} = 20 \times 1.732 = 34.64 \text{ m.}$$

$$\text{Check: } \tan 30^\circ = \frac{20}{34.64} = 0.577 = \frac{1}{\sqrt{3}}. \text{ Correct.}$$

$$\text{And } \tan 60^\circ = \frac{60}{34.64} = 1.732 = \sqrt{3}. \text{ Correct.}$$

Answer: Original height 20 m; horizontal distance $20\sqrt{3} \approx 34.64$ m

Example 38 [Google · pattern]

How many triangles are there whose three sides are whole numbers and whose perimeter is exactly 20?

SOLUTION

Insight: fix an order $a \leq b \leq c$ so each triangle is counted once, then use the triangle inequality to bound the largest side before listing.

Let $a \leq b \leq c$ with $a + b + c = 20$.

Triangle inequality: $a + b > c$.

Since $a + b = 20 - c$, this says $20 - c > c \rightarrow 20 > 2c \rightarrow c < 10$.

Also $c \geq \frac{20}{3} = 6.67$, because c is the largest of three numbers summing to 20.

So $c \in \{7, 8, 9\}$. Now list.

$c = 9$: $a + b = 11$ with $a \leq b \leq 9$, so $b \geq 5.5$, giving $b \in \{6, 7, 8, 9\}$:

$(2, 9, 9), (3, 8, 9), (4, 7, 9), (5, 6, 9)$ — 4 triangles.

$c = 8$: $a + b = 12$ with $a \leq b \leq 8$, so $b \geq 6$, giving $b \in \{6, 7, 8\}$:

$(4, 8, 8), (5, 7, 8), (6, 6, 8)$ — 3 triangles.

$c = 7$: $a + b = 13$ with $a \leq b \leq 7$, so $b \geq 6.5$, giving $b = 7$ only:

$(6, 7, 7)$ — 1 triangle.

Total = $4 + 3 + 1 = 8$.

Answer: 8 triangles

PRACTICE · TIER 3 target: 4 min/Q

1. Triangle ABC has area 192 cm^2 . Midpoints are joined three times in a row. Find the area of the third midpoint triangle.

2. An ant crawls on the surface of a box $8 \times 6 \times 5$ cm, from one corner to the opposite corner. Find the shortest surface path, and say why the space diagonal is not the answer.
3. A cylinder is packed snugly inside a cube of edge 10 cm (the cylinder stands upright and touches all four side walls, the floor and the lid). What fraction of the cube is empty? Take $\pi = 3.1416$.
4. Find the point on the y -axis that is equidistant from $(2, 3)$ and $(-1, 5)$.
5. Two circles of radii 8 cm and 3 cm have centres 13 cm apart. Find both the direct and the transverse common tangent lengths.
6. A cone is cut by a plane parallel to the base at one third of the height, measured from the apex. Find the ratio of the small cone's volume to the frustum's volume, and the ratio of their curved surface areas.
7. From a point on level ground the elevation of a tower top is 30° . Walking 40 m directly towards the tower, the elevation becomes 60° . Find the tower's height.
8. How many triangles have whole-number sides and perimeter exactly 24?

ANSWER KEY

1. 3 cm^2 . Each midpoint step scales lengths by $\frac{1}{2}$, so area by $\frac{1}{4}$. Three steps give $192 \div 4^3 = \frac{192}{64} = 3$. The rule area ratio = (side ratio) 2 is what makes the halving compound so fast.
2. $\sqrt{185} \approx 13.60$ cm. The three unfoldings give $\sqrt{8^2 + (6+5)^2} = \sqrt{185}$, $\sqrt{(8+5)^2 + 6^2} = \sqrt{205}$, and $\sqrt{(8+6)^2 + 5^2} = \sqrt{221}$. The smallest pairs the largest edge alone. The space diagonal $\sqrt{64 + 36 + 25} = \sqrt{125} \approx 11.18$ is shorter but passes through the inside of the box, which the ant cannot do.
3. About 21.5% empty. Cylinder radius 5, height 10: volume = $3.1416 \times 25 \times 10 = 785.4$; cube = 1000; empty = 214.6, so 21.46%. In general the ratio is $\frac{\pi}{4} \approx 0.7854$, independent of the edge length.
4. $(0, \frac{13}{4})$. Let $P(0, k)$. Then $2^2 + (k-3)^2 = 1^2 + (k-5)^2$, so $4 + k^2 - 6k + 9 = 1 + k^2 - 10k + 25$, giving $13 - 6k = 26 - 10k$, so $4k = 13$ and $k = 3.25$. Only the y -coordinate is unknown, which is why one equation is enough.
5. Direct = $\sqrt{13^2 - (8-3)^2} = \sqrt{169 - 25} = 12$ cm. Transverse = $\sqrt{13^2 - (8+3)^2} = \sqrt{169 - 121} = \sqrt{48} = 4\sqrt{3} \approx 6.93$ cm. The transverse tangent is always shorter, because it must cross between the circles.
6. Volumes 1 : 26; curved surface areas 1 : 8. Scale factor $k = \frac{1}{3}$, so volume ratio to the whole is $\frac{1}{27}$ and the frustum takes the other $\frac{26}{27}$. Curved SA scales as $k^2 = \frac{1}{9}$, so the small cone is $\frac{1}{9}$ of the whole and the frustum's slant band is $\frac{8}{9}$. Note volumes and areas scale by different powers — never reuse one ratio for both.
7. $20\sqrt{3} \approx 34.64$ m. Let the height be h and the nearer distance be d . Then $\tan 60^\circ = \frac{h}{d} \rightarrow d = \frac{h}{\sqrt{3}}$, and $\tan 30^\circ = \frac{h}{d+40} \rightarrow d+40 = h\sqrt{3}$. Subtract: $h\sqrt{3} - \frac{h}{\sqrt{3}} = 40 \rightarrow \frac{h(3-1)}{\sqrt{3}} = 40 \rightarrow h = \frac{40\sqrt{3}}{2} = 20\sqrt{3}$.
8. 12 triangles. With $a \leq b \leq c$ and $a+b+c = 24$: $c < 12$ and $c \geq 8$, so $c \in \{8, 9, 10, 11\}$. Listing gives $(8, 8, 8)$; $(7, 8, 9)$, $(6, 9, 9)$; $(4, 10, 10)$, $(5, 9, 10)$, $(6, 8, 10)$, $(7, 7, 10)$; $(2, 11, 11)$, $(3, 10, 11)$, $(4, 9, 11)$, $(5, 8, 11)$, $(6, 7, 11)$. Count: $1 + 2 + 4 + 5 = 12$.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 90 s/Q · 18 questions · 27 min

1. Two angles of a triangle are 52° and 61° . Find the exterior angle at the third vertex.
2. A right triangle has hypotenuse 26 cm and one leg 10 cm. Find the other leg and the area.
3. A circle has area 616 cm^2 . Find its circumference ($\pi = \frac{22}{7}$).
4. A regular polygon has 18 sides. Find each interior angle and the number of diagonals.
5. Find the area of a rhombus whose diagonals are 24 cm and 10 cm, and its perimeter.
6. A cone and a cylinder have the same radius 7 cm and the same height 12 cm. Find the ratio of their volumes.
7. A cube has total surface area 600 cm^2 . Find its volume and its space diagonal.
8. Find the slope of the line through $(2, -3)$ and $(6, 5)$, and the slope of any line perpendicular to it.
9. A sector of a circle of radius 14 cm has an angle of 90° . Find its area and perimeter ($\pi = \frac{22}{7}$).
10. A tower casts a 25 m shadow when the sun's elevation is 45° . Find the tower's height.
11. Two similar triangles have corresponding sides 6 cm and 15 cm. If the smaller has area 48 cm^2 , find the larger area.

12. A hemisphere has radius 10.5 cm. Find its volume and total surface area ($\pi = \frac{22}{7}$).
13. A rectangular sheet 22 cm by 14 cm is rolled up so that the 22 cm side becomes the circumference of an open cylinder. Find the radius, the height and the volume ($\pi = \frac{22}{7}$).
14. Find the perpendicular distance from (3, 4) to the line $5x - 12y + 9 = 0$.
15. A wire of length 132 cm is bent into a circle. Find the area enclosed ($\pi = \frac{22}{7}$).
16. The angle of elevation of a cliff top from a boat is 60° and the boat is 50 m from the base. Find the cliff's height ($\sqrt{3} = 1.732$).
17. A closed cuboid tank is $3 \times 2 \times 1.5$ m. Find the cost of painting its outside at Rs. 120 per m^2 .
18. Triangle PQR has area 120 cm^2 . S is the midpoint of QR . Find the area of triangle PQS .

ANSWER KEY

1. 113° . The exterior angle equals the sum of the two opposite interior angles: $52 + 61$.
2. Other leg = 24 cm ($26^2 - 10^2 = 676 - 100 = 576$); area = $\frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2$.
3. 88 cm . $r^2 = 616 \times \frac{7}{22} = 196 \rightarrow r = 14$; $2 \times \frac{22}{7} \times 14 = 88$.
4. Interior = $\frac{(18-2)180}{18} = 160^\circ$; diagonals = $\frac{18 \times 15}{2} = 135$.
5. Area = $\frac{1}{2} \times 24 \times 10 = 120 \text{ cm}^2$; side = $\sqrt{12^2 + 5^2} = 13$, so perimeter = 52 cm .
6. $1 : 3$. A cone is always one third of the cylinder on the same base and height.
7. $6a^2 = 600 \rightarrow a = 10$; volume = 1000 cm^3 ; diagonal = $10\sqrt{3} \approx 17.32 \text{ cm}$.
8. Slope = $\frac{5-(-3)}{6-2} = \frac{8}{4} = 2$; perpendicular slope = $-\frac{1}{2}$.
9. Quarter circle. Area = $\frac{1}{4} \times \frac{22}{7} \times 196 = 154 \text{ cm}^2$; perimeter = $\frac{1}{4} \times 88 + 14 + 14 = 22 + 28 = 50 \text{ cm}$.
10. 25 m . $\tan 45^\circ = 1$, so height = shadow.
11. 300 cm^2 . Side ratio $6 : 15 = 2 : 5$, so area ratio $4 : 25$; $48 \times \frac{25}{4}$.
12. Volume = $\frac{2}{3} \times \frac{22}{7} \times 1157.625 = 2425.5 \text{ cm}^3$; total SA = $3\pi r^2 = 3 \times \frac{22}{7} \times 110.25 = 1039.5 \text{ cm}^2$.
13. $2\pi r = 22 \rightarrow r = 3.5 \text{ cm}$; height = 14 cm ; volume = $\frac{22}{7} \times 12.25 \times 14 = 539 \text{ cm}^3$.
14. $\frac{24}{13} \approx 1.85$. $\frac{|5(3)-12(4)+9|}{\sqrt{25+144}} = \frac{|-24|}{13} = \frac{24}{13}$.
15. 1386 cm^2 . $2 \times \frac{22}{7} \times r = 132 \rightarrow r = 21$; area = $\frac{22}{7} \times 441$.
16. $50\sqrt{3} \approx 86.6 \text{ m}$. $\tan 60^\circ = \frac{h}{50}$.
17. Rs. 3,240. SA = $2(3 \times 2 + 2 \times 1.5 + 1.5 \times 3) = 2(6 + 3 + 4.5) = 27 \text{ m}^2$; $27 \times 120 = 3240$.
18. 60 cm^2 . A median always splits a triangle into two equal areas.

ONE-PAGE REVISION CARD

Angles — straight line 180° , point 360° , triangle 180° . Exterior angle = sum of the two opposite interiors. Co-interior angles across parallels add to 180° .

Triangles — Area = $\frac{1}{2}bh$; Heron = $\sqrt{s(s-a)(s-b)(s-c)}$ with $s = \frac{\text{perimeter}}{2}$. Equilateral: $\frac{\sqrt{3}}{4}a^2$. Right-triangle inradius = $\frac{a+b-c}{2}$, circumradius = $\frac{c}{2}$. Triples: 3, 4, 5 · 5, 12, 13 · 8, 15, 17 · 7, 24, 25 · 9, 40, 41 · 20, 21, 29.

Similarity — sides $k \rightarrow$ perimeters k , areas k^2 , volumes k^3 . Midpoint triangle: $\frac{1}{4}$ the area.

Circles — $C = 2\pi r$, $A = \pi r^2$. Chord: $r^2 = h^2 + \left(\frac{L}{2}\right)^2$. Sector: arc = $\frac{\theta}{360} \times 2\pi r$, area = $\frac{\theta}{360} \times \pi r^2$, perimeter = arc + $2r$. Tangents: direct = $\sqrt{d^2 - (r_1 - r_2)^2}$, transverse = $\sqrt{d^2 - (r_1 + r_2)^2}$.

Polygons — interior sum $(n-2)180$; exterior = $\frac{360}{n}$; diagonals = $\frac{n(n-3)}{2}$.

Solids — Cylinder $\pi r^2 h$, CSA $2\pi r h$. Cone $\frac{1}{3}\pi r^2 h$, CSA $\pi r l$, $l = \sqrt{r^2 + h^2}$. Sphere $\frac{4}{3}\pi r^3$, SA $4\pi r^2$. Hemisphere $\frac{2}{3}\pi r^3$, TSA $3\pi r^2$. Frustum $\frac{1}{3}\pi h(R^2 + Rr + r^2)$.

Coordinates — distance $\sqrt{(\Delta x)^2 + (\Delta y)^2}$; midpoint = averages; slope = $\frac{\Delta y}{\Delta x}$; perpendicular slopes multiply to -1 ; triangle area = $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$; point-to-line = $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

Trig — $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 45^\circ = 1$, $\tan 60^\circ = \sqrt{3}$. Elevation: $\tan \theta = \frac{\text{height}}{\text{distance}}$. At 45° height = distance.

Shortcuts worth memorising

- Use $\pi = \frac{22}{7}$ whenever a radius is a multiple of 7 or of 3.5 — the fraction cancels and no decimals appear.
- Melting or recasting: set volumes equal and cancel π and any common factors **before** multiplying. The answer is usually a ratio of cubes.
- Interior-angle polygon questions: go via the exterior angle, $n = 360 \div \text{exterior}$.
- “Touch a line and return” (shortest path): reflect one point, draw the straight line, read the crossing point.
- Ant on a box: unfold. Pair the **largest** edge alone against the sum of the other two.
- Two elevation angles from the same line: the horizontal distance is shared. Write it twice and equate.

Top 5 traps

1. The distance from the centre reaches the chord’s **midpoint**: double the half-chord at the end.
2. A sector’s perimeter is arc + $2r$, not just the arc.
3. A path “all round” a plot adds its width **twice** to each dimension.
4. Equal perimeters do not mean equal areas — a circle beats a square every time.
5. On a box’s **surface**, the space diagonal is not available. Unfold instead.

Series

Find the rule, then the term.

What you need to know

WHAT YOU NEED TO KNOW

The 5-step attack (use it in this order, every time)

1. Write the **differences** between consecutive terms. Constant? → add that number.
2. Differences not constant? Write the **second differences**. Constant? → the series is quadratic; keep extending the difference row.
3. Try **ratios** (divide each term by the one before). Constant or a neat pattern ($\times 2$, $\times 3$, $\times 4$...)?
4. Compare with n^2 , n^3 , $n!$, primes: is each term $n^2 \pm c$, $n^3 \pm c$, or $n(n+1)$?
5. Still stuck? The series is **alternating** (two rules used turn by turn) or **interleaved** (two separate series zipped together). Split the odd positions from the even positions.

Standard rules you must recognise on sight

Pattern	Example
Add a constant (AP)	3, 7, 11, 15, 19 ($d = 4$)
Multiply by a constant (GP)	2, 6, 18, 54 ($r = 3$)
Growing difference	7, 12, 19, 28, 39 (diffs 5, 7, 9, 11)
$\times k \pm c$	5, 11, 23, 47 ($\times 2 + 1$)
$\times k$, k itself grows	3, 6, 18, 72 ($\times 2, \times 3, \times 4$)
Perfect squares	1, 4, 9, 16, 25
Squares ± 1	2, 5, 10, 17, 26 ($n^2 + 1$)
Perfect cubes	1, 8, 27, 64, 125
Cubes $\pm c$	2, 9, 28, 65 ($n^3 + 1$)
$n(n+1)$ (products)	2, 6, 12, 20, 30
Factorials	1, 2, 6, 24, 120
Fibonacci type	2, 3, 5, 8, 13, 21
Alternating rules	5, 10, 8, 16, 14 ($\times 2, -2, \times 2, -2$)
Interleaved	3, 20, 6, 17, 9, 14 (two series)

Formulas

- AP: n -th term $= a + (n-1)d$. Sum $= \frac{n}{2}[2a + (n-1)d]$.
- GP: n -th term $= ar^{n-1}$. Sum of n terms $= \frac{a(r^n-1)}{r-1}$, $r \neq 1$.
- Infinite GP, $|r| < 1$: sum $= \frac{a}{1-r}$.
- $1 + 2 + \dots + n = \frac{n(n+1)}{2}$
- $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$
- $1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$
- Triangular numbers $T_k = \frac{k(k+1)}{2}$; $T_1 + T_2 + \dots + T_n = \frac{n(n+1)(n+2)}{6}$

Numbers you should know cold

- Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400
- Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728
- Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47
- Factorials: 1, 2, 6, 24, 120, 720, 5040

Letters

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

- Memory hook: **E**5, **J**10, **O**15, **T**20, **Y**25 (“EJOTY”).
- Letter series = number series in disguise. Convert to numbers, solve, convert back.
- Wrap-around: after Z, start again at A. Position 27 → A, 28 → B.

Warm-up

WARM-UP

build the reflex

Example 1

Find the next term: 3, 7, 11, 15, ?

SOLUTION

Differences: $7 - 3 = 4$, $11 - 7 = 4$, $15 - 11 = 4$. Constant.

Next = $15 + 4 = 19$.

Answer: 19

Example 2

Find the next term: 2, 6, 18, 54, ?

SOLUTION

Ratios: $6 \div 2 = 3$, $18 \div 6 = 3$, $54 \div 18 = 3$. Constant.

Next = $54 \times 3 = 162$.

Answer: 162

Example 3

Find the next term: 1, 4, 9, 16, ?

SOLUTION

$1 = 1^2$, $4 = 2^2$, $9 = 3^2$, $16 = 4^2$.

Next = $5^2 = 25$.

Answer: 25

Example 4

Find the next term: 2, 5, 10, 17, 26, ?

SOLUTION

Compare with squares: $1 + 1 = 2$, $4 + 1 = 5$, $9 + 1 = 10$, $16 + 1 = 17$, $25 + 1 = 26$.
So the rule is $n^2 + 1$. Next: $6^2 + 1 = 36 + 1 = 37$.

Answer: 37**Example 5**

Find the next term: 1, 8, 27, 64, ?

SOLUTION

$1 = 1^3$, $8 = 2^3$, $27 = 3^3$, $64 = 4^3$.
Next $= 5^3 = 125$.

Answer: 125**Example 6**

Find the next letter: B, D, F, H, ?

SOLUTION

Positions: B = 2, D = 4, F = 6, H = 8. Add 2 each time.
Next position $= 8 + 2 = 10$. Letter number 10 is J.

Answer: J**Example 7**

Find the next term: 5, 11, 23, 47, ?

SOLUTION

$5 \times 2 = 10$, $10 + 1 = 11$. $11 \times 2 = 22$, $22 + 1 = 23$. $23 \times 2 = 46$, $46 + 1 = 47$.
Rule: $\times 2 + 1$. Next $= 47 \times 2 + 1 = 94 + 1 = 95$.

Answer: 95**Example 8**

Find the next term: 120, 60, 20, 5, ?

SOLUTION

$120 \div 60 = 2$, so divide by 2. $60 \div 20 = 3$, divide by 3. $20 \div 5 = 4$, divide by 4.
The divisor grows: 2, 3, 4, 5. Next $= 5 \div 5 = 1$.

Answer: 1**Tier 1 — Service companies**

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Find the next term: 7, 12, 19, 28, 39, ?

SOLUTION

Differences: $12 - 7 = 5$, $19 - 12 = 7$, $28 - 19 = 9$, $39 - 28 = 11$.
The differences are 5, 7, 9, 11 — they grow by 2.

Next difference = $11 + 2 = 13$.

Next term = $39 + 13 = 52$.

Answer: 52

SHORTCUT

When first differences form their own AP, the series is quadratic. You never need the formula — just keep the difference row going. It is faster and it cannot go wrong.

Example 10 [Infosys pattern]

Find the next term: 4, 9, 20, 43, 90, ?

SOLUTION

Test $\times 2$ plus something:

$$4 \times 2 = 8, \text{ and } 9 - 8 = 1.$$

$$9 \times 2 = 18, \text{ and } 20 - 18 = 2.$$

$$20 \times 2 = 40, \text{ and } 43 - 40 = 3.$$

$$43 \times 2 = 86, \text{ and } 90 - 86 = 4.$$

The added number is 1, 2, 3, 4 — so next we add 5.

$$\text{Next term} = 90 \times 2 + 5 = 180 + 5 = 185.$$

Answer: 185

Example 11 [Wipro pattern]

Find the next term: 3, 6, 18, 72, ?

SOLUTION

$$\text{Ratios: } 6 \div 3 = 2, 18 \div 6 = 3, 72 \div 18 = 4.$$

The multiplier grows: 2, 3, 4, so next is 5.

$$\text{Next term} = 72 \times 5 = 360.$$

Answer: 360

Example 12 [Accenture pattern]

One term is wrong. Which one? 2, 5, 10, 17, 26, 38, 50

SOLUTION

Check against $n^2 + 1$:

$$1^2 + 1 = 2 \checkmark \quad 2^2 + 1 = 5 \checkmark \quad 3^2 + 1 = 10 \checkmark \quad 4^2 + 1 = 17 \checkmark$$

$$5^2 + 1 = 26 \checkmark \quad 6^2 + 1 = 37, \text{ but the series shows } 38 \times \quad 7^2 + 1 = 50 \checkmark$$

Only 38 breaks the rule. It should be 37.

Answer: 38 (it should be 37)

TRAP

A “wrong term” question asks **which term is wrong**, not **what it should be**. The two answers are different numbers (here 38 and 37). Read the last line of the question before you tick an option.

Example 13 [Capgemini pattern]

Find the missing term: 6, 13, 27, ?, 111

SOLUTION

$$6 \times 2 = 12, 12 + 1 = 13. \quad 13 \times 2 = 26, 26 + 1 = 27.$$

So the rule is $\times 2 + 1$.

Missing term = $27 \times 2 + 1 = 54 + 1 = 55$.

Check forward: $55 \times 2 + 1 = 110 + 1 = 111$ ✓ matches the last term.

Answer: 55

SHORTCUT

With a missing **middle** term, always check both ways: build it from the left, then test it against the term on the right. If both agree, your rule is certainly correct.

Example 14 [TCS NQT pattern]

Find the next letter: C, F, I, L, ?

SOLUTION

Positions: C = 3, F = 6, I = 9, L = 12.

Differences: $6 - 3 = 3$, $9 - 6 = 3$, $12 - 9 = 3$. Add 3.

Next position = $12 + 3 = 15$. Letter number 15 is O.

Answer: O

Example 15 [Cognizant pattern]

Find the next term: A1, C4, E9, G16, ?

SOLUTION

Letters: A = 1, C = 3, E = 5, G = 7. Add 2 each time.

Next letter position = $7 + 2 = 9 \rightarrow$ I.

Numbers: 1, 4, 9, 16 are $1^2, 2^2, 3^2, 4^2$.

Next number = $5^2 = 25$.

Answer: I25

Example 16 [TCS NQT pattern]

Find the next term: 1, 2, 6, 24, 120, ?

SOLUTION

Ratios: $2 \div 1 = 2$, $6 \div 2 = 3$, $24 \div 6 = 4$, $120 \div 24 = 5$.

The multiplier is 2, 3, 4, 5, so next is 6.

Next term = $120 \times 6 = 720$.

(These are the factorials $1!, 2!, 3!, 4!, 5!, 6!$.)

Answer: 720

Example 17 [Wipro pattern]

Find the next term: 2, 3, 5, 8, 13, 21, ?

SOLUTION

Test “add the two terms before”:

$2 + 3 = 5$ ✓ $3 + 5 = 8$ ✓ $5 + 8 = 13$ ✓ $8 + 13 = 21$ ✓

Next term = $13 + 21 = 34$.

Answer: 34

Example 18 [Accenture pattern]

Find the next term: 5, 10, 8, 16, 14, 28, ?

SOLUTION

The series goes up, down, up, down. Check step by step:

$$5 \times 2 = 10 \quad 10 - 2 = 8 \quad 8 \times 2 = 16 \quad 16 - 2 = 14 \quad 14 \times 2 = 28$$

Two rules take turns: $\times 2$, then -2 .

The last step used was $\times 2$, so now we subtract 2.

$$\text{Next term} = 28 - 2 = 26.$$

Answer: 26

Example 19 [Infosys pattern]

Find the next term: 3, 20, 6, 17, 9, 14, 12, ?

SOLUTION

The terms swing high and low, so split them.

Positions 1, 3, 5, 7: 3, 6, 9, 12 — add 3 each time.

Positions 2, 4, 6, 8: 20, 17, 14, ? — subtract 3 each time.

The asked term is at position 8, so it belongs to the second series.

$$\text{Next term} = 14 - 3 = 11.$$

Answer: 11

SHORTCUT

Interleaved series have a give-away shape: the terms zig-zag, and the number of terms is often even. Number the positions 1, 2, 3, ... and read the odd ones and even ones as two separate lists.

Example 20 [TCS NQT pattern]

One pair of letters is wrong: AZ, CX, EV, GT, IS

SOLUTION

First letters: A = 1, C = 3, E = 5, G = 7, I = 9. Add 2. All correct.

Second letters: Z = 26, X = 24, V = 22, T = 20, S = 19.

Differences: $26 - 24 = 2$, $24 - 22 = 2$, $22 - 20 = 2$, but $20 - 19 = 1 \neq 2$

The rule is “subtract 2”, so the fifth second letter should be $20 - 2 = 18$, which is R.

So IS is wrong; it should be IR.

Answer: IS (it should be IR)

TRAP

Never count letters in your head past about J. Write the position numbers down. One slip of a single position turns a 20-second question into a wrong answer, and you will never notice it.

PRACTICE · TIER 1 target: 60 s/Q

1. 8, 15, 22, 29, ?
2. 3, 9, 27, 81, ?
3. 4, 5, 9, 18, 34, ?
4. 1, 3, 7, 15, 31, ?
5. 100, 96, 88, 76, 60, ?
6. One term is wrong: 6, 12, 21, 33, 46, 66. Which one?
7. Z, W, T, Q, ?
8. B2, D8, F18, H32, ?
9. 7, 14, 28, 56, ?
10. 2, 9, 28, 65, ?

11. Find the missing term: 11, 13, 17, 19, ?, 29
12. 144, 121, 100, 81, ?
13. 5, 6, 9, 14, 21, ?
14. AC, EG, IK, MO, ?
15. 1, 5, 14, 30, 55, ?

ANSWER KEY

1. **36** — constant difference $+7$; $29 + 7 = 36$.
2. **243** — constant ratio $\times 3$; $81 \times 3 = 243$.
3. **59** — differences are 1, 4, 9, 16 (squares); next $+25$; $34 + 25 = 59$.
4. **63** — rule $\times 2 + 1$; $31 \times 2 + 1 = 63$.
5. **40** — differences $-4, -8, -12, -16$; next -20 ; $60 - 20 = 40$.
6. **46** — differences should be $+6, +9, +12, +15, +18$; the fifth term must be $33 + 15 = 48$, not 46.
7. **N** — positions 26, 23, 20, 17, subtract 3; $17 - 3 = 14 \rightarrow N$.
8. **J50** — letters B, D, F, H, J ($+2$); numbers $2n^2$ give 2, 8, 18, 32, 50.
9. **112** — constant ratio $\times 2$; $56 \times 2 = 112$.
10. **126** — rule $n^3 + 1$; $5^3 + 1 = 125 + 1 = 126$.
11. **23** — consecutive primes: 11, 13, 17, 19, 23, 29.
12. **64** — squares going down: $12^2, 11^2, 10^2, 9^2, 8^2$; $8^2 = 64$.
13. **30** — differences 1, 3, 5, 7 (odd numbers); next $+9$; $21 + 9 = 30$.
14. **QS** — letter positions (1, 3), (5, 7), (9, 11), (13, 15), (17, 19) $\rightarrow Q, S$.
15. **91** — running sums of squares: 1, $1\{+\}4, 1\{+\}4\{+\}9, \dots$; $55 + 36 = 91$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A ride-hailing hub logs completed rides for five days: 120, 132, 156, 192, 240. If the pattern holds, how many rides on day 6?

SOLUTION

Differences: $132 - 120 = 12$, $156 - 132 = 24$, $192 - 156 = 36$, $240 - 192 = 48$.

The differences are 12, 24, 36, 48 — they grow by 12.

Next difference $= 48 + 12 = 60$.

Day 6 $= 240 + 60 = 300$.

Answer: 300 rides

Example 22 [Shopee · pattern]

Orders taken by a new seller in the first five weeks: 5, 11, 24, 51, 106. Predict week 6.

SOLUTION

Try $\times 2$ and see what is left over:

$5 \times 2 = 10$; $11 - 10 = 1$

$11 \times 2 = 22$; $24 - 22 = 2$

$24 \times 2 = 48$; $51 - 48 = 3$

$51 \times 2 = 102$; $106 - 102 = 4$

The leftover is 1, 2, 3, 4, so next it is 5.

Week 6 $= 106 \times 2 + 5 = 212 + 5 = 217$.

Answer: 217 orders

Example 23 [DBS · pattern]

Find the next term: 4, 7, 12, 14, 36, 21, ?

SOLUTION

The terms do not move smoothly, so split by position.

Odd positions (1, 3, 5, 7): 4, 12, 36, ?

$$12 \div 4 = 3, \quad 36 \div 12 = 3 \rightarrow \text{multiply by 3.}$$

Even positions (2, 4, 6): 7, 14, 21 $\rightarrow$ add 7.

The asked term sits at position 7, which is odd.

$$\text{Next term} = 36 \times 3 = 108.$$

Answer: 108

TRAP

In an interleaved series, count the position of the **missing** term before you compute. Students who read “the last given term was 21, so add 7” get 28 — a distractor that is always offered.

Example 24 [Agoda · pattern]

Weekly bookings were recorded as 3, 8, 18, 38, 78, 157, 318. One figure was typed wrongly. Which one?

SOLUTION

Test the rule $\times 2 + 2$:

$$3 \times 2 + 2 = 8 \checkmark$$

$$8 \times 2 + 2 = 18 \checkmark$$

$$18 \times 2 + 2 = 38 \checkmark$$

$$38 \times 2 + 2 = 78 \checkmark$$

$$78 \times 2 + 2 = 158, \text{ but the list shows } 157 \times$$

$$158 \times 2 + 2 = 318 \checkmark \text{ — the last figure fits the } \textbf{corrected} \text{ value, which confirms the rule.}$$

Answer: 157 (it should be 158)

SHORTCUT

In a wrong-term question, check the **term after** the suspect using the corrected value. If it fits, you have found the single error. If it does not, your rule is wrong, not the data.

Example 25 [GIC · pattern]

Find the next term: 2, 12, 36, 80, 150, ?

SOLUTION

The terms grow fast, so divide each one by n^2 and see what is left.

$$2 \div 1^2 = 2 \div 1 = 2 \quad 12 \div 2^2 = 12 \div 4 = 3 \quad 36 \div 3^2 = 36 \div 9 = 4$$

$$80 \div 4^2 = 80 \div 16 = 5 \quad 150 \div 5^2 = 150 \div 25 = 6$$

The leftovers are 2, 3, 4, 5, 6 — that is $n + 1$. So the rule should be $n^2(n + 1)$. Test it:

$$n = 1: 1^2 \times 2 = 2 \checkmark$$

$$n = 2: 2^2 \times 3 = 4 \times 3 = 12 \checkmark$$

$$n = 3: 3^2 \times 4 = 9 \times 4 = 36 \checkmark$$

$$n = 4: 4^2 \times 5 = 16 \times 5 = 80 \checkmark$$

$$n = 5: 5^2 \times 6 = 25 \times 6 = 150 \checkmark$$

Rule: $n^2(n + 1)$.

$$n = 6: 6^2 \times 7 = 36 \times 7 = 252.$$

Answer: 252

Example 26 [SCB · pattern]

Find the next term: 5, 7, 12, 19, 31, 50, ?

SOLUTION

Differences: $7 - 5 = 2$, $12 - 7 = 5$, $19 - 12 = 7$, $31 - 19 = 12$, $50 - 31 = 19$.

From the second one on, the differences 5, 7, 12, 19 are the earlier terms themselves. That is the sign of a Fibonacci rule, so test it.

$5 + 7 = 12$ ✓ $7 + 12 = 19$ ✓ $12 + 19 = 31$ ✓ $19 + 31 = 50$ ✓

Next term = $31 + 50 = 81$.

Answer: 81

Example 27 [LINE MAN · pattern]

Find the next term: J3, M9, P27, S81, ?

SOLUTION

Letters: J = 10, M = 13, P = 16, S = 19. Add 3 each time.

Next position = $19 + 3 = 22 \rightarrow V$.

Numbers: 3, 9, 27, 81; each is $\times 3$.

Next number = $81 \times 3 = 243$.

Answer: V243

Example 28 [Razer · pattern]

Find the next term: 1, 3, 8, 19, 42, 89, ?

SOLUTION

Double each term and see the leftover:

$1 \times 2 = 2$; $3 - 2 = 1$

$3 \times 2 = 6$; $8 - 6 = 2$

$8 \times 2 = 16$; $19 - 16 = 3$

$19 \times 2 = 38$; $42 - 38 = 4$

$42 \times 2 = 84$; $89 - 84 = 5$

Leftovers 1, 2, 3, 4, 5, so next add 6.

Next term = $89 \times 2 + 6 = 178 + 6 = 184$.

Answer: 184

Example 29 [Sea · pattern]

A server's error count each hour follows 6, 11, 21, 41, 81, In which hour does the count first go above 1000?

SOLUTION

Find the rule. $6 \times 2 = 12$, $12 - 1 = 11$ ✓. Test again: $11 \times 2 - 1 = 21$ ✓, $21 \times 2 - 1 = 41$ ✓, $41 \times 2 - 1 = 81$ ✓.

Rule: $\times 2 - 1$. Now list the terms with their hour numbers.

hour 1: 6

hour 2: 11

hour 3: 21

hour 4: 41

hour 5: 81

hour 6: $81 \times 2 - 1 = 162 - 1 = 161$

hour 7: $161 \times 2 - 1 = 322 - 1 = 321$

hour 8: $321 \times 2 - 1 = 642 - 1 = 641$

hour 9: $641 \times 2 - 1 = 1282 - 1 = 1281$
1281 is the first value above 1000, and it happens in hour 9.

Answer: Hour 9 (count 1281)

Example 30 [Agoda · pattern]

Two terms are missing: 2, 4, 10, ?, 82, ?

SOLUTION

Find the rule from the terms you have.

$$2 \times 3 = 6, \quad 6 - 2 = 4 \checkmark$$

$$4 \times 3 = 12, \quad 12 - 2 = 10 \checkmark$$

Rule: $\times 3 - 2$.

Fourth term = $10 \times 3 - 2 = 30 - 2 = 28$.

Check: $28 \times 3 - 2 = 84 - 2 = 82 \checkmark$ — matches the given fifth term.

Sixth term = $82 \times 3 - 2 = 246 - 2 = 244$.

Answer: 28 and 244

PRACTICE · TIER 2 target: 100 s/Q

- 9, 19, 40, 83, ?
- Daily app installs: 50, 65, 95, 140, 200, ? (next day)
- 4, 6, 12, 30, 90, ?
- One figure is wrong: 5, 16, 49, 148, 447, 1336. Which one?
- 2, 3, 7, 16, 32, ?
- C5, F10, I17, L26, ?
- 1, 2, 5, 26, ?
- Monthly revenue in S\$ thousands: 12, 18, 27, 40.5, ?
- 3, 12, 27, 48, 75, ?
- 7, 2, 14, 6, 28, 18, 56, ?
- Find the missing term: 6, 13, ?, 59, 122
- One pair is wrong: BD, FH, JL, NQ, RT. Which one?

ANSWER KEY

- 170** — rule $\times 2$ then add 1, 2, 3, 4; $83 \times 2 + 4 = 166 + 4 = 170$.
- 275** — differences 15, 30, 45, 60; next +75; $200 + 75 = 275$.
- 315** — multipliers 1.5, 2, 2.5, 3, 3.5; $90 \times 3.5 = 315$.
- 447** — rule is $\times 3 + 1$; $148 \times 3 + 1 = 445$, not 447. (And $445 \times 3 + 1 = 1336 \checkmark$.)
- 57** — differences 1, 4, 9, 16 (squares); next +25; $32 + 25 = 57$.
- O37** — letters +3: C, F, I, L, O; numbers $n^2 + 1$ for $n = 2, 3, 4, 5, 6$: 5, 10, 17, 26, 37.
- 677** — each term is (previous)² + 1; $26^2 + 1 = 676 + 1 = 677$.
- 60.75** — constant ratio $\times 1.5$; $40.5 \times 1.5 = 60.75$.
- 108** — terms are $3n^2$; $3 \times 6^2 = 3 \times 36 = 108$.
- 54** — interleaved: odd places $\times 2$ (7, 14, 28, 56), even places $\times 3$ (2, 6, 18, 54); position 8 is even.
- 28** — rule $\times 2$ then add 1, 2, 3, 4; $13 \times 2 + 2 = 28$, and $28 \times 2 + 3 = 59 \checkmark$.
- NQ** — inside each pair the gap is +2 and pairs step by +4: B,D; F,H; J,L; N,**P**; R,T. So NQ should be NP.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Google · pattern]

Find the next term: 1, 11, 21, 1211, 111221, ?

SOLUTION

Insight: the rule is not arithmetic at all. Each term describes the previous term out loud.

Read term 1: “one 1” → write 11 → term 2 is 11 ✓

Read term 2 (11): “two 1s” → write 21 → term 3 is 21 ✓

Read term 3 (21): “one 2, one 1” → write 12, 11 → term 4 is 1211 ✓

Read term 4 (1211): “one 1, one 2, two 1s” → 11, 12, 21 → term 5 is 111221 ✓

Now read term 5: 111221.

Break it into equal runs: 111 | 22 | 1.

- three 1s → 3 1
- two 2s → 2 2
- one 1 → 1 1

Write them together: 31 22 11 → 312211.

Answer: 312211

TRAP

No difference or ratio will ever crack this series. If the term lengths jump wildly (1 digit, 2, 2, 4, 6) and the digits stay small, stop doing arithmetic and start **describing**.

Example 32 [Amazon · pattern]

The series is 1, 2, 4, 8, 15, 26, 42, Find the 10th term.

SOLUTION

Insight: it looks like doubling but breaks at 15. Take differences twice; when a difference row becomes an AP, the series is fixed forever and you only need to extend rows.

Terms: 1, 2, 4, 8, 15, 26, 42

First differences: $2 - 1 = 1$, $4 - 2 = 2$, $8 - 4 = 4$, $15 - 8 = 7$, $26 - 15 = 11$, $42 - 26 = 16$

So: 1, 2, 4, 7, 11, 16

Second differences: $2 - 1 = 1$, $4 - 2 = 2$, $7 - 4 = 3$, $11 - 7 = 4$, $16 - 11 = 5$

So: 1, 2, 3, 4, 5 — a clean AP. Extend it: 6, 7, 8.

Rebuild the first-difference row:

after 16 comes $16 + 6 = 22$, then $22 + 7 = 29$, then $29 + 8 = 37$.

First differences: 1, 2, 4, 7, 11, 16, 22, 29, 37.

Rebuild the terms:

term 8 = $42 + 22 = 64$

term 9 = $64 + 29 = 93$

term 10 = $93 + 37 = 130$

Cross-check with a formula. Constant third differences mean the terms fit a cubic. The cubic through the first four terms is $a_n = \frac{n^3 - 3n^2 + 8n}{6}$.

Test $n = 5$: $\frac{125 - 75 + 40}{6} = \frac{90}{6} = 15$ ✓

Test $n = 10$: $\frac{1000 - 300 + 80}{6} = \frac{780}{6} = 130$ ✓

Answer: 130

Example 33 [Goldman Sachs · pattern]

The series 1, 3, 6, 10, 15, ... continues in the same way. Find the sum of its first 20 terms.

SOLUTION

Insight: adding up 20 terms one by one is slow and error-prone. Recognise the terms as triangular numbers and use the closed sum.

Differences: $3 - 1 = 2$, $6 - 3 = 3$, $10 - 6 = 4$, $15 - 10 = 5$. So term $k = 1 + 2 + \dots + k$, the triangular number

$$T_k = \frac{k(k+1)}{2}$$

Now sum them:

$$\sum_{k=1}^n T_k = \sum_{k=1}^n \frac{k^2 + k}{2} = \frac{1}{2} \sum_{k=1}^n k^2 + \frac{1}{2} \sum_{k=1}^n k$$

Put in the two standard sums:

$$= \frac{1}{2} \cdot \frac{n(n+1)(2n+1)}{6} + \frac{1}{2} \cdot \frac{n(n+1)}{2} = \frac{n(n+1)}{4} \left[\frac{2n+1}{3} + 1 \right]$$

Inside the bracket,

$$\frac{2n+1}{3} + 1 = \frac{2n+1+3}{3} = \frac{2n+4}{3} = \frac{2(n+2)}{3}$$

So

$$\frac{n(n+1)}{4} \cdot \frac{2(n+2)}{3} = \frac{2n(n+1)(n+2)}{12} = \frac{n(n+1)(n+2)}{6}$$

Now put $n = 20$:

$$\frac{20 \times 21 \times 22}{6}$$

$20 \times 21 = 420$. $420 \times 22 = 9240$. $9240 \div 6 = 1540$.

Answer: 1540

SHORTCUT

Memorise the three-in-a-row: sum of the first n whole numbers is $\frac{n(n+1)}{2}$; sum of the first n triangular numbers is $\frac{n(n+1)(n+2)}{6}$. The pattern of factors just grows by one more bracket.

Example 34 [Microsoft · pattern]

A series starts $a_1 = 3$ and follows $a_{n+1} = a_n^2 - 2a_n + 2$. Find a_5 .

SOLUTION

Insight: the rule is almost a perfect square. Shift the whole series by 1 and the mess disappears.

Write the rule and subtract 1 from both sides:

$$a_{n+1} - 1 = a_n^2 - 2a_n + 1 = (a_n - 1)^2$$

Let $b_n = a_n - 1$. Then $b_{n+1} = b_n^2$ — just keep squaring.

$$b_1 = a_1 - 1 = 3 - 1 = 2$$

$$b_2 = 2^2 = 4$$

$$b_3 = 4^2 = 16$$

$$b_4 = 16^2 = 256$$

$$b_5 = 256^2 = 65536$$

$$\text{So } a_5 = b_5 + 1 = 65536 + 1 = 65537.$$

Check the first few directly:

$$a_2 = 3^2 - 2(3) + 2 = 9 - 6 + 2 = 5, \text{ and } b_2 + 1 = 4 + 1 = 5 \checkmark$$

$$a_3 = 5^2 - 2(5) + 2 = 25 - 10 + 2 = 17, \text{ and } b_3 + 1 = 16 + 1 = 17 \checkmark$$

$$a_4 = 17^2 - 2(17) + 2 = 289 - 34 + 2 = 257, \text{ and } b_4 + 1 = 256 + 1 = 257 \checkmark$$

Answer: 65537

Example 35 [D. E. Shaw · pattern]

A series has $a_1 = 5$, $a_2 = 7$, and $a_{n+2} = a_{n+1} - a_n$. Find a_{100} .

SOLUTION

Insight: a “subtract the earlier term” rule always repeats with period 6. Never compute 100 terms — find the cycle, then use a remainder.

$$a_1 = 5$$

$$a_2 = 7$$

$$a_3 = 7 - 5 = 2$$

$$a_4 = 2 - 7 = -5$$

$$a_5 = -5 - 2 = -7$$

$$a_6 = -7 - (-5) = -7 + 5 = -2$$

$$a_7 = -2 - (-7) = -2 + 7 = 5 \leftarrow \text{same as } a_1$$

$$a_8 = 5 - (-2) = 5 + 2 = 7 \leftarrow \text{same as } a_2$$

So the block 5, 7, 2, -5, -7, -2 repeats every 6 terms.

Now reduce 100 modulo 6. $6 \times 16 = 96$, and $100 - 96 = 4$.

So $a_{100} = a_4 = -5$.

Answer: -5

Example 36 [Adobe · pattern]

A letter series runs A, B, D, G, K, P, Find the 7th term, and decide whether the letter Z ever appears.

SOLUTION

Insight: turn letters into positions immediately. The “gap” row is the real series.

Positions: A = 1, B = 2, D = 4, G = 7, K = 11, P = 16.

Gaps: $2 - 1 = 1$, $4 - 2 = 2$, $7 - 4 = 3$, $11 - 7 = 4$, $16 - 11 = 5$.

The gaps are 1, 2, 3, 4, 5, so the next gap is 6.

7th position = $16 + 6 = 22 \rightarrow V$.

Now the general term. Position of the n -th letter is

$$p_n = 1 + (1 + 2 + \cdots + (n - 1)) = 1 + \frac{(n - 1)n}{2}$$

Check: $n = 6$ gives $1 + \frac{5 \times 6}{2} = 1 + 15 = 16 \checkmark$ (P).

Z sits at position 26. Set $p_n = 26$:

$$1 + \frac{(n - 1)n}{2} = 26 \Rightarrow (n - 1)n = 50$$

Consecutive whole numbers with product 50? $6 \times 7 = 42$ and $7 \times 8 = 56$. Nothing gives 50.

So no term lands on 26. The series steps from $n = 7$ (position 22, V) straight to $n = 8$ (position $1 + \frac{7 \times 8}{2} = 1 + 28 = 29$), which has already run past Z.

Answer: 7th term is V; Z never appears

Example 37 [Uber · pattern]

A driver's bonus, in dollars, for successive trips is $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$. (a) If this went on for ever, what is the total? (b) How many trips are needed for the running total to go above 1.99?

SOLUTION

Insight: this is a GP with $r = \frac{1}{2}$. The partial sum is always “the limit minus one more halving”, so part (b) becomes a powers-of-2 question.

(a) $a = 1, r = \frac{1}{2}, |r| < 1$, so

$$S_{\infty} = \frac{a}{1-r} = \frac{1}{1-\frac{1}{2}} = \frac{1}{\frac{1}{2}} = 2$$

(b) Sum of the first n terms:

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{1 - \left(\frac{1}{2}\right)^n}{\frac{1}{2}} = 2\left(1 - \frac{1}{2^n}\right) = 2 - \frac{2}{2^n} = 2 - \frac{1}{2^{n-1}}$$

We need $S_n > 1.99$:

$$2 - \frac{1}{2^{n-1}} > 1.99 \Rightarrow \frac{1}{2^{n-1}} < 0.01 \Rightarrow 2^{n-1} > 100$$

Powers of 2: $2^6 = 64$ (not enough), $2^7 = 128 > 100$ ✓

So $n-1 \geq 7$, giving $n = 8$.

Verify both sides of the boundary:

$n = 7$: $S_7 = 2 - \frac{1}{2^6} = 2 - \frac{1}{64} = 2 - 0.015625 = 1.984375$ — below 1.99.

$n = 8$: $S_8 = 2 - \frac{1}{2^7} = 2 - \frac{1}{128} = 2 - 0.0078125 = 1.9921875$ — above 1.99 ✓

Answer: (a) 2 (b) 8 trips

Example 38 [Amazon · pattern]

A series is built by $a_1 = 4$ and $a_{n+1} = 3a_n + 2^n$. Find a_6 , and show that every term is even.

SOLUTION

Insight: the rule mixes a multiplier with a moving add-on. Compute term by term, and prove the “even” claim by induction, not by checking a few terms.

$$a_1 = 4$$

$$a_2 = 3(4) + 2^1 = 12 + 2 = 14$$

$$a_3 = 3(14) + 2^2 = 42 + 4 = 46$$

$$a_4 = 3(46) + 2^3 = 138 + 8 = 146$$

$$a_5 = 3(146) + 2^4 = 438 + 16 = 454$$

$$a_6 = 3(454) + 2^5 = 1362 + 32 = 1394$$

Why every term is even.

Base step: $a_1 = 4$, which is even.

Inductive step: suppose a_n is even, say $a_n = 2m$ for a whole number m . Then

$$a_{n+1} = 3(2m) + 2^n = 6m + 2^n = 2(3m + 2^{n-1})$$

For every $n \geq 1$, 2^{n-1} is a whole number, so a_{n+1} is 2 times a whole number — even.

Since a_1 is even and “even $\rightarrow$ next is even”, **all** terms are even.

Answer: $a_6 = 1394$; all terms are even

PRACTICE · TIER 3 target: 4 min/Q

- Find the next term: 1, 11, 21, 1211, 111221, 312211, ?
- For the series 1, 2, 4, 8, 15, 26, 42, ... find the 12th term.
- A series has $a_1 = 2, a_2 = 5$ and $a_{n+2} = a_{n+1} - a_n$. Find a_{2025} .
- Find the sum of the first 15 terms of 1, 3, 6, 10, 15, ...
- A series has $a_1 = 5$ and $a_{n+1} = a_n^2 - 4a_n + 6$. Find a_4 .
- Find the next letter: A, C, F, J, O, ?
- For the series $1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots$, how many terms are needed before the running total first goes above 1.49?
- Find the 10th term of 1, 4, 10, 20, 35, 56, ...

ANSWER KEY

1. **13112221** — read 312211 in runs: $3|1|22|11 \rightarrow$ “one 3, one 1, two 2s, two 1s” $\rightarrow 13, 11, 22, 21$.
2. **232** — extend the difference rows. First differences continue 22, 29, 37, 46, 56; terms continue 64, 93, 130, 176, 232. Check with $a_n = \frac{n^3-3n^2+8n}{6}$: $\frac{1728-432+96}{6} = \frac{1392}{6} = 232 \checkmark$.
3. **3** — the rule repeats with period 6: 2, 5, 3, -2, -5, -3, then 2 again. Since $2025 = 6 \times 337 + 3$, $a_{2025} = a_3 = 3$. And $a_3 = 5 - 2 = 3$.
4. **680** — these are triangular numbers, and $T_1 + \dots + T_n = \frac{n(n+1)(n+2)}{6}$. With $n = 15$: $\frac{15 \times 16 \times 17}{6} = \frac{4080}{6} = 680$.
5. **6563** — subtract 2: $a_{n+1} - 2 = a_n^2 - 4a_n + 4 = (a_n - 2)^2$. With $b_n = a_n - 2$: $b_1 = 3, b_2 = 9, b_3 = 81, b_4 = 6561$. So $a_4 = 6561 + 2 = 6563$.
6. **U** — positions 1, 3, 6, 10, 15 with gaps 2, 3, 4, 5; next gap 6 gives $15 + 6 = 21 \rightarrow$ U.
7. **5 terms** — $S_n = \left(\frac{3}{2}\right)\left(1 - \frac{1}{3^n}\right)$. Need $1 - \frac{1}{3^n} > 0.99333$, i.e. $3^n > 150$, so $n = 5$ ($3^4 = 81, 3^5 = 243$). Check: $S_4 = 1.5 \times \frac{80}{81} = 1.4815$ (too small); $S_5 = 1.5 \times \frac{242}{243} = 1.4938 > 1.49 \checkmark$.
8. **220** — differences are the triangular numbers 3, 6, 10, 15, 21, so term n is $\frac{n(n+1)(n+2)}{6}$. With $n = 10$: $\frac{10 \times 11 \times 12}{6} = \frac{1320}{6} = 220$.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 25 minutes for all 20

1. 12, 17, 23, 30, ?
2. 3, 4, 8, 17, 33, ?
3. D, H, L, P, ?
4. 6, 12, 24, 48, ?
5. One term is wrong: 7, 14, 28, 55, 112. Which one?
6. 1, 4, 27, 256, ?
7. A2, D6, G12, J20, ?
8. 1000, 200, 40, 8, ?
9. 5, 12, 26, 54, ?
10. Find the missing term: 4, 9, ?, 34, 54
11. 2, 6, 12, 20, 30, ?
12. A series has $a_1 = 1, a_2 = 1$ and $a_{n+2} = a_{n+1} - a_n$. Find a_{50} .
13. 13, 16, 22, 31, 43, ?
14. ZY, WV, TS, QP, ?
15. 3, 7, 16, 35, 74, ?
16. Find the sum of the first 12 terms of 1, 3, 6, 10, 15, ...
17. 1, 11, 21, 1211, ?
18. 2, 100, 4, 95, 8, 90, 16, ?
19. 121, 144, 169, 196, ?
20. One term is wrong: 2, 6, 12, 20, 30, 44. Which one?

ANSWER KEY

1. **38** — differences 5, 6, 7; next +8; $30 + 8 = 38$.
2. **58** — differences 1, 4, 9, 16 (squares); next +25; $33 + 25 = 58$.
3. **T** — positions 4, 8, 12, 16 (+4); $16 + 4 = 20 \rightarrow$ T.
4. **96** — constant ratio $\times 2$; $48 \times 2 = 96$.
5. **55** — the rule is $\times 2$: 7, 14, 28, 56, 112. The fourth term should be 56.
6. **3125** — terms are n^n : $1^1, 2^2, 3^3, 4^4, 5^5 = 3125$.
7. **M30** — letters +3 (A, D, G, J, M); numbers $n(n+1)$: 2, 6, 12, 20, 30.
8. **1.6** — constant ratio $\div 5$; $8 \div 5 = 1.6$.
9. **110** — rule $\times 2 + 2$; $54 \times 2 + 2 = 110$.
10. **19** — differences 5, 10, 15, 20; $9 + 10 = 19$, and $19 + 15 = 34 \checkmark$.

11. **42** — terms are $n(n + 1)$; $6 \times 7 = 42$.
12. **1** — period 6: 1, 1, 0, -1, -1, 0. $50 = 6 \times 8 + 2$, so $a_{50} = a_2 = 1$.
13. **58** — differences 3, 6, 9, 12; next +15; $43 + 15 = 58$.
14. **NM** — pairs are (26, 25), (23, 22), (20, 19), (17, 16), so next is (14, 13) $\rightarrow$ N, M.
15. **153** — rule $\times 2$ then add 1, 2, 3, 4, 5; $74 \times 2 + 5 = 153$.
16. **364** — triangular numbers; $\frac{12 \times 13 \times 14}{6} = \frac{2184}{6} = 364$.
17. **111221** — read 1211 aloud: “one 1, one 2, two 1s”.
18. **85** — interleaved: odd places $\times 2$ (2, 4, 8, 16); even places -5 (100, 95, 90, 85); position 8 is even.
19. **225** — squares of 11, 12, 13, 14, 15; $15^2 = 225$.
20. **44** — terms should be $n(n + 1)$: 2, 6, 12, 20, 30, 42. The sixth term should be 42.

ONE-PAGE REVISION CARD

The 5-step attack

1. Differences constant? → AP.
2. Second differences constant? → quadratic; keep extending the difference row.
3. Ratios constant, or growing as $\times 2, \times 3, \times 4$? → GP or growing-multiplier.
4. Near $n^2, n^3, n!, n(n+1)$, or the primes?
5. Zig-zag shape? → alternating rules, or two interleaved series (split odd and even positions).

Rules to recognise on sight

$+d / \times r$ constant	3, 7, 11, 15 / 2, 6, 18, 54
growing difference / multiplier	7, 12, 19, 28, 39 / 3, 6, 18, 72
$\times 2 + c$ with c growing	4, 9, 20, 43, 90
$n^2 / n^2 \pm 1$	1, 4, 9, 16 / 2, 5, 10, 17
$n^3 / n^3 \pm 1$	1, 8, 27, 64 / 2, 9, 28, 65
$n(n+1)$	2, 6, 12, 20, 30
$n^2(n+1)$	2, 12, 36, 80, 150
factorial	1, 2, 6, 24, 120, 720
Fibonacci type	5, 7, 12, 19, 31, 50
square previous	1, 2, 5, 26, 677
look-and-say	1, 11, 21, 1211, 111221

Formulas

- AP: $a_n = a + (n-1)d$; $S_n = \frac{n}{2}[2a + (n-1)d]$
- GP: $a_n = ar^{n-1}$; $S_n = \frac{a(r^n-1)}{r-1}$; $S_\infty = \frac{a}{1-r}$ for $|r| < 1$
- $\sum n = \frac{n(n+1)}{2}$; $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$; $\sum n^3 = \left(\frac{n(n+1)}{2}\right)^2$
- Triangular: $T_k = \frac{k(k+1)}{2}$; $\sum_{k=1}^n T_k = \frac{n(n+1)(n+2)}{6}$
- Periodic rule $a_{n+2} = a_{n+1} - a_n$ always repeats with period 6.

Shortcuts

- Extend the difference row instead of fitting a formula. Faster, safer.
- Missing middle term: build it from the left, then test it against the right.
- Wrong-term question: correct the suspect, then check that the **next** term now fits.
- Letters → numbers (EJOTY: E5, J10, O15, T20, Y25) → solve → back to letters.
- Huge term index (like the 100th or 2025th term)? Look for a repeating cycle, then divide and take the remainder.

Top 5 traps

1. “Which term is wrong” $\neq$ “what should it be”. Two different numbers.
2. In an interleaved series, check the **position number** of the asked term first.
3. 1, 4, 9, 16 may be the terms themselves **or** the differences. Test both.
4. Counting letters in your head. Write the positions down, always.
5. Remainder work: for period 6, a remainder of 0 means the **6th** term, not the 1st.

Coding–Decoding & Cryptarithmic

Letters hide numbers. Digits hide letters.

What you need to know

WHAT YOU NEED TO KNOW

The alphabet, both ways

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
From left	1	2	3	4	5	6	7	8	9	10	11	12	13
From right	26	25	24	23	22	21	20	19	18	17	16	15	14

Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
From left	14	15	16	17	18	19	20	21	22	23	24	25	26
From right	13	12	11	10	9	8	7	6	5	4	3	2	1

- Memory hook **EJOTY**: E = 5, J = 10, O = 15, T = 20, Y = 25. Count a step or two from the nearest of these.
- **Position from the right** = 27 − (position from the left).
- **Opposite letter** (A ↔ Z, B ↔ Y, C ↔ X, ...): position 27 − *n*. So M ↔ N, and that is the only pair in the middle.
- **Shift with wrap**: new position = ((*p* + *k* − 1) mod 26) + 1. After Z comes A again.

The six coding types you will meet

1. Letter shift	Every letter moves the same number of steps. CAT → DBU is +1.
2. Word pattern	The whole word is rearranged: reversed, halves swapped, letters paired.
3. Number coding	A word becomes a number: sum of positions, positions listed, digit sums, position × rank.
4. Symbol / operator coding	Symbols are swapped for + − × ÷. Translate, then apply BODMAS.
5. Substitution (message) coding	Whole words get code words. Match sentences and take the common code for the common word.
6. Conditional coding	A base table plus 2–3 “if ...” rules that override it for the first or last item.

Cryptarithmic — the rules of the game

- Each letter stands for exactly one digit, 0–9. Different letters, different digits (unless the question says otherwise).
- A **leading digit is never 0**.
- When you add two numbers, the carry out of any column is **0 or 1**. For three numbers it is 0, 1 or 2.
- Start at the **units column**. It is the only column with no carry coming in.

Reversal identities — memorise these, they solve half the puzzles

$$\overline{AB} + \overline{BA} = 11(A + B)$$

$$\overline{AB} - \overline{BA} = 9(A - B)$$

$$\overline{ABA} + \overline{BAB} = 111(A + B)$$

$$\overline{ABC} + \overline{CBA} = 101(A + C) + 20B$$

$$\overline{AA} = 11A, \quad \overline{AAA} = 111A, \quad \overline{AAAA} = 1111A$$

- Useful factorisations: $111 = 3 \times 37$, $1111 = 11 \times 101$, $11111 = 41 \times 271$.
- Units-digit facts: a number $\times 4$ or $\times 6$ is always even, so any digit it produces in the units place is even.

Warm-up

WARM-UP

build the reflex

Example 1

If CAT is written as DBU, how is DOG written?

SOLUTION

C $\rightarrow$ D, A $\rightarrow$ B, T $\rightarrow$ U. Each letter moves 1 step forward.
D $\rightarrow$ E, O $\rightarrow$ P, G $\rightarrow$ H.

Answer: EPH

Example 2

What is the position of R from the left, and which letter is the opposite of R?

SOLUTION

From EJOTY, O = 15, so P = 16, Q = 17, R = 18.
Opposite letter position = $27 - 18 = 9$. Letter 9 is I.

Answer: 18; opposite is I

Example 3

If A = 1, B = 2, ..., Z = 26, find the value of CAB.

SOLUTION

C = 3, A = 1, B = 2.
 $3 + 1 + 2 = 6$.

Answer: 6

Example 4

Code SUN by moving each letter 2 steps back.

SOLUTION

S = 19; $19 - 2 = 17 \rightarrow$ Q.
U = 21; $21 - 2 = 19 \rightarrow$ S.
N = 14; $14 - 2 = 12 \rightarrow$ L.

Answer: QSL

Example 5

In a code the word is simply written backwards. Code LAMP.

SOLUTION

| LAMP read from the end: P, M, A, L.

Answer: PMAL

Example 6

If FIVE is coded 6-9-22-5, code NINE.

SOLUTION

The rule is “replace each letter by its position”.

N = 14, I = 9, N = 14, E = 5.

Answer: 14-9-14-5

Example 7

In an addition puzzle the units column reads $B + B$ and the answer shows 4 with a carry of 1. Find B.

SOLUTION

The column result is 14, since the written digit is 4 and 1 is carried.

$B + B = 14$, so $2B = 14$ and $B = 7$.

Check: $7 + 7 = 14$ ✓ — write 4, carry 1.

Answer: $B = 7$

Example 8

Code CAB by replacing each letter with its opposite letter.

SOLUTION

C = 3; $27 - 3 = 24 \rightarrow X$.

A = 1; $27 - 1 = 26 \rightarrow Z$.

B = 2; $27 - 2 = 25 \rightarrow Y$.

Answer: XZY

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

In a code TIGER is written as UJHFS. How is LION written?

SOLUTION

Compare letter by letter.

T = 20 $\rightarrow$ U = 21: +1

I = 9 $\rightarrow$ J = 10: +1

G = 7 $\rightarrow$ H = 8: +1

E = 5 $\rightarrow$ F = 6: +1

R = 18 $\rightarrow$ S = 19: +1

The shift is +1 for all. Apply it to LION:

L = 12 $\rightarrow$ 13 = M I = 9 $\rightarrow$ 10 = J O = 15 $\rightarrow$ 16 = P N = 14 $\rightarrow$ 15 = O

Answer: MJPO

SHORTCUT

Never check all the letters. Check **one** letter to guess the shift, then check **one more** to confirm it. Two checks are enough, and they take four seconds.

Example 10 [Infosys pattern]

If MOUSE is coded OQWUG, code PLANT.

SOLUTION

$M = 13 \rightarrow O = 15: +2$. Confirm with $E = 5 \rightarrow G = 7: +2$ ✓

Apply +2 to PLANT:

$P = 16 \rightarrow 18 = R$

$L = 12 \rightarrow 14 = N$

$A = 1 \rightarrow 3 = C$

$N = 14 \rightarrow 16 = P$

$T = 20 \rightarrow 22 = V$

Answer: RNCPV

Example 11 [Wipro pattern]

If BAT is coded YZG, how is CAP coded?

SOLUTION

$B = 2 \rightarrow Y = 25$. Note $2 + 25 = 27$.

$A = 1 \rightarrow Z = 26$. And $1 + 26 = 27$ ✓

$T = 20 \rightarrow G = 7$. And $20 + 7 = 27$ ✓

So the rule is the opposite letter: new position $= 27 - n$.

$C = 3: 27 - 3 = 24 \rightarrow X$

$A = 1: 27 - 1 = 26 \rightarrow Z$

$P = 16: 27 - 16 = 11 \rightarrow K$

Answer: XZK

TRAP

A shift and an opposite-letter code can look alike for a word full of early letters. Always test a letter from the **second half** of the alphabet (here $T \rightarrow G$). A shift moves it a little; an opposite code throws it across.

Example 12 [Accenture pattern]

In a code $RAIN = 42$. Using the same rule, what is SUN?

SOLUTION

Check the sum-of-positions rule for RAIN:

$R = 18, A = 1, I = 9, N = 14$.

$18 + 1 = 19; 19 + 9 = 28; 28 + 14 = 42$ ✓

Now SUN: $S = 19, U = 21, N = 14$.

$19 + 21 = 40; 40 + 14 = 54$.

Answer: 54

Example 13 [Capgemini pattern]

In a code DELHI is written as JIMFE. How is MUMBAI written?

SOLUTION

The code has the same length, so look at the **last** code letter first.
 Code letter E = 5; the first word letter is D = 4. That is +1.
 Code letter J = 10; the last word letter is I = 9. That is +1.
 So the rule is: shift every letter +1, then write the word backwards.
 Check fully: D → E, E → F, L → M, H → I, I → J gives EFMIJ; reversed it is JIMFE ✓

Now MUMBAI.

Step 1 — shift +1:

M = 13 → 14 = N

U = 21 → 22 = V

M = 13 → 14 = N

B = 2 → 3 = C

A = 1 → 2 = B

I = 9 → 10 = J

That gives NVNCBJ.

Step 2 — reverse it: J, B, C, N, V, N.

Answer: JBCNVN

Example 14 [TCS NQT pattern]

In a certain language:

- “pit na so” means “you are good”
- “na re ma” means “they are smart”
- “so ma de” means “good smart boy”

What is the code for “smart”?

SOLUTION

Find which sentences share the word “smart”: the 2nd and the 3rd.

Codes in sentence 2: pit? no — they are na, re, ma.

Codes in sentence 3: so, ma, de.

The only code appearing in both lists is **ma**. So “smart” = ma.

Cross-check the rest:

“are” is in sentences 1 and 2. Sentence 1 codes: pit, na, so. Sentence 2 codes: na, re, ma. Common: na. So “are” = na.

“good” is in sentences 1 and 3. Common code of {pit, na, so} and {so, ma, de} is so. So “good” = so.

Everything is consistent.

Answer: ma

SHORTCUT

For substitution puzzles, do not try to decode the whole language. Circle the **one word** the question asks about, find the **two sentences** that contain it, and take the code common to both. Thirty seconds, done.

Example 15 [Cognizant pattern]

In a code 2 = C, 3 = F, 4 = I. What does 6 stand for?

SOLUTION

C is letter 3, F is letter 6, I is letter 9.

So the number 2 gives position 3, 3 gives 6, 4 gives 9.

Positions go 3, 6, 9 — they rise by 3 as the number rises by 1.

Rule: position = $3 \times (\text{number} - 1)$.

Check: number 4 → $3 \times 3 = 9$ ✓ (I).

Number 6 → $3 \times (6 - 1) = 3 \times 5 = 15$. Letter 15 is O.

Answer: O

Example 16 [TCS NQT pattern]

If FLOWER is coded 6-12-15-23-5-18, decode 7-1-18-4-5-14.

SOLUTION

Check the rule: F = 6, L = 12, O = 15, W = 23, E = 5, R = 18 ✓ — plain positions.

Now read the numbers back as letters:

7 → G 1 → A 18 → R 4 → D 5 → E 14 → N

Answer: GARDEN

Example 17 [Accenture pattern]

If CHAIR is coded FKDLU, code TABLE.

SOLUTION

C = 3 → F = 6: +3. Confirm with R = 18 → U = 21: +3 ✓

T = 20 → 23 = W

A = 1 → 4 = D

B = 2 → 5 = E

L = 12 → 15 = O

E = 5 → 8 = H

Answer: WDEOH

Example 18 [Wipro pattern]

In a code CAT is written DCW. Using the same rule, code BIRD.

SOLUTION

C = 3 → D = 4: moved +1.

A = 1 → C = 3: moved +2.

T = 20 → W = 23: moved +3.

So each letter moves forward by **its own place in the word**: 1st letter +1, 2nd +2, 3rd +3, and so on.

BIRD:

B = 2, place 1: $2 + 1 = 3 \rightarrow C$

I = 9, place 2: $9 + 2 = 11 \rightarrow K$

R = 18, place 3: $18 + 3 = 21 \rightarrow U$

D = 4, place 4: $4 + 4 = 8 \rightarrow H$

Answer: CKUH

TRAP

Wrap-around kills marks. If a shift pushes you past 26, subtract 26 — do not stop at Z. Example: X = 24 with +4 gives $28 - 26 = 2 \rightarrow B$, not “off the end”.

Example 19 [Infosys pattern]

If the sign + means $\times$, - means $\div$, $\times$ means -, and $\div$ means +, find the value of

$$16 - 4 \div 2 + 3 \times 5$$

SOLUTION

Replace every sign by what it really means:

- $16 - 4$ becomes $16 \div 4$
- $\div 2$ becomes $+2$

- $+3$ becomes $\times 3$
- $\times 5$ becomes -5

The real expression is

$$16 \div 4 + 2 \times 3 - 5$$

Now BODMAS. Division and multiplication first:

$$16 \div 4 = 4 \quad \text{and} \quad 2 \times 3 = 6$$

The expression becomes $4 + 6 - 5$.

$$4 + 6 = 10; \quad 10 - 5 = 5.$$

Answer: 5

Example 20 [TCS NQT pattern]

$\overline{AB}$ is a two-digit number and $\overline{BA}$ is the same digits reversed. If $\overline{AB} + \overline{BA} = 99$, find $A + B$.

SOLUTION

Write the numbers out in full.

$$\overline{AB} = 10A + B \quad \text{and} \quad \overline{BA} = 10B + A$$

Add them:

$$(10A + B) + (10B + A) = 11A + 11B = 11(A + B)$$

So $11(A + B) = 99$, giving $A + B = 99 \div 11 = 9$.

Check with one example: $A = 5, B = 4$ gives $54 + 45 = 99$ ✓

Answer: 9

PRACTICE · TIER 1 target: 60 s/Q

1. If FROG is coded GSPH, code BIRD.
2. If LAMP is coded MBNQ, decode TVO.
3. Code MOON using the opposite-letter rule ($A \leftrightarrow Z, B \leftrightarrow Y, \dots$).
4. If $A = 1, B = 2, \dots$, find the value of CODE.
5. If SKY is coded PHV, code SUN.
6. “kal bat ro” means “she is nice”; “bat ti la” means “is he there”. What is the code for “is”?
7. If $5 = E, 10 = J, 15 = O$, what does 20 stand for?
8. If the sign $+$ means $-$, $-$ means $\times$, $\times$ means $\div$, and $\div$ means $+$, find $18 \times 3 + 6 - 2 \div 4$.
9. In a code the word is first written backwards, then every letter is moved 1 step forward (so TEAM $\rightarrow$ NBFU). Code PLAY.
10. If $\overline{AB} + \overline{BA} = 143$, find $A + B$.
11. If RED is coded IVW, code BLUE.
12. In a code each letter moves forward by its place in the word (1st letter $+1$, 2nd $+2, \dots$). Code DEAL.
13. If MONDAY = 72 under the sum-of-positions rule, what is FRIDAY?
14. Decode 8-15-14-5-25.
15. In the addition $\overline{A8} + \overline{3B} = 105$, find A and B .

ANSWER KEY

1. **CJSE** — the shift is $+1$ ($F \rightarrow G, R \rightarrow S$); $B \rightarrow C, I \rightarrow J, R \rightarrow S, D \rightarrow E$.
2. **SUN** — the code shift is $+1$, so decoding is -1 : $T \rightarrow S, V \rightarrow U, O \rightarrow N$.
3. **NLLM** — $27 - 13 = 14$ (N), $27 - 15 = 12$ (L), L again, $27 - 14 = 13$ (M).
4. $27 - 3 + 15 + 4 + 5 = 27$.
5. **PRK** — $S = 19 \rightarrow P = 16$ is -3 ; $S \rightarrow P, U = 21 \rightarrow R = 18, N = 14 \rightarrow K = 11$.
6. **bat** — “is” is the only word in both sentences; “bat” is the only code in both.
7. **T** — the rule is simply the alphabet position; letter 20 is T.

8. **-2** — the real sum is $18 \div 3 - 6 \times 2 + 4 = 6 - 12 + 4 = -2$.
9. **ZBMQ** — reverse PLAY to get YALP, then +1: $Y \rightarrow Z, A \rightarrow B, L \rightarrow M, P \rightarrow Q$.
10. **13** — $11(A + B) = 143$, so $A + B = 143 \div 11 = 13$.
11. **YOFV** — opposite letters: $27 - 2 = 25$ (Y), $27 - 12 = 15$ (O), $27 - 21 = 6$ (F), $27 - 5 = 22$ (V).
12. **EGDP** — $D+1 = E, E+2 = G, A+3 = D, L+4 = P$.
13. **63** — $6 + 18 + 9 + 4 + 1 + 25 = 63$.
14. **HONEY** — letters 8, 15, 14, 5, 25.
15. **A = 6, B = 7** — units: $8 + B$ must end in 5, so $B = 7$ (carry 1); tens: $A + 3 + 1 = 10$, so $A = 6$. Check $68 + 37 = 105 \checkmark$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A code moves every **vowel** 2 steps forward and every **consonant** 1 step back. Code MANGO.

SOLUTION

Split the word: vowels are A and O; consonants are M, N, G.

M = 13, consonant: $13 - 1 = 12 \rightarrow L$

A = 1, vowel: $1 + 2 = 3 \rightarrow C$

N = 14, consonant: $14 - 1 = 13 \rightarrow M$

G = 7, consonant: $7 - 1 = 6 \rightarrow F$

O = 15, vowel: $15 + 2 = 17 \rightarrow Q$

Answer: LCMFQ

Example 22 [Shopee · pattern]

In a seller's private language:

- “ho ra mi” means “fast cheap delivery”
- “mi lo ka” means “delivery is late”
- “ra se lo” means “cheap and is”

Find the codes for “cheap” and for “delivery”.

SOLUTION

“**cheap**” appears in sentences 1 and 3.

Sentence 1 codes: {ho, ra, mi}. Sentence 3 codes: {ra, se, lo}.

Codes in both lists: ra. So “cheap” = ra.

“**delivery**” appears in sentences 1 and 2.

Sentence 1 codes: {ho, ra, mi}. Sentence 2 codes: {mi, lo, ka}.

Codes in both lists: mi. So “delivery” = mi.

Cross-check with a third word. “is” appears in sentences 2 and 3.

{mi, lo, ka} and {ra, se, lo} share lo. So “is” = lo.

Now sentence 1 has ho left over for “fast”, sentence 2 has ka left for “late”, sentence 3 has se left for “and”.

Nothing clashes, so the answers are safe.

Answer: cheap = ra, delivery = mi

Example 23 [DBS · pattern]

In a code, each letter is first replaced by its alphabet position, and then that number is replaced by the sum of its digits. So FARE becomes 6195. Code TRIP.

SOLUTION

First confirm the rule on FARE.

$F = 6 \rightarrow$ single digit already $\rightarrow 6$

$A = 1 \rightarrow 1$

$R = 18 \rightarrow 1 + 8 = 9$

$E = 5 \rightarrow 5$

Written together: 6, 1, 9, 5 $\rightarrow 6195$ ✓

Now TRIP.

$T = 20 \rightarrow 2 + 0 = 2$

$R = 18 \rightarrow 1 + 8 = 9$

$I = 9 \rightarrow 9$

$P = 16 \rightarrow 1 + 6 = 7$

Written together: 2, 9, 9, 7.

Answer: 2997

TRAP

“Digit sum” means add the digits **once**, not until you reach a single digit, unless the question says so. Here $T = 20$ gives 2 in one step; do not keep going and turn 9 into 9, or 18 into 9 into 9. Follow the worked example the question gives you.

Example 24 [Agoda · pattern]

A booking system codes words by these rules:

- If the word begins with a **consonant**: move every letter 2 steps forward.
- If the word begins with a **vowel**: write the word backwards first, then move every letter 1 step back.

Code ORANGE and PLANE.

SOLUTION

ORANGE begins with O, a vowel. Use rule 2.

Step 1 — write it backwards: E, G, N, A, R, O $\rightarrow$ EGNARO

Step 2 — move each letter 1 step back:

$E = 5 \rightarrow 4 = D$

$G = 7 \rightarrow 6 = F$

$N = 14 \rightarrow 13 = M$

$A = 1 \rightarrow 0$? Position 0 does not exist, so wrap: $1 - 1 + 26 = 26 \rightarrow Z$

$R = 18 \rightarrow 17 = Q$

$O = 15 \rightarrow 14 = N$

Result: DFMZQN

PLANE begins with P, a consonant. Use rule 1: move every letter 2 forward.

$P = 16 \rightarrow 18 = R$

$L = 12 \rightarrow 14 = N$

$A = 1 \rightarrow 3 = C$

$N = 14 \rightarrow 16 = P$

$E = 5 \rightarrow 7 = G$

Result: RNCPG

Answer: ORANGE $\rightarrow$ DFMZQN, PLANE $\rightarrow$ RNCPG

Example 25 [GIC · pattern]

In the addition below, A, B, C are digits and $\overline{ABC}$ is a three-digit number.

$$\overline{ABC} + \overline{ABC} + \overline{ABC} = \overline{CCC}$$

Find A , B and C .

SOLUTION

Write both sides as numbers.

Left side $= 3 \times \overline{ABC}$.

Right side $= \overline{CCC} = 111 \times C$.

So

$$3 \times \overline{ABC} = 111C \Rightarrow \overline{ABC} = \frac{111C}{3} = 37C$$

Now use the units digit. The units digit of $\overline{ABC}$ is C . The units digit of $37C$ is the units digit of $7C$. So

$$7C \text{ and } C \text{ must end in the same digit} \Rightarrow 7C - C = 6C \text{ ends in } 0$$

$6C$ ends in 0 only when $C = 0$ or $C = 5$.

- $C = 0$ gives $\overline{ABC} = 0$, not a three-digit number. Reject.
- $C = 5$ gives $\overline{ABC} = 37 \times 5 = 185$.

So $A = 1$, $B = 8$, $C = 5$.

Check: $185 + 185 + 185 = 555$ ✓ and $555 = \overline{CCC}$ with $C = 5$ ✓

Answer: $A = 1$, $B = 8$, $C = 5$

SHORTCUT

Repeated-digit numbers factor beautifully: $\overline{CC} = 11C$, $\overline{CCC} = 111C = 3 \times 37 \times C$, $\overline{CCCC} = 1111C = 11 \times 101 \times C$. The moment you see one in a puzzle, replace it and the algebra collapses to one line.

Example 26 [SCB · pattern]

A , B , C are digits with $A \neq 0$ and $C \neq 0$. Given

$$\overline{ABC} + \overline{CBA} = 1049$$

find B , and count how many ordered pairs (A, C) are possible.

SOLUTION

Expand both numbers.

$$\overline{ABC} = 100A + 10B + C$$

$$\overline{CBA} = 100C + 10B + A$$

Add:

$$101A + 20B + 101C = 101(A + C) + 20B = 1049$$

Find $A + C$. $A + C$ is a whole number at most 18, and $101(A + C) \leq 1049$ gives $A + C \leq 10.3$, so $A + C \leq 10$.

Try the largest values:

- $A + C = 10$: $1010 + 20B = 1049 \Rightarrow 20B = 39$. Not a whole number. Reject.
- $A + C = 9$: $909 + 20B = 1049 \Rightarrow 20B = 140 \Rightarrow B = 7$ ✓
- $A + C = 8$: $808 + 20B = 1049 \Rightarrow 20B = 241$. Reject.

Only $A + C = 9$ works, and then $B = 7$.

Count the pairs. Both A and C lead a three-digit number, so both are from 1 to 9.

$(A, C) = (1, 8), (2, 7), (3, 6), (4, 5), (5, 4), (6, 3), (7, 2), (8, 1)$.

$(9, 0)$ and $(0, 9)$ are not allowed because a leading digit cannot be 0.

That is 8 ordered pairs.

Answer: $B = 7$; 8 ordered pairs

Example 27 [LINE MAN · pattern]

In a code, P means +, Q means −, R means $\times$ and S means $\div$. Find the value of

$$36 \text{ S } 6 \text{ R } 5 \text{ Q } 12 \text{ P } 8$$

SOLUTION

Translate every letter:

$$36 \div 6 \times 5 - 12 + 8$$

BODMAS: do $\div$ and $\times$ from left to right first.

$$36 \div 6 = 6$$

$$6 \times 5 = 30$$

Now the expression is $30 - 12 + 8$.

$$30 - 12 = 18; 18 + 8 = 26.$$

Answer: 26

TRAP

After translating, work **left to right** among $\times$ and $\div$. Doing 6×5 before $36 \div 6$ gives $36 \div 30$, which is wrong. Same for + and −: $30 - 12 + 8$ is 26, not $30 - 20 = 10$.

Example 28 [Razer · pattern]

In a code, each letter is replaced by (its alphabet position) $\times$ (its place in the word).

(a) Code GAME. (b) Decode 2-18-33-20.

SOLUTION**(a) GAME.**

$$G = 7, \text{ place 1: } 7 \times 1 = 7$$

$$A = 1, \text{ place 2: } 1 \times 2 = 2$$

$$M = 13, \text{ place 3: } 13 \times 3 = 39$$

$$E = 5, \text{ place 4: } 5 \times 4 = 20$$

Code: 7-2-39-20

(b) Decode 2-18-33-20. To undo the rule, divide each number by its place.

$$\text{Place 1: } 2 \div 1 = 2 \rightarrow B$$

$$\text{Place 2: } 18 \div 2 = 9 \rightarrow I$$

$$\text{Place 3: } 33 \div 3 = 11 \rightarrow K$$

$$\text{Place 4: } 20 \div 4 = 5 \rightarrow E$$

Answer: (a) 7-2-39-20 (b) BIKE

Example 29 [Shopee · pattern]

Digits are coded by this table:

Digit	1	2	3	4	5	6	7	8	9	0
Code	\$	?	#	@	!	%	&	*	=	:

Two extra rules override the table:

- (i) If the first digit is odd and the last digit is even, swap the codes of the first and last digits.
- (ii) If the first and last digits are both odd, code both of them as ©.

Code the numbers 58347 and 39482.

SOLUTION

58347. First digit 5 (odd), last digit 7 (odd). Both odd $\rightarrow$ rule (ii) applies.

Plain codes: $5 \rightarrow !$, $8 \rightarrow *$, $3 \rightarrow \#$, $4 \rightarrow @$, $7 \rightarrow \&$

Rule (ii) replaces the first and last codes with ©.

Answer: © * # @ ©

39482. First digit 3 (odd), last digit 2 (even). Rule (i) applies.

Plain codes: $3 \rightarrow \#$, $9 \rightarrow =$, $4 \rightarrow @$, $8 \rightarrow *$, $2 \rightarrow ?$

Swap the first and last codes: the # and the ? change places.

Answer: $? = @ * \#$

Answer: 58347 $\rightarrow$ © * # @ © 39482 $\rightarrow$? = @ * #

Example 30 [Grab · pattern]

$\overline{AB}$ is a two-digit number with $A > B$. When it is multiplied by 3 the answer is a three-digit number whose three digits are all the same, $\overline{CCC}$. Find $\overline{AB}$.

SOLUTION

Set up. $3 \times \overline{AB} = \overline{CCC} = 111C$.

So

$$\overline{AB} = \frac{111C}{3} = 37C$$

$\overline{AB}$ must have two digits, so $10 \leq 37C \leq 99$.

$$37 \times 1 = 37 \quad (2 \text{ digits, ok})$$

$$37 \times 2 = 74 \quad (2 \text{ digits, ok})$$

$$37 \times 3 = 111 \quad (3 \text{ digits, too big})$$

So $C = 1$ giving 37, or $C = 2$ giving 74.

Use the condition $A > B$.

• 37: $A = 3$, $B = 7$, so $A < B$. Reject.

• 74: $A = 7$, $B = 4$, so $A > B$ ✓

Check: $74 \times 3 = 222$, and 222 is $\overline{CCC}$ with $C = 2$ ✓

Answer: $\overline{AB} = 74$ (and $C = 2$)

PRACTICE · TIER 2 target: 100 s/Q

1. A code moves every vowel 1 step forward and every consonant 3 steps forward. Code TIGER.
2. “ma tu li” means “buy fresh fish”; “li po ka” means “fish and rice”; “tu ka ne” means “fresh rice today”. What is the code for “rice”?
3. Using the “position, then digit sum” rule of Example 23, code MONK.
4. If $\overline{ABC} + \overline{CBA} = 1191$, find B .
5. If P means $\div$, Q means $\times$, R means $+$ and S means $-$, find the value of $48 \text{ P } 8 \text{ Q } 5 \text{ R } 9 \text{ S } 14$.
6. A code works like this: if the word starts with a consonant, write it backwards and then move each letter 1 step forward; if it starts with a vowel, just move each letter 2 steps forward. Code TABLE and EAGLE.
7. Using “position $\times$ place in the word”, code DUST.
8. Using the same rule, decode 3-16-27-64.
9. A two-digit number $\overline{AB}$ with $A < B$ is multiplied by 6 and gives $\overline{CCC}$. Find $\overline{AB}$.
10. Using the digit table and the two rules of Example 29, plus a third rule — (iii) if the first and last digits are both even, code both as § — code the number 47296.
11. In a code every letter is replaced by its opposite letter ($A \leftrightarrow Z$, ...) and then the whole word is written backwards. Code LAMP.
12. $\overline{AA} + \overline{BB} = 132$, where A and B are different non-zero digits. How many ordered pairs (A, B) are possible?

ANSWER KEY

1. **WJJFU** — $T+3 = W$, $I(\text{vowel})+1 = J$, $G+3 = J$, $E(\text{vowel})+1 = F$, $R+3 = U$.
2. **ka** — “rice” is in sentences 2 and 3; the code common to {li, po, ka} and {tu, ka, ne} is ka.
3. **4652** — $M = 13 \rightarrow 4$, $O = 15 \rightarrow 6$, $N = 14 \rightarrow 5$, $K = 11 \rightarrow 2$.
4. **$B = 4$** — $101(A + C) + 20B = 1191$; $A + C = 11$ gives $1111 + 20B = 1191$, so $20B = 80$ and $B = 4$.
5. **25** — the real sum is $48 \div 8 \times 5 + 9 - 14 = 6 \times 5 + 9 - 14 = 30 + 9 - 14 = 25$.
6. **FMCBU and GCING** — TABLE starts with a consonant: reverse to ELBAT, then $+1 \rightarrow$ FMCBU. EAGLE starts with a vowel: $+2 \rightarrow$ GCING.
7. **4-42-57-80** — $4 \times 1 = 4$, $21 \times 2 = 42$, $19 \times 3 = 57$, $20 \times 4 = 80$.
8. **CHIP** — $3 \div 1 = 3$ (C), $16 \div 2 = 8$ (H), $27 \div 3 = 9$ (I), $64 \div 4 = 16$ (P).
9. **37** — $6 \times \overline{AB} = 111C$, so $\overline{AB} = 18.5C$; two-digit whole answers need $C = 2$ (37) or $C = 4$ (74). Only 37 has $A < B$. Check $37 \times 6 = 222 \checkmark$.
10. **$\S \& ? = \S$** — 4 and 6 are both even, so rule (iii) fires. Plain codes were $@ \& ? = \%$; the first and last become $\S$.
11. **KNZO** — opposites give O, Z, N, K; reversed that is K, N, Z, O.
12. **6** — $11A + 11B = 132$ so $A + B = 12$; the ordered pairs of different non-zero digits are $(3, 9), (4, 8), (5, 7), (7, 5), (8, 4), (9, 3)$. $(6, 6)$ is barred because the digits must differ.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Google · pattern]

Find the smallest six-digit number $\overline{ABCDEF}$ such that

$$\overline{ABCDEF} \times 3 = \overline{BCDEFA}$$

That is, multiplying by 3 moves the first digit to the end.

SOLUTION

Insight: do not think about six unknown digits. Name the leading digit A and call everything after it a single block X . Moving a digit from front to back is just arithmetic on A and X .

Let $X = \overline{BCDEF}$, the last five digits (it may start with 0). Then

$$\overline{ABCDEF} = 100000A + X$$

$$\overline{BCDEFA} = 10X + A$$

The condition says

$$3(100000A + X) = 10X + A$$

$$300000A + 3X = 10X + A$$

$$300000A - A = 10X - 3X$$

$$299999A = 7X$$

$$X = \frac{299999A}{7}$$

Divide: $299999 \div 7 = 42857$ exactly ($7 \times 42857 = 299999$). So

$$X = 42857A$$

Now bound A . X has at most five digits, so $X \leq 99999$:

$$42857A \leq 99999 \Rightarrow A \leq 2.33$$

And $A \geq 1$ (leading digit). So $A = 1$ or $A = 2$.

- $A = 1$: $X = 42857$, number = 142857.

Check: $142857 \times 3 = 428571$ ✓ (and 428571 is 42857 followed by 1 ✓)
 • $A = 2$: $X = 85714$, number = 285714.
 Check: $285714 \times 3 = 857142$ ✓
 Both work. The smaller is 142857.

Answer: 142857

SHORTCUT

Whenever a puzzle says “move the first digit to the end” or “move the last digit to the front”, write the number as $10^k A + X$ or $10X + d$. One equation replaces a whole page of digit-by-digit carrying.

Example 32 [Amazon · pattern]

A four-letter word is called **self-coded** if replacing each letter by its opposite letter ($A \leftrightarrow Z$, $B \leftrightarrow Y$, ...) gives exactly the word written backwards. Letters may repeat and the “word” need not be a real English word. How many self-coded four-letter words are there?

SOLUTION

Insight: the condition links letter 1 with letter 4 and letter 2 with letter 3. Once you pick the first half, the second half is forced. So count the free choices.

Write the word as $w_1 w_2 w_3 w_4$. Let $\text{opp}(w)$ be the opposite letter.

The coded word is $\text{opp}(w_1) \text{opp}(w_2) \text{opp}(w_3) \text{opp}(w_4)$.

The reversed word is $w_4 w_3 w_2 w_1$.

Setting them equal, position by position:

$$\text{opp}(w_1) = w_4, \quad \text{opp}(w_2) = w_3, \quad \text{opp}(w_3) = w_2, \quad \text{opp}(w_4) = w_1$$

The third condition is the same as the second (opp applied twice returns the original letter, since $27 - (27 - n) = n$). The fourth is the same as the first.

So only two real conditions remain:

$$w_4 = \text{opp}(w_1) \quad \text{and} \quad w_3 = \text{opp}(w_2)$$

Count. w_1 is free: 26 choices. w_2 is free: 26 choices. Then w_3 and w_4 are fixed.

$$26 \times 26 = 676$$

Example check: take $w_1 = C$, $w_2 = B$. Then $w_3 = \text{opp}(B) = Y$ and $w_4 = \text{opp}(C) = X$, giving CBYX.

Code it: $C \rightarrow X$, $B \rightarrow Y$, $Y \rightarrow B$, $X \rightarrow C$ gives XYBC. Reverse CBYX: X, Y, B, C ✓ They match.

Answer: 676

Example 33 [Goldman Sachs · pattern]

A, B, C are three **different** non-zero digits and

$$\overline{AB} + \overline{BC} + \overline{CA} = 209$$

How many different sets $\{A, B, C\}$ satisfy this?

SOLUTION

Insight: the three numbers use each letter once in the tens place and once in the units place. That symmetry collapses the sum to a multiple of 11.

Expand:

$$\overline{AB} = 10A + B$$

$$\overline{BC} = 10B + C$$

$$\overline{CA} = 10C + A$$

Add:

$$(10A + A) + (10B + B) + (10C + C) = 11A + 11B + 11C = 11(A + B + C)$$

So

$$11(A + B + C) = 209 \Rightarrow A + B + C = \frac{209}{11} = 19$$

Now count. We need sets of three different digits from 1 to 9 adding to 19.

Organise by the largest digit.

- Largest = 9: the other two are different, below 9, adding to $19 - 9 = 10$: (2, 8), (3, 7), (4, 6). ((5, 5) repeats, (1, 9) reuses 9.) $\rightarrow$ 3 sets
- Largest = 8: other two below 8, adding to $19 - 8 = 11$: (4, 7), (5, 6). ((3, 8) reuses 8.) $\rightarrow$ 2 sets
- Largest = 7: other two below 7, adding to $19 - 7 = 12$: the biggest possible is $5 + 6 = 11 < 12$. $\rightarrow$ 0 sets
- Largest = 6 or less: the biggest total is $4 + 5 + 6 = 15 < 19$. $\rightarrow$ 0 sets

The sets are {2, 8, 9}, {3, 7, 9}, {4, 6, 9}, {4, 7, 8}, {5, 6, 8}.

Answer: 5 sets

Example 34 [Microsoft · pattern]

A coding rule replaces every letter by another letter, using a fixed one-to-one table (no two letters share a code). How many different code words can the word LEVEL produce?

SOLUTION

Insight: the code depends only on the distinct letters. Repeated letters must always map to the same code letter, so they cost nothing extra.

LEVEL uses which distinct letters?

L (places 1 and 5), E (places 2 and 4), V (place 3). That is **3 distinct letters**.

The table is one-to-one, so the codes of L, E and V must be three **different** letters.

- Choose the code for L: 26 ways.
- Choose the code for E: it must differ from L's code: 25 ways.
- Choose the code for V: it must differ from both: 24 ways.

$$26 \times 25 \times 24$$

$$26 \times 25 = 650$$

$$650 \times 24 = 15600$$

Note the shape of the code word is always $xyzyx$, so no two of these 15600 choices give the same code word — every count is distinct.

Answer: 15600

TRAP

Under a one-to-one substitution the **pattern of repeats never changes**. LEVEL must code to something of the shape $xyzyx$. If an option shows five different letters, it is impossible, no matter how clever it looks.

Example 35 [D. E. Shaw · pattern]

A and B are different non-zero digits and C is a digit. In the addition

$$\overline{ABA} + \overline{BAB} = \overline{CCC}$$

how many ordered pairs (A, B) are possible?

SOLUTION

Insight: the palindrome shape makes both numbers a multiple of 101 and 10. Expand once and the whole puzzle becomes “ $A + B$ is a single digit”.

$$\overline{ABA} = 100A + 10B + A = 101A + 10B$$

$$\overline{BAB} = 100B + 10A + B = 101B + 10A$$

Add:

$$111A + 111B = 111(A + B)$$

And $\overline{CCC} = 111C$. So

$$111(A + B) = 111C \Rightarrow A + B = C$$

Conditions.

- C is a single digit, so $A + B \leq 9$.
- $A \geq 1$ and $B \geq 1$ (both lead a three-digit number, and the question says non-zero).
- $A \neq B$.
- $C \neq 0$ is automatic, since $A + B \geq 1 + 2 = 3$.

Count ordered pairs (A, B) with $A, B \geq 1$, $A \neq B$, $A + B \leq 9$.

First count all ordered pairs with $A, B \geq 1$ and $A + B = s$: there are $s - 1$ of them. $s = 2$ gives 1 pair; $s = 3$ gives 2; $s = 4$ gives 3; $s = 5$ gives 4.

$s = 6$ gives 5; $s = 7$ gives 6; $s = 8$ gives 7; $s = 9$ gives 8. Total = $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36$.

Now remove the pairs with $A = B$: these are $(1, 1), (2, 2), (3, 3), (4, 4)$ — only these four keep $A + B \leq 9$. $((5, 5)$ gives 10, too big.)

$$36 - 4 = 32$$

Spot check: $(1, 2)$ gives $121 + 212 = 333$ ✓ with $C = 3$. $(6, 3)$ gives $636 + 363 = 999$ ✓ with $C = 9$.

Answer: 32 ordered pairs

Example 36 [Adobe · pattern]

Code X moves every letter k steps forward (with wrap-around). Code Y replaces every letter by its opposite letter.

(a) With $k = 7$, is “ Y then X ” ever the same as “ X then Y ” for any letter?

(b) For which k do the two orders agree for **every** letter?

SOLUTION

Insight: write both codes as formulas on the position number, then compose them. Two codes agree exactly when their formulas agree modulo 26.

Let a letter have position p . Work modulo 26 throughout (position 27 means 1, and so on).

$$X : p \rightarrow p + k$$

$$Y : p \rightarrow 27 - p, \text{ and } 27 \equiv 1 \pmod{26}, \text{ so } Y : p \rightarrow 1 - p.$$

“ **Y then X** ”: first $p \rightarrow 1 - p$, then add k :

$$p \rightarrow (1 - p) + k = 1 + k - p$$

“ **X then Y** ”: first $p \rightarrow p + k$, then apply Y :

$$p \rightarrow 1 - (p + k) = 1 - k - p$$

(a) Put $k = 7$.

“ Y then X ” gives $8 - p$. “ X then Y ” gives $-6 - p \equiv 20 - p \pmod{26}$.

These are equal only if $8 \equiv 20 \pmod{26}$, i.e. $20 - 8 = 12$ is a multiple of 26. It is not.

So the two orders **never** agree, for any letter.

(b) General k . Set the two formulas equal:

$$1 + k - p \equiv 1 - k - p \pmod{26}$$

$$k \equiv -k \pmod{26}$$

$$2k \equiv 0 \pmod{26}$$

$$k \equiv 0 \pmod{13}$$

Within one full turn ($0 \leq k \leq 25$) this gives $k = 0$ (no shift at all) or $k = 13$.

So the only real answer is $k = 13$.

Check with $k = 13$ on the letter C ($p = 3$):

“Y then X”: $C \rightarrow X$ ($27 - 3 = 24$), then $24 + 13 = 37$, $37 - 26 = 11 \rightarrow K$.

“X then Y”: $C \rightarrow 3 + 13 = 16 \rightarrow P$, then $27 - 16 = 11 \rightarrow K$ ✓ Same.

Answer: (a) never (b) $k = 13$

Example 37 [Uber · pattern]

A dispatch system codes each digit d of a trip ID as $c = (3d + 7) \bmod 10$.

(a) Show the code can always be undone, and write the decoding rule.

(b) Would the rule $c = (4d + 3) \bmod 10$ also work?

SOLUTION

Insight: a rule $c = (md + b) \bmod 10$ is reversible exactly when m has an inverse modulo 10, which happens only when m shares no factor with 10.

(a) We need a number m' with $3m' \equiv 1 \pmod{10}$.

Try: $3 \times 1 = 3$, $3 \times 2 = 6$, $3 \times 3 = 9$, $3 \times 7 = 21 \equiv 1$ ✓

So $m' = 7$. Now undo the rule:

$$c \equiv 3d + 7 \pmod{10}$$

$$c - 7 \equiv 3d$$

$$7(c - 7) \equiv 21d \equiv d \pmod{10}$$

$$d \equiv 7c - 49 \equiv 7c + 1 \pmod{10} \quad (\text{since } -49 + 50 = 1)$$

Decoding rule: $d = (7c + 1) \bmod 10$.

Test it on every kind of digit:

$d = 0 \rightarrow c = 7$; decode $7(7) + 1 = 50 \rightarrow 0$ ✓

$d = 4 \rightarrow c = 19 \bmod 10 = 9$; decode $7(9) + 1 = 64 \rightarrow 4$ ✓

$d = 9 \rightarrow c = 34 \bmod 10 = 4$; decode $7(4) + 1 = 29 \rightarrow 9$ ✓

(b) With $m = 4$, look at two different digits d and $d + 5$:

$$4(d + 5) + 3 = 4d + 20 + 3 \equiv 4d + 3 \pmod{10}$$

So d and $d + 5$ always get the **same** code. For example $d = 1$ gives 7, and $d = 6$ gives $27 \bmod 10 = 7$ too.

Two different digits sharing one code means the message cannot be decoded. The rule fails.

The reason: 4 and 10 share the factor 2, so 4 has no inverse modulo 10.

Answer: (a) $d = (7c + 1) \bmod 10$ (b) No — d and $d + 5$ collide

Example 38 [Goldman Sachs · pattern]

Find the five-digit number $\overline{ABCDE}$ such that

$$\overline{ABCDE} \times 4 = \overline{EDCBA}$$

SOLUTION

Insight: squeeze the unknowns with size and parity before touching algebra. Multiplying by 4 must keep the answer at five digits, and the answer must be even — those two facts alone pin down A and E .

Step 1 — find A . The product still has five digits, so

$$\overline{ABCDE} \times 4 \leq 99999 \Rightarrow \overline{ABCDE} \leq 24999 \Rightarrow A = 1 \text{ or } 2$$

The product $\overline{EDCBA}$ is $4 \times$ (something), so it is even. Its units digit is A , so A is even.

Therefore $A = 2$.

Step 2 — find E . The first digit of the product is E . Since $\overline{ABCDE} \geq 20000$,

$$\overline{EDCBA} \geq 4 \times 20000 = 80000 \Rightarrow E = 8 \text{ or } 9$$

Size alone leaves both 8 and 9. So use the units column of the multiplication instead: $4 \times E$ must end in $A = 2$.
 $4 \times 8 = 32$ ends in 2 ✓ $4 \times 9 = 36$ ends in 6 ✗
 So $E = 8$.

Step 3 — algebra for B, C, D . The number is $2BCD8$ and the product is $8DCB2$.

$$\overline{ABCDE} = 20000 + 1000B + 100C + 10D + 8$$

$$4 \times \text{that} = 80000 + 4000B + 400C + 40D + 32$$

$$\overline{EDCBA} = 80000 + 1000D + 100C + 10B + 2$$

Set them equal and cancel the 80000:

$$4000B + 400C + 40D + 32 = 1000D + 100C + 10B + 2$$

$$4000B - 10B + 400C - 100C + 40D - 1000D + 32 - 2 = 0$$

$$3990B + 300C - 960D + 30 = 0$$

Divide every term by 30:

$$133B + 10C - 32D + 1 = 0 \Rightarrow 32D = 133B + 10C + 1$$

Step 4 — test B . Since the number is at most 24999 and starts with 2, $B \leq 4$. Also $D \leq 9$, so $32D \leq 288$.

- $B = 0$: $32D = 10C + 1$, which is odd. But $32D$ is even. Impossible.
- $B = 1$: $32D = 133 + 10C + 1 = 134 + 10C$. Since $10C \geq 0$, we need $32D \geq 134$, so $D \geq 134 \div 32 = 4.2$, i.e. $D \geq 5$. Try each: $32 \times 5 = 160 \rightarrow 10C = 26$ ✗; $32 \times 6 = 192 \rightarrow 10C = 58$ ✗; $32 \times 7 = 224 \rightarrow 10C = 90 \rightarrow C = 9$ ✓; $32 \times 8 = 256 \rightarrow 10C = 122$ ✗; $32 \times 9 = 288 \rightarrow 10C = 154$ ✗.

So $B = 1, C = 9, D = 7$.

- $B = 2$: $32D = 266 + 10C + 1 = 267 + 10C$, which is odd. Impossible.
- $B = 3$: $32D = 399 + 10C + 1 = 400 + 10C \geq 400 > 288$. Impossible.
- $B = 4$: even larger. Impossible.

The number is 21978.

Check: $21978 \times 4 = 87912$; $20000 \times 4 = 80000$; $1978 \times 4 = 7912$; $80000 + 7912 = 87912$.

And 21978 written backwards is 87912 ✓

Answer: 21978

PRACTICE · TIER 3 target: 4 min/Q

1. In Example 31 two six-digit numbers worked. Find the larger one.
2. Under a one-to-one letter substitution, how many different code words can SUCCESS produce?
3. In $\overline{ABA} + \overline{BAB} = \overline{CCC}$ with A, B different non-zero digits, how many ordered pairs (A, B) give $C = 9$?
4. A code shifts every letter k steps forward, $1 \leq k \leq 25$. For which k does applying the code **twice** return every word to its original form?
5. A digit is coded as $c = (7d + 4) \bmod 10$. Write the decoding rule.
6. Find the four-digit number $\overline{ABCD}$ with $\overline{ABCD} \times 9 = \overline{DCBA}$.
7. A five-letter word is **self-coded** if replacing each letter by its opposite letter gives the word written backwards. How many such five-letter words are there?
8. A, B, C are different non-zero digits with $\overline{AB} + \overline{BC} + \overline{CA} = 176$. How many sets $\{A, B, C\}$ are possible?

ANSWER KEY

1. **285714** — from $X = 42857A$, the case $A = 2$ gives $X = 85714$ and the number 285714. Check: $285714 \times 3 = 857142$, which is 85714 followed by 2 ✓.
2. **358800** — SUCCESS uses the distinct letters S, U, C, E — four of them. A one-to-one table gives $26 \times 25 \times 24 \times 23 = 358800$. The repeated S's and C's add no freedom.
3. **8** — the identity gives $A + B = C = 9$. Ordered pairs of different non-zero digits summing to 9: $(1, 8), (2, 7), (3, 6), (4, 5), (5, 4), (6, 3), (7, 2), (8, 1)$.

4. $k = 13$ — two shifts of k give a shift of $2k$, and that must be a whole number of turns: $2k \equiv 0 \pmod{26}$, so k is a multiple of 13. In the range 1 to 25 only $k = 13$ works (ROT13).
5. $d = (3c + 8) \bmod 10$ — the inverse of 7 modulo 10 is 3, since $7 \times 3 = 21 \equiv 1$. From $c \equiv 7d + 4$ we get $d \equiv 3(c - 4) = 3c - 12 \equiv 3c + 8$. Test $d = 5$: $c = 39 \bmod 10 = 9$, and $3(9) + 8 = 35 \rightarrow 5$ ✓.
6. **1089** — $9 \times \overline{ABCD}$ is still four digits, so $\overline{ABCD} \leq 1111$ and $A = 1$. The product is then at least 9000, so $D = 9$. Expanding $9(1000 + 100B + 10C + 9) = 9000 + 100C + 10B + 1$ gives $89B = C - 8$, so $B = 0$ and $C = 8$. Check $1089 \times 9 = 9801$ ✓.
7. **0** — the condition forces $\text{opp}(w_3) = w_3$ for the middle letter, i.e. $27 - p = p$, so $p = 13.5$. No letter has that position, so no such word exists.
8. **8** — $11(A + B + C) = 176$ gives $A + B + C = 16$. By largest digit: 9 with $(1, 6), (2, 5), (3, 4)$; 8 with $(1, 7), (2, 6), (3, 5)$; 7 with $(3, 6), (4, 5)$; nothing below. Total $3 + 3 + 2 = 8$.

Mixed set — exam conditions

PRACTICE • TIER 1 target: 30 minutes for all 20

1. If PEN is coded QFO, code BOOK.
2. Code FLY using the opposite-letter rule.
3. If A = 1, B = 2, ..., find the value of TEAM.
4. “ho pa ni” means “big red car”; “pa ti lo” means “red fast bike”. What is the code for “red”?
5. If @ means +, # means −, \$ means $\times$ and & means $\div$, find $24 \ \& \ 6 \ \$ \ 5 \ \@ \ 8 \ \# \ 3$.
6. If $\overline{AB} + \overline{BA} = 154$, find $A + B$.
7. In a code every letter moves forward by its place in the word. Code DESK.
8. Decode 3-1-18-4.
9. If $\overline{ABC} + \overline{ABC} + \overline{ABC} = \overline{CCC}$, find $A + B + C$.
10. A code moves every vowel 3 steps forward and every consonant 1 step forward. Code CHAIR.
11. In a code the word is first written backwards, then every letter moves 2 steps forward. Code FROG.
12. A code shifts every letter k steps forward, $1 \leq k \leq 25$. For how many values of k does the coded form of ZOO contain the letter A?
13. If $\overline{ABA} + \overline{BAB} = 777$ with A, B different non-zero digits, how many ordered pairs (A, B) are there?
14. Each digit d is coded as $(3d + 1) \bmod 10$. Code the number 482.
15. Find the four-digit number $\overline{ABCD}$ with $\overline{ABCD} \times 4 = \overline{DCBA}$.
16. Under a one-to-one letter substitution, how many code words can DIGIT produce?
17. If MANGO is coded NBOHP, decode TVHBS.
18. In the addition $\overline{A7} + \overline{5B} = 132$, find $A + B$.
19. Using “position $\times$ place in the word”, code FAN.
20. Which letter is 8th from the right end of the alphabet?

ANSWER KEY

1. **CPPL** — the shift is +1; B $\rightarrow$ C, O $\rightarrow$ P, O $\rightarrow$ P, K $\rightarrow$ L.
2. **UOB** — $27 - 6 = 21$ (U), $27 - 12 = 15$ (O), $27 - 25 = 2$ (B).
3. **39** — $20 + 5 + 1 + 13 = 39$.
4. **pa** — “red” is the only shared word; “pa” is the only shared code.
5. **25** — the real sum is $24 \div 6 \times 5 + 8 - 3 = 4 \times 5 + 8 - 3 = 20 + 8 - 3 = 25$.
6. **14** — $11(A + B) = 154$, so $A + B = 14$.
7. **EGVO** — D+1 = E, E+2 = G, S+3 = V, K+4 = O.
8. **CARD** — letters 3, 1, 18, 4.
9. **14** — $3 \times \overline{ABC} = 111C$ gives $\overline{ABC} = 37C$; the units digit forces $C = 5$, so the number is 185 and $1 + 8 + 5 = 14$.
10. **DIDLS** — C+1 = D, H+1 = I, A (vowel) +3 = D, I (vowel) +3 = L, R+1 = S.
11. **IQTH** — reverse FROG to GORF, then +2: G $\rightarrow$ I, O $\rightarrow$ Q, R $\rightarrow$ T, F $\rightarrow$ H.

12. **2** — A appears if some letter lands on position 1. Z = 26 needs $k = 1$; O = 15 needs $15 + k \equiv 1 \pmod{26}$, i.e. $k = 12$. So $k \in \{1, 12\}$.
13. **6** — $111(A + B) = 777$ gives $A + B = 7$; ordered pairs of different non-zero digits: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$.
14. **357** — $4 \rightarrow 13 \pmod{10} = 3$; $8 \rightarrow 25 \pmod{10} = 5$; $2 \rightarrow 7$.
15. **2178** — the product stays four digits, so $\overline{ABCD} \leq 2499$ and $A = 1$ or 2 ; the product is even and ends in A , so $A = 2$. Then the number is at least 2000, so the product is at least 8000 and $D \geq 8$; $4D$ must end in 2, so $D = 8$. Expanding $4(2000 + 100B + 10C + 8) = 8000 + 100C + 10B + 2$ gives $2C = 13B + 1$, so $B = 1$ and $C = 7$. Check $2178 \times 4 = 8712$ ✓.
16. **358800** — DIGIT has four distinct letters D, I, G, T; $26 \times 25 \times 24 \times 23 = 358800$.
17. **SUGAR** — the code shift is $+1$, so decode with -1 : T $\rightarrow$ S, V $\rightarrow$ U, H $\rightarrow$ G, B $\rightarrow$ A, S $\rightarrow$ R.
18. **12** — units: $7 + B$ ends in 2, so $B = 5$ with carry 1; tens: $A + 5 + 1 = 13$, so $A = 7$. Check $77 + 55 = 132$ ✓. $A + B = 7 + 5 = 12$.
19. **6-2-42** — $6 \times 1 = 6$, $1 \times 2 = 2$, $14 \times 3 = 42$.
20. **S** — $27 - 8 = 19$, and letter 19 is S.

ONE-PAGE REVISION CARD

Alphabet facts

- EJOTY: E = 5, J = 10, O = 15, T = 20, Y = 25. Count from the nearest one.
- Position from the right = $27 - \text{position from the left}$.
- Opposite letter position = $27 - n$. Middle pair: M $\leftrightarrow$ N.
- Shift with wrap: new position = $((p + k - 1) \bmod 26) + 1$. Past 26? Subtract 26.
- Applying the same shift twice = a shift of $2k$. A shift of 13 is its own undo (ROT13).

The six coding types

1. **Letter shift** — test one letter, confirm with a second, then apply.
2. **Word pattern** — reversed or paired. Match the code's **last** letter to the word's **first**.
3. **Number coding** — sum of positions; positions listed; digit sums; position $\times$ place.
4. **Symbol / operator** — translate every sign, then BODMAS, then left to right.
5. **Substitution message** — the code shared by the two sentences containing the asked word.
6. **Conditional** — plain codes first, then the “if” rules — first and last item only.

Cryptarithmic toolkit

$$\overline{AB} + \overline{BA} = 11(A + B) \quad \overline{AB} - \overline{BA} = 9(A - B)$$

$$\overline{ABA} + \overline{BAB} = 111(A + B) \quad \overline{ABC} + \overline{CBA} = 101(A + C) + 20B$$

$$\overline{AA} = 11A \quad \overline{AAA} = 111A = 3 \cdot 37 \cdot A \quad \overline{AAAA} = 1111A = 11 \cdot 101 \cdot A$$

- Leading digit is never 0. Carry from any column of a two-number addition is 0 or 1.
- Start at the units column — nothing is carried into it.
- “Move the first digit to the end”: write the number as $10^k A + X$, the result as $10X + A$.
- “Move the last digit to the front”: write the number as $10X + d$, the result as $10^k d + X$.
- $\times 4, \times 6, \times 9$ puzzles: size limit first ($4N$ must still fit), parity second (product is even).
- $c = (md + b) \bmod 10$ reverses only when m is 1, 3, 7 or 9; undo by multiplying by its partner ($3 \leftrightarrow 7, 9 \leftrightarrow 9$).

Counting under a one-to-one substitution

- Count only the **distinct** letters, say r of them: $26 \times 25 \times \dots$ (r factors).
- The pattern of repeated letters can never change.

Top 5 traps

1. Testing the shift on early letters only. Always test a letter past M.
2. Forgetting wrap-around. Past Z go back to A; before A go back to Z.
3. In operator coding, breaking BODMAS or working right to left. $30 - 12 + 8 = 26$, never 10.
4. Conditional coding: the “if” rules usually look only at the **first and last** item. Do not apply them to the middle.
5. Cryptarithmic: assuming letters must be different when the question never said so — and assuming they may repeat when it did say so. Read that line twice.

Blood Relations, Directions & Ranking

Draw the tree, draw the map, count the row

What you need to know

QUICK REFERENCE

1. Blood relations — the three symbols you draw

- Write a **male** with a small + after the name, a **female** with a small –.
- Join a married couple with = on the same line (same generation).
- Drop a short vertical line from the couple to their children. Children sit on one horizontal line (they are siblings).
- **Same generation sits on the same level.** If two people are on the same level they are siblings, cousins or spouses — never parent and child.

2. Relation dictionary (learn this cold)

Father's / Mother's father	Grandfather	Husband's or wife's brother	Brother-in-law
Father's / Mother's mother	Grandmother	Husband's or wife's sister	Sister-in-law
Father's brother	Paternal uncle	Sister's husband	Brother-in-law
Father's sister	Paternal aunt	Brother's wife	Sister-in-law
Mother's brother	Maternal uncle	Son's wife	Daughter-in-law
Mother's sister	Maternal aunt	Daughter's husband	Son-in-law
Brother's / Sister's son	Nephew	Son's / Daughter's son	Grandson
Brother's / Sister's daughter	Niece	Son's / Daughter's daughter	Granddaughter
Uncle's / Aunt's child	Cousin	Wife's / Husband's parent	Parent-in-law

3. Coded relations. Solve one symbol at a time, left to right. Write each new fact on the tree before reading the next symbol. Never try to hold three symbols in your head.

4. Directions — the compass wheel

Clockwise order, 45° apart each step:

$$N \rightarrow NE \rightarrow E \rightarrow SE \rightarrow S \rightarrow SW \rightarrow W \rightarrow NW \rightarrow N$$

Turn table (memorise, do not re-derive):

Facing	Turn left	Turn right	About turn
North	West	East	South
East	North	South	West
South	East	West	North
West	South	North	East

- One left turn = 90° anticlockwise. One right turn = 90° clockwise.
- Two same-side turns = about turn = 180°.
- Angles: take North = 0° and count **clockwise**. E = 90°, S = 180°, W = 270°, NE = 45°, SE = 135°, SW = 225°, NW = 315°. Add for clockwise, subtract for anticlockwise, then reduce modulo 360°.

5. Net displacement. Add all North legs, subtract all South legs $\rightarrow$ vertical net v . Add all East legs, subtract all West legs $\rightarrow$ horizontal net h .

$$\text{distance} = \sqrt{h^2 + v^2}$$

Useful triples: (3, 4, 5), (5, 12, 13), (6, 8, 10), (8, 15, 17), (9, 12, 15), (7, 24, 25), (20, 21, 29), (9, 40, 41). If $|h| = |v|$ the answer is $h\sqrt{2}$ and the point is on an exact 45° corner line.

- **Naming the direction.** A corner name here means the **quarter**, not the exact 45° line. “North-East of the start” only means North of it and East of it. So $v = 3$ North with $h = 4$ East is still reported as North-East, even though the exact corner line needs $|h| = |v|$.
- Read the two nets as a pair of words: v positive = North, v negative = South; h positive = East, h negative = West. If one net is 0, say “due North”, “due South”, “due East” or “due West”.

6. Shadows

- Just after sunrise the sun is in the **East**, so every shadow points **West**.
- Just before sunset the sun is in the **West**, so every shadow points **East**.
- At about noon the shadow is too short to give a direction. Any question set “at 12 noon” has no usable direction.
- Left / right rule: a person facing direction D has the **right** hand pointing at D turned 90° clockwise, and the **left** hand at D turned 90° anticlockwise.

7. Ranking (a single row, no ties)

$$\text{Total} = (\text{rank from left}) + (\text{rank from right}) - 1$$

$$\text{rank from right} = \text{Total} - (\text{rank from left}) + 1$$

- People strictly between A and B (with A left of B), both ranks counted from the left:

$$\text{between} = (B \text{ from left}) - (A \text{ from left}) - 1$$

- “There are x people between A and B ” means their positions differ by $x + 1$.
- If k people scored **above** you, your rank from the top is $k + 1$.
- If k people scored **below** you, your rank from the bottom is $k + 1$.

Warm-up

WARM-UP

build the reflex

Example 1

Pointing to a boy, Ravi said, “He is the son of my father’s only son.” How is the boy related to Ravi?

SOLUTION

Ravi’s father’s only son = Ravi himself.
So the boy is the son of Ravi.

Answer: Son

Example 2

A is the sister of B. C is the mother of B. D is the father of C. How is A related to D?

SOLUTION

C is the mother of B, and A is B’s sister, so C is also A’s mother.
D is C’s father, so D is A’s grandfather.
A is female, so A is D’s granddaughter.

Answer: Granddaughter

Example 3

Meera faces North. She turns right, then turns right again. Which way does she face now?

SOLUTION

Facing North, turn right $\rightarrow$ East.

Facing East, turn right $\rightarrow$ South.

Answer: South

Example 4

A man walks 6 km East, then 8 km North. How far is he from his starting point?

SOLUTION

$h = 6$ (East), $v = 8$ (North).

distance $= \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ km.

Answer: 10 km

Example 5

In a row of 40 students, Arun is 12th from the left. What is his rank from the right?

SOLUTION

rank from right $= 40 - 12 + 1 = 29$.

Answer: 29th from the right

Example 6

Nisha is 9th from the top and 14th from the bottom of a merit list. How many students are on the list?

SOLUTION

Total $= 9 + 14 - 1 = 22$.

Answer: 22 students

Example 7

Somchai stands facing the rising sun. Which direction is his left hand pointing?

SOLUTION

The rising sun is in the East, so he faces East.

Facing East, the left hand points to East turned 90° anticlockwise = North.

Answer: North

Example 8

'P & Q' means P is the mother of Q. 'P $\times$ Q' means P is the brother of Q.

What does 'A & B $\times$ C' tell you about A and C?

SOLUTION

A & B: A is the mother of B.

B $\times$ C: B is the brother of C, so B and C have the same mother.

That mother is A.

Answer: A is the mother of C

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Pointing to a lady, Ramesh said, “She is the daughter of the only son of my grandfather.” How is the lady related to Ramesh?

SOLUTION

Grandfather’s only son = Ramesh’s father.

So the lady is the daughter of Ramesh’s father.

A daughter of Ramesh’s father is Ramesh’s sister.

Answer: Sister

SHORTCUT

Peel the sentence from the **inside out**, never from the start. Find the innermost phrase (“my grandfather”), replace it with a real person, then move one layer out. Write each layer on paper as a short chain: grandfather → his only son = father → his daughter = sister.

Example 10 [Infosys pattern]

In a code:

‘A % B’ means A is the father of B.

‘A ÷ B’ means A is the daughter of B.

‘A × B’ means A is the brother of B.

How is P related to S in ‘P ÷ Q % R × S’?

SOLUTION

P ÷ Q: P is the daughter of Q.

Q % R: Q is the father of R.

R × S: R is the brother of S.

So R and S have the same father. That father is Q.

Therefore P, R and S are all children of Q.

P is female, so P is the sister of S.

Answer: P is the sister of S

Example 11 [Accenture pattern]

Kavya leaves home, walks 3 km North, turns right and walks 4 km, then turns right and walks 6 km. How far is she from home, and in which direction?

SOLUTION

Leg 1: 3 km North.

Facing North, turn right → East. Leg 2: 4 km East.

Facing East, turn right → South. Leg 3: 6 km South.

Vertical net = $3 - 6 = -3$, i.e. 3 km South.

Horizontal net = 4 km East.

distance = $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ km.

She is South and East of home, so the direction is South-East.

Answer: 5 km, South-East of home

Example 12 [Wipro pattern]

In a row, Priya is 14th from the left and Rahul is 17th from the right. They swap seats. After the swap Priya is 21st from the left. How many people are in the row?

SOLUTION

After the swap Priya is sitting in Rahul's old seat.

So Rahul's old seat is 21st from the left.

That same seat is 17th from the right.

Total = $21 + 17 - 1 = 37$.

Answer: 37 people

❗ Priya's original 14th position is not needed. Extra data is normal in this pattern.

Example 13 [Capgemini pattern]

In a row of 45 children, Anil is 18th from the left and Bala is 21st from the right. How many children sit between them?

SOLUTION

Bring both ranks to the same side.

Bala from the left = $45 - 21 + 1 = 25$.

Anil is at 18, Bala is at 25, and $25 > 18$.

Between = $25 - 18 - 1 = 6$.

Answer: 6 children

TRAP

Do **not** subtract the two given ranks when they are counted from **opposite** ends. Here $21 - 18 = 3$ is meaningless. Always convert both to "from the left" first.

Example 14 [Cognizant pattern]

Introducing a man, a woman said, "His wife is the only daughter of my mother." How is the man related to the woman?

SOLUTION

The only daughter of the woman's mother must be the woman herself (she is her mother's only daughter).

So the man's wife is the woman.

Therefore the man is her husband.

Answer: Husband

TRAP

"The only daughter of my mother" and "the only son of my father" very often point **back at the speaker**. Before you invent a new person, check whether the speaker fits the description. Here inventing a sister would give the wrong answer "brother-in-law".

Example 15 [TCS NQT pattern]

One evening, just before sunset, Suresh and Tanvi stood facing each other. Tanvi's shadow fell exactly to the right of Suresh. Which direction was Suresh facing?

SOLUTION

Just before sunset the sun is in the West, so every shadow points East.

So East is on Suresh's right.

A person whose right hand points East is facing North (North turned 90° clockwise is East).

Answer: North

Example 16 [Infosys pattern]

Meena is the mother of Karan. Karan is the brother of Deepa. Deepa is the daughter of Suresh. Gopal is the father of Suresh. How is Gopal related to Karan?

SOLUTION

Deepa is the daughter of Suresh, so Suresh is Deepa's father.
Karan is Deepa's brother, so Suresh is also Karan's father.
Gopal is Suresh's father, so Gopal is Karan's father's father.

Answer: Gopal is Karan's grandfather (paternal)

Example 17 [Accenture pattern]

In a class, Ravi's rank is 9th from the top. Sita's rank is 12th from the bottom. Exactly 5 students rank between them, and Ravi ranks above Sita. How many students are in the class?

SOLUTION

Ravi is 9th from the top.
5 students sit between them, so Sita is $9 + 5 + 1 = 15$ th from the top.
Sita is also 12th from the bottom.
Total = $15 + 12 - 1 = 26$.

Answer: 26 students

TRAP

"5 students between them" is **not** "Sita is 5 places below Ravi". Positions differ by $5 + 1 = 6$, not by 5. Miscounting this by one is the single most common ranking error.

Example 18 [Wipro pattern]

A delivery rider starts at the depot and rides 5 km South, then 12 km West, then 5 km North, then 6 km East. How far is he from the depot and in which direction?

SOLUTION

Vertical: 5 South and 5 North cancel, net = 0.
Horizontal: 12 West and 6 East give net = $12 - 6 = 6$ km West.
distance = $\sqrt{0^2 + 6^2} = 6$ km.

Answer: 6 km, due West of the depot

SHORTCUT

Never draw the path leg by leg on a map. Make two running totals only: a **North–South** total and an **East–West** total. Five legs collapse into two numbers in about ten seconds.

Example 19 [Capgemini pattern]

In a queue of 60 people, Nithya is 24th from the front. Vikram stands 5 places behind Nithya. What is Vikram's position from the back?

SOLUTION

Vikram from the front = $24 + 5 = 29$.
Vikram from the back = $60 - 29 + 1 = 32$.

Answer: 32nd from the back

PRACTICE · TIER 1 target: 60 s/Q

1. Pointing to a man, Leela said, "His mother is the only daughter of my mother." How is the man related to Leela?
2. A is the brother of B. B is the sister of C. C is the son of D. How is A related to D?
3. Aarav faces South. He turns left, then right, then right again. Which direction does he face?
4. A man walks 10 m North, 6 m East, then 10 m South. How far is he from the start and in which direction?
5. In a row of 32 children, Kavir is 11th from the right. What is his position from the left?
6. Mei is 7th from the top and 23rd from the bottom of a merit list. How many students are on the list?
7. In a queue, Deepak is 15th from the front and 19th from the back. How many people are in the queue?
8. Pointing to a boy, Farida said, "He is the son of my grandfather's only son." How is the boy related to Farida?
9. 'A ÷ B' means A is the son of B. 'A × B' means A is the sister of B. What does 'P ÷ Q × R' tell you about P and R?
10. Rekha walks 4 km West, turns right and walks 3 km, then turns right and walks 4 km. How far is she from the start and in which direction?
11. In a row of 50 people, Suri is 20th from the left and Tina is 20th from the right. How many people are between them?
12. Early in the morning, just after sunrise, Rohit stood facing a pole. The pole's shadow fell exactly to his left. Which direction was Rohit facing?
13. V is the daughter of W. W is the son of X. X is the mother of Y. Y is the brother of W. How is V related to Y?
14. A man walks 5 km South, turns left and walks 5 km, turns left and walks 5 km, then turns right and walks 3 km. How far is he from the start and in which direction?
15. In a class of 40, Anu's rank from the top is 16 and Bala's rank from the top is 25. How many students rank between them?

ANSWER KEY

1. **Son.** The only daughter of Leela's mother is Leela, so Leela is his mother.
2. **A is the son of D.** C is D's son; B is C's sibling; A is B's brother; so all three are D's children and A is male.
3. **West.** South → left → East → right → South → right → West.
4. **6 m, due East.** North 10 and South 10 cancel; only the 6 m East survives.
5. **22nd from the left.** $32 - 11 + 1 = 22$.
6. **29 students.** $7 + 23 - 1 = 29$.
7. **33 people.** $15 + 19 - 1 = 33$.
8. **Brother.** Grandfather's only son is Farida's father; his son is Farida's brother.
9. **P is the nephew of R.** Q is R's sister, so R is P's uncle or aunt; P is a son, so P is male.
10. **3 km, due North.** West 4 then North 3 then East 4; the two 4s cancel.
11. **10 people.** Tina from left = $50 - 20 + 1 = 31$; between = $31 - 20 - 1 = 10$.
12. **North.** After sunrise shadows point West; West on his left means he faces North.
13. **V is the niece of Y.** W and Y are both children of X, so they are siblings; V is W's daughter.
14. **8 km, due East.** South 5, East 5, North 5, East 3: vertical cancels, horizontal = $5 + 3 = 8$.
15. **8 students.** $25 - 16 - 1 = 8$.

Tier 2 — Singapore & Thailand**TIER 2**

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 20 [Grab · pattern]

A rider leaves the hub and rides 9 km North, 12 km East, 3 km South and 4 km West.

(a) What is his straight-line distance from the hub?

(b) What fraction of the distance he rode is that straight-line distance?

SOLUTION

(a) Vertical net = $9 - 3 = 6$ km North.

Horizontal net = $12 - 4 = 8$ km East.

distance = $\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ km.

(b) Distance ridden = $9 + 12 + 3 + 4 = 28$ km.

Fraction = $\frac{10}{28}$. Divide top and bottom by 2: $10 \div 2 = 5$ and $28 \div 2 = 14$, so the fraction is $\frac{5}{14}$.

Answer: 10 km (North-East of the hub); 5/14 of the distance ridden

Example 21 [Sea/Shopee · pattern]

Six people are in one family: Anong, Boon, Chai, Dara, Esha, Farid.

- Anong and Boon are a married couple, and Boon is the father in the family.
- Chai is the son of Anong.
- Dara is the sister of Chai.
- Esha is the daughter of Dara.
- Farid is the brother of Esha.

(a) How is Farid related to Anong?

(b) How many grandchildren does Boon have?

SOLUTION

Chai is Anong's son and Dara is Chai's sister, so Dara is also a child of Anong and Boon.

Generation 1: Anong = Boon.

Generation 2: Chai and Dara.

Esha is Dara's daughter, and Farid is Esha's brother, so Farid is also Dara's child.

Generation 3: Esha and Farid.

(a) Farid is the son of Anong's daughter, so Farid is Anong's grandson.

(b) The only third-generation people named are Esha and Farid. Nothing says Chai has children, so Boon has 2 grandchildren.

Answer: (a) Grandson (b) 2 grandchildren

TRAP

Count only the people the question actually names. Chai **may** have children in real life, but the puzzle never says so, so he contributes zero grandchildren. Reasoning "he is grown up, so he must have kids" loses the mark.

Example 22 [DBS · pattern]

In a batch of 250 trainees, 36 percent of the trainees scored above Meera. No two scores are equal. How many trainees scored below her?

SOLUTION

Trainees above Meera = $\frac{36}{100} \times 250 = \frac{36 \times 250}{100} = \frac{9000}{100} = 90$.

So Meera's rank from the top = $90 + 1 = 91$.

Trainees below her = $250 - 91 = 159$.

Answer: 159 trainees

TRAP

36% of 250 is 90 **people above her**, not her rank. Her rank is 91. Answering 160 (that is, $250 - 90$) forgets to exclude Meera herself.

Example 23 [GIC · pattern]

Kishore faces North. He turns 135° clockwise, then 180° anticlockwise, then 90° clockwise. Which direction does he face?

SOLUTION

Use bearings: North = 0° , measured clockwise.

Start: 0° .

After 135° clockwise: $0 + 135 = 135^\circ$.

After 180° anticlockwise: $135 - 180 = -45^\circ$. Add 360° : 315° .

After 90° clockwise: $315 + 90 = 405^\circ$. Subtract 360° : 45° .

45° is North-East.

Answer: North-East

SHORTCUT

Turn every turn into a bearing sum. Clockwise is $+$, anticlockwise is $-$, then reduce modulo 360. Three turns become one addition. Chasing the compass step by step is where mistakes happen.

Example 24 [Agoda · pattern]

Pim and Wichai start from the same point. Pim walks 8 km East, then 6 km North. Wichai walks 4 km West, then 6 km South, then 12 km East. How far apart are they now, and where is Pim relative to Wichai?

SOLUTION

Put the start at $(0, 0)$, with East as $+x$ and North as $+y$.

Pim: East 8 gives $(8, 0)$; North 6 gives $(8, 6)$.

Wichai: West 4 gives $(-4, 0)$; South 6 gives $(-4, -6)$; East 12 gives $(-4 + 12, -6) = (8, -6)$.

Both have $x = 8$, so they are on the same North–South line.

Gap = $6 - (-6) = 12$ km.

Pim's y is larger, so Pim is to the North.

Answer: 12 km apart; Pim is due North of Wichai

Example 25 [SCB · pattern]

In a code:

'A + B' means A is the father of B.

'A - B' means A is the mother of B.

'A $\times$ B' means A is the brother of B.

'A $\div$ B' means A is the sister of B.

Which expression shows that D is the **paternal aunt** of E?

(a) $D \div F + E$

(b) $D \div F - E$

(c) $D \times F + E$

(d) $F + E \div D$

SOLUTION

Paternal aunt = father's sister.

So we need two links: D is the **sister** of someone, and that someone is the **father** of E.

"D is the sister of F" is written $D \div F$.

"F is the father of E" is written $F + E$.

Joining them: $D \div F + E$.

Answer: (a) $D \div F + E$

(b) $D \div F - E$ makes F the **mother** of E, so D would be a maternal aunt.

(c) $D \times F + E$ makes D a brother, so D would be a paternal uncle.

(d) $F + E \div D$ says E is D's sister, which is the wrong generation.

Example 26 [LINE MAN · pattern]

In a queue of riders, Anucha is 16th from the front and 22nd from the back. Boonmee then joins the queue standing immediately behind Anucha, and everybody behind shifts back by one. What is Boonmee's position from the back?

SOLUTION

Original total = $16 + 22 - 1 = 37$ riders.

After Boonmee joins, total = $37 + 1 = 38$.

Anucha is still 16th from the front (nobody ahead of him changed).

Boonmee is immediately behind Anucha, so Boonmee is 17th from the front.

Boonmee from the back = $38 - 17 + 1 = 22$.

Answer: 22nd from the back

Example 27 [Razer · pattern]

A drone starts at base. It flies North at 12 km/h for 40 minutes, then East at 16 km/h for 45 minutes, then South at 9 km/h for 20 minutes. How far is it from base, and in which direction?

SOLUTION

Convert each leg to a distance.

North: $12 \times \frac{40}{60} = 12 \times \frac{2}{3} = 8$ km.

East: $16 \times \frac{45}{60} = 16 \times \frac{3}{4} = 12$ km.

South: $9 \times \frac{20}{60} = 9 \times \frac{1}{3} = 3$ km.

Vertical net = $8 - 3 = 5$ km North.

Horizontal net = 12 km East.

distance = $\sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ km.

North and East, so the direction is North-East.

Answer: 13 km, North-East of base

Example 28 [Grab · pattern]

Six people: Gita, Hari, Ivan, Jaya, Kiran, Lalit.

- Gita is the grandmother of Kiran.
- Hari is the only son of Gita.
- Jaya is the daughter of Gita.
- Ivan is the husband of Jaya.
- Lalit is the son of Hari.
- Kiran is a boy and is not the child of Hari.

(a) How is Ivan related to Lalit?

(b) How is Kiran related to Hari?

SOLUTION

Gita's children: Hari (her only son) and Jaya.

Jaya is married to Ivan.

Lalit is Hari's son, so Lalit is Gita's grandchild.

Kiran is Gita's grandchild but not Hari's child, so Kiran must be a child of Jaya and Ivan.

(a) Lalit's father is Hari. Hari's sister is Jaya. Jaya's husband is Ivan.

So Ivan is the husband of Lalit's paternal aunt, that is, Lalit's uncle.

(b) Kiran's mother is Jaya. Jaya's brother is Hari.

So Hari is Kiran's maternal uncle, and Kiran (a boy) is Hari's nephew.

Answer: (a) Ivan is Lalit's uncle (b) Kiran is Hari's nephew

PRACTICE · TIER 2 target: 100 s/Q

1. A courier leaves the warehouse and rides 20 km North, 15 km East, 8 km South and 3 km West. How far is he from the warehouse, and in which direction?
2. In a Bangkok office of 180 staff, 45 percent of the staff scored below Pim in an internal test. No scores tie. What is Pim's rank from the top?
3. A drone flies East at 24 km/h for 25 minutes, then North at 18 km/h for 80 minutes. How far is it from the launch point?
4. In a code, 'A © B' means A is the brother of B, 'A @ B' means A is the mother of B, 'A £ B' means A is the daughter of B, and 'A ¥ B' means A is the wife of B. Which expression means "P is the maternal uncle of Q"?
 (a) P © R @ Q (b) P © R £ Q (c) P £ R @ Q (d) R @ Q © P
5. In a row of riders, Boonmee is 18th from the left and Chai is 27th from the right. Exactly 5 riders stand between them, and Boonmee is to the left of Chai. How many riders are in the row?
6. Ken is the son of Lily. Lily is the only daughter of Mano. Nina is the wife of Mano. Oscar is the brother of Lily. Priya is the daughter of Oscar. How is Priya related to Ken?
7. Two cyclists start from the same point. One rides 30 km North then 40 km East. The other rides 40 km South then 30 km West. How far apart are they? (Leave the answer in surd form and also to the nearest km.)
8. Anong faces North-East. She turns 135° anticlockwise, then 90° clockwise. Which direction does she face?
9. In a test of 400 candidates with no ties, Arjun's rank from the top is 3 less than three times his rank from the bottom. What is his rank from the top?
10. A man starts walking towards the setting sun. After 8 km he turns left and walks 6 km, then turns left and walks 14 km. How far is he from the start, and in which direction?
11. Tarun is the brother of Uma. Vidya is the mother of Uma. Wasim is the father of Tarun. Xu is the mother of Wasim. How is Xu related to Uma?

ANSWER KEY

1. $12\sqrt{2} \approx 17.0$ km, **North-East**. Vertical = $20 - 8 = 12$ N, horizontal = $15 - 3 = 12$ E, so $\sqrt{144 + 144} = 12\sqrt{2}$.
2. **99th from the top**. Below her = $0.45 \times 180 = 81$, so her rank from the bottom is 82, and $180 - 82 + 1 = 99$.
3. **26 km**. East = $24 \times \frac{25}{60} = 10$ km; North = $18 \times \frac{80}{60} = 24$ km; $\sqrt{100 + 576} = \sqrt{676} = 26$.
4. **(a) P © R @ Q**. Maternal uncle = mother's brother: P is the brother of R, and R is the mother of Q. In (b) R is Q's daughter, in (c) P is female, in (d) the links point the wrong way.
5. **50 riders**. Chai from the left = $18 + 5 + 1 = 24$; total = $24 + 27 - 1 = 50$.
6. **Priya is Ken's cousin**. Lily and Oscar are both children of Mano and Nina; Ken is Lily's son and Priya is Oscar's daughter.
7. **$70\sqrt{2} \approx 99$ km**. First is at (40, 30), second at (-30, -40); gaps are 70 and 70, so $\sqrt{4900 + 4900} = 70\sqrt{2} \approx 98.99$.
8. **North**. NE = 45°; $45 - 135 = -90 \rightarrow 270^\circ$ (West); $270 + 90 = 360 \rightarrow 0^\circ$ (North).
9. **300th from the top**. Let the rank from the bottom be b ; then $(3b - 3) + b = 401$, so $4b = 404$, $b = 101$ and the rank from the top is $3(101) - 3 = 300$.
10. **$6\sqrt{2} \approx 8.49$ km, South-East**. West 8, then South 6, then East 14: horizontal = $-8 + 14 = 6$ E, vertical = 6 S.
11. **Grandmother (paternal)**. Wasim fathers Tarun, and Uma is Tarun's sister, so Wasim is Uma's father too; Xu is Wasim's mother.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 29 [Google · pattern]

In a class of 35 students with no tied scores, exactly 12 students scored more than Ria, and exactly 18 students scored less than Sam. How many students rank strictly between Sam and Ria?

SOLUTION

Insight: turn every clue into a single number — the rank from the top. Then the counting is one subtraction.

12 scored above Ria, so Ria's rank from the top = $12 + 1 = 13$.

18 scored below Sam, so Sam's rank from the bottom = $18 + 1 = 19$.

Convert Sam to a rank from the top: $35 - 19 + 1 = 17$.

Ria is 13th, Sam is 17th, and $17 > 13$.

Students strictly between = $17 - 13 - 1 = 3$.

Answer: 3 students

Example 30 [Amazon · pattern]

A robot starts at the origin facing North. On move k it drives forward exactly k metres, then turns 90° to its right. After 100 moves, how far is it from the origin, in which direction, and which way is it facing?

SOLUTION

Insight: the robot repeats its heading every 4 moves, so group the moves in blocks of 4 and add the blocks.

Headings cycle: move 1 North, move 2 East, move 3 South, move 4 West, move 5 North again, and so on.

Take East as $+x$ and North as $+y$.

Block 1 (moves 1 to 4):

y : $+1$ then -3 , so y changes by $1 - 3 = -2$.

x : $+2$ then -4 , so x changes by $2 - 4 = -2$.

Block 2 (moves 5 to 8):

y : $+5 - 7 = -2$. x : $+6 - 8 = -2$.

Every block of 4 changes the position by exactly $(-2, -2)$, because the two lengths in each direction always differ by 2.

Number of blocks = $100 \div 4 = 25$.

Final position = $(25 \times (-2), 25 \times (-2)) = (-50, -50)$.

distance = $\sqrt{50^2 + 50^2} = \sqrt{2500 + 2500} = \sqrt{5000} = 50\sqrt{2} \approx 70.7$ m.

x negative is West and y negative is South, so the direction is South-West.

Facing: 100 is a multiple of 4, so the heading has come back to the start.

Answer: $50\sqrt{2} \approx 70.7$ m South-West of the origin, facing North

SHORTCUT

Any “turn the same way every step” walk closes its heading after 4 steps. So never simulate 100 moves. Simulate 4, read off the block shift, and multiply.

Example 31 [Goldman Sachs · pattern]

Thirty children stand in a row. Arjun stands somewhere to the left of Bhavna, with at least 5 children between them. Arjun is not in the first 4 positions from the left. What is the largest possible rank of Bhavna counted from the right?

SOLUTION

Insight: “rank from the right” gets bigger as a person moves left. So push Bhavna as far left as the rules allow.

Number the positions 1 to 30 from the left.

Arjun is not in positions 1 to 4, so Arjun's position $a \geq 5$.

At least 5 children between them means $b - a - 1 \geq 5$, so $b \geq a + 6$.

The smallest possible b comes from the smallest possible a :

$a = 5$ gives $b \geq 5 + 6 = 11$.

So Bhavna's leftmost possible position is 11.

Bhavna's rank from the right = $30 - 11 + 1 = 20$.

Check: Arjun at 5, Bhavna at 11, children in positions 6, 7, 8, 9, 10 lie between them — that is exactly 5. Valid.

Answer: 20th from the right

Example 32 [D. E. Shaw · pattern]

A hiker walks 10 km due North. She then walks 10 km in a straight line and finds that she is now exactly North-East of her starting point. In which direction did she walk the second leg, and how far is she from the start?

SOLUTION

Insight: “exactly North-East” is not vague — it is the precise condition $x = y$ with both positive.

Put the start at $(0, 0)$, East as $+x$, North as $+y$.

After the first leg she is at $(0, 10)$.

Let the end point be (x, y) . North-East of the start means $x = y$ and $x > 0$.

The second leg has length 10, so

$$x^2 + (y - 10)^2 = 100$$

Substitute $y = x$:

$$x^2 + (x - 10)^2 = 100$$

$$x^2 + x^2 - 20x + 100 = 100$$

$$2x^2 - 20x = 0$$

$$2x(x - 10) = 0$$

So $x = 0$ or $x = 10$. $x = 0$ is rejected because then she is due North, not North-East.

So $x = 10$ and $y = 10$; the end point is $(10, 10)$.

The second leg went from $(0, 10)$ to $(10, 10)$: the y value did not change and x increased, so she walked **due East**.

distance from start $= \sqrt{10^2 + 10^2} = \sqrt{200} = 10\sqrt{2} \approx 14.14$ km.

Answer: She walked due East; she is $10\sqrt{2} \approx 14.14$ km from the start

Example 33 [Adobe · pattern]

At a family dinner the guests included: 1 grandfather, 1 grandmother, 2 fathers, 2 mothers, 4 children, 3 grandchildren, 1 brother, 2 sisters, 2 sons, 2 daughters, 1 father-in-law, 1 mother-in-law and 1 daughter-in-law. What is the smallest number of people who could have been at the dinner?

SOLUTION

Insight: the counts are roles, not people. One person can fill many roles at once, so build the smallest family tree that covers every role.

Try three generations along one line.

Generation 1: a grandfather (G) and a grandmother (H), married.

Generation 2: their son S, and S's wife W.

Generation 3: three children of S and W — one boy B and two girls C and D.

That is $2 + 2 + 3 = 7$ people. Now check every role.

- grandfather: G. Count 1. ✓
- grandmother: H. Count 1. ✓
- fathers: G (father of S) and S (father of B, C, D). Count 2. ✓
- mothers: H and W. Count 2. ✓
- children (people who are somebody's child at the table): S, B, C, D. Count 4. ✓ (W's parents are not present, so W is not counted.)
- grandchildren: B, C, D. Count 3. ✓
- brother: B is the brother of C and D. S is an only child, so he is nobody's brother. Count 1. ✓
- sisters: C and D. Count 2. ✓
- sons: S and B. Count 2. ✓
- daughters: C and D. Count 2. ✓ (W is not anybody's daughter at this table.)
- father-in-law: G, to W. Count 1. ✓
- mother-in-law: H, to W. Count 1. ✓

- daughter-in-law: W. Count 1. ✓

Every role matches exactly. And 7 cannot be beaten: the list needs 3 distinct grandchildren, plus at least one parent of them, plus that parent's spouse, plus a grandfather and a grandmother, which is already $3 + 2 + 2 = 7$.

Answer: 7 people

Example 34 [Uber · pattern]

Two riders leave the same junction at the same moment. Rider A goes due North at 30 km/h. Rider B goes due East at 40 km/h.

- (a) After how long are they exactly 25 km apart?
 (b) At that moment, in which direction is B from A?

SOLUTION

Insight: two perpendicular constant velocities separate at a constant speed, namely the resultant. So the gap grows linearly and you never need to solve a quadratic.

(a) Separation speed = $\sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50$ km/h.

Time = $\frac{25}{50} = 0.5$ h = 30 minutes.

Check directly: in 0.5 h, A covers $30 \times 0.5 = 15$ km North; B covers $40 \times 0.5 = 20$ km East.

Gap = $\sqrt{15^2 + 20^2} = \sqrt{225 + 400} = \sqrt{625} = 25$ km. ✓

(b) With East as $+x$ and North as $+y$: A is at (0, 15), B is at (20, 0).

From A to B the change is $x : 0 \rightarrow 20$ (East) and $y : 15 \rightarrow 0$ (South).

Both changes are 20 and 15, both non-zero, so B lies South and East of A.

Answer: (a) 30 minutes (b) B is South-East of A

Example 35 [Google · pattern]

Forty students sit in a single row. Every student is given a different score. Call a student a **peak** if their score is higher than the score of every student sitting next to them (a student at either end has only one neighbour). What is the largest possible number of peaks?

SOLUTION

Insight: two peaks can never sit next to each other, so the peaks form a set of seats with no two adjacent. That upper bound is the whole problem.

Step 1 — the bound. Suppose seats i and $i + 1$ were both peaks. Then the student at i must score higher than the student at $i + 1$, and the student at $i + 1$ must score higher than the student at i . That is impossible. So no two peaks are adjacent.

In a row of 40 seats, the largest set of seats with no two adjacent has size 20 — pair the seats as (1, 2), (3, 4), ..., (39, 40); that is 20 pairs and each pair can contribute at most one peak.

So the number of peaks is at most 20.

Step 2 — reach the bound. Give the 20 odd seats 1, 3, 5, ..., 39 the twenty highest scores (21 to 40) and give the 20 even seats the twenty lowest scores (1 to 20).

Every odd seat then holds a score of at least 21, and each of its neighbours (even seats) holds a score of at most 20.

So every one of the 20 odd seats is a peak.

The bound 20 is reached, so the maximum is exactly 20.

Answer: 20 peaks

PRACTICE · TIER 3 target: 4 min/Q

1. A robot starts at the origin facing East. On move k it drives forward k metres and then turns 90° to its left. After 200 moves, how far is it from the origin, in which direction, and which way is it facing?
2. Twenty-five students sit in a row, all with different scores. Call a student a **valley** if their score is lower than every neighbour's score (an end student has one neighbour). What is the largest possible number of valleys?

3. In a class of 30 with no tied scores, exactly 8 students scored above Mei and exactly 14 scored below Nat. How many students rank strictly between Nat and Mei?
4. Aarti and Bilal start from the same point at the same time. Aarti cycles due North at 5 km/h and Bilal cycles due East at 12 km/h. After how many minutes are they exactly 39 km apart?
5. At a family lunch the guests included 1 grandfather, 1 grandmother, 2 fathers, 2 mothers, 3 children, 2 grandchildren, 1 brother, 1 sister, 2 sons, 1 daughter, 1 father-in-law, 1 mother-in-law and 1 daughter-in-law. What is the smallest number of people who could have been there?
6. Eight runners finish a race with no ties. Exactly 3 finish before Hari. Exactly 2 runners finish between Hari and Ikram. Jia finishes immediately after Ikram, and Jia is not 2nd. In what position did Jia finish?
7. Seven people P, Q, R, S, T, U, V stand in a row of seven places. Exactly two people stand between P and Q, with Q to the right of P. R stands exactly three places to the right of Q. In how many different ways can the seven of them stand?

ANSWER KEY

1. $100\sqrt{2} \approx 141.4$ m South-West of the origin, facing East. Headings cycle E, N, W, S. In moves 1 to 4 the x change is $+1 - 3 = -2$ and the y change is $+2 - 4 = -2$, and every later block of 4 repeats this. With $200 \div 4 = 50$ blocks the position is $(-100, -100)$, and 200 is a multiple of 4 so the heading returns to East.
2. **13 valleys.** If seats i and $i + 1$ were both valleys, each score would have to be lower than the other, which is impossible; so no two valleys are adjacent and at most 13 of the 25 seats qualify (seats 1, 3, ..., 25). Giving the 13 odd seats the 13 lowest scores and the 12 even seats the 12 highest makes all 13 valleys real.
3. **6 students.** Mei's rank from the top $= 8 + 1 = 9$. Nat's rank from the bottom $= 14 + 1 = 15$, so Nat's rank from the top $= 30 - 15 + 1 = 16$. Between $= 16 - 9 - 1 = 6$.
4. **180 minutes.** The velocities are perpendicular, so the gap grows at $\sqrt{5^2 + 12^2} = 13$ km/h. Time $= \frac{39}{13} = 3$ h = 180 min.
5. **6 people.** Grandfather, grandmother, their son, his wife, and the son's two children (one boy, one girl). Fathers: grandfather and son. Mothers: grandmother and wife. Children at the table: son, boy, girl — that is 3. Daughter: only the granddaughter, since the wife's parents are absent. Every role in the list is matched exactly.
6. **8th (last).** 3 finish before Hari, so Hari is 4th. Two runners between Hari and Ikram puts Ikram at $4 + 3 = 7$ or $4 - 3 = 1$. If Ikram were 1st then Jia would be 2nd, which is ruled out. So Ikram is 7th and Jia is 8th.
7. **24 ways.** Two people between P and Q gives $Q = P + 3$; R is three places right of Q gives $R = Q + 3 = P + 6$. With 7 places, $P \geq 1$ and $R \leq 7$ force $P = 1, Q = 4, R = 7$. The remaining four people fill places 2, 3, 5, 6 in $4! = 24$ ways.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 25 questions in 30 min — here, 20 in 24 min

1. Pointing to a photograph, Sunita said, "She is the mother of my son's only sister." How is the woman in the photograph related to Sunita?
2. In a row of 36 students, Hema is 14th from the left and Ismail is 9th from the right. How many students sit between them?
3. Vikram faces West. He turns 90° clockwise, then 180° , then 90° anticlockwise. Which direction does he face?
4. A man walks 4 km North, 3 km East, 8 km South and 3 km West. How far is he from the start, and in which direction?
5. 'A @ B' means A is the father of B. 'A © B' means A is the sister of B. What does 'P @ Q © R' tell you about P and R?
6. In a class, Rani is 12th from the top. Three times as many students rank below her as above her. How many students are in the class?
7. Just after sunrise, Hiro walks towards the sun for 2 km, turns right and walks 3 km, then turns right and walks 2 km. How far is he from the start, and in which direction?
8. Q is the son of P. R is the sister of Q. S is the mother of R. T is the father of S. How is T related to Q?

9. In a queue of 45 people, Ganesh is 19th from the front. Two people standing ahead of him leave the queue. What is his new position from the back?
10. A car travels 15 km South, then 20 km West, then 15 km North. How far is it from the start, and in which direction?
11. Introducing a boy, Ahmed said, "He is the only son of the only brother of my mother." How is the boy related to Ahmed?
12. A rank list has 60 names. Lin is 22nd from the top and Mala is 34th from the bottom. How many names lie between them?
13. A drone flies 9 km East, then 40 km North. What is its straight-line distance from the launch point?
14. Six runners finish with no ties. Exactly 2 finish before Nina and exactly 2 finish after Ravi. How many runners finish between Nina and Ravi?
15. A boy walks 10 m North, turns left and walks 10 m, turns left and walks 10 m, then turns right and walks 10 m. How far is he from the start, and in which direction?
16. 'P – Q' means P is the mother of Q. 'P ÷ Q' means P is the father of Q. 'P × Q' means P is the brother of Q. Which expression shows that A is the paternal grandfather of D?
 (a) $A \div B \div D$ (b) $A \div B - D$ (c) $A - B \div D$ (d) $A \times B \div D$
17. In a merit list, Sara is 15th from the top and Tariq is 15th from the bottom. Exactly 4 students rank between them, and Sara ranks above Tariq. How many students are on the list?
18. One evening, just before sunset, Nok stood facing a tower whose shadow fell exactly behind her. Which direction was Nok facing?
19. A man is 8th from the left and 12th from the right in a row. Three more people then join at the right end. What is his new position from the right?
20. Rahul is the brother of Priya. Sonal is the mother of Rahul. Tarun is the son of Priya. Umesh is the husband of Sonal. How is Tarun related to Umesh?

ANSWER KEY

1. **She is Sunita herself.** Sunita's son's only sister is Sunita's daughter, and that daughter's mother is Sunita.
2. **13 students.** Ismail from the left = $36 - 9 + 1 = 28$; between = $28 - 14 - 1 = 13$.
3. **East.** West = 270° ; $270 + 90 = 360 \rightarrow 0^\circ$; $0 + 180 = 180^\circ$; $180 - 90 = 90^\circ = \text{East}$.
4. **4 km, due South.** Vertical = $4 - 8 = -4$; horizontal = $3 - 3 = 0$.
5. **P is the father of R.** Q is R's sister, so they share a father, and that father is P.
6. **45 students.** Above her = 11; below her = $3 \times 11 = 33$; total = $11 + 1 + 33 = 45$.
7. **3 km, due South.** East 2, then South 3, then West 2: the two 2s cancel.
8. **Maternal grandfather.** S mothers R, and Q is R's brother, so S is Q's mother; T is S's father.
9. **27th from the back.** New total = 43; his position from the front = $19 - 2 = 17$; $43 - 17 + 1 = 27$.
10. **20 km, due West.** South 15 and North 15 cancel.
11. **Cousin.** His mother's only brother is Ahmed's maternal uncle; that uncle's son is Ahmed's cousin.
12. **4 names.** Mala from the top = $60 - 34 + 1 = 27$; between = $27 - 22 - 1 = 4$.
13. **41 km.** $\sqrt{9^2 + 40^2} = \sqrt{81 + 1600} = \sqrt{1681} = 41$.
14. **None — they finish next to each other.** Nina is 3rd; Ravi is $6 - 2 = 4\text{th}$; $4 - 3 - 1 = 0$.
15. **20 m, due West.** North 10, West 10, South 10, West 10: vertical cancels, horizontal = 20 W.
16. **(a) $A \div B \div D$.** Father of the father is the paternal grandfather. In (b) B is D's mother, which gives a maternal grandfather; in (c) A is female; in (d) A is an uncle.
17. **34 students.** Tariq from the top = $15 + 4 + 1 = 20$; total = $20 + 15 - 1 = 34$.
18. **West.** Before sunset shadows point East; the shadow was behind her, so East was behind her and she faced West.
19. **15th from the right.** Original total = $8 + 12 - 1 = 19$; new total = 22; his place from the left is still 8, so $22 - 8 + 1 = 15$.
20. **Grandson.** Umesh and Sonal are the parents of Rahul and Priya; Tarun is Priya's son.

ONE-PAGE REVISION CARD

Blood relations

- Draw the tree. Same generation on the same line, children hanging below.
- Peel a long sentence from the innermost phrase outward: grandfather → his only son = father → his daughter = sister.
- Resolve coded relations one symbol at a time and write each fact down before reading the next symbol.
- Key words: paternal = father's side, maternal = mother's side; nephew/niece = sibling's child; cousin = uncle's or aunt's child.

Directions

- Clockwise: N, NE, E, SE, S, SW, W, NW.
- Facing N: left = W, right = E. Facing E: left = N, right = S. Facing S: left = E, right = W. Facing W: left = S, right = N.
- Bearings: N = 0° and count clockwise. Clockwise turns add, anticlockwise turns subtract, then reduce modulo 360°.
- displacement = $\sqrt{h^2 + v^2}$, with h = East minus West and v = North minus South.
- Triples to recognise on sight: (3, 4, 5), (5, 12, 13), (6, 8, 10), (8, 15, 17), (7, 24, 25), (9, 40, 41), (20, 21, 29).
- $|h| = |v|$ gives $h\sqrt{2}$ and a corner direction.
- Sunrise → shadows point West. Sunset → shadows point East. Noon gives no direction.

Ranking

- Total = left rank + right rank - 1.
- right rank = Total - left rank + 1.
- Both ranks from the same end first, then between = larger - smaller - 1.
- “ x people between” means the positions differ by $x + 1$.
- k people above you → your rank from the top is $k + 1$.

Shortcuts

- Collapse a multi-leg walk into two running totals, not a drawing.
- Sum all turns as one bearing instead of stepping around the compass.
- A “turn the same way every step” walk repeats its heading every 4 steps; simulate 4 and multiply.
- Perpendicular velocities separate at the resultant speed, so the gap is linear in time.
- For a maximum count of peaks or valleys in a row, use “no two can be adjacent”, then build the alternating example.

Top 5 traps

1. Subtracting two ranks that are counted from **opposite** ends. Convert to one end first.
2. Reading “ x people between” as a gap of x . The gap is $x + 1$.
3. Forgetting that “the only daughter of my mother” is usually the **speaker**.
4. Turning $p\%$ of a class into a rank. If $p\%$ scored above you, your rank is that count **plus one**.
5. Flipping left and right. A person **facing you** has their left on your right — but a shadow or a compass direction never flips.

Seating Arrangement & Puzzles

Fix the certain seats first, then squeeze

What you need to know

QUICK REFERENCE

1. Always draw the same picture. For a **row**, draw the seats left to right on the page and number them 1, 2, 3, ... from the **West** end. For a **circle**, draw 1 at the top and number **clockwise**. Keep this habit and every clue reads the same way every time.

2. Left and right in a row

Everyone faces	A person's LEFT is	A person's RIGHT is
North	West side — towards seat 1	East side — towards seat n
South	East side — towards seat n	West side — towards seat 1

3. Left and right in a circle

Person faces	LEFT means move	RIGHT means move
the centre	clockwise	anticlockwise
outward	anticlockwise	clockwise

Why: stand at the top of the circle facing the centre. You are looking **down** the page, so your left hand points to the **right** of the page — and that is the clockwise side.

4. Opposite seats. In a circle of n seats with n even, the person opposite seat k is at seat $k + \frac{n}{2}$ (wrap around by subtracting n if you go past n). If n is odd, **nobody** is exactly opposite anybody.

5. Square and rectangular tables. Number the seats clockwise as usual. With 8 people at a square table — 4 at the corners and 4 at the middles of the sides — the seats alternate: corner, middle, corner, middle, ... So if the corners are seats 1, 3, 5, 7 then the middles are 2, 4, 6, 8. Corner people and middle people are often told to face opposite ways, so check each person's own facing before you use left or right.

6. Floors, stacks and boxes. Number floor 1 at the **bottom**. "Above" means a larger number, "below" a smaller one. A stack of boxes works exactly like a building: bottom = 1, top = n .

7. Phrase dictionary — learn it exactly

A is immediately to the left of B	A is exactly 1 step to B 's left
A is second to the left of B	A is exactly 2 steps to B 's left
A is to the left of B	somewhere to the left — position not fixed
Exactly x people between A and B	their seat numbers differ by $x + 1$
A and B are neighbours / adjacent	seat numbers differ by 1
A sits at an end / extreme	seat 1 or seat n
A sits exactly in the middle of n seats	seat $(n + 1)/2$, only if n is odd

8. Double rows. The two rows look at each other, so one row faces South and the other faces North. **Read the facing the question gives you — it is not always Row 1 that faces South.** Draw both rows with West

on the left. Then **seat k of Row 1 faces seat k of Row 2**. Remember that left and right are **flipped** between the two rows.

9. Scheduling grids (days, months, cities, prices). Draw a table: one row per slot (Monday to Friday, or January to December in the order given), one column per attribute. Fill only what is certain. “Immediately after” means the very next slot in the list.

10. The order to use the clues

1. First: clues that fix one seat with certainty — ends, exact middle, a named floor, “opposite”.
2. Second: clues that chain off a fixed seat — “immediately left of”, “second to the right of”.
3. Third: clues with two possible cases — branch, and test each branch.
4. Last: negative clues (“not adjacent to”, “does not sit at an end”). These kill wrong branches instead of building the answer.

Warm-up

WARM-UP

build the reflex

Example 1

Five friends A, B, C, D, E sit in a row facing North. C is at the extreme left end. B is at the extreme right end. D sits immediately to the right of C. A sits immediately to the left of B. Who sits in the middle?

SOLUTION

Number the seats 1 to 5 from the left.

$C = 1$, $B = 5$.

All face North, so “right” means a larger seat number. $D = 1 + 1 = 2$.

“Left” means a smaller seat number. $A = 5 - 1 = 4$.

The only seat left is 3, so E sits there.

Answer: E

Example 2

Six people sit around a circular table facing the centre. P sits third to the left of Q. How many people sit between Q and P when you count clockwise from Q?

SOLUTION

Facing the centre, “left” means clockwise.

So from Q, move 3 seats clockwise to reach P.

The seats passed on the way are 2 seats, holding 2 people.

Answer: 2 people

Example 3

In a 6-floor building (floor 1 at the bottom), Riya lives on floor 4 and Sam lives 2 floors above Riya. Which floor is Sam on?

SOLUTION

“Above” means add. $4 + 2 = 6$.

Answer: Floor 6

Example 4

Eight people sit around a circular table facing the centre. The seats are numbered 1 to 8 clockwise. Who sits exactly opposite the person at seat 3?

SOLUTION

$n = 8$, so opposite means add $8/2 = 4$.
 $3 + 4 = 7$.

Answer: Seat 7

Example 5

Five boxes P, Q, R, S, T are stacked one above another. P is at the bottom, Q is immediately above P, R is at the top and S is immediately below R. Which box is in the middle?

SOLUTION

Number the positions 1 (bottom) to 5 (top).
 $P = 1, Q = 2, R = 5, S = 4$.
The only position left is 3, so T is there — and 3 is the middle of 5.

Answer: T

Example 6

Four people sit in a row facing South. The row is drawn with West on the left. M sits immediately to the right of N. Who sits further West?

SOLUTION

Facing South, a person's right hand points West.
So M is one seat to the West of N.

Answer: M

Example 7

Seven people sit in a row facing North, with West on the left. X sits 4th from the West end. How many people sit to X's right?

SOLUTION

Facing North, right means the East side.
X is at seat 4, so seats 5, 6 and 7 are to his right.
That is $7 - 4 = 3$ people.

Answer: 3 people

Example 8

Five meetings are held from Monday to Friday, one each day. The audit is on Wednesday. The demo is two days after the audit. Which day is the demo?

SOLUTION

Wednesday + 2 days = Friday.

Answer: Friday

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Seven friends A, B, C, D, E, F, G sit in a row facing North.

- A sits at the extreme left end.
- B sits at the extreme right end.
- D sits exactly in the middle.
- F sits immediately to the left of D.
- C sits fourth to the right of A.
- E does not sit next to A.

Who sits between C and B?

SOLUTION

Number the seats 1 to 7 from the left. All face North, so left = smaller number, right = larger number.

A = 1 and B = 7.

Exact middle of 7 seats is seat $(7 + 1)/2 = 4$, so D = 4.

F is immediately to the left of D: $F = 4 - 1 = 3$.

C is fourth to the right of A: $C = 1 + 4 = 5$.

Seats still empty: 2 and 6, for E and G.

E must not sit next to A. A is at seat 1, and seat 2 is next to it, so E is not at 2.

Therefore E = 6 and G = 2.

Final row: A(1), G(2), F(3), D(4), C(5), E(6), B(7).

C is at 5 and B is at 7, so seat 6 lies between them.

Answer: E

SHORTCUT

Use the clues in this order: ends first, then the exact middle, then “immediately next to” chains, and **only at the end** the negative clue. Starting from “E does not sit next to A” rules out **nothing**, because A’s seat is not known yet. Once A is pinned to seat 1, the same clue kills exactly one seat in one second.

Example 10 [Infosys pattern]

Six people P, Q, R, S, T, U sit around a circular table facing the centre.

- S sits immediately to the left of Q.
- P sits second to the left of Q.
- R sits immediately to the right of Q.
- T sits exactly opposite S.

(a) Who sits between P and T? (b) Who sits exactly opposite P?

SOLUTION

Number the seats 1 to 6 clockwise and put Q at seat 1 (a circle has no fixed start, so we may choose).

Everyone faces the centre, so left = clockwise and right = anticlockwise.

S is immediately to Q’s left: $1 + 1 = 2$. S = 2.

P is second to Q’s left: $1 + 2 = 3$. P = 3.

R is immediately to Q’s right: $1 - 1 = 0$, which wraps to seat 6. R = 6.

T is opposite S: $2 + 6/2 = 2 + 3 = 5$. T = 5.

The only seat left is 4, so U = 4.

Final clockwise: Q(1), S(2), P(3), U(4), T(5), R(6).

(a) P is at 3 and T is at 5, so seat 4 lies between them.

(b) Opposite P = $3 + 3 = 6$.

Answer: (a) U (b) R

Example 11 [Accenture pattern]

Six people — Nisha, Omar, Priya, Rakesh, Sunil and Tara — live on six floors of a building. Floor 1 is the bottom and floor 6 is the top. One person per floor.

- Priya lives on the topmost floor.
- Rakesh lives immediately below Priya.
- Sunil lives on floor 1.
- Nisha lives on an even-numbered floor.
- Omar lives above Nisha, and exactly one person lives between them.

Who lives on floor 3?

SOLUTION

Priya = 6 and Rakesh = 5 (immediately below 6).

Sunil = 1.

Floors still free: 2, 3, 4.

Nisha is on an even floor, so Nisha = 2 or Nisha = 4.

“Exactly one person between them” and “Omar above Nisha” give Omar = Nisha + 2.

If Nisha = 4 then Omar = 6, but floor 6 is Priya’s. Rejected.

If Nisha = 2 then Omar = 4, which is free. Accepted.

The only floor left is 3, so Tara lives there.

Final: Sunil(1), Nisha(2), Tara(3), Omar(4), Rakesh(5), Priya(6).

Answer: Tara

TRAP

“Exactly one person lives between them” means the floors differ by 2, not by 1. Reading it as 1 gives Nisha = 2 and Omar = 3, so you would answer “Omar” for floor 3 instead of “Tara”. The danger is that the wrong reading still fills every floor without a clash, so nothing warns you. Always convert a “between” clue into a floor **gap** of $x + 1$ before you place anybody.

Example 12 [Wipro pattern]

Eight people sit in two rows of four, facing each other.

Row 1 — A, B, C, D — all face South.

Row 2 — P, Q, R, S — all face North.

Both rows are drawn with West on the left, and seat k of Row 1 faces seat k of Row 2.

- B sits at an end of Row 1 and faces Q.
- C sits second to the right of B.
- P faces C.
- A sits at an end of Row 1.
- S sits immediately to the right of P.

Who faces D?

SOLUTION

Row 1 faces South, so **right** means a smaller seat number and **left** a larger one.

Row 2 faces North, so **right** means a larger seat number.

B is at an end, so B = 1 or B = 4.

C is second to B’s right = $B - 2$. If B = 1 this is seat -1 , which does not exist.

So B = 4, and $C = 4 - 2 = 2$.

A is at an end of Row 1; seat 4 is taken, so A = 1.

The last Row-1 seat is 3, so D = 3.

Row 1: A(1), C(2), D(3), B(4).

B faces Q and B is at seat 4, so Q = 4.

P faces C and C is at seat 2, so P = 2.

S is immediately to P’s right = $2 + 1 = 3$. S = 3.

The last Row-2 seat is 1, so R = 1.

Row 2: R(1), P(2), S(3), Q(4).

D is at seat 3, so D faces the Row-2 person at seat 3.

Answer: S

TRAP

The meaning of left and right **flips** between the two rows. In Row 1 (facing South) “second to the right” subtracts 2; in Row 2 (facing North) the same words add 2. Students who use one rule for both rows get half the puzzle wrong and still feel confident.

Example 13 [Capgemini pattern]

Five students — Tanvi, Ujjwal, Vikas, Wendy and Xu — visit a library on five different days, Monday to Friday.

- Tanvi visits on Thursday.
- Wendy visits on Friday.
- Ujjwal visits on the day immediately after Vikas.
- Vikas does not visit on Monday.

Who visits on Monday?

SOLUTION

Thursday = Tanvi, Friday = Wendy.

Days still free: Monday, Tuesday, Wednesday, for Ujjwal, Vikas and Xu.

Ujjwal is on the day right after Vikas, so we need two free days in a row:

Monday–Tuesday works, Tuesday–Wednesday works, Wednesday–Thursday fails (Thursday is taken).

Vikas is not on Monday, so Monday–Tuesday is out.

So Vikas = Tuesday and Ujjwal = Wednesday.

The only day left is Monday, for Xu.

Answer: Xu

Example 14 [Cognizant pattern]

Five boxes — black, blue, green, red and white — are stacked one above another.

- The black box is at the bottom.
- Exactly two boxes lie between the black box and the blue box.
- The green box is immediately above the red box.
- The white box lies somewhere above the blue box.

How many boxes lie between the green box and the white box?

SOLUTION

Number the positions 1 (bottom) to 5 (top).

Black = 1.

Two boxes between black and blue means the positions differ by $2 + 1 = 3$, so blue = $1 + 3 = 4$.

Free positions: 2, 3, 5.

Green is immediately above red, so red and green take two positions in a row.

From 2, 3, 5 the only pair in a row is 2 and 3. So red = 2 and green = 3.

White takes the last free position, 5. Check: 5 is above blue at 4. ✓

Final: black(1), red(2), green(3), blue(4), white(5).

Green is at 3 and white is at 5, so position 4 lies between them — that is 1 box.

Answer: 1 box (the blue box)

Example 15 [TCS NQT pattern]

Five people A, B, C, D, E sit around a circular table facing **outward**.

- A sits second to the right of B.

- C sits immediately to the left of A.
- D is not an immediate neighbour of B.

Who sits immediately to the right of E?

SOLUTION

Number the seats 1 to 5 clockwise and put B at seat 1.

They face **outward**, so left = anticlockwise and right = clockwise.

A is second to B's right: $1 + 2 = 3$. $A = 3$.

C is immediately to A's left: $3 - 1 = 2$. $C = 2$.

Free seats: 4 and 5, for D and E.

B is at seat 1; its neighbours are seats 2 and 5.

D is not a neighbour of B, so D is not at 5. Therefore $D = 4$ and $E = 5$.

Final clockwise: B(1), C(2), A(3), D(4), E(5).

E faces outward, so E's right is clockwise: $5 + 1 = 6$, which wraps to seat 1.

Answer: B

SHORTCUT

For a circle, always fix one named person at seat 1 before you do anything else. A circle has no real starting point, so this costs nothing and turns every clue into simple addition and subtraction.

Example 16 [Infosys pattern]

Six people L, M, N, O, P, Q sit in a row facing North.

- N sits third from the left.
- M sits at one of the two ends.
- Exactly three people sit between M and O.
- L sits immediately to the right of N.
- O sits to the right of L.
- Q does not sit at either end.

Where does P sit?

SOLUTION

Number the seats 1 to 6 from the left. Facing North, right = larger number.

$N = 3$.

L is immediately to N's right: $L = 3 + 1 = 4$.

Three people between M and O means their seat numbers differ by $3 + 1 = 4$.

M is at an end, so $M = 1$ or $M = 6$.

If $M = 1$ then $O = 5$. If $M = 6$ then $O = 2$.

But O must be to the right of L, and L is at seat 4, so $O > 4$.

That kills $O = 2$. So $M = 1$ and $O = 5$.

Free seats: 2 and 6, for P and Q.

Q is not at an end, so Q is not at 6. Therefore $Q = 2$ and $P = 6$.

Final: M(1), Q(2), N(3), L(4), O(5), P(6).

Answer: P sits at seat 6, the extreme right end

Example 17 [Wipro pattern]

Five people — Aarti, Bibek, Chandan, Dipa and Elias — have birthdays in five different months: January, March, May, August and November (listed in calendar order).

- Aarti's birthday is in January.
- Exactly two of them have birthdays after Bibek.
- Chandan's birthday is in the month immediately after Bibek's in this list.
- Dipa's birthday is not in November.

Whose birthday is in March?

SOLUTION

Number the months in order: January(1), March(2), May(3), August(4), November(5).

Aarti = January = 1.

Exactly two people come after Bibek, so Bibek has 2 months of the list after him: Bibek = $5 - 2 = 3$ = May.

Chandan is in the month right after Bibek: $3 + 1 = 4$ = August.

Free months: March(2) and November(5), for Dipa and Elias.

Dipa is not in November, so Dipa = March and Elias = November.

Final: Aarti–January, Dipa–March, Bibek–May, Chandan–August, Elias–November.

Answer: Dipa

Example 18 [Accenture pattern]

Four people W, X, Y, Z sit at the four corners of a square table, all facing the centre.

- W sits diagonally opposite Y.
- X sits immediately to the left of W.

Who sits immediately to the right of Y?

SOLUTION

Number the four corners 1, 2, 3, 4 clockwise and put W at corner 1.

Diagonally opposite in a 4-seat ring means add $4/2 = 2$: $Y = 1 + 2 = 3$.

All face the centre, so left = clockwise. $X = 1 + 1 = 2$.

The last corner, 4, goes to Z.

Y is at corner 3 and faces the centre, so Y's right is anticlockwise: $3 - 1 = 2$.

Answer: X

Example 19 [TCS NQT pattern]

Five students stand in a line in order of height.

- Deepa is taller than Farah but shorter than Gita.
- Harsh is taller than Gita.
- Imran is the shortest.

Who is the second tallest?

SOLUTION

Write each clue as a chain, tallest on the left.

“Deepa taller than Farah, shorter than Gita” gives $Gita > Deepa > Farah$.

“Harsh taller than Gita” puts Harsh on the far left: $Harsh > Gita > Deepa > Farah$.

Imran is the shortest, so Imran goes at the far right:

$Harsh > Gita > Deepa > Farah > Imran$.

All five are now in one chain, so the order is fixed. The second name is Gita.

Answer: Gita

PRACTICE · TIER 1 target: 60 s/Q

Questions 1 to 3. Six friends — Ajay, Bina, Chetan, Divya, Esha and Farhan — sit in a row facing North, drawn with West on the left. Chetan sits at the extreme right end. Ajay sits third from the left. Bina sits immediately to the left of Ajay. Divya sits at the extreme left end. Esha does not sit next to Chetan.

1. Who sits immediately to the right of Esha?
2. How many people sit between Bina and Farhan?
3. Who sits exactly between Divya and Ajay?

Questions 4 to 6. Eight people — Karan, Latika, Mohan, Nisha, Omar, Pooja, Qadir and Rita — sit around a circular table facing the centre. Mohan sits third to the left of Karan. Nisha sits exactly opposite Mohan. Omar sits immediately to the left of Karan. Pooja sits second to the right of Nisha. Latika sits immediately to the left of Mohan. Qadir sits immediately to the right of Mohan.

4. Who sits exactly opposite Karan?
5. Who sits second to the left of Pooja?
6. How many people sit between Qadir and Rita, counting clockwise from Qadir?

Questions 7 to 9. Five people — Tara, Umesh, Vani, Wasim and Yash — live on five floors of a building, floor 1 at the bottom. Vani lives on the top floor. Tara lives immediately below Vani. Umesh lives on floor 1. Wasim lives above Yash.

7. Who lives on floor 3?
8. How many people live between Umesh and Tara?
9. Who lives immediately above Yash?
10. Six people sit around a circular table facing **outward**. A sits second to the left of B, and C sits immediately to the right of B. How many people sit between C and A, counting clockwise from C?
11. Four boxes W, X, Y, Z are stacked. Z is at the bottom, X is immediately above Z, W is somewhere above X, and Y is not at the top. Which box is at the top?
12. Seven meetings are held from Monday to Sunday, one each day. The review is on Tuesday. The demo is three days after the review. The audit is on the day immediately before the demo. On which day is the audit?
13. Five students are ranked by height. Ravi is taller than Sunil but shorter than Tina. Uma is taller than Tina. Vikas is the shortest. Who is the second shortest?

ANSWER KEY

Seats for questions 1–3, from the left: Divya(1), Bina(2), Ajay(3), Esha(4), Farhan(5), Chetan(6). Esha cannot be at 5 because that seat is next to Chetan.

1. **Farhan**. Facing North, Esha's right is seat 5.
2. **2 people**. Bina is at 2 and Farhan at 5, so seats 3 and 4 lie between.
3. **Bina**. She is at seat 2, between seats 1 and 3.

Seats for questions 4–6, clockwise: Karan(1), Omar(2), Qadir(3), Mohan(4), Latika(5), Pooja(6), Rita(7), Nisha(8).

4. **Latika**. Opposite Karan = $1 + 4 = 5$.
5. **Nisha**. Facing the centre, left is clockwise: $6 + 2 = 8$.
6. **3 people**. From seat 3 clockwise to seat 7 you pass seats 4, 5 and 6 — Mohan, Latika and Pooja.

Floors for questions 7–9: Umesh(1), Yash(2), Wasim(3), Tara(4), Vani(5).

7. **Wasim**. Vani is on 5 and Tara on 4, Umesh on 1, so Yash and Wasim take 2 and 3; Wasim is above Yash.
8. **2 people**. Floors 2 and 3 lie between floors 1 and 4.
9. **Wasim**. Yash is on floor 2, so floor 3 is immediately above.
10. **2 people**. Facing outward, left is anticlockwise and right is clockwise. Put B at seat 1: $A = 1 - 2 = -1$, which wraps round to seat $-1 + 6 = 5$; and $C = 1 + 1 = 2$. From seat 2 clockwise to seat 5 you pass seats 3 and 4.
11. **W**. Z(1), X(2); W is above X, and Y is not at the top, so $Y = 3$ and $W = 4$.
12. **Thursday**. Review Tuesday, demo Tuesday $+3 =$ Friday, audit the day before Friday.
13. **Sunil**. The chain is Uma $>$ Tina $>$ Ravi $>$ Sunil $>$ Vikas.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 20 [Grab · pattern]

Eight analysts A, B, C, D, E, F, G, H sit around a circular table facing the centre.

- C sits third to the left of A.
- B sits exactly opposite C.
- E sits second to the right of B.
- D sits immediately to the left of E.
- H sits immediately to the right of C.
- F is not an immediate neighbour of C.

(a) Who sits exactly opposite E? (b) Who sits between C and E, counting clockwise from C?

SOLUTION

Number the seats 1 to 8 clockwise and put A at seat 1.

All face the centre: left = clockwise (+), right = anticlockwise (−).

C is third to A's left: $1 + 3 = 4$. $C = 4$.

B is opposite C: $4 + 8/2 = 4 + 4 = 8$. $B = 8$.

E is second to B's right: $8 - 2 = 6$. $E = 6$.

D is immediately to E's left: $6 + 1 = 7$. $D = 7$.

H is immediately to C's right: $4 - 1 = 3$. $H = 3$.

Free seats: 2 and 5, for F and G.

C is at seat 4, so C's neighbours are seats 3 and 5. F is not a neighbour of C, so F is not at 5.

Therefore $F = 2$ and $G = 5$.

Final clockwise: A(1), F(2), H(3), C(4), G(5), E(6), D(7), B(8).

(a) Opposite E = $6 + 4 = 10$, and $10 - 8 = 2$.

(b) From seat 4 clockwise to seat 6 you pass seat 5 only.

Answer: (a) F (b) G

Example 21 [Sea/Shopee · pattern]

Ten people sit in two rows of five, facing each other.

Row 1 — Aisha, Ben, Chen, Dara, Eko — all face South.

Row 2 — Fajar, Gita, Hanif, Intan, Joko — all face North.

Both rows are drawn with West on the left, and seat k of Row 1 faces seat k of Row 2.

- Dara sits exactly in the middle of Row 1.
- Chen sits at the extreme West end of Row 1.
- Ben sits second to the left of Dara.
- Eko sits immediately to the left of Dara.
- Aisha faces Gita.
- Fajar sits at the extreme West end of Row 2.
- Hanif faces Eko.
- Joko sits at the extreme East end of Row 2.

(a) Who faces Chen? (b) Who sits second to the right of Ben?

SOLUTION

Row 1 faces South, so **left** = larger seat number and **right** = smaller seat number.

Row 2 faces North, so **left** = smaller seat number and **right** = larger seat number.

Row 1. Exact middle of 5 seats is seat 3, so $Dara = 3$.

Chen is at the West end: $Chen = 1$.

Ben is second to Dara's left: $3 + 2 = 5$. $Ben = 5$.

Eko is immediately to Dara's left: $3 + 1 = 4$. $Eko = 4$.

The only Row-1 seat left is 2, so $Aisha = 2$.

Row 1: Chen(1), Aisha(2), Dara(3), Eko(4), Ben(5).

Row 2. Fajar = 1 and Joko = 5.

Hanif faces Eko, and Eko is at seat 4, so $Hanif = 4$.

Aisha faces Gita, and Aisha is at seat 2, so Gita = 2.
 The only Row-2 seat left is 3, so Intan = 3.
 Row 2: Fajar(1), Gita(2), Intan(3), Hanif(4), Joko(5).
(a) Chen is at seat 1, so Chen faces the Row-2 person at seat 1.
(b) Ben is at seat 5 and faces South, so his right is $5 - 2 = 3$.

Answer: (a) Fajar (b) Dara

Example 22 [DBS · pattern]

Seven people — Farah, Gopal, Hui, Ivan, Jaya, Kamal and Lin — live on seven floors of a building, floor 1 at the bottom and floor 7 at the top. One person per floor.

- Farah lives on floor 6.
- Exactly three people live between Farah and Gopal.
- Hui lives immediately above Gopal.
- Jaya lives immediately below Farah.
- Ivan lives on an odd-numbered floor above Hui.
- Kamal does not live on floor 1.

(a) How many people live between Kamal and Ivan? **(b)** Who lives on floor 3?

SOLUTION

Farah = 6.

Three people between Farah and Gopal means the floors differ by $3 + 1 = 4$.

So Gopal = $6 - 4 = 2$, or Gopal = $6 + 4 = 10$ which does not exist. Gopal = 2.

Hui is immediately above Gopal: Hui = $2 + 1 = 3$.

Jaya is immediately below Farah: Jaya = $6 - 1 = 5$.

Ivan is on an odd floor above 3: the odd floors above 3 are 5 and 7. Floor 5 is Jaya's, so Ivan = 7.

Free floors: 1 and 4, for Kamal and Lin.

Kamal is not on floor 1, so Kamal = 4 and Lin = 1.

Final: Lin(1), Gopal(2), Hui(3), Kamal(4), Jaya(5), Farah(6), Ivan(7).

(a) Kamal is on 4 and Ivan on 7, so floors 5 and 6 lie between: 2 people.

(b) Floor 3 is Hui's.

Answer: (a) 2 people (b) Hui

Example 23 [GIC · pattern]

Eight people sit at a square table. Four — A, B, C, D — sit at the corners and **face the centre**. Four — P, Q, R, S — sit at the middles of the sides and **face outward**. The eight seats are numbered 1 to 8 clockwise, and seats 1, 3, 5, 7 are the corners.

- A sits at seat 1.
- P sits immediately to the left of A.
- B sits exactly opposite A.
- Q sits second to the right of P.
- C sits immediately to the left of Q.
- R sits exactly opposite Q.

(a) Who sits second to the right of B? **(b)** Who sits exactly opposite S?

SOLUTION

Two different rules are in use here, so check each person's facing before moving.

Corner people face the centre: left = clockwise (+), right = anticlockwise (−).

Middle people face outward: left = anticlockwise (−), right = clockwise (+).

A = 1 (a corner).

P is immediately to A's left. A faces the centre, so $1 + 1 = 2$. P = 2, which is a middle seat. ✓

B is opposite A: $1 + 4 = 5$, a corner. ✓

Q is second to P's right. P faces outward, so $2 + 2 = 4$, a middle seat. Q = 4. ✓

C is immediately to Q's left. Q faces outward, so $4 - 1 = 3$, a corner. $C = 3$. ✓

R is opposite Q: $4 + 4 = 8$, a middle seat. $R = 8$. ✓

Seats left: 6 (a middle) and 7 (a corner). S is a middle person, so $S = 6$; D is a corner person, so $D = 7$.

Final clockwise: A(1), P(2), C(3), Q(4), B(5), S(6), D(7), R(8).

(a) B is at seat 5 and faces the centre, so B's right is anticlockwise: $5 - 2 = 3$.

(b) Opposite $S = 6 + 4 = 10$, and $10 - 8 = 2$.

Answer: (a) C (b) P

TRAP

At a mixed-facing table, "second to the right" has **two** meanings depending on who is speaking. Apply the rule of the person named in the clue, not the rule of the person you are trying to place.

Example 24 [Agoda · pattern]

Five consultants — Anan, Bua, Chai, Dara and Ek — each fly to a different city (Bangkok, Hanoi, Jakarta, Manila, Seoul) on five different days, Monday to Friday.

- The Seoul trip is on Tuesday.
- Chai travels to Manila.
- Anan travels on the day immediately before Chai.
- Bua goes to Seoul.
- The Jakarta trip is on Friday.
- Dara travels on Monday, but not to Hanoi.

Who travels to Hanoi, and on which day?

SOLUTION

Bua goes to Seoul and Seoul is on Tuesday, so $Bua = Tuesday$.

$Dara = Monday$.

Dara's city cannot be Seoul (that is Tuesday), cannot be Jakarta (that is Friday), and is not Hanoi.

So Dara goes to Bangkok or Manila. Manila belongs to Chai, and Chai is not on Monday because Dara is.

Therefore Dara goes to **Bangkok** on Monday.

Days left: Wednesday, Thursday, Friday, for Anan, Chai and Ek.

Anan is the day right before Chai, so the pairs to test are Wednesday–Thursday and Thursday–Friday.

Friday is the Jakarta trip, but Chai's city is Manila, so Chai is not on Friday.

That kills Thursday–Friday. So $Anan = Wednesday$ and $Chai = Thursday$.

Ek takes the last day, Friday, and therefore the Jakarta trip.

The only city left is Hanoi, and the only person left is Anan, on Wednesday.

Final: Dara–Bangkok–Monday, Bua–Seoul–Tuesday, Anan–Hanoi–Wednesday, Chai–Manila–Thursday, Ek–Jakarta–Friday.

Answer: Anan, on Wednesday

SHORTCUT

In a grid puzzle, write the **day** column first and the **city** column second. Days are usually pinned harder than attributes, and once the days are fixed each remaining city is forced by elimination in one pass.

Example 25 [LINE MAN · pattern]

Eight riders A, B, C, D, E, F, G, H stand in a row facing North, drawn with West on the left. Each made a different number of deliveries today, from the set 10, 12, 15, 18, 20, 24, 28, 30.

Positions. D stands fourth from the left. Exactly three riders stand between D and G. B stands immediately to the left of G. A stands at the extreme left end. E stands immediately to the right of A. F stands immediately to the left of B. H stands between D and F.

Deliveries. The rider at the extreme left made 20. G made 30. D made exactly twice as many as E. The rider immediately to the right of D made 15. C made 10. B made more than F.

(a) Who made 18 deliveries? (b) How many deliveries did the rider standing third from the left make?

SOLUTION

Step 1 — the row. Number the seats 1 to 8 from the left. Facing North: right = larger, left = smaller.

$D = 4$.

Three riders between D and G means the seats differ by $3 + 1 = 4$, so $G = 4 + 4 = 8$ (seat 0 does not exist).

B is immediately to G's left: $B = 8 - 1 = 7$.

$A = 1$, and $E = 1 + 1 = 2$.

F is immediately to B's left: $F = 7 - 1 = 6$.

H stands between D(4) and F(6), so $H = 5$.

The last seat, 3, goes to C.

Row: A(1), E(2), C(3), D(4), H(5), F(6), B(7), G(8).

Step 2 — the deliveries.

A is at the extreme left, so $A = 20$.

$G = 30$.

$C = 10$.

The rider immediately to D's right is at seat 5, which is H, so $H = 15$.

D is twice E. Test every value of E in the set:

$E = 10$ gives $D = 20$, already taken by A. Rejected.

$E = 12$ gives $D = 24$, still free. Possible.

$E = 15$ gives $D = 30$, taken by G. Rejected.

No other value doubles into the set. So $E = 12$ and $D = 24$.

Used so far: 20, 30, 10, 15, 12, 24. Left: 18 and 28, for F and B.

B made more than F, so $B = 28$ and $F = 18$.

(a) 18 belongs to F.

(b) Third from the left is seat 3, which is C.

Answer: (a) F (b) 10 deliveries

Example 26 [Razer · pattern]

Six boxes P, Q, R, S, T, U are stacked one above another. Each has a different weight from 5, 8, 12, 15, 20 and 25 kg.

- Exactly two boxes lie above Q.
- S lies immediately below Q.
- R is at the bottom.
- P is at the top.
- T lies above U.
- Q weighs 20 kg.
- The two boxes below S weigh 13 kg in total.
- U is heavier than R.
- S weighs 25 kg.
- T weighs more than P.

(a) How many boxes lie between U and T? (b) What does the topmost box weigh?

SOLUTION

Step 1 — the stack. Number the positions 1 (bottom) to 6 (top).

Two boxes lie above Q, so Q is third from the top: $Q = 6 - 2 = 4$.

S is immediately below Q: $S = 4 - 1 = 3$.

$R = 1$ and $P = 6$.

Free positions: 2 and 5, for T and U. T is above U, so $U = 2$ and $T = 5$.

Stack from the bottom: R(1), U(2), S(3), Q(4), T(5), P(6).

Step 2 — the weights.

$Q = 20$ and $S = 25$. Left to place: 5, 8, 12, 15.

The boxes below S are at positions 1 and 2, that is R and U, and they total 13.

From 5, 8, 12, 15 the pairs are $5 + 8 = 13$, $5 + 12 = 17$, $5 + 15 = 20$, $8 + 12 = 20$, $8 + 15 = 23$, $12 + 15 = 27$.

Only $5 + 8 = 13$ works, so R and U are 5 and 8 in some order.

U is heavier than R, so $U = 8$ and $R = 5$.

Left: 12 and 15, for T and P. T is heavier than P, so $T = 15$ and $P = 12$.

Final: R 5 kg, U 8 kg, S 25 kg, Q 20 kg, T 15 kg, P 12 kg.

(a) U is at 2 and T is at 5, so positions 3 and 4 lie between: 2 boxes.

(b) The top box is P.

Answer: (a) 2 boxes (b) 12 kg

Example 27 [Agoda · pattern]

Six colleagues — Nadia, Omar, Priya, Rafi, Sita and Tan — each took leave in a different month of the same year: January, March, April, July, September and November (listed in calendar order). Each took a different number of days off: 3, 5, 7, 10, 12 and 15.

- Exactly two of them took leave in months earlier than Nadia's.
- Omar took leave in the month immediately after Nadia's in this list.
- Tan took leave in the last of these months.
- Sita took leave in the month immediately before Tan's in this list.
- Priya took leave before Rafi.
- Nadia took 12 days.
- The person who took leave in January took 5 days.
- Rafi took 3 days.
- Tan took more days than Sita, and Sita took more days than Omar.

(a) Who took leave in March, and for how many days? (b) How many days did the person who took leave in September take?

SOLUTION

Step 1 — the months. Number them: January(1), March(2), April(3), July(4), September(5), November(6).

Two people are earlier than Nadia, so Nadia is third in the list: $Nadia = 3 = \text{April}$.

Omar is the month right after: $Omar = 4 = \text{July}$.

Tan is last: $Tan = 6 = \text{November}$.

Sita is right before Tan: $Sita = 5 = \text{September}$.

Months left: January(1) and March(2), for Priya and Rafi. Priya is before Rafi, so $Priya = \text{January}$ and $Rafi = \text{March}$.

Step 2 — the days.

$Nadia = 12$.

The January person is Priya, so $Priya = 5$.

$Rafi = 3$.

Left: 7, 10, 15, for Omar, Sita and Tan.

The chain $Tan > Sita > Omar$ forces the largest to Tan, the middle to Sita and the smallest to Omar:

$Tan = 15$, $Sita = 10$, $Omar = 7$.

Final: Priya–January–5, Rafi–March–3, Nadia–April–12, Omar–July–7, Sita–September–10, Tan–November–15.

(a) March belongs to Rafi, with 3 days.

(b) September belongs to Sita.

Answer: (a) Rafi, 3 days (b) 10 days

Example 28 [Grab · pattern]

Six people sit around a circular table. A, C and E face the centre. B, D and F face outward.

- C sits third to the left of A.
- B sits immediately to the right of C.

- D sits second to the right of B.
- F sits immediately to the left of A.

(a) Who sits second to the right of E? (b) How many people sit between F and D, counting clockwise from F?

SOLUTION

Number the seats 1 to 6 clockwise and put A at seat 1.

Facing the centre: left = +, right = -. Facing outward: left = -, right = +.

C is third to A's left. A faces the centre, so $1 + 3 = 4$. C = 4.

B is immediately to C's right. C faces the centre, so $4 - 1 = 3$. B = 3.

D is second to B's right. B faces **outward**, so $3 + 2 = 5$. D = 5.

F is immediately to A's left. A faces the centre, so $1 + 1 = 2$. F = 2.

The last seat, 6, goes to E.

Final clockwise: A(1), F(2), B(3), C(4), D(5), E(6).

(a) E faces the centre, so E's right is anticlockwise: $6 - 2 = 4$.

(b) From seat 2 clockwise to seat 5 you pass seats 3 and 4 — B and C.

Answer: (a) C (b) 2 people

PRACTICE • TIER 2 target: 100 s/Q

Questions 1 to 4. Ten people sit in two rows of five, facing each other. Row 1 — Nita, Omar, Pavi, Quan, Ratna — all face North. Row 2 — Surya, Tavi, Udin, Vina, Wati — all face South. Both rows are drawn with West on the left, and seat k of Row 1 faces seat k of Row 2. Quan sits exactly in the middle of Row 1. Omar sits at the extreme East end of Row 1. Nita sits immediately to the left of Quan. Pavi sits at the extreme West end of Row 1. Surya faces Quan. Tavi sits immediately to the left of Surya. Vina sits at the extreme East end of Row 2. Udin faces Nita.

1. Who faces Ratna?
2. Who sits second to the right of Nita?
3. How many people sit between Wati and Tavi?
4. Who faces the person sitting immediately to the right of Udin?

Questions 5 to 8. Seven people — Arun, Bella, Cheng, Divya, Ethan, Farah and Gopal — live on seven floors of a building, floor 1 at the bottom. Cheng lives on floor 5. Exactly two people live between Cheng and Arun. Bella lives immediately above Arun. Divya lives on the topmost floor. Ethan lives immediately below Divya. Farah does not live on floor 1.

5. Who lives on floor 4?
6. How many people live between Bella and Ethan?
7. Who lives immediately below Cheng?
8. If Gopal and Divya swap floors, who then lives on floor 1?
9. Eight people sit around a circular table. A, B, C and D face the centre; P, Q, R and S face outward. B sits third to the left of A. P sits immediately to the right of B. C sits second to the right of P. Q sits exactly opposite P. R sits immediately to the left of A. S sits exactly opposite R. Who sits exactly opposite A?
10. Six boxes L, M, N, O, P, Q are stacked, with different weights of 4, 7, 9, 11, 14 and 18 kg. N is at the bottom. Exactly three boxes lie between N and M. O lies immediately above M. L lies immediately above N. P lies below Q. The bottom box weighs 9 kg. M weighs twice as much as L. O weighs 4 kg. Q is heavier than P. What is the total weight of the boxes immediately above and immediately below Q?
11. Six people took a course in six different months: February, April, May, August, October and December. Exactly three of them took it after Reza. Sofia took it in the month immediately before Reza's. Tan took it in the last of these months. Umi took it in February. Wira took it before Yusuf. Who took the course in October?

ANSWER KEY

Questions 1–4. Row 1 from the West: Pavi(1), Nita(2), Quan(3), Ratna(4), Omar(5). Row 2 from the West: Wati(1), Udin(2), Surya(3), Tavi(4), Vina(5). Row 1 faces North, so its “left” is a smaller seat number; Row 2 faces South, so its “left” is a larger one.

1. **Tavi.** Ratna is at seat 4, so she faces Row 2 seat 4.
2. **Ratna.** Nita is at 2 and faces North, so her right is $2 + 2 = 4$.
3. **2 people.** Wati is at 1 and Tavi at 4, so seats 2 and 3 lie between.
4. **Pavi.** Udin is at 2 and faces South, so his right is seat $2 - 1 = 1$, which is Wati; Wati faces Row 1 seat 1.

Questions 5–8. Floors: Gopal(1), Arun(2), Bella(3), Farah(4), Cheng(5), Ethan(6), Divya(7). Two people between Cheng(5) and Arun means the floors differ by 3, so Arun is on 2 (floor 8 does not exist).

5. **Farah.** Floors 1 and 4 are the last two free, and Farah cannot be on floor 1.
6. **2 people.** Floors 4 and 5 lie between floors 3 and 6.
7. **Farah.** Floor 4 is immediately below floor 5.
8. **Divya.** The swap puts her on floor 1 and Gopal on floor 7.
9. **C.** Put A at seat 1. $B = 1 + 3 = 4$. $P = 4 - 1 = 3$ (B faces the centre). $C = 3 + 2 = 5$ (P faces outward). $Q = 3 + 4 = 7$. $R = 1 + 1 = 2$. $S = 2 + 4 = 6$, so D takes seat 8. Opposite $A = 1 + 4 = 5$.
10. **25 kg.** Stack from the bottom: N(1), L(2), P(3), Q(4), M(5), O(6). $N = 9$ kg and $O = 4$ kg. For $M = 2L$ the only pair left in the set is $L = 7$ and $M = 14$ (the pair 9 and 18 is blocked because 9 is already N’s). The remaining 11 and 18 go to P and Q with Q heavier, so $Q = 18$ and $P = 11$. Immediately above Q is M (14 kg) and immediately below is P (11 kg): $14 + 11 = 25$.
11. **Yusuf.** Three people come after Reza, so Reza is third: May. Sofia is April. Tan is December, Umi is February. The months left are August and October for Wira and Yusuf, and Wira is earlier, so Wira is August.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 29 [Google · pattern]

Five friends sit in a row of five chairs. Two of them, Ravi and Sneha, refuse to sit next to each other. In how many different orders can the five of them sit?

SOLUTION

Insight: counting “not together” directly means many cases. Count everything, then subtract the arrangements where they ARE together.

Step 1 — all arrangements. Five people in five chairs:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120.$$

Step 2 — arrangements with Ravi and Sneha together. Glue them into one block.

The block plus the other 3 people make 4 objects, arranged in $4! = 24$ ways.

Inside the block, Ravi and Sneha can swap: 2 ways.

$$\text{Together} = 24 \times 2 = 48.$$

Step 3 — subtract.

$$120 - 48 = 72.$$

Answer: 72 arrangements

Example 30 [Amazon · pattern]

Six people sit around a round table. Two seatings count as the same if one is a rotation of the other. In how many ways can they sit if two particular people, X and Y, must sit exactly opposite each other?

SOLUTION

Insight: “rotations are the same” is handled by nailing one person to one seat. After that the table behaves like a plain list of seats.

Number the seats 1 to 6 clockwise.

Fix X at seat 1. Every rotation of a seating now has exactly one version with X at seat 1, so no seating is counted twice.

Y must sit opposite X: $1 + 6/2 = 1 + 3 = 4$. That is exactly 1 choice for Y.

The remaining 4 people go into the remaining 4 seats (2, 3, 5, 6) in

$4! = 4 \times 3 \times 2 \times 1 = 24$ ways.

Total = $1 \times 24 = 24$.

Check: without the opposite rule the answer would be $(6 - 1)! = 120$. Of the 5 seats Y could take, exactly 1 is opposite X, so $120 \times \frac{1}{5} = 24$. ✓

Answer: 24 ways

SHORTCUT

For any round-table count with “rotations are the same”, fix one named person first. The count then drops from $n!$ to $(n - 1)!$ automatically and you never divide at the end.

Example 31 [Microsoft · pattern]

Seven people P, Q, R, S, T, U, V stand in a row of seven places, numbered 1 to 7 from the left.

- P stands to the left of Q, with exactly one person between them.
- R stands three places to the right of Q.
- S stands at one of the two ends.

In how many different ways can the seven of them stand?

SOLUTION

Insight: P, Q and R are locked into a rigid chain. Slide the whole chain along the row and see how few places it actually fits.

One person between P and Q means the places differ by $1 + 1 = 2$, and P is on the left, so

$$Q = P + 2$$

R is three places to Q's right, so

$$R = Q + 3 = P + 5$$

P must be at least 1 and R at most 7, so $P + 5 \leq 7$, giving $P \leq 2$.

So $P = 1$ or $P = 2$. Only two cases.

Case $P = 1$: $Q = 3$, $R = 6$. Places left: 2, 4, 5, 7.

S must be at an end. Place 1 is taken by P, so $S = 7$.

The other three people, T, U and V, fill places 2, 4, 5 in $3! = 6$ ways.

Case $P = 2$: $Q = 4$, $R = 7$. Places left: 1, 3, 5, 6.

S must be at an end. Place 7 is taken by R, so $S = 1$.

T, U and V fill places 3, 5, 6 in $3! = 6$ ways.

Total = $6 + 6 = 12$.

Answer: 12 ways

Example 32 [Goldman Sachs · pattern]

Ten people sit around a circular table facing the centre. Each of them is either a Knight, who always tells the truth, or a Knave, who always lies. Every single person says the same sentence: “**The person immediately to my left is a Knave.**” How many Knights are at the table?

SOLUTION

Insight: each sentence links a person to exactly ONE neighbour, so the types must alternate around the whole circle.

Take any person X.

If X is a Knight, the sentence is true, so X's left neighbour is a Knave.

If X is a Knave, the sentence is false, so X's left neighbour is **not** a Knave — the left neighbour is a Knight.
 Either way: **X and X's left neighbour are always of opposite types.**
 Now walk around the table in the “left” direction. The type flips at every single step:
 Knight, Knave, Knight, Knave, ...
 After 10 steps you are back at the person you started from.
 Ten flips bring you back to the type you started with, so the pattern is consistent. ✓
 Because the types strictly alternate around 10 seats, exactly half the seats are Knights:
 $10 \div 2 = 5$.

Answer: 5 Knights

The same walk with 11 people would need 11 flips to return to the start, which changes the type — a contradiction. So this table is **impossible** with any odd number of people.

Example 33 [D. E. Shaw · pattern]

A row has 15 chairs, all empty. Some people sit down. What is the smallest number of people who must be seated so that **every** empty chair has at least one occupied chair next to it — that is, nobody arriving later can sit without sitting beside somebody?

SOLUTION

Insight: each seated person can “cover” at most two empty chairs — the one on each side. That gives a hard lower bound before you try any pattern.

Step 1 — the bound. Let m people be seated.

Empty chairs = $15 - m$.

Each seated person has at most 2 neighbouring chairs, so the m people together touch at most $2m$ chairs.

Every empty chair must be touched, so

$$15 - m \leq 2m$$

$$15 \leq 3m$$

$$m \geq 5$$

So 4 people can never be enough.

Step 2 — show that 5 is enough. Seat people at chairs 2, 5, 8, 11 and 14.

Chair 1 and chair 3 touch the person at 2. ✓

Chairs 4 and 6 touch the person at 5. ✓

Chairs 7 and 9 touch the person at 8. ✓

Chairs 10 and 12 touch the person at 11. ✓

Chairs 13 and 15 touch the person at 14. ✓

That is all 10 empty chairs covered, using 5 people.

The bound says at least 5, and the pattern reaches 5.

Answer: 5 people

SHORTCUT

The pattern “empty, person, empty, empty, person, empty, ...” — a person every third chair, starting at chair 2 — is the general answer. For n chairs the minimum is the smallest whole number that is at least $n/3$.

Example 34 [Uber · pattern]

Four married couples — eight people — sit around a round table with eight seats. Each husband must sit exactly opposite his own wife. Two seatings count as the same if one is a rotation of the other. In how many ways can they be seated?

SOLUTION

Insight: “opposite” ties the 8 seats into 4 fixed seat-pairs. Choose which couple gets which pair, then who in the couple sits where.

Number the seats 1 to 8 clockwise. The opposite pairs are (1, 5), (2, 6), (3, 7) and (4, 8).

Step 1 — kill the rotations. Fix the first husband at seat 1. Now every seating is counted exactly once. His wife is then forced into seat 5. That is 1 way.

Step 2 — the other three couples. Three seat-pairs are still free: (2, 6), (3, 7), (4, 8).

Choose which of the three remaining couples goes to which pair:

$$3! = 3 \times 2 \times 1 = 6 \text{ ways.}$$

Step 3 — inside each pair. For each couple, the husband may take either of the two seats in their pair, and the wife takes the other: 2 ways per couple.

For three couples: $2 \times 2 \times 2 = 2^3 = 8$ ways.

$$\text{Total} = 6 \times 8 = 48.$$

Answer: 48 ways

Example 35 [Google · pattern]

Six teams play a round robin: every pair of teams plays exactly once and no match is drawn. Team A wins all of its matches. Teams B, C and D each win exactly 3 matches. Team E wins exactly 1 match. How many matches does team F win?

SOLUTION

Insight: every match produces exactly one win, so the total number of wins equals the total number of matches. That single invariant answers the question.

Step 1 — count the matches. With 6 teams, each pair plays once:

$$\binom{6}{2} = \frac{6 \times 5}{2} = 15$$

So there are 15 matches, and therefore 15 wins in total.

Step 2 — add the known wins.

A: 5 (it plays 5 matches and wins them all).

B, C, D: $3 + 3 + 3 = 9$.

E: 1.

Known total = $5 + 9 + 1 = 15$.

Step 3 — F's wins.

$$F = 15 - 15 = 0$$

Step 4 — check that this is actually possible.

F loses all 5 of its matches, and A wins all 5 of its own.

Each of B, C, D has lost to A and beaten F, giving 1 win each so far; each needs 2 more.

E has beaten F — that is already E's single win — so E must lose to B, C and D. Those three games give B, C, D one more win each: now 2 each.

The last three matches are B–C, C–D and D–B. Let B beat C, C beat D and D beat B. Each of B, C, D picks up exactly 1 more win, reaching 3 each. ✓

Everything matches the story, so the answer stands.

Answer: 0 matches

PRACTICE · TIER 3 target: 4 min/Q

1. Six people sit in a row of six chairs. Two of them, Meera and Nikhil, must not sit next to each other. In how many different orders can the six sit?
2. Eight people sit around a round table, where two seatings are the same if one is a rotation of the other. In how many ways can they sit if two particular people must sit next to each other?
3. Five people P, Q, R, S, T sit in a row of five chairs. Exactly one person sits between P and Q. In how many different ways can the five sit?

- A row has 21 chairs, all empty. What is the smallest number of people who must be seated so that every empty chair has at least one occupied chair next to it?
- Twelve people sit around a circular table. Each one says: **“The person immediately to my right is a liar.”** Truth-tellers always tell the truth and liars always lie. How many truth-tellers are at the table?
- Five teams play a round robin — every pair plays once, no draws. Team X wins all 4 of its matches. Teams Y and Z each win exactly 2. Team W wins exactly 1. How many matches does team V win?
- Seven people sit in a row of seven chairs. Three of them — A, B and C — must appear in that order from left to right, though not necessarily next to each other. In how many ways can the seven sit?

ANSWER KEY

- 480 orders.** All orders = $6! = 720$. Together: glue the pair into a block, giving $5! = 120$ arrangements of 5 objects, times 2 for the swap inside the block = 240. Then $720 - 240 = 480$.
- 1440 ways.** Glue the pair into one block. Seven objects around a round table give $(7 - 1)! = 720$ seatings, and the pair can swap inside the block in 2 ways: $720 \times 2 = 1440$.
- 36 ways.** Chairs for the pair must differ by 2: $(1, 3), (2, 4), (3, 5)$ — 3 chair-pairs, and P and Q can swap in each, so $3 \times 2 = 6$ placements. The other three people fill the rest in $3! = 6$ ways: $6 \times 6 = 36$.
- 7 people.** With m seated, $21 - m \leq 2m$ gives $m \geq 7$. Chairs 2, 5, 8, 11, 14, 17 and 20 cover every empty chair, so 7 is enough.
- 6 truth-tellers.** A truth-teller's right neighbour is a liar; a liar's statement is false, so a liar's right neighbour is a truth-teller. Types therefore alternate all the way round, and 12 is even, so exactly half — 6 — are truth-tellers.
- 1 match.** Total matches = $\binom{5}{2} = 10$, so total wins = 10. Known wins = $4 + 2 + 2 + 1 = 9$, leaving $10 - 9 = 1$ for V. It is possible: X beats everyone; Y beats Z and W; Z beats V and W; V beats Y; W beats V.
- 840 ways.** Choose the 3 chairs for A, B, C and the 4 chairs for the others: every arrangement of all 7 people can be paired with the $3! = 6$ orders of A, B, C, and exactly one of those six is the required order. So the answer is $7!/3! = 5040/6 = 840$.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 20 questions in 26 min

Questions 1 to 4. Eight students — A, B, C, D, E, F, G, H — sit in a row facing North, drawn with West on the left. D sits fourth from the right end. A sits at the extreme left end. Exactly two students sit between A and B. C sits immediately to the right of D. G sits at the extreme right end. F sits immediately to the left of B. E does not sit next to A.

- Who sits exactly between F and D?
- How many students sit between C and G?
- Who sits third to the left of E?
- If A and G swap seats, who then sits at the extreme left end?

Questions 5 to 8. Six people — Ila, Jay, Kim, Lee, Mira and Noor — sit around a circular table facing the centre. Jay sits second to the left of Ila. Kim sits exactly opposite Jay. Lee sits immediately to the left of Jay. Mira sits immediately to the right of Kim.

- Who sits exactly opposite Ila?
- Who sits immediately to the right of Noor?
- How many people sit between Lee and Kim, counting clockwise from Lee?
- Who sits third to the right of Mira?
- Five boxes — black, blue, green, red and white — are stacked one above another. The red box is at the top, the blue box is immediately below the red box, and the green box is at the bottom. Exactly one box lies between the green box and the white box. Which box is in the middle of the stack?
- In a 7-floor building, Amit lives on floor 3, Bhavna lives 3 floors above Amit, and Chetan lives immediately below Bhavna. Which floor does Chetan live on?

11. Seven people sit in a row facing South, drawn with West on the left. Rani sits third from the West end. How many people sit to her right?
12. Five people sit around a circular table facing outward. X sits second to the right of Y. Who sits second to the left of X?
13. Six meetings are held from Monday to Saturday, one each day. The review is on Monday and the demo is on Thursday. The audit is on the day immediately after the review. On which day is the audit?
14. Four friends sit at the four corners of a square table facing the centre. Priya sits diagonally opposite Sanjay, and Tarun sits immediately to the left of Priya. Who sits immediately to the right of Sanjay?
15. Nine people sit in a row of nine chairs facing North. Mohan sits 4th from the left and Nita sits 3rd from the right. How many chairs lie between them?
16. Six students are ranked by marks, with no ties. Priya scored more than Qadir but less than Rohit. Sana scored more than Rohit. Tarun scored less than Qadir. Umi scored the lowest. Who ranks third?
17. Eight people sit around a circular table facing the centre, in seats numbered 1 to 8 clockwise. How many people sit between the person at seat 2 and the person at seat 7, counting clockwise from seat 2?
18. Five boxes P, Q, R, S, T are stacked. Exactly two boxes lie above Q. S lies immediately below Q. T is at the top. P lies below S. Which box is at the bottom?
19. Seven people sit in a row of seven chairs. A sits at the extreme left end and exactly three people sit between A and B. What is B's position counted from the right end?
20. Six people sit around a circular table facing the centre. Deepa sits third to the left of Farhan. Who sits third to the right of Deepa?

ANSWER KEY

Questions 1–4. Seats from the left: A(1), H(2), F(3), B(4), D(5), C(6), E(7), G(8). D is fourth from the right of 8 seats, so $D = 8 - 4 + 1 = 5$; two students between A(1) and B means $B = 4$; $F = 3$; $C = 6$; $G = 8$; E cannot take seat 2 because that seat touches A, so $E = 7$ and $H = 2$.

1. **B.** F is at 3 and D at 5, so seat 4 lies between.
2. **1 student.** C is at 6 and G at 8, so only seat 7 lies between.
3. **B.** Facing North, left is a smaller number: $7 - 3 = 4$.
4. **G.** The swap puts G at seat 1.

Questions 5–8. Clockwise: Ila(1), Noor(2), Jay(3), Lee(4), Mira(5), Kim(6). Facing the centre, left is clockwise and right is anticlockwise.

5. **Lee.** Opposite Ila $= 1 + 3 = 4$.
6. **Ila.** Noor is at 2, so his right is $2 - 1 = 1$.
7. **1 person.** From seat 4 clockwise to seat 6 you pass seat 5 only.
8. **Noor.** Mira is at 5, so third to her right is $5 - 3 = 2$.
9. **The white box.** Positions from the bottom: green(1), black(2), white(3), blue(4), red(5). One box between green(1) and white puts white at 3, which is the middle of 5.
10. **Floor 5.** Bhavna is on $3 + 3 = 6$, so Chetan is on 5.
11. **2 people.** Facing South, right means the West side, so seats 1 and 2 are to her right.
12. **Y.** Facing outward, right is clockwise and left is anticlockwise. Put Y at seat 1; then $X = 1 + 2 = 3$, and second to X's left is $3 - 2 = 1$.
13. **Tuesday.** The review is on Monday, and "immediately after" is the very next day.
14. **Tarun.** Put Priya at corner 1; Sanjay $= 1 + 2 = 3$; facing the centre, left is clockwise so Tarun $= 2$; Sanjay's right is $3 - 1 = 2$.
15. **2 chairs.** Nita from the left $= 9 - 3 + 1 = 7$, and $7 - 4 - 1 = 2$.
16. **Priya.** The chain is Sana > Rohit > Priya > Qadir > Tarun > Umi.
17. **4 people.** Seats 3, 4, 5 and 6 lie between.
18. **P.** Two boxes above Q puts Q at position 3; $S = 2$; $T = 5$; P is below S, so $P = 1$ and $R = 4$.

19. **3rd from the right.** $A = 1$, and three people between gives $B = 1 + 4 = 5$; then $7 - 5 + 1 = 3$.
20. **Farhan.** Put Farhan at seat 1; $Deepa = 1 + 3 = 4$; third to Deepa's right is $4 - 3 = 1$. (In a circle of 6, "third to the left" and "third to the right" both land on the opposite seat.)

ONE-PAGE REVISION CARD

Draw it the same way every time

- Row: seats 1 to n , left to right, with West on the left.
- Circle: seat 1 at the top, numbered clockwise.
- Floors and stacks: 1 at the bottom, n at the top.

Left and right

- Row, facing North: left = smaller seat number, right = larger.
- Row, facing South: left = larger seat number, right = smaller.
- Circle, facing the centre: left = clockwise (+), right = anticlockwise (−).
- Circle, facing outward: left = anticlockwise (−), right = clockwise (+).
- Double row: the two rows use opposite rules, and seat k faces seat k .

Key counts

- Opposite in a circle of n (even only): seat $k + n/2$, wrapping by $-n$.
- Exactly x people between A and B : seat numbers differ by $x + 1$.
- Exact middle of n seats: seat $(n + 1)/2$, and only when n is odd.
- Position from the right in a row of n : $n - (\text{position from the left}) + 1$.
- Square table with 8 seats: corners and middles alternate, so corners are 1, 3, 5, 7.

Shortcuts

- Fix one named person at seat 1 of a circle before anything else — a circle has no real start.
- Use clues in this order: ends and exact middle, then “immediately next to” chains, then two-case clues, then negative clues last.
- In a grid puzzle, pin the day or floor column first; the attribute column then falls out by elimination.
- Counting arrangements with “must not sit together”: count all, subtract the together case.
- Counting round-table arrangements with “rotations are the same”: fix one person, and $n!$ becomes $(n - 1)!$.
- Every match has exactly one winner, so total wins = total matches = $\binom{n}{2}$.

Top 5 traps

1. Reading “two people between A and B ” as a gap of 2. The gap is 3.
2. Using one left/right rule for both rows of a double row. They are opposite.
3. Using the **reader’s** left instead of the **seated person’s** left in a circle.
4. Saying somebody is “opposite” in a circle with an odd number of seats. Nobody is.
5. Starting from a negative clue (“X does not sit next to Y”). It removes options; it never places anybody. Save it for last.

Syllogisms & Logical Deduction

Draw the circles. Never argue from the words.

What you need to know

WHAT YOU NEED TO KNOW

Rule 0 — the only rule that matters. A conclusion “follows” only if it is true in **every** diagram that fits the statements. One diagram where it fails kills it.

Rule 0b — the exam convention. Treat every statement as true, even if it is silly in real life. Treat every class named as non-empty (it has at least one member).

The four statement types.

Code	Form	Covers all of the subject?	Covers all of the predicate?
A	All S are P	Yes	No
E	No S is P	Yes	Yes
I	Some S are P	No	No
O	Some S are not P	No	Yes

Conversion (flipping one statement).

- All S are P $\Rightarrow$ Some P are S. (Never “All P are S”.)
- No S is P $\Rightarrow$ No P is S $\Rightarrow$ Some S are not P, Some P are not S.
- Some S are P $\Rightarrow$ Some P are S.
- Some S are not P $\Rightarrow$ **nothing**. Cannot be flipped.

The combination table. Statement 1 links A to B, statement 2 links B to C.

Statement 1	Statement 2	Conclusion about A and C
All A are B	All B are C	All A are C
All A are B	No B is C	No A is C
Some A are B	All B are C	Some A are C
Some A are B	No B is C	Some A are not C
No A is B	All B are C	Some C are not A

Nothing at all follows from: All + Some, Some + Some, No + No, or any pair whose first statement is “Some A are not B”.

The five validity checks.

1. The middle term (B) must cover a whole class in at least one statement.
2. Two negative statements give nothing.
3. One negative statement $\Rightarrow$ the conclusion must be negative.
4. Two “some” statements give nothing.
5. One “some” statement $\Rightarrow$ the conclusion must be a “some” statement.

Either–or rule. Write “Either I or II follows” only when **both** of these hold:

1. Neither I nor II follows on its own, and

2. I and II are a complementary pair on the **same two terms**: (All S are P / Some S are not P), (No S is P / Some S are P), or (Some S are P / Some S are not P).

Possibility conclusions. “X being Y is a possibility” is TRUE unless the statements make it impossible. So look for **one** legal diagram, not for proof.

New-pattern words. “Only a few A are B” and “Almost all A are B” both mean Some A are B **and** Some A are not B. “Only A are B” means All B are A. “At least some A are B” means Some A are B.

Reverse syllogism. You are given the conclusions and must pick the statement set. Test each set forward with the combination table. Keep the set that forces **all** conclusions, not just one.

Warm-up

WARM-UP

build the reflex

Example 1

Statement: All pens are blue. Which follows — (a) All blue things are pens, (b) Some blue things are pens?

SOLUTION

“All S are P” flips only to “Some P are S”. The pens sit inside the blue circle, so part of blue is pen.

Answer: (b) Some blue things are pens.

Example 2

Statement: No cat is a dog. Write two things that follow.

SOLUTION

“No S is P” flips fully: No dog is a cat. It also weakens to a “some not”: Some cats are not dogs.

Answer: No dog is a cat; Some cats are not dogs.

Example 3

Statement: Some mangoes are ripe. What follows?

SOLUTION

“Some S are P” flips to “Some P are S”. The overlap belongs to both circles.

Answer: Some ripe things are mangoes.

Example 4

Statement: Some books are not heavy. What follows?

SOLUTION

The O type cannot be flipped. Knowing part of “books” sits outside “heavy” tells you nothing about the inside of “heavy”.

Answer: Nothing follows.

Example 5

Statements: All A are B. All B are C. Conclusion: All A are C.

SOLUTION

A sits inside B. B sits inside C. So A sits inside C.

Answer: It follows.

Example 6

Statements: Some A are B. Some B are C. Conclusion: Some A are C.

SOLUTION

The A-part of B and the C-part of B can be two different pieces of B. Diagram: B has a left piece touching A and a right piece touching C, with no contact between A and C. Both statements hold, the conclusion fails.

Answer: Does not follow (two “some” statements give nothing).

Example 7

Are these a complementary pair: “All X are Y” and “Some X are not Y”?

SOLUTION

Same subject X, same predicate Y. A-type against O-type. If all X are Y is false, then at least one X sits outside Y, so some X are not Y. Exactly one of them must be true.

Answer: Yes — A/O complementary pair.

Example 8

Statements: All fans are lights. No light is a switch. Conclusion: No fan is a switch.

SOLUTION

Fans sit inside lights. Lights and switches do not touch at all. So fans cannot touch switches either.

Answer: It follows.

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Statements: All pens are books. All books are papers.

Conclusions: I. All pens are papers. II. Some papers are pens.

SOLUTION

Pens $\subset$ books $\subset$ papers. So pens $\subset$ papers. I follows. “All pens are papers” flips to “Some papers are pens”. II follows.

Answer: Both I and II follow.

Example 10 [Accenture pattern]

Statements: All roses are flowers. No flower is a stone.

Conclusions: I. No rose is a stone. II. Some stones are roses.

SOLUTION

Roses $\subset$ flowers, and the flower circle never touches the stone circle. So no rose touches stone. I follows. II says roses and stones overlap. That directly contradicts I, which is forced. So II is false.

Answer: Only I follows.

Example 11 [Infosys pattern]

Statements: Some doctors are singers. All singers are artists.

Conclusions: I. Some doctors are artists. II. All artists are doctors.

SOLUTION

Take one person who is both doctor and singer. Every singer is an artist, so that person is an artist. That person is a doctor and an artist. I follows. II asks the whole artist circle to sit inside doctors. Nothing forces that — an artist may be neither singer nor doctor. II fails.

Answer: Only I follows.

Example 12 [Wipro pattern]

Statements: Some tables are chairs. No chair is a bed.

Conclusions: I. Some tables are not beds. II. No table is a bed.

SOLUTION

Some + No $\Rightarrow$ “Some are not”. The tables that are chairs cannot be beds, because no chair is a bed. So some tables are not beds. I follows. II claims the **whole** table circle avoids beds. The tables that are not chairs are free to be beds. II fails.

Answer: Only I follows.

TRAP

“Some A are not B” is a **weak** claim, and “No A is B” is a **strong** claim. A “some” statement in the premises can never give you a “No” conclusion. Rule 5: one particular premise $\Rightarrow$ a particular conclusion.

Example 13 [Capgemini pattern]

Statements: All apples are fruits. Some fruits are sweet.

Conclusions: I. Some apples are sweet. II. Some apples are not sweet.

SOLUTION

Check I alone. Draw the sweet circle overlapping the fruit circle **outside** the apple part. All apples are fruits: true. Some fruits are sweet: true. Some apples are sweet: false. So I does not follow.

Check II alone. Now draw apple = fruit = sweet, all three the same circle. All apples are fruits: true. Some fruits are sweet: true. Some apples are not sweet: false. So II does not follow.

Now the either–or test. I is “Some apples are sweet” (I type). II is “Some apples are not sweet” (O type). Same subject, same predicate. Apples exist, so at least one of the two must be true.

Answer: Either I or II follows.

SHORTCUT

The either–or 5-second test. Only try it when **both** conclusions already failed. Then check just two things: same two terms? and one of the three pairs (All / Some-not), (No / Some), (Some / Some-not)? If yes, write “Either I or II”. If the terms differ, stop — it is “Neither follows”.

Example 14 [TCS NQT pattern]

Statements: All fans are lights. No light is a switch.

Conclusions: I. No fan is a switch. II. Some switches are not lights.

SOLUTION

I: fans $\subset$ lights, lights $\cap$ switches = $\emptyset$. So fans $\cap$ switches = $\emptyset$. I follows. II: “No light is a switch” flips to “No switch is a light”. Switches exist, so take one. It is a switch and it is not a light. II follows.

Answer: Both I and II follow.

Example 15 [Cognizant pattern]

Statements: All cars are vehicles. All vehicles are machines. Some machines are toys.

Conclusions: I. Some cars are toys. II. All cars are machines.

SOLUTION

II first, because it is a clean chain: cars $\subset$ vehicles $\subset$ machines. So all cars are machines. II follows.

I: the toy overlap sits somewhere in the machine circle. Draw it in the part of machines that is outside vehicles.

All three statements still hold and no car is a toy. I fails.

Answer: Only II follows.

Example 16 [Infosys pattern]

Statements: All apples are fruits. Some fruits are sweet.

Conclusions: I. All apples being sweet is a possibility. II. Some apples are sweet.

SOLUTION

II was killed in Example 13. It does not follow.

I is a **possibility** conclusion. I only need one legal diagram. Put the whole apple circle inside the sweet region, and put the sweet region inside fruits. Check: all apples are fruits — yes. Some fruits are sweet — yes, the apples are. Nothing is broken, so the possibility is real.

Answer: Only I follows.

SHORTCUT

Possibility questions are the easy ones. A normal conclusion needs to survive every diagram. A possibility conclusion needs only one diagram. So ask the opposite question: “do the statements **forbid** this?” If not forbidden, it is possible.

Example 17 [Accenture pattern]

Statements: No book is a pen. All pens are inks.

Conclusions: I. Some inks are not books. II. No ink is a book.

SOLUTION

Pens exist. Every pen is an ink. No pen is a book (flip of “No book is a pen”). So take a pen: it is an ink and it is not a book. I follows.

II: inks may also include things that are books. Draw the ink circle big enough to swallow some books while keeping pens away from books. Both statements hold, II fails.

Answer: Only I follows.

Example 18 [TCS NQT pattern]

Statements: Some keys are locks. Some locks are doors.

Conclusions: I. Some keys are doors. II. No key is a door.

SOLUTION

Two “some” statements, so neither conclusion can be forced. Diagram 1: keys touch the left of locks, doors touch the right of locks, keys and doors apart. Then I is false. Diagram 2: keys, locks and doors all overlap in one shared patch. Then II is false.

Either–or test: I is “Some keys are doors” (I type), II is “No key is a door” (E type), same two terms. That is a complementary pair.

Answer: Either I or II follows.

Example 19 [Wipro pattern]

Statements: All singers are dancers. Some dancers are actors. No actor is a writer.

Conclusions: I. Some dancers are not writers. II. Some singers are writers.

SOLUTION

I: take a dancer who is an actor (exists, from statement 2). No actor is a writer, so this dancer is not a writer. So some dancers are not writers. I follows.

II: singers sit inside dancers, but nothing pushes them into writers. Draw writers touching only the dancer region that has no singers. All three statements hold and no singer is a writer. II fails.

Answer: Only I follows.

TRAP

The chain-break trap. “All singers are dancers” plus “Some dancers are actors” is All + Some. The combination table says: **no conclusion**. The “some” part of dancers may miss the singers completely. Students who read left-to-right and say “so some singers are actors” lose this mark every time.

PRACTICE · TIER 1 target: 60 s/Q

In each question, take the statements as true even if they differ from real facts. Decide which conclusion(s) follow.

1. **Statements:** All cups are mugs. All mugs are jars.
Conclusions: I. All cups are jars. II. Some jars are cups.
2. **Statements:** All dogs are pets. No pet is wild.
Conclusions: I. No dog is wild. II. Some wild things are pets.
3. **Statements:** Some pens are red. All red things are bright.
Conclusions: I. Some pens are bright. II. All pens are bright.
4. **Statements:** Some files are folders. No folder is a disk.
Conclusions: I. Some files are not disks. II. No file is a disk.
5. **Statements:** All trains are buses. Some buses are taxis.
Conclusions: I. Some trains are taxis. II. Some trains are not taxis.
6. **Statements:** No shirt is a shoe. All shoes are leather.
Conclusions: I. Some leather things are not shirts. II. No leather thing is a shirt.
7. **Statements:** All bricks are stones. All stones are hard.
Conclusions: I. All bricks are hard. II. Some hard things are bricks.
8. **Statements:** Some birds are crows. Some crows are black.
Conclusions: I. Some birds are black. II. No bird is black.
9. **Statements:** All keys are metals. Some metals are gold.
Conclusions: I. Some keys are gold. II. All keys being gold is a possibility.
10. **Statements:** All fruits are sweet. No sweet thing is sour. Some sour things are bitter.
Conclusions: I. No fruit is sour. II. Some bitter things are not sweet.
11. **Statements:** Some cards are paper. All paper is thin.
Conclusions: I. Some cards are thin. II. Some thin things are cards.
12. **Statements:** All lamps are bulbs. Some bulbs are LEDs. No LED is cheap.
Conclusions: I. Some bulbs are not cheap. II. Some lamps are not cheap.
13. **Statements:** Only a few phones are cameras. All cameras are lenses.
Conclusions: I. Some phones are lenses. II. Some phones are not cameras.
14. **Statements:** No river is a lake. Some lakes are ponds.
Conclusions: I. Some ponds are not rivers. II. All ponds being rivers is a possibility.

15. **Statements:** All servers are machines. All machines are assets. Some assets are leased.
Conclusions: I. Some servers are leased. II. All servers are assets.

ANSWER KEY

1. **Both follow.** Chain cups $\subset$ mugs $\subset$ jars gives I; flip of I gives II.
2. **Only I.** Dogs sit inside pets, pets avoid wild, so dogs avoid wild. II contradicts the second statement.
3. **Only I.** Some + All = Some. II needs all pens inside red, which is not given.
4. **Only I.** Some + No = Some-not. Files outside folders may still be disks, so II fails.
5. **Either I or II.** All + Some gives nothing; I and II are the (Some / Some-not) pair on trains and taxis.
6. **Only I.** Shoes are leather and no shoe is a shirt, so some leather is not shirt. Leather may also contain shirts, so II fails.
7. **Both follow.** Chain gives I; flip of I gives II.
8. **Either I or II.** Two “some” statements force nothing; I and II are the (Some / No) pair on birds and black.
9. **Only II.** Keys need not touch gold, so I fails. Putting keys inside gold inside metals breaks nothing, so the possibility stands.
10. **Both follow.** Fruits inside sweet, sweet avoids sour, so I. The bitter-and-sour things are not sweet, so II.
11. **Both follow.** Some + All = Some cards are thin; flipping gives II.
12. **Only I.** The LED-bulbs are not cheap, so I. Lamps may sit in the non-LED part of bulbs, so II fails.
13. **Both follow.** “Only a few phones are cameras” = some are + some are not. Some phones are cameras, and all cameras are lenses, so I; the second half gives II.
14. **Only I.** Pond-lakes are not rivers, so I. Those same ponds can never be rivers, so “all ponds are rivers” is impossible and II fails.
15. **Only II.** Chain servers $\subset$ machines $\subset$ assets gives II. The leased part of assets may avoid servers, so I fails.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 20 [Grab · pattern]

Statements: All drivers are partners. Some partners are owners. All owners are insured.

Conclusions: I. Some partners are insured. II. Some drivers are insured.

SOLUTION

Work on the pair that actually links up: “Some partners are owners” + “All owners are insured” is Some + All = Some. So some partners are insured. I follows.

II needs drivers to reach insured. Drivers sit inside partners, but the owner-overlap can sit in the partner region that has no drivers. Legal diagram: partners is a big circle, drivers a small circle on the left, owners a small circle on the right, insured wrapped around owners. Every statement is true and no driver is insured. II fails.

Answer: Only I follows.

Example 21 [Shopee · pattern]

Statements: Some riders are students. All students are young.

Conclusions: I. All riders being young is a possibility. II. Some riders are young.

SOLUTION

II: Some + All = Some. The rider-students are young, so some riders are young. II follows.

I: possibility test. Draw the rider circle fully inside “young”, with the student circle also inside “young” and overlapping riders. Some riders are students — true. All students are young — true. Nothing is broken, so the possibility holds. I follows.

Answer: Both I and II follow.

Example 22 [DBS · pattern]

Statements: No merchant is a courier. All couriers are staff. Some staff are managers.

Conclusions: I. Some staff are not merchants. II. Some managers are not merchants. III. All merchants being staff is a possibility.

SOLUTION

I: couriers exist, every courier is staff, and no courier is a merchant (flip of statement 1). So take a courier: staff, not merchant. I follows.

II: managers are a slice of staff. Nothing stops that slice from lying fully inside merchants — merchants only have to avoid couriers, and merchants are allowed to be staff. Diagram: staff contains couriers on the left and a merchant-manager block on the right. Every statement holds, and every manager is a merchant. II fails.

III: the same diagram already puts all merchants inside staff. So it is possible. III follows.

Answer: I and III follow.

SHORTCUT

Reuse one diagram for several conclusions. If a single legal diagram kills a “must-follow” conclusion, that very diagram usually **proves** the matching possibility conclusion. In Example 22 the picture that broke II is the picture that confirmed III. One drawing, two answers.

Example 23 [Agoda · pattern]

Statements: All apps are tools. No tool is free. Some free items are useful.

Conclusions: I. No app is free. II. Some useful things are not tools.

SOLUTION

I: All + No = No. Apps sit inside tools, tools avoid free, so apps avoid free. I follows.

II: “No tool is free” flips to “No free item is a tool”. Statement 3 gives an item that is free and useful. That item is free, so it is not a tool. So it is useful and not a tool. II follows.

Answer: Both I and II follow.

Example 24 [Sea · pattern]

Statements: Some sellers are vendors. All vendors are verified. No verified account is banned.

Conclusions: I. Some sellers are not banned. II. All sellers being verified is a possibility. III. Some verified accounts are sellers.

SOLUTION

Follow one element through the chain. There is a seller who is a vendor (statement 1). Every vendor is verified, so that seller is verified. No verified account is banned, so that seller is not banned.

I: that seller is a seller and not banned. I follows. III: that seller is verified and is a seller. So some verified accounts are sellers. III follows. II: possibility. Push the whole seller circle inside verified. Statement 1 still holds (the vendor-sellers are still there), statement 2 untouched, statement 3 untouched. Possible. II follows.

Answer: I, II and III all follow.

Example 25 [SCB · pattern]

Statements: All buses are trains. Some trains are cars. All cars are jeeps.

Conclusions: I. Some buses are jeeps. II. No bus is a jeep.

SOLUTION

I alone: put the car-overlap in the part of trains with no buses, and put jeeps around cars only. Then no bus is a jeep, so I fails.

II alone: now let some buses also be cars. Buses are still inside trains — fine. Some trains are cars — fine. All cars are jeeps — fine. Those buses are jeeps, so II fails.

Both failed. Either–or test: I is “Some buses are jeeps”, II is “No bus is a jeep”. Same two terms, and they are the (Some / No) pair.

Answer: Either I or II follows.

Example 26 [GIC · pattern]

Reverse syllogism. Which pair of statements makes **both** conclusions definitely true?

Conclusions: I. Some pilots are not cadets. II. Some cadets are officers.

- (a) All pilots are cadets. All cadets are officers.
- (b) All cadets are officers. No pilot is an officer.
- (c) Some pilots are cadets. No cadet is an officer.
- (d) All officers are cadets. Some pilots are officers.

SOLUTION

Test (a): all pilots are cadets, so “some pilots are not cadets” is false. Reject.

Test (b): “All cadets are officers” flips to “Some officers are cadets”, which is the same as “Some cadets are officers”. II holds. For I: no pilot is an officer, and every cadet is an officer. So a pilot can never be a cadet, i.e. no pilot is a cadet. Pilots exist, so some pilots are not cadets. I holds. Keep (b).

Test (c): no cadet is an officer, so II is false. Reject.

Test (d): all officers are cadets gives II. But some pilots are officers, and officers are cadets, so those pilots **are** cadets; nothing forces any pilot to sit outside cadets. I is not guaranteed. Reject.

Answer: (b) All cadets are officers. No pilot is an officer.

Example 27 [LINE MAN · pattern]

Statements: Only a few engineers are managers. All managers are leaders.

Conclusions: I. Some engineers are not managers. II. Some engineers are leaders. III. All engineers being leaders is a possibility.

SOLUTION

First translate. “Only a few engineers are managers” means two things at once: Some engineers are managers, **and** Some engineers are not managers.

I is the second half, given directly. I follows. II: some engineers are managers, all managers are leaders. Some + All = Some. II follows. III: possibility. The engineers who are not managers are free to be leaders anyway. Draw the whole engineer circle inside leaders, with the manager circle also inside leaders and cutting engineers partly. Both statements survive. III follows.

Answer: I, II and III all follow.

TRAP

“Only a few” is not “a few”. It carries a hidden negative half. “Only a few X are Y” = Some X are Y AND Some X are not Y. Students who use only the positive half lose the free “Some X are not Y” mark. And “Only X are Y” is different again — it means **All Y are X**.

Example 28 [Razer · pattern]

Statements: Some contracts are leases. All leases are documents. No document is verbal.

Conclusions: I. Some contracts are not verbal. II. No contract is verbal. III. Some documents are contracts.

SOLUTION

Chain one element. A contract that is a lease exists. It is a document (statement 2). It is not verbal (statement 3).

I: that contract is not verbal. I follows. III: that same thing is a document and a contract. III follows. II: the contracts that are **not** leases have no restriction on them. Draw half of the contract circle inside “verbal”, far from documents. All three statements hold and II is false. II fails.

Answer: I and III follow.

PRACTICE · TIER 2 target: 100 s/Q

No options. Write which conclusions follow, and for possibility conclusions say possible or impossible.

1. **Statements:** All riders are partners. Some partners are drivers. All drivers are insured.
Conclusions: I. Some partners are insured. II. Some riders are insured. III. All riders being insured is a possibility.
2. **Statements:** Some orders are refunds. No refund is a coupon. All coupons are discounts.
Conclusions: I. Some orders are not coupons. II. Some discounts are not refunds. III. No order is a coupon.
3. **Statements:** Only a few merchants are exporters. All exporters are licensed.
Conclusions: I. Some merchants are licensed. II. Some merchants are not exporters. III. All merchants being licensed is a possibility.
4. **Statements:** All wallets are accounts. Some accounts are frozen. No frozen account is active.
Conclusions: I. Some accounts are not active. II. Some wallets are not active. III. Some accounts are not frozen.
5. **Statements:** No courier is a manager. All managers are staff. Some staff are trainees.
Conclusions: I. Some staff are not couriers. II. All couriers being staff is a possibility. III. Some trainees are not couriers.
6. **Statements:** All apps are tools. Some tools are paid. All paid items are licensed.
Conclusions: I. Some tools are licensed. II. Some apps are licensed. III. No app is licensed.
7. **Statements:** All flights are journeys. No journey is free. Some free things are useful.
Conclusions: I. No flight is free. II. Some useful things are not journeys. III. Some journeys are useful.
8. **Statements:** Some hotels are resorts. All resorts are properties. Some properties are rented.
Conclusions: I. Some hotels are properties. II. Some hotels are rented. III. Some properties are hotels.
9. **Statements:** All bookings are records. Some records are cancelled.
Conclusions: I. Some bookings are cancelled. II. No booking is cancelled. III. All records being bookings is a possibility.
10. **Statements:** No vendor is a buyer. Some buyers are agents. All agents are members.
Conclusions: I. Some members are not vendors. II. Some agents are not vendors. III. All vendors being members is a possibility.
11. **Statements:** Only a few cards are credit cards. No credit card is free.
Conclusions: I. Some cards are not free. II. Some cards are not credit cards. III. All cards being free is a possibility.

ANSWER KEY

1. **I and III.** Some partners are drivers and all drivers are insured, so I. Riders may sit in the driver-free part of partners, so II fails; but pushing riders inside insured breaks nothing, so III is possible.
2. **I and II.** The order-refunds are not coupons, so I. Coupons are discounts and are not refunds, so II. Non-refund orders may be coupons, so III fails.
3. **All three.** “Only a few” gives II directly; its positive half plus “all exporters are licensed” gives I; the non-exporter merchants may also be licensed, so III is possible.
4. **Only I.** Frozen accounts are not active, so I. Wallets may avoid the frozen slice, so II fails. Every account, wallets included, may be frozen, so III is not forced.
5. **I and II.** Managers are staff, and no manager is a courier, so some staff are not couriers, giving I. Couriers are barred only from being managers, so every courier may still sit inside staff, which makes II possible. Trainees are staff, and nothing stops all of them being couriers, so III fails.
6. **I follows, and either II or III follows.** Paid tools are licensed, so I. Apps may or may not touch the licensed region, so neither II nor III is forced; they are the (Some / No) pair on apps and licensed.
7. **I and II.** Flights sit inside journeys and journeys avoid free, so I. Free things are not journeys, and some free things are useful, so II. Journeys and useful things need not meet, so III fails.

8. **I and III.** The hotel-resorts are properties, so I, and flipping gives III. The rented slice of properties may avoid hotels, so II fails.
9. **III follows, and either I or II follows.** All + Some forces nothing about bookings and cancelled, and I, II are the (Some / No) pair. Setting records = bookings keeps both statements true, so III is possible.
10. **All three.** The buyer-agents are not vendors, so II; they are members too, so I. Vendors are only barred from being buyers, so they may all be members, giving III.
11. **I and II.** “Only a few” gives II; its positive half plus “no credit card is free” gives I. Those same credit cards can never be free, so III is impossible.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 29 [Google · pattern]

Statements: All P are Q. Some Q are not R.

Question: Does “Some P are not R” follow? Prove your answer.

SOLUTION

Insight: the “some Q” in statement 2 need not be any of the P’s. The two statements touch Q in possibly disjoint places.

Build a counterexample with real sets. Let $P = \{1\}$, $Q = \{1, 2\}$, $R = \{1\}$.

Check statement 1: is every member of P in Q? $1 \in Q$. Yes. Check statement 2: is some member of Q outside R? $2 \in Q$ and $2 \notin R$. Yes. Check the conclusion: is some member of P outside R? P has only 1, and $1 \in R$. No.

Both statements are true and the conclusion is false, so it cannot follow.

Note also what rule 1 says: the middle term is Q. Statement 1 “All P are Q” covers all of P but not all of Q. Statement 2 “Some Q are not R” covers all of R but not all of Q. Q is never covered fully, so no valid conclusion exists at all.

Answer: It does not follow.

Example 30 [Amazon · pattern]

Statements: No P is Q. All Q are R.

Question: List every valid conclusion that links P and R.

SOLUTION

Insight: the Q block is a wedge that lives inside R and is banned from P. Only R can “see” that wedge, so only R-first conclusions can survive.

Q is non-empty, so pick $q \in Q$. $q \in R$ (statement 2). $q \notin P$ (flip of statement 1: no Q is P). So q is an R that is not a P. **“Some R are not P” is valid.**

Now test the other candidates one by one.

“No P is R”: counterexample — put P inside R but outside Q. No P is Q: true. All Q are R: true. But every P is R. Fails.

“Some P are not R”: same diagram, P sits fully inside R. Fails.

“All R are P”: Q is inside R and Q avoids P, so R has members outside P. Fails.

“Some P are R”: counterexample — put P completely outside R. No P is Q: true (Q is inside R). All Q are R: true. No P is R. Fails.

So only one conclusion survives, plus the things it is equivalent to.

Answer: “Some R are not P” — and nothing else. (It does not flip into a P-first form.)

Example 31 [Microsoft · pattern]

Statements: All X are Y. Some Y are Z.

Question: Which of these are possible?

- (a) All Z are X (b) No Z is X (c) All Y are Z (d) Some X are not Y

SOLUTION

Insight: for a possibility, you do not argue — you construct. Build the most extreme legal picture you can and see if the statements survive.

(a) All Z are X. Construct: let $Z \subseteq X \subseteq Y$, with Z non-empty. All X are Y: true. Some Y are Z: the Z members are in X, hence in Y. True. Survives. **Possible.**

(b) No Z is X. Construct: Y is a big circle. X sits on the left of Y. Z sits on the right, half inside Y, half outside, never touching X. All X are Y: true. Some Y are Z: true (the inside half). No Z is X: true. Survives. **Possible.**

(c) All Y are Z. Construct: let $X \subseteq Y = Z$. All X are Y: true. Some Y are Z: true (all of them). Survives. **Possible.**

(d) Some X are not Y. This directly contradicts statement 1, which says every X is in Y. A statement given as true cannot be contradicted. **Impossible.**

Answer: (a), (b) and (c) are possible; (d) is impossible.

SHORTCUT

The possibility killer list. A possibility conclusion is impossible in exactly two situations, and no others:

1. it contradicts a given statement outright (“Some X are not Y” against “All X are Y”), or
2. it contradicts a conclusion that **definitely** follows (“All ponds are rivers” when some ponds have already been proved not to be rivers).

Everything else is possible. Do not hunt for more.

Example 32 [Goldman Sachs · pattern]

Reverse syllogism, both directions. Which single pair of statements forces **both** “Some A are not C” and “Some C are not A”?

- (a) Some A are B; No B is C (b) All A are B; No B is C
(c) All B are A; No B is C (d) Some A are B; All B are C

SOLUTION

Insight: a “some-not” in both directions needs the two circles to be fully separated. Only a pair that forces “No A is C” can deliver that.

(a) Some A are B, No B is C. The A-part of B avoids C, so **some A are not C** — good. But the rest of A is free to sit inside C, and C may be fully inside A’s remainder, so “Some C are not A” is not forced. Reject.

(b) All A are B, No B is C. All + No = **No A is C**. From “No A is C”: A is non-empty, so some A are not C. First conclusion holds. Flip it: No C is A. C is non-empty, so some C are not A. Second conclusion holds. Keep.

(c) All B are A, No B is C. B is inside A and avoids C, so some A are not C — good. But C may sit entirely inside A (just outside B), so “Some C are not A” fails. Reject.

(d) Some A are B, All B are C. This gives “Some A are C”, a positive conclusion. Nothing negative is forced. Reject.

Answer: (b) All A are B; No B is C.

Example 33 [Adobe · pattern]

Statements: Some singers are dancers. Some dancers are actors. Some actors are writers.

Question: How many of these four must be true?

- I. Some singers are actors II. Some dancers are writers III. Some singers are writers IV. Some actors are dancers

SOLUTION

Insight: every statement here is an “I” type. Rule 4 says two particular statements give nothing. So the only true conclusions are flips of the statements themselves, not new links.

Build one single diagram that breaks as much as possible. Place four circles in a row: singers — dancers — actors — writers. Let singers overlap dancers only on the far left of dancers. Let dancers overlap actors only on the far right of dancers. Let actors overlap writers only on the far right of actors. Give the overlaps no shared members at all.

Check the statements in this diagram: some singers are dancers — yes. Some dancers are actors — yes. Some actors are writers — yes. Diagram is legal.

I: singers touch only dancers here, never actors. I is false in this diagram. Not forced. II: dancers touch actors, but the actor-writer overlap is elsewhere. False here. Not forced. III: singers are three steps from writers. False here. Not forced. IV: this is just “Some dancers are actors” flipped. An I-type statement always flips. Always true.

Answer: Exactly one — only IV.

Example 34 [Uber · pattern]

Chain problem. Statements: All A_1 are A_2 . All A_2 are A_3 All A_9 are A_{10} . Also: No A_{10} is B.

Which follow? I. No A_3 is B II. Some A_{10} are A_2 III. No B is A_1 IV. Some A_1 are not A_5

SOLUTION

Insight: nine “All” statements nest into one another, so the whole tower collapses to $A_1 \subseteq A_2 \subseteq \dots \subseteq A_{10}$. After that, one negative statement at the top wipes out B for every level.

I: $A_3 \subseteq A_4 \subseteq \dots \subseteq A_{10}$, so every A_3 is an A_{10} . No A_{10} is B, so no A_3 is B. **Follows.**

II: $A_2 \subseteq A_{10}$, and A_2 is non-empty. Take $a \in A_2$. Then $a \in A_{10}$ and $a \in A_2$. So some A_{10} are A_2 . **Follows.**

III: “No A_{10} is B” flips to “No B is A_{10} ”. $A_1 \subseteq A_{10}$, so anything in A_1 is in A_{10} , and B avoids all of A_{10} . So no B is A_1 . **Follows.**

IV: $A_1 \subseteq A_5$ by the chain, so **every** A_1 is an A_5 . “Some A_1 are not A_5 ” is therefore false, not merely unproved. **Fails.**

Answer: I, II and III follow. IV does not.

Example 35 [D. E. Shaw · pattern]

Statements: Some A are B. All B are C. Some C are D. No D is A.

Conclusions: I. Some C are not D II. Some B are not D III. Some C are A IV. No B is D

SOLUTION

Insight: pick the one element that all four statements can talk about — a thing that is both A and B. Statement 4 protects it from D, statement 2 pushes it into C. Everything true here is true because of that single element.

Statement 1 gives an element x with $x \in A$ and $x \in B$. Statement 2: $x \in B \subseteq C$, so $x \in C$. Statement 4: No D is A, and $x \in A$, so $x \notin D$.

So x is at once: A, B, C, and not D.

I: $x \in C$ and $x \notin D$. So some C are not D. **Follows.** II: $x \in B$ and $x \notin D$. So some B are not D. **Follows.** III: $x \in C$ and $x \in A$. So some C are A. **Follows.** IV: this claims **no** B is D. Only the A-part of B is protected. Counter-diagram: let $B = \{x, y\}$ where y is in B and in D but not in A. Some A are B: true (x). All B are C: put both x, y inside C — true. Some C are D: true (y). No D is A: true (y is not in A). Yet y is a B that is a D. **Fails.**

Answer: I, II and III follow. IV does not.

TRAP

“Some are not” versus “No”. In Example 35, conclusions II and IV look like the same idea in different words. They are not. II needs **one** safe element; IV needs **every** element to be safe. Almost every hard syllogism hides this gap. When a conclusion starts with “No”, ask: have I protected the **whole** circle, or only the part I traced?

PRACTICE • TIER 3 target: 4 min/Q

1. Statements: All A are B. Some B are not C. Does “Some A are not C” follow? Give a counterexample with real sets, or a proof.
2. Statements: No P is Q. All Q are R. Write down every conclusion that links P and R, and show why each rejected candidate fails.
3. Statements: Some X are Y. All Y are Z. No Z is W. Which are possible? (a) All X are W (b) Some X are W (c) All Z are Y (d) No X is Z
4. Which single pair forces both “Some A are not C” and “Some C are not A”? (a) All B are A; No B is C (b) Some A are B; No B is C (c) No A is B; All B are C (d) All A are B; No B is C
5. Statements: Some singers are dancers. Some dancers are actors. Some actors are writers. How many of the following must be true? I. Some writers are actors. II. Some dancers are singers. III. Some singers are writers. IV. Some actors are singers.
6. Statements: All B_1 are B_2 , All B_2 are B_3 , ..., All B_7 are B_8 , and Some B_8 are K. Which follow? I. Some B_1 are K. II. All B_1 are B_8 . III. Some B_8 are B_4 . IV. All B_8 being K is a possibility.
7. Statements: All P are Q. Some Q are R. No R is S. All S are T. Which follow? I. Some Q are not S. II. Some T are not R. III. All P being R is a possibility. IV. Some P are not S.

ANSWER KEY

1. **Does not follow.** Take $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$. All A are B holds; 2 is a B that is not a C, so statement 2 holds; but the only A, namely 1, is a C. The middle term B is never fully covered, so by rule 1 no conclusion is possible at all.
2. **Only “Some R are not P”.** Q is non-empty; each Q is an R (statement 2) and no Q is a P (flip of statement 1), so that Q is an R outside P. “No P is R” and “Some P are not R” both die to the diagram with P inside R but outside Q. “Some P are R” dies to the diagram with P wholly outside R. “All R are P” dies because the Q-wedge is R but not P.
3. **(b) and (c) only.** (a) is impossible: some X are Y, all Y are Z, and no Z is W, so those X are not W — “all X are W” is blocked. (b) is possible: the X’s outside Y may be W. (c) is possible: set $Z = Y$; both remaining statements still hold. (d) is impossible: the X’s that are Y are Z, so X and Z must touch.
4. **(d) All A are B; No B is C.** It forces “No A is C”, and a full separation gives a “some-not” in both directions. (a) gives “Some A are not C” but lets C sit entirely inside A. (b) also gives “Some A are not C” only, since C may sit entirely inside the non-B part of A. (c) gives “Some C are not A” but not the reverse, since all of A may lie inside C.
5. **Exactly two — I and II.** Both are simple flips of given “some” statements, and an I-type statement always flips. III and IV need a link across two steps, and two particular statements give nothing; the row-of-circles diagram with disjoint overlaps kills them.
6. **II, III and IV.** The chain gives $B_1 \subseteq B_2 \subseteq \dots \subseteq B_8$, so II holds; $B_4 \subseteq B_8$ and B_4 is non-empty, so some B_8 are B_4 , giving III. I fails because the K-overlap may sit in the part of B_8 outside B_1 . IV is possible: putting all of B_8 inside K breaks nothing.
7. **I, II and III.** I: some Q are R and no R is S, so those Q are not S. II: S is non-empty, every S is T, and no S is R (flip of statement 3), so some T are not R. III: put P inside the $Q \cap R$ overlap — all four statements survive, so it is possible. IV fails: put P inside S inside Q, with R a separate slice of Q; then every P is an S.

Mixed set — exam conditions

PRACTICE • TIER 1 target: 25 min for 18 questions

Take the statements as true. State which conclusion(s) follow.

1. **Statements:** All chairs are wooden. All wooden things are heavy.
Conclusions: I. All chairs are heavy. II. Some heavy things are chairs.
2. **Statements:** Some cats are dogs. No dog is a rat.
Conclusions: I. Some cats are not rats. II. No cat is a rat.
3. **Statements:** All gold is metal. Some metal is costly.
Conclusions: I. Some gold is costly. II. No gold is costly.
4. **Statements:** No pen is a pencil. All pencils are erasers.
Conclusions: I. Some erasers are not pens. II. All pens being erasers is a possibility.
5. **Statements:** Some laptops are tablets. All tablets are devices.
Conclusions: I. Some laptops are devices. II. All laptops are devices.
6. **Statements:** All coders are testers. Some testers are leads. No lead is an intern.
Conclusions: I. Some testers are not interns. II. Some coders are not interns.
7. **Statements:** Only a few papers are journals. All journals are indexed.
Conclusions: I. Some papers are indexed. II. All papers are indexed.
8. **Statements:** All A are B. No B is C.
Conclusions: I. No A is C. II. No C is A. III. Some C are not A.
9. **Statements:** Some roads are lanes. Some lanes are streets.
Conclusions: I. Some roads are streets. II. No road is a street.
10. **Statements:** All shops are stores. Some stores are malls. All malls are big.
Conclusions: I. Some stores are big. II. Some shops are big.
11. **Statements:** No fan is a cooler. Some coolers are air conditioners.
Conclusions: I. Some air conditioners are not fans. II. All air conditioners being fans is a possibility.
12. **Statements:** All X are Y. All Y are Z. Some Z are W.
Conclusions: I. Some X are W. II. All X are Z. III. All Z being X is a possibility.
13. **Statements:** Some pens are inks. All inks are fluids. No fluid is solid.
Conclusions: I. Some pens are not solids. II. Some fluids are pens.
14. **Statements:** All P are Q. Some Q are not R.
Conclusions: I. Some P are not R. II. All P are R.
15. **Statements:** Some books are novels. All novels are stories. Some stories are long.
Conclusions: I. Some books are stories. II. Some books are long.
16. **Statements:** All buses are heavy. No heavy thing is cheap. Some cheap things are useful.
Conclusions: I. No bus is cheap. II. Some useful things are not heavy.
17. **Statements:** Only a few staff are interns. No intern is permanent.
Conclusions: I. Some staff are not permanent. II. All staff being permanent is a possibility.
18. **Statements:** All A are B. Some B are C. No C is D.
Conclusions: I. Some B are not D. II. Some A are not D. III. All A being C is a possibility.

ANSWER KEY

1. **Both.** Chain gives I; flipping I gives II.
2. **Only I.** The cat-dogs are not rats. Cats outside dogs may be rats, so II fails.
3. **Either I or II.** All + Some forces nothing; I and II are the (Some / No) pair on gold and costly.
4. **Both.** Pencils are erasers and are not pens, so I. Pens are only barred from being pencils, so they may all be erasers, giving II.
5. **Only I.** Some + All = Some. The laptops outside tablets need not be devices, so II fails.
6. **Only I.** The tester-leads are not interns, so I. Coders may sit in the lead-free part of testers, so II fails.
7. **Only I.** "Only a few" gives some papers are journals, hence indexed. Its negative half says some papers are not journals, so "all papers are indexed" is not forced.
8. **All three.** All + No = No A is C, giving I; flipping gives II; C is non-empty so III follows from II.
9. **Either I or II.** Two "some" statements force nothing, and the two conclusions are the (Some / No) pair on roads and streets.

10. **Only I.** The mall-stores are big, so I. Shops may avoid the mall slice, so II fails.
11. **Only I.** The cooler-ACs are not fans, so I. Those same ACs can never be fans, so II is impossible.
12. **II and III.** Chain X inside Y inside Z gives II. The W overlap may miss X , so I fails. Setting $X = Y = Z$ keeps every statement true, so III is possible.
13. **Both.** The pen-inks are fluids, hence not solids, giving I; those same things are fluids and pens, giving II.
14. **Either I or II.** Neither is forced (put the non- R part of Q outside P for II to fail, and put P inside R for I to fail), and they are the (All / Some-not) pair on P and R .
15. **Only I.** The book-novels are stories, so I. The “long” slice of stories may miss the books, so II fails.
16. **Both.** Buses sit inside heavy and heavy avoids cheap, so I. The cheap-and-useful things are not heavy, so II.
17. **Only I.** Some staff are interns and no intern is permanent, so I. Those same staff can never be permanent, so II is impossible.
18. **I and III.** The B ’s that are C are not D , giving I. A may avoid that slice, so II fails. Pushing all of A into the $B \cap C$ overlap breaks nothing, so III is possible.

ONE-PAGE REVISION CARD

THE FOUR TYPES

Code	Form	Flips to
A	All S are P	Some P are S
E	No S is P	No P is S; Some S are not P; Some P are not S
I	Some S are P	Some P are S
O	Some S are not P	nothing

COMBINATION TABLE (A links to B, B links to C)

All A are B	All B are C	All A are C
All A are B	No B is C	No A is C
Some A are B	All B are C	Some A are C
Some A are B	No B is C	Some A are not C
No A is B	All B are C	Some C are not A
All A are B	Some B are C	nothing
Some A are B	Some B are C	nothing

THE FIVE CHECKS — middle term must be fully covered once; two negatives give nothing; one negative $\Rightarrow$ negative conclusion; two “some” give nothing; one “some” $\Rightarrow$ “some” conclusion.

EITHER–OR — allowed only when both conclusions fail alone AND they form (All / Some-not), (No / Some) or (Some / Some-not) on the **same two terms**.

POSSIBILITY — true unless it contradicts a statement or contradicts a conclusion that definitely follows. Construct one legal diagram, do not argue.

NEW-PATTERN WORDS — “Only a few A are B” = Some are + Some are not. “Only A are B” = All B are A. “At least some” = Some.

SHORTCUTS

1. Trace one element through the chain instead of drawing the whole picture.
2. One counter-diagram kills a “must-follow” conclusion — and usually proves the matching possibility conclusion. Reuse the drawing.
3. For reverse syllogism, run each option **forward** through the combination table.
4. “No” conclusions need the whole circle protected, not just the traced element.

TOP 5 TRAPS

1. Flipping “All A are B” into “All B are A”. Only “Some B are A” is allowed.
2. Trying to flip “Some A are not B”. It flips to nothing.
3. All + Some (in that order) — students force a conclusion where there is none.
4. Writing “Either I or II” when the two conclusions use different terms.
5. Marking a possibility conclusion false just because it is not proved. Possibility only needs to be **not forbidden**.

Critical Reasoning

Find the gap between the evidence and the claim.

What you need to know

WHAT YOU NEED TO KNOW

Every argument has three parts.

- **Evidence** — the facts given.
- **Conclusion** — the claim being pushed. Look for: so, hence, therefore, thus, clearly, this shows, we must.
- **Gap** — the unstated jump from evidence to conclusion. Every question in this chapter attacks the gap.

The seven question types and the one test for each.

Type	Test
Statement–Assumption	Is it needed ? Negate it. If the argument collapses, it is an assumption.
Statement–Conclusion	Can it be got from the given words alone ? No outside knowledge, no guessing.
Statement–Inference	Is it probably true given the facts, without over-reaching beyond the data set?
Statement–Argument	Strong = relevant + substantial + tied to the exact question. Weak otherwise.
Course of Action	Three filters: does it hit the cause , is it practical , is it inside the actor's power ?
Cause and Effect	Which event came first and could stand alone? That one is the cause.
Strengthen / Weaken	Find the gap. Strengthen = close it or rule out a rival cause. Weaken = supply a rival cause or a counter-case.

The negation test, in three steps.

1. Write the candidate assumption.
2. Flip it to its opposite (“some” → “none”, “will” → “will not”).
3. Re-read the statement. If the statement now makes no sense, the candidate **is** an assumption. If the statement still stands, it is not.

Assumption versus inference.

	Assumption	Inference
Where it sits	Before the statement	After the statement
Status	Taken for granted, unsaid	Drawn out, follows
Question it answers	Why does this make sense?	What else must be true now?

Strong argument checklist. An argument is **weak** if it is any of these: vague, trivial, a single personal example, a re-statement of the question, a pure appeal to emotion, based on a false comparison, or about a side issue.

Course of action — reject if: it is extreme (ban / shut down / expel / shift everyone), it costs far more than the cheaper fix that is also on offer, it is outside the actor's power, or it treats a symptom while ignoring a named cause.

Cause and effect — the four labels.

1. I is the cause and II is its effect.
2. II is the cause and I is its effect.
3. Both are effects of one common cause.
4. Both are independent causes.

Percent versus absolute. Growth rate $\times$ base = the actual increase. A 40% rise on a base of 120 adds 48. A 10% rise on a base of 900 adds 90. The smaller rate can add much more.

Before-and-after averages. An average can rise with nobody improving, if the low members were removed from the group. Always ask: **did the group itself change?**

Warm-up

WARM-UP

build the reflex

Example 1

Split into evidence and conclusion: “Sales fell 20% right after the price rise. So the price rise caused the fall.”

SOLUTION

The word “So” marks the conclusion. Evidence: sales fell 20% after the price rise. Conclusion: the price rise caused the fall. Gap: something else in that period could have cut sales.

Answer: Evidence = sales fell after the price rise. Conclusion = the price rise caused it.

Example 2

Advertisement: “Buy the Shakti helmet — it keeps you safe.” Is “people want to be safe” an assumption?

SOLUTION

Negate it: “People do not want to be safe.” Then a helmet that keeps you safe is not a selling point, and the advertisement is pointless. The argument collapses.

Answer: Yes, it is an assumption.

Example 3

Statement: “The school has started evening classes for weak students.” Label each as assumption or inference: (i) Some students in the school are weak. (ii) Weak students may improve.

SOLUTION

(i) sits **before** the decision. If no student were weak, starting the classes would make no sense. That is taken for granted — an assumption. (ii) sits **after** the decision. It is a likely result, drawn out of the statement.

Answer: (i) assumption, (ii) inference.

Example 4

Argument: “City X put speed cameras on Ring Road. Accidents fell 30%. So cameras cut accidents.” Which weakens it? (a) The cameras were costly. (b) Ring Road was shut for repairs for half the period.

SOLUTION

(a) is about money, not about whether cameras cut accidents. Irrelevant. (b) gives a rival cause: fewer cars on the road means fewer accidents, cameras or not.

Answer: (b).

Example 5

Same argument. Which strengthens it? (a) Traffic volume on Ring Road stayed the same through the period. (b) The city also bought new ambulances.

SOLUTION

(a) shuts down the main rival cause, “there were simply fewer cars”. That leaves the cameras standing. (b) is about treating injuries, not preventing crashes.

Answer: (a).

Example 6

Problem: “Many students reach school late because they miss the single bus trip.” Which course of action follows? (a) Run a second bus trip 30 minutes later. (b) Expel students who come late.

SOLUTION

(a) hits the named cause (one trip only), costs little, and is inside the school’s power. (b) punishes the student for a problem the school created. Extreme, and it does not fix the cause.

Answer: (a) only.

Example 7

Event I: The state cut bus fares in July. Event II: Bus ridership rose 25% in August. Label the pair.

SOLUTION

I happened first, and a fare cut can happen on its own for policy reasons. Cheaper travel pulls riders in. The order and the mechanism both run $I \rightarrow II$.

Answer: I is the cause and II is its effect.

Example 8

Question: “Should coding tests be compulsory for all engineering students?” Which argument is strong? (a) Yes — most campus recruiters screen on coding, so the test matches what students will actually face. (b) No — tests are hard.

SOLUTION

(a) is relevant and substantial: it links the rule to a real outcome for the student. (b) is vague and trivial. Every test is hard; that says nothing about this one.

Answer: (a) is strong; (b) is weak.

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Statement: A bank notice says, “Customers can now update their address through our mobile app.”

Assumptions: I. Some customers have a smartphone with the app installed. II. No customer will ever visit a branch again.

SOLUTION

I — negate it: “No customer has a smartphone with the app.” Then the notice reaches nobody and is useless. The notice collapses. I is an assumption.

II — negate it: “Some customers will still visit a branch.” The notice is still perfectly sensible; it offers an extra channel, not the only one. The notice survives. II is not an assumption.

Answer: Only I is an assumption.

TRAP

Extreme words kill an assumption. “No customer will ever”, “all”, “only”, “never”, “always” — an assumption has to be the **minimum** the argument needs. The argument in Example 9 needs **some** app users, not **every** customer. If an option is stronger than the argument needs, it is not an assumption.

Example 10 [Accenture pattern]

Statement: In the last term, 60 of the 200 students of Class 10 scored above 80%.

Conclusions: I. 140 students scored 80% or below. II. The class average was above 80%.

SOLUTION

I: total students = 200. Students above 80% = 60. Remaining = $200 - 60 = 140$. Every one of those 140 is “not above 80%”, i.e. 80% or below. I follows.

II: 60 students scored high; the other 140 could all have scored 30%. Rough check with those numbers: average $\approx \frac{60 \times 85 + 140 \times 30}{200} = (5100 + 4200) \div 200 = 9300 \div 200 = 46.5\%$. So the class average need not be above 80%. II does not follow.

Answer: Only I follows.

Example 11 [Infosys pattern]

Statement: The landfill serving a city will be full in six months.

Courses of action: I. The city should start separating wet and dry waste at the household level. II. The city should stop collecting waste.

SOLUTION

I — three filters. Cause: separated dry waste can be recycled, so less goes to the landfill. Hits the cause. Practical: yes, many cities do it. Power: waste collection is the city’s own job. I follows.

II — stopping collection does not reduce waste; it moves it to the streets. It makes the problem worse and is outside any sane duty of the city. II does not follow.

Answer: Only I follows.

Example 12 [Wipro pattern]

Question: Should voting be made compulsory?

Arguments: I. Yes — when only a third of voters turn out, the winning side may represent a small slice of the people. II. No — many people are busy on polling day.

SOLUTION

I: it names a real, sizeable defect (a government chosen by a minority) and ties it directly to the proposal. Relevant and substantial. Strong.

II: polling day is a declared holiday in most places, and “busy” is vague — no size, no group, no number. It is also an argument against **any** voting arrangement, not this one. Weak.

Answer: Only argument I is strong.

Example 13 [Capgemini pattern]

Statement I: The municipal body doubled parking fees in the central market in March.

Statement II: Shops in the central market reported 15% lower footfall in April.

SOLUTION

Order: March comes before April, so II cannot have caused I. Mechanism: a higher parking fee raises the cost of a shopping trip, so fewer people drive in. That runs $I \rightarrow II$. Independence: the fee change was a decision of the municipal body; it did not need the footfall drop to happen.

Answer: I is the cause and II is its effect.

SHORTCUT

The cause-effect 3-question drill. Run the pair through these in order and stop at the first one that decides it.

1. **Which came first?** The later event can never be the cause. March before April settles Example 13 in one look.
2. **Can one happen on its own?** A policy decision, a rain spell, a rule change — these do not wait for the other event. That one is the cause.
3. **If neither of the above works,** ask whether a third thing (a festival, a new law, a price shock) could produce both. If yes, the label is “both are effects of a common cause”.

Example 14 [Cognizant pattern]

Statement: Company Z says its new training programme works: attrition fell from 18% to 11% in one year.

Which weakens this most? (a) The programme cost Rs. 40 lakh. (b) In the same year the company raised salaries by 20%. (c) Attrition in the industry is normally about 15%.

SOLUTION

The gap: “attrition fell” → “the training caused it”.

(a) cost says nothing about cause. (b) gives a rival cause that is well known to cut attrition, and it happened in the same window. The fall may be entirely the pay rise. This attacks the gap directly. (c) tells you 11% is good, which if anything helps the claim.

Size check: the fall is $18 - 11 = 7$ percentage points, i.e. $7 \div 18 = 0.3888 \approx 38.9\%$ of the old rate. A 20% pay rise can easily explain a fall of that size.

Answer: (b).

Example 15 [TCS NQT pattern]

Statement: A survey of 1,000 office workers in Pune found that 62% skip breakfast on working days.

Which is a valid inference? (a) Most Indians skip breakfast. (b) More than half of the workers surveyed skip breakfast on working days. (c) Skipping breakfast harms health.

SOLUTION

(a) jumps from 1,000 office workers in one city to a whole country. The sample does not cover farmers, students, or other cities. Over-reach. (b) $62\% > 50\%$, and it stays inside the surveyed group. Safe. (c) brings in outside medical knowledge. The statement says nothing about health.

Answer: (b).

SHORTCUT

The scope ruler. For any inference, underline the group in the statement (“1,000 office workers in Pune”) and the group in the option (“Indians”). If the option’s group is **bigger**, or the option’s time period is **longer**, reject it without reading further. This single check settles most inference questions in under 20 seconds.

Example 16 [Accenture pattern]

Statement: A college has raised the hostel fee by 10% in order to improve the quality of hostel food.

Assumptions: I. Better food costs more money. II. Students will pay the raised fee.

SOLUTION

I — negate: “Better food does not cost more money.” Then raising the fee is not needed to improve food, and the stated reason falls apart. Collapses. I is an assumption.

II — negate: “Students will not pay the raised fee.” Then no extra money arrives, and the food cannot be improved. The plan cannot work. Collapses. II is an assumption.

Answer: Both I and II are assumptions.

Example 17 [Infosys pattern]

Statement: Of 500 candidates who appeared, 320 cleared the aptitude round, and 96 of those 320 cleared the interview.

Conclusions: I. 180 candidates did not clear the aptitude round. II. 30% of those who cleared aptitude also cleared the interview.

SOLUTION

I: appeared = 500, cleared = 320. Not cleared = $500 - 320 = 180$. I follows.

II: interview clears = 96 out of 320. $96 \div 320 = 0.3$. $0.3 \times 100 = 30\%$. II follows.

Answer: Both I and II follow.

TRAP

Percentage of what? In Example 17, $96 \div 500 = 19.2\%$, and $96 \div 320 = 30\%$. Both are true sentences about different bases. A conclusion is right only if it names the **same** base the question used. Always read the words after “of”.

Example 18 [Wipro pattern]

Statement: Sector 12 floods every monsoon because its storm drains choke with plastic waste.

Courses of action: I. The drains should be cleared before the monsoon. II. The sale of thin plastic bags in the area should be banned. III. Residents of Sector 12 should be shifted elsewhere.

SOLUTION

I: removes the blockage that is named as the reason. Cheap, practical, within the municipality’s power. Follows.

II: cuts the **supply** of the material that chokes the drains. It attacks the cause at the source and is a normal municipal power. Follows.

III: moving an entire population to avoid clearing drains is extreme and costs enormously more than the two fixes already on the table. Does not follow.

Answer: I and II follow.

Example 19 [Capgemini pattern]

Statement: An online grocer says its same-day delivery lifted repeat orders. Evidence: in the cities that have same-day delivery, the repeat-order rate is 1.4 times the rate in the other cities.

Which strengthens the claim most? (a) Same-day delivery is popular with customers. (b) Before same-day delivery started, those cities had the same repeat-order rate as the rest. (c) The grocer plans to expand same-day delivery.

SOLUTION

The gap: the two sets of cities may simply have been different all along.

(a) restates the claim in softer words. It adds no new fact. (b) removes exactly the rival explanation: the cities started level, so the 1.4 times gap opened **after** the change. Strongest. (c) is about the future. It is not evidence for the past.

Answer: (b).

PRACTICE · TIER 1 target: 60 s/Q

1. **Statement:** “Buy Sunrise Atta — get 1 kg free with every 5 kg pack.”

Assumptions: I. Customers respond to free offers. II. Sunrise Atta is of poor quality.

2. **Statement:** The train from Chennai to Coimbatore will now run 6 days a week instead of 3.
Conclusions: I. The railway expects more passengers on this route than it can carry in 3 trips. II. All other trains on this route have been stopped.
3. **Statement:** A factory's power bill rose 40% although production stayed the same.
Courses of action: I. The factory should have its wiring and machines checked for leakage. II. The factory should shut down.
4. **Statement I:** The government cut the import duty on edible oil in May. **Statement II:** Retail edible oil prices fell in June.
Label the pair.
5. **Statement:** Office notice — "All employees must reach the office by 9:30 a.m. from Monday."
Assumptions: I. Employees are able to reach by 9:30 a.m. II. Some employees were arriving later than this earlier.
6. **Statement:** Of 1,200 people surveyed in a town, 900 own a two-wheeler.
Conclusions: I. 75% of those surveyed own a two-wheeler. II. 300 of those surveyed own no vehicle at all.
7. **Statement:** A coaching centre says its course works, because 70% of its students cleared the exam.
Which weakens it? I. The centre admits only students who already scored in the top 10% of a screening test. II. The course fee is high.
8. **Question:** Should schools ban mobile phones inside classrooms?
Arguments: I. Yes — phones pull attention away and a teacher cannot watch forty screens at once. II. No — phones are expensive.
9. **Statement:** A bus depot has added live tracking for all 200 of its buses.
Assumptions: I. Passengers want to know where their bus is. II. Each of the 200 buses carries a working tracking device.
10. **Statement:** Many first-year students drop out because they cannot follow lectures given in English.
Courses of action: I. The college should run a six-week English bridge course before the first semester. II. The college should teach every subject in the local language from now on.
11. **Statement:** A shop sold 240 units in January and 300 units in February.
Conclusions: I. Sales rose by 25%. II. Sales will rise again in March.
12. **Statement I:** A new flyover opened on the highway in April. **Statement II:** Average travel time on that highway fell by 18 minutes in May.
Label the pair.
13. **Statement:** Hospital notice — "Visitors are allowed only between 5 p.m. and 7 p.m."
Assumptions: I. Visits outside these hours disturb patients or staff. II. Visitors will follow the notice.
14. **Statement:** The state has appointed 1,500 new government school teachers this year.
Label each: (i) There was a shortage of teachers. (ii) The number of students per teacher will fall.
15. **Statement:** A food app says its new 10-minute delivery raised order volume: in the 3 cities that have it, orders rose 30%.
Which strengthens and which weakens? I. In the other 12 cities orders rose only 4% in the same period. II. The app also spent far more on advertising in those 3 cities.

ANSWER KEY

1. **Only I.** Negating I ("customers ignore free offers") makes the offer pointless. II is not needed at all — good products run offers too.
2. **Only I.** More trips make sense only if more demand is expected. II is an extra fact nobody stated.
3. **Only I.** A 40% jump with flat production points to leakage or a faulty meter, so checking hits the cause. Shutting down is extreme.
4. **I is the cause, II is its effect.** The duty cut came first and lowers the landed cost, which feeds into retail price.
5. **Both.** Negate I and the rule is impossible to obey; negate II and there is no reason to issue the notice.
6. **Only I.** $900 \div 1200 = 0.75 = 75\%$. II is wrong: the other 300 may own cars or cycles.

7. **Only I weakens.** If only top-decile students are admitted, the 70% pass rate may be the intake, not the teaching. The fee says nothing about whether the course works.
8. **Only I is strong.** It names a real classroom mechanism. “Phones are expensive” is not a reason to allow them in class.
9. **Both.** Negate I and the feature has no user; negate II and tracking cannot cover all 200 buses as claimed.
10. **Only I.** A bridge course attacks the named cause at low cost. Switching every subject’s language is extreme and ignores that later exams and jobs still use English.
11. **Only I.** $\text{Rise} = 300 - 240 = 60$; $60 \div 240 = 0.25 = 25\%$. II predicts the future, which the statement cannot support.
12. **I is the cause, II is its effect.** The flyover opened first and removes a bottleneck, which shortens travel time.
13. **Both.** Negate I and there is no reason for a window; negate II and the notice achieves nothing.
14. **(i) assumption, (ii) inference.** A shortage is the unstated reason behind the appointments; a better student-teacher ratio is a result that follows from them.
15. **I strengthens, II weakens.** I shows the other cities did not rise the same way, so the difference tracks the feature. II supplies a rival cause for the same 30%.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 20 [Grab · pattern]

Statement: A ride-hailing firm raised surge pricing during rain in Jakarta. Completed trips during rain went from 6,200 per hour to 7,400 per hour. The firm concludes that surge pricing brings more drivers online and so serves riders better.

Which weakens the conclusion most?

- (a) Driver earnings per trip rose by 22%.
- (b) Ride requests during rain went from 9,000 per hour to 14,000 per hour.
- (c) Some riders complained about the higher fare.

SOLUTION

The claim is “riders are served better”. The right measure is the share of requests that actually got a ride, not the raw count of trips.

Before: $6,200 \div 9,000 = 0.6888\ldots$ $0.6888 \times 100 \approx 68.9\%$.

After: $7,400 \div 14,000 = 0.5285\ldots$ $0.5285 \times 100 \approx 52.9\%$.

So the fulfilment rate fell by about $68.9 - 52.9 = 16$ percentage points. More trips were completed, but a **smaller** share of riders got a car.

(a) supports the firm’s story about drivers, not against it. (c) is a complaint, not a measure of service.

Answer: (b) — the share of requests served fell from about 68.9% to about 52.9%.

SHORTCUT

Raw count up does not mean rate up. Whenever a claim rests on a count (trips, defects, returns, accidents), hunt for the denominator (requests, units made, orders, kilometres driven). If the denominator grew faster than the count, the rate moved the other way. This single move answers a large share of Tier 2 weaken questions.

Example 21 [Shopee · pattern]

Statement: A seller’s record: Quarter 1 — 12,000 orders and 960 returns. Quarter 2 — 15,000 orders and 1,050 returns.

Conclusions: I. The return rate fell. II. The number of returns fell.

| **SOLUTION**

Return rate in Q1: $960 \div 12,000 = 0.08$, so 8%. Return rate in Q2: $1,050 \div 15,000 = 0.07$, so 7%. $8\% > 7\%$, so the rate fell. I follows.

Number of returns: 960 in Q1 and 1,050 in Q2. $1,050 > 960$, so the count **rose** by $1,050 - 960 = 90$. II does not follow.

Answer: Only I follows.

Example 22 [DBS · pattern]

Statement: A bank announces that from next month any transfer above S\$10,000 to a new payee will be held for 12 hours. The bank says this will cut scam losses.

Which are assumptions?

- I. A meaningful share of scam losses comes from transfers to payees the customer has never paid before.
- II. Victims can notice and report a scam within 12 hours of sending.
- III. No genuine customer ever needs to send a large amount urgently.

SOLUTION

I — negate: “Almost no scam loss involves a new payee.” Then the hold catches nothing and the measure cannot cut losses. Collapses. Assumption.

II — negate: “Victims cannot notice within 12 hours.” Then the money is released before anyone reports it, and the hold saves nothing. Collapses. Assumption.

III — negate: “Some genuine customers do need urgent large transfers.” The bank can still run the policy and still cut scam losses; it just inconveniences those customers. The argument survives. Not an assumption — it is far stronger than the argument needs.

Answer: I and II are assumptions; III is not.

Example 23 [Agoda · pattern]

Statement: Mobile bookings rose 35% in the month after the app redesign shipped. The product team credits the redesign.

Which most weakens the credit given to the redesign?

- (a) Desktop bookings were flat in the same month.
- (b) The redesign shipped in the same week a three-day national holiday sale began.
- (c) The redesign took four months to build.

SOLUTION

Gap: “bookings rose after X” → “X caused the rise”. The way to attack it is a rival cause that sits in the same window.

(a) actually helps the redesign story, because the redesign was mobile-only and only mobile moved. (b) is a rival cause with the right timing and the right size: a national holiday sale lifts bookings on its own, redesign or not.

(c) is about effort spent. Effort is not evidence of effect.

Answer: (b).

Example 24 [SCB · pattern]

Statement: A bank’s Bangkok call centre has an average wait of 11 minutes. An audit finds that 40% of all calls are simple balance enquiries.

Courses of action: I. Offer balance enquiries through an automated self-service menu. II. Double the number of agents. III. Stop taking calls after 6 p.m.

SOLUTION

I: the audit names a cause — 4 calls in every 10 need no human at all. Moving them to self-service removes about 40% of the load at low cost. Hits the cause, practical, inside the bank’s power. Follows.

II: doubling agents would cut the wait, but it doubles the largest cost line to handle work that a menu can absorb for almost nothing. With the cheaper fix named in the same statement, this is a weak action.

III: refusing calls does not shorten the wait for the calls you do take; it pushes customers into the next day's queue. Does not follow.

Answer: Only I clearly follows. II is weak (costly, ignores the named cause); III does not follow.

Example 25 [Sea · pattern]

Statement: In a survey of 2,400 online shoppers in Singapore, 1,560 named free returns as the top reason they choose a platform. The company concludes: "So we must offer free returns."

Part (a): Which is a safe inference? **Part (b):** What does the conclusion assume?

SOLUTION

(a) Compute the share: $1,560 \div 2,400 = 0.65$, so 65%. Safe inference: 65% of the shoppers surveyed named free returns as their top reason. It stays inside the surveyed group and adds no new claim.

Unsafe: "65% of all Singaporeans want free returns" — the survey covered online shoppers, not everyone.

(b) The jump is from **these 2,400 shoppers** to **our customers**. Assumption: the company's own target customers behave like the surveyed group. Negate it — if the company's customers are nothing like this sample, the survey gives no reason to change anything, and the conclusion falls.

Answer: (a) 65% of the surveyed shoppers named free returns. (b) That the company's target customers resemble the surveyed group.

Example 26 [LINE MAN · pattern]

Statement: A delivery firm pilots "batching" — two orders carried on one rider trip. Over two weeks the cost per order fell from THB 38 to THB 33. The firm says batching cuts cost.

Which strengthens, and which weakens?

- I. Average delivery time and cancellation rate were unchanged over the two weeks.
- II. Rider pay per trip was cut by 15% at the start of the same two weeks.

SOLUTION

Size of the effect: fall = $38 - 33 = 5$ baht. As a share: $5 \div 38 = 0.1315\dots$, so about 13.2%.

I strengthens. The obvious counter to batching is "you saved money by making service worse". I shuts that door: the same speed, the same cancellations, lower cost.

II weakens. A 15% cut in rider pay lowers cost per order all by itself, with no batching at all. It is a rival cause of the right size to explain a 13.2% fall.

Answer: I strengthens; II weakens.

Example 27 [GIC · pattern]

Question: Should a pension fund move 10% of its equity holding into infrastructure debt?

Arguments: I. Yes — over the last ten years infrastructure debt returned about 8.2% a year with smaller year-to-year swings than equity, and the fund pays out over a 20-year horizon. II. No — infrastructure projects can be delayed.

SOLUTION

I is strong. It gives a number (8.2%), a comparison on the thing that matters to a pension fund (swings versus payout horizon), and it is tied to this exact decision.

II is weak **as written**. Every asset carries risk; the argument gives no size, no frequency, and no comparison with the risk of the equity it would replace. An argument that would apply equally to every alternative cannot decide between them.

Answer: Only argument I is strong.

TRAP

A true statement can still be a weak argument. “Infrastructure projects can be delayed” is true. It is weak because it is unquantified and applies to every option on the table. Strength is about **relevance and size**, not about truth. Students who mark every true sentence “strong” lose most of this section.

Example 28 [Razer · pattern]

Statement: A gaming hardware firm reports: 2023 revenue S\$480 million, of which 12% came from software. 2024 revenue S\$540 million, of which 18% came from software.

Conclusions: I. Software revenue grew by more than 60%. II. Hardware revenue fell.

SOLUTION

Software 2023 = $0.12 \times 480 = 57.6$ million. Software 2024 = $0.18 \times 540 = 97.2$ million.

Growth = $97.2 - 57.6 = 39.6$ million. As a share: $\frac{39.6}{57.6} = 0.6875$, so 68.75%. $68.75\% > 60\%$. I follows.

Hardware 2023 = $480 - 57.6 = 422.4$ million. Hardware 2024 = $540 - 97.2 = 442.8$ million. $442.8 > 422.4$, so hardware **rose** by 20.4 million. II does not follow.

Answer: Only I follows.

PRACTICE · TIER 2 target: 100 s/Q

- Statement:** Rider cancellations fell from 9% to 6% after a new “confirm your pickup point” screen was added.
Conclusions: I. Cancellations fell by 3 percentage points. II. Cancellations fell by one-third.
- Statement:** An e-wallet will require face verification for all first-time users, to cut fraud.
Assumptions: I. A meaningful share of fraud comes from newly opened accounts. II. Most first-time users have a phone with a working camera.
- Statement:** A logistics firm says its new routing software cut fuel cost per parcel from THB 14 to THB 11.
Which weakens the causal claim? I. Diesel prices fell 20% during the pilot. II. The software licence cost THB 2 million.
- Statement:** A hotel chain in Phuket finds that 42% of guests skip the paid breakfast and eat outside.
Courses of action: I. Lower the breakfast price and bundle it into the room rate. II. Lock the hotel gates during breakfast hours.
- Statement:** A marketplace had 2.5 million monthly buyers in 2023 and 3.0 million in 2024. GMV rose from S\$600 million to S\$690 million.
Conclusions: I. Buyers grew by 20%. II. GMV per buyer fell.
- Statement I:** A city raised the minimum taxi fare in June. **Statement II:** App-based ride bookings in that city rose 14% in July.
Label the pair.
- Statement:** Of 5,000 parcels sampled, 150 were delivered late, and 90 of those 150 came from a single zone.
Inferences: I. The late rate in the sample was 3%. II. 60% of the late parcels came from one zone.
- Question:** Should a bank allow accounts to be opened fully online?
Arguments: I. Yes — branch onboarding costs the bank about S\$40 per account against about S\$6 online, and digital identity checks are already legal here. II. No — older customers may not like the change.
- Statement:** A grocery app says its “repeat basket” feature raised order frequency from 2.4 to 3.0 orders per user per month.
Which strengthens and which weakens? I. Users were randomly assigned to see the feature or not. II. The company ran a cashback campaign for feature users in the same period.

10. **Statement:** An airline will remove free seat selection so that it can sell seat choice as an add-on.
Assumptions: I. Enough passengers will pay for a seat they used to get free. II. Passengers will stop flying this airline.
11. **Statement:** Fund A returned 9% in year 1 and -4% in year 2. Fund B returned 3% in each of the two years.
Conclusions: I. Over the two years Fund A's total return was higher than Fund B's. II. Fund A's returns swung more than Fund B's.

ANSWER KEY

- Both.** $9 - 6 = 3$ percentage points, and $\frac{3}{9} = \frac{1}{3}$, a one-third fall. The two sentences describe the same drop in two correct ways.
- Both.** Negate I and the check targets the wrong place; negate II and most first-time users cannot complete it, so the rule blocks genuine sign-ups instead of fraud.
- Only I.** A 20% diesel fall cuts fuel cost per parcel without any software; $14 \rightarrow 11$ is a fall of $3 \div 14 \approx 21.4\%$, well inside what fuel prices alone could do. II attacks whether the software is worth buying, not whether it cut fuel cost.
- Only I.** Guests eat outside because the paid breakfast is not worth its price; bundling it attacks that cause. Locking gates is coercive and would damage the business.
- Both.** Buyer growth $= 0.5 \div 2.5 = 0.20 = 20\%$. GMV per buyer: $600 \div 2.5 = \$240$ in 2023 and $690 \div 3.0 = \$230$ in 2024, so it fell by \$10.
- I is the cause, II is its effect.** The fare rise came first and made taxis dearer relative to app rides, pushing riders across.
- Both.** $150 \div 5,000 = 0.03 = 3\%$. $90 \div 150 = 0.6 = 60\%$. Both stay inside the sample.
- Only I is strong.** It gives the cost gap per account and confirms the legal path. "Older customers may not like it" has no size and does not say the online option must be the only one.
- I strengthens, II weakens.** Random assignment makes the two groups alike, so the rise tracks the feature. A cashback aimed at feature users is a rival cause for the same rise.
- Only I.** Negate I and there is no revenue, so the plan fails. II is not needed — the airline is assuming the opposite of II.
- Only II.** Fund A over two years: $1.09 \times 0.96 = 1.0464$, a gain of 4.64%. Fund B: $1.03 \times 1.03 = 1.0609$, a gain of 6.09%. So Fund A is **lower**, and I is false. A moved from $+9\%$ to -4% , a far wider swing than $+3\%$ twice, so II is true.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 29 [Google · pattern]

Statement: A study of 300 successful startups found that 78% of them had a "fail fast" culture. The study concludes that a fail-fast culture drives success.

Name the flaw, and show with numbers how the conclusion could be exactly backwards.

SOLUTION

Insight: the sample was chosen by the outcome. A rate measured only among winners tells you nothing until you know the same rate among losers.

The number 78% answers: "among successes, how many had the culture?" The conclusion needs the reverse: "among firms with the culture, how many succeeded?"

Build a world where the study's fact is true and the conclusion is false. Suppose the period had 300 successes and 5,000 failures.

Successes with the culture $= 0.78 \times 300 = 234$. Suppose 80% of the failures also had it: $0.80 \times 5,000 = 4,000$.

Now compute success rate **by culture**. With the culture: total $= 234 + 4,000 = 4,234$; successes $= 234$. Rate $= 234 \div 4,234 = 0.0552... \approx 5.5\%$.

Without the culture: successes $= 300 - 234 = 66$; failures $= 5,000 - 4,000 = 1,000$; total $= 1,066$. Rate $= 66 \div 1,066 = 0.0619... \approx 6.2\%$.

So in this world the fail-fast firms succeed **less** often ($5.5\% < 6.2\%$), while the study's 78% figure is still perfectly true.

Answer: Survivorship bias. The study needs the culture rate among failed startups too; without it the 78% figure is consistent with the culture making success less likely.

Example 30 [Amazon · pattern]

Statement: A fraud filter flags 1 transaction in every 50. It catches 95% of all fraudulent transactions. Fraud is 0.4% of all transactions. A manager says: “Since it catches 95% of fraud, a flagged transaction is almost certainly fraud.”
Is the manager right? Compute.

SOLUTION

Insight: “catches 95% of fraud” is about the fraud column. “A flagged item is fraud” is about the flagged row. With a tiny base rate, those two numbers are miles apart.

Work with 100,000 transactions.

Fraudulent transactions = $0.4\% \times 100,000 = 0.004 \times 100,000 = 400$. Caught by the filter = $95\% \times 400 = 0.95 \times 400 = 380$.

Total flagged = $100,000 \div 50 = 2,000$.

Of those 2,000 flags, only 380 are actually fraud. Share = $380 \div 2,000 = 0.19$, so 19%.

The other $2,000 - 380 = 1,620$ flags are clean transactions — that is 81% of all flags.

So a flagged transaction is fraud about 1 time in 5, not “almost certainly”.

Answer: No. Only about 19% of flagged transactions are fraud; 81% are false alarms.

SHORTCUT

The 100,000 table. For any “the test caught X% so a positive means Y” argument, write four numbers for 100,000 cases: true cases, caught cases, total positives, and positives that are false. You never need probability notation. Two multiplications and one division settle it, and it works for fraud filters, medical tests, spam filters and security alerts alike.

Example 31 [Goldman Sachs · pattern]

Statement: A trading desk put its 10 worst-performing analysts into a coaching programme. The next quarter, that group's average profit improved by 30%, while the top 10 analysts, who got no coaching, slipped by 8%. The head of desk concludes that coaching works and should be extended.

What is wrong, and what evidence would settle it?

SOLUTION

Insight: the two groups were picked by their extreme results in one quarter. Extreme results contain luck, and luck does not repeat. Both groups would drift toward the middle next quarter with no coaching at all.

Look at what the data itself shows. The top 10 got **no** coaching and still moved — and they moved **downward**, toward the middle. The bottom 10 moved upward, toward the middle. Both groups moved in the direction the “drift to the average” story predicts, and only one of them was coached.

So the desk's own control group already behaves as if the effect is drift, not coaching. The 30% figure is measured against a quarter that was unusually bad, which inflates it.

What would settle it: take the bottom 20 analysts, and assign 10 at random to coaching and 10 at random to none. Both groups then drift equally, so any extra gain in the coached half is the coaching. Random assignment **inside** the extreme group is the key — comparing an extreme group with a non-extreme group cannot work.

Answer: Regression to the average. The fix is a randomised control group drawn from the same bottom-10 pool.

Example 32 [Microsoft · pattern]

Argument: “Our app now loads in under one second. So users will not abandon it.”

Sort these into necessary assumption, sufficient assumption, and neither:

- (a) Load time is one reason users abandon apps.
- (b) Any app that loads in under one second is never abandoned.
- (c) Before the change, the app took three seconds to load.
- (d) No competitor’s app loads in under one second.

SOLUTION

Insight: a necessary assumption is the smallest thing the argument cannot live without. A sufficient assumption is a bigger claim that, if handed to you free, finishes the argument. Tests differ: negate for necessary, add for sufficient.

(a) Negation test: “Load time is **not** a reason users abandon apps.” Then making the app fast cannot stop abandonment, and the conclusion dies. The argument cannot live without (a). **Necessary.**

(b) Negation test: “Some app that loads under one second is abandoned.” Does the argument die? Not necessarily — our app could still be one of the ones that is not abandoned. So (b) is not necessary. Addition test: if (b) is handed to us as true, then our sub-second app is never abandoned, and the conclusion follows with no gap left. **Sufficient, not necessary.**

(c) Negation: “Before the change, the app did not take three seconds to load.” Say it took 1.5 seconds before. The argument still reads the same: it is now under one second, so users will not abandon it. The argument survives, so (c) is not necessary. Addition test: knowing the old load time still leaves the gap open — nothing links load time to abandonment. **Neither.**

(d) Negation: “Some competitor also loads under one second.” The conclusion is about **abandonment**, not about being fastest. The argument survives. Adding (d) also does not close the gap. **Neither.**

Answer: (a) necessary; (b) sufficient; (c) and (d) neither.

Example 33 [Uber · pattern]

Statement: A city capped the number of ride-hailing licences. Average driver earnings then rose from Rs. 1,150 to Rs. 1,380 per day. The regulator says the cap made drivers better off.

Build a numeric case in which the figures are exactly right and yet not one driver earned a single rupee more.

SOLUTION

Insight: an average is computed over whoever is still in the group. Remove the weakest members and the average jumps, even if every individual is frozen.

Take 100 drivers before the cap, and split them into two groups. Let 75 drivers earn Rs. 1,380 a day, and 25 drivers earn Rs. 460 a day.

Check the before-average. Earnings of the 75 = $75 \times 1,380 = 103,500$. Earnings of the 25 = $25 \times 460 = 11,500$. Total = $103,500 + 11,500 = 115,000$. Average = $115,000 \div 100 = 1,150$. That matches the “before” figure exactly.

Now apply the cap. Licences are cut, and the 25 lowest earners — the part-timers with the fewest trips — lose theirs and leave the market.

After: 75 drivers remain, each still earning Rs. 1,380 a day, exactly as before. Total = $75 \times 1,380 = 103,500$. Average = $103,500 \div 75 = 1,380$. That matches the “after” figure exactly.

Not one remaining driver earned more than before, and 25 drivers lost their livelihood entirely. The average rose only because the group changed.

To settle it properly the regulator must track the **same** drivers before and after, not the average of whoever holds a licence at the time.

Answer: Selection effect. Dropping the 25 lowest earners raises the average from 1,150 to 1,380 with every surviving driver’s pay unchanged.

TRAP

Ratios and averages both hide a changing denominator. “Cost per order fell”, “average GPA rose”, “defects per unit dropped”, “earnings per driver rose” — before you accept any of them as an improvement, ask two questions: **did the top line change, or did the bottom line change?** and **is the group being measured the same group as before?** An average that rises because the weakest left is not an improvement.

Example 34 [Adobe · pattern]

Statement: A design tool’s team reports that average time to first export fell from 22 minutes to 14 minutes after a redesign. They conclude designers are more productive.

Part (a): Name the gap. **Part (b):** Show numerically how a single extra fact could shrink the gain to almost nothing.

Part (c): What fact would make the claim solid?

SOLUTION

Insight: the team measured a milestone, not the job. A milestone can move earlier without the job finishing any sooner.

(a) The gap is between “time to first export” and “productivity”. Productivity is output per unit of total time. First export is one checkpoint inside the work, and a redesign can move that checkpoint without changing the finish line.

(b) Headline gain: $22 - 14 = 8$ minutes, i.e. $8 \div 22 = 0.3636... \approx 36.4\%$.

Now add this fact: in the new tool, users export a rough draft early and then spend an average of 6 more minutes editing **after** that first export.

True time to a finished asset, new tool = $14 + 6 = 20$ minutes. Gain against the old 22 minutes = $22 - 20 = 2$ minutes. As a share: $2 \div 22 = 0.0909... \approx 9.1\%$.

So a headline “36.4% faster” becomes a real “9.1% faster” — and if the post-export edit were 8 minutes or more, the gain would vanish completely.

(c) The claim needs the end-to-end number: time from starting a project to delivering the finished asset, measured on the same kind of projects. For example, “median project completion fell from 3.2 hours to 2.1 hours” — a fall of $1.1 \div 3.2 = 0.34375 \approx 34.4\%$ on the measure that actually matters.

Answer: (a) milestone is not the same as total output; (b) with 6 minutes of post-export editing the real gain drops from 36.4% to 9.1%; (c) end-to-end project completion time.

Example 35 [D. E. Shaw · pattern]

Statement: At an exchange, the retail segment’s volume grew 40% in 2024 while the institutional segment grew 10%. Base volumes at the start of 2024 were: retail 120 units, institutional 900 units. An internal note says: “Retail is now adding more new volume than institutional.”

Part (a): Is the note right? **Part (b):** Below what institutional base would it be right?

SOLUTION

Insight: a growth rate is a ratio; new volume is a rate times a base. Comparing rates answers a different question from comparing increases.

(a) Retail increase = $40\% \times 120 = 0.40 \times 120 = 48$ units. Institutional increase = $10\% \times 900 = 0.10 \times 900 = 90$ units.

$90 > 48$. Institutional added 90 units of new volume against retail’s 48 — almost double. The note is wrong.

Check the totals to be sure. Retail end = $120 + 48 = 168$. Institutional end = $900 + 90 = 990$. Total start = $120 + 900 = 1,020$. Total end = $168 + 990 = 1,158$. Total increase = $1,158 - 1,020 = 138$, and $48 + 90 = 138$. Consistent.

Note also the share of total growth: retail contributed $48 \div 138 = 0.3478 \approx 34.8\%$, institutional $90 \div 138 \approx 65.2\%$.

(b) Let the institutional base be I , with the same rates. Retail adds $0.40 \times 120 = 48$. Institutional adds $0.10 \times I$. Retail adds more when $48 > 0.10I$. Divide both sides by 0.10: $480 > I$.

So retail would add more only if the institutional base were below 480 units. The actual base, 900 units, is nearly twice that.

Answer: (a) No — institutional added 90 units against retail’s 48. (b) Only if the institutional base were below 480 units.

PRACTICE · TIER 3 target: 4 min/Q

1. A fraud filter flags 3% of all transactions and catches 90% of all fraud. Fraud is 0.5% of transactions. What share of flagged transactions is actually fraud? Show the working for 100,000 transactions.
2. “Nine of the ten best-performing funds of the decade used strategy S, so S is superior.” Which single extra fact would most weaken this, and why? (a) The tenth fund used strategy T. (b) Of the 400 funds that used S, 380 closed after losses. (c) Strategy S needs expensive software.
3. A manager notes that the 12 branches with the worst service scores improved most after a new scoring dashboard was installed, while the 12 best branches slipped. She concludes the dashboard works. Name the flaw and describe the study design that would settle it.
4. A college raised its entry cut-off. Average first-year GPA rose from 6.4 to 7.1, and the principal credits better teaching. Using 200 students, construct numbers in which the two averages are exactly right and no student’s GPA improved at all.
5. A marketplace reports Category A grew 50% and Category B grew 12%. Base GMV: A was S\$200 million, B was S\$950 million. A manager concludes A contributed more to total growth. Evaluate, and find the base of B at which the two contributions would be equal.
6. Argument: “Our checkout now takes two taps instead of five. So cart abandonment will fall.” Which is a **necessary** assumption and which is a **sufficient** one? (a) The number of taps is one reason people abandon carts. (b) Any two-tap checkout has zero abandonment. (c) Competing stores also use a two-tap checkout.
7. “Districts with more bookstores have higher average household income. So opening bookstores raises income.” Give the two strongest alternative explanations, and say what evidence would separate them from the stated one.

ANSWER KEY

1. **15%.** Out of 100,000 transactions: fraud = $0.005 \times 100,000 = 500$; caught = $0.90 \times 500 = 450$; total flagged = $0.03 \times 100,000 = 3,000$. So $450 \div 3,000 = 0.15 = 15\%$ of flags are fraud, and 2,550 flags (85%) are false alarms. A high catch rate says nothing about precision when the base rate is 0.5%.
2. **(b).** The claim is chosen from the winners only. If 400 funds used S and 380 of them closed after losses, then S has a success rate of about $20 \div 400 = 5\%$, and its presence among the top ten is what you would expect from a strategy that is simply high-variance. (a) is irrelevant; (c) is about cost, not performance.
3. **Regression to the average.** Both groups were selected on one quarter’s extreme scores, and extremes contain luck that does not repeat, so the worst drift up and the best drift down with no treatment at all. Note that the best branches slipped **without** the dashboard doing anything to them — the control behaviour is already visible in her own data. Fix: take the 24 worst-scoring branches and randomly give the dashboard to 12; compare the two halves.
4. Let 160 students have GPA 7.1 and 40 have GPA 3.6. Total = $160 \times 7.1 + 40 \times 3.6 = 1,136 + 144 = 1,280$, so the average is $1,280 \div 200 = 6.4$ — the “before” figure. Raising the cut-off simply stops admitting the 40 weakest. The remaining 160 still have GPA 7.1, so the new average is exactly 7.1 with nobody improving by a single point. The intake changed, not the teaching.
5. **The manager is wrong.** A adds $0.50 \times 200 = \text{S\$}100$ million; B adds $0.12 \times 950 = \text{S\$}114$ million. B contributes more. For equality, solve $0.12B = 100$, so $B = 100 \div 0.12 = 833.33$ million. Below a B base of about S\$833 million, A would contribute more; at S\$950 million it does not.
6. **(a) is necessary, (b) is sufficient, (c) is neither.** Negate (a) — taps are not a reason people abandon — and cutting taps cannot reduce abandonment, so the argument dies; that is the necessity test. (b) is far more than needed, and is not necessary (our checkout could still do well even if some two-tap checkouts fail), but if granted it closes the gap completely. (c) is about rivals, not about our abandonment: negate it and the argument is untouched, and adding it closes nothing.

7. **Reverse causation** — richer districts can afford to buy books, so bookstores open where income already is high; and **common cause** — both bookstore count and income are driven by a third factor such as population density and education level. To separate these from “bookstores raise income”, look at the time order and at a shock: find districts where bookstores opened for an outside reason (a rent subsidy, say) and check whether income rose **afterwards** relative to similar districts with no new stores.

Mixed set — exam conditions

PRACTICE • TIER 1 target: 28 min for 18 questions

1. **Statement:** Advertisement — “Learn data science in 12 weeks. No coding background needed.”
Assumptions: I. Some people without a coding background want to learn data science. II. Coding cannot be learned.
2. **Statement:** A store sold 180 units at Rs. 250 each in week 1, and 150 units at Rs. 300 each in week 2.
Conclusions: I. Revenue rose in week 2. II. The number of units sold fell.
3. **Statement:** Students in a hostel report frequent stomach illness.
Courses of action: I. Have the hostel’s water tank and kitchen inspected. II. Close the hostel.
4. **Statement I:** A company moved to a four-day week in January. **Statement II:** Its job applications tripled in February.
Label the pair.
5. **Statement:** A firm says its new packaging cut breakage from 5% to 2%.
Which weakens the causal claim? I. The firm switched to a different courier in the same month. II. The new packaging costs 8% more.
6. **Question:** Should aptitude tests be dropped from campus hiring?
Arguments: I. No — a 60-minute test screens 5,000 applicants that 20 interviewers cannot possibly meet. II. Yes — some good coders are weak at arithmetic.
7. **Statement:** Of 800 commuters surveyed in Bengaluru, 520 use a two-wheeler.
Inference: 65% of the surveyed commuters use a two-wheeler. Valid or not?
8. **Statement:** A hospital will send SMS reminders to reduce missed appointments.
Assumptions: I. Patients read the SMS they receive. II. Some appointments are missed because patients forget them.
9. **Statement:** Fund X fell 20% in year 1 and rose 25% in year 2.
Conclusions: I. Fund X is back at its starting value. II. Fund X is above its starting value.
10. **Statement:** A bus route runs 90% empty after 10 p.m., but night-shift workers depend on it.
Courses of action: I. Replace the full-size bus with a minibus after 10 p.m. II. Cancel the route.
11. **Statement:** Districts with more police officers report more crime, so posting police raises crime.
What is the flaw?
12. **Statement:** A factory’s defect count fell from 240 to 180 while output rose from 8,000 to 9,000 units.
Conclusions: I. The defect rate fell. II. The defect rate halved.
13. **Statement I:** Heavy rain fell over the catchment area for three days. **Statement II:** The city reservoir level rose by 2 metres.
Label the pair.
14. **Statement:** A college will publish every teacher’s student-feedback score in order to improve teaching.
Assumptions: I. Teachers care how their published score looks. II. Students give feedback honestly.
15. **Statement:** The state has made helmets compulsory for pillion riders.
Label each: (i) Pillion riders were being injured. (ii) Helmet sales will rise.

16. **Question:** Should a company allow permanent work from home?
Arguments: I. Yes — the company's own 18-month data shows output per engineer unchanged while office rent fell 35%. II. No — employees may feel lonely.
17. **Statement:** 80% of road accidents in a district happen within 10 km of the victim's home. Therefore driving near home is more dangerous than driving far away.
 Evaluate.
18. **Statement:** A bank's app crashes for users on Android 9 and older, who are 14% of its users.
Courses of action: I. Release a patched build tested on Android 9. II. Ask all users to buy newer phones.

ANSWER KEY

1. **Only I.** Negate I and the course has no market, so the advertisement makes no sense. II is the opposite of what a teaching business assumes.
2. **Only II.** Week 1 revenue = $180 \times 250 = 45,000$; week 2 = $150 \times 300 = 45,000$. Revenue was **equal**, so I is false. Units fell from 180 to 150, so II is true.
3. **Only I.** Water and food are the usual causes of stomach illness in a hostel, and inspection is cheap and within the warden's power. Closing the hostel is extreme and leaves students with nowhere to live.
4. **I is the cause, II is its effect.** The four-day week came first and is a strong draw for applicants.
5. **Only I.** A courier change in the same window is a rival cause of the breakage fall. Cost does not bear on whether the packaging worked.
6. **Only I is strong.** It gives the scale problem in numbers. "Some good coders are weak at arithmetic" has no size and does not weigh the cost of screening 5,000 people another way.
7. **Valid.** $520 \div 800 = 0.65 = 65\%$, and it stays inside the surveyed group.
8. **Both.** Negate I and the reminder is never seen; negate II and forgetting is not a cause, so reminders cannot help.
9. **Only I.** Take 100 as the start. After year 1: $100 \times 0.80 = 80$. After year 2: $80 \times 1.25 = 100$. Exactly back, so II is false.
10. **Only I.** A minibus keeps the service the night workers need at a much lower cost, which is the actual problem. Cancelling strands them.
11. **Reverse causation.** Police are posted **to** districts that already have high crime, so crime drives officer numbers rather than the other way round. Comparing crime before and after a posting, in similar districts, would separate the two.
12. **Only I.** Old rate = $240 \div 8,000 = 0.03 = 3\%$; new rate = $180 \div 9,000 = 0.02 = 2\%$. It fell, so I holds. A fall from 3% to 2% is a one-third fall, not a half, so II is false.
13. **I is the cause, II is its effect.** Rain in the catchment area feeds the reservoir; the order and the mechanism both run I to II.
14. **Both.** Negate I and publication changes no behaviour; negate II and the published scores carry no useful signal.
15. **(i) assumption, (ii) inference.** Injuries to pillion riders are the unstated reason for the rule; a rise in helmet sales is a consequence that follows from it.
16. **Only I is strong.** It cites the company's own long-run data on both output and cost. "Employees may feel lonely" is unmeasured and offers no comparison with the gains.
17. **Does not follow.** Almost all driving is done near home, so most accidents will be near home even if every kilometre is equally risky. The claim needs accidents **per kilometre driven** at each distance, not the share of accidents.
18. **Only I.** A patched build fixes the named cause for the 14% affected, at ordinary engineering cost and fully within the bank's control. Telling customers to buy new phones is outside the bank's power and would lose those users.

ONE-PAGE REVISION CARD

THE THREE PARTS — Evidence → **gap** → Conclusion. Every question attacks the gap. Conclusion markers: so, hence, therefore, thus, this shows, we must.

THE SEVEN TYPES AND THEIR TESTS

Assumption	Negate it. Argument collapses $\Rightarrow$ it is an assumption.
Conclusion	Derivable from the given words alone? No outside facts.
Inference	Stays inside the stated group and period.
Argument	Strong = relevant + sized + tied to this exact question.
Course of action	Hits the cause? Practical? Within the actor's power?
Cause-effect	Which came first and could stand alone?
Strengthen / weaken	Close the gap, or supply a rival cause.

ASSUMPTION vs INFERENCE — assumption sits **before** the statement and is unsaid; inference sits **after** it and is drawn out.

SHORTCUTS

1. **Scope ruler** — if the option's group or time period is bigger than the statement's, reject the inference at once.
2. **Raw count vs rate** — for any count claim, find the denominator. If the denominator grew faster, the rate moved the other way.
3. **The 100,000 table** — for any "test catches X%" claim, write: true cases, caught cases, total positives, false positives. Two multiplications and one division.
4. **Rate $\times$ base = increase** — a 40% rise on 120 adds 48; a 10% rise on 900 adds 90.
5. **Necessary vs sufficient** — negate for necessary, add for sufficient.

TOP 5 TRAPS

1. **Extreme words.** "All", "never", "only", "no one" almost always break an assumption. An assumption is the **minimum** the argument needs.
2. **Percentage of what?** $96 \div 500 = 19.2\%$ and $96 \div 320 = 30\%$ are both true. Check the base.
3. **True but weak.** A true sentence is a weak argument if it has no size or applies equally to every option.
4. **After therefore because.** "B followed A" is not "A caused B". Hunt for the rival cause sitting in the same window.
5. **The group changed.** An average rises when the weakest leave. A rate falls when the denominator grows. Ask whether the thing measured is still the same thing.

Analogy, Classification & Non-Verbal Reasoning

Name the rule, then apply it to the new pair.

What you need to know

WHAT YOU NEED TO KNOW

Letter positions — learn these five anchors (EJOTY)

Letter	E	J	O	T	Y	A	M	N	Z
Position	5	10	15	20	25	1	13	14	26

- To find any letter fast: jump to the nearest anchor, then step. $R = T - 2 = 20 - 2 = 18$.
- Opposite-letter rule:** letter value + opposite value = 27. $A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, $M \leftrightarrow N$. To get the opposite of a letter with value v , compute $27 - v$.

Analogy — the 4 questions to ask, in this order

- Is the second term the first **shifted by a fixed number** of letters? (+1, +2, +3, -1 ...)
- Is it the **opposite letter** ($27 - v$)?
- Is it the first term **reversed**, or the letters **re-ordered**?
- For numbers: test n^2 , n^3 , $n^2 \pm c$, $n^3 \pm c$, $n(n+1)$, $n(n-1)$, $\frac{n(n+1)}{2}$, digit operations.

Classification (odd one out) — the 4 tests

- Same **number family**? (all squares, all cubes, all primes, all multiples of k)
- Same **gap pattern** inside each group? (BD, FH, JL all have gap 2)
- Same **category**? (metals vs alloy, cities vs river, tools vs product)
- Same **shape property**? (sides, symmetry, closed vs open)

Mirror image (a vertical mirror, placed to the right of the word)

- The whole word is **read in reverse order**, and each letter is flipped left $\leftrightarrow$ right.
- Block capitals unchanged by a vertical flip: **A H I M O T U V W X Y**.
- Digits unchanged by a vertical flip: **0** and **8**.
- A word looks **exactly the same** in a mirror only if (i) every letter is in the list above, and (ii) the word is a palindrome. Example: MUM, TOOT, MOM, WOW.
- Mirror image of a clock time: mirror time = $11 : 60 - \text{actual time}$ (use $23 : 60$ if the hour reads 12).

Water image (a horizontal mirror, placed below the word)

- The **order of letters does not change**. Each letter is flipped top $\leftrightarrow$ bottom.
- Block capitals unchanged by a horizontal flip: **B C D E H I K O X**.
- Digits unchanged by a horizontal flip: **0, 3, 8**.

Paper folding and punching

- Fold a sheet n times, then punch k holes clear of every fold line: holes on unfolding = $k \times 2^n$.
- A punch that sits **exactly on a fold line** gives only **half** as many holes (the two mirrored holes merge into one).
- A cut at the folded corner that was the **centre** of the sheet opens into **one** shape at the centre, not four.

Painted cube: side n , cut into n^3 unit cubes

Painted faces	Number of unit cubes
exactly 3 (corners)	8

exactly 2 (edges)	$12(n - 2)$
exactly 1 (faces)	$6(n - 2)^2$
exactly 0 (inside)	$(n - 2)^3$
at least 1	$n^3 - (n - 2)^3$

Check: $8 + 12(n - 2) + 6(n - 2)^2 + (n - 2)^3 = n^3$.

Painted cuboid $a \times b \times c$, cut into unit cubes. Write $p = a - 2$, $q = b - 2$, $r = c - 2$.

- 3 faces: 8 2 faces: $4(p + q + r)$
- 1 face: $2(pq + qr + rp)$ 0 faces: pqr

Dice

- Standard die: opposite faces add to 7. So $1 \leftrightarrow 6$, $2 \leftrightarrow 5$, $3 \leftrightarrow 4$.
- If a face is seen next to **four different** numbers across the views, the **fifth** unseen number is opposite it.
- Two views with the **same face in the same slot**: the die was turned about that axis. The face that moves into the right-hand slot is the face that was at the **back**, i.e. the opposite of the old front.
- Net rule: in a **straight strip of four faces**, every second face is opposite ($1\text{st} \leftrightarrow 3\text{rd}$, $2\text{nd} \leftrightarrow 4\text{th}$). Flaps hanging above and below the strip are opposite each other.

Counting figures

- Squares of all sizes in an $m \times n$ grid of unit squares: $\sum_{k=1}^{\min(m,n)} (m - k + 1)(n - k + 1)$.
- Squares in an $n \times n$ grid = $\frac{n(n+1)(2n+1)}{6}$. For $n = 8$: 204.
- Rectangles in an $m \times n$ grid = $\binom{m+1}{2} \times \binom{n+1}{2}$.
- Triangle with every side cut into n parts and lines drawn parallel to all three sides:
 - ▶ upward triangles of size k : $(n - k + 1)(n - k + 2) \div 2$
 - ▶ downward triangles of size k : $(n - 2k + 1)(n - 2k + 2) \div 2$ (only while $n \geq 2k$)
 - ▶ totals: $n = 1 \rightarrow 1$, $2 \rightarrow 5$, $3 \rightarrow 13$, $4 \rightarrow 27$, $5 \rightarrow 48$, $6 \rightarrow 78$
- Squares (including tilted ones) with corners on an $m \times m$ grid of **dots**: $\sum_{k=1}^{m-1} k(m - k)^2$. For $m = 5$: 50.
- Triangle ABC with n points marked on BC and all joined to A : number of triangles = $\binom{n+2}{2}$.

Symmetry of block capitals (used for figure odd-one-out)

- Vertical axis only: A M T U V W Y
- Horizontal axis only: B C D E K
- Both axes: H I O X
- Neither: F G J L N P Q R S Z

Warm-up

WARM-UP

build the reflex

Example 1

What is the position number of Q? Which letter sits at position 12?

SOLUTION

Anchor O = 15. Q is 2 letters after O, so $Q = 15 + 2 = 17$.

Anchor J = 10. Position 12 is 2 letters after J: $K = 11$, $L = 12$.

Answer: Q = 17; position 12 = L

Example 2

Find the opposite letter of F.

SOLUTION

F = 6. Opposite value = $27 - 6 = 21$. Anchor T = 20, so 21 = U.

Answer: U

Example 3

BD : FH :: JL : ?

SOLUTION

Inside each pair the gap is +2: B(2) D(4), F(6) H(8), J(10) L(12).

Across the pairs the first letters go B(2) → F(6) → J(10), a jump of +4.

Next first letter = $10 + 4 = 14 = \text{N}$. Second letter = $14 + 2 = 16 = \text{P}$.

Answer: NP

Example 4

6 : 42 :: 9 : ?

SOLUTION

Test $n(n + 1)$: $6 \times 7 = 42$. The rule fits.

So for 9: $9 \times 10 = 90$.

Answer: 90

Example 5

Odd one out: 25, 36, 48, 64.

SOLUTION

$25 = 5^2$, $36 = 6^2$, $64 = 8^2$. And $48: 6^2 = 36$, $7^2 = 49$, so 48 is not a perfect square.

Answer: 48

Example 6

Mirror image of MAT written in block capitals.

SOLUTION

Step 1 — reverse the order: T A M.

Step 2 — flip each letter left ↔ right. T, A and M are all in the vertical-symmetry list, so each one looks unchanged.

The image reads TAM.

Answer: TAM

Example 7

Water image of CODE written in block capitals.

SOLUTION

A water image does **not** reverse the order. It flips each letter top ↔ bottom.

C, O, D and E are all in the horizontal-symmetry list, so every letter looks unchanged.

Answer: CODE (it looks the same)

Example 8

A square sheet is folded in half 3 times, then one hole is punched clear of every fold. How many holes are there when the sheet is opened?

SOLUTION

Each fold doubles the number of layers: $1 \rightarrow 2 \rightarrow 4 \rightarrow 8$ layers.

The punch goes through all 8 layers, so $1 \times 2^3 = 8$ holes.

Answer: 8 holes

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

MASK : NBTL :: FIRE : ?

(a) GJSF

(b) GJSE

(c) GKSF

(d) FJSG

SOLUTION

Compare the letters one by one.

M(13) $\rightarrow$ N(14): +1. A(1) $\rightarrow$ B(2): +1. S(19) $\rightarrow$ T(20): +1. K(11) $\rightarrow$ L(12): +1.

The rule is +1 on every letter. Apply it to FIRE.

F(6) $\rightarrow$ 7 = G. I(9) $\rightarrow$ 10 = J. R(18) $\rightarrow$ 19 = S. E(5) $\rightarrow$ 6 = F.

Answer: (a) GJSF

Why the others are wrong: (b) GJSE forgets to shift the last letter. (c) GKSF uses +2 on the second letter. (d) FJSG shifts the wrong letters.

Example 10 [Infosys pattern]

CAT : XZG :: DOG : ?

SOLUTION

Test the opposite-letter rule ($27 - v$).

C = 3, $27 - 3 = 24 = X$. ✓

A = 1, $27 - 1 = 26 = Z$. ✓

T = 20, $27 - 20 = 7 = G$. ✓

The rule holds. Now DOG:

D = 4, $27 - 4 = 23 = W$.

O = 15, $27 - 15 = 12 = L$.

G = 7, $27 - 7 = 20 = T$.

Answer: WLT

SHORTCUT

Opposite letters in one second. Learn the five pairs $A \leftrightarrow Z$, $E \leftrightarrow V$, $I \leftrightarrow R$, $O \leftrightarrow L$, $U \leftrightarrow F$ (the vowels). Any other letter is only a step or two away from one of these. Check: $E = 5$, $27 - 5 = 22 = V$. ✓ $I = 9$, $27 - 9 = 18 = R$. ✓

Example 11 [Accenture pattern]

4 : 63 :: 6 : ?

SOLUTION

Test the cube family first. $4^3 = 64$, and $64 - 1 = 63$. So the rule is $n^3 - 1$.

For $n = 6$: $6^3 = 216$, so $216 - 1 = 215$.

Answer: 215

Example 12 [Wipro pattern]

12 : 78 :: 16 : ?

SOLUTION

$12^2 = 144$, too big. $12 \times 6 = 72$, close but not a clean rule.

Test the triangular-number rule $\frac{n(n+1)}{2}$:

$$\frac{12 \times 13}{2} = \frac{156}{2} = 78.$$

It fits.

For $n = 16$:

$$\frac{16 \times 17}{2} = \frac{272}{2} = 136.$$

Answer: 136

Example 13 [TCS NQT pattern]

Odd one out: 27, 64, 125, 196, 343.

SOLUTION

Test cubes: $3^3 = 27$, $4^3 = 64$, $5^3 = 125$, $7^3 = 343$. Four of the five are perfect cubes.

196: $5^3 = 125$ and $6^3 = 216$, so 196 is not a cube. (It is 14^2 , a square — that is the bait.)

Answer: 196

TRAP

64 belongs to two families. $64 = 8^2$ and $64 = 4^3$. In a list of cubes it is a cube; in a list of squares it is a square. Never mark 64 as the odd one out until you have tested every other number.

Example 14 [Capgemini pattern]

Odd one out: ACE, BDF, CEG, DGI.

SOLUTION

Write the gaps inside each group.

ACE: A(1) C(3) E(5) → gaps 2, 2.

BDF: B(2) D(4) F(6) → gaps 2, 2.

CEG: C(3) E(5) G(7) → gaps 2, 2.

DGI: D(4) G(7) I(9) → gaps 3, 2.

DGI breaks the pattern.

Answer: DGI

Example 15 [Cognizant pattern]

Doctor : Hospital :: Pilot : ?

(a) Airport

(b) Aeroplane

(c) Runway

(d) Passenger

SOLUTION

Name the relation in one sentence: **a doctor does his job inside a hospital.**

Now test each option with the same sentence.

- Airport — a pilot does not do his job inside the airport; he passes through it.
- Aeroplane — a pilot does his job inside the aeroplane. ✓
- Runway — that is where the plane moves, not where the pilot works.
- Passenger — that is a person, not a workplace.

Answer: (b) Aeroplane

SHORTCUT

Say the relation out loud in a full sentence before you look at the options. If your sentence works for exactly one option, that option is the answer. If it works for two, your sentence was too loose — make it tighter (“works inside”, not “is connected with”).

Example 16 [Accenture pattern]

Which word, in block capitals, looks **exactly the same** in a vertical mirror?

- (a) MOTOR (b) TOOT (c) HOUSE (d) WATER

SOLUTION

Two conditions must both hold: every letter must be vertically symmetric, and the word must read the same backwards.

MOTOR — R is not vertically symmetric. Fails test 1.

TOOT — T, O, O, T are all in the list A H I M O T U V W X Y. ✓ Reversed, TOOT reads TOOT. ✓

HOUSE — S and E are not vertically symmetric. Fails test 1.

WATER — R and E are not vertically symmetric. Fails test 1.

Answer: (b) TOOT

Example 17 [Infosys pattern]

Which word, in block capitals, is unchanged in its water image?

- (a) BOOK (b) CHAIR (c) DEEP (d) HOUSE

SOLUTION

A water image keeps the order of the letters. So we only need every letter to be horizontally symmetric. The list is B C D E H I K O X.

BOOK — B ✓, O ✓, O ✓, K ✓. All in the list.

CHAIR — A and R are not in the list.

DEEP — P is not in the list.

HOUSE — U and S are not in the list.

Answer: (a) BOOK

TRAP

Mirror reverses the order; water does not. Students lose easy marks by reversing the letters for a water image. Vertical mirror → read backwards. Water → same order, letters flipped top to bottom.

Example 18 [TCS NQT pattern]

A square sheet is folded in half twice (left over right, then top over bottom). Two holes are punched, both clear of every fold line. How many holes appear when the sheet is opened?

SOLUTION

Layers after fold 1: 2. Layers after fold 2: $2 \times 2 = 4$.

Each punch pierces all 4 layers, so each punch gives 4 holes.

Two punches give $2 \times 4 = 8$ holes.

Using the formula: $k \times 2^n = 2 \times 2^2 = 2 \times 4 = 8$.

Answer: 8 holes

Example 19 [Wipro pattern]

A die is shown in two positions. In position 1 the top is 1, the front is 2 and the right is 3. In position 2 the top is 1, the front is 3 and the right is 4. Which number is opposite 2?

SOLUTION

The top face is 1 in both positions, so the die was only spun about the vertical axis.

In position 2 the front face is 3 — and 3 was the **right** face in position 1. So the die was turned a quarter turn, bringing the right face round to the front.

With that same quarter turn, the face that moves into the right-hand slot is the face that was at the **back**.

The back face in position 1 is the face opposite the front face, which was 2.

In position 2 the right face is 4. So the back face in position 1 was 4.

Therefore 4 is opposite 2.

Answer: 4

Example 20 [Cognizant pattern]

How many squares of all sizes are there in a 4×4 grid of unit squares?

SOLUTION

Count size by size.

Size 1×1 : $4 \times 4 = 16$.

Size 2×2 : the top-left corner can sit in 3 columns and 3 rows, so $3 \times 3 = 9$.

Size 3×3 : $2 \times 2 = 4$.

Size 4×4 : $1 \times 1 = 1$.

Total = $16 + 9 + 4 + 1 = 30$.

Answer: 30 squares

SHORTCUT

Squares in an $n \times n$ grid are just $1^2 + 2^2 + \dots + n^2$, read backwards. So a 5×5 grid has $25 + 16 + 9 + 4 + 1 = 55$, and an 8×8 chessboard has 204. Learn 30, 55, 91, 140, 204 for $n = 4, 5, 6, 7, 8$.

PRACTICE · TIER 1 target: 60 s/Q

1. RT : SU :: LN : ?
2. 9 : 90 :: 12 : ?
3. Odd one out: 49, 81, 100, 128.
4. What is the opposite letter of M?

5. Which word, in block capitals, looks exactly the same in a vertical mirror? (a) MUM (b) NOON (c) DOOR (d) LOOP
6. Write the water image of CHOICE (block capitals).
7. A sheet is folded in half 4 times and one hole is punched clear of every fold. How many holes on unfolding?
8. A standard die shows 4 on top. What is on the bottom face?
9. How many squares of all sizes are there in a 3×3 grid of unit squares?
10. How many rectangles of all sizes are there in a 2×3 grid of unit squares?
11. $AZ : BY :: CX : ?$
12. $7 : 48 :: 10 : ?$
13. Odd one out: BD, DF, FH, HK.
14. How many rectangles of all sizes are there in a 3×3 grid of unit squares?
15. A clock reads 3:40. What time does its mirror image show?

ANSWER KEY

1. **MO.** $R(18)T(20) \rightarrow S(19)U(21)$ is $+1$ on each letter; $L(12)N(14) \rightarrow M(13)O(15)$.
2. **156.** Rule $n(n+1)$: $9 \times 10 = 90$, so $12 \times 13 = 156$.
3. **128.** $49 = 7^2$, $81 = 9^2$, $100 = 10^2$; $11^2 = 121$ and $12^2 = 144$, so 128 is not a square.
4. **N.** $M = 13$; $27 - 13 = 14 = N$.
5. **(a) MUM.** M and U are vertically symmetric and MUM is a palindrome. NOON fails because N is not symmetric; DOOR and LOOP fail on R and P and are not palindromes.
6. **CHOICE.** Every letter (C, H, O, I, C, E) is horizontally symmetric, and the order does not change.
7. **16 holes.** $1 \times 2^4 = 16$.
8. **3.** Opposite faces of a standard die add to 7, and $7 - 4 = 3$.
9. **14.** $9 + 4 + 1 = 14$.
10. **18.** $\binom{3}{2} \times \binom{4}{2} = 3 \times 6 = 18$.
11. **DW.** $A(1)Z(26) \rightarrow B(2)Y(25)$; first letter $+1$, second letter -1 . $C(3)X(24) \rightarrow D(4)W(23)$.
12. **99.** Rule $n^2 - 1$: $7^2 - 1 = 48$, so $10^2 - 1 = 99$.
13. **HK.** The gaps are $BD = 2$, $DF = 2$, $FH = 2$, but $HK = 3$ (H is 8, K is 11).
14. **36.** $\binom{4}{2} \times \binom{4}{2} = 6 \times 6 = 36$.
15. **8:20.** Mirror time = $11 : 60 - 3 : 40 = 8 : 20$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A cube of side 5 cm is painted red on all six faces and then cut into 125 cubes of side 1 cm. How many of the small cubes have exactly 3, exactly 2, exactly 1, and no painted faces?

SOLUTION

Here $n = 5$, so $n - 2 = 3$.

3 painted faces — only the corners of the big cube: 8.

2 painted faces — the cubes along the edges but not at the corners. Each edge has $n - 2 = 3$ such cubes, and there are 12 edges: $12 \times 3 = 36$.

1 painted face — the cubes in the middle of each face. Each face has a $(n - 2) \times (n - 2) = 3 \times 3 = 9$ block of them, and there are 6 faces: $6 \times 9 = 54$.

0 painted faces — the solid block hidden inside: $(n - 2)^3 = 3^3 = 27$.

Check the total: $8 + 36 + 54 + 27 = 125$. ✓

Answer: 3 faces: 8; 2 faces: 36; 1 face: 54; 0 faces: 27

Example 22 [Shopee · pattern]

A wooden block measuring $6 \times 4 \times 3$ units is painted on every outside face and cut into 72 unit cubes. How many unit cubes have exactly 2 painted faces, and how many have none?

SOLUTION

Write $p = 6 - 2 = 4$, $q = 4 - 2 = 2$, $r = 3 - 2 = 1$.

Exactly 2 faces (edge cubes that are not corners):

$$4(p + q + r) = 4(4 + 2 + 1) = 4 \times 7 = 28.$$

No paint (the inner block):

$$pqr = 4 \times 2 \times 1 = 8.$$

Check with the other two counts. One face: $2(pq + qr + rp) = 2(8 + 2 + 4) = 2 \times 14 = 28$. Three faces: 8.

Total: $8 + 28 + 28 + 8 = 72$. ✓

Answer: 28 cubes with exactly 2 faces; 8 cubes with no paint

TRAP

A “2” in a dimension kills the inner block. If any side of a cuboid is 1 or 2 units, then p , q or r is 0 or negative, and the number of unpainted cubes is **zero**. Always compute pqr before you answer, never assume it is positive.

Example 23 [Agoda · pattern]

A cube of side 4 cm stands on a table. Every face **except the bottom** is painted. The cube is cut into 64 unit cubes. How many have no paint at all, and how many have exactly one painted face?

SOLUTION

Give each unit cube coordinates (x, y, z) with $x, y, z \in \{1, 2, 3, 4\}$ and $z = 1$ at the table.

The painted outer surfaces are $x = 1$, $x = 4$, $y = 1$, $y = 4$ and $z = 4$ (the top). The bottom $z = 1$ is **not** painted.

No paint. The cube must avoid all five painted surfaces. So $x \in \{2, 3\}$, $y \in \{2, 3\}$, and $z \in \{1, 2, 3\}$ (any layer except the top).

$$2 \times 2 \times 3 = 12.$$

Exactly one painted face. Count the two kinds separately.

Kind A — on one side wall only. Take $x = 1$. To avoid the other walls and the top: $y \in \{2, 3\}$ (2 ways) and $z \in \{1, 2, 3\}$ (3 ways), giving $2 \times 3 = 6$. There are 4 side walls, so $4 \times 6 = 24$.

Kind B — on the top only. $z = 4$, with $x \in \{2, 3\}$ and $y \in \{2, 3\}$: $2 \times 2 = 4$.

Total with exactly one painted face = $24 + 4 = 28$.

Answer: 12 with no paint; 28 with exactly one painted face

For completeness: exactly 2 faces = 20 and exactly 3 faces = 4 (the four **top** corners only — the four bottom corners touch just two painted walls). Check: $12 + 28 + 20 + 4 = 64$. ✓

Example 24 [DBS · pattern]

A die is shown in two positions. Position 1: top 5, front 1, right 2. Position 2: top 3, front 1, right 5. Which number is opposite 2?

SOLUTION

The **front** face is 1 in both positions. So the die was rolled about the front-to-back axis, like a wheel seen face-on.

In position 2, the face 5 sits on the right — and 5 was on the **top** in position 1. So the die rolled a quarter turn clockwise as seen from the front (top $\rightarrow$ right).

Under that same quarter turn, the face that moves into the **top** slot is the face that was on the **left**.
 The left face in position 1 is the face opposite the right face, which was 2.
 In position 2 the top is 3. So the left face in position 1 was 3.
 Therefore 3 is opposite 2.

Answer: 3

SHORTCUT

One common face, two positions — use the “carousel” rule. Fix the repeated face. The three remaining visible slots rotate in a ring. Whichever face steps into a slot came from the slot **before** it in the ring; the face that steps in from the hidden side is the **opposite** of the face that just left the ring.

Example 25 [Sea · pattern]

In a 4×5 grid of unit squares, how many rectangles of all sizes are there that are **not** squares?

SOLUTION

Step 1 — all rectangles. A rectangle is fixed by choosing 2 of the 5 horizontal lines and 2 of the 6 vertical lines.

$$\binom{5}{2} \times \binom{6}{2} = 10 \times 15 = 150.$$

Step 2 — the squares. Count size by size (4 rows, 5 columns).

Size 1: $4 \times 5 = 20$. Size 2: $3 \times 4 = 12$. Size 3: $2 \times 3 = 6$. Size 4: $1 \times 2 = 2$.

Squares = $20 + 12 + 6 + 2 = 40$.

Step 3 — subtract. $150 - 40 = 110$.

Answer: 110

Example 26 [GIC · pattern]

$(4, 9, 25) : (2, 3, 5) :: (36, 64, 121) : ?$

SOLUTION

Look at the first triple against the second.

$4 \rightarrow 2, 9 \rightarrow 3, 25 \rightarrow 5$. Each output is the **square root** of the input.

Apply the same rule.

$\sqrt{36} = 6, \sqrt{64} = 8, \sqrt{121} = 11$.

Answer: (6, 8, 11)

Example 27 [SCB · pattern]

The value of a word is the sum of the position numbers of its letters. Given MATH : 42 :: FACE : ?, find the missing value. Then say which of CAB, BAD, ACE — if any — has the same value as FACE.

SOLUTION

Check the given pair. M = 13, A = 1, T = 20, H = 8.

$$13 + 1 + 20 + 8 = 42. \checkmark$$

Apply to FACE. F = 6, A = 1, C = 3, E = 5.

$$6 + 1 + 3 + 5 = 15.$$

Now the three test words.

CAB = $3 + 1 + 2 = 6$.

BAD = $2 + 1 + 4 = 7$.

$$ACE = 1 + 3 + 5 = 9.$$

None equals 15. The closest, ACE, is short by 6.

Answer: FACE = 15; none of CAB (6), BAD (7), ACE (9) matches it

Example 28 [LINE MAN · pattern]

A square sheet is folded left over right, then top over bottom. A small square notch is cut from the corner of the folded packet that was the **centre** of the original sheet. What is seen when the sheet is opened?

SOLUTION

After the two folds the packet is one quarter of the sheet. Its four corners come from four different places:

- one corner is the **centre** of the original sheet,
- two corners are **midpoints** of the sheet's edges,
- one corner is a **corner** of the sheet.

The notch is cut at the centre corner. All 4 layers are cut at once, and all 4 of those quarter-notches meet at the same point — the centre.

So the four quarter-holes join into **one** hole, sitting exactly at the centre of the sheet. Its side is twice the side of the notch that was cut.

Answer: A single square hole at the centre of the sheet, of side twice the cut notch

TRAP

A cut at the centre corner does not give four holes. Cuts at that corner **merge**. Cuts at the far corner (a real corner of the sheet) give four separate holes, one in each corner. Always ask: does this corner of the packet sit on a fold line?

Example 29 [Razer · pattern]

In a figure series, figure 1 is a triangle containing 1 dot, with an arrow pointing north. In each next figure the polygon gains one side, the number of dots increases by 2, and the arrow turns 90° clockwise. Describe figure 5.

SOLUTION

Treat each attribute as its own little series, and use $n = 5$, so there are $5 - 1 = 4$ steps from figure 1.

Sides: start 3, add 1 each step. $3 + 4 \times 1 = 7 \rightarrow$ a heptagon.

Dots: start 1, add 2 each step. $1 + 4 \times 2 = 1 + 8 = 9$ dots.

Arrow: start north, turn 90° clockwise each step. Total turn = $4 \times 90^\circ = 360^\circ$, which is one full turn. The arrow points north again.

Answer: A heptagon (7 sides) with 9 dots inside, arrow pointing north

SHORTCUT

Split a figure series into separate columns, one per attribute. Shape, count of elements, shading, rotation, position. Solve each column on its own like a number series. Never try to read the whole picture at once.

Example 30 [Grab · pattern]

An equilateral triangle has each side divided into 3 equal parts, and lines are drawn parallel to all three sides. How many triangles of all sizes are there?

SOLUTION

Count upward triangles and downward triangles separately. Here $n = 3$.

Upward, size 1: the small triangles sit in 3 rows. Row 1 holds 1, row 2 holds 2, row 3 holds 3. Total $1 + 2 + 3 = 6$.

Upward, size 2: such a triangle needs 2 rows of room, so its apex can only sit in the top $3 - 2 + 1 = 2$ rows. Row 1 holds 1 apex, row 2 holds 2. Total $1 + 2 = 3$.

Upward, size 3: the whole triangle: 1.

Upward total $= 6 + 3 + 1 = 10$.

Downward, size 1: the inverted unit triangles fill the gaps between the upward ones. Row 2 holds 1, row 3 holds 2. Total $1 + 2 = 3$.

Downward, size 2: such a triangle needs $2 \times 2 = 4$ rows, but there are only 3. So none.

Downward total $= 3$.

Grand total $= 10 + 3 = 13$.

Answer: 13 triangles

Same answer straight from the formulas. Upward, $k = 1, 2, 3$: $\frac{3 \times 4}{2} + \frac{2 \times 3}{2} + \frac{1 \times 2}{2} = 6 + 3 + 1 = 10$. Downward, $k = 1$: $\frac{(3-2+1)(3-2+2)}{2} = \frac{2 \times 3}{2} = 3$. Total $= 13$.

PRACTICE · TIER 2 target: 100 s/Q

1. A cube of side 5 is painted on all faces and cut into 125 unit cubes. How many have exactly one painted face?
2. A block $4 \times 3 \times 2$ is painted on all faces and cut into 24 unit cubes. How many have exactly two painted faces?
3. A die is shown twice. Position 1: top 2, front 4, right 5. Position 2: top 2, front 5, right 3. Which number is opposite 4?
4. How many squares of all sizes are in a 6×6 grid of unit squares?
5. In a 4×4 grid of unit squares, how many rectangles are not squares?
6. GATE : HBUF :: MOCK : ?
7. 1234 : 2468 :: 3102 : ?
8. A square sheet is folded left over right, then top over bottom. A small square is cut at the corner of the packet that was the centre of the sheet. Describe what is seen on unfolding.
9. A clock reads 9:45. What time does its mirror image show?
10. Odd one out among block capitals by symmetry: A, M, T, H.
11. A triangle has each side divided into 4 equal parts, with lines parallel to all three sides. How many triangles of all sizes?
12. Figure 1 is a square with 2 dots and an arrow pointing north. Each next figure adds 3 dots and turns the arrow 45° clockwise. Describe figure 6.

ANSWER KEY

1. **54.** $6(n-2)^2 = 6 \times 3^2 = 54$.
2. **12.** $p = 2, q = 1, r = 0$; $4(p+q+r) = 4 \times 3 = 12$. (Note $pqr = 0$, so **no** cube is unpainted.)
3. **3.** The top stays 2, so the die spun about the vertical axis; the new front (5) was the old right, so the new right (3) was the old back, which is the opposite of the old front 4.
4. **91.** $36 + 25 + 16 + 9 + 4 + 1 = 91$.
5. **70.** Rectangles $= \binom{5}{2}^2 = 100$; squares $= 16 + 9 + 4 + 1 = 30$; $100 - 30 = 70$.
6. **NPDL.** Every letter shifts +1: M $\rightarrow$ N, O $\rightarrow$ P, C $\rightarrow$ D, K $\rightarrow$ L.
7. **6204.** Each digit is doubled: $3 \rightarrow 6, 1 \rightarrow 2, 0 \rightarrow 0, 2 \rightarrow 4$.
8. **One square hole at the centre,** of side twice the cut. The four quarter-cuts meet at the centre and merge.
9. **2:15.** $11 : 60 - 9 : 45 = 2 : 15$.
10. **H.** A, M and T have a vertical axis of symmetry only; H has both a vertical and a horizontal axis.
11. **27.** Upward: $10 + 6 + 3 + 1 = 20$; downward: $6 + 1 = 7$; total 27.
12. **A square with 17 dots, arrow pointing south-west.** Dots $= 2 + 5 \times 3 = 17$; turn $= 5 \times 45^\circ = 225^\circ$ clockwise from north, which points south-west. (The shape does not change — only two attributes were given a rule.)

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Google · pattern]

A cube of side n (an integer, $n > 2$) is painted on all six faces and cut into n^3 unit cubes. For which n is the number of unit cubes with exactly **one** painted face equal to the number with **no** painted face?

SOLUTION

Insight: both counts are powers of the same quantity $n - 2$, so the equation collapses to a single power.

One painted face: $6(n - 2)^2$.

No painted face: $(n - 2)^3$.

Set them equal:

$$6(n - 2)^2 = (n - 2)^3.$$

Since $n > 2$, $(n - 2)^2 \neq 0$, so divide both sides by $(n - 2)^2$:

$$6 = n - 2 \Rightarrow n = 8.$$

Check with $n = 8$.

One face: $6(8 - 2)^2 = 6 \times 36 = 216$.

No face: $(8 - 2)^3 = 6^3 = 216$. ✓

Answer: $n = 8$

Example 32 [Amazon · pattern]

A cube of side n is painted on all six faces and cut into n^3 unit cubes. One unit cube is picked at random. What is the expected number of painted faces on it? Evaluate for $n = 5$.

SOLUTION

Insight: do not classify the cubes at all. Count painted unit-squares on the whole surface, then divide.

The big cube has 6 faces. Each face is an $n \times n$ square, so it carries n^2 painted unit-squares.

Total painted unit-squares = $6n^2$.

Every one of those unit-squares sits on exactly one unit cube, so the **sum** of painted faces over all n^3 unit cubes is $6n^2$.

$$\text{Expected value} = \frac{\text{total}}{\text{number of cubes}} = \frac{6n^2}{n^3} = \frac{6}{n}.$$

For $n = 5$: $\frac{6}{5} = 1.2$.

Answer: $\frac{6}{n}$; for $n = 5$ it is 1.2 painted faces

Sanity check for $n = 3$: $\frac{6}{3} = 2$. By direct count: $8 \times 3 + 12 \times 2 + 6 \times 1 + 1 \times 0 = 24 + 24 + 6 = 54$, and $54 \div 27 = 2$. ✓

SHORTCUT

Counting the surface beats classifying the pieces. Any question of the form “total painted faces” or “average painted faces” is answered by $6n^2$ in one line. Only questions that ask for a **specific class** (exactly 2, exactly 0) need the four formulas.

Example 33 [Goldman Sachs · pattern]

On a standard 8×8 chessboard, how many rectangles can be drawn along the grid lines, and how many of them are **not** squares?

SOLUTION

Insight: a rectangle is not a shape you hunt for — it is a choice of 2 horizontal lines and 2 vertical lines.

The board has $8 + 1 = 9$ horizontal lines and 9 vertical lines.

All rectangles.

$$\binom{9}{2} \times \binom{9}{2} = 36 \times 36 = 1296.$$

(Check $\binom{9}{2} = \frac{9 \times 8}{2} = 36$.)

Squares. A $k \times k$ square has its top-left corner in $(9 - k)$ positions across and $(9 - k)$ down, so there are $(9 - k)^2$ of them.

$k = 1 : 64, k = 2 : 49, k = 3 : 36, k = 4 : 25, k = 5 : 16, k = 6 : 9, k = 7 : 4, k = 8 : 1.$

Sum = $64 + 49 = 113; 113 + 36 = 149; 149 + 25 = 174; 174 + 16 = 190; 190 + 9 = 199; 199 + 4 = 203; 203 + 1 = 204.$

Non-squares. $1296 - 204 = 1092.$

Answer: 1296 rectangles in all; 1092 of them are not squares

Example 34 [Microsoft · pattern]

An equilateral triangle has each side divided into 6 equal parts, and lines are drawn parallel to all three sides. How many triangles of all sizes are there?

SOLUTION

Insight: upward triangles and downward triangles obey two different counting laws. Count them apart, then add.

Upward triangles of size k . Their apexes form a triangular block with $n - k + 1$ rows, so the count is $\frac{(n-k+1)(n-k+2)}{2}$ with $n = 6$.

$$k = 1 : \frac{6 \times 7}{2} = 21$$

$$k = 2 : \frac{5 \times 6}{2} = 15$$

$$k = 3 : \frac{4 \times 5}{2} = 10$$

$$k = 4 : \frac{3 \times 4}{2} = 6$$

$$k = 5 : \frac{2 \times 3}{2} = 3$$

$$k = 6 : \frac{1 \times 2}{2} = 1$$

Upward total = $21 + 15 + 10 + 6 + 3 + 1 = 56.$

Downward triangles of size k . A downward triangle of size k needs $2k$ rows of room, so the count is $\frac{(n-2k+1)(n-2k+2)}{2}$, and only while $n \geq 2k$.

$$k = 1 : \frac{5 \times 6}{2} = 15$$

$$k = 2 : \frac{3 \times 4}{2} = 6$$

$$k = 3 : \frac{1 \times 2}{2} = 1$$

Downward total = $15 + 6 + 1 = 22.$

Grand total = $56 + 22 = 78.$

Answer: 78 triangles

TRAP

Downward triangles are the ones everybody forgets. For $n = 6$ they are 22 out of 78 — more than a quarter of the answer. Write two separate columns on your rough sheet before you count anything.

Example 35 [Adobe · pattern]

A square sheet is folded in half 4 times. Three holes are then punched. Two of them are clear of every fold line; the third sits exactly on the last fold line. How many holes appear when the sheet is opened?

SOLUTION

Insight: a punch on a fold line pierces the same number of layers, but the unfolded holes come in touching pairs that merge into one.

After 4 folds the packet has $2^4 = 16$ layers.

Punch A (clear of folds): pierces 16 layers, and the 16 holes land at 16 separate points. $\rightarrow$ 16 holes.

Punch B (clear of folds): the same. $\rightarrow$ 16 holes.

Punch C (on a fold line): it still pierces 16 layers. But the fold line is a mirror line, so the holes come out in 8 mirror-pairs, and each pair sits astride the crease, touching. Each pair opens as a **single** hole. $\rightarrow$ 8 holes.

Total = $16 + 16 + 8 = 40$.

Answer: 40 holes

Example 36 [Uber · pattern]

A cube net is laid flat as a straight strip of four faces A, B, C, D (in that order). Face E is attached above C, and face F is attached below A. List the three pairs of opposite faces, and name every face that touches face B.

SOLUTION

Insight: folding a straight strip of four faces wraps a belt around the cube, so faces two apart in the strip end up back to back.

The belt. A, B, C, D fold into a ring around the cube. Going round the ring, A sits next to B and D; C also sits next to B and D. A face is opposite whatever it does **not** sit next to, so:

$$A \text{ opposite } C, \quad B \text{ opposite } D.$$

The flaps. The belt leaves exactly two faces open — the top and the bottom. E and F are the only faces left, so they must fill those two. Therefore

$$E \text{ opposite } F.$$

(It does not matter that E hangs off C and F hangs off A. Once the belt is wrapped, any flap on the strip folds onto the top or the bottom, and the two flaps cannot land on the same one.)

Faces touching B. Every face of a cube touches all four faces except its own opposite. B's opposite is D. So B touches A, C, E and F.

Answer: Opposite pairs: A–C, B–D, E–F. Face B touches A, C, E and F.

Example 37 [D. E. Shaw · pattern]

A cube of side n is painted and cut into n^3 unit cubes. What is the smallest n for which **more than half** of the unit cubes have no paint at all?

SOLUTION

Insight: the unpainted fraction is $(1 - \frac{2}{n})^3$, so the question is about one cube root, not about n itself.

$$\text{Unpainted fraction} = \frac{(n-2)^3}{n^3} = \left(1 - \frac{2}{n}\right)^3.$$

We need

$$\left(1 - \frac{2}{n}\right)^3 > \frac{1}{2}.$$

Take cube roots. $\sqrt[3]{\frac{1}{2}} \approx 0.7937$.

So we need $1 - \frac{2}{n} > 0.7937$, that is $\frac{2}{n} < 0.2063$, that is $n > \frac{2}{0.2063} \approx 9.69$.

The smallest integer is $n = 10$.

Check $n = 10$: unpainted = $8^3 = 512$ out of 1000. $\frac{512}{1000} = 0.512 > 0.5$. ✓

Check $n = 9$: unpainted = $7^3 = 343$ out of 729. $\frac{343}{729} \approx 0.470 < 0.5$. (too small)

Answer: $n = 10$

Example 38 [Google · pattern]

Dots are arranged in a 5×5 square grid. How many squares can be formed with all four corners on grid dots, counting **tilted** squares as well?

SOLUTION

Insight: every tilted square sits snugly inside exactly one axis-aligned square whose corners are also grid dots. So group the squares by that bounding box.

Take a bounding box of side k (measured in grid steps). Inside it you can draw:

- the box itself (1 square), and
- a tilted square for each way of sliding the corner along the box's side: the corner can sit at distance 1, 2, ..., $k - 1$ from the box's own corner.

So each $k \times k$ bounding box holds exactly k squares.

How many $k \times k$ bounding boxes are there? On a 5×5 dot grid, the top-left dot of a $k \times k$ box has $(5 - k)$ choices across and $(5 - k)$ down, so $(5 - k)^2$ boxes.

Now total up:

$$\sum_{k=1}^4 k(5 - k)^2.$$

$$k = 1 : 1 \times 4^2 = 1 \times 16 = 16$$

$$k = 2 : 2 \times 3^2 = 2 \times 9 = 18$$

$$k = 3 : 3 \times 2^2 = 3 \times 4 = 12$$

$$k = 4 : 4 \times 1^2 = 4 \times 1 = 4$$

$$\text{Total} = 16 + 18 + 12 + 4 = 50.$$

Of these, the axis-aligned ones are $16 + 9 + 4 + 1 = 30$, so 20 squares are tilted.

Answer: 50 squares (30 straight, 20 tilted)

SHORTCUT

“Squares on a dot grid” versus “squares in a grid of cells”. If the question shows **dots**, tilted squares count and the answer uses $\sum k(m - k)^2$. If it shows a **grid of little boxes**, only straight squares count and the answer uses $\sum (m - k + 1)(n - k + 1)$. Read which one is drawn before you start.

PRACTICE · TIER 3 target: 4 min/Q

1. A cube of side n is painted and cut into n^3 unit cubes. For which n is the number with no paint exactly 3 times the number with exactly 2 painted faces?
2. A cube of side 6 is painted and cut into 216 unit cubes. One is picked at random. What is the expected number of painted faces on it?
3. In a 6×6 grid of unit squares, how many rectangles are not squares?

- A triangle has each side divided into 5 equal parts, with lines parallel to all three sides. How many triangles of all sizes?
- A square sheet is folded in half 3 times. Two holes are punched: one clear of every fold, one exactly on the last fold line. How many holes on unfolding?
- Odd one out: 1, 5, 14, 30, 55, 90.
- Dots are arranged in a 4×4 square grid. How many squares (including tilted ones) have all four corners on grid dots?
- A cube is painted with 3 colours, each colour covering one pair of opposite faces. It is cut into 64 unit cubes. How many unit cubes show 3 different colours?

ANSWER KEY

- $n = 8$. We need $(n - 2)^3 = 3 \times 12(n - 2) = 36(n - 2)$. Divide by $(n - 2)$ (non-zero since $n > 2$): $(n - 2)^2 = 36$, so $n - 2 = 6$ and $n = 8$. Check: no paint = $6^3 = 216$; exactly 2 faces = $12 \times 6 = 72$; $3 \times 72 = 216$. ✓
1. Total painted unit-squares = $6 \times 6^2 = 216$. Divide by 216 unit cubes: $\frac{216}{216} = 1$. (Same as $\frac{6}{n} = \frac{6}{6} = 1$.)
350. All rectangles = $\binom{7}{2}^2 = 21^2 = 441$. Squares = $36 + 25 + 16 + 9 + 4 + 1 = 91$. $441 - 91 = 350$.
48. Upward: $15 + 10 + 6 + 3 + 1 = 35$. Downward ($n = 5$): $k = 1 : \frac{4 \times 5}{2} = 10$; $k = 2 : \frac{2 \times 3}{2} = 3$. Downward = 13. Total $35 + 13 = 48$.
12. Three folds give $2^3 = 8$ layers. The free punch gives 8 holes; the punch on the fold gives 8 pierced layers whose holes merge in mirror-pairs, giving 4 holes. $8 + 4 = 12$.
90. The list is the count of squares in a $k \times k$ grid: 1, 5, 14, 30, 55, 91. The sixth term should be 91, not 90.
20. $\sum_{k=1}^3 k(4 - k)^2 = 1 \times 9 + 2 \times 4 + 3 \times 1 = 9 + 8 + 3 = 20$. (Straight: 14; tilted: 6.)
8. Only the 8 corner cubes have 3 painted faces. The three faces meeting at a corner are mutually adjacent, so they belong to three **different** opposite-pairs and therefore carry three different colours. No other unit cube has 3 painted faces at all.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 30 min for all 18

- CAT : XZG :: BAT : ?
- A cube of side 3 is painted on all faces and cut into 27 unit cubes. How many have no paint?
- Odd one out: 121, 144, 169, 180.
- How many squares of all sizes are in a 5×5 grid of unit squares?
- Write the mirror image of MOM (block capitals).
- A sheet is folded in half 5 times and one hole is punched clear of every fold. How many holes on unfolding?
- 6 : 216 :: 9 : ?
- How many rectangles of all sizes are in a 3×4 grid of unit squares?
- A die is shown twice. Position 1: top 6, front 1, right 2. Position 2: top 6, front 2, right 4. Which number is opposite 1?
- Write the water image of BIKE (block capitals).
- PRIME : QSJNF :: SOLID : ?
- A block $5 \times 4 \times 3$ is painted on all faces and cut into 60 unit cubes. How many have no paint?
- Odd one out: ACE, EGI, IKM, MOR.
- A triangle has each side divided into 2 equal parts, with lines parallel to all three sides. How many triangles of all sizes?
- A clock reads 7:15. What time does its mirror image show?
- Dots form a 4×4 square grid. How many squares, including tilted ones, have all four corners on dots?
- A cube of side n is painted and cut into n^3 unit cubes. For which n are the “exactly 2 faces” and “exactly 1 face” counts equal?
- 15 : 240 :: 18 : ?

ANSWER KEY

1. **YZG.** Opposite-letter rule: $B(2) \rightarrow 27 - 2 = 25 = Y$; $A \rightarrow Z$; $T(20) \rightarrow 7 = G$.
2. **1.** $(3 - 2)^3 = 1^3 = 1$.
3. **180.** $121 = 11^2$, $144 = 12^2$, $169 = 13^2$; $13^2 = 169$ and $14^2 = 196$, so 180 is not a square.
4. **55.** $25 + 16 + 9 + 4 + 1 = 55$.
5. **MOM.** M and O are vertically symmetric and MOM is a palindrome, so the image is identical.
6. **32.** $1 \times 2^5 = 32$.
7. **729.** Rule n^3 : $6^3 = 216$, so $9^3 = 729$.
8. **60.** $\binom{4}{2} \times \binom{5}{2} = 6 \times 10 = 60$.
9. **4.** Top stays 6, so the die spun about the vertical axis. The new front (2) was the old right, so the new right (4) was the old back — the opposite of the old front 1.
10. **BIKE.** B, I, K and E are all horizontally symmetric, and a water image keeps the order.
11. **TPMJE.** Every letter shifts +1: $S \rightarrow T$, $O \rightarrow P$, $L \rightarrow M$, $I \rightarrow J$, $D \rightarrow E$.
12. **6.** $p = 3$, $q = 2$, $r = 1$; $pqr = 3 \times 2 \times 1 = 6$.
13. **MOR.** Gaps are 2, 2 in ACE, EGI and IKM; MOR has M(13) O(15) R(18), gaps 2, 3.
14. **5.** Upward: $3 + 1 = 4$; downward: 1. Total 5.
15. **4:45.** $11 : 60 - 7 : 15 = 4 : 45$.
16. **20.** $\sum_{k=1}^3 k(4 - k)^2 = 9 + 8 + 3 = 20$.
17. **$n = 4$.** $12(n - 2) = 6(n - 2)^2 \rightarrow 2 = n - 2 \rightarrow n = 4$. Check: $12 \times 2 = 24$ and $6 \times 2^2 = 24$. ✓
18. **342.** Rule $n(n + 1)$: $15 \times 16 = 240$, so $18 \times 19 = 342$.

ONE-PAGE REVISION CARD

Letters. EJOTY: E 5, J 10, O 15, T 20, Y 25. Opposite letter: $27 - v$. Vowel pairs A ↔ Z, E ↔ V, I ↔ R, O ↔ L, U ↔ F.

Analogy order of tests. fixed shift → opposite letter → reversal / re-order → number family ($n^2, n^3, n^2 \pm c, n^3 \pm c, n(n+1), \frac{n(n+1)}{2}$) → digit operation.

Classification tests. number family / gap pattern / category / shape property.

Mirror (vertical). Reverse the order, flip each letter side to side. Unchanged letters A H I M O T U V W X Y; digits 0, 8. Clock: 11 : 60 – time.

Water (horizontal). Keep the order, flip each letter top to bottom. Unchanged letters B C D E H I K O X; digits 0, 3, 8.

Symmetry of capitals. Vertical only: A M T U V W Y. Horizontal only: B C D E K. Both: H I O X.

Paper folding. n folds, k free punches → $k \times 2^n$ holes. A punch on a fold line gives half as many. A cut at the packet corner that was the **centre** opens as one shape at the centre.

Painted cube, side n . 3 faces = 8; 2 faces = $12(n-2)$; 1 face = $6(n-2)^2$; 0 faces = $(n-2)^3$; at least one = $n^3 - (n-2)^3$. Expected painted faces = $\frac{6}{n}$.

Painted cuboid, $p = a - 2, q = b - 2, r = c - 2$: 3 faces = 8; 2 faces = $4(p + q + r)$; 1 face = $2(pq + qr + rp)$; 0 faces = pqr .

Dice. Standard die: opposite faces sum to 7. Seen next to four different numbers → the fifth is opposite. Two views sharing a slot: the face entering the right-hand slot came from the back, i.e. opposite the old front. Straight strip of four in a net: 1st ↔ 3rd, 2nd ↔ 4th; the two flaps are opposite each other.

Counting figures.

- Squares in $m \times n$ cells = $\sum_k (m - k + 1)(n - k + 1)$; for $n \times n$ this is $\frac{n(n+1)(2n+1)}{6} \rightarrow 5, 14, 30, 55, 91, 140, 204$.
- Rectangles in $m \times n$ cells = $\binom{m+1}{2} \binom{n+1}{2}$.
- Triangle cut into n parts per side: totals 1, 5, 13, 27, 48, 78 for $n = 1 \dots 6$. Upward size k : $(n - k + 1)(n - k + 2) \div 2$. Downward size k : $(n - 2k + 1)(n - 2k + 2) \div 2$ while $n \geq 2k$.
- Squares on an $m \times m$ dot grid, tilted included = $\sum_{k=1}^{m-1} k(m - k)^2 \rightarrow 1, 6, 20, 50$ for $m = 2, 3, 4, 5$.

The top 5 traps

- Water image does not reverse the letters.** Only the mirror image does.
- 64 is both a square and a cube** — decide the family from the other numbers first.
- Downward triangles.** Forgetting them costs you a quarter of the answer.
- A zero in pqr .** If a cuboid has a side of 1 or 2, no cube is unpainted.
- Folded-corner cuts merge.** A cut at the centre corner gives one hole, not four; a punch on a fold gives half the expected number.

Clocks, Calendars & Classic Puzzles

One angle formula, one odd-day rule, and a clear head.

What you need to know

WHAT YOU NEED TO KNOW

Clock — speeds

- Minute hand: $360^\circ \div 60 = 6^\circ$ per minute.
- Hour hand: $360^\circ \div 720 = 0.5^\circ$ per minute (it moves 30° in one hour).
- Relative speed of minute hand over hour hand: $6 - 0.5 = 5.5^\circ$ per minute.
- In 60 minutes the minute hand gains 55 minute-spaces on the hour hand.

Clock — the one formula you need

$$\theta = |30H - 5.5M|$$

H = hour reading, M = minutes past the hour. If $\theta > 180^\circ$, the answer is $360^\circ - \theta$.

Clock — when is the angle exactly θ ? Between H and $H + 1$ o'clock:

$$M = \frac{2}{11}(30H \pm \theta)$$

Keep any value with $0 \leq M < 60$. Special cases: $\theta = 0$ (hands together) gives $M = \frac{60H}{11}$; $\theta = 180$ gives the opposite position.

Clock — how often

Event	In 12 hours	In 24 hours
Hands coincide (0°)	11	22
Hands opposite (180°)	11	22
Hands in a straight line	22	44
Hands at right angles	22	44
Hands at any other fixed angle	22	44

- Gap between two coincidences: $\frac{720}{11} = 65\frac{5}{11}$ minutes.
- All three hands (hour, minute, second) coincide only at 12 o'clock: **twice a day**.
- Mirror image of a clock time = $11 : 60 - \text{time}$ (use $23 : 60$ when the hour reads 12).

Faulty clocks

- Going forward: true time known $\rightarrow$ add the gain, or subtract the loss.
- Going backward: the clock shows C minutes while true time is T minutes, and

$$\frac{T}{C} = \frac{\text{true minutes per hour}}{\text{clock minutes per hour}}.$$

A clock gaining g min per hour shows $60 + g$ clock-minutes per 60 true minutes.

- A clock that gains or loses d minutes per day next shows the correct time after $720 \div d$ days (it must drift a whole 12 hours).
- Two clocks drifting apart: if one gains and the other loses, the gap grows at the **sum** of the two rates; if both gain (or both lose), it grows at the **difference**.

Calendar — odd days

- “Odd days” = days left over after taking out whole weeks.
- Ordinary year (365 days) = 1 odd day. Leap year (366 days) = 2 odd days.
- Leap year: divisible by 4, but a century year must be divisible by 400. So 1900 is not a leap year; 2000 is.
- 100 years = 5 odd days. 200 years = 3. 300 years = 1. 400 years = 0.
- 400 years = $400 \times 365 + 97 = 146097$ days = 20871 whole weeks. The calendar repeats exactly every 400 years.
- Day code: 0 = Sunday, 1 = Monday, 2 = Tuesday, 3 = Wednesday, 4 = Thursday, 5 = Friday, 6 = Saturday.

Calendar — days in each month (running totals, ordinary year)

Jan	Feb	Mar	Apr	May	Jun
31	59	90	120	151	181
Jul	Aug	Sep	Oct	Nov	Dec
212	243	273	304	334	365

Add 1 to every total from Feb onward in a leap year.

Calendar — method for “what day was DD-MM-YYYY?”

1. Odd days from the completed centuries (use the 0 / 1 / 3 / 5 table).
2. Odd days from the completed years of the current century: (ordinary years) $\times 1$ + (leap years) $\times 2$, then take mod 7.
3. Odd days from the current year: day-of-year number, mod 7.
4. Add all three, take mod 7, read off the day code.

Calendar — repeating years

- An ordinary year’s calendar repeats after 6 or 11 years (12 if a non-leap century year such as 2100 falls in between); a leap year’s repeats after 28 years (again, as long as no non-leap century year falls in between).
- A month of 31 days has 5 of the first three weekdays; a month of 30 days has 5 of the first two.
- Probability a random year has 53 Mondays: ordinary $\frac{1}{7}$, leap $\frac{2}{7}$, any year $\frac{71}{400}$.

Ages

- Fix one unknown as x = the **present** age of the smaller person. Every other age is then $x \pm$ something.
- “ n years ago” means subtract n from **both** ages. “In n years” means add n to **both**.
- The **difference** of two ages never changes. Use it to check your answer.

Weighing and balance puzzles

- Weights on **one pan only**: powers of 2 (1, 2, 4, 8, 16, ...) reach every whole number up to $2^n - 1$.
- Weights allowed on **both pans**: powers of 3 (1, 3, 9, 27, 81, ...) reach every whole number up to $\frac{3^n - 1}{2}$. Four weights reach 40; five reach 121.
- One odd coin among N , and you know whether it is heavy or light: split into 3 near-equal groups. Minimum weighings = $\lceil \log_3 N \rceil$. So $N \leq 3$ needs 1, $N \leq 9$ needs 2, $N \leq 27$ needs 3.
- One odd coin among 12 with **unknown** direction: 3 weighings. Counting bound: 3 weighings give $3^3 = 27$ outcome-patterns, and N coins with 2 directions need $2N$ patterns, so $N \leq 13$. Twelve is the largest N for which 3 weighings always name the coin **and** say heavy or light.

Crossing and measuring puzzles

- Farmer, wolf, goat, cabbage: 7 crossings.
- Torch on a bridge, only 2 may cross at a time, the pair moves at the slower speed. Move the two **slowest** people first. With the two slowest $a < b$ and the two fastest $f_1 < f_2$, the cost of getting a and b across (and leaving f_1, f_2 at the start) is

$$\min(f_1 + 2f_2 + b, 2f_1 + a + b).$$

Then repeat on the people who are left.

- Two jugs of capacity p and q can measure exactly v litres if and only if v is a multiple of $\gcd(p, q)$ and $v \leq \max(p, q)$.

Matchsticks

- A single row of n unit squares needs $3n + 1$ sticks.
- An $n \times n$ grid of unit squares needs $2n(n + 1)$ sticks.

Warm-up

WARM-UP

build the reflex

Example 1

What is the angle between the hands at 3:00?

SOLUTION

$$H = 3, M = 0.$$

$$\theta = |30 \times 3 - 5.5 \times 0| = |90 - 0| = 90^\circ.$$

Answer: 90°

Example 2

What is the angle between the hands at 4:20?

SOLUTION

$$H = 4, M = 20.$$

$$30 \times 4 = 120, \quad 5.5 \times 20 = 110.$$

$$\theta = |120 - 110| = 10^\circ.$$

Answer: 10°

Example 3

How many times in 24 hours do the two hands point in the same direction?

SOLUTION

They coincide 11 times in 12 hours (not 12, because the 12 o'clock overlap is shared by the two halves).

In 24 hours: $11 \times 2 = 22$.

Answer: 22 times

Example 4

Was 1900 a leap year?

SOLUTION

$1900 \div 4 = 475$, so it passes the first test.

But 1900 is a century year, so it must also be divisible by 400. $1900 \div 400 = 4.75$ — not a whole number.

So 1900 was an ordinary year of 365 days. (2000 **was** a leap year: $2000 \div 400 = 5$.)

Answer: No

Example 5

1 January 2026 is a Thursday. What day is 31 December 2026?

SOLUTION

2026 is not divisible by 4, so it is an ordinary year: 365 days.

1 January is day 1 and 31 December is day 365. The gap between them is $365 - 1 = 364$ days.

$364 = 52 \times 7$, which is exactly 52 weeks with nothing left over.

So 31 December falls on the same weekday as 1 January.

Answer: Thursday

Example 6

A clock loses 5 minutes every hour. It is set right at 6:00 a.m. What does it show at 10:00 a.m. true time?

SOLUTION

True time passed = $10 - 6 = 4$ hours.

Loss = $4 \times 5 = 20$ minutes.

The clock is behind by 20 minutes: $10 : 00 - 0 : 20 = 9 : 40$.

Answer: 9:40 a.m.

Example 7

A father is 3 times as old as his son. The sum of their ages is 48. How old is the son?

SOLUTION

Let the son be x years. Then the father is $3x$.

$$x + 3x = 48 \Rightarrow 4x = 48 \Rightarrow x = 12.$$

Check: son 12, father 36, sum = 48. ✓

Answer: 12 years

Example 8

Nine identical-looking balls; one is heavier. Using a pan balance, what is the smallest number of weighings that always finds it?

SOLUTION

Split 9 into three groups of 3.

Weighing 1: put group A against group B. Either one side sinks (the heavy ball is there), or they balance (the heavy ball is in C). Either way you are left with 3 balls.

Weighing 2: take 2 of those 3 and weigh them. If one sinks, that is it; if they balance, it is the third.

$9 \leq 3^2$, so 2 weighings are enough.

Answer: 2 weighings

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Find the angle between the hands of a clock at 5:40.

SOLUTION

$$H = 5, M = 40.$$

$$30H = 30 \times 5 = 150.$$

$$5.5M = 5.5 \times 40 = 220.$$

$$\theta = |150 - 220| = 70^\circ.$$

$70 < 180$, so no adjustment is needed.

Answer: 70°

Example 10 [Accenture pattern]

Find the angle between the hands at 10:10.

SOLUTION

$$H = 10, M = 10.$$

$$30H = 300. \quad 5.5M = 55.$$

$$\theta = |300 - 55| = 245^\circ.$$

This is more than 180° , so take the other way round:

$$360 - 245 = 115^\circ.$$

Answer: 115°

Answer choices in these tests almost always give the **smaller** angle. If your number is above 180, subtract it from 360 before you look at the options.

TRAP

$5.5 \times M$, **not** $6 \times M$. The 5.5 already allows for the hour hand creeping forward while the minutes pass. Using $6M$ gives an answer that is wrong by $0.5M$ degrees — at 40 minutes past, that is a 20° error.

Example 11 [Infosys pattern]

At what time between 4 and 5 o'clock are the two hands together?

SOLUTION

Use $M = \frac{2}{11}(30H \pm \theta)$ with $H = 4$ and $\theta = 0$.

$$M = \frac{2}{11}(30 \times 4) = \frac{2}{11} \times 120 = \frac{240}{11}.$$

$240 \div 11 = 21$ remainder 9, so $M = 21\frac{9}{11}$ minutes.

Check: $\theta = |30 \times 4 - 5.5 \times \frac{240}{11}| = |120 - \frac{5.5 \times 240}{11}| = |120 - \frac{1320}{11}| = |120 - 120| = 0. \checkmark$

Answer: 4:21 and $\frac{9}{11}$ minutes past

Example 12 [Wipro pattern]

At what time between 8 and 9 o'clock are the hands exactly opposite each other?

SOLUTION

Opposite means $\theta = 180^\circ$. Here $H = 8$, so $30H = 240$.

Try the minus sign first:

$$M = \frac{2}{11}(240 - 180) = \frac{2}{11} \times 60 = \frac{120}{11} = 10\frac{10}{11}.$$

This lies between 0 and 60, so it is valid.

Try the plus sign: $M = \frac{2}{11}(240 + 180) = \frac{840}{11} = 76.36$, which is more than 60. Reject it.

| **Check:** $5.5 \times \frac{120}{11} = \frac{660}{11} = 60$, and $|240 - 60| = 180$. ✓

Answer: 8:10 and $\frac{10}{11}$ minutes past

Example 13 [Capgemini pattern]

At what times between 3 and 4 o'clock are the hands at right angles?

SOLUTION

$\theta = 90^\circ$, $H = 3$, so $30H = 90$.

Minus sign:

$$M = \frac{2}{11}(90 - 90) = 0.$$

So the first time is exactly 3:00.

Plus sign:

$$M = \frac{2}{11}(90 + 90) = \frac{2}{11} \times 180 = \frac{360}{11}.$$

$360 \div 11 = 32$ remainder 8, so $M = 32\frac{8}{11}$.

Check: $5.5 \times \frac{360}{11} = \frac{1980}{11} = 180$, and $|90 - 180| = 90$. ✓

Answer: 3:00 and 3:32 and $\frac{8}{11}$ minutes past

SHORTCUT

Two answers per hour, almost always. For any angle strictly between 0° and 180° , both signs usually give a valid minute value, so there are two such times in that hour. Compute both, then throw away any M outside $0 \leq M < 60$.

Example 14 [TCS NQT pattern]

A watch gains 10 minutes every hour. It is set right at 9:00 a.m. What is the **true** time when the watch shows 4:00 p.m.?

SOLUTION

The watch shows 9 a.m. to 4 p.m. = 7 hours = 420 watch-minutes.

Rate: in 60 true minutes the watch shows $60 + 10 = 70$ watch-minutes.

So true minutes = watch-minutes $\times \frac{60}{70}$:

$$420 \times \frac{60}{70} = 420 \times \frac{6}{7} = 360 \text{ minutes} = 6 \text{ hours.}$$

True time = 9 : 00 a.m. + 6 h = 3 : 00 p.m.

Answer: 3:00 p.m.

TRAP

“Gains 10 minutes” does not mean “subtract 10 minutes”. The gain builds up with every hour that passes. Set up the ratio 60 : 70 and scale — never subtract a flat amount unless the true elapsed time is what you were given.

Example 15 [Cognizant pattern]

A clock loses 8 minutes every day. If it is set right now, after how many days will it show the correct time again?

SOLUTION

A 12-hour clock face shows the correct time again only when it has drifted a whole 12 hours.

12 hours = $12 \times 60 = 720$ minutes.

Days needed = $720 \div 8 = 90$.

Answer: 90 days

Example 16 [Accenture pattern]

What day of the week is 15 August 2026?

SOLUTION

Step 1 — completed centuries. 2000 completed years = 1600 years (0 odd days) + 400 years (0 odd days) = 0 odd days.

Step 2 — completed years 2001 to 2025. That is 25 years.

Leap years: 2004, 2008, 2012, 2016, 2020, 2024 $\rightarrow$ 6 leap years.

Ordinary years: $25 - 6 = 19$.

Odd days = $19 \times 1 + 6 \times 2 = 19 + 12 = 31$.

$31 \div 7 = 4$ remainder 3, so 3 odd days.

Step 3 — days inside 2026 up to 15 August. 2026 is ordinary. Using the running totals, the end of July is day 212.

$212 + 15 = 227$.

$227 \div 7 = 32$ remainder 3, so 3 odd days.

Step 4 — add. $0 + 3 + 3 = 6$ odd days.

Day code 6 = Saturday.

Answer: Saturday

Example 17 [Infosys pattern]

What day of the week was 26 January 1950?

SOLUTION

Step 1 — completed centuries. 1900 completed years = 1600 (0 odd days) + 300 (1 odd day) = 1 odd day.

Step 2 — completed years 1901 to 1949. That is 49 years.

Leap years: 1904, 1908, ..., 1948. Count = $\frac{1948-1904}{4} + 1 = \frac{44}{4} + 1 = 11 + 1 = 12$.

Ordinary years: $49 - 12 = 37$.

Odd days = $37 \times 1 + 12 \times 2 = 37 + 24 = 61$.

$61 \div 7 = 8$ remainder 5, so 5 odd days.

Step 3 — days inside 1950 up to 26 January. That is simply 26.

$26 \div 7 = 3$ remainder 5, so 5 odd days.

Step 4 — add. $1 + 5 + 5 = 11$. $11 \div 7 = 1$ remainder 4.

Day code 4 = Thursday.

Answer: Thursday

SHORTCUT

Never add up 365s. Convert each block to odd days first (century block, whole years, days in the current year), add three small numbers, take mod 7 once at the end. The whole computation fits in four lines.

Example 18 [Wipro pattern]

A father is 4 times as old as his son. In 12 years the father will be twice as old as the son. Find their present ages.

SOLUTION

Let the son be s years now. Then the father is $4s$.

In 12 years: son = $s + 12$, father = $4s + 12$.

The condition gives

$$4s + 12 = 2(s + 12).$$

$$4s + 12 = 2s + 24$$

$$4s - 2s = 24 - 12$$

$$2s = 12 \Rightarrow s = 6.$$

$$\text{Father} = 4 \times 6 = 24.$$

Check: in 12 years the son is 18 and the father is 36, and $36 = 2 \times 18$. ✓

Answer: Son 6 years, father 24 years

Example 19 [Capgemini pattern]

The ages of Aarav and Bhavna are in the ratio 4 : 5. Six years from now the ratio will be 5 : 6. Find their present ages.

SOLUTION

Let the ages be $4x$ and $5x$.

In 6 years: $4x + 6$ and $5x + 6$.

$$\frac{4x + 6}{5x + 6} = \frac{5}{6}$$

Cross-multiply:

$$6(4x + 6) = 5(5x + 6)$$

$$24x + 36 = 25x + 30$$

$$36 - 30 = 25x - 24x$$

$$6 = x.$$

$$\text{Aarav} = 4 \times 6 = 24, \text{ Bhavna} = 5 \times 6 = 30.$$

Check: in 6 years they are 30 and 36, and $\frac{30}{36} = \frac{5}{6}$. ✓

Answer: 24 years and 30 years

Example 20 [TCS NQT pattern]

You have 8 coins that look the same. One is heavier than the other seven, which all weigh the same. Using only a pan balance, find the heavy coin in 2 weighings. Describe the plan.

SOLUTION

Weighing 1. Split as $3 + 3 + 2$. Put the two groups of 3 on the pans.

- If one side sinks, the heavy coin is among those 3.
- If they balance, the heavy coin is among the 2 set aside.

Weighing 2, case A (3 suspects). Weigh coin 1 against coin 2. If one sinks it is the heavy one; if they balance it is coin 3.

Weighing 2, case B (2 suspects). Weigh them against each other. The side that sinks is the heavy coin.

Two weighings always finish the job because $8 \leq 3^2 = 9$.

Answer: 2 weighings, splitting $3 + 3 + 2$

PRACTICE · TIER 1 target: 60 s/Q

- Find the angle between the hands at 2:30.
- Find the angle between the hands at 6:40.
- At what time between 7 and 8 o'clock are the hands together?
- How many times in 12 hours are the hands at right angles?
- A clock gains 6 minutes a day and is set right at noon. What will it show 5 days later at noon, true time?
- A watch loses 10 minutes every hour and is set right at 7:00 a.m. What is the true time when the watch shows 12 noon?
- How many leap years are there from 2001 to 2100 inclusive?
- 1 January 2025 is a Wednesday. What day is 1 January 2026?
- What day of the week was 1 May 2005?
- How many odd days are there in 300 years?
- A month of 31 days begins on a Friday. How many Saturdays does it have?
- A son's age is one third of his father's. In 12 years it will be one half. Find both present ages.
- A mother and her daughter are 60 years old in total. Ten years ago the mother was 7 times as old as the daughter. Find their present ages.
- What is the least number of pan-balance weighings that always finds the one heavier ball among 27?
- The day after tomorrow is Sunday. What was the day before yesterday?

ANSWER KEY

- 105°.** $|30 \times 2 - 5.5 \times 30| = |60 - 165| = 105$.
- 40°.** $|30 \times 6 - 5.5 \times 40| = |180 - 220| = 40$.
- 7:38 and $\frac{2}{11}$ min.** $M = \frac{2}{11}(30 \times 7) = \frac{420}{11} = 38\frac{2}{11}$.
- 22.** Two right angles in most hours, but the 3 o'clock and 9 o'clock overlaps are shared, so it is 22, not 24.
- 12:30 p.m.** Gain = $5 \times 6 = 30$ minutes, and the clock runs fast, so it reads half an hour ahead.
- 1:00 p.m.** The watch shows 5 h = 300 watch-min; it shows only 50 watch-min per 60 true min, so true = $300 \times \frac{60}{50} = 360$ min = 6 h. 7 : 00 + 6 = 13 : 00.
- 24.** 2004, 2008, ..., 2096 is $\frac{2096-2004}{4} + 1 = 24$ years. 2100 is a century year not divisible by 400, so it is not a leap year.
- Thursday.** 2025 is an ordinary year, so it carries 1 odd day.
- Sunday.** Centuries: 0. Years 2001–2004: 3 ordinary + 1 leap = 3 + 2 = 5. Days in 2005 to 1 May: 120 + 1 = 121, and 121 = 17 × 7 + 2, so 2. Total 5 + 2 = 7 → 0 → Sunday.
- 1.** 100 years give 5, so 300 give 15, and 15 ÷ 7 leaves 1.
- 5.** 31 = 4 × 7 + 3, so the first three weekdays (Friday, Saturday, Sunday) each occur 5 times.
- Son 12, father 36.** $s = \frac{f}{3}$ and $s + 12 = \frac{f+12}{2}$ give $2s + 24 = 3s + 12$, so $s = 12$. Check: 24 and 48. ✓
- Mother 45, daughter 15.** $m + d = 60$ and $m - 10 = 7(d - 10)$ give $m = 7d - 60$, so $8d = 120$ and $d = 15$. Check: 35 and 5, and 35 = 7 × 5. ✓
- 3.** $27 = 3^3$, so three three-way splits are enough and nothing smaller works.
- Wednesday.** Day after tomorrow is Sunday, so today is Friday, and the day before yesterday was Wednesday.

Tier 2 — Singapore & Thailand**TIER 2**

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A wall clock in a Bangkok dispatch office gains 12 minutes every hour. It was set correct at 9:00 a.m. A supervisor looks at it and reads 5:00 p.m. What is the true time?

SOLUTION

What the clock shows: 9:00 a.m. to 5:00 p.m. = 8 hours = 480 clock-minutes.

Rate: in 60 true minutes this clock runs through $60 + 12 = 72$ clock-minutes.

So every 72 clock-minutes correspond to 60 true minutes:

$$\text{true minutes} = 480 \times \frac{60}{72}.$$

$$480 \times 60 = 28800. \quad 28800 \div 72 = 400.$$

True elapsed = 400 minutes = 6 hours 40 minutes.

True time = 9 : 00 a.m. + 6 h 40 min = 3 : 40 p.m.

Answer: 3:40 p.m.

Example 22 [DBS · pattern]

A clock is set right at 8:00 a.m. It gains 10 minutes in 24 hours. What is the true time when the clock shows 1:00 p.m. on the next day?

SOLUTION

What the clock shows: 8:00 a.m. on day 1 to 1:00 p.m. on day 2.

8 a.m. to 8 a.m. is 24 hours; 8 a.m. to 1 p.m. is 5 hours more. Total = 29 hours.

Rate: in $24 \times 60 = 1440$ true minutes this clock shows $1440 + 10 = 1450$ clock-minutes.

$$\text{true time} = 29 \times \frac{1440}{1450} = 29 \times \frac{144}{145}.$$

$$145 = 5 \times 29, \text{ so } \frac{29}{145} = \frac{1}{5}:$$

$$29 \times \frac{144}{145} = \frac{144}{5} = 28.8 \text{ hours.}$$

$$0.8 \text{ hour} = 0.8 \times 60 = 48 \text{ minutes.}$$

True elapsed = 28 hours 48 minutes.

8 : 00 a.m. + 24 h = 8 : 00 a.m. next day. Then +4 h 48 min = 12 : 48 p.m.

Answer: 12:48 p.m. on the next day

SHORTCUT

Cancel before you divide. In $29 \times \frac{1440}{1450}$ notice $1450 = 5 \times 290 = 5 \times 10 \times 29$. The 29 cancels straight away and the sum becomes $\frac{144}{5}$. Hunting for a common factor first turns a messy division into one line.

Example 23 [Shopee · pattern]

Two clocks in a warehouse are set to the correct time at 12 noon. One gains 1 minute per hour; the other loses 2 minutes per hour. At what time will the two clocks be exactly 1 hour apart?

SOLUTION

Insight: work with the gap between them, not with either clock alone.

In 1 true hour, clock A moves ahead of correct time by 1 minute and clock B falls behind by 2 minutes.

So the gap between A and B grows by $1 + 2 = 3$ minutes every hour.

We need a gap of 1 hour = 60 minutes:

$$\text{hours} = 60 \div 3 = 20.$$

12 noon + 20 hours = 8 : 00 a.m. the next day.

Answer: 8:00 a.m. the next day

Example 24 [Agoda · pattern]

A hotel runs a maintenance cycle every 45 days. The first cycle starts on Monday, 3 March 2025. On what date and weekday does the second cycle start?

SOLUTION

The date. 2025 is not a leap year. Day-of-year for 3 March = $59 + 3 = 62$.

Add 45 days: $62 + 45 = 107$.

Day 107 of an ordinary year: end of March is day 90, so $107 - 90 = 17$ April.

The weekday. $45 \div 7 = 6$ remainder 3.

Monday + 3 days = Thursday.

Check by the odd-day method. Completed centuries: 0. Years 2001–2024: 24 years with 6 leap years (2004–2024), so $18 + 12 = 30$, and $30 \div 7$ leaves 2. Days in 2025 to 17 April: $90 + 17 = 107$, and $107 = 15 \times 7 + 2$, so 2. Total $2 + 2 = 4 \rightarrow$ Thursday. ✓

Answer: Thursday, 17 April 2025

Example 25 [SCB · pattern]

Which is the first year after 2025 whose calendar is exactly the same as that of 2025?

SOLUTION

Two conditions must hold: the new year must start on the same weekday, and it must be the same type (ordinary or leap). 2025 is ordinary, so we need an ordinary year starting on the same weekday.

Add odd days year by year and stop when the running total is a multiple of 7.

Year passed	Type	Odd days	Running total mod 7
2025	ordinary	1	1
2026	ordinary	1	2
2027	ordinary	1	3
2028	leap	2	5
2029	ordinary	1	6
2030	ordinary	1	0

After 2030 has passed, the total is 0. So 1 January 2031 falls on the same weekday as 1 January 2025.

Is 2031 ordinary? $2031 \div 4 = 507.75$, so yes.

Both conditions hold.

Answer: 2031

TRAP

Same starting weekday is not enough. A leap year and an ordinary year that start on the same day still differ from 1 March onwards. Always check the leap status too.

Example 26 [GIC · pattern]

Priya is 8 years older than Wei Ming. Five years ago Priya was 3 times as old as Wei Ming. What will the sum of their ages be 4 years from now?

SOLUTION

Let Wei Ming be w years now. Then Priya is $w + 8$.

Five years ago: Wei Ming = $w - 5$, Priya = $w + 8 - 5 = w + 3$.

The condition:

$$w + 3 = 3(w - 5)$$

$$w + 3 = 3w - 15$$

$$3 + 15 = 3w - w$$

$$18 = 2w \Rightarrow w = 9.$$

So Wei Ming is 9 and Priya is $9 + 8 = 17$.

Check: five years ago they were 4 and 12, and $12 = 3 \times 4$. ✓

In 4 years:

$$(9 + 4) + (17 + 4) = 13 + 21 = 34.$$

Answer: 34 years

Example 27 [LINE MAN · pattern]

Three years ago the average age of a family of 5 was 29 years. A baby has been born since then, and the average age of the 6 members today is 27 years. How old is the baby?

SOLUTION

Three years ago. Total age of the 5 members $= 5 \times 29 = 145$ years.

Today, those same 5 members. Each one is 3 years older, so the total rises by $5 \times 3 = 15$:

$$145 + 15 = 160 \text{ years.}$$

Today, all 6 members. Total $= 6 \times 27 = 162$ years.

The baby.

$$162 - 160 = 2 \text{ years.}$$

Sanity check. The baby was born in the last 3 years, so its age must lie between 0 and 3. 2 passes. ✓

Answer: 2 years

The trap is to subtract $162 - 145 = 17$ and call the baby 17 years old. That forgets that the five original members also aged 3 years each — and it fails the sanity check, because a baby born in the last 3 years cannot be 17.

Example 28 [Razer · pattern]

A shopkeeper wants a set of weights so that, using a two-pan balance, he can weigh every whole number of kilograms from 1 to 40. Weights may be placed on **either** pan. What is the smallest number of weights, and what are they?

SOLUTION

Insight: putting a weight on the other pan lets it count as -1 as well as $+1$, so each weight has three states: left pan, right pan, or unused. Three states point to powers of 3.

Take the weights 1, 3, 9, 27 kg.

With signs $+1, 0, -1$ on each, the reachable totals run from $-(1 + 3 + 9 + 27) = -40$ to $+40$, and every whole number in between can be written exactly once.

Some checks:

- $2 = 3 - 1$ (put 3 with the goods' pan opposite, and 1 beside the goods)
- $5 = 9 - 3 - 1$
- $11 = 9 + 3 - 1$
- $20 = 27 - 9 + 3 - 1$
- $40 = 27 + 9 + 3 + 1$

Could 3 weights do it? Three weights give at most $\frac{3^3-1}{2} = \frac{26}{2} = 13$ different positive totals, which is less than 40. So 3 is impossible.

Answer: 4 weights: 1 kg, 3 kg, 9 kg, 27 kg

Example 29 [Grab · pattern]

Four engineers must cross a narrow footbridge at night. They have one torch. At most two may cross at a time, and anyone crossing must carry the torch. They walk at 1, 2, 5 and 10 minutes for the crossing; a pair moves at the slower person's pace. What is the least total time?

SOLUTION

Insight: the two slow people should cross together, so that the big number is paid only once. Someone fast must be waiting on the far side to bring the torch back.

The plan:

Move	Who crosses	Time
1	1 and 2 go over	2
2	1 comes back	1
3	5 and 10 go over	10
4	2 comes back	2
5	1 and 2 go over	2

Total = $2 + 1 + 10 + 2 + 2 = 17$ minutes.

Compare with the obvious plan (the fastest person escorts everyone):

1 and 10 go over (10), 1 comes back (1), 1 and 5 go over (5), 1 comes back (1), 1 and 2 go over (2). Total = $10 + 1 + 5 + 1 + 2 = 19$ minutes.

17 is better.

The box formula agrees. It prices only the two slowest, $a = 5$ and $b = 10$, with helpers $f_1 = 1$ and $f_2 = 2$:

$$\min(f_1 + 2f_2 + b, 2f_1 + a + b) = \min(1 + 4 + 10, 2 + 5 + 10) = \min(15, 17) = 15.$$

That 15 covers moves 1, 2, 3 and 4 and leaves 1 and 2 back at the start. They still have to cross, which costs 2. Total $15 + 2 = 17$. ✓

Answer: 17 minutes

Example 30 [Sea · pattern]

You have a 5-litre jug and a 3-litre jug, no markings, and a tap. Measure out exactly 4 litres.

SOLUTION

First check it is possible. $\gcd(5, 3) = 1$, and $4 \leq 5$. So yes.

Step	Action	5-L jug / 3-L jug
start	both empty	0 / 0
1	fill the 5-L jug	5 / 0
2	pour into the 3-L jug until it is full	2 / 3
3	empty the 3-L jug	2 / 0
4	pour the 2 L across	0 / 2
5	fill the 5-L jug	5 / 2
6	top up the 3-L jug (it takes 1 L)	4 / 3

The 5-litre jug now holds exactly 4 litres.

Answer: 4 litres in the 5-L jug after 6 moves

SHORTCUT

Jug puzzles have only two strategies. Either keep filling the big jug and emptying into the small one, or keep filling the small jug and pouring into the big one. Try the first; if it stalls, try the second. One of them always reaches every achievable amount.

PRACTICE · TIER 2 target: 100 s/Q

1. Find the angle between the hands at 7:35.
2. At what times between 5 and 6 o'clock are the hands 60° apart?
3. A clock loses 12 minutes every hour and is set right at 8:00 a.m. What is the true time when it shows 4:00 p.m.?
4. A clock gains 5 minutes a day. After how many days will it show the correct time again?
5. Two clocks are set right at 6:00 a.m. One gains 2 minutes per hour, the other loses 3 minutes per hour. When will they be 50 minutes apart?
6. What day of the week is 15 June 2030?
7. Which is the first year after 2024 with exactly the same calendar as 2024?
8. 15 March 2024 was a Friday. What day was 20 August 2024?
9. The ages of a mother and her son are in the ratio 7 : 2. In 10 years the ratio will be 9 : 4. Find their present ages.
10. What is the smallest set of weights that can weigh every whole number of kilograms from 1 to 121 on a two-pan balance, weights allowed on both pans?
11. Four people must cross a bridge at night with one torch, two at a time, moving at the slower pace. Their times are 1, 3, 6 and 8 minutes. What is the least total time?
12. Using a 7-litre jug and a 4-litre jug, measure exactly 5 litres. List the moves.

ANSWER KEY

1. **17.5°.** $|30 \times 7 - 5.5 \times 35| = |210 - 192.5| = 17.5$.
2. **5:16 and $\frac{4}{11}$ min, and 5:38 and $\frac{2}{11}$ min.** $M = \frac{2}{11}(150 - 60) = \frac{180}{11} = 16\frac{4}{11}$ and $M = \frac{2}{11}(150 + 60) = \frac{420}{11} = 38\frac{2}{11}$.
3. **6:00 p.m.** The clock shows 8 h = 480 clock-min, at 48 clock-min per 60 true min. True = $480 \times \frac{60}{48} = 600$ min = 10 h, so 8 : 00 + 10 = 18 : 00.
4. **144 days.** $720 \div 5 = 144$.
5. **4:00 p.m. the same day.** The gap grows at $2 + 3 = 5$ min per hour, so $50 \div 5 = 10$ hours after 6 a.m.
6. **Saturday.** Centuries 0. Years 2001–2029: 29 years, 7 leap (2004–2028), so $22 + 14 = 36$, and $36 \div 7$ leaves 1. Days in 2030 to 15 June: $151 + 15 = 166$, and $166 = 23 \times 7 + 5$, so 5. Total $1 + 5 = 6 \rightarrow$ Saturday.
7. **2052.** A leap year's calendar repeats after 28 years, and no non-leap century year lies between 2024 and 2052.
8. **Tuesday.** 2024 is a leap year. Day-of-year: 15 March = $60 + 15 = 75$; 20 August = $213 + 20 = 233$. Gap = $233 - 75 = 158$ days, and $158 = 22 \times 7 + 4$, so Friday +4 = Tuesday.
9. **Mother 35, son 10.** $\frac{7x+10}{2x+10} = \frac{9}{4}$ gives $28x + 40 = 18x + 90$, so $10x = 50$ and $x = 5$. Check: $\frac{45}{20} = \frac{9}{4}$. ✓
10. **Five weights: 1, 3, 9, 27, 81 kg.** They reach $\frac{3^5-1}{2} = \frac{242}{2} = 121$. Four weights reach only 40.
11. **18 minutes.** 1 and 3 over (3), 1 back (1), 6 and 8 over (8), 3 back (3), 1 and 3 over (3): $3 + 1 + 8 + 3 + 3 = 18$. The escort plan costs $8 + 1 + 6 + 1 + 3 = 19$.
12. **5 L in the 7-L jug.** Fill 4 and pour into 7 (7 has 4). Fill 4 again and top up the 7-L jug, which takes 3, leaving 1 in the 4-L jug. Empty the 7-L jug, pour the 1 L in, then fill the 4-L jug and empty it into the 7-L jug: $1 + 4 = 5$.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 31 [Google · pattern]

How many times in 24 hours are the hour and minute hands exactly at right angles? Explain, do not just quote the number.

SOLUTION

Insight: stop thinking about two hands. Think about one imaginary hand whose position is the gap between them. That gap hand turns at a steady 5.5° per minute.

In 12 hours = 720 minutes, the gap grows by

$$5.5 \times 720 = 3960 \text{ degrees.}$$

$3960 \div 360 = 11$. So in 12 hours the gap makes exactly **11 complete turns**.

During one complete turn of the gap, the gap passes through 90° once and through 270° once — and 270° also looks like a right angle, measured the short way.

So each turn gives 2 right angles, and $11 \times 2 = 22$ right angles in 12 hours.

In 24 hours: $22 \times 2 = 44$.

Why the same logic gives 22 for “hands together”. The gap equals 0° only once per turn, so it happens 11 times in 12 hours and 22 times a day. Same for 180° .

Answer: 44 times

This one argument answers the whole family of questions. Any angle strictly between 0° and 180° occurs $2 \times 11 \times 2 = 44$ times a day; the two end values 0° and 180° occur 22 times a day each.

Example 32 [Amazon · pattern]

A clock has an hour hand, a minute hand and a second hand. How many times in 24 hours do all three point in exactly the same direction?

SOLUTION

Insight: each pair of hands has its own regular meeting timetable. Write down the hour-and-minute timetable, write down the minute-and-second timetable, and ask whether any moment appears on both lists.

Step 1 — when do the hour and minute hands meet? From Example 31, 11 times in 12 hours, evenly spaced, so at

$$t = k \times \frac{720}{11} \text{ minutes after 12:00, } k = 0, 1, 2, \dots, 10.$$

Step 2 — when do the minute and second hands meet? In t minutes the second hand makes t full turns and the minute hand makes $t/60$ full turns. They point the same way exactly when the second hand has gained a **whole number** of turns:

$$t - \frac{t}{60} = j \Rightarrow \frac{59}{60}t = j \Rightarrow t = \frac{60j}{59}, \quad j = 0, 1, 2, \dots$$

Step 3 — can both conditions hold at once? Set the two lists equal:

$$\frac{60j}{59} = \frac{720k}{11}.$$

Multiply both sides by $11 \times 59 = 649$:

$$660j = 42480k.$$

Divide both sides by 60:

$$11j = 708k.$$

11 is prime. Does 11 divide 708? $11 \times 64 = 704$, and $708 - 704 = 4$, so no. Therefore 11 must divide k .

In 12 hours k runs only over 0, 1, ..., 10, so the only multiple of 11 available is $k = 0$.

$k = 0$ is 12 o'clock itself. At 12:00 all three hands point at the 12.

So in 12 hours it happens once, and in 24 hours: at 12 midnight and at 12 noon.

Answer: 2 times a day

TRAP

“The hands meet 22 times a day, so all three must meet 22 times” is wrong. The second hand is a much faster wheel; it only lines up with the other two at the one moment when every hand resets together. Check the arithmetic of the meeting times before you assume.

Example 33 [Goldman Sachs · pattern]

A year is picked at random. What is the probability that it contains 53 Sundays?

SOLUTION

Insight: 52 Sundays are guaranteed. The question is only about the 1 or 2 leftover days.

Case A — an ordinary year. $365 = 52 \times 7 + 1$. There are 52 of each weekday, plus 1 extra day. That extra day is equally likely to be any of the 7 weekdays, so

$$P(53 \text{ Sundays} \mid \text{ordinary}) = \frac{1}{7}.$$

Case B — a leap year. $366 = 52 \times 7 + 2$. The 2 extra days are a consecutive pair: (Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat) or (Sat, Sun) — 7 equally likely pairs. Two of them contain a Sunday, so

$$P(53 \text{ Sundays} \mid \text{leap}) = \frac{2}{7}.$$

How likely is a leap year? Over one full 400-year cycle there are 97 leap years.

$$P(\text{leap}) = \frac{97}{400}, \quad P(\text{ordinary}) = \frac{303}{400}.$$

Combine.

$$P = \frac{303}{400} \times \frac{1}{7} + \frac{97}{400} \times \frac{2}{7} = \frac{303 + 194}{400 \times 7} = \frac{497}{2800}.$$

Divide top and bottom by 7: $497 \div 7 = 71$ and $2800 \div 7 = 400$.

$$P = \frac{71}{400} = 0.1775.$$

Answer: $\frac{71}{400} = 0.1775$

Example 34 [Microsoft · pattern]

A 12-hour digital clock displays the time as H:MM or HH:MM, with no leading zero on the hour (so 1:05, 10:30, 12:45). Ignoring the colon, how many times in 24 hours is the display a palindrome?

SOLUTION

Insight: split by the number of digits in the hour, because a 3-character string and a 4-character string have different palindrome conditions.

Case A — single-digit hours 1 to 9. The string is hm_1m_2 (3 characters). A palindrome needs the first and last to match:

$$m_2 = h.$$

The middle digit m_1 is free, but it is the tens digit of the minutes, so $m_1 \in \{0, 1, 2, 3, 4, 5\}$ — 6 choices.

For each hour h there are 6 palindromes. Example for $h = 1$: 1:01, 1:11, 1:21, 1:31, 1:41, 1:51.

Total = $9 \times 6 = 54$.

Case B — two-digit hours 10, 11, 12. The string is $h_1h_2m_1m_2$ (4 characters). A palindrome needs

$$m_2 = h_1 \quad \text{and} \quad m_1 = h_2.$$

Here $h_1 = 1$ always, so $m_2 = 1$.

- Hour 10: $h_2 = 0$, so $m_1 = 0$ and the time is 10:01. Minutes 01 is valid. ✓
- Hour 11: $h_2 = 1$, so $m_1 = 1$ and the time is 11:11. Valid. ✓
- Hour 12: $h_2 = 2$, so $m_1 = 2$ and the time is 12:21. Valid. ✓

Total = 3.

In 12 hours: $54 + 3 = 57$.

In 24 hours: $57 \times 2 = 114$.

Answer: 114 times

Example 35 [D. E. Shaw · pattern]

You have 12 coins. Exactly one is counterfeit, and it may be heavier **or** lighter than the rest; you do not know which. With a two-pan balance and only 3 weighings, find the fake coin and say whether it is heavy or light.

SOLUTION

Insight: count the outcomes before you design anything. Each weighing has 3 results, so 3 weighings give $3^3 = 27$ outcome-patterns. There are $12 \times 2 = 24$ possible answers. $24 \leq 27$, so it is just possible — which means no weighing may be wasted; every weighing must split the possibilities into three nearly equal parts.

Label the coins 1...12.

Weighing 1: 1, 2, 3, 4 against 5, 6, 7, 8.

Branch A — they balance. The fake is among 9, 10, 11, 12 (8 possibilities: 4 coins $\times$ 2 directions).

Weighing 2: 9, 10, 11 against 1, 2, 3 (three coins known to be good).

- Balance $\rightarrow$ the fake is 12. Weighing 3: 12 against a good coin tells you heavy or light.
- Left heavy $\rightarrow$ one of 9, 10, 11 is heavy. Weighing 3: 9 against 10. Heavier side is the fake; if they balance it is 11.
- Left light $\rightarrow$ one of 9, 10, 11 is light. Weighing 3: 9 against 10. Lighter side is the fake; if they balance it is 11.

Branch B — left side heavy. Then either one of 1, 2, 3, 4 is heavy, or one of 5, 6, 7, 8 is light (8 possibilities).

Weighing 2: 1, 2, 5 against 3, 4, 6.

- **Balance** $\rightarrow$ none of 1, 2, 3, 4, 5, 6 is the fake, so it is 7 or 8, and it must be **light**. Weighing 3: 7 against 8; the lighter one is the fake.
- **Left heavy** $\rightarrow$ the cause is 1 heavy, 2 heavy, or 6 light. Weighing 3: 1 against 2. Heavier is the fake; if they balance, 6 is light.
- **Left light** $\rightarrow$ the cause is 3 heavy, 4 heavy, or 5 light. Weighing 3: 3 against 4. Heavier is the fake; if they balance, 5 is light.

Branch C — right side heavy. Mirror image of Branch B with the roles of the two groups swapped.

Every branch ends after weighing 3 with one coin named and its direction known.

Answer: Yes — 3 weighings are enough, using the 4 vs 4 opening and the mixed regrouping above

Where the limit sits. With 13 coins, $13 \times 2 = 26$ answers still fit inside 27 outcomes, and 3 weighings can indeed **name** the fake — but in one branch they cannot also say whether it is heavy or light. Twelve is the largest number for which 3 weighings give you both facts every time.

Example 36 [Uber · pattern]

You have two ropes. Each one burns from end to end in exactly 60 minutes, but neither burns at a steady rate — half the rope may take 50 minutes and the other half 10. You have a lighter. Measure exactly 45 minutes.

SOLUTION

Insight: you cannot trust any point along a rope, but you can trust a whole rope. Lighting both ends halves the time, whatever the unevenness.

Step 1. At time 0, light rope A at **both** ends and rope B at **one** end.

Rope A now has two flames eating towards each other. Together they consume the whole 60-minutes-worth of rope at double speed, so they meet after exactly $60 \div 2 = 30$ minutes — no matter where they meet.

Step 2. The moment rope A is finished, 30 minutes have passed. Rope B has been burning from one end for those 30 minutes, so what is left of rope B is worth exactly $60 - 30 = 30$ minutes of one-ended burning.

Step 3. At that instant, light the **other** end of rope B as well.

The remaining 30-minutes-worth now burns from both ends, so it is gone in $30 \div 2 = 15$ minutes.

Total. $30 + 15 = 45$ minutes.

Answer: 45 minutes: both ends of A and one end of B at the start; at 30 minutes light B's second end

The same two ropes also give 30 minutes (both ends of one rope), 60 minutes (one end of one rope) and 15 minutes (the stretch from the 30-minute mark to the 45-minute mark). The one move that does all the work is **lighting the second end**: it halves whatever burning time is left.

Example 37 [Adobe · pattern]

Twelve matchsticks are laid out to make a 2×2 grid of four unit squares. (a) Show that 12 sticks is right. (b) Remove exactly 2 sticks so that exactly 2 squares remain. (c) How many sticks would a 5×5 grid need?

SOLUTION

(a) A 2×2 grid has 3 horizontal lines, each 2 units long, and 3 vertical lines, each 2 units long.

$$3 \times 2 + 3 \times 2 = 6 + 6 = 12. \checkmark$$

The general rule: an $n \times n$ grid has $(n + 1)$ lines each way, each n sticks long, so $2n(n + 1)$ sticks. For $n = 2$: $2 \times 2 \times 3 = 12. \checkmark$

(b) **Insight: the figure holds 5 squares, not 4 — the big outer square counts too. Removing an outer stick destroys the big square straight away, which is wasteful. Removing an interior stick kills two small squares at once and leaves the big square alone.**

There are 4 interior sticks: the upper and lower halves of the middle vertical line, and the left and right halves of the middle horizontal line.

Remove the **upper half of the middle vertical line** — that kills the top-left and top-right unit squares.

Remove the **right half of the middle horizontal line** — that kills the top-right (already gone) and the bottom-right unit squares.

Squares destroyed: top-left, top-right, bottom-right. Squares surviving: the bottom-left unit square, and the big 2×2 square, whose 8 outer sticks are all untouched.

Exactly 2 squares remain.

(c) $2n(n + 1)$ with $n = 5$:

$$2 \times 5 \times 6 = 60 \text{ sticks.}$$

Answer: (a) $2n(n + 1) = 12$; (b) remove two interior sticks, leaving 1 small square and the big square; (c) 60 sticks

SHORTCUT

Count the big square. In matchstick square puzzles, always list squares of every size before you start removing. A 2×2 grid has 5 squares, a 3×3 grid has 14. Most wrong answers come from counting only the unit squares.

Example 38 [Google · pattern]

Show that the whole Gregorian calendar repeats exactly every 400 years, and use that fact to find the day of the week for 15 August 2347, given that 15 August 1947 was a Friday.

SOLUTION

Insight: the calendar repeats only if the 400-year block contains a whole number of weeks. Count the days.

Step 1 — leap years in 400 years. Multiples of 4: $400 \div 4 = 100$. Remove the century years: $400 \div 100 = 4$ of them. Add back those divisible by 400: $400 \div 400 = 1$.

$$100 - 4 + 1 = 97 \text{ leap years.}$$

Step 2 — total days.

$$400 \times 365 + 97 = 146000 + 97 = 146097.$$

Step 3 — is that a whole number of weeks?

$$146097 \div 7 = 20871 \text{ exactly (since } 7 \times 20871 = 146097).$$

Remainder 0. So a 400-year block contains 0 odd days.

Step 4 — the consequence. Every date shifts by 0 weekdays after 400 years, and the leap pattern also repeats (2000 and 2400 are both leap years, 2100 / 2200 / 2300 and 2500 / 2600 / 2700 are all non-leap). So the calendar is identical.

Step 5 — apply it. $2347 - 1947 = 400$.

So 15 August 2347 falls on the same weekday as 15 August 1947.

Answer: Friday

Quick verification of the 1947 date by the odd-day method: centuries $1600 \rightarrow 0$, $300 \rightarrow 1$. Years 1901–1946: 46 years, 11 leap (1904 to 1944), so $35 + 22 = 57$, and $57 \div 7$ leaves 1. Days in 1947 to 15 August: $212 + 15 = 227$, and $227 \div 7$ leaves 3. Total $0 + 1 + 1 + 3 = 5 \rightarrow$ Friday. ✓

PRACTICE · TIER 3 target: 4 min/Q

- How many times in 24 hours are the hour and minute hands exactly opposite each other? Justify the count.
- At what times between 3 and 4 o'clock is the angle between the hands exactly 50° ?
- The minute hand of a clock is 7 cm long. How far does its tip travel in 45 minutes? (Take $\pi = \frac{22}{7}$.)
- A leap year is picked at random. What is the probability that it has 53 Fridays?
- How many times in 12 hours is the angle between the hands exactly 0° ?
- Five people must cross a bridge at night with one torch, two at a time, moving at the slower pace. Their times are 1, 2, 4, 8 and 16 minutes. What is the least total time?
- How many matchsticks are needed for a 4×4 grid of unit squares, and how many squares of all sizes does it contain?
- A 12-hour digital clock shows H:MM or HH:MM with no leading zero. How many times in 24 hours does the display use only one distinct digit (for example 1:11)?

ANSWER KEY

- 22.** The gap between the hands turns at 5.5° per minute, so in 12 hours it makes $5.5 \times 720 \div 360 = 11$ complete turns. The gap equals 180° exactly once per turn, giving 11 times in 12 hours and 22 in a day. (It is 22, not 24, because the gap makes 11 turns, not 12.)
- 3:07 and $\frac{3}{11}$ min, and 3:25 and $\frac{5}{11}$ min.** $M = \frac{2}{11}(90 - 50) = \frac{80}{11} = 7\frac{3}{11}$ and $M = \frac{2}{11}(90 + 50) = \frac{280}{11} = 25\frac{5}{11}$. Both lie in $0 \leq M < 60$, so both count.
- 33 cm.** In 60 minutes the tip covers one full circle, $2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm. In 45 minutes it covers $\frac{3}{4}$ of that: $44 \times \frac{3}{4} = 33$ cm.
- $\frac{2}{7}$.** A leap year is 52 weeks plus 2 extra days, which form one of 7 equally likely consecutive pairs. Two of those pairs — (Thu, Fri) and (Fri, Sat) — contain a Friday.
- 11.** The gap makes 11 complete turns in 12 hours and is 0° once per turn. The 12:00 overlap is counted once, at the start.

6. **28 minutes.** Handle the two slowest, 8 and 16, with helpers $f_1 = 1$, $f_2 = 2$: $\min(1 + 4 + 16, 2 + 8 + 16) = \min(21, 26) = 21$. That leaves 1, 2, 4 at the start; three people cost $4 + 1 + 2 = 7$ (1 and 4 over, 1 back, 1 and 2 over). Total $21 + 7 = 28$.
7. **40 sticks and 30 squares.** Sticks $= 2n(n + 1) = 2 \times 4 \times 5 = 40$. Squares $= 16 + 9 + 4 + 1 = 30$.
8. **12.** Single-digit hours need $h = m_1 = m_2$ with $m_1 \leq 5$, giving 1:11, 2:22, 3:33, 4:44, 5:55 — that is 5. Two-digit hours need all four digits equal: only 11:11 works (10 and 12 have two different digits). So $5 + 1 = 6$ in 12 hours, and $6 \times 2 = 12$ in a day.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 30 min for all 18

- Find the angle between the hands at 9:30.
- At what time between 6 and 7 o'clock are the hands together?
- A clock gains 15 minutes a day. After how many days will it show the correct time again?
- What day of the week was 15 August 1947?
- How many odd days are there in 100 years?
- A watch gains 5 minutes every hour and is set right at 8:00 a.m. What is the true time when it shows 9:00 p.m.?
- A father is 30 years older than his son. In 6 years the father will be 3 times as old as the son. Find their present ages.
- How many times in 24 hours are the two hands in a straight line?
- What day of the week was 1 January 2000?
- What is the smallest set of weights that weighs every whole number of kilograms from 1 to 13 on a two-pan balance, weights allowed on both pans?
- A farmer must ferry a wolf, a goat and a cabbage across a river. The boat holds the farmer plus one item. The wolf must not be left alone with the goat, nor the goat with the cabbage. What is the least number of crossings?
- Find the angle between the hands at 12:25.
- A month of 30 days begins on a Sunday. How many Mondays does it have?
- Which is the first year after 2023 with exactly the same calendar as 2023?
- Twenty-seven coins look alike; one is lighter. What is the least number of pan-balance weighings that always finds it?
- If today is Tuesday, what day will it be 61 days from now?
- You have two ropes, each burning end to end in exactly 1 hour but at an uneven rate. How do you measure 45 minutes?
- At what time between 2 and 3 o'clock are the hands at right angles?

ANSWER KEY

- 105°.** $|30 \times 9 - 5.5 \times 30| = |270 - 165| = 105$.
- 6:32 and $\frac{8}{11}$ min.** $M = \frac{2}{11}(30 \times 6) = \frac{360}{11} = 32\frac{8}{11}$.
- 48 days.** $720 \div 15 = 48$.
- Friday.** Centuries $1600 \rightarrow 0, 300 \rightarrow 1$. Years $1901\text{--}1946$: $35 + 22 = 57 \rightarrow 1$. Days to 15 Aug: $212 + 15 = 227 \rightarrow 3$. Total $0 + 1 + 1 + 3 = 5 \rightarrow$ Friday.
- 5.** 76 ordinary years and 24 leap years give $76 + 48 = 124$, and $124 \div 7$ leaves 5.
- 8:00 p.m.** The watch shows 13 h = 780 watch-min at 65 watch-min per 60 true min, so true = $780 \times \frac{60}{65} = 720$ min = 12 h. 8 : 00 a.m. + 12 = 8 : 00 p.m.
- Son 9, father 39.** $f = s + 30$ and $s + 36 = 3(s + 6)$ give $2s = 18$, so $s = 9$. Check: 15 and 45, and $45 = 3 \times 15$. ✓
- 44.** The gap makes 11 turns per 12 hours; a straight line means 0° or 180° , which is twice per turn. $11 \times 2 \times 2 = 44$.

9. **Saturday.** Centuries $1600 \rightarrow 0, 300 \rightarrow 1$. Years 1901–1999: 99 years, 24 leap, so $75 + 48 = 123 \rightarrow 4$. 1 January is 1 day. Total $0 + 1 + 4 + 1 = 6 \rightarrow$ Saturday.
10. **Three weights: 1, 3, 9 kg.** They reach $\frac{3^3-1}{2} = 13$ exactly. Two weights reach only 4.
11. **7 crossings.** Goat over; return empty; wolf over; goat back; cabbage over; return empty; goat over.
12. **137.5°.** Treat 12 as $H = 0$: $|0 - 5.5 \times 25| = 137.5$.
13. **5.** $30 = 4 \times 7 + 2$, so the first two weekdays (Sunday and Monday) occur 5 times each.
14. **2034.** Add odd days year by year and stop at a multiple of 7. Running total after 2023 is 1; after 2024 (leap) $1 + 2 = 3$; 2025 $\rightarrow 4$; 2026 $\rightarrow 5$; 2027 $\rightarrow 6$; 2028 (leap) $\rightarrow 8 \rightarrow 1$; 2029 $\rightarrow 2$; 2030 $\rightarrow 3$; 2031 $\rightarrow 4$; 2032 (leap) $\rightarrow 6$; 2033 $\rightarrow 0$. So 1 January 2034 falls on the same weekday, and 2034 is ordinary like 2023.
15. **3.** $27 = 3^3$, so three three-way splits suffice, and $3^2 = 9 < 27$ rules out 2.
16. **Sunday.** $61 = 8 \times 7 + 5$, so Tuesday $+5 =$ Sunday.
17. **Light both ends of rope A and one end of rope B.** A is gone at 30 minutes; at that moment light B's other end, and B's remaining 30-minutes-worth burns out in 15. Total $30 + 15 = 45$.
18. **2:27 and $\frac{3}{11}$ min.** $M = \frac{2}{11}(60 + 90) = \frac{300}{11} = 27\frac{3}{11}$. The minus branch gives $\frac{2}{11}(60 - 90) < 0$, so it is rejected — this hour has only one right angle.

ONE-PAGE REVISION CARD

Clock speeds. Minute hand $6^\circ/\text{min}$, hour hand $0.5^\circ/\text{min}$, gap grows at $5.5^\circ/\text{min}$. The gap makes 11 complete turns every 12 hours.

Angle. $\theta = |30H - 5.5M|$; if over 180° , use $360 - \theta$.

When is the angle θ ? $M = \frac{2}{11}(30H \pm \theta)$, keeping $0 \leq M < 60$.

Counts in 24 hours. Together 22. Opposite 22. Straight line 44. Right angle 44. Any other fixed angle 44. All three hands together: 2.

Gap between coincidences. $\frac{720}{11} = 65\frac{5}{11}$ minutes.

Mirror of a clock. $11 : 60 - \text{time}$ (use $23 : 60$ if the hour reads 12).

Faulty clocks. A clock gaining g min/hour shows $60 + g$ clock-minutes per 60 true minutes, so

$$\text{true} = \text{shown} \times \frac{60}{60 + g}.$$

A clock drifting d min/day is next correct after $720 \div d$ days. Two clocks: use the **sum** of the rates if one gains and one loses, the **difference** if both gain.

Odd days. Ordinary year 1, leap year 2. $100 \rightarrow 5$, $200 \rightarrow 3$, $300 \rightarrow 1$, $400 \rightarrow 0$. Day code: 0 Sun, 1 Mon, 2 Tue, 3 Wed, 4 Thu, 5 Fri, 6 Sat.

Leap rule. Divisible by 4; a century year must be divisible by 400.

Month running totals (ordinary year). Jan 31, Feb 59, Mar 90, Apr 120, May 151, Jun 181, Jul 212, Aug 243, Sep 273, Oct 304, Nov 334, Dec 365. Add 1 from February onwards in a leap year.

Day-of-week recipe. century odd days + year odd days ($1 \times \text{ordinary} + 2 \times \text{leap}$) + day-of-year, all mod 7.

Calendar repeats. Ordinary year: after 6 or 11 years (12 across a non-leap century year). Leap year: after 28 years. Everything: after 400 years (146097 days = 20871 weeks exactly). 31-day month: 5 of the first three weekdays. 30-day month: 5 of the first two.

Probability of 53 of a given weekday. Ordinary $\frac{1}{7}$, leap $\frac{2}{7}$, a random year $\frac{71}{400}$.

Ages. Let x be the present age of the younger person. Shift **both** ages by the same amount. The difference of two ages is constant — use it to check.

Weighing. One pan only: 1, 2, 4, 8, ... reaches $2^n - 1$. Both pans: 1, 3, 9, 27, 81 reaches $\frac{3^n - 1}{2}$, so 4 weights reach 40 and 5 reach 121. Finding one odd item of known direction among N takes $\lceil \log_3 N \rceil$ weighings; unknown direction among 12 takes 3.

Crossings and measuring. Wolf-goat-cabbage: 7 crossings. Torch bridge: send the two slowest together, cost $\min(f_1 + 2f_2 + b, 2f_1 + a + b)$, then repeat. Two jugs p and q reach v if $\gcd(p, q)$ divides v and $v \leq \max(p, q)$. Two uneven 60-minute ropes: both ends of one plus one end of the other gives 45 minutes.

Matchsticks. A row of n squares: $3n + 1$ sticks. An $n \times n$ grid: $2n(n + 1)$ sticks, and $1^2 + \dots + n^2$ squares.

The top 5 traps

1. **Use $5.5M$, never $6M$.** The hour hand keeps moving while the minutes pass.
2. **Angle above 180° ?** Subtract from 360 before matching an option.
3. **“Gains 10 minutes” is a rate, not a one-off shift.** Scale with the ratio $60 : (60 + g)$ whenever you are going from clock time back to true time.
4. **Same start weekday is not the same calendar.** The leap status must match too.
5. **In an average-age problem, everybody ages.** Add $n \times (\text{number of people})$ before comparing totals.

Data Interpretation: Tables, Bar, Line, Pie

Read the chart, pick the base, divide once

What you need to know

WHAT YOU NEED TO KNOW

The only four operations in DI. Add a row or column. Take a percent. Take a ratio. Compare two numbers. Everything else is dressing.

Percent change.

$$\% \text{ change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

The **old** value is always the base. A rise of $x\%$ means multiply by $(1 + \frac{x}{100})$.

Share of total.

$$\text{share of } A = \frac{A}{\text{total}} \times 100 \quad (\text{read as a percent})$$

Pie chart.

- value of a sector = $\frac{\text{percent}}{100} \times \text{total}$
- central angle = percent $\times 3.6$ degrees
- percent = $\frac{\text{angle}}{3.6}$
- Full circle = $360^\circ = 100\%$

Bar chart. Each bar is one number. A grouped bar chart is a table drawn sideways. Read the axis scale once, then treat it as a table.

Line chart. Two different things get drawn as lines, and you must know which one you have:

- a line of **values** (revenue, users) — the value falls when the line falls;
- a line of **growth rates** (percent over previous year) — the value still **rises** while the line stays above zero.

Linking two charts. If the total grows by multiplier M and a part's share goes from s_1 to s_2 , then

$$\text{part multiplier} = \frac{s_2}{s_1} \times M$$

So the part can grow even when its share falls.

Weighted growth. If parts with shares $w_1, w_2, \dots$ (written as fractions that add to 1) grow by $g_1, g_2, \dots$ percent, the total grows by

$$w_1 g_1 + w_2 g_2 + \dots \text{ percent}$$

Average from a table. Average = (sum of entries) $\div$ (number of entries). But an **average rate** (price per unit, fare per trip) is

$$\frac{\text{total value}}{\text{total quantity}}$$

never the plain average of the rates.

Index numbers. If a base year is set to 100, then any two index values divide like the real values do. Index $138 \rightarrow 120.75$ is a change of $\frac{120.75-138}{138} = -12.5\%$, whatever the real numbers are.

Approximation discipline.

- Round to 2 significant figures **only** when the options are far apart.

- $\frac{1}{3} \approx 33\%$, $\frac{1}{6} \approx 16.7\%$, $\frac{1}{7} \approx 14.3\%$, $\frac{1}{8} = 12.5\%$, $\frac{1}{9} \approx 11.1\%$, $\frac{1}{11} \approx 9.1\%$, $\frac{1}{12} \approx 8.3\%$.
- To find A as a percent of B : first find 10% of B , then scale.
- Never round a base before dividing. Round the answer, not the input.

Ratio comparison without dividing. To compare $\frac{a}{b}$ and $\frac{c}{d}$, cross-multiply: ad versus cb . The bigger product sits over the bigger fraction.

Warm-up

WARM-UP

build the reflex

Warm-ups 1–5 use this small table. Units sold, in thousands.

Store	2022	2023
A	40	50
B	60	54
C	25	30
D	75	90

Example 1

Find the total for 2022.

SOLUTION

$$40 + 60 = 100. 100 + 25 = 125. 125 + 75 = 200.$$

Answer: 200 thousand units

Example 2

Find the percent change for store A.

SOLUTION

$$\text{Rise} = 50 - 40 = 10. \text{ Base is the old value, } 40. \left(\frac{10}{40}\right) \times 100 = 0.25 \times 100 = 25.$$

Answer: +25%

Example 3

Find the percent change for store B.

SOLUTION

$$\text{Fall} = 60 - 54 = 6. \text{ Base is } 60. \left(\frac{6}{60}\right) \times 100 = 10. \text{ The value went down.}$$

Answer: -10%

Example 4

Find the ratio of C to D in 2023.

SOLUTION

$$30 : 90. \text{ Divide both by } 30: 1 : 3.$$

Answer: 1 : 3

Example 5

Find the average of the four 2023 values.

SOLUTION

$$50 + 54 = 104, 104 + 30 = 134, 134 + 90 = 224, 224 \div 4 = 56.$$

Answer: 56 thousand units

Example 6

A pie chart shows exports of Rs. 7,200 crore. Electronics is 25%. Find the value of the electronics sector.

SOLUTION

$$25\% = \frac{25}{100} = \frac{1}{4}, 7200 \div 4 = 1800.$$

Answer: Rs. 1,800 crore

Example 7

A pie chart of a total of 6,000 has a sector of 54° . Find its value.

SOLUTION

$$\text{Fraction of the circle} = \frac{54}{360} \cdot \frac{54}{360} = \frac{27}{180} = \frac{3}{20}, 6000 \times \frac{3}{20} = (6000 \div 20) \times 3 = 300 \times 3 = 900.$$

Answer: 900

Example 8

A sector is 18% of a pie chart. Find its central angle.

SOLUTION

$$\text{Angle} = \text{percent} \times 3.6, 18 \times 3.6 = 18 \times 3 + 18 \times 0.6 = 54 + 10.8 = 64.8.$$

Answer: 64.8°

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Data set 1 — table

▮ Laptops sold by five stores, in units.

Store	2021	2022	2023
Ahmedabad	1,200	1,500	1,800
Bengaluru	2,400	2,200	2,640
Chennai	1,600	2,000	2,200
Delhi	3,000	3,300	3,960
Pune	1,800	2,000	2,400

Example 9 [TCS NQT pattern]

Find the total number of laptops sold by all five stores in 2022.

▮ **SOLUTION**

Add the 2022 column. $1500 + 2200 = 3700$. $3700 + 2000 = 5700$. $5700 + 3300 = 9000$. $9000 + 2000 = 11000$.

Answer: 11,000 units

Two totals you will reuse: 2021 total = $1200 + 2400 + 1600 + 3000 + 1800 = 10,000$, and 2023 total = $1800 + 2640 + 2200 + 3960 + 2400 = 13,000$.

Example 10 [Infosys pattern]

Find the percent rise in Delhi's sales from 2021 to 2023.

SOLUTION

Old = 3000. New = 3960. Rise = $3960 - 3000 = 960$. $\left(\frac{960}{3000}\right) \times 100$. $\frac{960}{3000} = \frac{96}{300} = \frac{32}{100} = 0.32$. $0.32 \times 100 = 32$.

Answer: +32%

Example 11 [Accenture pattern]

Which store had the highest percent growth from 2021 to 2023?

SOLUTION

Do all five. Each time, divide the new value by the old value. Ahmedabad: $\frac{1800}{1200} = 1.5 \rightarrow +50\%$. Bengaluru: $\frac{2640}{2400} = 1.1 \rightarrow +10\%$. Chennai: $\frac{2200}{1600} = 1.375 \rightarrow +37.5\%$. Delhi: $\frac{3960}{3000} = 1.32 \rightarrow +32\%$. Pune: $\frac{2400}{1800} = \frac{4}{3} = 1.3333 \rightarrow +33.33\%$. The largest multiplier is 1.5.

Answer: Ahmedabad, +50%

SHORTCUT

Do not compute "rise, then divide by old". Compute **new $\div$ old** in one step and read the growth straight off the multiplier. 1.32 means +32%. 0.88 means -12%. One division instead of a subtraction plus a division.

TRAP

Delhi rose by 960 units and Ahmedabad rose by only 600 units, yet Ahmedabad grew faster in percent. A bigger **absolute** rise is not a bigger **percent** rise. Read the question and see which one it asks for.

Example 12 [Wipro pattern]

In 2023, Chennai's sales are what percent of the five-store total?

SOLUTION

Chennai 2023 = 2200. Total 2023 = 13000. $\left(\frac{2200}{13000}\right) \times 100$. Cancel 100 from top and bottom: $\frac{2200}{13000} = \frac{22}{130} = \frac{11}{65}$. $11 \div 65$: $65 \times 0.16 = 10.4$. Remainder $11 - 10.4 = 0.6$. $\frac{0.6}{65} = 0.00923$. So $\frac{11}{65} = 0.16923$. $0.16923 \times 100 = 16.923$.

Answer: About 16.92%

Example 13 [TCS NQT pattern]

Find the ratio of Bengaluru's 2023 sales to Pune's 2023 sales.

SOLUTION

2640 : 2400. Divide both by 10: 264 : 240. Divide both by 24: $264 \div 24 = 11$ and $240 \div 24 = 10$.

Answer: 11 : 10

Data set 2 — grouped bar chart

Revenue of a food-delivery app, in Rs. crore. Two bars per quarter.

Quarter	2022	2023
Q1	240	300
Q2	280	322
Q3	320	416
Q4	360	450

Example 14 [Capgemini pattern]

Find the total 2023 revenue and the percent growth over 2022.

SOLUTION

2023 total: $300 + 322 = 622$. $622 + 416 = 1038$. $1038 + 450 = 1488$. 2022 total: $240 + 280 = 520$; $520 + 320 = 840$; $840 + 360 = 1200$. Rise = $1488 - 1200 = 288$. $\left(\frac{288}{1200}\right) \times 100$. $\frac{288}{1200} = \frac{24}{100} = 0.24$. $0.24 \times 100 = 24$.

Answer: Rs. 1,488 crore; +24%

Example 15 [Cognizant pattern]

Which quarter had the highest percent growth from 2022 to 2023?

SOLUTION

Q1: $\frac{300}{240} = 1.25 \rightarrow +25\%$. Q2: $\frac{322}{280} = 1.15 \rightarrow +15\%$. Q3: $\frac{416}{320} = 1.30 \rightarrow +30\%$. Q4: $\frac{450}{360} = 1.25 \rightarrow +25\%$. The largest is Q3.

Answer: Q3, +30%

Example 16 [Accenture pattern]

Q4 of 2023 is what percent of the 2023 yearly revenue?

SOLUTION

Q4 = 450. Year = 1488. $\left(\frac{450}{1488}\right) \times 100$. First take 30% of 1488: $1488 \times 0.30 = 446.4$. $450 - 446.4 = 3.6$ is left over. $\frac{3.6}{1488} = 0.00242$, which is 0.242%. So the share is $30 + 0.242 = 30.242$.

Answer: About 30.24%

SHORTCUT

For “A is what percent of B”, anchor on an easy percent of B first. Here 30% of 1488 was 446.4, already very close to 450. The remaining gap is tiny, so you only fine-tune. This beats long division every time.

Data set 3 — pie chart

A company of 7,200 employees, split by department.

Dept	Engineering	Sales	Support	Operations	Finance	HR
Share	35%	20%	15%	18%	7%	5%

Example 17 [TCS NQT pattern]

How many employees work in Operations?

SOLUTION

18% of 7,200. 1% of 7200 = 72. $18 \times 72 = 18 \times 70 + 18 \times 2 = 1260 + 36 = 1296$.

Answer: 1,296 employees

Example 18 [Infosys pattern]

Find the central angle of the Support sector.

SOLUTION

$$\text{Angle} = \text{percent} \times 3.6. \quad 15 \times 3.6 = 54.$$

Answer: 54°

SHORTCUT

Percent to degrees: multiply by 3.6. Degrees to percent: divide by 3.6. Memorise four anchors: $10\% = 36^\circ$, $25\% = 90^\circ$, $50\% = 180^\circ$, $1\% = 3.6^\circ$.

Example 19 [Wipro pattern]

Engineering has how many percent more employees than Sales?

SOLUTION

Both sectors come from the same total, so you may compare the **percents** directly and never compute the headcounts. Difference = $35 - 20 = 15$ percentage points. Base is Sales, which is 20. $\left(\frac{15}{20}\right) \times 100 = 0.75 \times 100 = 75$. Check with real numbers: Engineering = $0.35 \times 7200 = 2520$, Sales = $0.20 \times 7200 = 1440$. $\frac{2520 - 1440}{1440} = \frac{1080}{1440} = 0.75$. Same answer.

Answer: 75% more

TRAP

This shortcut works **only** when both sectors come from the same pie. If the two sectors sit in two different pies with different totals, you must convert to real values first. Two pies never share a base.

Data set 4 — line chart

I Average daily active users of an app, in lakh.

Month	Jan	Feb	Mar	Apr	May	Jun
Users	12	15	18	16.2	21.6	27

Example 20 [Capgemini pattern]

(a) Find the percent change from March to April. (b) Find the percent change from January to June.

SOLUTION

(a) Old = 18, new = 16.2. Fall = $18 - 16.2 = 1.8$. $\left(\frac{1.8}{18}\right) \times 100 = 0.1 \times 100 = 10$. The line fell, so this is a drop. (b) Old = 12, new = 27. $\frac{27}{12} = \frac{9}{4} = 2.25$. A multiplier of 2.25 means $2.25 - 1 = 1.25$, that is 125% growth.

Answer: (a) -10% (b) +125%

PRACTICE · TIER 1 target: 60 s/Q

Questions 1–6. Units sold, in hundreds, by four dealers.

Dealer	2022	2023
K	250	325

L	400	360
M	320	400
N	280	336

- Find the total units sold in 2023 (in hundreds).
- Find the percent change for dealer L.
- Which dealer had the largest percent growth?
- Which dealer had the largest rise in absolute units?
- Find the ratio of M to L in 2023.
- M's 2023 sales are what percent of the 2023 total?

Questions 7–10. A household budget of Rs. 60,000 a month.

Head	Rent	Food	Transport	School fees	Utilities	Savings
Share	30%	22%	12%	18%	8%	10%

- Find the amount spent on food.
- Find the central angle of the school-fees sector.
- Rent is how many percent more than school fees?
- Rent and food together take what amount?

Questions 11–14. A startup's headcount at each year end.

Year	2019	2020	2021	2022	2023
Staff	40	60	90	81	108

- Find the percent rise from 2019 to 2021.
- In which year did headcount fall, and by what percent?
- Find the percent rise from 2022 to 2023.
- Find the average headcount over the five years.

ANSWER KEY

- 1,421 hundred, i.e. 1,42,100 units.** $325 + 360 = 685$; $685 + 400 = 1085$; $1085 + 336 = 1421$.
- −10%.** $\frac{360}{400} = 0.90$, so a fall of 10%.
- K.** $\frac{325}{250} = 1.30$ (+30%); $\frac{400}{320} = 1.25$ (+25%); $\frac{336}{280} = 1.20$ (+20%); L fell.
- M.** Rises are K +75, L −40, M +80, N +56. M wins on units but K wins on percent — the two questions have different answers on purpose.
- 10 : 9.** $400 : 360$; divide both by 40.
- About 28.15%.** $(\frac{400}{1421}) \times 100$. 28% of 1421 = 397.88; the gap 2.12 is 0.15% of 1421; so $28 + 0.15 = 28.15$.
- Rs. 13,200.** 1% of 60000 = 600, so $22 \times 600 = 13200$.
- 64.8°.** $18 \times 3.6 = 64.8$.
- 66 $\frac{2}{3}$ %.** Same pie, so compare shares: $\frac{30-18}{18} = \frac{12}{18} = \frac{2}{3}$.
- Rs. 31,200.** $30 + 22 = 52\%$, and $52 \times 600 = 31200$.
- +125%.** $\frac{90}{40} = 2.25$, so a rise of 1.25 times the original.
- 2022, a fall of 10%.** $\frac{81}{90} = 0.90$.
- +33 $\frac{1}{3}$ %.** $\frac{108}{81} = \frac{4}{3}$.
- 75.8.** $40 + 60 + 90 + 81 + 108 = 379$, and $379 \div 5 = 75.8$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Data set 5 — table with two columns that must be multiplied

I A ride-hailing firm. Trips are in thousands; the fare is an average per trip.

City	Trips ('000)	Average fare (S\$)
Singapore	480	12.50
Bangkok	900	5.00
Jakarta	1,500	3.20
Manila	1,200	3.75
Hanoi	600	2.80

Example 21 [Grab · pattern]

Find each city's revenue and the total revenue.

SOLUTION

Revenue = trips $\times$ fare. Trips are in thousands, so each product is in thousands of S\$. Singapore: $480 \times 12.50 = 480 \times 12 + 480 \times 0.5 = 5760 + 240 = 6000$. Bangkok: $900 \times 5 = 4500$. Jakarta: $1500 \times 3.20 = 1500 \times 3 + 1500 \times 0.2 = 4500 + 300 = 4800$. Manila: $1200 \times 3.75 = 1200 \times 3 + 1200 \times 0.75 = 3600 + 900 = 4500$. Hanoi: $600 \times 2.80 = 600 \times 2 + 600 \times 0.8 = 1200 + 480 = 1680$. Total: $6000 + 4500 = 10500$; $+4800 = 15300$; $+4500 = 19800$; $+1680 = 21480$.

Answer: Singapore S\$6.00 m, Bangkok S\$4.50 m, Jakarta S\$4.80 m, Manila S\$4.50 m, Hanoi S\$1.68 m; total S\$21.48 m

Example 22 [Sea/Shopee · pattern]

Singapore's revenue is what percent of the total?

SOLUTION

$\left(\frac{6000}{21480}\right) \times 100$. Anchor: 28% of 21480 = $21480 \times 0.28 = 6014.4$. That is slightly above 6000. Gap = $6014.4 - 6000 = 14.4$. $\frac{14.4}{21480} = 0.00067$, which is 0.067%. So the share is $28 - 0.067 = 27.933$.

Answer: About 27.93%

TRAP

Singapore has only $\frac{480}{4680} \approx 10.3\%$ of the trips but nearly 28% of the revenue. A share of one column is never a share of another column. Always ask: share **of what**?

Example 23 [Grab · pattern]

The firm keeps 20% of every fare as commission. Find the total commission, and the commission earned in Jakarta.

SOLUTION

Total commission = $0.20 \times 21480 = 4296$ (thousand S\$). Jakarta commission = $0.20 \times 4800 = 960$ (thousand S\$).

Answer: S\$4.296 million in all; S\$0.96 million from Jakarta

Data set 6 — two pie charts, two different totals

I A Bangkok logistics firm's cost split. Totals differ between the two years.

Head	Fuel	Wages	Vehicle lease	Warehouse	Admin
2022 (total THB 48 m)	30%	25%	20%	15%	10%

2023 (total THB 60 m)	24%	28%	20%	16%	12%
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Example 24 [LINE MAN · pattern]

Find the percent change in the fuel bill from 2022 to 2023.

SOLUTION

2022 fuel = $0.30 \times 48 = 14.4$ million THB. 2023 fuel = $0.24 \times 60 = 14.4$ million THB. Change = $14.4 - 14.4 = 0$.

Answer: No change, 0%

TRAP

The fuel **share** dropped from 30% to 24%, a fall of 6 percentage points, and yet the fuel **bill** did not fall at all. A falling share only means the head grew slower than the total. It never by itself means the value fell.

Example 25 [SCB · pattern]

Find the percent change in wages.

SOLUTION

2022 wages = $0.25 \times 48 = 12$ million. 2023 wages = $0.28 \times 60 = 16.8$ million. Rise = $16.8 - 12 = 4.8$. $\left(\frac{4.8}{12}\right) \times 100 = 0.4 \times 100 = 40$.

Answer: +40%

Example 26 [Agoda · pattern]

Which cost head grew the fastest in percent terms?

SOLUTION

Use the link rule. Total multiplier $M = \frac{60}{48} = 1.25$. Part multiplier = $\frac{\text{new share}}{\text{old share}} \times M$. Fuel: $\left(\frac{24}{30}\right) \times 1.25 = 0.8 \times 1.25 = 1.00 \rightarrow 0\%$. Wages: $\left(\frac{28}{25}\right) \times 1.25 = 1.12 \times 1.25 = 1.40 \rightarrow +40\%$. Lease: $\left(\frac{20}{20}\right) \times 1.25 = 1 \times 1.25 = 1.25 \rightarrow +25\%$. Warehouse: $\left(\frac{16}{15}\right) \times 1.25 = 1.0667 \times 1.25 = 1.3333 \rightarrow +33.33\%$. Admin: $\left(\frac{12}{10}\right) \times 1.25 = 1.2 \times 1.25 = 1.50 \rightarrow +50\%$. Check one by hand: Admin 2022 = $0.10 \times 48 = 4.8$; 2023 = $0.12 \times 60 = 7.2$; $\frac{7.2}{4.8} = 1.5$. Matches.

Answer: Admin, +50%

SHORTCUT

The link rule is the whole of two-chart DI.

$$\text{part multiplier} = \frac{\text{new share}}{\text{old share}} \times \frac{\text{new total}}{\text{old total}}$$

You never compute a single rupee value. Five growth rates in five one-line divisions.

Data set 7 — bar chart with a line drawn over it

■ Bars: units shipped, in thousands. Line: cost per unit, in S\$.

Month	Units ('000)	Cost per unit (S\$)
Jul	50	8.00
Aug	60	7.50
Sep	75	7.20
Oct	80	7.50

Example 27 [Shopee · pattern]

Find the total cost each month. Which month cost the most, and what was the percent rise in total cost from July to October?

SOLUTION

Total cost = units $\times$ cost per unit, in thousands of S\$. Jul: $50 \times 8.00 = 400$. Aug: $60 \times 7.50 = 450$. Sep: $75 \times 7.20 = 75 \times 7 + 75 \times 0.2 = 525 + 15 = 540$. Oct: $80 \times 7.50 = 600$. Highest is October at 600 thousand S\$. Rise from July: $600 - 400 = 200$. $(\frac{200}{400}) \times 100 = 50$.

Answer: October, S\$600,000; a rise of 50% over July

Example 28 [DBS · pattern]

Find the average cost per unit over the four months.

SOLUTION

An average **rate** is total value $\div$ total quantity. Total cost = $400 + 450 = 850$; $850 + 540 = 1390$; $1390 + 600 = 1990$ (thousand S\$). Total units = $50 + 60 = 110$; $110 + 75 = 185$; $185 + 80 = 265$ (thousand). Average = $\frac{1990}{265}$. $265 \times 7 = 1855$. Remainder $1990 - 1855 = 135$. $\frac{135}{265} = 0.5094$. So $\frac{1990}{265} = 7.5094$.

Answer: About S\$7.51 per unit

TRAP

The plain average of the four rates is $(8.00 + 7.50 + 7.20 + 7.50) \div 4 = 30.20 \div 4 = 7.55$. That is **wrong**. It treats July (50,000 units) as equally important as October (80,000 units). An average rate must always be total $\div$ total.

Data set 8 — a line chart of growth rates, not values

I The line shows the percent growth of a fintech's revenue over the previous year. Revenue in 2019 was S\$40 million.

Year	2020	2021	2022	2023
Growth over previous year	+20%	+25%	−10%	+30%

Example 29 [GIC · pattern]

(a) Find the revenue in each year. (b) In which year was revenue the highest before 2023? (c) Find the net percent change from 2019 to 2023.

SOLUTION

(a) Multiply the multipliers one year at a time. 2020: $40 \times 1.20 = 48$. 2021: $48 \times 1.25 = 48 + 48 \times 0.25 = 48 + 12 = 60$. 2022: $60 \times 0.90 = 54$. 2023: $54 \times 1.30 = 54 + 54 \times 0.30 = 54 + 16.2 = 70.2$. (b) The values are 40, 48, 60, 54, 70.2. Before 2023 the highest is 60, in 2021. (c) Net multiplier = $\frac{70.2}{40} = 1.755$. $1.755 - 1 = 0.755 \rightarrow +75.5\%$. Check by multiplying the rates: $1.20 \times 1.25 = 1.50$; $1.50 \times 0.90 = 1.35$; $1.35 \times 1.30 = 1.755$. Matches.

Answer: (a) 48, 60, 54, 70.2 (S\$ million) (b) 2021 (c) +75.5%

TRAP

The four plotted points are +20, +25, −10, +30. Read as a **value** line, 2022 looks like the worst year of the four and looks negative. Both readings are false. Revenue in 2022 was 54, which is above 2020's 48 and above 2019's 40. Revenue **rose** in 2020, 2021 and 2023 and fell only in 2022 — the one year the line dipped below zero. Check the axis label before you read a single point.

PRACTICE · TIER 2 target: 100 s/Q**Questions 1–5.** A logistics firm's markets.

Market	Orders ('000)	Average order value (\$)
Singapore	120	45
Malaysia	250	28
Thailand	300	22
Vietnam	180	18
Philippines	200	20

- Find the total value of goods sold across all five markets.
- Which market produces the highest value, and how much?
- Thailand's value is what percent of the total?
- Find the overall average order value across all markets.
- If Vietnam's orders grow 25% and its average order value rises to S\$20, find the percent rise in Vietnam's value.

Questions 6–8. A Thai retailer's revenue split.

Division	Apparel	Grocery	Electronics	Home
2022 (total THB 250 m)	36%	28%	20%	16%
2023 (total THB 300 m)	30%	30%	24%	16%

- Find the percent change in apparel revenue.
- Find the percent change in electronics revenue.
- Which division grew the fastest?

Questions 9–11. A bank's fee income grew as follows. Fee income in 2020 was S\$80 million.

Year	2021	2022	2023
Growth over previous year	+25%	+20%	−12.5%

- Find the fee income in 2023.
- Find the net percent change from 2020 to 2023.
- In which year was fee income the highest?

ANSWER KEY

- S\$26.24 million.** $120 \times 45 = 5400$; $250 \times 28 = 7000$; $300 \times 22 = 6600$; $180 \times 18 = 3240$; $200 \times 20 = 4000$. Sum = 26240 thousand.
- Malaysia, S\$7.0 million.** Thailand has more orders but a lower value per order.
- About 25.15%.** $\frac{6600}{26240}$: 25% of 26240 is 6560; the extra 40 is 0.15% of 26240.
- About S\$24.99.** Total orders = $120 + 250 + 300 + 180 + 200 = 1050$ thousand. $\frac{26240}{1050}$: $1050 \times 25 = 26250$, which is 10 too high, and $\frac{10}{1050} = 0.0095$, so $25 - 0.01 = 24.99$. The plain average of the five rates, $133 \div 5 = 26.60$, is the wrong answer.
- About +38.89%.** Multiplier = $1.25 \times \left(\frac{20}{18}\right) = 1.25 \times 1.1111 = 1.3889$. Values: $3240 \rightarrow 225 \times 20 = 4500$.
- 0%.** $0.36 \times 250 = 90$ and $0.30 \times 300 = 90$. Link rule: $\left(\frac{30}{36}\right) \times \left(\frac{300}{250}\right) = 0.8333 \times 1.2 = 1.00$.
- +44%.** $\left(\frac{24}{20}\right) \times 1.2 = 1.2 \times 1.2 = 1.44$. Values: $50 \rightarrow 72$.
- Electronics, +44%.** Grocery $\left(\frac{30}{28}\right) \times 1.2 = 1.2857$ (+28.57%); Home $\left(\frac{16}{16}\right) \times 1.2 = 1.2$ (+20%); Apparel 0%.
- S\$105 million.** $80 \times 1.25 = 100$; $100 \times 1.20 = 120$; $120 \times 0.875 = 105$.
- +31.25%.** $\frac{105}{80} = 1.3125$. Or $1.25 \times 1.20 \times 0.875 = 1.3125$.
- 2022, at S\$120 million.** The growth line is still positive in 2022, so income kept rising that year; it fell only in 2023.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 30 [Goldman Sachs · pattern]

Two firms report the regional split of their revenue.

Firm	Revenue	Asia	Europe
X	\$250 m	30%	26%
Y	\$400 m	20%	35%

(a) Whose Asia revenue is larger? (b) Taking the two firms together, what percent of combined revenue comes from Asia?

SOLUTION

Insight: a percent is meaningless until you know what total it sits on. Convert to money first, then re-divide.

(a) Firm X Asia = $0.30 \times 250 = 75$ million. Firm Y Asia = $0.20 \times 400 = 80$ million. So Y is larger, even though Y has the **smaller** Asia percentage.

(b) Combined Asia = $75 + 80 = 155$ million. Combined revenue = $250 + 400 = 650$ million. $\left(\frac{155}{650}\right) \times 100$. Anchor: 23% of 650 = 149.5. Gap = $155 - 149.5 = 5.5$. $\frac{5.5}{650} = 0.00846$, that is 0.846%. Share = $23 + 0.846 = 23.846$.

Cross-check with the weighted-average form: $\frac{250 \times 30 + 400 \times 20}{650} = \frac{7500 + 8000}{650} = \frac{15500}{650} = 23.846$. Matches.

Answer: (a) Firm Y, \$80 m against \$75 m (b) About 23.85%

TRAP

The plain average of 30% and 20% is 25%, and it is wrong. Percentages only average when the totals behind them are equal. Here Y is 1.6 times the size of X, so Y's 20% pulls the combined figure down towards 20, giving 23.85 rather than 25.

Example 31 [Google · pattern]

A region's share of a company's revenue fell from 25% to 22% in one year. For which total-growth rates g did the region's own revenue still rise?

SOLUTION

Insight: the part's multiplier is the share ratio times the total's multiplier. Ask when that product exceeds 1.

Let last year's total be T and this year's total be $T(1 + \frac{g}{100})$. Region last year = $0.25T$. Region this year = $0.22 \times T(1 + \frac{g}{100})$. Region multiplier $m = \frac{0.22T(1 + \frac{g}{100})}{0.25T} = 0.88(1 + \frac{g}{100})$. The region rose when $m > 1$: $0.88(1 + \frac{g}{100}) > 1$ $1 + \frac{g}{100} > \frac{1}{0.88}$ $\frac{1}{0.88} = \frac{100}{88} = \frac{25}{22} = 1.136363...$ $\frac{g}{100} > 0.136363...$ $g > 13.6363...$ As a fraction, $g > 13\frac{7}{11}$ percent. Check at $g = 20$: $m = 0.88 \times 1.20 = 1.056$, a rise of 5.6%. Yes. Check at $g = 10$: $m = 0.88 \times 1.10 = 0.968$, a fall of 3.2%. Correct, no rise. Check the boundary: at $1 + \frac{g}{100} = \frac{25}{22}$ we get $m = 0.88 \times \frac{25}{22} = (\frac{22}{25}) \times (\frac{25}{22}) = 1$ exactly, so the region is flat there. Just above it, the region grows.

Answer: $g > 13\frac{7}{11}\%$, that is, above about 13.64%

SHORTCUT

Turn every share question into two multipliers and stop thinking about money.

$$\text{share ratio} \times \text{total multiplier} = \text{part multiplier}$$

The share ratio here is $\frac{22}{25} = 0.88$. Any total growth that beats $\frac{1}{0.88}$ saves the region.

Example 32 [Amazon · pattern]

Three warehouses A, B and C shipped 84,000 parcels last month. This month A rose 20%, B fell 10%, C rose 25%, and the total rose to 95,600. Last month B shipped 4,000 more parcels than A. Find last month's figures for all three.

SOLUTION

Insight: three unknowns, three facts. Write the total, the growth total and the link, then eliminate down to one letter.

Let A, B, C be last month's parcels. Fact 1: $A + B + C = 84000$. Fact 2: $1.20A + 0.90B + 1.25C = 95600$. Fact 3: $B = A + 4000$.

Put Fact 3 into Fact 1: $A + (A + 4000) + C = 84000$ $2A + C = 80000$ $C = 80000 - 2A$.

Put Fact 3 and this into Fact 2: $1.20A + 0.90(A + 4000) + 1.25(80000 - 2A) = 95600$ $1.20A + 0.90A + 3600 + 100000 - 2.50A = 95600$ Collect the A terms: $1.20 + 0.90 - 2.50 = -0.40$. Collect the constants: $3600 + 100000 = 103600$. $-0.40A + 103600 = 95600$ $-0.40A = -8000$ $A = \frac{8000}{0.40} = 20000$. Then $B = 20000 + 4000 = 24000$. Then $C = 80000 - 2 \times 20000 = 40000$.

Check Fact 1: $20000 + 24000 + 40000 = 84000$. Correct. Check Fact 2: $1.20 \times 20000 = 24000$; $0.90 \times 24000 = 21600$; $1.25 \times 40000 = 50000$. $24000 + 21600 = 45600$; $45600 + 50000 = 95600$. Correct.

Answer: A = 20,000; B = 24,000; C = 40,000

Example 33 [Microsoft · pattern]

A line chart gives an index of quarterly revenue with Q1 set to 100.

Quarter	Q1	Q2	Q3	Q4
Index	100	115	138	120.75

Actual Q3 revenue was Rs. 82.8 crore. Find (a) Q1 revenue, (b) the percent change from Q3 to Q4, (c) the total revenue for the year.

SOLUTION

Insight: an index is the real series divided by a constant. So every ratio and every percent change can be read straight off the index, with no real numbers at all.

(b) first, because it needs nothing else. $\frac{120.75 - 138}{138} = -\frac{17.25}{138}$. $138 \times 0.125 = 17.25$, so $\frac{17.25}{138} = 0.125$. Change = -12.5% .

(a) Index 138 corresponds to Rs. 82.8 crore, so 1 index point is $\frac{82.8}{138}$. $138 \times 0.6 = 82.8$, so $\frac{82.8}{138} = 0.6$. Q1 has index 100, so Q1 revenue = $100 \times 0.6 = 60$ crore.

(c) Multiply each index by 0.6. Q2 = $115 \times 0.6 = 69$. Q3 = $138 \times 0.6 = 82.8$. Matches the given value, so the scale is right. Q4 = $120.75 \times 0.6 = 72.45$. Total = $60 + 69 = 129$; $129 + 82.8 = 211.8$; $211.8 + 72.45 = 284.25$.

Answer: (a) Rs. 60 crore (b) -12.5% (c) Rs. 284.25 crore

Example 34 [D. E. Shaw · pattern]

Five funds report their return and their fee for the year, both in thousands of S\$. Rank them by return per unit of fee, doing as little division as possible.

Fund	P	Q	R	S	T
Return	840	960	1,150	1,344	725

Fee	210	256	250	336	145
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SOLUTION

Insight: you do not need the value of a ratio to rank ratios. You only need to know which side of a round benchmark each one falls on.

Try the benchmark 4. P: $4 \times 210 = 840$. Return is exactly 840, so $P = 4.0$. Q: $4 \times 256 = 1024$. Return is 960, which is less, so $Q < 4$. Fine-tune Q: $3.75 \times 256 = 960$ exactly, so $Q = 3.75$. R: $4 \times 250 = 1000$. Return is 1,150, which is more, so $R > 4$. Fine-tune R: $4.6 \times 250 = 1150$, so $R = 4.6$. S: $4 \times 336 = 1344$. Return is exactly 1,344, so $S = 4.0$. T: $4 \times 145 = 580$. Return is 725, much more, so $T > 4$. Fine-tune T: $5 \times 145 = 725$, so $T = 5.0$.

Ranking: $T (5.0) > R (4.6) > P = S (4.0) > Q (3.75)$.

Answer: T is best, then R, then P and S tied, then Q

SHORTCUT

Never compute a ratio you only have to **compare**. Multiply the denominator by a round number and see which side the numerator falls. Two multiplications beat one long division, and they are exact.

Example 35 [Uber · pattern]

A bar chart groups a week's drivers by earnings band.

Band (\$\$)	0–200	200–400	400–600	600–800	800–1000
Drivers	120	260	340	180	100

(a) How many drivers are there? (b) What is the largest the total weekly payout could be? (c) What is the smallest the average earning per driver could be? (d) Give the usual midpoint estimate of the average.

SOLUTION

Insight: grouped data does not give you a number. It gives you an interval. Push every driver to the top of the band for the maximum, to the bottom for the minimum.

(a) $120 + 260 = 380$; $380 + 340 = 720$; $720 + 180 = 900$; $900 + 100 = 1000$ drivers.

(b) Put every driver at the top of the band. $120 \times 200 = 24000$. $260 \times 400 = 104000$. $340 \times 600 = 204000$. $180 \times 800 = 144000$. $100 \times 1000 = 100000$. Sum: $24000 + 104000 = 128000$; $+204000 = 332000$; $+144000 = 476000$; $+100000 = 576000$.

(c) Put every driver at the bottom of the band. $120 \times 0 = 0$. $260 \times 200 = 52000$. $340 \times 400 = 136000$. $180 \times 600 = 108000$. $100 \times 800 = 80000$. Sum: $0 + 52000 = 52000$; $+136000 = 188000$; $+108000 = 296000$; $+80000 = 376000$. Smallest possible average = $\frac{376000}{1000} = 376$.

(d) Midpoints are 100, 300, 500, 700, 900. $120 \times 100 = 12000$; $260 \times 300 = 78000$; $340 \times 500 = 170000$; $180 \times 700 = 126000$; $100 \times 900 = 90000$. Sum: $12000 + 78000 = 90000$; $+170000 = 260000$; $+126000 = 386000$; $+90000 = 476000$. Estimate = $\frac{476000}{1000} = 476$.

Answer: (a) 1,000 (b) S\$576,000 (c) S\$376 (d) about S\$476

TRAP

A grouped bar chart can never give an exact mean. Here the true average lies anywhere from S\$376 to S\$576. If a question asks for “the average” from grouped bands, it wants the midpoint estimate — and the honest answer is a range.

Example 36 [Adobe · pattern]

A media firm's 2023 revenue of Rs. 480 crore was split Subscriptions 45%, Ads 30%, Licensing 25%. In 2024 subscriptions grew 20%, ads fell 10%, and licensing grew 8%.

Find (a) the 2024 total, (b) the overall growth rate, (c) the 2024 share of each stream, (d) the change in the subscriptions share in percentage points.

SOLUTION

Insight: the total's growth is the weighted average of the parts' growths, with last year's shares as the weights. Compute that first — it checks all the rest.

(b) Weighted growth = $0.45 \times 20 + 0.30 \times (-10) + 0.25 \times 8$. $0.45 \times 20 = 9$. $0.30 \times (-10) = -3$. $0.25 \times 8 = 2$. $9 - 3 + 2 = 8$. So the total grew 8%.

(a) 2024 total = $480 \times 1.08 = 480 + 480 \times 0.08 = 480 + 38.4 = 518.4$ crore.

(c) 2023 values: Subscriptions = $0.45 \times 480 = 216$; Ads = $0.30 \times 480 = 144$; Licensing = $0.25 \times 480 = 120$. Check: $216 + 144 + 120 = 480$. 2024 values: Subscriptions = $216 \times 1.20 = 216 + 43.2 = 259.2$. Ads = $144 \times 0.90 = 129.6$. Licensing = $120 \times 1.08 = 120 + 9.6 = 129.6$. Check the total: $259.2 + 129.6 = 388.8$; $388.8 + 129.6 = 518.4$. Matches part (a). Shares: Subscriptions = $\frac{259.2}{518.4} = 0.50 \rightarrow 50\%$ (because $518.4 \times 0.5 = 259.2$). Ads = $\frac{129.6}{518.4} = 0.25 \rightarrow 25\%$ (because $518.4 \times 0.25 = 129.6$). Licensing = $\frac{129.6}{518.4} = 25\%$. Check: $50 + 25 + 25 = 100$.

(d) $50 - 45 = 5$, so the subscriptions share rose by 5 percentage points.

Answer: (a) Rs. 518.4 crore (b) +8% (c) 50%, 25%, 25% (d) up 5 percentage points

TRAP

"Rose by 5 percentage points" and "rose by 5 percent" are different sentences. The share went from 45 to 50, which is 5 **percentage points** but $\left(\frac{5}{45}\right) \times 100 = 11.1$ **percent**. Use the phrase the question uses.

PRACTICE · TIER 3 target: 4 min/Q

1. A product's share of a firm's sales fell from 40% to 36%, while total sales rose 15%. Find the percent change in that product's sales.
2. Firm A earns Rs. 600 crore with 25% from exports. Firm B earns Rs. 900 crore with 40% from exports. Taking both firms together, what percent of revenue comes from exports?
3. A bar chart groups delivery riders by orders completed: 0–10: 30 riders; 10–20: 50 riders; 20–30: 80 riders; 30–40: 40 riders. Find the smallest and the largest possible mean number of orders per rider.
4. Two branches together sold 1,80,000 units last year. This year branch P rose 15%, branch Q fell 5%, and the combined total rose to 1,94,000. Find last year's figures for P and Q.
5. An index of monthly output has January = 100 and April = 132. February's index was 120 and February's output was 4,500 units. Find April's output and the percent rise from February to April.
6. Rank these four firms by revenue per employee: A, Rs. 96 crore with 1,200 staff; B, Rs. 105 crore with 1,400 staff; C, Rs. 130 crore with 1,300 staff; D, Rs. 84 crore with 1,050 staff.
7. A firm's 2023 revenue of Rs. 800 crore was 50% India, 30% SE Asia, 20% Europe. In 2024 India grew 12%, SE Asia grew 25%, Europe fell 5%. Find the overall growth rate and SE Asia's 2024 share.

ANSWER KEY

1. **+3.5%**. Part multiplier = $\frac{\text{new share}}{\text{old share}} \times (\text{total multiplier}) = \left(\frac{36}{40}\right) \times 1.15 = 0.90 \times 1.15 = 1.035$. The share fell by 4 percentage points and the product still grew, because 15% growth beats the $\frac{1}{0.9} = 11.1\%$ needed to stay level.
2. **34%**. Exports: $0.25 \times 600 = 150$ and $0.40 \times 900 = 360$, so $150 + 360 = 510$. Combined revenue = 1500. $\frac{510}{1500} = 0.34$. Averaging 25 and 40 to get 32.5 is wrong, because B is the larger firm and pulls the answer up.
3. **Smallest 16.5, largest 26.5**. Total riders = $30 + 50 + 80 + 40 = 200$. Bottom of each band: $0 + 500 + 1600 + 1200 = 3300$, and $\frac{3300}{200} = 16.5$. Top of each band: $300 + 1000 + 2400 + 1600 = 5300$, and $\frac{5300}{200} = 26.5$. Grouped data gives a range, never a single mean.

4. **P = 1,15,000 and Q = 65,000.** From $P + Q = 180000$ and $1.15P + 0.95Q = 194000$: substitute $Q = 180000 - P$ to get $1.15P + 171000 - 0.95P = 194000$, so $0.20P = 23000$ and $P = 115000$. Check: $1.15 \times 115000 = 132250$ and $0.95 \times 65000 = 61750$; together 1,94,000.
5. **4,950 units; a rise of 10%.** One index point = $\frac{4500}{120} = 37.5$ units. April = $132 \times 37.5 = 4950$. Percent rise = $\frac{132-120}{120} = \frac{12}{120} = 10\%$. You never needed January's actual output.
6. **C > A = D > B.** $\frac{96}{1200} = 0.08$; $\frac{105}{1400} = 0.075$; $\frac{130}{1300} = 0.10$; $\frac{84}{1050} = 0.08$ (all in crore per employee, i.e. 8, 7.5, 10 and 8 lakh). Benchmark test: is revenue more than $0.08 \times$ staff? For C, $0.08 \times 1300 = 104 < 130$, so C is above; for B, $0.08 \times 1400 = 112 > 105$, so B is below.
7. **+12.5% overall; SE Asia is $33\frac{1}{3}\%$ of 2024 revenue.** Weighted growth = $0.5 \times 12 + 0.3 \times 25 + 0.2 \times (-5) = 6 + 7.5 - 1 = 12.5$. Values: India 400 $\rightarrow$ 448, SE Asia 240 $\rightarrow$ 300, Europe 160 $\rightarrow$ 152, total 900. Share = $\frac{300}{900} = \frac{1}{3}$.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 90 s/Q · 18 questions in 27 minutes

Questions 1–6. Enrolment at four coaching centres.

Centre	2022	2023
P	600	750
Q	900	810
R	500	600
S	700	840

- Find the total 2023 enrolment.
- Find the percent change for centre Q.
- Which centre had the highest percent growth?
- Find the ratio of S to R in 2023.
- P's 2023 enrolment is what percent of the 2023 total?
- Find the overall percent growth in total enrolment.

Questions 7–11. A city uses 7,500 MW of power, split as below.

Sector	Industry	Households	Commercial	Transport	Others
Share	34%	28%	20%	10%	8%

- How many MW go to Commercial?
- Find the central angle of the Households sector.
- Industry uses how many percent more than Commercial?
- Industry and Households together use how many MW?
- Next year total use grows 20% to 9,000 MW and Transport's share rises to 12%. Find the percent rise in Transport's MW.

Questions 12–15. Quarterly revenue of a SaaS firm, in thousands of S\$.

Quarter	Q1	Q2	Q3	Q4
Revenue	400	460	552	496.8

- Find the percent change from Q2 to Q3.
- Find the percent change from Q1 to Q4.
- Find the total revenue for the year.
- Find the average quarterly revenue.

Questions 16–18.

1. A region's share of a firm's sales fell from 30% to 27% while total sales rose 25%. Find the percent change in that region's sales.
2. Firm X earns Rs. 500 crore with 20% from services. Firm Y earns Rs. 300 crore with 40% from services. Find the combined services share.
3. Express 4,853 as a percent of 19,720, to the nearest whole percent.

ANSWER KEY

1. **3,000.** $750 + 810 = 1560$; $1560 + 600 = 2160$; $2160 + 840 = 3000$.
2. **−10%.** $\frac{810}{900} = 0.90$.
3. **P, +25%.** $\frac{750}{600} = 1.25$; $\frac{600}{500} = 1.20$; $\frac{840}{700} = 1.20$; Q fell.
4. **7 : 5.** $840 : 600$; divide both by 120.
5. **25%.** $\frac{750}{3000} = \frac{1}{4}$.
6. **$11\frac{1}{9}\%$.** 2022 total = $600 + 900 + 500 + 700 = 2700$. $\frac{3000}{2700} = \frac{10}{9} = 1.1111$.
7. **1,500 MW.** 1% of 7500 = 75, so $20 \times 75 = 1500$.
8. **100.8°.** $28 \times 3.6 = 100.8$.
9. **70%.** Same pie, so compare shares: $\frac{34-20}{20} = \frac{14}{20} = 0.70$.
10. **4,650 MW.** $34 + 28 = 62\%$, and $62 \times 75 = 4650$.
11. **+44%.** Link rule: $(\frac{12}{10}) \times 1.20 = 1.2 \times 1.2 = 1.44$. Values: $750 \rightarrow 1080$.
12. **+20%.** $\frac{552}{460} = 1.20$.
13. **+24.2%.** $\frac{496.8}{400} = 1.242$. Or chain the quarters: $1.15 \times 1.20 \times 0.90 = 1.242$.
14. **1,908.8 thousand S\$.** $400 + 460 = 860$; $860 + 552 = 1412$; $1412 + 496.8 = 1908.8$.
15. **477.2 thousand S\$.** $1908.8 \div 4 = 477.2$.
16. **+12.5%.** $(\frac{27}{30}) \times 1.25 = 0.90 \times 1.25 = 1.125$. The share fell and the sales still rose.
17. **27.5%.** Services = $0.20 \times 500 + 0.40 \times 300 = 100 + 120 = 220$; total = 800; $\frac{220}{800} = 0.275$. Averaging 20 and 40 to get 30 is the trap.
18. **25%.** $19720 \times 0.25 = 4930$; 4,853 is 77 short, and $\frac{77}{19720} \approx 0.39\%$, so the true value is 24.61%, which rounds to 25%. Note that rounding 19,720 up to 20,000 **before** dividing would give $\frac{4853}{20000} = 24.3\%$ — round the answer, never the base.

ONE-PAGE REVISION CARD

The four moves. Add a row or column · take a percent · take a ratio · compare two numbers. Nothing else happens in DI.

Percent change = $\frac{\text{new} - \text{old}}{\text{old}} \times 100$. Faster: compute $\text{new} \div \text{old}$ and read the multiplier. $1.32 \rightarrow +32\%$. $0.88 \rightarrow -12\%$.

Share of total = $\frac{\text{part}}{\text{total}} \times 100$.

Pie. value = $\left(\frac{\text{percent}}{100}\right) \times \text{total} \cdot \text{angle} = \text{percent} \times 3.6 \cdot \text{percent} = \text{angle} \div 3.6$. Anchors: $1\% = 3.6^\circ \cdot 10\% = 36^\circ \cdot 25\% = 90^\circ \cdot 50\% = 180^\circ$.

Two sectors of the SAME pie: compare the percents directly, skip the values. Two sectors of **different** pies: you must convert to real values first.

The link rule (two charts).

$$\text{part multiplier} = \frac{\text{new share}}{\text{old share}} \times \frac{\text{new total}}{\text{old total}}$$

A share can fall while the value rises. $\text{Share} \times 0.9$ needs total growth above $\frac{1}{0.9} = 11.1\%$ to break even.

Weighted growth. Total growth = $w_1g_1 + w_2g_2 + \dots$, weights = the **old** shares. Combining two firms' percentages means weighting by their **sizes**, never a plain average.

Average rate. $\frac{\text{total value}}{\text{total quantity}}$. Never the plain average of the rates.

Index numbers. Any two index values divide exactly like the real values. Percent changes need no real number at all. One index point = $\frac{\text{known value}}{\text{its index}}$.

Grouped bars. You get a range, not a mean. Bottom of every band gives the minimum; top of every band gives the maximum; midpoints give the usual estimate.

Approximation. Anchor on 10% of the base, then adjust. Round the **answer**, never the base. Compare ratios by cross-multiplying, not by dividing.

Top 5 traps.

1. **Wrong base.** Percent change always divides by the old value. "A is x% more than B" divides by B.
2. **Absolute versus percent.** The biggest rise in units is often not the biggest rise in percent. Two different questions.
3. **A falling share is not a falling value.** Check the total's growth before you say anything fell.
4. **Averaging percentages.** Only legal when the totals behind them are equal. Otherwise weight by size — this kills the "average of two firms' margins" question every time.
5. **Line-chart type.** A line of growth rates is not a line of values. Revenue keeps rising while a growth line falls, as long as the line stays above zero.

Data Interpretation: Caselets, Mixed & Missing Data

Turn the paragraph into a grid, then fill the blanks

What you need to know

WHAT YOU NEED TO KNOW

Caselet method — do this every time, before reading any question.

1. Read the paragraph once and write down the **grand total**.
2. Draw a tree or a small table. One branch per split.
3. Fill every box in absolute numbers, not percentages.
4. Add a check line: the boxes must add back to the grand total.
5. Only now read the questions. They will all be one look-up each.

Watch the base. “30% of the rest”, “half of those who failed”, “two-thirds of the remainder” — each phrase takes a **new** base. Write down which box the percent sits on before you multiply.

Two-set Venn.

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- neither = total - $n(A \cup B)$
- only $A = n(A) - n(A \cap B)$
- exactly one = $n(A \cup B) - n(A \cap B)$

Three-set Venn.

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

With $x = n(A \cap B \cap C)$:

- exactly two = $[n(A \cap B) - x] + [n(B \cap C) - x] + [n(A \cap C) - x]$
- exactly one = $n(A \cup B \cup C) - (\text{exactly two}) - x$
- only $A = n(A) - n(A \cap B) - n(A \cap C) + x$

Venn bounds (when the overlap is not given).

- largest possible $n(A \cap B) = \min(n(A), n(B))$
- smallest possible $n(A \cap B) = n(A) + n(B) - \text{total}$ (or 0 if that is negative)
- smallest possible $n(A \cap B \cap C) = n(A) + n(B) + n(C) - 2 \times \text{total}$ (or 0 if negative)

Missing-value grid.

- Every row adds to its row total. Every column adds to its column total. The row totals and the column totals both add to the same grand total.
- **Find a row or column with exactly one blank, and fill it.** Every fill creates a new such row or column. Repeat until done.
- Always cross-check the grand total both ways at the end.

Averages hide totals. “The average of n items is a ” means $\text{sum} = na$. Convert every average into a sum before you do anything else.

Radar chart. One axis per attribute, all on the same scale. Read each axis as a separate number. The **total score** is the sum of the axis values. Never compare the visual area.

Bubble chart. x -axis, y -axis, and bubble **size** is a third variable. Three numbers per bubble. A big bubble far left is not a big x .

Combining two groups' percent changes.

$$\text{combined \% change} = \frac{w_1 g_1 + w_2 g_2}{w_1 + w_2}$$

where w_1, w_2 are the **old** sizes. Never the plain average of g_1 and g_2 .

Ratio splits. To split T in the ratio $a : b : c$, one part $= \frac{T}{a+b+c}$, then multiply.

Warm-up

WARM-UP

build the reflex

I Warm-ups 1–3 use this: in a class of 120, 60 play cricket, 45 play football, 20 play both.

Example 1

How many play at least one game?

SOLUTION

$$n(C \cup F) = 60 + 45 - 20. \quad 60 + 45 = 105. \quad 105 - 20 = 85.$$

Answer: 85

Example 2

How many play neither game?

SOLUTION

$$120 - 85 = 35.$$

Answer: 35

Example 3

How many play only cricket?

SOLUTION

$$\text{Only cricket} = 60 - 20 = 40.$$

Answer: 40

I Warm-ups 4–5 use this grid. One entry per cell; the blanks are marked with a dash.

	Mon	Tue	Total
A	40	—	95
B	35	50	85
Total	75	—	180

Example 4

Find A's Tuesday value.

SOLUTION

$$\text{Row A: } 40 + ? = 95. \quad ? = 95 - 40 = 55.$$

Answer: 55

Example 5

Find the Tuesday column total.

SOLUTION

Column Tue: $55 + 50 = 105$. Check the grand total: $75 + 105 = 180$. Correct.

Answer: 105

Example 6

Given $n(A) = 30$, $n(B) = 25$, $n(C) = 20$, $n(A \cap B) = 10$, $n(B \cap C) = 8$, $n(A \cap C) = 6$ and $n(A \cap B \cap C) = 3$, find $n(A \cup B \cup C)$.

SOLUTION

$30 + 25 + 20 = 75$. $10 + 8 + 6 = 24$. $75 - 24 = 51$. $51 + 3 = 54$.

Answer: 54

Example 7

Split 720 in the ratio 5 : 4.

SOLUTION

Total parts = $5 + 4 = 9$. One part = $720 \div 9 = 80$. First share = $5 \times 80 = 400$. Second share = $4 \times 80 = 320$.
Check: $400 + 320 = 720$.

Answer: 400 and 320

Example 8

A radar chart scores a phone out of 10 on five axes: Speed 8, Price 6, Design 7, Support 5, Battery 9. Find the total and the average score.

SOLUTION

$8 + 6 = 14$. $14 + 7 = 21$. $21 + 5 = 26$. $26 + 9 = 35$. Average = $35 \div 5 = 7$.

Answer: Total 35, average 7

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Caselet 1

A bookstore sold 1,800 books in a week. 40% of them were fiction. Of the fiction books, 25% were hardcover. Of the non-fiction books, 30% were hardcover.

Example 9 [TCS NQT pattern]

How many non-fiction books were sold?

SOLUTION

Fiction = 40% of 1800 = $0.40 \times 1800 = 720$. Non-fiction = $1800 - 720 = 1080$.

Answer: 1,080 books

SHORTCUT

Build the full grid **once**, then answer every question by looking it up.

	Hardcover	Paperback	Total
Fiction	180	540	720
Non-fiction	324	756	1,080
Total	504	1,296	1,800

Fiction hardcover = $0.25 \times 720 = 180$, so paperback = $720 - 180 = 540$. Non-fiction hardcover = $0.30 \times 1080 = 324$, so paperback = $1080 - 324 = 756$. Check: $180 + 324 = 504$, $540 + 756 = 1296$, and $504 + 1296 = 1800$. The grid is consistent.

Example 10 [Infosys pattern]

How many hardcover books were sold in all?

SOLUTION

Fiction hardcover = 0.25×720 . $720 \div 4 = 180$. Non-fiction hardcover = 0.30×1080 . $1080 \times 0.3 = 324$.
Total = $180 + 324 = 504$.

Answer: 504 books

Example 11 [Accenture pattern]

Hardcover books are what percent of all books sold?

SOLUTION

$\left(\frac{504}{1800}\right) \times 100$. $\frac{504}{1800}$: divide both by 18, giving $\frac{28}{100} \cdot \frac{28}{100} = 0.28$, so 28%.

Answer: 28%

TRAP

Do not average the two hardcover rates. $(25 + 30) \div 2 = 27.5\%$, which is **not** the answer. The two groups are different sizes: non-fiction is 1,080 books against fiction's 720, so the 30% rate carries more weight and pulls the answer up to 28%.

Example 12 [Wipro pattern]

Find the ratio of fiction paperbacks to non-fiction paperbacks.

SOLUTION

540 : 756. Divide both by 4: 135 : 189. Divide both by 27: $135 \div 27 = 5$ and $189 \div 27 = 7$. Check: $5 \times 108 = 540$ and $7 \times 108 = 756$. Correct.

Answer: 5 : 7

Venn data set

In a survey of 500 commuters: 260 use the metro, 210 use a bus, 180 use a cab. 90 use both metro and bus, 70 use both bus and cab, 80 use both metro and cab, and 40 use all three.

Example 13 [Capgemini pattern]

How many commuters use none of the three?

SOLUTION

$n(M \cup B \cup C) = 260 + 210 + 180 - 90 - 70 - 80 + 40$. Add the singles: $260 + 210 = 470$; $470 + 180 = 650$. Add the pairs: $90 + 70 = 160$; $160 + 80 = 240$. $650 - 240 = 410$. $410 + 40 = 450$. None = $500 - 450 = 50$.

Answer: 50 commuters

Example 14 [TCS NQT pattern]

How many use only the metro?

SOLUTION

Only metro = $n(M) - n(M \cap B) - n(M \cap C) + n(M \cap B \cap C)$. = $260 - 90 - 80 + 40$. $260 - 90 = 170$. $170 - 80 = 90$. $90 + 40 = 130$.

Answer: 130 commuters

SHORTCUT

Fill the Venn from the centre outwards and you never need a formula again. Centre (all three) = 40. Metro-and-bus only = $90 - 40 = 50$. Bus-and-cab only = $70 - 40 = 30$. Metro-and-cab only = $80 - 40 = 40$. Metro only = $260 - 50 - 40 - 40 = 130$. Bus only = $210 - 50 - 30 - 40 = 90$. Cab only = $180 - 30 - 40 - 40 = 70$. Now all seven regions are known: $130 + 90 + 70 + 50 + 30 + 40 + 40 = 450$, so 50 use none. Every question is a sum of regions.

Example 15 [Cognizant pattern]

How many use exactly two of the three modes?

SOLUTION

From the filled diagram, the three “exactly two” regions are metro-and-bus only = $90 - 40 = 50$, bus-and-cab only = $70 - 40 = 30$, metro-and-cab only = $80 - 40 = 40$. $50 + 30 = 80$. $80 + 40 = 120$.

Answer: 120 commuters

TRAP

“90 use both metro and bus” means 90 are in the overlap **including** the 40 who use all three. It does not mean 90 use exactly those two. Subtract the centre before you call a region “exactly two”.

Missing-value grid

I Sales by branch and product, in Rs. lakh. Dashes are missing.

Branch	Product X	Product Y	Product Z	Total
North	120	—	90	350
South	—	160	110	420
East	100	130	—	360
Total	370	430	—	1,130

Example 16 [Infosys pattern]

Find North’s product Y sales and South’s product X sales.

SOLUTION

Row North has exactly one blank. $120 + ? + 90 = 350$ $210 + ? = 350$ $? = 350 - 210 = 140$. Column X now has exactly one blank. $120 + ? + 100 = 370$ $220 + ? = 370$ $? = 370 - 220 = 150$. Check row South: $150 + 160 + 110 = 420$. Correct. Check column Y: $140 + 160 + 130 = 430$. Correct.

Answer: North-Y = 140, South-X = 150

Example 17 [Accenture pattern]

Find East's product Z sales and the product Z column total.

SOLUTION

Row East: $100 + 130 + ? = 360$. $230 + ? = 360$, so $? = 130$. Column Z: $90 + 110 + 130 = 330$. Cross-check the grand total two ways. By columns: $370 + 430 + 330 = 1130$. By rows: $350 + 420 + 360 = 1130$. Both agree.

Answer: East-Z = 130; column Z total = 330

SHORTCUT

Never solve a grid with algebra. Scan for the row or column that has **exactly one** dash and fill it by subtraction. Filling it always turns another line into a one-dash line. Three subtractions finish most grids.

Example 18 [Wipro pattern]

Product Y is what percent of total sales?

SOLUTION

$\left(\frac{430}{1130}\right) \times 100$. Anchor: 38% of $1130 = 1130 \times 0.38 = 429.4$. Gap = $430 - 429.4 = 0.6$. $\frac{0.6}{1130} = 0.00053$, that is 0.053% . Share = $38 + 0.053 = 38.053$.

Answer: About 38.05%

Radar and bubble charts

I A radar chart scores two phones out of 10 on five axes.

Axis	Camera	Battery	Display	Speed	Price value
Phone A	8	6	9	7	5
Phone B	6	9	7	8	8

Example 19 [Capgemini pattern]

Which phone has the higher total score, and by what percent?

SOLUTION

Phone A: $8 + 6 = 14$; $14 + 9 = 23$; $23 + 7 = 30$; $30 + 5 = 35$. Phone B: $6 + 9 = 15$; $15 + 7 = 22$; $22 + 8 = 30$; $30 + 8 = 38$. B is higher by $38 - 35 = 3$. Base is A, which is 35. $\left(\frac{3}{35}\right) \times 100$. $35 \times 0.085 = 2.975$. Gap = $3 - 2.975 = 0.025$, and $\frac{0.025}{35} = 0.0007$. So $\frac{3}{35} = 0.08571$, that is 8.571% .

Answer: Phone B, by about 8.57%

TRAP

A radar chart looks like a shape, and a bigger shape looks like a better product. Do not judge by area. Phone A's shape is wider on camera and display, but B wins on the sum. Read every axis as a number and add.

I A bubble chart of four ad campaigns. The x -axis is spend, the y -axis is leads, and the bubble size is the number of conversions.

Campaign	Spend (Rs. lakh)	Leads ('00)	Conversions
C1	20	15	300
C2	35	21	420

C3	50	40	600
C4	25	20	500

Example 20 [Cognizant pattern]

Which campaign has the lowest cost per conversion?

SOLUTION

Cost per conversion = spend in rupees $\div$ conversions. One lakh = 100{,}000. C1: $20\{,000\} \div 300 = 6666.67$. C2: $35\{,000\} \div 420 = 8333.33$. C3: $50\{,000\} \div 600 = 8333.33$. C4: $25\{,000\} \div 500 = 5000$. The smallest is C4.

Answer: C4, at Rs. 5,000 per conversion

PRACTICE · TIER 1 target: 60 s/Q

Questions 1–4. A factory made 4,800 units last month. 35% failed the first quality check. Of those that failed, 60% were repaired and then passed; the rest were scrapped.

1. How many units failed the first check?
2. How many units were scrapped?
3. How many good units did the factory end up with?
4. Scrapped units are what percent of production?

Questions 5–8. Of 300 students, 180 study Python, 150 study Java, and 60 study both.

1. How many study at least one language?
2. How many study neither?
3. How many study only Python?
4. How many study only Java?

Questions 9–12. Staff on duty, by shift and day.

Shift	Mon	Tue	Wed	Total
Morning	45	50	—	150
Evening	—	40	60	135
Total	80	90	—	285

1. Find the morning shift's Wednesday figure.
2. Find the evening shift's Monday figure.
3. Find the Wednesday column total.
4. The morning shift is what percent of the whole week's total?

Questions 13–14. A radar chart scores three cities out of 10.

Axis	Bangkok	Manila	Hanoi
Cost	7	8	9
Talent	8	6	5
Infrastructure	6	5	6
Market	9	7	6

1. Which city has the highest total score?
2. Bangkok's total exceeds Manila's by what percent?

ANSWER KEY

1. **1,680.** $0.35 \times 4800 = 1680$.
2. **672.** Repaired $= 0.60 \times 1680 = 1008$; scrapped $= 1680 - 1008 = 672$.
3. **4,128.** Passed first time $= 4800 - 1680 = 3120$; add the 1,008 repaired: $3120 + 1008 = 4128$.
4. **14%.** $\left(\frac{672}{4800}\right) \times 100 = 14$. Note $0.35 \times 0.40 = 0.14$ directly — 40% of the 35% that failed.
5. **270.** $180 + 150 - 60 = 270$.
6. **30.** $300 - 270 = 30$.
7. **120.** $180 - 60$.
8. **90.** $150 - 60$.
9. **55.** Row morning: $150 - 45 - 50 = 55$.
10. **35.** Column Mon: $80 - 45 = 35$. Check evening row: $35 + 40 + 60 = 135$. Correct.
11. **115.** $55 + 60 = 115$. Check: $80 + 90 + 115 = 285$. Correct.
12. **About 52.63%.** $\left(\frac{150}{285}\right) \times 100$: 52% of 285 is 148.2, the gap 1.8 is 0.63% of 285, so 52.63.
13. **Bangkok, 30.** Bangkok $= 7 + 8 + 6 + 9 = 30$; Manila $= 8 + 6 + 5 + 7 = 26$; Hanoi $= 9 + 5 + 6 + 6 = 26$.
14. **About 15.38%.** $\frac{30-26}{26} = \frac{4}{26} = 0.1538$.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Arithmetic-heavy caselet

A Bangkok cloud kitchen runs three brands. Last month it took 24,000 orders. Thai-Bowl took $\frac{3}{8}$ of them, Grill took $\frac{5}{12}$, and the rest went to Sweet. The average order values are THB 180, THB 220 and THB 150 in that order.

Example 21 [LINE MAN · pattern]

Find each brand's revenue and the total revenue.

SOLUTION

Orders first. Thai-Bowl $= 24000 \times \frac{3}{8} = (24000 \div 8) \times 3 = 3000 \times 3 = 9000$. Grill $= 24000 \times \frac{5}{12} = (24000 \div 12) \times 5 = 2000 \times 5 = 10000$. Sweet $= 24000 - 9000 - 10000 = 5000$. Check: $9000 + 10000 + 5000 = 24000$. Correct.

Revenue next. Thai-Bowl $= 9000 \times 180 = 9 \times 180 \times 1000 = 1620 \times 1000 = 1\{, \}620\{, \}000$. Grill $= 10000 \times 220 = 2\{, \}200\{, \}000$. Sweet $= 5000 \times 150 = 750\{, \}000$. Total $= 1620000 + 2200000 = 3820000$; $3820000 + 750000 = 4570000$.

Answer: THB 1.62 million, THB 2.20 million and THB 0.75 million; total THB 4.57 million

Example 22 [Grab · pattern]

Find the overall average order value.

SOLUTION

An average rate is total value $\div$ total quantity. $4\{, \}570\{, \}000 \div 24\{, \}000$. $24000 \times 190 = 45\{, \}60\{, \}000$. Remainder $= 4570000 - 4560000 = 10000$. $\frac{10000}{24000} = 0.4167$. So the average is $190 + 0.4167 = 190.4167$.

Answer: About THB 190.42

TRAP

The plain average of 180, 220 and 150 is $550 \div 3 = 183.33$. That is wrong. Grill, the most expensive brand, also has the most orders, so it carries the most weight and drags the true average up to 190.42. Weight by order count, always.

Example 23 [Sea/Shopee · pattern]

Grill's revenue is what percent of the total?

SOLUTION

$(2\{, \}200\{, \}\frac{000}{4}\{, \}570\{, \}000) \times 100$. Cancel the zeros: $\frac{220}{457}$. Anchor: 48% of 457 = 219.36. Gap = $220 - 219.36 = 0.64$. $\frac{0.64}{457} = 0.0014$, that is 0.14%. Share = $48 + 0.14 = 48.14$.

Answer: About 48.14%

Venn with an unknown centre

In a survey of 1,500 Singapore shoppers: 720 use GrabPay, 660 use PayNow, 480 use a credit card. 300 use GrabPay and PayNow, 180 use PayNow and card, 240 use GrabPay and card, and 270 use none of the three.

Example 24 [DBS · pattern]

How many use all three?

SOLUTION

At least one = $1500 - 270 = 1230$. Let x be the number using all three. $1230 = 720 + 660 + 480 - 300 - 180 - 240 + x$. Singles: $720 + 660 = 1380$; $1380 + 480 = 1860$. Pairs: $300 + 180 = 480$; $480 + 240 = 720$. $1860 - 720 = 1140$. $1230 = 1140 + x$ $x = 1230 - 1140 = 90$.

Answer: 90 shoppers

Example 25 [Razer · pattern]

How many use exactly one of the three?

SOLUTION

Fill the diagram from the centre. Centre = 90. GrabPay and PayNow only = $300 - 90 = 210$. PayNow and card only = $180 - 90 = 90$. GrabPay and card only = $240 - 90 = 150$. Exactly two = $210 + 90 + 150 = 450$. Exactly one = $1230 - 450 - 90 = 690$.

Check each "only" region separately. GrabPay only = $720 - 210 - 150 - 90 = 270$. PayNow only = $660 - 210 - 90 - 90 = 270$. Card only = $480 - 90 - 150 - 90 = 150$. $270 + 270 + 150 = 690$. Matches.

Answer: 690 shoppers

Missing grid with outside conditions

Shipments in thousands, by region and month. You are also told that February's total was 25% more than January's, and that the West region's three-month total was 310.

Region	Jan	Feb	Mar	Total
North	60	—	90	230
West	80	100	—	310
East	60	—	70	—
Total	—	—	—	—

Example 26 [Agoda · pattern]

Fill in every missing value.

SOLUTION

Step 1 — January total. All three January cells are known. $60 + 80 + 60 = 200$.

Step 2 — February total. It is 25% more than January. $200 \times 1.25 = 250$.

Step 3 — North in February. Row North now has one blank. $60 + ? + 90 = 230$ $150 + ? = 230$, so $? = 80$.

Step 4 — East in February. Column February now has one blank. $80 + 100 + ? = 250$ $180 + ? = 250$, so $? = 70$.

Step 5 — West in March. Row West now has one blank. $80 + 100 + ? = 310$ $180 + ? = 310$, so $? = 130$.

Step 6 — East's row total. $60 + 70 + 70 = 200$.

Step 7 — March total. $90 + 130 + 70 = 290$.

Step 8 — grand total, both ways. By columns: $200 + 250 + 290 = 740$. By rows: $230 + 310 + 200 = 740$. They agree, so the grid is right.

Region	Jan	Feb	Mar	Total
North	60	80	90	230
West	80	100	130	310
East	60	70	70	200
Total	200	250	290	740

Answer: North-Feb 80, East-Feb 70, West-Mar 130; totals 200, 250, 290 and 740

Example 27 [Shopee · pattern]

(a) Find the percent rise in total shipments from February to March. (b) Which region grew fastest from January to March?

SOLUTION

(a) Old = 250, new = 290. Rise = $290 - 250 = 40$. $\left(\frac{40}{250}\right) \times 100$. $\frac{40}{250} = \frac{16}{100} = 0.16$. So +16%.

(b) Divide March by January for each region. North: $\frac{90}{60} = 1.5 \rightarrow +50\%$. West: $\frac{130}{80} = 1.625 \rightarrow +62.5\%$. East: $\frac{70}{60} = 1.1667 \rightarrow +16.67\%$.

Answer: (a) +16% (b) West, +62.5%

SHORTCUT

When a grid comes with a sentence outside it ("February was 25% more than January"), that sentence is usually the **first** thing you use. It converts one unknown total into a number, and that number unlocks a one-blank line.

Bubble chart

Four rider-acquisition campaigns. x is the spend, y is new riders, and the bubble size is riders still active after 30 days.

Campaign	Spend (S\$'000)	New riders	Retained
A	40	2,500	750
B	60	4,500	1,350
C	75	5,000	1,250
D	50	4,000	1,600

Example 28 [Grab · pattern]

Find each campaign's retention rate and its cost per retained rider. Which campaign is best?

SOLUTION

Retention rate = retained $\div$ new. A: $\frac{750}{2500} = 0.30 \rightarrow 30\%$. B: $\frac{1350}{4500} = 0.30 \rightarrow 30\%$. C: $\frac{1250}{5000} = 0.25 \rightarrow 25\%$. D: $\frac{1600}{4000} = 0.40 \rightarrow 40\%$.

Cost per retained rider = spend in S\\$ $\div$ retained. A: $40000 \div 750$. $750 \times 53 = 39750$; remainder 250; $\frac{250}{750} = 0.333$. So 53.33. B: $60000 \div 1350$. $1350 \times 44 = 59400$; remainder 600; $\frac{600}{1350} = 0.444$. So 44.44. C: $75000 \div$

$1250 = 60.00$ exactly, since $1250 \times 60 = 75000$. D: $50000 \div 1600 = 31.25$ exactly, since $1600 \times 31.25 = 50000$. The lowest cost is D.

Answer: D is best: 40% retention at S\$31.25 per retained rider

TRAP

C has the biggest x (spend) and the biggest y (new riders), so it sits top-right on the chart and looks like the winner. But its bubble is smaller than D's, and D spent S\$25,000 less. C has the worst retention of the four (25%) and the highest cost per retained rider (S\$60.00). The bubble is what pays. Always read all three numbers of a bubble before ranking.

Comparing two data sources

A firm reports monthly active users two ways. A bar chart gives the total MAU in lakh: Jan 120, Feb 132, Mar 145.2. A table gives the Android share: Jan 75%, Feb 72%, Mar 70%.

Example 29 [GIC · pattern]

Find the percent growth from January to March for (a) iOS users, (b) Android users.

SOLUTION

iOS share = $100 - \text{Android share}$, so Jan 25%, Feb 28%, Mar 30%.

(a) iOS Jan = $0.25 \times 120 = 30$ lakh. iOS Mar = $0.30 \times 145.2 = 43.56$ lakh. Rise = $43.56 - 30 = 13.56$. $\left(\frac{13.56}{30}\right) \times 100 = 0.452 \times 100 = 45.2$. Link-rule check: $\left(\frac{30}{25}\right) \times \left(\frac{145.2}{120}\right) = 1.2 \times 1.21 = 1.452$. Matches.

(b) Android Jan = $0.75 \times 120 = 90$ lakh. Android Mar = $0.70 \times 145.2 = 101.64$ lakh. Rise = $101.64 - 90 = 11.64$. $\left(\frac{11.64}{90}\right) \times 100 = \frac{11.64}{90} = 0.129333$. So +12.93%. Link-rule check: $\left(\frac{70}{75}\right) \times 1.21 = 0.93333 \times 1.21 = 1.129333$. Matches.

Answer: (a) iOS +45.2% (b) Android about +12.93%

PRACTICE · TIER 2 target: 100 s/Q

Questions 1–4. A Singapore gym chain has 4,500 members. 40% are on the annual plan, 35% on the quarterly plan, and the rest on the monthly plan. Per month they pay S\$60, S\$75 and S\$95 respectively.

1. How many members are on the monthly plan?
2. Find the chain's total monthly revenue.
3. Find the average revenue per member per month.
4. The quarterly plan brings in what percent of revenue?

Questions 5–8. Of 800 trainees, 520 cleared the written test and 460 cleared the interview. 300 cleared both.

1. How many cleared at least one stage?
2. How many cleared neither?
3. How many cleared exactly one stage?
4. How many cleared only the interview?

Questions 9–11. Quarterly units sold by two stores.

Store	Q1	Q2	Q3	Q4	Total
A	150	—	180	200	700
B	120	140	—	190	610
Total	270	310	340	390	1,310

1. Find store A's Q2 figure.
2. Find store B's Q3 figure.

3. Find the percent rise in the combined total from Q1 to Q4.

ANSWER KEY

1. **1,125.** Monthly plan share = $100 - 40 - 35 = 25\%$, and $0.25 \times 4500 = 1125$.
2. **\$333,000.** Annual = $0.40 \times 4500 = 1800$, quarterly = $0.35 \times 4500 = 1575$. Then $1800 \times 60 = 108000$, $1575 \times 75 = 118125$, $1125 \times 95 = 106875$; sum = 333000.
3. **\$74.** $333000 \div 4500 = 74$, because $4500 \times 74 = 333000$. The plain average of 60, 75 and 95 is 76.67, which is wrong.
4. **About 35.47%.** $\frac{118125}{333000}$: 35% of 333,000 is 116,550; the gap 1,575 is 0.47% of 333,000.
5. **680.** $520 + 460 - 300 = 680$.
6. **120.** $800 - 680 = 120$.
7. **380.** $680 - 300 = 380$. Or $(520 - 300) + (460 - 300) = 220 + 160$.
8. **160.** $460 - 300 = 160$.
9. **170.** Column Q2: $310 - 140 = 170$. Check row A: $150 + 170 + 180 + 200 = 700$. Correct.
10. **160.** Column Q3: $340 - 180 = 160$. Check row B: $120 + 140 + 160 + 190 = 610$. Correct.
11. **About 44.44%.** $\frac{390-270}{270} = \frac{120}{270} = \frac{4}{9} = 0.4444$.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 30 [Google · pattern]

In a group of 200 people, 130 like tea and 110 like coffee. (a) Find the smallest and the largest possible number who like both. (b) If exactly 20 people like neither, find the number who like both.

SOLUTION

Insight: the overlap is squeezed between two extremes — one set sitting entirely inside the other, and the two sets together just filling the group.

(a) **Largest.** The overlap can never be bigger than the smaller set. Coffee has 110 people, so at most 110 like both. That happens when every coffee drinker also likes tea. Check it is legal: then tea-only = $130 - 110 = 20$ and coffee-only = 0, so the union is 130, which fits inside 200. Legal.

Smallest. Push the sets apart as far as the group allows. $n(T \cup C) = 130 + 110 - n(T \cap C) \leq 200$ $240 - n(T \cap C) \leq 200$ $n(T \cap C) \geq 240 - 200 = 40$. Check at 40: union = $240 - 40 = 200$, exactly filling the group, so nobody likes neither. Legal.

(b) If 20 like neither, the union is fixed. $n(T \cup C) = 200 - 20 = 180$. $180 = 130 + 110 - n(T \cap C)$ $180 = 240 - n(T \cap C)$ $n(T \cap C) = 240 - 180 = 60$. The answer is now a single number, not a range, because knowing “neither” fixes the union.

Answer: (a) between 40 and 110 (b) exactly 60

SHORTCUT

Read the pattern in part (a): $130 + 110 - 200 = 40$. **Sum of the sets minus the total** is the forced minimum overlap. If that number is negative, the minimum is simply 0.

Example 31 [Amazon · pattern]

Of 100 users, 80 use app A, 95 use app B and 90 use app C. Find the smallest possible number who use all three, and the largest.

SOLUTION

Insight: count the people who are MISSING from each app. Somebody fails to use all three only if they are missing from at least one, so the misses are the budget.

Missing from A = $100 - 80 = 20$. Missing from B = $100 - 95 = 5$. Missing from C = $100 - 90 = 10$. Total misses = $20 + 5 + 10 = 35$.

Each person who does **not** use all three uses up at least one miss. So at most 35 people fail to use all three. Users of all three $\geq 100 - 35 = 65$.

Can 65 be reached? Only if no person is missing from two different apps, i.e. the three “missing” groups are disjoint. They need $20 + 5 + 10 = 35$ distinct people, and there are 100 people, so yes. Then exactly 35 people miss one app each, and $100 - 35 = 65$ use all three. So 65 is achievable.

Largest. All-three can never exceed the smallest set, which is A with 80. Check 80 is legal: let every one of the 80 A-users use B and C too. Then B has 15 more users outside A and C has 10 more, all fine inside 100.

Answer: Smallest 65, largest 80

The same idea in formula form: minimum $n(A \cap B \cap C) = n(A) + n(B) + n(C) - 2 \times \text{total}$, and here $80 + 95 + 90 - 200 = 65$. If it comes out negative, the minimum is 0.

Example 32 [Goldman Sachs · pattern]

A revenue grid by region and product is almost blank. You are told: the grand total is Rs. 1,200 crore; the North region is 30% of the total; product P is 40% of the total; North’s product-P revenue is one-third of North’s total; South’s product-Q revenue is Rs. 150 crore; and the East region’s total is twice the South region’s total.

Find every cell, and then find East’s product-P revenue as a percent of all product-P revenue.

SOLUTION

Insight: every sentence is one equation. Convert all of them to numbers first, put them into the grid, and only then start subtracting — the grid does the algebra for you.

Step 1 — the margins. North = $0.30 \times 1200 = 360$. Product P total = $0.40 \times 1200 = 480$. Product Q total = $1200 - 480 = 720$.

Step 2 — South and East. South + East = $1200 - 360 = 840$. East = $2 \times$ South, so South + $2 \times$ South = 840. $3 \times$ South = 840, so South = 280 and East = 560.

Step 3 — North’s row. North-P = $360 \div 3 = 120$. North-Q = $360 - 120 = 240$.

Step 4 — South’s row. South-Q = 150 (given). South-P = $280 - 150 = 130$.

Step 5 — East’s row, from the P column. $120 + 130 + \text{East-P} = 480$ $250 + \text{East-P} = 480$, so East-P = 230. East-Q = $560 - 230 = 330$.

Step 6 — check the Q column. $240 + 150 + 330 = 720$. Correct.

Region	P	Q	Total
North	120	240	360
South	130	150	280
East	230	330	560
Total	480	720	1,200

Step 7 — the percent asked. $\left(\frac{230}{480}\right) \times 100$. $480 \times 0.47 = 225.6$. Gap = $230 - 225.6 = 4.4$. $\frac{4.4}{480} = 0.00917$, that is 0.917%. Share = $47 + 0.917 = 47.917$.

Answer: Grid as shown; East’s P revenue is about 47.92% of all product-P revenue

Example 33 [D. E. Shaw · pattern]

A fund reports that 60% of its assets returned 12% for the year, and the other 40% is split half in bonds and half in property. The fund’s overall return was 9.6%, and the bonds returned 5%. Find the property return.

SOLUTION

Insight: an overall return is a weighted average, and a weighted average can be peeled one layer at a time. Solve the outer layer first, then use its answer as the total for the inner layer.

Outer layer. Let r be the return on the whole 40% block. $0.60 \times 12 + 0.40 \times r = 9.6$ $7.2 + 0.40r = 9.6$ $0.40r = 9.6 - 7.2 = 2.4$ $r = \frac{2.4}{0.40} = 6$. So the 40% block returned 6%.

Inner layer. That block is half bonds (5%) and half property (c). $0.5 \times 5 + 0.5 \times c = 6$ $2.5 + 0.5c = 6$ $0.5c = 3.5$ $c = 7$.

Check from the top. The three pieces are 60% at 12%, 20% at 5%, 20% at 7%. $0.60 \times 12 = 7.2$. $0.20 \times 5 = 1.0$. $0.20 \times 7 = 1.4$. $7.2 + 1.0 + 1.4 = 9.6$. Correct.

Answer: Property returned 7%

Example 34 [Microsoft · pattern]

A report says: “Our Bangkok store’s sales rose 15% and our Chiang Mai store’s sales rose 20%, so together we grew 17.5%.” Last year Bangkok sold THB 30 million and Chiang Mai THB 10 million.

(a) Is the claim right? (b) What is the true combined growth? (c) For what split of last year’s sales would 17.5% have been correct?

SOLUTION

Insight: a combined percent change is a weighted average of the two changes, and the weights are LAST YEAR’s sizes. A plain average is only right when the two sizes are equal.

(a) and (b). Compute the real numbers. Bangkok this year = $30 \times 1.15 = 30 + 4.5 = 34.5$. Chiang Mai this year = $10 \times 1.20 = 12$. Total last year = $30 + 10 = 40$. Total this year = $34.5 + 12 = 46.5$. Rise = $46.5 - 40 = 6.5$. $\left(\frac{6.5}{40}\right) \times 100 = 0.1625 \times 100 = 16.25$. So the claim of 17.5% is too high. The truth is 16.25%.

Weighted-average check: $\frac{30 \times 15 + 10 \times 20}{40} = \frac{450 + 200}{40} = \frac{650}{40} = 16.25$. Matches.

(c) Let a fraction w of last year’s sales sit in Bangkok. $15w + 20(1 - w) = 17.5$ $15w + 20 - 20w = 17.5$ $-5w = -2.5$ $w = 0.5$. So 17.5% would be correct only if the two stores had been exactly equal in size last year. The report averaged the two rates as if that were true.

Answer: (a) No (b) +16.25% (c) only a 50–50 split

TRAP

This is the single most common error in mixed DI: averaging growth rates, margins, or pass percentages across groups of different sizes. Ask “different sizes?” before you ever average a percentage. If the sizes differ, weight by size.

Example 35 [Uber · pattern]

A bubble chart shows five city-launch options. The x -axis is the one-off setup cost and the bubble size is the expected monthly profit, both in thousands of S\$.

City	V	W	X	Y	Z
Setup cost	40	60	50	30	70
Monthly profit	9	15	11	6	16

The budget is S\$150,000 and you may launch any set of cities. Which set maximises the monthly profit?

SOLUTION

Insight: the obvious move — rank by profit per S\$ of cost and take the best ones — is wrong here, because the budget must be spent as fully as possible. You have to check the subsets.

First see why the greedy rule fails. Profit per unit cost: V: $\frac{9}{40} = 0.225$. W: $\frac{15}{60} = 0.250$. X: $\frac{11}{50} = 0.220$. Y: $\frac{6}{30} = 0.200$. Z: $\frac{16}{70} = 0.2286$. Greedy takes W (0.250), then Z (0.2286): cost $60 + 70 = 130$, profit $15 + 16 = 31$. Only \$20,000 is left and nothing costs that little. Greedy stops at 31.

Can four cities fit? The four cheapest cost $30 + 40 + 50 + 60 = 180 > 150$. No. So at most three cities.

List every three-city set with cost at most 150. V W X: $40 + 60 + 50 = 150$; profit $9 + 15 + 11 = 35$. V W Y: $40 + 60 + 30 = 130$; profit $9 + 15 + 6 = 30$. V X Y: $40 + 50 + 30 = 120$; profit $9 + 11 + 6 = 26$. V Y Z: $40 + 30 + 70 = 140$; profit $9 + 6 + 16 = 31$. W X Y: $60 + 50 + 30 = 140$; profit $15 + 11 + 6 = 32$. X Y Z: $50 + 30 + 70 = 150$; profit $11 + 6 + 16 = 33$. Over budget and therefore dropped: V W Z (170), V X Z (160), W X Z (180), W Y Z (160).

The best is V, W, X at exactly \$150,000 for \$35,000 a month. Two-city sets can only do worse: the best two are W and Z at 31.

Answer: Launch V, W and X: cost \$150,000, profit \$35,000 per month

SHORTCUT

Whenever a budget question has few items, do not reason — enumerate. With five items there are only ten three-item sets, and writing them out takes 60 seconds and cannot be wrong. Save cleverness for problems with no small list.

Example 36 [Adobe · pattern]

A table of monthly downloads, in thousands, has three smudged entries.

Month	Jan	Feb	Mar	Apr	May	Jun
Downloads	80	—	120	—	150	—

You are told the six-month average was 125, that February was 25% more than January, and that June was twice April. Find the three missing values, and the percent rise from May to June.

SOLUTION

Insight: an average is a total wearing a disguise. Turn it into a sum on the first line, and the puzzle becomes ordinary arithmetic.

Step 1 — the sum. Average $\times$ count = sum. $125 \times 6 = 750$.

Step 2 — February. $80 \times 1.25 = 80 + 20 = 100$.

Step 3 — what is left for April and June. Known so far: $80 + 100 + 120 + 150$. $80 + 100 = 180$; $180 + 120 = 300$; $300 + 150 = 450$. April + June = $750 - 450 = 300$.

Step 4 — use the June condition. June = $2 \times$ April, so April + $2 \times$ April = 300. $3 \times$ April = 300 April = 100, and June = 200.

Step 5 — check. $80 + 100 + 120 + 100 + 150 + 200$. $80 + 100 = 180$; $+120 = 300$; $+100 = 400$; $+150 = 550$; $+200 = 750$. $750 \div 6 = 125$. Correct.

Step 6 — May to June. Rise = $200 - 150 = 50$. $\left(\frac{50}{150}\right) \times 100 = \left(\frac{1}{3}\right) \times 100 = 33.33$.

Answer: Feb 100, Apr 100, Jun 200 (thousand); May to June is $+33\frac{1}{3}\%$

PRACTICE · TIER 3 target: 4 min/Q

- In a group of 150 people, 95 own a bike and 80 own a car. Find the least and the greatest possible number who own both.
- Of 120 employees, 100 know Excel, 95 know SQL and 90 know Python. Find the smallest possible number who know all three.

- A revenue grid has two regions and two products and totals Rs. 900 crore. North is 40% of the total, product A is 55% of the total, and North's product-A revenue is Rs. 240 crore. Find South's product-B revenue.
- Division P's costs rose 8% and division Q's rose 18%. A manager reports a 13% rise overall. Last year P spent Rs. 60 crore and Q spent Rs. 40 crore. Find the true rise and say what the manager did wrong.
- Five weeks of sales average 620 units. Week 1 was 500, week 2 was 680, week 4 was 660, and week 5 was twice week 3. Find week 3 and week 5.
- Four projects cost Rs. 25, 30, 40 and 45 lakh and return profits of Rs. 11, 13, 17 and 20 lakh respectively. With a budget of Rs. 70 lakh, which set gives the highest profit?
- In a survey of 400 people, 55% read the Times, 45% read the Post and 20% read both. How many read neither?

ANSWER KEY

- Least 25, greatest 80.** Least = $95 + 80 - 150 = 25$; at 25 the union is exactly 150, filling the group. Greatest = $\min(95, 80) = 80$; then every car owner also owns a bike, and the union is 95, which fits inside 150.
- 45.** Misses are $120 - 100 = 20$, $120 - 95 = 25$, $120 - 90 = 30$, and $20 + 25 + 30 = 75$. At most 75 people can fail to know all three, so at least $120 - 75 = 45$ know all three. It is achievable because the 75 "missing" slots fit inside 120 people without overlapping.
- Rs. 285 crore.** North = $0.40 \times 900 = 360$, so South = 540. Product A = $0.55 \times 900 = 495$, so product B = 405. North-A = 240, so North-B = $360 - 240 = 120$. South-B = $405 - 120 = 285$. Check South-A = $495 - 240 = 255$, and $255 + 285 = 540$. Correct.
- The true rise is 12%; the manager averaged the two rates.** Weighted: $\frac{60 \times 8 + 40 \times 18}{100} = \frac{480 + 720}{100} = 12$. Values: $60 \times 1.08 = 64.8$ and $40 \times 1.18 = 47.2$, so 112 against 100. The plain average $(8 + 18) \div 2 = 13$ would be right only if the two divisions had spent equally.
- Week 3 = 420, week 5 = 840.** Sum = $620 \times 5 = 3100$. Known = $500 + 680 + 660 = 1840$. So week 3 + week 5 = 1260, and since week 5 = $2 \times$ week 3, $3 \times$ week 3 = 1260. Check: $500 + 680 + 420 + 660 + 840 = 3100$.
- Take the Rs. 25 lakh and Rs. 45 lakh projects, for Rs. 31 lakh profit.** Sets within budget: (25, 45) costs 70 and yields $11 + 20 = 31$; (30, 40) costs 70 and yields $13 + 17 = 30$; (25, 40) costs 65 and yields 28; (25, 30) costs 55 and yields 24; no three projects fit, since the cheapest three cost $25 + 30 + 40 = 95$. So 31 is the best.
- 80 people.** At least one = $55 + 45 - 20 = 80\%$, so neither = 20%, and $0.20 \times 400 = 80$. Working in percentages first and converting once at the end saves three multiplications.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 90 s/Q · 18 questions in 27 minutes

Questions 1–4. A hospital saw 3,600 patients in a month. 45% went to the OPD, 30% to emergency, and the rest were admitted. Of the admitted patients, 25% needed surgery.

- How many patients were admitted?
- How many needed surgery?
- Surgery patients are what percent of all patients?
- Find the ratio of OPD patients to emergency patients.

Questions 5–8. Of 500 job applicants, 280 passed the written test and 220 passed the interview. 150 passed both.

- How many passed at least one stage?
- How many passed neither?
- How many passed only the interview?
- How many passed exactly one stage?

Questions 9–12. Tickets handled by two teams.

Team	Jan	Feb	Mar	Total
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Red	90	—	120	320
Blue	—	100	130	330
Total	190	210	250	650

- Find Red's February figure.
- Find Blue's January figure.
- Find the percent rise in the combined total from January to March.
- Red's tickets are what percent of the grand total?

Questions 13–14. A radar chart scores three laptops out of 10.

Axis	L1	L2	L3
Speed	8	9	7
Battery	7	5	9
Weight	6	7	8
Price value	9	6	7

- Which laptop has the highest total score?
- L3's total exceeds L2's by what percent?

Questions 15–18.

- In a class of 60, 40 passed Maths and 35 passed Science. Find the least possible number who passed both.
- Store A's sales rose 10% from Rs. 50 lakh and store B's rose 25% from Rs. 30 lakh. Find the combined percent rise.
- Four months of sales average 450 units. Three of them are 400, 500 and 380. Find the fourth.
- In a group of 90 people, 50 play chess, 45 play carrom and 40 play badminton; 20 play chess and carrom, 18 play carrom and badminton, 22 play chess and badminton, and 10 play all three. How many play at least one game?

ANSWER KEY

- 900.** Admitted share = $100 - 45 - 30 = 25\%$, and $0.25 \times 3600 = 900$.
- 225.** $0.25 \times 900 = 225$.
- 6.25%.** $\left(\frac{225}{3600}\right) \times 100 = 6.25$. Or $0.25 \times 0.25 = 0.0625$ directly.
- 3 : 2.** OPD = $0.45 \times 3600 = 1620$ and emergency = $0.30 \times 3600 = 1080$; $1620 : 1080$, divide both by 540. Or just compare the shares, 45 : 30.
- 350.** $280 + 220 - 150 = 350$.
- 150.** $500 - 350 = 150$.
- 70.** $220 - 150 = 70$.
- 200.** $350 - 150 = 200$. Or $(280 - 150) + (220 - 150) = 130 + 70$.
- 110.** Column Feb: $210 - 100 = 110$. Check row Red: $90 + 110 + 120 = 320$. Correct.
- 100.** Column Jan: $190 - 90 = 100$. Check row Blue: $100 + 100 + 130 = 330$. Correct.
- About 31.58%.** $\frac{250-190}{190} = \frac{60}{190} = 0.3158$.
- About 49.23%.** $\left(\frac{320}{650}\right) \times 100$: 49% of 650 is 318.5, the gap 1.5 is 0.23% of 650.
- L3, with 31.** L1 = $8 + 7 + 6 + 9 = 30$; L2 = $9 + 5 + 7 + 6 = 27$; L3 = $7 + 9 + 8 + 7 = 31$.
- About 14.81%.** $\frac{31-27}{27} = \frac{4}{27} = 0.1481$.
- 15.** Least overlap = $n(M) + n(S) - \text{total} = 40 + 35 - 60 = 15$. The two pass lists name 75 passes between them, but there are only 60 students, so at least $75 - 60 = 15$ students must be named on both lists.
- +15.625%.** Weighted: $\frac{50 \times 10 + 30 \times 25}{80} = \frac{500 + 750}{80} = \frac{1250}{80} = 15.625$. Check: $55 + 37.5 = 92.5$ against 80, a rise of 12.5 on 80. The plain average $(10 + 25) \div 2 = 17.5$ is the trap.
- 520.** Sum = $450 \times 4 = 1800$; known = $400 + 500 + 380 = 1280$; $1800 - 1280 = 520$.
- 85.** $50 + 45 + 40 = 135$; $20 + 18 + 22 = 60$; $135 - 60 = 75$; $75 + 10 = 85$. So 5 people play none.

ONE-PAGE REVISION CARD

Caselet routine. Grand total → draw the grid or tree → fill in absolute numbers → check that the boxes re-add to the total → then read the questions. Never answer a caselet question straight from the paragraph.

Base discipline. “60% of those who failed” sits on the failed box, not the total. Write the base beside every percentage before multiplying.

Two-set Venn. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ · neither = total – $n(A \cup B)$ · only $A = n(A) - n(A \cap B)$ · exactly one = $n(A \cup B) - n(A \cap B)$.

Three-set Venn. $n(A \cup B \cup C) = \text{singles} - \text{pairs} + \text{centre}$. Fill from the centre out: centre, then each pair minus the centre, then each single minus the three pieces already placed. Seven regions, every question is a sum of them.

Venn bounds. $\max n(A \cap B) = \min(n(A), n(B))$ · $\min n(A \cap B) = n(A) + n(B) - \text{total}$ (0 if negative). $\min n(A \cap B \cap C) = n(A) + n(B) + n(C) - 2 \times \text{total}$ (0 if negative). Equivalent method: add up the people **missing** from each set; that sum is the most who can fail to be in all three.

Missing grid. Find the row or column with exactly one dash. Fill it. Repeat. Finish by adding the grand total both by rows and by columns — they must agree. A sentence printed outside the grid usually unlocks the first blank.

Averages. Average $\times$ count = sum. Convert every average to a sum on line one.

Average rate. total value $\div$ total quantity, never the plain average of the rates.

Combining groups. Combined percent change = $\frac{w_1 g_1 + w_2 g_2}{w_1 + w_2}$, with w the **old** sizes. Equal sizes is the only case where a plain average is right.

Radar. Sum the axes; ignore the shape. **Bubble.** Three numbers per point: x , y , and size. The size is usually what the question is really about.

Top 5 traps.

1. **Shifting base.** “Of the rest”, “of those who failed”, “of the remainder” — each phrase restarts the percentage on a new, smaller number.
2. **Only A is not A.** $n(A)$ includes everyone in the overlaps. Subtract them before calling a region “only” or “exactly”.
3. **Pairwise counts include the centre.** “80 use both metro and cab” already contains the 40 who use all three.
4. **Averaging percentages of unequal groups.** Pass rates, growth rates, margins, order values — weight by size or the answer is wrong.
5. **Grid not cross-checked.** Always add the grand total by rows and by columns. If they disagree, one subtraction was wrong, and you will find it in seconds.

Data Sufficiency

Decide if you can solve it. Do not solve it.

What you need to know

WHAT YOU NEED TO KNOW

The question is never “what is the answer”. The question is “do I have enough to get **one** answer?” Stop the moment you know that.

The standard five options (2-statement format). This book uses them everywhere.

(a)	Statement I alone is sufficient, but statement II alone is not.
(b)	Statement II alone is sufficient, but statement I alone is not.
(c)	Both statements together are sufficient, but neither alone is.
(d)	Each statement alone is sufficient.
(e)	Both together are still not sufficient.

The 3-statement format (I, II, III). The question becomes “which combinations are sufficient?” Typical options: “I and II only”, “II and III only”, “any two of the three”, “all three are needed”, “even all three are not enough”. Method is the same: test each combination separately.

The AD / BCE grid — the only procedure you need.

1. Read the question. Write down what a complete answer would look like (a number? a yes or a no?).
2. Test statement I **alone**. Cover statement II with your finger.
3. If I works $\Rightarrow$ the answer is **(a) or (d)**. Now test II alone. II works too $\Rightarrow$ (d). II fails $\Rightarrow$ (a).
4. If I fails $\Rightarrow$ the answer is **(b), (c) or (e)**. Test II alone. II works $\Rightarrow$ (b). II fails $\Rightarrow$ now, and only now, combine them. Works $\Rightarrow$ (c). Fails $\Rightarrow$ (e).

Sufficient means exactly one answer.

- A value question is settled when the unknown is pinned to **one** number.
- A yes/no question is settled when the answer is **always yes** or **always no**. A definite **no** is just as sufficient as a definite yes.
- Two possible values $\Rightarrow$ not sufficient. $x^2 = 49$ gives $x = 7$ or $x = -7$.

Counting equations (linear only).

- k unknowns need k **independent** linear equations for a unique solution.
- $2x + 3y = 12$ and $4x + 6y = 24$ are the **same** equation. That is one equation, not two.
- Non-linear equations break the count: $xy = 12$ and $x + y = 7$ give two answer pairs.

Things you often do NOT need.

- For a percentage, ratio or fraction answer, absolute values are usually unnecessary.
- For “is n divisible by d ”, the value of n is unnecessary.
- For a yes/no about an average, the individual items are unnecessary.

Standing assumptions of the format.

1. Both statements are **true**. They can never contradict each other. If they seem to, you made an arithmetic error.
2. The statements do not have to be independent of each other in meaning, but each one must be tested **on its own** first.
3. Unless stated, a “number” may be negative, zero or a fraction.

Warm-up

WARM-UP

build the reflex

Example 1

Find x . I: $x + 3 = 10$. II: $2x = 14$.

SOLUTION

I alone: $x = 10 - 3 = 7$. One value. Sufficient. II alone: $x = 14 \div 2 = 7$. One value. Sufficient.

Answer: (d) Each alone is sufficient.

Example 2

Find x . I: $x^2 = 49$. II: $x > 0$.

SOLUTION

I alone: $x = 7$ or $x = -7$. Two values. Not sufficient. II alone: any positive number. Not sufficient. Together: $x = 7$ or -7 , and $x > 0$, so $x = 7$. One value.

Answer: (c) Both together.

Example 3

Is the integer n even? I: n is a multiple of 6. II: n is a multiple of 3.

SOLUTION

I alone: $n = 6k = 2(3k)$, always even. Definite **yes**. Sufficient. II alone: $n = 3$ is odd, $n = 6$ is even. Two different answers. Not sufficient.

Answer: (a) I alone.

Example 4

Find the area of a rectangle. I: Its length is 12 cm. II: Its perimeter is 34 cm.

SOLUTION

I alone: breadth unknown. Not sufficient. II alone: length and breadth both unknown. Not sufficient. Together: $2(12 + b) = 34 \Rightarrow 12 + b = 17 \Rightarrow b = 5$. Area = $12 \times 5 = 60 \text{ cm}^2$.

Answer: (c) Both together.

Example 5

Find $x - y$. I: $x + y = 10$. II: $3x - 3y = 12$.

SOLUTION

I alone: gives the sum, not the difference. $x = 6, y = 4$ gives 2; $x = 9, y = 1$ gives 8. Not sufficient. II alone: divide by 3. $x - y = 12 \div 3 = 4$. One value. Sufficient.

Answer: (b) II alone.

I You never found x or y . You did not need to.

Example 6

Is $x > 5$? I: $x > 3$. II: x is an integer.

SOLUTION

I alone: $x = 4$ gives no, $x = 9$ gives yes. Not sufficient. II alone: any integer. Not sufficient. Together: $x = 4$ (integer, > 3) gives no; $x = 9$ gives yes. Still both answers.

Answer: (e) Not sufficient even together.

Example 7

How old is Ravi? I: Ravi is 5 years older than Meera. II: Meera is 18 years old.

SOLUTION

I alone: Meera's age unknown. Not sufficient. II alone: says nothing about Ravi. Not sufficient. Together: $18 + 5 = 23$ years.

Answer: (c) Both together.

Example 8

What is the speed of a car? I: It covers 180 km. II: It takes 3 hours.

SOLUTION

I alone: no time. Not sufficient. II alone: no distance. Not sufficient. Together: speed = $180 \div 3 = 60$ km/h.

Answer: (c) Both together.

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

What is the two-digit number N ? I: The sum of its digits is 11. II: The difference of its digits is 3.

SOLUTION

I alone: 29, 38, 47, 56, 65, 74, 83, 92. Eight numbers. Not sufficient. II alone: 14, 25, 41, 52, 63, ... many. Not sufficient. Together: digits add to 11 and differ by 3. Let the digits be a and b . $a + b = 11$ and $a - b = 3$ give $2a = 14$, $a = 7$, $b = 4$. So the digits are 7 and 4. The number is **74 or 47**. Two numbers.

Answer: (e) Not sufficient even together.

TRAP

Solving for the **digits** is not the same as solving for the **number**. The order of the digits is a second unknown. Always ask: have I pinned the exact thing the question named?

Example 10 [Infosys pattern]

What is the cost of one pen? I: 5 pens and 3 notebooks cost Rs. 145. II: 10 pens and 6 notebooks cost Rs. 290.

SOLUTION

Let a pen cost p and a notebook cost n . I: $5p + 3n = 145$. One equation, two unknowns. Not sufficient. II: $10p + 6n = 290$. Divide both sides by 2: $5p + 3n = 145$. Not sufficient. Together: statement II **is** statement I. You still have one equation and two unknowns. Check: $p = 20$, $n = 15$ gives $100 + 45 = 145$. Also $p = 26$, $n = 5$ gives $130 + 15 = 145$. Two different pen prices fit both statements.

Answer: (e) Not sufficient even together.

TRAP

The repeated equation. When statement II is statement I multiplied by a constant, the answer is (e), never (c). Test it fast: divide statement II by the factor that makes the first coefficient match, then compare.

Example 11 [Accenture pattern]

What simple interest does a sum earn in 2 years? I: The principal is Rs. 24,000. II: The rate is 7.5% per annum.

SOLUTION

I alone: no rate. Not sufficient. II alone: no principal. Not sufficient. Together: $SI = \frac{24000 \times 7.5 \times 2}{100}$. $24000 \times 7.5 = 180000$. $180000 \times 2 = 360000$. $360000 \div 100 = 3600$. $SI = \text{Rs. } 3,600$. One value.

Answer: (c) Both together.

Example 12 [TCS NQT pattern]

Is the positive integer x divisible by 12? I: x is divisible by 4 and by 6. II: x is divisible by 24.

SOLUTION

I alone: divisible by 4 and by 6 means divisible by their LCM. $4 = 2^2$, $6 = 2 \times 3$, so $LCM = 2^2 \times 3 = 12$. Definite **yes**. Sufficient. II alone: $x = 24k = 12(2k)$, always a multiple of 12. Definite **yes**. Sufficient.

Answer: (d) Each alone is sufficient.

TRAP

“Divisible by 4 and by 6” gives 12, not 24. Multiply the **LCM**, not the two numbers. $x = 12$ is divisible by 4 and 6 but not by 24.

Example 13 [Wipro pattern]

What is Priya’s monthly salary? I: She saves 20% of her salary. II: Her total monthly expenses are Rs. 32,000.

SOLUTION

I alone: a percentage with no rupee figure. Not sufficient. II alone: expenses alone say nothing about the salary. Not sufficient. Together: she saves 20%, so she spends $100 - 20 = 80\%$ of her salary. 80% of salary = 32000. $\text{Salary} = 32000 \times \frac{100}{80} = \frac{3200000}{80} = 40000$. Rs. 40,000.

Answer: (c) Both together.

Example 14 [Capgemini pattern]

What is the area of a square? I: Its perimeter is 48 cm. II: Its diagonal is $12\sqrt{2}$ cm.

SOLUTION

I alone: side = $48 \div 4 = 12$ cm. Area = $12 \times 12 = 144 \text{ cm}^2$. Sufficient. II alone: for a square, diagonal = side $\times \sqrt{2}$. So side = $12\sqrt{2} \div \sqrt{2} = 12$ cm. Area = 144 cm^2 . Sufficient.

Answer: (d) Each alone is sufficient.

I For a square, **any one** measurement fixes every other one. Side, perimeter, diagonal, area, inradius — one gives all.

Example 15 [Cognizant pattern]

What is the length of a train? I: It crosses a pole in 12 seconds. II: Its speed is 54 km/h.

SOLUTION

I alone: no speed. Not sufficient. II alone: no time. Not sufficient. Together: $54 \text{ km/h} = 54 \times \frac{5}{18} = \frac{270}{18} = 15 \text{ m/s}$. Crossing a pole means covering exactly the train’s own length. Length = $15 \times 12 = 180 \text{ m}$.

Answer: (c) Both together.

SHORTCUT

km/h to m/s: multiply by $\frac{5}{18}$. m/s to km/h: multiply by $\frac{18}{5}$. Memorise the ladder: 18 km/h = 5 m/s, 36 = 10, 54 = 15, 72 = 20, 90 = 25.

Example 16 [TCS NQT pattern]

Three-statement format. What is the father's present age? I: The father is 4 times as old as his son now. II: Five years ago the father was 7 times as old as his son. III: The sum of their present ages is 50 years.

SOLUTION

Let the son be s and the father be f today. I: $f = 4s$. II: $f - 5 = 7(s - 5)$. III: $f + s = 50$.

I and II: $4s - 5 = 7s - 35 \Rightarrow 35 - 5 = 7s - 4s \Rightarrow 30 = 3s \Rightarrow s = 10, f = 40$. Works.

I and III: $4s + s = 50 \Rightarrow 5s = 50 \Rightarrow s = 10, f = 40$. Works.

II and III: $f = 50 - s$, so $50 - s - 5 = 7s - 35 \Rightarrow 45 - s = 7s - 35 \Rightarrow 45 + 35 = 8s \Rightarrow 80 = 8s \Rightarrow s = 10, f = 40$. Works.

Every pair gives the same answer, 40 years. No single statement works alone: I alone has two unknowns, II alone has two unknowns, III alone has two unknowns.

Answer: Any two of the three statements are sufficient.

Example 17 [Accenture pattern]

Is the positive integer p a prime number? I: p lies between 30 and 40. II: p is odd and is divisible by neither 3 nor 5.

SOLUTION

I alone: 31 is prime, 32 is not. Two answers. Not sufficient. II alone: 7 is prime, 49 is not (it is 7×7). Not sufficient. Together: list 31 to 39 and strike out what II forbids.

- Even: 32, 34, 36, 38 — out.
- Divisible by 3: 33, 39 — out.
- Divisible by 5: 35 — out.

Left: 31 and 37. Both are prime. So the answer is **yes** either way.

Answer: (c) Both together.

I You never learned which number p is. For a yes/no question you do not have to.

SHORTCUT

A yes/no question can be settled without finding the unknown. Shrink the list of possibilities until every survivor gives the same yes or the same no. Stop there.

Example 18 [Infosys pattern]

What is the ratio of boys to girls in a class? I: There are 18 more boys than girls. II: There are 90 students in total.

SOLUTION

I alone: 28 boys and 10 girls give $28 : 10 = 14 : 5$. 54 boys and 36 girls give $3 : 2$. Different ratios. Not sufficient.

II alone: no split given. Not sufficient. Together: $b + g = 90$ and $b - g = 18$. Add: $2b = 108 \Rightarrow b = 54$. Then $g = 90 - 54 = 36$. Ratio = $54 : 36 = 3 : 2$ (divide both by 18).

Answer: (c) Both together.

Example 19 [TCS NQT pattern]

What is the shopkeeper's profit percent? I: He marks his goods 40% above cost and gives a 25% discount on the marked price. II: The cost price is Rs. 800.

SOLUTION

I alone: take cost = 100 (you may, because the answer is a percent). Marked price = 140. Discount = 25% of 140 = 35. Selling price = 140 - 35 = 105. Profit = 105 - 100 = 5 on a cost of 100, so profit = 5%. Sufficient.
II alone: no selling price. Not sufficient.

Answer: (a) I alone.

SHORTCUT

Percent answers are scale-free. When the question asks for a percentage or a ratio, set the base to 100 and work. If statement I fixes every **rate** in the chain, it is sufficient, and the rupee value in statement II is a decoy.

Example 20 [Wipro pattern]

What is the remainder when the positive integer N is divided by 5? I: N leaves remainder 3 when divided by 15. II: N leaves remainder 8 when divided by 10.

SOLUTION

I alone: $N = 15k + 3$. Now $15k$ is a multiple of 5, so $N = 5(3k) + 3$ and the remainder is 3. Sufficient. II alone: $N = 10k + 8 = 10k + 5 + 3 = 5(2k + 1) + 3$, so the remainder is again 3. Sufficient. Quick check with a number that fits both: $N = 18$. $18 = 15 + 3$ and $18 = 10 + 8$, and $18 = 5 \times 3 + 3$. Remainder 3.

Answer: (d) Each alone is sufficient.

■ You never found N . Remainder questions rarely need the number itself — only the part of it that survives the division.

PRACTICE · TIER 1 target: 60 s/Q

Use the five standard options (a)–(e) from the formula box.

1. What is the value of x ? I: $3x - 7 = 14$. II: $x^2 = 49$.
2. What is the perimeter of a rectangle? I: Its area is 72 cm^2 . II: Its length is twice its breadth and its area is 72 cm^2 .
3. Is the number n divisible by 9? I: n is divisible by 3. II: The sum of the digits of n is 18.
4. What is the average of five numbers? I: Their total is 80. II: The sum of the first three is 45.
5. What is the speed of a boat in still water? I: Its downstream speed is 18 km/h. II: Its upstream speed is 12 km/h.
6. What is the value of a ? I: $a + b = 10$. II: $2a + 2b = 20$.
7. What is the cost price of an article? I: Selling it at Rs. 720 gives a 20% profit. II: Selling it at Rs. 540 gives a 10% loss.
8. In how many days can A and B finish a job together? I: A alone takes 12 days. II: B alone takes 24 days.
9. Is a four-digit number divisible by 4? I: Its last two digits form the number 36. II: The sum of its digits is 12.
10. What will the population of a town be after 2 years? I: It is 20,000 today. II: It grows at 10% per annum.
11. What is the 10th term of an arithmetic progression? I: The 4th term is 17 and the 7th term is 29. II: The sum of the first three terms is 27.
12. What is the value of the integer x ? I: x is a prime number between 10 and 20. II: x is odd.
13. Is $x > y$? I: $x^2 > y^2$. II: Both x and y are positive.
14. How long does a train take to cross a 200 m platform? I: The train is 150 m long. II: The train travels at 25 m/s.
15. What is the discount percent on an article? I: Its marked price is Rs. 1,200 and it sells for Rs. 960. II: The discount is Rs. 240.

ANSWER KEY

- (a). I gives $3x = 21$, $x = 7$. II gives $x = 7$ or -7 — two values, so it fails.
- (b). II: $2b^2 = 72 \Rightarrow b^2 = 36 \Rightarrow b = 6$, $l = 12$, perimeter $= 2(12 + 6) = 36$ cm. I fixes only the product $lb = 72$, and 12×6 (perimeter 36) and 9×8 (perimeter 34) both fit.
- (b). Digit sum 18 is a multiple of 9, so n is. I only gives a multiple of 3, and $n = 12$ is not a multiple of 9.
- (a). $80 \div 5 = 16$. II leaves the last two numbers unknown.
- (c). Still water speed $= \frac{\text{down} + \text{up}}{2} = \frac{18+12}{2} = 15$ km/h. Each alone gives only one of the two.
- (e). II is I doubled. One equation, two unknowns: $a = 4$ or $a = 6$ both fit.
- (d). I: $720 \div 1.2 = 600$. II: $540 \div 0.9 = 600$. Both pin Rs. 600.
- (c). Together $\frac{1}{12} + \frac{1}{24} = \frac{3}{24} = \frac{1}{8}$, so 8 days. One rate alone is not enough.
- (a). A number is divisible by 4 when its last two digits are; $36 \div 4 = 9$, so yes. Digit sum tests 3 and 9, not 4.
- (c). $20000 \times 1.1 \times 1.1 = 24,200$. Rate without base, or base without rate, fails.
- (a). $3d = 29 - 17 = 12 \Rightarrow d = 4$, $a = 17 - 3(4) = 5$, $t_{10} = 5 + 9(4) = 41$. II only gives $a + d = 9$.
- (e). Primes 11, 13, 17, 19 are all odd, so II adds nothing and four values survive.
- (c). With both positive, $x^2 > y^2$ forces $x > y$. Alone, I allows $x = -5$, $y = 2$.
- (c). Length + platform $= 350$ m; $350 \div 25 = 14$ s. Each alone misses a piece.
- (a).

$$\frac{1200 - 960}{1200} = \frac{240}{1200} = 20\%$$

. A rupee discount with no marked price gives no percent.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

What was a driver's total fare income on Tuesday? I: He completed 24 trips at an average fare of S\$11.50. II: His Tuesday income was 15% more than his Monday income.

SOLUTION

I alone: total $=$ number of trips $\times$ average fare $= 24 \times 11.50$. $24 \times 11 = 264$ and $24 \times 0.5 = 12$, so total $= 264 + 12 = \text{S\$}276$. Sufficient. II alone: Monday's income is unknown, so 15% more than an unknown is unknown. Not sufficient.

Answer: (a) I alone.

Example 22 [Shopee · pattern]

What is the cost price of an item a seller shipped? I: The platform charges a 12% fee on the selling price, and the item sold for S\$250. II: After the fee, the seller's profit was S\$40.

SOLUTION

I alone: it tells you what the seller **receives**, not what the item **cost**. Receipt $= 250 \times \frac{100-12}{100} = 250 \times 0.88 = 220$. Cost still unknown. Not sufficient. II alone: profit without any price figure. Not sufficient. Together: cost $=$ receipt $-$ profit $= 220 - 40 = \text{S\$}180$.

Answer: (c) Both together.

Example 23 [DBS · pattern]

A customer qualifies for the premium tier if her average monthly balance over three months is at least S\$5,000. Does she qualify? I: Her three monthly balances were S\$4,200, S\$5,100 and S\$6,000. II: Her third-month balance was S\$6,000.

SOLUTION

I alone: $\text{sum} = 4200 + 5100 = 9300$; $9300 + 6000 = 15300$. $\text{Average} = 15300 \div 3 = 5100$. Since $5100 \geq 5000$, the answer is a definite **yes**. Sufficient. II alone: one month out of three. The other two could be zero (average 2,000, no) or 6,000 each (average 6,000, yes). Not sufficient.

Answer: (a) I alone.

TRAP

One month below the target does not disqualify anybody. The rule is about the **average**. Read what is being averaged before you judge any single number.

Example 24 [Agoda · pattern]

What was a hotel's occupancy rate last Friday? I: 84 rooms were occupied. II: 21 rooms were empty.

SOLUTION

I alone: occupancy rate needs the total room count. Not sufficient. II alone: same problem from the other side. Not sufficient. Together: total rooms $= 84 + 21 = 105$. $\text{Rate} = \frac{84}{105}$. Divide top and bottom by 21: $84 \div 21 = 4$, $105 \div 21 = 5$, so $\frac{4}{5} = 80\%$.

Answer: (c) Both together.

Example 25 [SCB · pattern]

Nat changes S\$400 into Thai baht at a bank. How many baht does she receive? I: The bank's rate is S\$1 = THB 26.5. II: The bank deducts a 1.5% commission from the converted amount.

SOLUTION

I alone: $\text{gross} = 400 \times 26.5 = 10,600$ baht, but the question asks what she **receives**, and a fee may apply. Not sufficient on its own. II alone: a fee percent with no rate. Not sufficient. Together: $\text{gross} = 400 \times 26.5$. $400 \times 26 = 10400$ and $400 \times 0.5 = 200$, so $\text{gross} = 10,600$ baht. $\text{Commission} = 1.5\% \text{ of } 10600 = 10600 \times 0.015 = 159$. $\text{Received} = 10600 - 159 = \text{THB } 10,441$.

Answer: (c) Both together.

Example 26 [Sea/Shopee · pattern]

Three-statement format. How many March orders were delivered on time? I: There were 12,500 orders in March. II: The on-time rate was 92%. III: 1,000 orders were late.

SOLUTION

I and II: $12500 \times \frac{92}{100} = 125 \times 92 = 11,500$. Works.

I and III: $\text{on time} = 12500 - 1000 = 11,500$. Works.

II and III: $\text{late rate} = 100 - 92 = 8\%$. So $8\% \text{ of total} = 1000$, $\text{total} = 1000 \times \frac{100}{8} = 12,500$. $\text{On time} = 12500 - 1000 = 11,500$. Works.

No single statement works alone: I gives no split, II gives no count, III gives no total.

Answer: Any two of the three statements are sufficient.

Example 27 [GIC · pattern]

Over two years, did a fund beat its benchmark? I: The fund returned 8% in year 1 and 5% in year 2; the benchmark returned 6% in year 1 and 7% in year 2. II: The benchmark's two-year total return was 13.42%.

SOLUTION

I alone: chain the growth factors, do not add the percents. Fund: 1.08×1.05 . $1.08 \times 1.05 = 1.08 + 0.054 = 1.134$, a 13.4% gain. Benchmark: $1.06 \times 1.07 = 1.06 + 0.0742 = 1.1342$, a 13.42% gain. $13.40\% < 13.42\%$,

so the fund did **not** beat the benchmark. A definite **no**. Sufficient. II alone: it gives the benchmark only, nothing about the fund. Not sufficient.

Answer: (a) I alone.

TRAP

$8 + 5 = 13$ and $6 + 7 = 13$ tempts you into “equal, so cannot decide”. Compounding breaks the tie: the pair that is **less spread out** compounds higher. Always multiply the factors.

Example 28 [LINE MAN · pattern]

A delivery app charges a fixed base fee plus a fixed rate per km. What is the rate per km? I: A 4 km delivery costs THB 62. II: A 7 km delivery costs THB 89.

SOLUTION

Let the base be B and the per-km rate be r . I alone: $B + 4r = 62$. One equation, two unknowns. Not sufficient. II alone: $B + 7r = 89$. Same problem. Not sufficient. Together: subtract the first from the second. $(B + 7r) - (B + 4r) = 89 - 62 \Rightarrow 3r = 27 \Rightarrow r = 9$. Check: $B = 62 - 4(9) = 62 - 36 = 26$; then $26 + 7(9) = 26 + 63 = 89$. Correct.

Answer: (c) Both together.

SHORTCUT

Two points on a straight-line price list give both the slope and the base. Subtract to kill the base, then back-substitute. If the question asks only for the per-km rate, you can stop right after the subtraction.

Example 29 [Razer · pattern]

Is the average product rating above 4.0? I: 60 reviews average 4.2 and the other 40 average 3.7. II: More than half the reviews give 5 stars.

SOLUTION

I alone: weighted average $= \frac{60 \times 4.2 + 40 \times 3.7}{100}$. $60 \times 4.2 = 252$. $40 \times 3.7 = 148$. $252 + 148 = 400$. $400 \div 100 = 4.0$ exactly. “Above 4.0” is therefore **no**. A definite answer. Sufficient. II alone: 51 fives and 49 ones give an average near 3.0; 51 fives and 49 fours give 4.5. Both answers possible. Not sufficient.

Answer: (a) I alone.

The answer is “no” and that still counts as sufficient. Students lose this mark by thinking a boundary result means “cannot say”.

PRACTICE · TIER 2 target: 100 s/Q

Options (a)–(e) as before, except where a question names its own options.

1. What is the price of 1 kg of rice at a stall? I: 3 kg of rice and 2 kg of dal cost THB 410. II: 6 kg of rice and 4 kg of dal cost THB 820.
2. What were a driver’s net earnings for the day? I: His gross fares were S\$320. II: The platform keeps a 20% commission.
3. Did more than 500 customers visit a store on Sunday? I: 600 customers visited across Saturday and Sunday together. II: Sunday’s count was 50% higher than Saturday’s.
4. What is a company’s profit margin on revenue? I: Revenue was S\$4.5 million and total cost was S\$3.6 million. II: Profit was S\$0.9 million.
5. What did Ploy pay for a phone including 7% VAT? I: The price before VAT was THB 18,000. II: The VAT charged was THB 1,260.

6. What is the value of an order in Singapore dollars? I: The order is THB 21,200. II: $\text{S\$1} = \text{THB } 26.5$.
7. **Three statements.** What is the monthly rent of a shop? I: Rent is 18% of monthly revenue. II: Monthly revenue is S\$25,000. III: Rent plus utilities is S\$5,000, and utilities alone are S\$500. Options: (a) I and II together, but III alone does not work (b) III alone, but I and II together do not work (c) III alone works, and I and II together also work (d) all three statements are needed
8. Is the discount on a laptop more than 15%? I: The marked price is S\$2,400 and it sells for S\$2,040. II: The discount is S\$360.
9. How long does an airport shuttle take to reach the hotel? I: The distance is 27 km. II: The shuttle averages 45 km/h.
10. How many people work at the Bangkok office? I: More than 120. II: Fewer than 150.
11. A fund grew exactly 25% during 2024. What was its value at the start of the year? I: It ended the year at S\$1.5 million. II: The gain was S\$300,000.
12. Is the average order value above THB 500? I: Total order value was THB 58,000. II: There were 120 orders.

ANSWER KEY

1. (e). II is I doubled, so it is the same equation. Rice at 90 with dal at 70, or rice at 100 with dal at 55, both fit.
2. (c). $\text{Net} = 320 \times 0.80 = \text{S\$}256$. A gross figure with no rate, or a rate with no gross, fails.
3. (c). Let Saturday = S . Then $S + 1.5S = 600 \Rightarrow 2.5S = 600 \Rightarrow S = 240$, Sunday = 360. That is not more than 500 — a definite **no**.
4. (a). Profit = $4.5 - 3.6 = 0.9$; margin = $\frac{0.9}{4.5} = 20\%$. II has no revenue to divide by.
5. (d). I: $18000 \times 1.07 = 19,260$. II: $1260 \div 0.07 = 18,000$ before VAT, so $18000 + 1260 = 19,260$.
6. (c). $21200 \div 26.5 = 800$. Either an amount with no rate, or a rate with no amount.
7. (c). III alone: $5000 - 500 = \text{S\$}4,500$. I and II together: $0.18 \times 25000 = \text{S\$}4,500$. Same answer by two routes.
8. (a). $\frac{2400 - 2040}{2400} = \frac{360}{2400} = 15\%$, which is **not** more than 15% — a definite no. A rupee-style amount alone gives no percent.
9. (c). $27 \div 45 = 0.6 \text{ h} = 36 \text{ minutes}$. Distance without speed, or speed without distance, fails.
10. (e). Together the count is any number from 121 to 149 — many values.
11. (d). I: $1.5 \div 1.25 = \text{S\$}1.2 \text{ million}$. II: $300000 \div 0.25 = \text{S\$}1.2 \text{ million}$.
12. (c). $58000 \div 120 = 483.33$, which is not above 500 — a definite no. Neither the total nor the count works alone.

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 30 [Google · pattern]

Is the positive integer n a perfect square? I: n has exactly 3 positive divisors. II: n is odd.

SOLUTION

Insight: divisors pair up as d and $\frac{n}{d}$, so the count of divisors is odd only when one divisor pairs with itself — that is, only for a perfect square.

I alone: exactly 3 divisors is an odd count, so n must be a perfect square. Concretely, if $n = p^2$ for a prime p , its divisors are $1, p, p^2$ — exactly 3. Any other shape gives a different count: p gives 2, pq gives 4, p^3 gives 4. So $n = p^2$, a perfect square. Definite **yes**. Sufficient. II alone: $n = 9$ is an odd perfect square; $n = 15$ is odd and not a perfect square. Not sufficient.

Answer: (a) I alone.

Example 31 [Goldman Sachs · pattern]

x and y are positive integers. Is the product xy even? I: $x + y$ is odd. II: $x^2 + y^2$ is odd.

SOLUTION

Insight: a product is even the moment one factor is even. So the whole question is “is at least one of them even?” — a parity question, not an arithmetic one.

I alone: a sum is odd only when one number is odd and the other is even (odd + odd = even, even + even = even). So one of them is even, and the product is even. Definite **yes**. Sufficient. II alone: squaring does not change parity. An even number squared is even ($4^2 = 16$); an odd number squared is odd ($3^2 = 9$). So $x^2 + y^2$ odd forces one of x^2, y^2 even and the other odd, hence one of x, y even and the other odd. The product is even. Definite **yes**. Sufficient.

Answer: (d) Each alone is sufficient.

Example 32 [Amazon · pattern]

A bag holds only red and blue balls. Two balls are drawn together (without replacement). Is the probability that both are red greater than $\frac{1}{2}$? I: There are 10 balls, of which 8 are red. II: More than 75% of the balls are red.

SOLUTION

Insight: drawing without replacement makes the second draw slightly worse than the first, so the probability is a little below p^2 . If p^2 clears $\frac{1}{2}$ with room to spare, the small penalty cannot drag it below.

I alone: $P = \frac{8}{10} \times \frac{7}{9} = \frac{56}{90} = \frac{28}{45} \cdot \frac{28}{45} = 0.622\dots$, and $0.622 > 0.5$. Definite **yes**. Sufficient.

II alone: let there be n balls with r red, $r > 0.75n$. Then

$$P = \frac{r}{n} \times \frac{r-1}{n-1}.$$

Test the tightest cases, where r is the smallest integer above $0.75n$:

- $n = 4, r = 4$: $P = 1$. Yes.
- $n = 5, r = 4$: $P = \frac{4}{5} \times \frac{3}{4} = \frac{3}{5} = 0.6$. Yes.
- $n = 8, r = 7$: $P = \frac{7}{8} \times \frac{6}{7} = \frac{6}{8} = 0.75$. Yes.
- $n = 9, r = 7$: $P = \frac{7}{9} \times \frac{6}{8} = \frac{42}{72} = 0.583$. Yes.
- $n = 100, r = 76$: $P = 0.76 \times \frac{75}{99} = 0.76 \times 0.7576 = 0.576$. Yes.
- Very large n : P approaches $0.75^2 = 0.5625$. Still above 0.5.

Now prove it instead of sampling. With $r > 0.75n$,

$$r(r-1) > 0.75n(0.75n-1) = 0.5625n^2 - 0.75n.$$

We need $P > \frac{1}{2}$, i.e. $r(r-1) > \frac{n(n-1)}{2} = 0.5n^2 - 0.5n$. It is enough that $0.5625n^2 - 0.75n \geq 0.5n^2 - 0.5n$, i.e. $0.0625n^2 \geq 0.25n$, i.e. $n \geq 4$. And n must be at least 4 anyway, because $r > 0.75n$ with $r \leq n$ needs at least 4 balls to leave room for a non-red one — and $n = 2, 3$ force $r = n$, giving $P = 1$. So the answer is always **yes**. Sufficient.

Answer: (d) Each alone is sufficient.

Sampling six cases is how you **believe** it. The inequality is how you **know** it. In an interview, do the sampling out loud, then produce the inequality.

Example 33 [Microsoft · pattern]

A jar holds black and white stones. Repeatedly, two stones are removed: if they match, one **black** stone is put back; if they differ, one **white** stone is put back. This runs until one stone is left. Is the last stone white? I: The jar starts with 12 white stones. II: The jar starts with 15 black stones.

SOLUTION

Insight: track the parity of the white count. Every move changes it by 0 or 2, never by 1 — so odd stays odd and even stays even for the whole game.

Check the three moves:

- Two black removed, one black added: white unchanged.
- Two white removed, one black added: white drops by 2.

- One of each removed, one white added: white loses 1 and gains 1, so white unchanged.

In every case the number of white stones keeps its parity.

I alone: white starts at 12, an even number, so it is even forever. At the end one stone remains. If it were white, the white count would be 1, which is odd — impossible. So the last stone is black: a definite **no**. Sufficient. II alone: the black count tells you nothing, because black is the colour whose parity is **not** protected. Start 15 black with 1 white: the pair (black, white) gives white back, and white parity is odd, so the last stone is white. Start 15 black with 2 white: white parity is even, so the last stone is black. Two different answers. Not sufficient.

Answer: (a) I alone.

SHORTCUT

Find the invariant, and one statement usually collapses to “irrelevant”. In any repeat-until-one puzzle, ask what each move does to a count’s parity, a total, or a sum modulo something. The quantity that never changes is the whole answer.

Example 34 [D. E. Shaw · pattern]

A biased coin shows heads with probability p . It is tossed until the first head appears. Is the expected number of tosses less than 4? I: $p = 0.3$. II: The probability of getting no head in the first two tosses is less than 0.5.

SOLUTION

Insight: for “toss until the first success”, the expected number of tosses is $\frac{1}{p}$. So the question “is $E < 4$?” is exactly the question “is $p > \frac{1}{4}$?”

Why $E = \frac{1}{p}$: let E be the expected number. The first toss always happens. With probability p you stop; with probability $1 - p$ you are back where you started.

$$E = 1 + (1 - p)E \Rightarrow E - (1 - p)E = 1 \Rightarrow pE = 1 \Rightarrow E = \frac{1}{p}.$$

And $\frac{1}{p} < 4 \Leftrightarrow p > \frac{1}{4} = 0.25$.

I alone: $p = 0.3 > 0.25$, so $E = \frac{1}{0.3} = 3.33$, which is less than 4. Definite **yes**. Sufficient. II alone: “no head in two tosses” has probability $(1 - p)^2$.

$$(1 - p)^2 < 0.5 \Rightarrow 1 - p < \sqrt{0.5} = 0.7071 \Rightarrow p > 0.2929.$$

Since $0.2929 > 0.25$, we get $E = \frac{1}{p} < \frac{1}{0.2929} = 3.41 < 4$. Definite **yes**. Sufficient.

Answer: (d) Each alone is sufficient.

Statement II never pins p down. It only pushes p past the threshold, and a threshold is all a yes/no question needs.

Example 35 [Adobe · pattern]

A set S has 8 distinct positive integers. Is the median of S greater than 20? I: The four smallest members are 3, 7, 11 and 19. II: The four largest members are 24, 30, 41 and 50.

SOLUTION

Insight: with 8 numbers the median is the average of the 4th and 5th. Statement I fixes the 4th, statement II fixes the 5th. Neither half is a median by itself.

I alone: the 4th smallest is 19. The 5th is some integer above 19, so at least 20.

- If the 5th is 20, median = $\frac{19+20}{2} = 19.5$, which is **not** above 20.
- If the 5th is 25, median = $\frac{19+25}{2} = 22$, which **is** above 20.

Two answers. Not sufficient.

II alone: the four largest are 24, 30, 41, 50, so the 5th smallest is 24 and the 4th is some value below 24.

- If the 4th is 10, median = $\frac{10+24}{2} = 17$. Not above 20.
- If the 4th is 23, median = $\frac{23+24}{2} = 23.5$. Above 20.

Two answers. Not sufficient.

Together: the list is 3, 7, 11, 19 then 24, 30, 41, 50 — all eight members. Median = $\frac{19+24}{2} = \frac{43}{2} = 21.5$, which is greater than 20. Definite **yes**.

Answer: (c) Both together.

Example 36 [Amazon · pattern]

A shop sold exactly 100 items today. Is the total revenue more than Rs. 50,000? I: The cheapest item sold for Rs. 460. II: The median selling price was Rs. 560.

SOLUTION

Insight: a minimum gives a floor for every item; a median gives a floor for the top half only. Add the two floors — if their sum already clears the target, the answer is yes no matter what the actual prices are.

I alone: every item is at least 460, so revenue $\geq 100 \times 460 = 46,000$. That is below 50,000, so “more than 50,000” could be false (all items at 460) or true (all at 1,000). Not sufficient.

II alone: with 100 sorted prices, median = $\frac{x_{50}+x_{51}}{2} = 560$. Since $x_{50} \leq x_{51}$, we get $x_{51} \geq 560$, so the top 50 items give at least $50 \times 560 = 28,000$. But the bottom 50 have no floor at all; they could be Rs. 1 each, giving a total near 28,050 — below 50,000. Or they could be huge. Not sufficient.

Together:

- Bottom 50 items: each at least 460, so at least $50 \times 460 = 23,000$.
- Top 50 items: each at least 560 (shown above), so at least $50 \times 560 = 28,000$.
- Minimum possible revenue = $23,000 + 28,000 = 51,000$.

Since $51,000 > 50,000$, the revenue is above 50,000 in every case. Definite **yes**.

Answer: (c) Both together.

The pattern to remember: **worst-case sum**. Build the cheapest world the statements allow. If even that world beats the target, you are done.

PRACTICE · TIER 3 target: 4 min/Q

Options (a)–(e) as before.

1. Is the positive integer n divisible by 6? I: n^2 is divisible by 36. II: n^3 is divisible by 8.
2. x, y, z are positive integers. Is the product xyz even? I: $x + y + z$ is even. II: xz is odd.
3. A box holds red and green counters. Repeatedly two are removed: if they match, a **green** counter is put back; if they differ, a **red** counter is put back. This runs until one counter is left. Is the last counter red? I: The box starts with 7 green counters. II: The box starts with 10 red counters.
4. A set has 7 distinct positive integers. Is the mean greater than the median? I: The largest member is 100. II: The seven numbers form an arithmetic progression.
5. Fifty students sat a test. Is the total of all their marks more than 1,900? I: The lowest mark was 35. II: The median mark was 45.
6. A bag holds only red and blue balls. Two are drawn without replacement. Is the probability that both are the same colour greater than $\frac{1}{2}$? I: The bag holds 10 balls. II: There are equally many red and blue balls.
7. A game costs Rs. 20 per play and pays Rs. 100 when a biased die shows a six. Is the expected profit per play positive? I: The probability of **not** getting a six is 0.78. II: The expected payout per play is Rs. 22.
8. n distinct points lie on a circle. Is the number of chords joining them more than 50? I: $n > 10$. II: The number of triangles with vertices at these points is 165.

ANSWER KEY

1. **(a).** $36 = 2^2 3^2$. If n^2 carries $2^2 3^2$, then n carries $2 \times 3 = 6$ (each prime's power in n^2 is double its power in n , so n has at least one 2 and one 3). Definite yes. II only forces n to be even: $n = 2$ has $n^3 = 8$ but is not a multiple of 6.

2. (a). Three odd numbers add to an odd total, so an even sum means all three are even, or exactly two are odd and one is even. Either way at least one factor is even, so the product is even — a definite yes. II says x and z are both odd but leaves y free: y even gives an even product, y odd gives an odd one.
3. (b). Check what each move does to the **red** count. Two reds out, one green in: red drops by 2. Two greens out, one green in: red unchanged. One of each out, one red in: red unchanged. So red parity never changes. Starting at 10 (even) it stays even, so the last counter cannot be a single red — a definite no. The green count in I is the unprotected one and decides nothing.
4. (b). In an arithmetic progression the terms are symmetric about the middle one, so the mean equals the median exactly — “greater” is a definite no. I fails: 1, 2, 3, 4, 5, 6, 100 has mean $\frac{121}{7} = 17.3$ against median 4 (yes), while 94, 95, 96, 97, 98, 99, 100 has mean = median = 97 (no).
5. (c). The bottom 25 marks are each at least 35, giving at least 875. The median 45 means the 26th mark is at least 45, so the top 25 give at least $25 \times 45 = 1,125$. Minimum total = $875 + 1,125 = 2,000 > 1,900$. Definite yes. Alone, I gives only 1,750 and II leaves the bottom half unbounded below.
6. (b). II alone settles it. Equal numbers means n red and n blue, so $2n$ balls. Favourable pairs = $\binom{n}{2} + \binom{n}{2} = n(n-1)$. Total pairs = $\binom{2n}{2} = (2n)\frac{2n-1}{2} = n(2n-1)$. So $P = \frac{n(n-1)}{n(2n-1)} = \frac{n-1}{2n-1}$, and this is always below $\frac{1}{2}$ because $2(n-1) = 2n-2$ is less than $2n-1$. So the answer is **no** for every n — definite. Check with $n = 5$: $P = \frac{4}{9} = 0.444$. I alone fails: 10 balls all red gives $P = 1$ (yes), 5 red and 5 blue gives 0.444 (no).
7. (d). Profit is positive when $100p > 20$, i.e. $p > 0.2$. I: $p = 1 - 0.78 = 0.22 > 0.2$, so expected profit = $22 - 20 = \text{Rs. } 2$, positive. II: payout $100p = 22$ gives $p = 0.22$, the same conclusion.
8. (d). Chords = $\binom{n}{2} = \frac{n(n-1)}{2}$, so chords > 50 needs $n(n-1) > 100$: true from $n = 11$ ($11 \times 10 = 110$) and false at $n = 10$ ($10 \times 9 = 90$). I: $n > 10$ means $n \geq 11$, so chords ≥ 55 — definite yes. II: $\binom{n}{3} = 165$. Try $n = 11$: $\frac{11 \times 10 \times 9}{6} = \frac{990}{6} = 165$. Fits, and $\binom{n}{3}$ grows with n , so $n = 11$ is the only value. Chords = $\binom{11}{2} = \frac{11 \times 10}{2} = 55 > 50$ — definite yes.

TRAP

Both statements are always true at once. They can never contradict each other. If you derive $x = 5$ from I and $x = 9$ from II, stop — you have made an arithmetic slip, not found a trick question. Re-check before you answer.

Mixed set — exam conditions

PRACTICE · TIER 1 target: 30 min for 18 questions

No tier labels. Options (a)–(e) as before, except where a question names its own options.

1. What is the value of y ? I: $4y - 9 = 11$. II: $y^2 - 25 = 0$.
2. What do 8 identical chairs cost? I: 3 chairs cost Rs. 4,200. II: 5 chairs cost Rs. 7,000.
3. Is the integer n a multiple of 15? I: n is a multiple of 10. II: n is a multiple of 6.
4. How much simple interest does a sum earn in 3 years? I: The sum doubles itself in 12 years at that rate. II: The sum is Rs. 20,000.
5. Is $x > 0$? I: $x^3 > 0$. II: $x^2 > 0$.
6. What is the perimeter of a right-angled triangle? I: The two legs are 6 cm and 8 cm. II: The hypotenuse is 10 cm and the area is 24 cm^2 .
7. What does a shirt sell for after two successive discounts? I: Its marked price is Rs. 2,000 and the discounts are 20% and then 10%. II: The total discount is Rs. 560.
8. How many students are in a class? I: The number is divisible by 7 and lies between 40 and 50. II: The number is a multiple of 14 and lies between 40 and 50.
9. What is the speed of the current? I: A boat covers 30 km downstream in 2 hours. II: The boat's downstream speed is 15 km/h.
10. Is a given quadrilateral a square? I: All four sides are equal. II: The diagonals are equal.
11. How many litres of milk are in a mixture? I: The mixture is 40 litres with milk : water = 3 : 2. II: The mixture has 16 litres of water, and the milk is 50% more than the water.

12. What is the compound interest for 2 years? I: The principal is Rs. 10,000 and the rate is 10% per annum. II: The rate is 10% per annum and the simple interest for 2 years is Rs. 2,000.
13. Is the average weight of a team above 60 kg? I: The 11 players weigh 660 kg in total. II: The heaviest player weighs 78 kg.
14. **Three statements.** What is a son's present age? I: His father is 30 years older than he is. II: In 6 years the father will be 3 times as old as the son. III: Four years ago the father was 7 times as old as the son. Options: (a) I and II only (b) I and III only (c) any two of the three (d) all three are needed
15. Is $2^n - 1$ a prime number, where n is a positive integer? I: $n = 11$. II: n is prime.
16. A jar holds 20 coins, each either a one-rupee or a two-rupee coin. How many are two-rupee coins? I: The coins are worth Rs. 32 in all. II: There are 8 one-rupee coins.
17. Is the product of the roots of $x^2 + bx + c = 0$ positive? I: $b = -5$. II: The two roots are equal.
18. A list has 100 numbers. Is their total at least 4,000? I: The smallest number is 30. II: The median is 50.

ANSWER KEY

- (a). $4y = 20$, $y = 5$. II gives $y = 5$ or $y = -5$.
- (d). I: $4200 \div 3 = 1400$ each, so $8 \times 1400 = \text{Rs. } 11,200$. II: $7000 \div 5 = 1400$ each, same total.
- (c). Together n is a multiple of $\text{LCM}(10, 6) = 30$, and 30 is a multiple of 15. Alone, $n = 10$ and $n = 6$ both fail.
- (c). Doubling in 12 years at simple interest means the interest equals the principal in 12 years, so the rate is $\frac{100}{12}\%$ per year and 3 years give 25% of the principal. With $P = 20,000$: $\text{SI} = 20000 \times 0.25 = \text{Rs. } 5,000$. Rate without principal, or principal without rate, fails.
- (a). A cube keeps the sign, so $x^3 > 0$ forces $x > 0$. $x^2 > 0$ only says $x \neq 0$; $x = -3$ works too.
- (d). I: hypotenuse $= \sqrt{36 + 64} = \sqrt{100} = 10$, perimeter $= 6 + 8 + 10 = 24$ cm. II: legs a, b with $ab = 48$ (from area $= a \frac{b}{2} = 24$) and $a^2 + b^2 = 100$, so $(a + b)^2 = 100 + 2(48) = 196$, $a + b = 14$, perimeter $= 14 + 10 = 24$ cm.
- (a). $2000 \times 0.8 \times 0.9 = 1,440$. II gives a rupee discount with no marked price — the same Rs. 560 off Rs. 5,000 would leave a different price.
- (b). Multiples of 7 between 40 and 50: 42 and 49 — two values. Multiples of 14 between 40 and 50: only 42.
- (e). Statement II just restates statement I ($30 \div 2 = 15$ km/h). Neither says anything about the upstream speed, so the current stays unknown.
- (c). Equal sides alone give a rhombus; equal diagonals alone give a rectangle or an isosceles trapezium. Both together force a square.
- (d). I: milk $= 40 \times \frac{3}{5} = 24$ litres. II: milk $= 16 \times 1.5 = 24$ litres.
- (d). I: $10000(1.1)^2 - 10000 = 12100 - 10000 = \text{Rs. } 2,100$. II: SI Rs. 2,000 over 2 years at 10% means $P \times 0.2 = 2000$, so $P = 10,000$, and the CI is again Rs. 2,100.
- (a). $660 \div 11 = 60$ exactly, which is **not** above 60 — a definite no. One player's weight settles nothing.
- (c). Let the son be s and the father f . I and II: $f = s + 30$ and $s + 36 = 3(s + 6) = 3s + 18$, so $2s = 18$, $s = 9$. I and III: $s + 26 = 7(s - 4) = 7s - 28$, so $6s = 54$, $s = 9$. II and III: $f = 3s + 12$ and $f = 7s - 24$, so $4s = 36$, $s = 9$. Every pair gives 9 years.
- (a). $2^{11} - 1 = 2047 = 23 \times 89$, so it is **not** prime — a definite no. II fails because $n = 3$ gives the prime 7 while $n = 11$ gives the composite 2047.
- (d). Let there be t two-rupee coins and o one-rupee coins, $o + t = 20$. I: $o + 2t = 32$, so $t = 12$. II: $o = 8$, so $t = 20 - 8 = 12$.
- (c). For $x^2 + bx + c$, the product of the roots is c . I gives b , not c . II alone allows a double root of 0 (product 0, not positive). Together, equal roots need $b^2 - 4c = 0$, so $25 - 4c = 0$, $c = 6.25 > 0$ — a definite yes.
- (c). Bottom 50 numbers are each at least 30, giving 1,500. The median 50 forces the 51st number to be at least 50, so the top 50 give at least 2,500. Minimum total $= 1500 + 2500 = 4,000$, so "at least 4,000" is a definite yes. Alone, I gives only 3,000 and II leaves the bottom half unbounded.

ONE-PAGE REVISION CARD

THE FIVE OPTIONS

(a)	I alone works, II alone does not
(b)	II alone works, I alone does not
(c)	Neither alone; both together work
(d)	Each one alone works
(e)	Even both together fail

THE AD / BCE GRID — Test I alone. Works $\Rightarrow$ answer is (a) or (d); now test II. Fails $\Rightarrow$ answer is (b), (c) or (e); now test II, and combine only if II also fails.

WHAT “SUFFICIENT” MEANS

- Value question: exactly **one** number survives.
- Yes/no question: the answer is **always yes** or **always no**. A definite no is sufficient.
- Two survivors $\Rightarrow$ insufficient. Always hunt for a second case before you commit.

COUNTING EQUATIONS — k unknowns need k **independent** linear equations. $4x + 6y = 24$ is not new information after $2x + 3y = 12$. Non-linear pairs ($x + y = 7$, $xy = 12$) often give two solutions, so they are not automatically enough.

SHORTCUTS

1. Set the base to 100 whenever the answer is a percent or a ratio. Rupee values are then decoys.
2. For a yes/no, shrink the candidate list until every survivor gives the same verdict. You never have to identify the unknown.
3. Two points on a linear price rule (base fee + rate) give the rate by subtraction.
4. For “is the total above T ?”, build the **cheapest legal world** from the floors the statements give. If even that beats T , the answer is yes.
5. In repeat-until-one puzzles, find the parity or sum that every move leaves unchanged. The statement that does not touch it is the useless one.
6. $\text{km/h} \rightarrow \text{m/s} : \times \frac{5}{18}$. Divisible by a and $b \Rightarrow$ divisible by $\text{LCM}(a, b)$, not by ab .

TOP 5 TRAPS

1. **Solving instead of deciding.** You get no marks for the number. Stop at “one answer exists”.
2. **The repeated equation.** If II is I times a constant, the answer is (e), never (c).
3. **Leaking statement I into statement II.** Cover I with your finger while you test II.
4. **Forgetting the second root or the negative case.** $x^2 = 49$, $|x| = 3$, even roots, and “digits found” versus “number found” all hide a second answer.
5. **Calling a boundary result “cannot say”.** An average of exactly 4.0 answers “is it above 4.0?” with a firm no — that is sufficient.

Full-Length Mock Papers

Five timed papers. Five different enemies.

What you need to know

WHAT YOU NEED TO KNOW

The five papers in this chapter.

Paper	Profile	Questions	Time
1	Service company, speed-first (TCS NQT pattern)	15	18 min
2	Service company, reasoning-heavy (Accenture pattern)	15	20 min
3	SE-Asia business analytics (Grab / Shopee / DBS pattern)	12	25 min
4	Product company, insight-first (Google / Amazon pattern)	8	35 min
5	Mixed final, all tiers shuffled	20	30 min

Exam arithmetic you must be able to do in your head.

- Seconds per question = (minutes $\times$ 60) $\div$ questions.
- Accuracy = correct $\div$ attempted. Score rate = correct $\div$ total.
- With +1 for a correct answer and $-m$ for a wrong one, a guess is worth

$$p \times 1 - (1 - p) \times m.$$

It pays when $p > \frac{m}{1+m}$. So $m = 0.25$ needs $p > 0.2$, and $m = \frac{1}{3}$ needs $p > 0.25$.

- Blind guess on 4 options: $p = 0.25$. On 5 options: $p = 0.20$.
- Marks still needed = target – marks banked. Divide by the marks per question left.

The three-pass rule. Pass 1: take every question you can finish in under the average time. Pass 2: return to the ones you flagged. Pass 3: guess the rest if there is no negative marking, or only where you eliminated two options if there is.

High-yield formula recall — the things these papers test again and again.

Percent change	net factor = $(1 \pm \frac{a}{100})(1 \pm \frac{b}{100})$
Profit	profit% = $\frac{SP - CP}{CP} \times 100$; $SP = MP(1 - \frac{d}{100})$
SI / CI	$SI = \frac{Prt}{100}$; $A = P(1 + \frac{r}{100})^t$; 2-year gap = $P(\frac{r}{100})^2$
Speed	$km/h \times \frac{5}{18} = m/s$; average speed = $\frac{\text{total distance}}{\text{total time}}$
Trains	crossing a pole: own length; a platform: length + platform
Boats	still = $\frac{\text{down} + \text{up}}{2}$; stream = $\frac{\text{down} - \text{up}}{2}$
Work	rate = $\frac{1}{\text{days}}$; add rates; $M_1 D_1 = M_2 D_2$ for the same job
Averages	weighted mean = $\frac{n_1 a_1 + n_2 a_2}{n_1 + n_2}$
Alligation	cheap : dear = $(d - m) : (m - c)$
Counting	$nPr = \frac{n!}{(n-r)!}$; $nCr = \frac{n!}{r!(n-r)!}$; circle: $(n - 1)!$
Probability	$P = \frac{\text{favourable}}{\text{total}}$; $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Numbers	unit digit cycles in 4; zeros in $n! = \lfloor \frac{n}{5} \rfloor + \lfloor \frac{n}{25} \rfloor + \dots$
Clocks	angle = $ 30H - 5.5M $ degrees; hands meet every $\frac{720}{11}$ minutes
Mensuration	circle πr^2 ; cylinder $\pi r^2 h$; cone $\frac{\pi r^2 h}{3}$; sphere $\frac{4\pi r^3}{3}$

Warm-up

WARM-UP

build the reflex

Example 1

A paper has 30 questions and 40 minutes. How many seconds per question?

SOLUTION

40 minutes = $40 \times 60 = 2400$ seconds. $2400 \div 30 = 80$ seconds.

Answer: 80 seconds per question.

Example 2

Marking is +1 for correct and -0.25 for wrong. You attempt 40 and get 32 right. What is your score?

SOLUTION

Wrong = $40 - 32 = 8$. Penalty = $8 \times 0.25 = 2$. Score = $32 - 2 = 30$.

Answer: 30 marks.

Example 3

With +1 and -0.25 , what accuracy makes guessing worth it?

SOLUTION

Expected value of a guess = $p(1) - (1 - p)(0.25)$. Set it above zero: $p - 0.25 + 0.25p > 0 \Rightarrow 1.25p > 0.25 \Rightarrow p > 0.2$. So you need better than 1 in 5. A blind guess among 4 options gives $p = 0.25$, which is above 0.2.

Answer: Guess whenever your chance beats 20%; blind 4-option guessing already does.

Example 4

A section has 25 questions and no negative marking. The cut-off is 60%. How many must be correct?

SOLUTION

60% of 25 = $25 \times 0.6 = 15$.

Answer: 15 questions.

Example 5

You spent 14 minutes on 10 questions. At that pace, can you finish 35 questions in 45 minutes?

SOLUTION

Current pace = $14 \div 10 = 1.4$ minutes per question. In seconds that is $1.4 \times 60 = 84$ seconds per question. At that pace, 35 questions need $35 \times 1.4 = 49$ minutes. $49 > 45$, so no. Required pace = $45 \div 35 = 1.286$ minutes = $1.286 \times 60 \approx 77$ seconds.

Answer: No. You must speed up from 84 s to about 77 s per question.

Example 6

Three sections have 20, 15 and 25 questions, with 60 minutes in total. Split the time so every question gets the same number of seconds.

SOLUTION

Total questions = $20 + 15 + 25 = 60$. Time per question = $60 \div 60 = 1$ minute. So the sections get 20, 15 and 25 minutes.

Answer: 20 min, 15 min, 25 min.

Example 7

You attempted 45 questions and 36 were correct. What is your accuracy?

SOLUTION

Accuracy = $\frac{36}{45}$. Divide top and bottom by 9: $\frac{4}{5} = 0.8 = 80\%$.

Answer: 80%.

Example 8

A test is out of 100 marks. You have banked 46 marks from sections A and B. Your target is 70. Section C has 30 one-mark questions. What accuracy do you need in section C?

SOLUTION

Marks still needed = $70 - 46 = 24$. Accuracy needed = $24 \div 30 = 0.8 = 80\%$.

Answer: 80% of section C.

SHORTCUT

Do the time arithmetic before question 1, not during question 7. Write the clock time at which you must be one-third and two-thirds done in the corner of your sheet. Checking against a fixed target costs 2 seconds; recomputing your pace mid-paper costs 30.

Tier 1 — Service companies

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 9 [TCS NQT pattern]

Find the unit digit of 7^{103} .

SOLUTION

Powers of 7 end in a repeating block of four: $7^1 = 7$, $7^2 = 49$, $7^3 = 343$, $7^4 = 2401$. So the cycle is 7, 9, 3, 1. Divide the exponent by 4: $103 = 4 \times 25 + 3$, so the remainder is 3. Remainder 3 means the third entry of the cycle, which is 3.

Answer: 3

■ Remainder 0 means the **fourth** entry, not the first.

Example 10 [Accenture pattern]

A price rises by 25% and then falls by 20%. What is the net change?

SOLUTION

Take the price as 100. After the rise: $100 \times 1.25 = 125$. After the fall: $125 \times 0.80 = 100$. Net change = $100 - 100 = 0$.

Answer: No change (0%).

SHORTCUT

Chain the factors, never add the percents. A rise of $a\%$ then a fall of $b\%$ gives the factor $(1 + \frac{a}{100})(1 - \frac{b}{100})$. Memorise the pairs that cancel exactly: $+25\%$ with -20% , $+20\%$ with $-16\frac{2}{3}\%$, $+50\%$ with $-33\frac{1}{3}\%$, $+100\%$ with -50% .

Example 11 [Infosys pattern]

The ages of Arun and Bala are in the ratio 4 : 5. After 6 years the ratio becomes 6 : 7. Find their present ages.

SOLUTION

Let the ages be $4x$ and $5x$. After 6 years: $\frac{4x+6}{5x+6} = \frac{6}{7}$. Cross-multiply: $7(4x + 6) = 6(5x + 6)$. $28x + 42 = 30x + 36$. $42 - 36 = 30x - 28x \Rightarrow 6 = 2x \Rightarrow x = 3$. Ages = $4(3) = 12$ and $5(3) = 15$. Check: $(12 + 6) : (15 + 6) = 18 : 21 = 6 : 7$. Correct.

Answer: 12 years and 15 years.

Example 12 [TCS NQT pattern]

A train 240 m long crosses a 360 m bridge in 30 seconds. Find its speed in km/h.

SOLUTION

Distance covered = train length + bridge length = $240 + 360 = 600$ m. Speed = $600 \div 30 = 20$ m/s. Convert: $20 \times \frac{18}{5} = \frac{360}{5} = 72$ km/h.

Answer: 72 km/h

Example 13 [Wipro pattern]

A does a job in 15 days and B does it in 10 days. How long do they take together?

SOLUTION

Use LCM units. Take the job as $\text{LCM}(15, 10) = 30$ units. A's rate = $30 \div 15 = 2$ units per day. B's rate = $30 \div 10 = 3$ units per day. Together = $2 + 3 = 5$ units per day. Time = $30 \div 5 = 6$ days.

Answer: 6 days

SHORTCUT

Never use $\frac{1}{a} + \frac{1}{b}$ under time pressure. Set the job equal to the LCM of the days, read off whole-number rates, add, divide. No fractions appear anywhere.

Example 14 [Capgemini pattern]

Find the difference between the compound interest and the simple interest on Rs. 12,000 at 10% per annum for 2 years.

SOLUTION

For exactly 2 years the gap is $P(\frac{r}{100})^2$. $= 12000 \times (\frac{10}{100})^2 = 12000 \times 0.01 = 120$. Long check: $\text{SI} = \frac{12000 \times 10 \times 2}{100} = 2400$. $\text{CI} = 12000(1.1)^2 - 12000 = 12000 \times 1.21 - 12000 = 14520 - 12000 = 2520$. Difference = $2520 - 2400 = 120$. Same.

Answer: Rs. 120

Example 15 [Cognizant pattern]

How many distinct arrangements are there of the letters of the word BANANA?

SOLUTION

BANANA has 6 letters: A appears 3 times, N appears 2 times, B appears once. Arrangements = $\frac{6!}{3! \times 2! \times 1!}$. $6! = 720$. $3! = 6$, $2! = 2$, so the denominator is $6 \times 2 = 12$. $720 \div 12 = 60$.

Answer: 60

Example 16 [TCS NQT pattern]

Two fair dice are rolled. What is the probability that the sum is 9?

SOLUTION

Total outcomes = $6 \times 6 = 36$. Sum 9 comes from (3, 6), (4, 5), (5, 4), (6, 3) — that is 4 outcomes. $P = \frac{4}{36} = \frac{1}{9}$.

Answer: $\frac{1}{9}$

TRAP

Dice outcomes are ordered. (3, 6) and (6, 3) are two different outcomes, not one. Students who list only the unordered pairs {3, 6} and {4, 5} get $\frac{2}{36} = \frac{1}{18}$ and lose the mark. Count out of 36 always, and write both orders unless the two dice show the same number.

Example 17 [Accenture pattern]

Find the next term: 2, 5, 11, 23, 47, ?

SOLUTION

Test “double and add 1”: $2 \times 2 + 1 = 5$. $5 \times 2 + 1 = 11$. $11 \times 2 + 1 = 23$. $23 \times 2 + 1 = 47$. The rule holds every time, so the next term is $47 \times 2 + 1 = 95$.

Answer: 95

Example 18 [Infosys pattern]

If MANGO is coded as NBOHP, how is APPLE coded?

SOLUTION

Compare letter by letter: M→N, A→B, N→O, G→H, O→P. Every letter moves forward by one place. APPLE: A→B, P→Q, P→Q, L→M, E→F.

Answer: BQQMF

Example 19 [TCS NQT pattern]

Pointing at a photograph, Rahul said, “She is the daughter of the only son of my grandmother.” How is she related to Rahul?

SOLUTION

Work outwards from the far end. The only son of Rahul’s grandmother is Rahul’s father. The daughter of Rahul’s father is Rahul’s sister.

Answer: His sister.

TRAP

Read relation puzzles **from the inside out**, and do not add people the sentence never named. “The only son of my grandmother” is the father — but only because Rahul himself is that son’s child. If the speaker were female, the answer would be the same; if the phrase said “a son”, an uncle would also be possible.

Example 20 [Accenture pattern]

A shop's sales were Rs. 480 lakh in 2022 and Rs. 600 lakh in 2023. Its costs were Rs. 400 lakh and Rs. 480 lakh. (i) By what percent did sales grow? (ii) Did the profit margin on sales rise or fall?

SOLUTION

(i) Increase = $600 - 480 = 120$. Growth = $\frac{120}{480} \times 100$. Since $\frac{120}{480} = \frac{1}{4}$, growth = 25%.

(ii) 2022 profit = $480 - 400 = 80$; margin = $\frac{80}{480} = \frac{1}{6} = 16.67\%$. 2023 profit = $600 - 480 = 120$; margin = $\frac{120}{600} = \frac{1}{5} = 20\%$. $20\% > 16.67\%$, so the margin rose.

Answer: (i) 25% growth. (ii) The margin rose, from 16.67% to 20%.

PRACTICE · TIER 1 target: PAPER 1 · TCS NQT pattern · 15 questions in 18 minutes

Marking: +1 correct, -0.25 wrong. Sit this in one unbroken block. No calculator.

- What is the unit digit of 13^{47} ?
- If 35% of a number is 168, what is the number?
- A sum grows to Rs. 8,400 in 4 years at 5% simple interest. Find the principal.
- The average of five consecutive even numbers is 36. What is the largest of them?
- A shopkeeper buys an article for Rs. 250, marks it 40% above cost and sells it at a 10% discount. Find his profit percent.
- Two pipes fill a tank in 12 minutes and 18 minutes. How long do they take together?
- A boat covers 24 km downstream in 2 hours and returns in 3 hours. Find its speed in still water.
- In how many ways can 3 boys and 2 girls sit in a row so that the two girls are always together?
- Find the HCF of 84 and 126.
- Find the next term: 5, 11, 23, 47, ?
- If $A : B = 2 : 3$ and $B : C = 4 : 5$, find $A : C$.
- A can finish a job in 20 days and B in 30 days. They work together for 6 days. What fraction of the job is left?
- In a code, TIGER is written as UJHFS. How is LION written?
- A man walks 5 km north, turns right and walks 12 km. How far is he from his starting point?
- A town's population rises 10% in one year and falls 10% the next. It is now 99,000. What was it at the start?

ANSWER KEY

7. Only the unit digit 3 matters. Cycle of 3: 3, 9, 7, 1. $47 = 4 \times 11 + 3$, so take the 3rd entry = 7.
480. $0.35N = 168 \Rightarrow N = 168 \div 0.35 = 16800 \div 35 = 480$.
- Rs. 7,000. In 4 years SI adds $4 \times 5 = 20\%$, so $1.2P = 8400$ and $P = 8400 \div 1.2 = 7,000$.
40. Five consecutive terms have their average in the middle, so the middle number is 36 and the list is 32, 34, 36, 38, 40.
- 26%. $MP = 250 \times 1.4 = 350$. $SP = 350 \times 0.9 = 315$. Profit = $315 - 250 = 65$. $\frac{65}{250} = 0.26 = 26\%$.
- 7.2 minutes. Take the tank as $LCM(12, 18) = 36$ units: rates 3 and 2, total 5 per minute, so $36 \div 5 = 7.2$ min.
- 10 km/h. Downstream = $24 \div 2 = 12$; upstream = $24 \div 3 = 8$; still water = $\frac{12+8}{2} = 10$.
48. Tie the girls into one block: 4 items arrange in $4! = 24$ ways, and the girls swap inside in $2! = 2$ ways. $24 \times 2 = 48$.
42. $84 = 2^2 \times 3 \times 7$ and $126 = 2 \times 3^2 \times 7$. Common part = $2 \times 3 \times 7 = 42$.
95. The rule is "double and add 1": $5 \rightarrow 11 \rightarrow 23 \rightarrow 47 \rightarrow 95$.
- 8 : 15. Make B common: $A : B = 8 : 12$ and $B : C = 12 : 15$, so $A : B : C = 8 : 12 : 15$ and $A : C = 8 : 15$.
- $\frac{1}{2}$. Combined rate = $\frac{1}{20} + \frac{1}{30} = \frac{3}{60} + \frac{2}{60} = \frac{5}{60} = \frac{1}{12}$ per day. In 6 days they finish $\frac{6}{12} = \frac{1}{2}$, so half is left.
- MJPO. Every letter shifts forward one place: $T \rightarrow U, I \rightarrow J, G \rightarrow H, E \rightarrow F, R \rightarrow S$. So $L \rightarrow M, I \rightarrow J, O \rightarrow P, N \rightarrow O$.
- 13 km. North then right is east, so the two legs are perpendicular: $\sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.
- 1,00,000. Net factor = $1.1 \times 0.9 = 0.99$, so $0.99P = 99000$ and $P = 99000 \div 0.99 = 100,000$.

Paper 1 scoring. 13–15 correct: strong, you will clear any service-company cut-off. 10–12: safe, but your speed is the thing to train. 7–9: you know the topics and lose marks to arithmetic slips — redo every wrong one without looking at the key. Below 7: go back to the chapter of each topic you missed before sitting Paper 2.

PRACTICE · TIER 1 target: PAPER 2 · Accenture pattern · 15 questions in 20 minutes

Marking: +1 correct, no negative marking. Reasoning-heavy.

- Statements: All pens are tools. Some tools are sharp. Conclusions: I. Some pens are sharp. II. Some tools are pens. Which follows?
- Find the odd one out: 121, 144, 169, 180, 196.
- Complete the letter series: B, D, G, K, P, ?
- A is B's father. B is C's sister. C is D's mother. How is A related to D?
- One term is wrong: 4, 9, 19, 39, 79, 160, 319. Which one?
- A student scoring 30% fails by 40 marks. Another scoring 45% gets 35 marks more than the pass mark. Find the maximum marks.
- A and B together earn Rs. 6,000 for a job. A alone could do it in 10 days and B alone in 15 days. What is A's share?
- What is the angle between the hands of a clock at 4:20?
- 1 January 2024 was a Monday. What day was 1 March 2024?
- 40 litres of a mixture has milk and water in the ratio 3 : 1. How much water must be added to make the ratio 3 : 2?
- A bag has 4 red, 5 green and 3 blue balls. One ball is drawn. Find the probability that it is not green.
- Find the value of 15% of 240 + 25% of 160.
- A cube painted on all faces is cut into 64 identical small cubes. How many small cubes have exactly two painted faces?
- P, Q, R, S, T sit in a row facing you. Q is immediately to the right of P. R is at the extreme left end. S sits between T and Q. Who is at the extreme right end?
- The incomes of A and B are in the ratio 5 : 4 and their expenditures in the ratio 3 : 2. If each saves Rs. 6,000, find A's income.

ANSWER KEY

- Only II.** "All pens are tools" flips to "Some tools are pens", so II follows. The sharp part of "tools" may lie entirely outside the pens, so I does not follow.
- 180.** The others are perfect squares: $11^2, 12^2, 13^2, 14^2$. 180 is not a square.
- V.** Positions: B(2), D(4), G(7), K(11), P(16) — the gaps are +2, +3, +4, +5. The next gap is +6: $16 + 6 = 22 = V$.
- Grandfather.** B is C's sister, so C is also A's child. C is D's mother, so D is A's grandchild and A is D's grandfather.
- 160.** The rule is "double and add 1": 4, 9, 19, 39, 79, then $79 \times 2 + 1 = 159$, not 160. (And $159 \times 2 + 1 = 319$ confirms it.)
- 500.** Let the total be T . Pass mark = $0.30T + 40$ and also = $0.45T - 35$. So $0.15T = 75$ and $T = 500$. (Pass mark = 190.)
- Rs. 3,600.** Wages split as the rates: $\frac{1}{10} : \frac{1}{15} = 3 : 2$. A gets $6000 \times \frac{3}{5} = 3,600$.
- 10 degrees.** Angle = $|30H - 5.5M| = |30(4) - 5.5(20)| = |120 - 110| = 10$.
- Friday.** Days from 1 Jan to 1 Mar = $31 + 29 = 60$ (2024 is a leap year). $60 = 7 \times 8 + 4$, so move 4 days on from Monday: Friday.
- 10 litres.** Milk = $40 \times \frac{3}{4} = 30$, water = 10. Milk does not change, so for 3 : 2 the water must be $30 \times \frac{2}{3} = 20$. Add $20 - 10 = 10$ litres.
- $\frac{7}{12}$.** Total = $4 + 5 + 3 = 12$; not green = $4 + 3 = 7$.
- 76.** $0.15 \times 240 = 36$ and $0.25 \times 160 = 40$; $36 + 40 = 76$.
- 24.** $64 = 4^3$, so the cube is $4 \times 4 \times 4$. Exactly two faces are painted on the edge cubes that are not corners: $12 \text{ edges} \times (4 - 2) = 12 \times 2 = 24$.
- T.** "S between T and Q" means the block is T, S, Q or Q, S, T. The order T, S, Q is impossible: it puts S immediately left of Q, but that seat belongs to P. So the block is Q, S, T, and with P immediately left of Q it grows to P, Q, S, T. R takes the extreme left seat, so the row is R, P, Q, S, T and T is on the extreme right.
- Rs. 15,000.** Incomes $5x, 4x$; expenditures $3y, 2y$. Then $5x - 3y = 6000$ and $4x - 2y = 6000$. The second gives $2x - y = 3000$, so $y = 2x - 3000$. Substituting: $5x - 6x + 9000 = 6000 \Rightarrow x = 3000$. A's income = $5(3000) = 15,000$.

Paper 2 scoring. There is no negative marking, so an unattempted question is a pure loss — mark something on all 15. 12–15: interview-ready on reasoning. 9–11: solid; drill clocks, calendars and seating. 6–8: your quant is carrying you and your reasoning is not — chapters 13 to 20. Below 6: rebuild from chapter 17 (syllogisms) upward.

Tier 2 — Singapore & Thailand

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 21 [Grab · pattern]

A ride costs a base fare of S\$3.20 plus S\$0.85 per km. During surge, the whole fare is multiplied by 1.4. What does a 12 km surge ride cost?

SOLUTION

Distance charge = 12×0.85 . $12 \times 0.85 = 12 \times 85 \div 100 = 1020 \div 100 = 10.20$. Normal fare = $3.20 + 10.20 = 13.40$. Surge fare = $13.40 \times 1.4 = 13.40 + 13.40 \times 0.4 = 13.40 + 5.36 = 18.76$.

Answer: S\$18.76

Example 22 [Shopee · pattern]

A seller lists an item at S\$45. The platform takes a 6% commission and a 2% payment fee, both on the selling price. The seller also pays S\$3 of shipping. The item cost the seller S\$28. What is the profit, and what is the margin on cost?

SOLUTION

Total percentage fees = $6\% + 2\% = 8\%$ of 45. 8% of 45 = $45 \times 0.08 = 3.60$. Net receipt = $45 - 3.60 - 3.00 = 38.40$. Profit = $38.40 - 28 = 10.40$. Margin on cost = $\frac{10.40}{28} \times 100$. $10.40 \div 28 = 0.3714\dots$, so about 37.14%.

Answer: Profit S\$10.40; margin on cost about 37.1%.

TRAP

Percentage fees sit on the selling price; the margin sits on the cost. Mixing the two bases is the single most common error in marketplace questions. Write the base next to every percent before you multiply.

Example 23 [DBS · pattern]

A deposit of S\$20,000 earns 3% per annum, compounded annually, for 3 years. What is the final amount?

SOLUTION

$A = 20000 \times (1.03)^3$. $(1.03)^2 = 1.03 \times 1.03 = 1.0609$. $(1.03)^3 = 1.0609 \times 1.03 = 1.0609 + 0.031827 = 1.092727$. $A = 20000 \times 1.092727 = 21,854.54$.

Answer: About S\$21,854.54

Example 24 [Agoda · pattern]

A hotel has 120 rooms. On the five weeknights it runs 85% occupancy at S\$180 a room; on the two weekend nights it runs 100% at S\$240 a room. Find the weekly room revenue.

SOLUTION

Weeknight rooms sold = $120 \times 0.85 = 102$. Weeknight revenue for one night = 102×180 . $102 \times 180 = 100 \times 180 + 2 \times 180 = 18000 + 360 = 18,360$. Five nights = $18360 \times 5 = 91,800$. Weekend revenue for one night = $120 \times 240 = 28,800$. Two nights = $28800 \times 2 = 57,600$. Total = $91800 + 57600 = 149,400$.

Answer: S\$149,400

Example 25 [SCB · pattern]

A trader converts S\$10,000 into baht at 26.4, invests it in Thailand for 6 months at 2.5% per annum simple interest, then converts back at 26.9. What is the gain or loss in Singapore dollars?

SOLUTION

Baht received = $10000 \times 26.4 = 264,000$. Interest for 6 months = $264000 \times \frac{2.5}{100} \times \frac{1}{2}$. $264000 \times 0.025 = 6,600$ for a full year, so half a year gives $6600 \div 2 = 3,300$. Total baht = $264000 + 3300 = 267,300$. Convert back: $267300 \div 26.9$. $26.9 \times 9,900 = 266,310$. Remainder = $267300 - 266310 = 990$. $990 \div 26.9 = 36.8$. So he gets back $9900 + 36.8 = \text{S\$}9,936.80$. Change = $9936.80 - 10000 = -63.20$.

Answer: A loss of about S\$63.20.

He earned 1.25% of interest but paid about 1.86% in exchange rate movement (26.9 baht per dollar buys back less than 26.4 did). Interest gains can be erased by the rate.

Example 26 [Sea/Shopee · pattern]

Each order earns a gross margin of S\$4.50. Acquiring a customer costs S\$18. (i) How many orders must one customer place before the company breaks even on them? (ii) If only 35% of customers ever place a second order (and nobody places a third), what is the average gross margin per acquired customer?

SOLUTION

(i) Break-even orders = $18 \div 4.50 = 4$.

(ii) Take 100 acquired customers. All 100 place a first order, and 35% of them, that is 35 customers, place a second. Nobody places a third. Total orders = $100 + 35 = 135$ from 100 customers. Average orders per customer = $135 \div 100 = 1.35$. Average gross margin = 1.35×4.50 . $1.35 \times 4.5 = 1.35 \times 4 + 1.35 \times 0.5 = 5.40 + 0.675 = 6.075$. Against a cost of 18, the company loses $18 - 6.075 = 11.925$ per customer.

Answer: (i) 4 orders. (ii) S\$6.075 per customer — a loss of about S\$11.93 each.

Example 27 [LINE MAN · pattern]

A rider averages 18 minutes per delivery. He works a 7-hour shift with a 30-minute break. How many deliveries does he complete?

SOLUTION

Working time = 7 hours – 30 minutes = 6.5 hours = $6.5 \times 60 = 390$ minutes. Deliveries = $390 \div 18 = 21.67$. He cannot finish a part delivery, so take the whole number below.

Answer: 21 deliveries.

TRAP

Round the way the situation rounds, not the way arithmetic rounds. Deliveries, buses, workers and packets round **down**; boxes needed, trips needed and staff required round **up**. 21.67 deliveries done is 21; 21.67 boxes needed is 22.

Example 28 [GIC · pattern]

A portfolio holds 60% equities, 30% bonds and 10% cash. Over the year equities returned 9%, bonds 4% and cash 1%. What was the portfolio return?

SOLUTION

Weighted return = $0.60(9) + 0.30(4) + 0.10(1)$. $0.60 \times 9 = 5.4$. $0.30 \times 4 = 1.2$. $0.10 \times 1 = 0.1$. Total = $5.4 + 1.2 + 0.1 = 6.7$.

Answer: 6.7%

Example 29 [Razer · pattern]

A product sells for S\$60 and costs S\$34 per unit to make. Fixed costs are S\$180,000. How many units must be sold to break even?

SOLUTION

Contribution per unit = $60 - 34 = 26$. Units = $180000 \div 26$. $26 \times 6,000 = 156,000$. Remainder = $180000 - 156000 = 24,000$. $26 \times 900 = 23,400$. Remainder = $24000 - 23400 = 600$. $26 \times 23 = 598$. Remainder 2. So $180000 \div 26 = 6,923.08$ approximately. You cannot sell part of a unit and you must **cover** the fixed cost, so round up.

Answer: 6,924 units.

PRACTICE · TIER 2 target: PAPER 3 · SE-Asia pattern · 12 questions in 25 minutes

No options. Write the number. Marking: +2 correct, no negative marking.

1. A ride costs THB 40 base + THB 6.50 per km + THB 2 per minute. A 9 km trip takes 22 minutes. What is the fare?
2. A shop's annual revenue rose from S\$84,000 to S\$113,400. What was the growth percent?
3. An invoice is USD 4,800. The bank quotes 1 USD = 1.35 SGD and adds a 0.8% margin against the customer. How many Singapore dollars are payable?
4. A warehouse holds 4,500 items. 8% are damaged. Of the rest, 25% are customer returns that cannot be resold. How many items are sellable?
5. Courier A handles 15 parcels per hour and courier B handles 12. Working together, how long do they take to clear 216 parcels?
6. A hotel room lists at THB 3,200 a night. A member discount of 15% applies, then a promo code takes 10% off, and 7% VAT is added at the end. What is the final price?
7. Six workers finish a task in 5 days. How long would 10 workers take?
8. At a sorting hub each parcel independently goes to lane 1 with probability 0.5, lane 2 with 0.3 and lane 3 with 0.2. Two parcels arrive. What is the probability they go to the same lane?
9. A fleet of 40 cars averages 11 km per litre. Ten new cars average 16 km per litre. What is the average of all 50 cars?
10. A subscription costs S\$12 monthly or S\$120 for a year paid upfront. What percent does the annual plan save?
11. A food stall pays THB 30,000 rent a month, spends THB 45 of ingredients per dish and sells each dish at THB 120. How many dishes a month does it need to break even?
12. An account grows 2% every quarter. What is the effective annual growth, to two decimal places?

ANSWER KEY

1. **THB 142.50.** Distance = $9 \times 6.5 = 58.50$. Time = $22 \times 2 = 44$. Fare = $40 + 58.50 + 44 = 142.50$.
2. **35%.** Increase = $113400 - 84000 = 29,400$. $\frac{29400}{84000} = 0.35$, so 35%.
3. **S\$6,531.84.** $4800 \times 1.35 = 6,480$. Margin against the customer means he pays 0.8% more: $6480 \times 1.008 = 6480 + 51.84 = 6,531.84$.
4. **3,105.** Damaged = $4500 \times 0.08 = 360$, leaving $4500 - 360 = 4,140$. Returns = $4140 \times 0.25 = 1,035$. Sellable = $4140 - 1035 = 3,105$.
5. **8 hours.** Combined rate = $15 + 12 = 27$ parcels per hour; $216 \div 27 = 8$.
6. **THB 2,619.36.** $3200 \times 0.85 = 2,720$. $2720 \times 0.90 = 2,448$. $2448 \times 1.07 = 2448 + 171.36 = 2,619.36$.
7. **3 days.** The job is $6 \times 5 = 30$ worker-days; $30 \div 10 = 3$.
8. **0.38.** Same lane = $0.5^2 + 0.3^2 + 0.2^2 = 0.25 + 0.09 + 0.04 = 0.38$.
9. **12 km per litre.** This is a weighted average of the **rates as given**: $\frac{40 \times 11 + 10 \times 16}{50} = \frac{440 + 160}{50} = \frac{600}{50} = 12$.
10. **16.67%.** Monthly for a year = $12 \times 12 = 144$. Saving = $144 - 120 = 24$. $\frac{24}{144} = \frac{1}{6} = 16.67\%$.

11. **400 dishes.** Contribution = $120 - 45 = 75$ per dish; $30000 \div 75 = 400$.
 12. **8.24%.** $(1.02)^2 = 1.0404$; $(1.0404)^2 = 1.08243216$. So the growth is $8.243...\% \approx 8.24\%$.

Paper 3 scoring, out of 24. 20–24: you are ready for a Grab or Shopee analytics screen. 14–19: the method is there and the decimals are not — redo every question writing each multiplication on its own line. Below 14: you are losing marks on multi-step chains, not on topics. Practise chaining factors (chapter 2) and weighted averages (chapter 4).

Tier 3 — Product companies

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 30 [Google · pattern]

How many four-digit numbers have all four digits different?

SOLUTION

Insight: fill the most restricted position first. The leading digit is the restricted one, because 0 is banned there.

First digit: 1 to 9, so 9 choices. Second digit: any of the 10 digits except the one already used — 9 choices (0 is allowed here). Third digit: 8 choices remain. Fourth digit: 7 choices remain. Total = $9 \times 9 \times 8 \times 7$. $9 \times 9 = 81$. $81 \times 8 = 648$. $648 \times 7 = 4,536$.

Answer: 4,536

Example 31 [Amazon · pattern]

You roll a fair die. You may keep the number shown, or reroll once and must keep the second roll. Playing best, what is your expected score?

SOLUTION

Insight: reroll exactly when the number in front of you is below the value of a fresh roll. A fresh roll is worth 3.5, so keep 4, 5, 6 and reroll 1, 2, 3.

A fresh roll has expectation $\frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$. Case A — the first roll is 4, 5 or 6 (probability $\frac{3}{6} = \frac{1}{2}$). You keep it, and its average value is $\frac{4+5+6}{3} = \frac{15}{3} = 5$. Case B — the first roll is 1, 2 or 3 (probability $\frac{1}{2}$). You reroll, and the reroll is worth 3.5. Expected score = $\frac{1}{2} \times 5 + \frac{1}{2} \times 3.5 = 2.5 + 1.75 = 4.25$.

Answer: 4.25

Check the rule itself: keeping a 3 would give 3, rerolling gives 3.5, so rerolling a 3 is right. Keeping a 4 gives 4, better than 3.5, so keeping is right. The cut is exactly where the insight said.

Example 32 [Goldman Sachs · pattern]

At a meetup of 25 people, can every person shake hands with exactly 3 others?

SOLUTION

Insight: every handshake adds 1 to two people's counts, so the total of all counts is always even — it is twice the number of handshakes.

If each of the 25 people shakes exactly 3 hands, the total of the counts is $25 \times 3 = 75$. But that total must equal $2 \times$ (number of handshakes), which is even. 75 is odd, so no such arrangement exists.

Answer: No. 75 is odd, and the sum of handshake counts must be even.

SHORTCUT

The handshake parity rule. In any group, the number of people who shake an odd number of hands is itself even. Whenever a puzzle asks “can every one of n people do exactly k ...”, check whether nk is even first. It settles many questions in five seconds.

Example 33 [Microsoft · pattern]

What is the highest power of 12 that divides $100!$?

SOLUTION

Insight: $12 = 2^2 \times 3$, so count the 2s and the 3s in $100!$ separately, then see how many complete bundles of (two 2s and one 3) you can build.

Count the 2s with repeated division: $\lfloor \frac{100}{2} \rfloor = 50$, $\lfloor \frac{100}{4} \rfloor = 25$, $\lfloor \frac{100}{8} \rfloor = 12$, $\lfloor \frac{100}{16} \rfloor = 6$, $\lfloor \frac{100}{32} \rfloor = 3$, $\lfloor \frac{100}{64} \rfloor = 1$, and $128 > 100$ stops it. Total 2s = $50 + 25 + 12 + 6 + 3 + 1 = 97$.

Count the 3s: $\lfloor \frac{100}{3} \rfloor = 33$, $\lfloor \frac{100}{9} \rfloor = 11$, $\lfloor \frac{100}{27} \rfloor = 3$, $\lfloor \frac{100}{81} \rfloor = 1$, and $243 > 100$ stops it. Total 3s = $33 + 11 + 3 + 1 = 48$.

Each 12 needs two 2s, so the 2s allow $\lfloor \frac{97}{2} \rfloor = 48$ of them. Each 12 needs one 3, so the 3s allow 48 of them. The smaller limit wins: $\min(48, 48) = 48$.

Answer: 12^{48}

Example 34 [D. E. Shaw · pattern]

Five people are in a room. Assuming birth months are equally likely and independent, what is the probability that at least two share a birth month?

SOLUTION

Insight: “at least two share” is hard to count directly. Count the opposite — all different — and subtract from 1.

Total possibilities = 12^5 . $12^2 = 144$, $12^3 = 1728$, $12^4 = 20736$, $12^5 = 248,832$.

All different: the first person has 12 choices, the second 11, then 10, 9, 8. $12 \times 11 = 132$. $132 \times 10 = 1320$. $1320 \times 9 = 11,880$. $11880 \times 8 = 95,040$.

$P(\text{all different}) = \frac{95040}{248832}$. Divide both by 48: $95040 \div 48 = 1980$ and $248832 \div 48 = 5184$, so $\frac{1980}{5184}$. Divide both by 12: $\frac{165}{432}$. Divide both by 3: $\frac{55}{144} = 0.3819$.

$P(\text{at least two share}) = 1 - 0.3819 = 0.6181$.

Answer: $\frac{89}{144}$, about 61.8%.

| Check the fraction: $1 - \frac{55}{144} = \frac{89}{144} = 0.6181$. Correct.

Example 35 [Uber · pattern]

A driver is paid S\$28 for every hour of his shift. Across the shift, 70% of his time is spent carrying passengers at an average 25 km/h and 30% is spent driving empty at 20 km/h. Running the car costs S\$0.35 per km. What is his net pay per hour?

SOLUTION

Insight: pay is per hour but cost is per km, so convert the hour into kilometres before comparing anything.

Take one hour of the shift. Carrying: 0.7 hours at 25 km/h = $0.7 \times 25 = 17.5$ km. Empty: 0.3 hours at 20 km/h = $0.3 \times 20 = 6$ km. Total distance per hour = $17.5 + 6 = 23.5$ km. Running cost = 23.5×0.35 . $23.5 \times 0.35 = 23.5 \times 35 \div 100 = 822.5 \div 100 = 8.225$. Net = $28 - 8.225 = 19.775$.

Answer: About S\$19.78 per hour.

| Notice how much the 30% empty time costs him: it adds 6 km, that is S\$2.10 an hour, with no fare attached. Cutting empty time is worth more than raising the fare.

Example 36 [Adobe · pattern]

A drawer holds socks of 4 colours, plenty of each. How many socks must you pull out in the dark to be **sure** of getting 3 of one colour?

SOLUTION

Insight: “be sure” means beat the worst possible luck. Build the unluckiest draw that still fails, then add one.

The unluckiest run that still has no colour appearing 3 times is 2 socks of each colour: $2 \times 4 = 8$ socks. At 8 socks it is still possible to have failed. The next sock must match one of the four colours, making a third of that colour. So $8 + 1 = 9$.

Answer: 9 socks.

■ The general rule: to guarantee k of one kind from c kinds, take $c(k - 1) + 1$. Here $4(3 - 1) + 1 = 9$.

PRACTICE · TIER 3 target: PAPER 4 · product-company pattern · 8 questions in 35 minutes

Show your reasoning. Marking: +3 correct, +1 for a correct method with an arithmetic slip, no negative marking.

1. How many integers from 1 to 1000 inclusive are divisible by neither 3 nor 5?
2. A fair coin is tossed until two heads appear in a row. What is the expected number of tosses?
3. Eight people sit at random around a round table. What is the probability that two particular people sit next to each other?
4. The numbers 1 to 50 are written on a board. Repeatedly, two numbers are erased and their positive difference is written back. This continues until one number is left. Can that last number be 0?
5. Thirty items in a shop have an average price of Rs. 400, and no item costs less than Rs. 250. What is the highest price the costliest item can possibly have?
6. In how many ways can 10 identical chocolates be given to 4 children so that each child gets at least one?
7. A city has 6 million people. Suppose 20% of them ever use taxis, and those who do average 0.3 taxi rides a day. If one taxi completes 20 rides a day, estimate the fleet size the city needs.
8. A screening test flags 98% of people who have a condition and wrongly flags 5% of people who do not. 1% of the population has the condition. A random person tests positive. What is the chance they actually have it?

ANSWER KEY

1. **533.** Divisible by 3: $\lfloor \frac{1000}{3} \rfloor = 333$. By 5: $\lfloor \frac{1000}{5} \rfloor = 200$. By both, i.e. by 15: $\lfloor \frac{1000}{15} \rfloor = 66$. By 3 or 5 = $333 + 200 - 66 = 467$. Neither = $1000 - 467 = 533$.
2. **6 tosses.** Let E_0 be the expectation from a fresh start and E_1 the expectation when the last toss was a head. $E_1 = 1 + \frac{1}{2}(0) + \frac{1}{2}E_0$ and $E_0 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0$. From the second, $\frac{1}{2}E_0 = 1 + \frac{1}{2}E_1$, so $E_0 = 2 + E_1$. Substituting into the first: $E_1 = 1 + \frac{1}{2}(2 + E_1)$, so $E_1 = 2 + \frac{1}{2}E_1$, giving $E_1 = 4$ and $E_0 = 6$.
3. $\frac{2}{7}$. Seat the first of the pair anywhere. The second person is equally likely to take any of the remaining 7 seats, and 2 of those 7 are next to the first. So $\frac{2}{7}$.
4. **No.** Replacing a and b by $|a - b|$ changes the total by $a + b - |a - b|$, which is $2 \times \min(a, b)$ — always even. So the parity of the total never changes. The starting total is $1 + 2 + \dots + 50 = (50)\frac{51}{2} = 1275$, which is odd. The final single number therefore must be odd, and 0 is even.
5. **Rs. 4,750.** Total = $30 \times 400 = 12,000$. To push one item as high as possible, hold the other 29 at the floor of 250: $29 \times 250 = 7,250$. Maximum = $12000 - 7250 = 4,750$.
6. **84.** Give each child one chocolate first; 6 remain to be split freely among 4 children. That is $\binom{6+4-1}{4-1} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = \frac{504}{6} = 84$.
7. **About 18,000 taxis.** Taxi users = $6,000,000 \times 0.20 = 1,200,000$. Rides per day = $1200000 \times 0.3 = 360,000$. Taxis = $360000 \div 20 = 18,000$. State the assumptions out loud — an estimation answer is judged on the chain, not the digits.

8. **About 16.5%.** Out of 10,000 people: 100 have it, and 98 of those test positive. 9,900 do not have it, and $9900 \times 0.05 = 495$ of those test positive. Total positives = $98 + 495 = 593$. True positives = $\frac{98}{593} = 0.1653$, about 16.5%.

TRAP

A very accurate test on a rare condition still gives mostly false alarms. The 5% error rate acts on the huge healthy group, so it produces 495 false positives against only 98 true ones. Always convert the percentages into counts out of 10,000 before judging.

Paper 4 scoring, out of 24. 18–24: you can hold your own in a product-company round. 12–17: your reasoning is sound and your first move is slow — for each question you missed, write down what the single insight was, in one line. Below 12: do not add speed yet. Redo Tier 3 of chapters 9, 10 and 20 and come back.

Mixed set — exam conditions

PRACTICE · TIER 1 target: PAPER 5 · mixed final · 20 questions in 30 minutes

All tiers, shuffled, no labels. Marking: +1 correct, −0.25 wrong. Sit it once, without stopping, and mark it strictly.

- Find 12.5% of 480.
- A train 180 m long passes a 270 m platform in 25 seconds. Find its speed in km/h.
- Two numbers have HCF 12 and LCM 180. One of them is 60. Find the other.
- Find the simple interest on Rs. 15,000 at 8% per annum for 2.5 years.
- A sells an article to B at a 20% profit, and B sells it to C at a 25% profit. C pays Rs. 900. What did A pay for it?
- Find the next term: 7, 14, 28, 56, ?
- Find the odd one out: 3, 5, 11, 14, 17, 23.
- In a code, FLOWER is written as REWOLF. How is GARDEN written?
- Statements: All doctors are graduates. No graduate is illiterate. Does “No doctor is illiterate” follow?
- At what time between 3 and 4 o’clock do the hands of a clock coincide?
- Two cards are drawn from a standard 52-card pack without replacement. What is the probability that both are aces?
- The average of 11 results is 50. The average of the first six is 49 and of the last six is 52. Find the sixth result.
- In what ratio must rice at Rs. 45 per kg be mixed with rice at Rs. 60 per kg to get a mixture worth Rs. 50 per kg?
- Find the area of the largest circle that fits inside a square of side 14 cm. (Take $\pi = \frac{22}{7}$.)
- How many 3-digit numbers can be formed from the digits 1 to 7 if no digit repeats?
- Data sufficiency.** Is the integer n divisible by 4? I: n is divisible by 8. II: n is even. Options: (a) I alone (b) II alone (c) both together (d) each alone (e) neither
- A man walks 10 m south, turns left and walks 10 m, then turns left and walks 10 m. Where is he now, and which way is he facing?
- Twelve men can finish a job in 18 days. After 6 days of work, 6 more men join them. How many more days does the job take?
- A’s salary is 25% more than B’s. By what percent is B’s salary less than A’s?
- Find the remainder when 2^{50} is divided by 7.

ANSWER KEY

- 60.** $12.5\% = \frac{1}{8}$, and $480 \div 8 = 60$.
- 64.8 km/h.** Distance = $180 + 270 = 450$ m in 25 s, so $450 \div 25 = 18$ m/s. $18 \times \frac{18}{5} = \frac{324}{5} = 64.8$ km/h.

3. **36.** For two numbers, $\text{HCF} \times \text{LCM} = \text{their product}$: $12 \times 180 = 2,160$. Other number $= 2160 \div 60 = 36$. (Check: $\text{HCF}(60,36) = 12$ and $\text{LCM} = 180$.)
4. **Rs. 3,000.** $\text{SI} = \frac{15000 \times 8 \times 2.5}{100} = \frac{15000 \times 20}{100} = 3,000$.
5. **Rs. 600.** The chain factor is $1.20 \times 1.25 = 1.5$, so A's cost $= 900 \div 1.5 = 600$. (Check: $600 \rightarrow 720 \rightarrow 900$.)
6. **112.** Each term doubles: 7, 14, 28, 56, 112.
7. **14.** Every other number is an odd prime; 14 is even and composite.
8. **NEDRAG.** The code simply reverses the word: GARDEN reversed is NEDRAG.
9. **Yes, it follows.** Doctors sit inside graduates, and graduates are entirely outside illiterates, so doctors are entirely outside illiterates.
10. **3 : 16 $\frac{4}{11}$.** At 3:00 the minute hand is 90 degrees behind the hour hand. The minute hand gains 5.5 degrees a minute, so it needs $90 \div 5.5 = \frac{180}{11} = 16\frac{4}{11}$ minutes.
11. **$\frac{1}{221}$.** $P = \frac{4}{52} \times \frac{3}{51} = \left(\frac{1}{13}\right) \times \left(\frac{1}{17}\right) = \frac{1}{221}$.
12. **56.** First six total $= 6 \times 49 = 294$. Last six total $= 6 \times 52 = 312$. All eleven total $= 11 \times 50 = 550$. The sixth result is counted in both groups, so it is $294 + 312 - 550 = 56$.
13. **2 : 1.** By alligation, cheap : dear $= (60 - 50) : (50 - 45) = 10 : 5 = 2 : 1$.
14. **154 cm².** The largest circle has diameter 14, so $r = 7$. Area $= \frac{22}{7} \times 7 \times 7 = 22 \times 7 = 154$.
15. **210.** $7 \times 6 \times 5 = 210$.
16. **(a).** $n = 8k = 4(2k)$, always a multiple of 4 — a definite yes. Statement II fails, because $n = 6$ is even and not a multiple of 4.
17. **10 m east of the start, facing north.** Walking south then turning left faces him east; 10 m east; turning left again faces him north; 10 m north cancels the 10 m south. Net displacement is 10 m east.
18. **8 days.** The job is $12 \times 18 = 216$ man-days. In 6 days, $12 \times 6 = 72$ man-days are done, leaving $216 - 72 = 144$. With $12 + 6 = 18$ men, $144 \div 18 = 8$ days.
19. **20%.** Let $B = 100$, so $A = 125$. B is short of A by 25, and that is measured against A: $\frac{25}{125} = \frac{1}{5} = 20\%$.
20. **4.** $2^3 = 8$ leaves remainder 1 on division by 7. Write $50 = 3 \times 16 + 2$, so $2^{50} = (2^3)^{16} \times 2^2$, which leaves $1^{16} \times 4 = 4$.

Paper 5 scoring, out of 20 with —0.25 per wrong answer.

16 and above	Placement-ready across all three tiers. Keep sitting timed papers weekly.
12 to 15.75	A good service-company score. Your gap is the last five questions, which are Tier 2 and Tier 3 — work those chapters' Tier 3 sections.
8 to 11.75	You know more than this score shows. Count your wrong answers: if more than 4, you are guessing too loosely for —0.25 marking.
Below 8	Stop timing yourself for two weeks. Work untimed until your accuracy is above 85%, then bring the clock back.

Also compute two numbers every time: **accuracy** (correct $\div$ attempted) and **coverage** (attempted $\div$ 20). Low coverage with high accuracy is a speed problem. High coverage with low accuracy is a discipline problem. They are fixed in opposite ways.

ONE-PAGE REVISION CARD

THE FIVE PAPERS — Paper 1 speed (15 Q / 18 min), Paper 2 reasoning (15 Q / 20 min), Paper 3 business analytics (12 Q / 25 min), Paper 4 insight (8 Q / 35 min), Paper 5 mixed (20 Q / 30 min).

EXAM ARITHMETIC

- Seconds per question = (minutes $\times$ 60) $\div$ questions.
- A guess pays when your chance beats $\frac{m}{1+m}$: with -0.25 that is 20%, with $-\frac{1}{3}$ it is 25%. With no negative marking, never leave a blank.
- Marks needed = target — banked, then divide by marks per remaining question.
- Accuracy = correct $\div$ attempted. Coverage = attempted $\div$ total. Diagnose with both.

THE THREE-PASS RULE — Pass 1 take the quick ones, pass 2 the flagged ones, pass 3 guess what is left under the rule above. Never let one question eat more than triple the average time.

FORMULAS THESE PAPERS KEEP ASKING FOR

Percent chain	factor = $(1 \pm \frac{a}{100})(1 \pm \frac{b}{100})$; +25% cancels -20%
Profit	SP = MP $(1 - \frac{d}{100})$; profit% on CP, discount% on MP, fees% on SP
SI / CI	SI = $\frac{Prt}{100}$; 2-year CI—SI gap = $P(\frac{r}{100})^2$
Speed	km/h $\times \frac{5}{18}$ = m/s; platform = train length + platform length
Boats	still = $\frac{\text{down}+\text{up}}{2}$; stream = $\frac{\text{down}-\text{up}}{2}$
Work	job = LCM of days; add the whole-number rates
Alligation	cheap : dear = $(d - m) : (m - c)$
Counting	nCr ; identical items to k people, each ≥ 1 : $\binom{n-1}{k-1}$
Probability	count the complement when the question says “at least”
Clock	angle = $ 30H - 5.5M $; the hands gain 5.5 degrees a minute
Numbers	unit digit cycles in 4; HCF $\times$ LCM = product of the two numbers

SHORTCUTS

1. Chain percentage factors; never add percents.
2. Set a job equal to the LCM of the given days so every rate is a whole number.
3. Fill the most restricted slot first in any counting problem.
4. In “at least two share”, compute the complement and subtract from 1.
5. Handshake parity: the sum of all counts is even, so nk odd is impossible.
6. Pigeonhole: to guarantee k of one kind out of c kinds, take $c(k - 1) + 1$.
7. Turn per-hour pay and per-km cost into the same unit before comparing them.

TOP 5 TRAPS

1. **Adding percents.** +25% then -20% is not +5%; it is zero.
2. **Mixing the base.** Profit is on cost, discount is on marked price, platform fees are on selling price. Write the base before you multiply.
3. **Rounding the wrong way.** Deliveries completed round down; units needed to cover a cost round up.
4. **Guessing loosely under negative marking.** With -0.25 , a blind guess is barely positive and a guess you feel bad about is negative. Eliminate one option first.
5. **Spending 4 minutes on one Tier 3 question inside a Tier 1 paper.** The marks are equal. Flag it and move.

Speed Arithmetic

Know these cold. There is no calculator on test day.

A.1 • Multiplication tables, 2 to 25

×	1	2	3	4	5	6	7	8	9	10	11	12
2	2	4	6	8	10	12	14	16	18	20	22	24
3	3	6	9	12	15	18	21	24	27	30	33	36
4	4	8	12	16	20	24	28	32	36	40	44	48
5	5	10	15	20	25	30	35	40	45	50	55	60
6	6	12	18	24	30	36	42	48	54	60	66	72
7	7	14	21	28	35	42	49	56	63	70	77	84
8	8	16	24	32	40	48	56	64	72	80	88	96
9	9	18	27	36	45	54	63	72	81	90	99	108
10	10	20	30	40	50	60	70	80	90	100	110	120
11	11	22	33	44	55	66	77	88	99	110	121	132
12	12	24	36	48	60	72	84	96	108	120	132	144
13	13	26	39	52	65	78	91	104	117	130	143	156
14	14	28	42	56	70	84	98	112	126	140	154	168
15	15	30	45	60	75	90	105	120	135	150	165	180
16	16	32	48	64	80	96	112	128	144	160	176	192
17	17	34	51	68	85	102	119	136	153	170	187	204
18	18	36	54	72	90	108	126	144	162	180	198	216
19	19	38	57	76	95	114	133	152	171	190	209	228
20	20	40	60	80	100	120	140	160	180	200	220	240
21	21	42	63	84	105	126	147	168	189	210	231	252
22	22	44	66	88	110	132	154	176	198	220	242	264
23	23	46	69	92	115	138	161	184	207	230	253	276
24	24	48	72	96	120	144	168	192	216	240	264	288
25	25	50	75	100	125	150	175	200	225	250	275	300

A.2 • Squares, 1 to 40

<i>n</i>	1	2	3	4	5	6	7	8	9	10
<i>n</i> ²	1	4	9	16	25	36	49	64	81	100
<i>n</i>	11	12	13	14	15	16	17	18	19	20
<i>n</i> ²	121	144	169	196	225	256	289	324	361	400
<i>n</i>	21	22	23	24	25	26	27	28	29	30
<i>n</i> ²	441	484	529	576	625	676	729	784	841	900
<i>n</i>	31	32	33	34	35	36	37	38	39	40
<i>n</i> ²	961	1024	1089	1156	1225	1296	1369	1444	1521	1600

Squares ending in 5 are instant: $35^2 = (3 \times 4) \mid 25 = 1225$. $85^2 = (8 \times 9) \mid 25 = 7225$. $115^2 = (11 \times 12) \mid 25 = 13225$.

A.3 • Cubes, 1 to 20

<i>n</i>	1	2	3	4	5	6	7	8	9	10
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n^3	1	8	27	64	125	216	343	512	729	1000
n	11	12	13	14	15	16	17	18	19	20
n^3	1331	1728	2197	2744	3375	4096	4913	5832	6859	8000

Unit digit of a cube gives back the unit digit of the number for 0, 1, 4, 5, 6, 9, and swaps $2 \leftrightarrow 8$, $3 \leftrightarrow 7$. So a cube ending in 3 came from a number ending in 7.

A.4 • Powers of 2, up to 2^{16}

n	0	1	2	3	4	5	6	7	8
2^n	1	2	4	8	16	32	64	128	256

n	9	10	11	12	13	14	15	16
2^n	512	1024	2048	4096	8192	16384	32768	65536

Also worth holding: $2^{20} = 1\ 048\ 576 \approx 10^6$, $2^{30} \approx 10^9$, $3^5 = 243$, $3^6 = 729$, $5^5 = 3125$, $7^4 = 2401$, $11^3 = 1331$.

A.5 • Fraction, percentage and decimal

Fraction	Percent	Decimal	Fraction	Percent	Decimal	Useful relatives
1/2	50%	0.5	1/12	81/3%	0.0833	5/12 = 412/3%, 7/12 = 581/3%
1/3	331/3%	0.3333	1/13	79/13%	0.0769	2/13 = 15.38%
1/4	25%	0.25	1/14	71/7%	0.0714	3/14 = 21.43%
1/5	20%	0.2	1/15	62/3%	0.0667	4/15 = 262/3%
1/6	162/3%	0.1667	1/16	6.25%	0.0625	3/16 = 18.75%, 5/16 = 31.25%
1/7	142/7%	0.1429	1/17	515/17%	0.0588	2/17 = 11.76%
1/8	12.5%	0.125	1/18	55/9%	0.0556	5/18 = 277/9%
1/9	111/9%	0.1111	1/19	55/19%	0.0526	3/19 = 15.79%
1/10	10%	0.1	1/20	5%	0.05	3/20 = 15%, 7/20 = 35%
1/11	91/11%	0.0909	1/24	41/6%	0.0417	1/25 = 4%, 1/40 = 2.5%, 1/50 = 2%

The ones that pay for themselves every single test: $2/3 = 662/3\%$, $3/4 = 75\%$, $2/5 = 40\%$, $3/5 = 60\%$, $4/5 = 80\%$, $3/8 = 37.5\%$, $5/8 = 62.5\%$, $7/8 = 87.5\%$, $5/6 = 831/3\%$.

A.6 • Square roots, 1 to 30

n	1	2	3	4	5	6	7	8	9	10
$\sqrt{n}$	1	1.414	1.732	2	2.236	2.449	2.646	2.828	3	3.162
n	11	12	13	14	15	16	17	18	19	20
$\sqrt{n}$	3.317	3.464	3.606	3.742	3.873	4	4.123	4.243	4.359	4.472
n	21	22	23	24	25	26	27	28	29	30
$\sqrt{n}$	4.583	4.69	4.796	4.899	5	5.099	5.196	5.292	5.385	5.477

The four you will actually use: $\sqrt{2} = 1.414$, $\sqrt{3} = 1.732$, $\sqrt{5} = 2.236$, $\sqrt{10} = 3.162$. And $1/\sqrt{2} = 0.707$, $1/\sqrt{3} = 0.577$, $\pi = 3.1416$, $22/7 = 3.1429$.

A.7 · HCF and LCM shortcuts

Product rule	For two numbers only: $\text{HCF} \times \text{LCM} = a \times b$. Find one, divide to get the other.
Euclid	$\text{HCF}(a, b) = \text{HCF}(b, a \bmod b)$. Keep replacing the bigger by the remainder until the remainder is 0. $\text{HCF}(84, 36) \rightarrow \text{HCF}(36, 12) \rightarrow \text{HCF}(12, 0) = 12$.
Coprime pairs	Two consecutive numbers are always coprime, so $\text{LCM}(n, n + 1) = n(n + 1)$ and $\text{HCF} = 1$. Same for two consecutive odd numbers.
Fractions	$\text{HCF} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$, $\text{LCM} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$.
Same remainder	Largest number that divides a, b, c leaving the same remainder $= \text{HCF}(b - a, c - b, c - a)$.
Given remainders	Largest number dividing a and b leaving $r_1, r_2 = \text{HCF}(a - r_1, b - r_2)$.
Common remainder r	Smallest number leaving remainder r with each of $a, b, c = \text{LCM}(a, b, c) + r$.
Short of a multiple	Leaves $a - k, b - k, c - k$ (same shortfall k) $= \text{LCM}(a, b, c) - k$.
Ratio form	If $\text{HCF} = h$, write the numbers as hx and hy with x, y coprime. Then $\text{LCM} = hxy$.

LCMs worth memorising

First n numbers	$\text{LCM}(1..5) = 60$, $\text{LCM}(1..6) = 60$, $\text{LCM}(1..8) = 840$, $\text{LCM}(1..10) = 2520$, $\text{LCM}(1..12) = 27720$
Clock / time sets	$\text{LCM}(2, 3, 4, 6) = 12$, $\text{LCM}(4, 6, 8, 12) = 24$, $\text{LCM}(6, 8, 9, 12) = 72$, $\text{LCM}(9, 12, 15) = 180$
Common pairs	$\text{LCM}(12, 18) = 36$, $(15, 20) = 60$, $(14, 21) = 42$, $(16, 24) = 48$, $(25, 30) = 150$, $(35, 49) = 245$

A.8 · Mental-math techniques — shown, not explained

$\times 5$	$68 \times 5 = \frac{68}{2} \times 10 = 34 \times 10 = 340$
$\times 25$	$68 \times 25 = \frac{68}{4} \times 100 = 17 \times 100 = 1700$
$\times 50$	$46 \times 50 = \frac{46}{2} \times 100 = 2300$
$\times 125$	$48 \times 125 = \frac{48}{8} \times 1000 = 6 \times 1000 = 6000$
$\div 5$	$340 \div 5 = 340 \times 2 \div 10 = 680 \div 10 = 68$
$\div 25$	$1700 \div 25 = 1700 \times 4 \div 100 = 6800 \div 100 = 68$
$\times 9$	$47 \times 9 = 470 - 47 = 423$
$\times 99$	$47 \times 99 = 4700 - 47 = 4653$
$\times 11$	47×11 : write 4 (4 + 7) 7 = 4 11 7 $\rightarrow$ 517 (carry the 1)
	$36 \times 11 = 3 (3 + 6) 6 = 396$
$\times 12$	$43 \times 12 = 430 + 86 = 516$ (ten times it, plus twice it)
$\times 15$	$28 \times 15 = 280 + 140 = 420$ (ten times it, plus half of that)
Squares of 5-enders	$65^2 = (6 \times 7) 25 = 4225$
Near 50	$53^2 = (25 + 3) 3^2 = 2809$. $47^2 = (25 - 3) 3^2 = 2209$.
Near 100	96×93 : $96 - 7 = 89$, $4 \times 7 = 28 \rightarrow 8928$
	108×104 : $108 + 4 = 112$, $8 \times 4 = 32 \rightarrow 11232$
Difference of squares	$48 \times 52 = 50^2 - 2^2 = 2500 - 4 = 2496$
	$37 \times 43 = 40^2 - 3^2 = 1600 - 9 = 1591$
Add by rounding	$297 + 458 = 300 + 458 - 3 = 755$
Subtract by rounding	$724 - 298 = 724 - 300 + 2 = 426$
Percent swap	8% of 25 = 25% of 8 = 2. 16% of 75 = 75% of 16 = 12.
Build a percent	15% of 840: 10% = 84, half of that = 42, total 126.
	35% of 60: 10% = 6, so 30% = 18, 5% = 3, total 21.

Percent of a percent	20% of 30% of $500 = 0.2 \times 0.3 \times 500 = 30$
Divide by 8	$1432 \div 8$: halve three times $\rightarrow 716 \rightarrow 358 \rightarrow 179$
Fraction first	$(\frac{3}{8}) \times 216 = 216 \div 8 \times 3 = 27 \times 3 = 81$ (never 216×3 first)
Check: digit sum	$427 \times 13 = 5551$? Digit sums: $4 \times 4 = 16 \rightarrow 7$; answer $5 + 5 + 5 + 1 = 16 \rightarrow 7$. Passes.
Check: last digit	37×48 must end in $7 \times 8 = 56 \rightarrow 6$. Any option not ending in 6 is dead.
Check: size	$37 \times 48 \approx 40 \times 50 = 2000$. The true value 1776 is near it. An option of 776 or 17760 is dead.

SHORTCUT

The two-second option killer. Before you compute anything, look at the options. Use the **last digit** and the **rough size**. On a service-company paper this alone answers roughly one question in six with no working at all.

Master Formula Index

Every formula in the book. The number on the left is the chapter.

How to revise from this page. Cover the right column. Read the chapter title. Write out every formula you remember. Uncover and compare. Whatever you missed is the chapter you re-open tonight.

Part I • Quantitative Aptitude

Chapter 1 • Numbers & Number System

- 1.1 Divisibility by 4: last 2 digits; by 8: last 3 digits; by 25: last 2 digits
- 1.2 Divisibility by 3 or 9: digit sum divisible by 3 or 9
- 1.3 Divisibility by 11: (sum of odd-place digits) – (sum of even-place digits) divisible by 11
- 1.4 Divisibility by 7: drop the last digit, subtract twice it, repeat
- 1.5 If $N = p^a q^b r^c$: number of factors = $(a+1)(b+1)(c+1)$
- 1.6 Sum of factors = $\left(\frac{p^{a+1}-1}{p-1}\right)\left(\frac{q^{b+1}-1}{q-1}\right)\left(\frac{r^{c+1}-1}{r-1}\right)$
- 1.7 Product of all factors = $N^{d/2}$, d = number of factors
- 1.8 Odd factors: drop the power of 2 first. Even factors = total – odd
- 1.9 $\text{HCF} \times \text{LCM} = a \times b$ (two numbers only)
- 1.10 $\text{HCF of fractions} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$; $\text{LCM of fractions} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$
- 1.11 Same remainder from a, b, c : $\text{HCF}(b-a, c-b, c-a)$
- 1.12 Remainders r_1, r_2 from a, b : $\text{HCF}(a-r_1, b-r_2)$
- 1.13 Smallest number leaving remainder r with a, b, c : $\text{LCM}(a, b, c) + r$
- 1.14 $(a \times b) \bmod n = ((a \bmod n) \times (b \bmod n)) \bmod n$
- 1.15 Fermat: p prime, $p \nmid a \rightarrow a^{p-1} \equiv 1 \pmod{p}$
- 1.16 Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ when $\text{HCF}(a, n) = 1$; $\varphi(n) = n \prod \left(1 - \frac{1}{p}\right)$
- 1.17 Unit digit: cycle length 4 for 2, 3, 7, 8; 2 for 4 and 9; 1 for 0, 1, 5, 6
- 1.18 Unit digit rule: power mod 4; if the remainder is 0 use the 4th term of the cycle
- 1.19 Last two digits: $76^k \equiv 76 \pmod{100}$; $2^{20} \equiv 76$; base ending in 1 $\rightarrow$ tens digit = (tens of base $\times$ unit of power) mod 10
- 1.20 Highest power of prime p in $n!$ = $\left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \dots$
- 1.21 Trailing zeros of $n!$ = $\left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{25} \right\rfloor + \left\lfloor \frac{n}{125} \right\rfloor + \dots$
- 1.22 Composite $m = p^\alpha q^\beta$ in $n!$: find each prime power, divide by its exponent, take the minimum
- 1.23 Base b : $(d_k \dots d_0)_b = d_k b^k + \dots + d_0$; convert by repeated division, read remainders bottom-up
- 1.24 Count of $1 \dots N$ divisible by a or b = $\left\lfloor \frac{N}{a} \right\rfloor + \left\lfloor \frac{N}{b} \right\rfloor - \left\lfloor \frac{N}{\text{LCM}(a, b)} \right\rfloor$

Chapter 2 • Percentages

- 2.1 $x\%$ of $N = (x/100)N$; A as a percent of $B = (A/B) \times 100$
- 2.2 Percentage change = $\frac{\text{new} - \text{old}}{\text{old}} \times 100$ — the **old** value is always the base
- 2.3 Increase $x\% \rightarrow$ multiply by $(1 + x/100)$; decrease $\rightarrow (1 - x/100)$
- 2.4 Two changes in a row: multiply the multipliers. Net = $a + b + (ab)/100$
- 2.5 A is $x\%$ more than $B \rightarrow B$ is $\frac{100x}{100+x}\%$ less than A
- 2.6 A is $x\%$ less than $B \rightarrow B$ is $\frac{100x}{100-x}\%$ more than A
- 2.7 Product constancy: price up $x\% \rightarrow$ quantity must fall $\frac{100x}{100+x}\%$
- 2.8 Growth for n years at $r\%$: $P(1 + r/100)^n$; depreciation: $P(1 - r/100)^n$
- 2.9 % error = $\frac{|\text{wrong} - \text{correct}|}{\text{correct}} \times 100$
- 2.10 Length $x\%$ too large $\rightarrow$ area $((1 + x/100)^2 - 1) \times 100\%$ too large; volume uses the cube
- 2.11 Savings = Income – Expenditure; set income = 100 to kill the fractions
- 2.12 Weighted percent: fraction w changes $a\%$, rest changes $b\% \rightarrow$ whole changes $wa + (1-w)b$
- 2.13 $n(A \text{ or } B) = n(A) + n(B) - n(A \text{ and } B)$; neither = $100\% - n(A \text{ or } B)$

- 2.14 Key conversions: $1/8 = 12.5\%$, $1/6 = 16\frac{2}{3}\%$, $1/3 = 33\frac{1}{3}\%$, $3/8 = 37.5\%$, $5/8 = 62.5\%$, $2/3 = 66\frac{2}{3}\%$

Chapter 3 • Ratio, Proportion & Partnership

- 3.1 Write $a : b$ as ak and bk — one unknown, not two
 3.2 Split N in $a : b : c$: one part $= \frac{N}{a+b+c}$, first share $= N \frac{a}{a+b+c}$
 3.3 Chain: $a : b = p : q$ and $b : c = r : s \rightarrow a : b : c = pr : qr : qs$
 3.4 Compounded ratio of $(a : b)$ and $(c : d) = ac : bd$
 3.5 Duplicate $a^2 : b^2$; triplicate $a^3 : b^3$; sub-duplicate $\sqrt{a} : \sqrt{b}$; sub-triplicate $\sqrt[3]{a} : \sqrt[3]{b}$
 3.6 $a : b = c : d \rightarrow ad = bc$ (product of extremes = product of means)
 3.7 Fourth proportional to a, b, c : $x = (bc)/a$
 3.8 Third proportional to a, b : $x = b^2/a$; mean proportional of a, b : $x = \sqrt{ab}$
 3.9 Continued proportion $a : b = b : c \rightarrow b^2 = ac$; if $a : b = b : c = c : d = k$ then $a/d = k^3$
 3.10 Componendo-dividendo: $\frac{a}{b} = \frac{c}{d} \rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$
 3.11 If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = k$ then $\frac{a+c+e}{b+d+f} = k$
 3.12 Direct variation: $x_1/y_1 = x_2/y_2$. Inverse: $x_1y_1 = x_2y_2$. Joint: $x = kyz/w$
 3.13 Partnership, same time: profit ratio = capital ratio
 3.14 Partnership, different times: profit ratio = capital $\times$ time (capital-months)
 3.15 Capital changed mid-year: add the pieces, $C_1t_1 + C_2t_2$
 3.16 Working partner: remove the salary or commission **first**, then split the rest by capital

Chapter 4 • Averages, Mixtures & Alligation

- 4.1 Average $= \frac{\text{Sum}}{\text{Count}}$; Sum = Average $\times$ Count — always convert to a sum first
 4.2 First n naturals: $(n+1)/2$. First n odds: n . First n evens: $n+1$
 4.3 Evenly spaced set: average = (first + last)/2
 4.4 Replace a by b among n values: change in average $= (b-a)/n$
 4.5 New member joins: new average $= \frac{nA+x}{n+1}$; if the average rises by d , then $x = A + (n+1)d$
 4.6 Weighted average $= \frac{n_1A_1+n_2A_2}{n_1+n_2}$
 4.7 Alligation: $\frac{\text{cheaper}}{\text{dearer}} = \frac{D}{M-C}$
 4.8 Repeated replacement, n times: pure left $= x(1-y/x)^n$
 4.9 Average speed $= \frac{\text{total distance}}{\text{total time}}$; equal distances at u, v : $2uv/(u+v)$
 4.10 Three equal distances at u, v, w : $3/(1/u + 1/v + 1/w)$
 4.11 Median is not pulled by extreme values; the mean is

Chapter 5 • Profit, Loss & Discount

- 5.1 Profit = SP – CP; Profit% $= \frac{\text{Profit}}{\text{CP}} \times 100$ (always on CP)
 5.2 SP = CP $\frac{100+p}{100}$; CP = SP $\times \frac{100}{100+p}$; use $-l$ for a loss
 5.3 Discount% $= \frac{\text{MP} - \text{SP}}{\text{MP}} \times 100$ (always on MP); SP $= \frac{\text{MP}(100-d)}{100}$
 5.4 Successive discounts: SP = MP $(1 - d_1/100)(1 - d_2/100)\dots$; two equal to $d_1 + d_2 - d_1d_2/100$
 5.5 Mark up $m\%$ then discount $d\%$: Profit% $= m - d - (md)/100$
 5.6 MP $(100 - d) = \text{CP}(100 + p)$; markup needed $= \text{MP}/\text{CP} = (100 + p)/(100 - d)$
 5.7 Same SP, one at $+x\%$ and one at $-x\%$: always a loss of $x^2/100$ per cent
 5.8 Mark up $m\%$ then discount $m\%$: always a loss of $m^2/100$ per cent
 5.9 CP of a articles = SP of b articles $\rightarrow$ Profit% $= (a-b)/b \times 100$
 5.10 False weight at cost price: Gain% $= \frac{\text{true} - \text{given}}{\text{given}} \times 100$
 5.11 False weight with markup: SP/CP $= \frac{100+m}{100} \times \frac{\text{true}}{\text{given}}$
 5.12 Profit as a share of SP $= \frac{p}{100+p} \times 100$ per cent
 5.13 Break-even units = fixed cost/ c , where c = SP per unit – variable cost per unit

Chapter 6 • Simple & Compound Interest

- 6.1 SI = $(PRT)/100$; $A = P(1 + RT/100)$
 6.2 $P = (100 \text{ SI})/(RT)$, $R = (100 \text{ SI})/(PT)$, $T = (100 \text{ SI})/(PR)$
 6.3 $A = P(1 + R/100)^n$; CI = $P[(1 + R/100)^n - 1]$

- 6.4 Half-yearly: $P(1 + R/200)^{2n}$; quarterly: $P(1 + R/400)^{4n}$; monthly: $P(1 + R/1200)^{12n}$
 6.5 $\text{EAR} = [(1 + R/(100k))^k - 1] \times 100$ per cent, k compoundings a year
 6.6 $\text{CI} - \text{SI}$ over 2 years = $P(R/100)^2$; over 3 years = $PR^2(300 + R)/10^6$
 6.7 2-year ratio $\text{CI}/\text{SI} = 1 + R/200$
 6.8 SI doubles when $RT = 100$; doubles in $T \rightarrow m$ times in $(m - 1)T$
 6.9 CI : m times in $T \rightarrow m^2$ times in $2T \rightarrow m^3$ times in $3T$
 6.10 Rule of 72: doubling time at $\text{CI} \approx 72/R$ years
 6.11 Equal instalments at CI : $P = x/v + x/v^2 + \dots + x/v^n$ with $v = 1 + R/100$
 6.12 $\text{EMI} = Pi(1 + i)^n / ((1 + i)^n - 1)$, $i = R/1200$, n months
 6.13 Flat-rate loan: $\text{EMI} = (P + \text{total interest})/\text{months}$; true rate $\approx$ twice the flat rate
 6.14 Real return: divide, never subtract — $(1 + R/100)/(1 + f/100)$

Chapter 7 • Time, Speed & Distance

- 7.1 speed = distance/time
 7.2 1 km/h = $5/18$ m/s; 1 m/s = $18/5$ km/h
 7.3 Fixed distance: speeds in ratio $a : b \rightarrow$ times in ratio $b : a$
 7.4 Average speed = total distance/total time; equal distances at u, v : $2uv/(u + v)$
 7.5 Three equal distances: $3uvw/(uv + vw + wu)$; equal times: $(u + v)/2$
 7.6 Late/early: $d = \frac{s_1 s_2}{s_2 - s_1} \times \Delta t$
 7.7 Relative speed: opposite $u + v$; same direction $|u - v|$
 7.8 Train past a pole: L/s . Past a platform: $(L + P)/s$. Two trains: $(L_1 + L_2)/\text{rel. speed}$
 7.9 Boats: down = $b + c$, up = $b - c$, $b = (\text{down} + \text{up})/2$, $c = (\text{down} - \text{up})/2$
 7.10 “A beats B by x m” $\rightarrow$ when A finishes, B has run (race $- x$) m
 7.11 Circular track, same direction: first meeting after $C/(u - v)$; opposite: $C/(u + v)$
 7.12 Distinct meeting points, speed ratio $a : b$ in lowest terms: $a + b$ opposite, $a - b$ same direction
 7.13 Meet then continue for t_1, t_2 more hours: time to meet = $\sqrt{t_1 t_2}$, speeds = $\sqrt{t_2} : \sqrt{t_1}$

Chapter 8 • Time & Work

- 8.1 Job in a days $\rightarrow$ one day's work = $1/a$. Add rates, never days
 8.2 Together: $\frac{ab}{a+b}$ days. Three: $\frac{abc}{ab+bc+ca}$ days
 8.3 LCM method: total work = LCM of the days, so every rate is a whole number
 8.4 Times in ratio $a : b \rightarrow$ efficiencies in ratio $b : a$
 8.5 A is $p\%$ more efficient than B $\rightarrow$ A's time = B's time $\div (1 + p/100)$
 8.6 $(M_1 D_1 H_1)/W_1 = (M_2 D_2 H_2)/W_2$
 8.7 Wages split in the ratio of work actually done
 8.8 Pipes: net rate = $\sum$ filling $- \sum$ emptying; fill time = $1 \div$ net rate
 8.9 Leak: pipe alone a h, with leak b h $\rightarrow$ leak empties alone in $\frac{ab}{b-a}$ hours
 8.10 Alternate days: compute one full 2-day block, count whole blocks, finish day by day
 8.11 Pairs p, q, r : $1/p + 1/q + 1/r = 2 \times$ (rate of all three)

Chapter 9 • Permutations & Combinations

- 9.1 AND $\rightarrow$ multiply. OR $\rightarrow$ add (cases must not overlap)
 9.2 ${}^n P_r = n!/(n - r)!$; ${}^n C_r = n!/(r!(n - r)!) = {}^n P_r/r!$
 9.3 ${}^n C_r = {}^n C_{n-r}$; ${}^n C_2 = n(n - 1)/2$; ${}^n C_3 = n(n - 1)(n - 2)/6$
 9.4 Pascal: ${}^n C_r + {}^n C_{r-1} = {}^{n+1} C_r$; $\sum_r {}^n C_r = 2^n$
 9.5 Subsets of an n -set = 2^n ; at least one item = $2^n - 1$
 9.6 Repetition allowed: r slots, n choices each $\rightarrow n^r$
 9.7 Alike objects in a row: $n!/(p!q!r! \dots)$
 9.8 Circular: $(n - 1)!$; necklace / garland: $(n - 1)!/2$
 9.9 Together $\rightarrow$ glue the group, multiply by its internal arrangements
 9.10 No two together $\rightarrow$ arrange the rest, then use the gaps; r from n in a row: ${}^{n-r+1} C_r$
 9.11 $x_1 + \dots + x_r = n$, each ≥ 1 : ${}^{n-1} C_{r-1}$; each ≥ 0 : ${}^{n+r-1} C_{r-1}$
 9.12 Distinct objects into distinct boxes: r^n ; into exactly 3, none empty: $3^n - 3 \cdot 2^n + 3$
 9.13 mn objects into m groups of n : labelled $(mn)!/(n!)^m$; unlabelled $(mn)!/((n!)^m m!)$
 9.14 $D_n = n!(1 - 1/1! + 1/2! - \dots)$; $D_2 = 1, D_3 = 2, D_4 = 9, D_5 = 44, D_6 = 265$

- 9.15 Exactly k in the right place: ${}^nC_k \cdot D_{n-k}$
 9.16 From n points (no 3 collinear): lines nC_2 , triangles nC_3
 9.17 Diagonals of an n -gon = $n(n-3)/2$
 9.18 Rectangles in an $a \times b$ grid of cells = ${}^{a+1}C_2 \times {}^{b+1}C_2$
 9.19 Grid paths, m east and n north: $(m+n)!/(m!n!)$
 9.20 Sum of all n -digit numbers from n distinct non-zero digits = $(n-1)! \times (\text{digit sum}) \times \underbrace{11\dots1}_n$

Chapter 10 • Probability

- 10.1 $P(E)$ = favourable/total, and $0 \leq P(E) \leq 1$
 10.2 $P(\text{not } E) = 1 - P(E)$ — use this for every “at least one”
 10.3 Odds in favour $a : b \rightarrow P = a/(a+b)$; odds against $a : b \rightarrow P = b/(a+b)$
 10.4 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 10.5 $P(A \cap B) = P(A) \cdot P(B | A)$; independent $\rightarrow P(A) \cdot P(B)$
 10.6 $P(A | B) = P(A \cap B)/P(B)$
 10.7 Total probability: $P(E) = \sum_i P(B_i)P(E | B_i)$
 10.8 Bayes: $P(B_i | E) = P(B_i)P(E | B_i)/\sum_j P(B_j)P(E | B_j)$
 10.9 $E[X] = \sum x_i p_i$; linearity $E[X+Y] = E[X] + E[Y]$ even when dependent
 10.10 Indicator trick: expected count = sum of the individual probabilities
 10.11 Binomial: $P(r) = {}^nC_r p^r q^{n-r}$; mean = np ; variance = npq
 10.12 Geometric: $P(\text{first success on try } k) = q^{k-1}p$; $E[\text{tries}] = 1/p$; memoryless
 10.13 Two dice: 36 outcomes; ways to make sums 2..12 are 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1
 10.14 Deck: 52 cards, 26 red, 4 suits of 13, 4 aces, 12 face cards, 6 red face cards
 10.15 n coins: 2^n outcomes; exactly r heads in nC_r ways
 10.16 $P(53 \text{ of a given weekday})$: ordinary year $1/7$, leap year $2/7$

Chapter 11 • Algebra Essentials

- 11.1 $(a \pm b)^2 = a^2 \pm 2ab + b^2$; $a^2 - b^2 = (a-b)(a+b)$
 11.2 $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$
 11.3 $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$
 11.4 $x + 1/x = k \rightarrow x^2 + 1/x^2 = k^2 - 2$, $x^3 + 1/x^3 = k^3 - 3k$
 11.5 $x - 1/x = k \rightarrow x^2 + 1/x^2 = k^2 + 2$, $x^3 - 1/x^3 = k^3 + 3k$
 11.6 Two linear equations: unique if $a_1/a_2 \neq b_1/b_2$; none if $a_1/a_2 = b_1/b_2 \neq c_1/c_2$
 11.7 $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$; $D = b^2 - 4ac$ decides the root type
 11.8 $\alpha + \beta = -b/a$, $\alpha\beta = c/a$; equation: $x^2 - (\text{sum})x + (\text{product}) = 0$
 11.9 $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$; $|\alpha - \beta| = \sqrt{D}/|a|$
 11.10 Extremum of $ax^2 + bx + c$ at $x = -b/(2a)$, value $c - b^2/(4a)$
 11.11 AP: $t_n = a + (n-1)d$; $S_n = n/2[2a + (n-1)d] = n/2(a+l)$
 11.12 GP: $t_n = ar^{n-1}$; $S_n = a(r^n - 1)/(r - 1)$; $S_\infty = a/(1-r)$ for $|r| < 1$
 11.13 HP: the reciprocals form an AP
 11.14 $\sum n = n(n+1)/2$; $\sum n^2 = n(n+1)(2n+1)/6$; $\sum n^3 = [n(n+1)/2]^2$
 11.15 $AM = (a+b)/2$, $GM = \sqrt{ab}$, $HM = 2ab/(a+b)$; $GM^2 = AM \times HM$; $AM \geq GM \geq HM$
 11.16 $a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, $a^{-n} = 1/a^n$, $a^{m/n} = \sqrt[n]{a^m}$
 11.17 Rationalise: $1/(\sqrt{a} - \sqrt{b}) = (\sqrt{a} + \sqrt{b})/(a-b)$
 11.18 $\log(mn) = \log m + \log n$; $\log(m/n) = \log m - \log n$; $\log(m^p) = p \log m$
 11.19 $\log_b a = (\log_c a)/(\log_c b) = 1/(\log_a b)$; digits in $N = \lfloor \log_{10} N \rfloor + 1$
 11.20 $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$, $\log_{10} 7 = 0.8451$
 11.21 $|x| < a \leftrightarrow -a < x < a$; $|x| > a \leftrightarrow x < -a$ or $x > a$
 11.22 For $x > 0$: $x + k/x \geq 2\sqrt{k}$, equality at $x = \sqrt{k}$
 11.23 $(f \circ g)(x) = f(g(x))$; remainder of $f(x)$ by $(x-a)$ is $f(a)$

Chapter 12 • Geometry & Mensuration

- 12.1 Straight line 180° , point 360° ; vertically opposite angles equal
 12.2 Triangle angle sum 180° ; exterior angle = sum of the two opposite interior angles
 12.3 Triangle inequality: $a + b > c$ for every pair

- 12.4 Pythagoras; triples $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, $(9, 40, 41)$, $(20, 21, 29)$
 12.5 Area $= 1/2 \times \text{base} \times \text{height}$; Heron $= \sqrt{s(s-a)(s-b)(s-c)}$, $s = (a+b+c)/2$
 12.6 Equilateral side a : area $= \sqrt{3}/4 a^2$, height $= \sqrt{3}/2 a$
 12.7 Similar triangles: sides $k \rightarrow$ perimeters k , areas k^2
 12.8 Centroid divides each median $2 : 1$ and makes 6 equal areas; midpoint triangle has $1/4$ the area
 12.9 Right triangle: inradius $r = (a+b-c)/2$, circumradius $R = c/2$
 12.10 Circle: $C = 2\pi r$, $A = \pi r^2$; chord: $r^2 = h^2 + (L/2)^2$
 12.11 Sector θ : arc $= \theta/360 \times 2\pi r$, area $= \theta/360 \times \pi r^2$
 12.12 Angle at centre $= 2 \times$ angle at circumference; angle in a semicircle $= 90^\circ$
 12.13 Tangent $\perp$ radius; two tangents from an outside point are equal
 12.14 Common tangents: direct $= \sqrt{d^2 - (r_1 - r_2)^2}$, transverse $= \sqrt{d^2 - (r_1 + r_2)^2}$
 12.15 Regular n -gon: interior sum $(n-2)180^\circ$, each exterior $360/n$, diagonals $n(n-3)/2$
 12.16 Square: a^2 , diagonal $a\sqrt{2}$. Rectangle: lb , diagonal $\sqrt{l^2 + b^2}$
 12.17 Rhombus $= 1/2 d_1 d_2$; trapezium $= 1/2(\text{sum of parallel sides}) \times h$
 12.18 Cube a^3 , TSA $6a^2$; cuboid lbh , TSA $2(lb + bh + hl)$
 12.19 Cylinder $\pi r^2 h$, CSA $2\pi r h$, TSA $2\pi r(r + h)$
 12.20 Cone $1/3 \pi r^2 h$, CSA $\pi r l$, TSA $\pi r(r + l)$, $l = \sqrt{r^2 + h^2}$
 12.21 Sphere $4/3 \pi r^3$, SA $4\pi r^2$; hemisphere $2/3 \pi r^3$, TSA $3\pi r^2$
 12.22 Frustum $V = 1/3 \pi h(R^2 + Rr + r^2)$; recasting keeps the volume, never the surface
 12.23 Distance $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$; midpoint $= ((x_1 + x_2)/2, (y_1 + y_2)/2)$
 12.24 Section formula $m : n$: $((mx_2 + nx_1)/(m+n), (my_2 + ny_1)/(m+n))$
 12.25 Slope $= (y_2 - y_1)/(x_2 - x_1)$; parallel $m_1 = m_2$; perpendicular $m_1 m_2 = -1$
 12.26 Triangle area $= 1/2 |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$
 12.27 Point to line: $|ax_0 + by_0 + c|/\sqrt{a^2 + b^2}$
 12.28 $\sin^2 \theta + \cos^2 \theta = 1$; $\tan \theta = \sin \theta / \cos \theta$
 12.29 $\sin 30 = 1/2$, $\sin 45 = 1/\sqrt{2}$, $\sin 60 = \sqrt{3}/2$; $\tan 30 = 1/\sqrt{3}$, $\tan 45 = 1$, $\tan 60 = \sqrt{3}$
 12.30 $\tan(\text{elevation}) = \text{height}/\text{horizontal distance}$; depression from the top $=$ elevation from the bottom

Part II • Logical Reasoning

Chapter 13 • Series

- 13.1 Order of attack: differences $\rightarrow$ second differences $\rightarrow$ ratios $\rightarrow n^2, n^3, n!$ $\rightarrow$ alternating or interleaved
 13.2 AP: $t_n = a + (n-1)d$, $S_n = n/2[2a + (n-1)d]$
 13.3 GP: $t_n = ar^{n-1}$, $S_n = a(r^n - 1)/(r - 1)$, $S_\infty = a/(1 - r)$ for $|r| < 1$
 13.4 $\sum n = n(n+1)/2$; $\sum n^2 = n(n+1)(2n+1)/6$; $\sum n^3 = [n(n+1)/2]^2$
 13.5 Triangular $T_k = k(k+1)/2$; $\sum_{k=1}^n T_k = n(n+1)(n+2)/6$
 13.6 $\times k \pm c$ pattern: 5, 11, 23, 47 is $\times 2 + 1$
 13.7 Growing multiplier: 3, 6, 18, 72 is $\times 2, \times 3, \times 4$
 13.8 Squares $\pm c$: 2, 5, 10, 17, 26 is $n^2 + 1$. Cubes $\pm c$: 2, 9, 28, 65 is $n^3 + 1$
 13.9 Products $n(n+1)$: 2, 6, 12, 20, 30. Factorials: 1, 2, 6, 24, 120
 13.10 Fibonacci type: each term is the sum of the two before it
 13.11 Interleaved: split the odd positions from the even positions and solve each separately
 13.12 Letters: A = 1 ... Z = 26; EJOTY anchors E5, J10, O15, T20, Y25; wrap past Z back to A

Chapter 14 • Coding--Decoding & Cryptarithmic

- 14.1 Position from the right $= 27 - (\text{position from the left})$
 14.2 Opposite letter of position n is position $27 - n$ (A $\leftrightarrow$ Z, M $\leftrightarrow$ N)
 14.3 Shift with wrap: new position $= ((p + k - 1) \bmod 26) + 1$
 14.4 Six coding types: letter shift, word pattern, number coding, symbol coding, substitution, conditional
 14.5 Symbol coding: translate every symbol first, then apply BODMAS
 14.6 Substitution coding: match two sentences and take the code common to both for the common word
 14.7 Cryptarithmic: one digit per letter, no leading zero, start at the units column
 14.8 Carry out of a 2-number column is 0 or 1; out of a 3-number column it is 0, 1 or 2
 14.9 $\overline{AB} + \overline{BA} = 11(A + B)$; $\overline{AB} - \overline{BA} = 9(A - B)$
 14.10 $\overline{ABA} + \overline{BAB} = 111(A + B)$; $\overline{ABC} + \overline{CBA} = 101(A + C) + 20B$
 14.11 $\overline{AA} = 11A$, $\overline{AAA} = 111A$, $\overline{AAAA} = 1111A$

14.12 $111 = 3 \times 37$, $1111 = 11 \times 101$, $11111 = 41 \times 271$

Chapter 15 • Blood Relations, Directions & Ranking

- 15.1 Draw the tree: + male, – female, = married, vertical line to children
- 15.2 Same level $\rightarrow$ siblings, cousins or spouses. Never parent and child
- 15.3 Coded relations: solve one symbol at a time, left to right, writing each fact down
- 15.4 Compass clockwise: N $\rightarrow$ NE $\rightarrow$ E $\rightarrow$ SE $\rightarrow$ S $\rightarrow$ SW $\rightarrow$ W $\rightarrow$ NW
- 15.5 Left turn = 90° anticlockwise; right turn = 90° clockwise; two same-side turns = about turn
- 15.6 Angles from North, clockwise: E 90° , S 180° , W 270° , NE 45° , SE 135° , SW 225° , NW 315°
- 15.7 Net displacement = $\sqrt{h^2 + v^2}$, v = North – South, h = East – West
- 15.8 If $|h| = |v|$ the distance is $h\sqrt{2}$
- 15.9 Shadows: just after sunrise they point West; just before sunset they point East
- 15.10 Facing D : right hand points at D turned 90° clockwise, left hand 90° anticlockwise
- 15.11 Total = (rank from left) + (rank from right) – 1
- 15.12 People strictly between A and B = (B from left) – (A from left) – 1
- 15.13 “ x people between A and B ” $\rightarrow$ their positions differ by $x + 1$

Chapter 16 • Seating Arrangement & Puzzles

- 16.1 Row: number seats 1 to n from the West end. Circle: seat 1 at the top, number clockwise
- 16.2 All facing North: left = towards seat 1. All facing South: left = towards seat n
- 16.3 Facing the centre: left = clockwise. Facing outward: left = anticlockwise
- 16.4 Circle of even n : opposite seat k is $k + n/2$ (wrap). Odd n : nobody is opposite anybody
- 16.5 Square table of 8: corners and side-middles alternate, so corners are 1, 3, 5, 7
- 16.6 Floors and stacks: number the bottom 1. “Above” means a larger number
- 16.7 “Immediately left” = 1 step. “Second to the left” = 2 steps. “To the left of” = position not fixed
- 16.8 “Exactly x between A and B ” $\rightarrow$ seat numbers differ by $x + 1$
- 16.9 “Exactly in the middle” of n seats $\rightarrow$ seat $(n + 1)/2$, and only when n is odd
- 16.10 Double rows face each other; seat k of Row 1 faces seat k of Row 2; left and right flip
- 16.11 Clue order: fixed seats first, then chained clues, then two-case branches, then negative clues

Chapter 17 • Syllogisms & Logical Deduction

- 17.1 A conclusion follows only if it is true in **every** diagram that fits. One counter-diagram kills it
- 17.2 Four types: A (All S are P), E (No S is P), I (Some S are P), O (Some S are not P)
- 17.3 All S are P $\rightarrow$ Some P are S. Never “All P are S”
- 17.4 No S is P $\rightarrow$ No P is S. Some S are P $\rightarrow$ Some P are S. “Some S are not P” cannot be flipped
- 17.5 All + All $\rightarrow$ All. All + No $\rightarrow$ No. Some + All $\rightarrow$ Some. Some + No $\rightarrow$ Some ... not
- 17.6 No A is B + All B are C $\rightarrow$ Some C are not A
- 17.7 Nothing follows from: All + Some, Some + Some, No + No, or a first statement of the form “Some A are not B”
- 17.8 Two negatives give nothing; one negative forces a negative conclusion
- 17.9 Two “some” statements give nothing; one “some” forces a “some” conclusion
- 17.10 Either–or: only when neither follows alone **and** the two form a complementary pair on the same terms
- 17.11 Possibility: true unless the statements make it impossible — find one legal diagram
- 17.12 “Only a few A are B” and “Almost all A are B” both mean Some are and Some are not
- 17.13 “Only A are B” means All B are A. “At least some A are B” means Some A are B

Chapter 18 • Critical Reasoning

- 18.1 Split every argument into evidence, conclusion and the unstated gap
- 18.2 Conclusion markers: so, hence, therefore, thus, clearly, this shows, we must
- 18.3 Assumption: negate it — if the argument collapses, it is an assumption
- 18.4 Conclusion: must come from the given words alone, with no outside knowledge
- 18.5 Inference: probably true from the facts, without over-reaching beyond the data given
- 18.6 Strong argument = relevant + substantial + tied to the exact question asked
- 18.7 Course of action: does it hit the cause, is it practical, is it in the actor’s power?
- 18.8 Reject a course of action that is extreme, disproportionately costly, or treats only a symptom
- 18.9 Cause and effect: the event that came first and could stand alone is the cause

- 18.10 Four labels: I causes II; II causes I; both from a common cause; both independent
 18.11 Strengthen = close the gap or rule out a rival cause. Weaken = supply a rival cause or a counter-case
 18.12 Assumption sits **before** the statement; inference comes **after** it
 18.13 Growth rate $\times$ base = actual increase — a small rate on a big base can win
 18.14 An average can rise with nobody improving, if the low members left the group

Chapter 19 • Analogy, Classification & Non-Verbal

- 19.1 EJOTY anchors: E5, J10, O15, T20, Y25; A1, M13, N14, Z26
 19.2 Opposite letter: value + opposite value = 27
 19.3 Analogy order of tests: fixed shift $\rightarrow$ opposite letter $\rightarrow$ reversal $\rightarrow$ number patterns
 19.4 Number patterns to test: $n^2, n^3, n^2 \pm c, n^3 \pm c, n(n+1), n(n-1), n(n+1)/2$
 19.5 Mirror image: reverse the order of the letters and flip each one left-right
 19.6 Letters unchanged by a vertical mirror: A H I M O T U V W X Y. Digits: 0 and 8
 19.7 Mirror time = 11 : 60 — actual time (use 23 : 60 if the hour reads 12)
 19.8 Water image: the order stays; each letter flips top-bottom. Unchanged: B C D E H I K O X. Digits: 0, 3, 8
 19.9 Fold n times, punch k holes clear of the folds $\rightarrow k \times 2^n$ holes
 19.10 A punch exactly on a fold line gives half as many holes
 19.11 Painted cube side n : 3 faces 8 ; 2 faces $12(n-2)$; 1 face $6(n-2)^2$; 0 faces $(n-2)^3$
 19.12 Painted cuboid, $p = a-2, q = b-2, r = c-2$: 2 faces $4(p+q+r)$, 1 face $2(pq+qr+rp)$, 0 faces pqr
 19.13 Standard die: opposite faces add to 7
 19.14 Dice net: in a straight strip of four, the 1st and 3rd are opposite, and the 2nd and 4th
 19.15 Squares in an $n \times n$ grid = $n(n+1)(2n+1)/6$; for $n = 8$ it is 204
 19.16 Rectangles in an $m \times n$ grid = $\binom{m+1}{2} \binom{n+1}{2}$
 19.17 Triangle cut into n parts a side: totals 1, 5, 13, 27, 48, 78 for $n = 1 \dots 6$
 19.18 Triangle ABC with n points on BC joined to A : $\binom{n+2}{2}$ triangles

Chapter 20 • Clocks, Calendars & Classic Puzzles

- 20.1 Minute hand $6^\circ/\text{min}$; hour hand $0.5^\circ/\text{min}$; relative $5.5^\circ/\text{min}$
 20.2 Angle $\theta = |30H - 5.5M|$; if $\theta > 180^\circ$ use $360^\circ - \theta$
 20.3 Angle exactly θ between H and $H+1$: $M = 2/11(30H \pm \theta)$
 20.4 Hands together: $M = 60H/11$; gap between coincidences = $720/11 = 65\frac{5}{11}$ minutes
 20.5 In 12 hours: coincide 11 times, opposite 11 times, right angles 22 times
 20.6 Mirror of a clock time = 11 : 60 — time (use 23 : 60 when the hour reads 12)
 20.7 A clock gaining g min/hour shows $60 + g$ clock-minutes per 60 true minutes
 20.8 A clock drifting d min/day is right again after $720 \div d$ days
 20.9 Ordinary year = 1 odd day; leap year = 2 odd days
 20.10 Leap year: divisible by 4, but a century year must be divisible by 400
 20.11 100 yr = 5 odd days, 200 = 3, 300 = 1, 400 = 0; the calendar repeats every 400 years
 20.12 Day codes: 0 Sunday, 1 Monday, ... 6 Saturday
 20.13 Month running totals (ordinary year): 31, 59, 90, 120, 151, 181, 212, 243, 273, 304, 334, 365
 20.14 Day of a date = (century odd days + year odd days + day-of-year) mod 7
 20.15 An ordinary year's calendar repeats after 6 or 11 years; a leap year's after 28
 20.16 $P(53 \text{ Mondays})$: ordinary $1/7$, leap $2/7$, a random year $71/400$
 20.17 Ages: the difference of two ages never changes — use it to check
 20.18 Weights on one pan: powers of 2 reach every whole number to $2^n - 1$
 20.19 Weights on both pans: powers of 3 reach every whole number to $(3^n - 1)/2$
 20.20 Odd coin, direction known: $\lceil \log_3 N \rceil$ weighings. Unknown direction, 12 coins: 3 weighings
 20.21 Bridge and torch: move the two slowest together; cost = $\min(f_1 + 2f_2 + b, 2f_1 + a + b)$
 20.22 Two jugs p, q measure v iff $\text{gcd}(p, q)$ divides v and $v \leq \max(p, q)$
 20.23 Matchsticks: a row of n squares needs $3n + 1$; an $n \times n$ grid needs $2n(n+1)$

Part III • Data Interpretation & Test Craft

Chapter 21 • DI: Tables, Bar, Line, Pie

- 21.1 Only four operations exist in DI: add a row, take a percent, take a ratio, compare two numbers

- 21.2 % change = $(\text{new} - \text{old})/\text{old} \times 100$ — the old value is always the base
- 21.3 share of $A = A/\text{total} \times 100$
- 21.4 Pie: value = $(\text{percent}/100) \times \text{total}$; central angle = $\text{percent} \times 3.6$ degrees
- 21.5 Pie: percent = $\text{angle}/3.6$; the full circle is $360^\circ = 100\%$
- 21.6 A line of **growth rates** still means the value is rising while the line sits above zero
- 21.7 Linking charts: part multiplier = $(s_2/s_1) \times M$ — a part can grow while its share falls
- 21.8 Weighted growth: total growth = $w_1g_1 + w_2g_2 + \dots$ with shares as fractions
- 21.9 Average **rate** = total value/total quantity, never the plain average of the rates
- 21.10 Index numbers divide exactly like the real values do
- 21.11 Compare a/b and c/d by cross-multiplying: ad versus cb
- 21.12 Approximation: round the answer, never the base, and only when the options are far apart

Chapter 22 • DI: Caselets, Mixed & Missing Data

- 22.1 Caselet method: grand total $\rightarrow$ tree or table $\rightarrow$ absolute numbers $\rightarrow$ check line $\rightarrow$ then read the questions
- 22.2 “30% of the rest” takes a **new** base — write down which box the percent sits on
- 22.3 $n(A \cup B) = n(A) + n(B) - n(A \cap B)$; neither = total $- n(A \cup B)$
- 22.4 only $A = n(A) - n(A \cap B)$; exactly one = $n(A \cup B) - n(A \cap B)$
- 22.5 $n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$
- 22.6 exactly two = $\sum [n(A \cap B) - x]$ over the three pairs, with $x = n(A \cap B \cap C)$
- 22.7 only $A = n(A) - n(A \cap B) - n(A \cap C) + x$
- 22.8 Largest $n(A \cap B) = \min(n(A), n(B))$; smallest = $n(A) + n(B) - \text{total}$, or 0 if negative
- 22.9 Smallest $n(A \cap B \cap C) = n(A) + n(B) + n(C) - 2 \times \text{total}$, or 0 if negative
- 22.10 Missing grid: find a row or column with exactly one blank, fill it, repeat. Cross-check both ways
- 22.11 “Average of n items is a ” means sum = na — convert every average to a sum first
- 22.12 Radar: read each axis separately; the total is the sum of the axis values, never the area
- 22.13 Bubble: three numbers per bubble — x , y and size. A big bubble is not a big x
- 22.14 Combined percent change = $(w_1g_1 + w_2g_2)/(w_1 + w_2)$ with w the **old** sizes
- 22.15 Split T in $a : b : c$: one part = $T/(a + b + c)$, then multiply

Chapter 23 • Data Sufficiency

- 23.1 The question is never “what is the answer” — it is “is this enough for exactly one answer?”
- 23.2 (a) I alone; (b) II alone; (c) both together, neither alone; (d) each alone; (e) not even together
- 23.3 AD/BCE grid: test I alone. Works $\rightarrow$ (a) or (d). Fails $\rightarrow$ (b), (c) or (e)
- 23.4 Combine the statements only after both have failed on their own
- 23.5 A value question is settled only when the unknown is pinned to one number
- 23.6 A yes/no question is settled by a definite **no** just as much as by a definite yes
- 23.7 Two possible values $\rightarrow$ not sufficient. $x^2 = 49$ gives $x = 7$ or -7
- 23.8 k unknowns need k **independent** linear equations
- 23.9 $2x + 3y = 12$ and $4x + 6y = 24$ are one equation, not two
- 23.10 Non-linear pairs break the count: $xy = 12$ with $x + y = 7$ gives two answer pairs
- 23.11 For a ratio, percentage or fraction answer, absolute values are usually not needed
- 23.12 Both statements are always true and can never contradict each other
- 23.13 Unless stated otherwise, a “number” may be negative, zero or a fraction

Chapter 24 • Full-Length Mock Papers

- 24.1 Seconds per question = $(\text{minutes} \times 60) \div \text{questions}$
- 24.2 Accuracy = correct $\div$ attempted. Score rate = correct $\div$ total
- 24.3 Value of a guess = $p \times 1 - (1 - p) \times m$ with $-m$ for a wrong answer
- 24.4 A guess pays when $p > m/(1 + m)$: $m = 0.25$ needs $p > 0.2$; $m = 1/3$ needs $p > 0.25$
- 24.5 Blind guess: 4 options $p = 0.25$; 5 options $p = 0.20$
- 24.6 Marks still needed = target — banked, divided by the marks per remaining question
- 24.7 Three-pass rule: take the quick ones, return to the flagged ones, then guess what is left
- 24.8 Guess freely with no negative marking; with negative marking only after eliminating two options

THE LAST WORD

A formula you can quote but cannot use is worth nothing on test day. The ones that will actually earn you marks are the ones you have used in at least five solved problems.

If a line on this page looks unfamiliar, do not memorise it. Go back to that chapter and work three examples that use it. Then come back. That is the whole method.